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Department of Defense Handbook Reliability Growth Management

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FOREWORD

1. This handbook is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense..

2. The government's materiel acquisition process for new military systems requiring development is invariably complex and difficult for many reasons. Generally, these systems require new technologies and represent a challenge to the state of the art. Moreover, the requirements for reliability, maintainability and other performance parameters are usually highly demanding. Consequently, striving to meet these requirements represents a significant portion of the entire acquisition process and, as a result, the setting of priorities and the allocation and reallocation of resources such as funds, manpower and time are often formidable management tasks.

3. Reliability growth management procedures have been developed for addressing the above problem. These techniques will enable the manager to plan, evaluate and control the reliability of a system during its development stage. The reliability growth concepts and methodologies presented in this handbook have evolved over the last decades by actual applications to Army, Navy and Air Force systems. Through these applications reliability growth management technology has been developed to the point where considerable payoffs in the effective management of the attainment of system reliability can now be achieved.

4. This handbook is written for use by both the manager and the analyst. Generally, the further into the handbook one reads, the more technical and detailed the material becomes. The fundamental concepts are covered early in the handbook and the details regarding implementing these concepts are discussed primarily in the latter sections. This format, together with an objective for as much completeness as possible within each section, have resulted in some concepts being repeated or discussed in more than one place in the handbook. This should help facilitate the use of this handbook for studying certain topics without extensively referring to previous material.

5. Comments, suggestions, or questions on this document should be addressed to the U.S. Army Materiel System Analysis Activity (AMSAA), Attn: AMSRD-AMS-LA 392 Hopkins Road Aberdeen Proving Ground MD, 21005-5071, or emailed to the AMSAA Webmaster, apgr-amsa-akwebmaster@us.army.mil , this will ensure that the information will be sent to the correct office. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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PARAGRAPH PAGE

Table of Contents

1 SCOPE 1

1.1 SCOPE. 1
1.2 APPLICATION. 1

2 APPLICABLE DOCUMENTS 2

2.1 GENERAL. 2
2.2 GOVERNMENT DOCUMENTS. 2
2.3 NON-GOVERNMENT PUBLICATIONS. 2

3 DEFINITIONS 3

3.1 RELIABILITY GROWTH TERMINOLOGY. 3
3.1.1 Reliability. 3
3.1.2 Operational Mode Summary/Mission Profile. 3
3.1.3 Reliability Growth. 3
3.1.4 Reliability Growth Management. 3
3.1.5 Repair. 3
3.1.6 Fix. 3
3.1.7 Failure Mode. 3
3.1.8 A-Mode. 3
3.1.9 B-Mode. 3
3.1.10 Fix Effectiveness Factor (FEF). 4
3.1.11 Growth Potential (GP). 4
3.1.12 Management Strategy (MS). 4
3.1.13 Growth rate. 4
3.1.14 Poisson Process. 4
3.1.15 Homogeneous Poisson Process (HPP). 4
3.1.16 Non-Homogeneous Poisson Process (NHPP). 4
3.1.17 Idealized Growth Curve (IGC). 4
3.1.18 Planned Growth Curve (PGC). 5
3.1.19 Reliability Growth Tracking Curve. 5
3.1.20 Reliability Growth Projection. 5
3.1.21 Exit Criterion (Milestone Threshold). 5

4 OVERVIEW 6

4.1 INTRODUCTION. 6
4.2 RELIABILITY GROWTH BACKGROUND 6
4.2.1 Reliability Growth Planning. 6
4.2.2 Reliability Growth Assessment (Evaluation). 6
4.2.3 Controlling Reliability Growth. 6
4.3 MANAGEMENT'S ROLE. 7
4.3.1 Basic Reliability Activities. 7
4.4 RELIABILITY GROWTH PROCESS. 10
4.4.1 Basic Process. 10
4.4.2 Additional elements. 10
4.4.3 Type A and Type B Failure Modes. 10

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

4.4.4 Achieving Growth. *11*
4.4.5 Attaining the Requirement. *11*
4.4.6 Growth Rate. *11*
4.4.7 Reliability Growth Management Control Processes. *12*
4.4.8 Basic Methods. *12*
4.4.9 Comparison of Methods. *13*
4.4.10 Assessment. *13*
4.4.11 Monitoring. *13*
4.5 FACTORS INFLUENCING GROWTH CURVE SHAPE. 14
4.5.1 Stages of the Development Program. *14*
4.5.2 Test Phases. *15*
4.5.3 System Configurations. *15*
4.5.4 Timing of Fixes. *15*
4.5.5 Test-Fix-Test. *15*
4.5.6 Pure Test-Fix-Test. *16*
4.5.7 Test-Find-Test. *16*
4.5.8 Test-Fix-Test with Delayed Fixes. *16*
4.6 COMBINED INFLUENCES OF FACTORS ON RELIABILITY GROWTH CURVE SHAPE. 17
4.6.1 Statistical Advantages of Test-Fix-Test. *18*
4.6.2 Growth Curve Re-Initialization. *18*
4.6.3 Shape Changes Due to Calendar Time. *18*
4.6.4 Reliability Growth Programmatic Concepts. *19*
4.6.5 Levels of Consideration for Growth. *19*
4.6.6 Analysis of Previous Programs. *20*
4.7 RELIABILITY GROWTH PLANNING, TRACKING AND PROJECTION. 20
4.7.1 Planning. *20*
4.7.2 Idealized Growth Curve. *20*
4.7.3 Planned Growth Curve. *21*
4.7.4 Other Planning Considerations. *22*
4.7.5 Reliability Growth Tracking. *23*
4.7.6 Demonstrated Reliability. *24*
4.7.7 Reliability Growth Tracking Curve. *24*
4.8 RELIABILITY GROWTH PROJECTION. 24
4.8.1 Projected Reliability. *24*
4.8.2 Extrapolated Reliability. *25*
4.9 THRESHOLD. 25
4.10 THRESHOLD PROGRAM. 25
4.11 RELIABILITY GROWTH MODELS. 25
4.12 REFERENCES. 25

5 RELIABILITY GROWTH 26

5.1 RELIABILITY GROWTH PLANNING. 26
5.1.1 Background. *26*
5.1.2 Planning Model Limitations. *26*
5.1.3 Demonstrate reliability requirement with statistical confidence. *27*
5.1.4 Management. *27*
5.1.5 Threshold Methodology. *27*
5.1.6 Planning Areas. *28*
5.1.6.1 Planning Models. 30
5.1.6.2 Planning Model based on AMSAA/Crow Model (Duane Postulate) Overview. 31
5.1.6.3 SPLAN Model Overview. 32
5.1.6.4 SSPLAN Model Overview. 33
5.1.6.5 PM2 Model Overview. 34

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

5.1.6.6 PM2 Discrete Model Overview. 35
5.1.6.7 Threshold Program Model Overview. 35
5.1.6.8 AMSAA/Crow or Original MIL-HDBK-189. 36
5.1.6.9 The AMSAA/Crow Growth Model. 38
5.1.7 Planning Model Issues. 41
5.1.8 Examples. 42
5.2 DETAILED STATEMENTS ON PLANNED GROWTH CURVE DEVELOPMENT MIL-HDBK-189 (1981). [2] 43
5.2.1 Planned Growth Curves. *43*
5.2.2 General Development of the Planned Growth Curve. 43
5.2.3 General Concepts for Construction. 43
5.2.4 Understanding the Development Program. 43
5.2.5 Portraying the Program in Total Test Units. 44
5.2.6 Determining the Starting Point. 45
5.2.7 Example of Determining a Starting Point. 45
5.2.8 Development of the Idealized Growth Curve. 45
5.2.9 Idealized Growth Model Based on Learning Curve Concept. 46
5.2.10 Summary of Method. 46
5.2.11 Basis of Model. 47
5.2.12 Procedures for Using Idealized Growth Curve Model. 53
5.2.13 Idealized Growth Model. 54
5.2.13.1 Case 1. How to Determine the Idealized Growth Curve. 54
5.2.13.2 Case 2. How to Determine the MTBF for a Test Phase. 57
5.2.13.3 Case 3. How to Determine how much Test Time is Needed. 58
5.2.13.4 Test Phase Reliabi1ity Growth. 59
Incorporate no Design Change. 62
5.2.14 Examples of Growth Curve Development. *62*
5.2.14.1 Example of Growth Curve Development for a Fire Control System. 62
5.2.14.2 Given Conditions. 62
5.2.14.3 Problem. 63
5.2.14.4 Construction of Idealized Curve. 63
5.2.14.5 Construction of Planned Curve. 65
5.3 SYSTEM LEVEL PLANNING MODEL (SPLAN). 67
5.3.1 Introduction. *67*
5.3.2 Background. *67*
5.3.3 Reliability Growth Operating Characteristic (OC) Analysis. *70*
5.3.4 Application. *75*
5.3.4.1 Example 1. 75
5.3.4.2 Example 2. 79
5.3.5 Summary. *79*
5.4 SUBSYSTEM LEVEL PLANNING MODEL (SSPLAN). 80
5.4.1 Subsystem Reliability Growth. *80*
5.4.1.1 Benefits and Special Considerations. 80
5.4.1.2 Overview of SSPLAN Approach. 80
5.4.1.3 List of Notations. 81
5.4.2 SSPLAN Methodology *83*
5.4.2.1 Model Assumptions. 83
5.4.2.2 Mathematical Basis for Growth Subsystems. 84
5.4.2.3 Mathematical Basis for Non-growth Subsystems. 85
5.4.2.4 Algorithm for Estimating Probability of Acceptance P_A 86
5.4.2.5 Calculation of Testing Costs. 90
5.4.2.6 Methodology for a Fixed Allocation of Subsystem Failure Intensities. 92
5.5 PLANNING MODEL BASED ON PROJECTION METHODOLOGY (PM2) 93
5.5.1 PM2 Overview of Approach. *93*
5.5.2 Background and Outline of PM2 Topics. *94*
5.5.3 Derived Reliability Growth Patterns. *95*
5.5.3.1 Assumptions. 95
5.5.3.2 Background Information. 96

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

5.5.3.3 Parsimonious Approximations. 97
5.5.4 Simulation. *101*
5.5.4.1 Simulation Overview. 101
5.5.4.1 Simulation Results. 101
5.5.5 Using Planning Parameters to Construct the Parsimonious *104*
5.5.6 MTBF Growth Curve. *104*
5.5.6.1 Methodology. 104
5.5.6.2 Comparisons of MTBF Approximations Using Planning Parameters. 106
5.5.7 Reliability Growth Potential. *114*
5.5.7.1 Growth Potential in Terms of Planning Parameters. 114
5.5.7.2 Planning Parameter β in Terms of Growth Potential. 114
5.5.7.3 Plausibility Metrics for Planning Parameters. 114
5.5.8 Generating a Planned Reliability Growth Path. *116*
5.6 PM2-DISCRETE. 118
5.6.1 Purpose. *118*
5.6.2 Impact. *118*
5.6.3 List of Notations. *119*
5.6.4 Model Assumptions. *120*
5.6.5 Management Metrics & Model Equations. *120*
5.6.5.1 Overview. 120
5.6.5.2 Expected Reliability. 121
5.6.5.3 Management Strategy. 122
5.6.5.4 Formulae for R_A, and R_B. 123
5.6.5.5 Reliability Growth Potential. 123
5.6.5.6 Formula for n. 124
5.6.5.7 Expected Number of Failures. 124
5.6.5.8 Expected Number of Failure Modes. 125
5.6.5.9 Expected Probability of Failure due to a New Mode. 126
5.6.5.10 Expected Fraction Surfaced of System Probability of Failure. 127
5.7 THRESHOLD PROGRAM. 128
5.7.1 Introduction. *128*
5.7.2 Background. *129*
5.7.3 Application *130*
5.7.4 Example. *130*
5.8 REFERENCES. 130

6 RELIABILITY GROWTH TRACKING. 132

6.1 INTRODUCTION. 132
6.1.1 Definition and Objectives of Reliability Growth Tracking. *133*
6.1.2 Managerial Role *133*
6.1.3 Types of Reliability Growth Tracking Models. *134*
6.1.4 Model Substitution. *134*
6.1.5 List of Notations. *134*
6.1.6 Some Practical Data Analysis Considerations. *136*
6.2 TRACKING MODELS OVERVIEW. 142
6.2.1 Reliability Growth Tracking Model – Continuous (RGTMC) Overview. *142*
6.2.1.1 RGTMC Purpose. 142
6.2.1.2 RGTMC Assumptions. 142
6.2.1.3 RGTMC Limitations. 142
6.2.1.4 RGTMC Benefits. 143
6.2.2 Reliability Growth Tracking Model – Discrete (RGTMD) Overview. *143*
6.2.2.1 RGTMD Purpose. 143
6.2.2.2 RGTMD Assumptions. 143
6.2.2.3 RGTMD Limitations. 143
6.2.2.4 RGTMD Benefits. 143

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

6.2.3 Subsystem Level Tracking Model (SSTRACK) Overview. 143
6.2.3.1 SSTRACK Purpose. 144
6.2.3.2 SSTRACK Assumptions. 144
6.2.3.3 SSTRACK Limitations. 144
6.2.3.4 SSTRACK Benefits. 144
6.3 TRACKING MODELS. 144
6.3.1 Reliability Growth Tracking Model – Continuous. 144
6.3.1.1 Basis for the Model. 144
6.3.1.2 Methodology. 148
6.3.1.3 Cumulative Number of Failures. 149
6.3.1.4 Number of Failures in an Interval. 149
6.3.1.5 Intensity Function. 150
6.3.1.6 Estimation Procedures for Individual Failure Time Data Model. 150
6.3.1.7 Option for Grouped Data. 159
6.3.2 Reliability Growth Tracking Model – Discrete (RGTMD). 163
6.3.2.1 Background. 163
6.3.2.2 Basis for Model. 163
6.3.2.3 List of Notations. 163
6.3.2.4 Model Development. 164
6.3.2.5 Estimation Procedures. 166
6.3.2.6 Point Estimation. 166
6.3.2.7 Interval Estimation. 167
6.3.2.8 Goodness-of-Fit. 168
6.3.2.9 Example. 168
6.3.3 Subsystem Level Tracking Model (SSTRACK). 171
6.3.3.1 Background and Conditions for Usage. 171
6.3.3.2 Lindström-Madden Method. 173
6.3.3.3 Example. 174
6.4 REFERENCES. 177

7 RELIAIBLITY GROWTH PROJECTION. 178

7.1 RELIABILITY PROJECTION BACKGROUND. 178
7.2 BASIC CONCEPTS, NOTATION AND ASSUMPTIONS. 180
7.2.1 List of Notation 183
7.2.2 Assumptions 183
7.3 PROJECTION MODELS 184
7.3.1 AMSAA/Crow Projection Model (ACPM). 184
7.3.1.1 Purpose. 184
7.3.1.2 Assumptions. 184
7.3.1.3 Limitations. 184
7.3.1.4 Benefits. 184
7.3.2 Crow Extended Reliability Projection Model 184
7.3.2.1 Purpose. 184
7.3.2.2 Assumptions. 185
7.3.2.3 Limitations. 185
7.3.2.4 Benefits. 185
7.3.3 AMSAA Maturity Projection Model (AMPM). 185
7.3.3.1 Purpose. 185
7.3.3.2 Assumptions. 185
7.3.3.3 Limitations. 185
7.3.3.4 Benefits. 185
7.3.4 AMPM based on Stein Estimation (AMPM-Stein). 186
7.3.4.1 Purpose. 186
7.3.4.2 Assumptions. 186
7.3.4.3 Limitations. 186
7.3.4.4 Benefits. 186
7.3.5 Discrete Projection Model (DPM). 186

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

7.3.5.1 Purpose. 186
7.3.5.2 Assumptions. 186
7.3.5.3 Limitations. 187
7.3.5.4 Benefits. 187
7.4 THE AMSAA/CROW PROJECTION MODEL (ACPM). 187
7.4.1 Background. *187*
7.4.2 AMSAA/Crow Model Notation and Additional Assumptions. *188*
7.4.3 List of Notations. *188*
7.4.4 Additional Assumptions for AMSAA/Crow. *189*
7.4.5 Methodology. *189*
7.4.6 Reliability Growth Potential. *196*
7.4.7 Maximum Likelihood Estimator versus the Unbiased Estimator for β. *196*
7.4.8 Example. *200*
7.5 THE CROW EXTENDED RELIABILITY PROJECTION MODEL. 202
7.5.1 Background. *202*
7.5.2 AMSAA/Crow Tracking Example. *206*
7.5.3 Estimation for the Extended Reliability Growth Model. *207*
7.5.4 Test-Fix-Find-Test Example. *207*
7.5.5 ACPM Example Using Crow Extended Data. *208*
7.5.6 Extended Reliability Growth Model with Pre-emptive Corrective Actions. *208*
Example. Test-Fix-Find-Test with Pre-emptive Corrective Actions. 209
7.5.7 Extended Model Management and Maturity Metrics. *209*
7.6 THE AMSAA MATURITY PROJECTION MODEL (AMPM). 209
7.6.1 Introduction. *209*
7.6.2 List of Notations *211*
7.6.3 Assumptions. *212*
7.6.4 AMPM Development. *213*
7.6.5 Limiting Behavior of AMPM. *218*
7.6.6 Estimation Procedure for AMPM. *220*
7.6.7 Example. *225*
7.6.8 AMPM Projection Using Crow Extended Data. *229*
Comparison of Projections for ACPM, Crow Extended and AMPM. 229
7.6.9 Analysis Considerations for Apparent Failure Mode Rates of Occurrence Changes. *230*
Gap Method 231
Segmented Fix Effectiveness Factor (FEF) Method. 238
Restart Method. 241
7.7 THE AMSAA MATURITY PROJECTION MODEL BASED ON STEIN ESTIMATION (AMPM-STEIN). 242
7.7.1 Differences in Technical Approach. *242*
7.7.2 Stein Approach to Projection using One Classification of Failure Modes. *245*
7.7.3 Stein Approach to Projection using Two Classifications of Failure Modes. *248*
7.7.4 Failure Rate due to Unobserved Modes as $k \rightarrow \infty$. *250*
7.7.5 AMPM-Stein Approximation using MLE. *252*
7.7.6 AMPM-Stein Approximation using MME. *256*
7.7.7 Cost versus Reliability Tradeoff Analysis. *260*
7.8 DISCRETE PROJECTION MODEL. 261
7.8.1 Introduction. *261*
7.8.1.1 Background and Motivation. 262
7.8.1.2 Overview. 262
7.8.1.3 List of Notations. 262
7.8.1.4 Model Assumptions. 263
7.8.1.5 Data Required. 263
7.8.1.6 Estimation of Failure Probabilities. 263
7.8.1.7 Reliability Growth Projection. 264
7.8.2 Estimation Procedures. 265

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

7.8.2.4 Goodness-of-Fit. 269
7.8.3 Monte-Carlo Simulation Study *269*
7.8.3.1 Overview. 269
7.8.2.2 Simulation Results. 270
7.8.2.2.1 Summary. 270
7.8.2.2.2 Accuracy of Moment-based Projections. 271
7.8.2.2.3 Accuracy of Likelihood-based Projections. 272
7.8.2.2.4 General Observations. 272
7.8.4 Concluding Remarks. *272*
7.9 REFERENCES. 273

8 NOTES. 274

8.1 INTENDED USE. 274
8.2 SUPERSEDING INFORMATION. 274
8.3 SUBJECT TERM (KEYWORD LISTING). 274
8.3.1 Reliability. *274*
8.3.2 Operational Mode Summary/Mission Profile. *274*
8.3.3 Reliability Growth. *274*
8.3.4 Reliability Growth Management. *274*
8.3.5 Repair. *274*
8.3.6 Fix. *274*
8.3.7 Failure Mode. *275*
8.3.8 A-Mode. *275*
8.3.9 B-Mode. *275*
8.3.10 Fix Effectiveness Factor (FEF). *275*
8.3.11 Growth Potential (GP). *275*
8.3.12 Management Strategy (MS). *275*
8.3.13 Growth rate. *275*
8.3.14 Poisson Process. *275*
8.3.15 Homogeneous Poisson Process (HPP). *276*
8.3.16 Non-Homogeneous Poisson Process (NHPP). *276*
8.3.17 Idealized Growth Curve (IGC). *276*
8.3.18 Planned Growth Curve (PGC). *276*
8.3.19 Reliability Growth Tracking Curve. *276*
8.3.20 Reliability Growth Projection. *276*
8.3.21 Exit Criterion (Milestone Threshold). *276*

APPENDIX A ENGINEERING ANALYSIS 277

A.1 SCOPE 277
A.1.1 Purpose *277*
A.1.2 Application *277*
A.2 ASSESSMENT AND SHORT TERM PROJECTION 277
A.2.1 Application. *278*
A.2.2 Objective. *278*
A.2.3 Design Changes. *278*
A.2.4 Significant Factors. *278*
A.2.5 Explanation of Factors. *279*
A.2.5.1 What is the failure rate being experienced in similar applications? 279
A.2.5.2 What is the failure rate of components to be left unchanged? 279
A.2.5.3 What is the analytically predicted failure rate? 279
A.2.5.4 How successful has the design group involved been in previous redesign efforts? 279
A.2.5.5 Is the failure of cause known? 280
A.2.5.6 Is the likelihood of introducing or enhancing other failure modes small? 280
A.2.5.7 Are there other failure modes indirect competition with the failure mode under consideration? 280

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

A.2.5.8 Have there been previous unsuccessful design changes for the failure mode under consideration? 280
A.2.5.9 Is the design change evolutionary rather than revolutionary? 280
A.2.5.10 Does the design group have confidence in the redesign effort? 280
A.2.6 Methodology. *281*
A.2.7 Example *281*
A.2.7.1 Objective 281
A.2.7.2 Problem statement. 281
A.2.7.3 Analysis 281
A.3 PLANNING AND LONG TERM PROJECTION 283
A.3.1 Purpose. *283*
A.3.2 Approach. *283*
A.3.3 Organization or program characteristics. *284*
A.3.4 Program-related questions. *284*
A.3.5 Synthesis. *287*
A.3.6 Analysis *287*
A.3.7 Example *287*
A.3.7.1 Objective. 287
A.3.7.2 Problem Statement. 287
A.3.7.3 Analysis of improvement in existing failure modes. 287
A.3.7.4 Analysis of new failure modes anticipated. 287

APPENDIX B RELIABILITY CASE PLAN OUTLINE 289

B.1 SCOPE 289
B.1.1 Purpose *289*
B.1.2 Application. *289*
B.1.3 References *291*

APPENDIX C RELIABILITY GROWTH MODELS 292

C.1 SCOPE 292
C.2 OVERVIEW 292
C.3 RELIABILITY GROWTH PLANNING 292
C.3.1 Duane's Model (1964). *292*
C.3.2 Selby-Miller RPM Model (1970). *293*
C.3.3 MIL-HDBK-189 Planning Model (1981). *294*
C.3.4 AMSAA System-Level Planning Model (1992). *295*
C.3.5 Ellner's Subsystem Planning Model (1992). *295*
C.3.6 Mioduski's Threshold Program (1992). *296*
C.3.7 Ellner-Hall PM2 Model (2006). *296*
C.4 RELIABILITY GROWTH TRACKING. 297
C.4.1 Weiss' Model (1956). *297*
C.4.2 Aroef's Model (1957). *297*
C.4.3 Rosner's IBM Model (1961). *298*
C.4.4 Lloyd-Lipow Model (1962). *298*
C.4.5 Chernoff-Woods Model (1962). *298*
C.4.6 Wolman's Model (1963). *298*
C.4.7 Cox-Lewis Model (1966). *298*
C.4.8 Barlow-Scheuer Model (1966). *299*
C.4.9 Virene's Gompertz Model (1968). *299*
C.4.10 Pollock's Model (1968). *299*
C.4.11 Crow's Continuous Tracking Model (1974). *299*
C.4.12 Lewis-Shedler Model (1976). *301*
C.4.13 Singpurwalla's Model (1978). *301*
C.4.14 Crow's Discrete Tracking Model (1983). *301*
C.4.15 Robinson-Dietrich Model (1988). *302*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

C.4.16 Kaplan-Cunha-Dykes-Shaver Model (1990).....302
C.4.17 Mazzuchi-Soyer Model (1991).....302
C.4.18 Heimann-Clark PR-NHPP Model (1992).....303
C.4.19 Fries' Discrete Learning-Curve Model (1993).....303
C.4.20 Modified-Gompertz Model (1994).....303
C.4.20 Ellner's Subsystem Tracking Model (1996).....304
C.4.21 Sen's Alternative to the NHPP (1998).....304
C.4.22 Donovan-Murphy Model (1999).....305
C.4.23 Pulcini's Model (2001).....305
C.4.24 Gaver-Jacobs-Glazenbrook-Seglie Model (2003).....305
C.5 RELIABILITY GROWTH PROJECTION.....305
C.5.1 Corcoran-Weingarten-Zehna Model (1964).....305
C.5.2 AMSAA/Crow Model (1982).....306
C.5.3 Ellner-Wald AMPM Model (1995).....306
C.5.4 Clark's Model (1999).....307
C.5.5 Ellner-Hall AMPM-Stein Model (2004).....308
C.5.6 Crow-Extended Model (2005).....308
C.6 RELIABILITY GROWTH SURVEYS AND HANDBOOKS.....308
C.6.1 Crow's Abbreviated Literature Review (1972).....308
C.6.2 DoD's First Military Handbook on Reliability Growth (1981).....308
C.6.3 Fries-Sen Survey on Discrete Reliability Growth Models (1996).....309
C.6.4 DoD's Guide for Achieving RAM (2005).....309
C.7 OTHER LITERATURE (I.E., THEORETICAL RESULTS, PERSPECTIVES AND APPLICATIONS).....309
C.7.1 Corcoran-Read Simulation Study (1967).....309
C.7.2 Barr's Paper (1970).....309
C.7.3 Read's Remark on Barlow-Scheuer Estimation Scheme (1971).....310
C.7.4 The AMSAA Reliability Growth Symposium (1972).....310
C.7.5 Langberg-Proschan Theoretical Paper (1979).....310
C.7.6 Jewell on Learning-Curve Models (1984).....310
C.7.7 Wong's Letter to the Editor (1988).....311
C.7.8 Wronka's Application of the RGTMC (1988).....311
C.7.9 Benton and Crow on Integrated Reliability Growth Testing (1989).....311
C.7.10 Frank's Corollary of the Duane's Postulate (1989).....311
C.7.11 Gibson-Crow Estimation Method for Fix Effectiveness (1989).....311
C.7.12 Woods' Study on the Effect of Discounting Failures (1990).....312
C.7.13 Higgins-Constantinides Application (1991).....312
C.7.14 IEC International Standards for Reliability Growth (1991).....312
C.7.15 Coolas' Application (1991).....313
C.7.16 Bieda's Application (1991).....313
C.7.17 Ellis' Robustness Study (1992).....313
C.7.18 Calabria-Guida-Pulcini Bayes Procedure for the NHPP (1992).....313
C.7.19 Meth's OSD Perspective on Reliability Growth (1992).....314
C.7.20 Demko on Non-Linear Reliability Growth (1993).....314
C.7.21 Farquhar and Mosleh on Growth Effectiveness (1995).....314
C.7.22 Demko on Reliability Growth Testing (1995).....314
C.7.23 Fries-Maillart Stopping Rules (1996).....315
C.7.24 Ebrahimi's MLE for the NHPP (1996).....315
C.7.25 Huang-McBeth-Vardeman One-Shot DT Programs (1996).....315
C.7.26 Xie-Zhao Monitoring Approach (1996).....315
C.7.27 Seglie's OSD Perspective on Reliability Growth (1998).....315
C.7.28 Hodge-Quigley-James-Marshal Framework (2001).....316
C.7.29 Crow's Methods to Reduce LCC (2003).....316
C.7.30 Gurunatha-Siegel Six-Sigma Process (2003).....316

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

C.7.31 Yadav-Singh-Goel Approach (2003). *317*
C.7.32 Quigley-Walls CI Procedures (2003). *317*
C.7.33 Smith on Planning (2004). *317*
C.7.34 Krasich and Quigley on the Design Phase (2004). *318*
C.7.35 Mortin-Ellner Paper (2005). *318*
C.7.36 Acevedo-Jackson-Kotlowitz Application (2006). *318*
C.8 RELIABILITY GROWTH SOFTWARE. 318
C.8.1 U.S. AMSAA. *318*
C.8.2 ReliaSoft Corporation. *319*
C.8.3 Relex Corporation. *319*
C.9 SUMMARY. 319
C.10 REFERENCES 320

APPENDIX D DERIVATIONS FOR OC ANALYSIS PROPOSITIONS 329

D.1 PROPOSITION 1. 329
D.1.1 Proof. *329*
D.2 PROPOSITION 2. 330
D.2.1 Proof. *330*
D.3 PROPOSITION 3. 331
D.3.1 Proof. *331*
D.4 PROPOSITION 4. 332
D.4.1 Proof. *332*
D.5 PROPOSITION 5. 333
D.5.1 Proof. *333*
D.6 PROPOSITION 6. 334
D.6.1 Proof. *334*

APPENDIX E THRESHOLD DERIVATIONS 336

APPENDIX F TABLES: APPROXIMATION OF THE PROBABILITY OF ACCEPTANCE 338

APPENDIX G ANNEX 380

G.1 ANNEX 1 380
G.2 ANNEX 2 380
G.3 ANNEX 3 382
G.3.1 Maximum Likelihood Estimates for AMPM *382*

APPENDIX H BIBLIOGRAPHY 386

Table of Figures

Figure **Page**

FIGURE 4-1. RACAS Data and Communication Flow 9
FIGURE 4-2. Reliability Growth Feedback Model 10
FIGURE 4-3. Reliability Growth Feedback Model with Hardware 10
FIGURE 4-4. Reliability Growth Management Model (Assessment) 13
FIGURE 4-5. Example of Planned Growth and Assessments 13
FIGURE 4-6. Reliability Growth Management Model (Monitoring) 14
FIGURE 4-7. Graph of Reliability in a Test-Fix-Test Program 16

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

FIGURE 4-8. Graph of Reliability in a Test-Find-Test Program 16
FIGURE 4-9. Graph of Reliability in a Test-Fix-Test Program with Delayed Fixes 17
FIGURE 4-10. The Nine Possible General Growth Patterns for Two Test Phases 17
FIGURE 4-11. Comparison of Growth Curves Based on Test Duration Vs Calendar Time 19
FIGURE 4-12. Global Analysis Determination of Planned Growth Curve 21
FIGURE 4-13. Development of Planned Growth Curve on a Phase by Phase Basis 22
FIGURE 4-14. Probability of demonstrating TR w/% Confidence as a Function of M(T)/TR and Expected Number of Failures 23
FIGURE 4-15. Reliability Growth Tracking Curve 24
FIGURE 4-16. Extrapolated and Projected Reliabilities 25
FIGURE 5-1. Planned Growth Curves with Milestone Threshold 28
FIGURE 5-2. Development Program Portrayed in Calendar Time 44
FIGURE 5-3. Development Program Portrayed in Test Units 45
FIGURE 5-4. Idealized Growth Curve 47
FIGURE 5-5. Example of Log-Log Plot at Ends of Test Phases 48
FIGURE 5-6. Average Failure Rates over Test Phases 49
FIGURE 5-7. Average MTBF's over Test Phases 50
FIGURE 5-8. Average MTBF's and Modified $\bar{\boldsymbol{m}}(t)$ Curve 50
FIGURE 5-9. Average MBF's and $\bar{\mathbf{m}}(t)$ Curve 51
FIGURE 5-10. Average MTBF's and Modified $\bar{\boldsymbol{m}}(\boldsymbol{t})$ Curve 51
FIGURE 5-11. Idealized Growth Curve 52
FIGURE 5-12. Log-Log Plot of Idealized Growth Curve *M(t)* 52
FIGURE 5-13. Average MTBF over *i*-th Test Phase 53
FIGURE 5-14. Idealized Growth Curve 56
FIGURE 5-15. Example of Idealized Growth Curve 57
FIGURE 5-16. Example of Average MTBF's 58
FIGURE 5-17. Effect of Deferring Redesign 60
FIGURE 5-18. Accounting for Calendar Time Required for Redesign 61
FIGURE 5-19. Accounting for Calendar Time Required for Redesign 62
FIGURE 5-20. Idealized Growth Curve 64
FIGURE 5-21. Idealized Growth Curve on Log-Log Scale 65
FIGURE 5-22. Planned Growth Curve 66
FIGURE 5-23. Example OC Curve for Reliability Demonstration Test 70
FIGURE 5-24. Idealized Reliability Growth Curve 76
FIGURE 5-25. Program and Alternate Idealized Growth Curves 77
FIGURE 5-26. Operating Characteristic (OC) Curve 78
FIGURE 5-27. System Architecture 83
FIGURE 5-28. Reliability Growth based on AMSAA Continuous Tracking Model 85
FIGURE 5-29. Average Number of Surfaced Modes (Loglogistic) 103
FIGURE 5-30. Reciprocal of the Failure Intensity (Loglogistic) 103
FIGURE 5-31. Average Number of Surfaced Modes (Geometric) 104
FIGURE 5-32. Reciprocal of the Failure Intensity (Geometric) 104
FIGURE 5-33. Reciprocal of the Failure Intensity (Gamma) 109
FIGURE 5-34. Reciprocal of the Failure Intensity (Log Normal) 109
FIGURE 5-35. Top W Modes (Log Normal) 110
FIGURE 5-36. Reciprocal of the Failure Intensity (Log Normal) 111
FIGURE 5-37. Top W Modes (Log Normal) 111
FIGURE 5-38. Reciprocal of the Failure Intensity (Geometric) 113
FIGURE 5-39. Top W Modes (Geometric) 113
FIGURE 5-40. PM2 Reliability Growth Planning Curve 117
FIGURE 6-1. Reliability Evaluation Flowchart 137
FIGURE 6-2. Pareto Chart of Failure Modes 138
FIGURE 6-3. System Failures by Major Subsystem 138
FIGURE 6-4. Cumulative Failures Vs Cumulative Operating Time 140
FIGURE 6-5. Planned Growth Curve 142
FIGURE 6-6. Failure Rates Between Modifications 146

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

FIGURE 6-7. Timeline for Phase 2 (*t* in first time interval) 146
FIGURE 6-8. Timeline for Phase 2 (t in second time interval) 146
FIGURE 6-9. Parametric Approximation to Failure Rates Between Modifications 148
FIGURE 6-10. Test Phase Reliability Growth based on AMSAA/Crow Continuous Tracking Model 149
FIGURE 6-11. Estimated Intensity Function 158
FIGURE 6-12. Estimated MTBF Function with 90% Interval Estimate at *T*=300 Hours 159
FIGURE 6-13. Test Data for Grouped Data Option 168
FIGURE 6-14. Estimated Failure Rate by Configuration 169
FIGURE 6-15. Estimated Reliability by Configuration 170
FIGURE 7-1. Observed Versus Estimate of Expected Number of B-Modes 227
FIGURE 7-2. Extrapolation of Estimated Expected Number of B-Modes as 227
FIGURE 7-3. Projected MTBF for Different K's. 228
FIGURE 7-4. Estimated Fraction of Expected Initial B-Mode Failure 228
FIGURE 7-5. Expected (Smooth) vs. Observed (Pts) Number of B-Modes 229
FIGURE 7-6. Example Curve for Illustrating the Gap Method 232
FIGURE 7-7. Model Results for AMPM 233
FIGURE 7-8. MTBF Projection Increases as Fix Effectiveness Improves 233
FIGURE 7-9. Estimated Expected Rate of Occurrence of New B-Modes 234
FIGURE 7-10. MTBF Projection Curve 235
FIGURE 7-11. AMPM Results Using the Gap Method 236
FIGURE 7-12. Visual Goodness-of-Fit with AMPM (Gap Method, v = 250 Hours) 237
FIGURE 7-13. Plot of MTBF Projections for AMPM (Gap Option, v = 250 Hours) 237
FIGURE 7-14. Plot of MTBF Projections for AMPM (Gap Option, v = 250 Hours) 238
FIGURE 7-15. AMPM Method Using Two FEFs 240
FIGURE 7-16. Moderate Improvement in MTBF Projection Using Segmented FSF Approach 241
FIGURE 7-17. "v" Should Be Chosen Based On Engineering Analysis 241
FIGURE 7-18. Relative Error of Moment-based Projection 271
FIGURE A-1. Defining and Refining Estimates. 283
FIGURE A-2. Feedback Model. 284
FIGURE C-1. Duane Reliability Growth Plot. 293
FIGURE C-2. L-HDBK-189 Planning Curve. 294
FIGURE C-3. SPLAN Planning Curve. 295
FIGURE C-4. PM2 Planning Curve. 297
FIGURE C-5. MVF vs. Test Time against Number of Failures. 300
FIGURE C-6. RGTMD Reliability. 301
FIGURE C-7. SSTRACK LCB on MTBF. 304
FIGURE C-8. Expected No. Modes. FIGURE C-9. Percent λB Observed. 307
FIGURE C-10. ROC of New Modes FIGURE C-11. Reliability Growth 307

TABLE **Page**

TABLE I. AMSAA Reliability Growth Data Study Summary: Historical Growth Parameter Estimates 29
TABLE II. Average number of Failures for H_i, M_i 58
TABLE III. Example Planning Data 78
TABLE IV. Example Planning Data 79
TABLE V. System Arrival Times 139
TABLE VI. Lower (L) And Upper (U) Coefficients for Confidence Intervals for MTBF from 152
TABLE VII. Lower Confidence Interval Coefficients for MTBF From 154
TABLE VIII. Critical Values for Cramer-Von Mises Goodness-Of-Fit Test 156
TABLE IX. Test Data for Individual Failure Time Option 157
TABLE X. Test Data for Grouped Option 162
TABLE XI. Observed Versus Expected Number of Failures 163
TABLE XII. Estimated Failure Rate and Estimated Reliability By Configuration 169
TABLE XIII. Table of Approximate Lower Confidence Bounds (LCB's) For Final Configuration 170
TABLE XIV. Subsystem Statistics 175
TABLE XV. System Approximate Lower Confidence Bounds 176

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

TABLE XVI. ACPM Projection Example Data 200
TABLE XVII. Test-Fix-Find Test Failure Times and Failure Mode Designations 205
TABLE XVIII. BD Failure Mode Data and Effectiveness Factors 206
TABLE XIX. Projected MTBFs 230
TABLE XX. Reliability Projections 270
TABLE A.I. Reference Values 281
TABLE A.II. Design Change Features 282
TABLE F.I. FOR 70 PERCENT CONFIDENCE 340
TABLE F.II. FOR 80 PERCENT CONFIDENCE 353
TABLE F.III. FOR 90 PERCENT CONFIDENCE 367

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1 SCOPE

1.1 SCOPE.

This handbook provides procuring activities and development contractors with an understanding of the concepts and principles of reliability growth, advantages of managing reliability growth, and guidelines and procedures to be used in managing reliability growth. It should be noted that this handbook is not intended to serve as a reliability growth plan to be applied to a program without any tailoring. This handbook, when used in conjunction with knowledge of the system and its development program, will allow the development of a reliability growth management plan that will aid in developing a final system that meets its requirements and lowers the life cycle cost of the fielded systems. It should be pointed out that this handbook is not intended to cover software reliability growth testing and planning, rather the intent only is to include software failures or incidents coincident as they occur and apply to the failure definition/scoring criteria from testing applicable to addressing reliability growth tracking.

1.2 APPLICATION.

The guide is intended for use on systems/equipment during their development phase by both producer and customer personnel.

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2 APPLICABLE DOCUMENTS

2.1 General.

The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government Documents.

The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DOD Guide for Achieving Reliability, Availability, and Maintainability, August 3, 2005.

2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein.

GEIA-STD-0009, "Reliability Program Standard for Systems Design, Development, and Manufacturing," August 01, 2008.

IEEE Std 1332-1998, "IEEE standard reliability program for the development and production of electronic systems and equipment," 1998.

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3 DEFINITIONS

3.1 Reliability Growth Terminology.

3.1.1 Reliability.
Reliability is the probability that an item will perform its intended function for a specified time and under stated conditions, which are consistent with that of the Operations Mode Summary/Mission Profile (OMS/MP).

3.1.2 Operational Mode Summary/Mission Profile.
Defines the concept of deployment, mission profile or details as to how equipment utilized, per cent operating time/mileage in various operating modes and percent of operating time/mileage, etc in operational environment or conditions (temperature, vibration, percent miles on terrain/road types, etc) under which equipment is utilized.

3.1.3 Reliability Growth.
Reliability growth is the positive improvement in a reliability parameter over a period of time due to implementation of corrective actions to system design, operation and maintenance procedures, or the associated manufacturing process.

3.1.4 Reliability Growth Management.
Reliability growth management is the management process associated with planning for reliability achievement as a function of time and other resources, and controlling the ongoing rate of achievement by reallocation of resources based on comparisons between planned and assessed reliability values.

3.1.5 Repair.
A repair is the repair of a failed part or replacement of a failed item with an identical unit in order to restore the system to be fully mission capable.

3.1.6 Fix.
A fix is a corrective action that results in a change to the design, operation and maintenance procedures, or to the manufacturing process of the item for the purpose of improving its reliability.

3.1.7 Failure Mode.
A failure mode is an individual failure for which a failure mechanism is determined. Individual failure modes may exhibit a given failure rate until a change is made in the design, operation and maintenance, or manufacturing process.

3.1.8 A-Mode.
An A-mode is a failure mode that will not be addressed via corrective action.

3.1.9 B-Mode.
A B-mode is a failure mode that will be addressed via corrective action, if exposed during testing. One caution with regard to B-mode failure correction action is during the test program,

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fixes may be developed that address the failure mode but are not fully compliant with the planned production model. While such fixes may appear to improve the reliability in test, the final production fix would need to be tested to assure adequacy of the corrective action.

3.1.10 Fix Effectiveness Factor (FEF).
A FEF is a fraction representing the fraction reduction in an individual initial mode failure rate due to implementation of a corrective action.

3.1.11 Growth Potential (GP).
Growth potential is a theoretical upper limit on reliability which corresponds to the reliability that would result if all B-modes were surfaced and fixed with an assessed FEF.

3.1.12 Management Strategy (MS).
MS is the fraction of the initial system failure intensity due to failure modes that would receive corrective action if surfaced during the developmental test program.

3.1.13 Growth rate.
A growth rate is the negative of the slope of the cumulative failure rate for an individual system plotted on log-log scale. This quantity is representative of the rate at which the system's reliability is improving as a result of implementation of corrective actions. A growth rate between (0,1) implies improvement in reliability, a growth rate of 1 implies no growth, and a growth rate greater than 1 implies reliability decay.

3.1.14 Poisson Process.
A Poisson process is a counting process for the number of events, $N(t)$, that occur during test interval $[0,t]$, where t is a measure of test duration. The counting process is required to have the following properties: (1) the number of events in non-overlapping intervals are stochastically independent; (2) the probability that exactly one event occurs in the interval $[t.\ t+\Delta t]$ equals $\lambda_t * \Delta t + o(\Delta t)$ where λ_t is a positive constant, which may depend on t, and $o(\Delta t)$ denotes an expression of $\Delta t \succ 0$ that becomes negligible in size compared to Δt as Δt approaches zero; and (3) the probability that more than one event occurs in an interval of length Δt equals $o(\Delta t)$. The above three properties can be shown to imply that $N(t)$ has a Poisson distribution with mean equal to $\int_0^t \lambda_s\ ds$, provided λ_s is an integrable function of s.

3.1.15 Homogeneous Poisson Process (HPP).
A HPP is a Poisson process such that the rate of occurrence of events is a constant with respect to test duration t.

3.1.16 Non-Homogeneous Poisson Process (NHPP).
A NHPP is a Poisson process with a non-constant recurrence rate with respect to test duration t.

3.1.17 Idealized Growth Curve (IGC).
An IGC is a planned growth curve that consists of a single smooth curve portraying the expected overall reliability growth pattern across test phases and is based on initial conditions, assumed growth rate, and/or planned management strategy.

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3.1.18 Planned Growth Curve (PGC).
A PGC is a plot of the anticipated system reliability versus test duration during the development program. The PGC is constructed on a phase-by-phase basis and as such may consist of more than one growth curve.

3.1.19 Reliability Growth Tracking Curve.
A reliability growth tracking curve is a plot of the best statistical representation of system reliability to demonstrated reliability data versus total test duration. This curve is the best statistical representation in comparison to the family of growth curves assumed for the overall reliability growth of the system.

3.1.20 Reliability Growth Projection.
Reliability growth projection is an assessment of reliability that can be anticipated at some future point in the development program. The rate of improvement in reliability is determined by (1) the on-going rate at which new problem modes are being surfaced, (2) the effectiveness and timeliness of the fixes, and (3) the set of failure modes that are addressed by fixes.

3.1.21 Exit Criterion (Milestone Threshold).
Reliability value that needs be exceeded to enter the next test phase. Threshold values are computed at particular points in time, referred to as milestones, which are major decision points that may be specified in terms of cumulative hours, miles, etc. Specifically, a threshold value is a reliability value that corresponds to a particular percentile point of an order distribution of reliability values. A reliability point estimate based on test failure data that falls at or below a threshold value (in the rejection region) indicates that the achieved reliability is statistically not in conformance with the idealized growth curve.

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4 OVERVIEW

4.1 Introduction.

This update to MIL-HDBK-189 includes several of the advances to reliability growth methodology that have been developed since the handbook was first published in 1981. This evolution has better defined three areas within the field of reliability growth management: (1) reliability growth planning; (2) reliability growth tracking; (3) and reliability growth projection.

4.2 Reliability Growth Background

Initial prototypes of complex weapon systems will invariably have reliability and performance deficiencies that generally could not be foreseen and eliminated in early design stages. To uncover and mitigate these deficiencies, early prototypes and later more mature units are subjected to a series of developmental and operational tests. The tests are specifically designed to expose the system components to the range of stresses that they are expected to encounter during the system's life cycle. Failures are analyzed, corrective actions are implemented, and modifications are tested to verify the effectiveness of the corrective actions. This developmental approach has been referred to as the Test-Analyze-Fix-Test (TAFT) procedure. In such a fashion one attempts to increase, or grow, the reliability of the prototypes to the stated reliability requirement, and the process is often referred to as a reliability growth program.

4.2.1 Reliability Growth Planning.

Reliability growth planning addresses program schedules, amount of testing, resources available and the realism of the test program in achieving the requirements. The planning is qualified and reflected in the construction of a reliability growth program plan curve(s). This curve(s) establishes interim reliability goals throughout the program. The curve(s) would show graphically how the reliability would be expected to grow over the entire program and/or over several test phases of the program, and may be shown in cumulative "test time" or by calendar time. Of concern to management would be those planning factors that they would have control of and commitment to in order that requirements can be achieved. These include: test and manpower resources, corrective action turnaround time, fix effectiveness of failure modes, and management strategy or fraction of failure modes addressed with fixes.

4.2.2 Reliability Growth Assessment (Evaluation).

To achieve program goals, it is important that the program manager be aware of reliability problems during the conduct of the program so that he can effect whatever changes are necessary, e.g., increased reliability emphasis. It is, therefore, essential that periodic assessments of reliability be made during the test program (usually, but not exclusively, at the end of a test phase) and compared to the planned reliability growth values. While most assessments would occur at the end of test phases, there may be occasions where interim assessments may be appropriate. These could include situations where clusters of failures might occur that warrant investigation and assessment. For such occasions, additional metrics or statistics may be appropriate.

4.2.3 Controlling Reliability Growth.

These assessments provide visibility of achievements and focus on deficiencies while there is still time to affect the system design. By making appropriate decisions with regard to the timely

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incorporation of effective fixes into the system commensurate with attaining the milestones and requirements, management can control the growth process.

4.3 Management's Role.

The various techniques associated with reliability growth management do not, in themselves, manage. They simply make reliability a more visible and manageable characteristic. Every level of management can take advantage of this visibility by requesting reliability growth plans and progress reports for review. Without this implementation, reliability growth cannot truly be managed. In addition to how appropriately the system is tested, there are at least four planning elements under management control including:

a. Management Strategy or the fraction of system initial failure rate addressed by fixes.
b. Rate at which failure modes are surfaced.
c. Turnaround time for analyzing and implementing corrective actions (fixes).
d. Fix effectiveness or the percent reduction in the rate of occurrence of fixed modes.

The planned growth curve and milestones are only targets. They do not imply that reliability will automatically grow to these values. On the contrary, these values will be attained only with the incorporation of an adequate number of effective design fixes into the system. This requires dedicated management attention to reliability growth. The methods in this guide are for the purpose of assisting management in making timely and appropriate decisions to ensure sufficient support of the reliability engineering design effort throughout the development testing program.
High level management involvement and commitment to reliability growth is necessary in order to have available all the options for difficult program decisions. For example, high level decisions in the following areas may be necessary in order to ensure that reliability goals are achieved:

a. Revise the program schedule;
b. Increase testing;
c. Fund additional development efforts;
d. Add or reallocate program resources;
e. Stop the program until interim reliability goals have been demonstrated.

Although some of these options may result in severe program delay or significant increase in development costs, they may have to be exercised in order to field equipment that meets user needs and has acceptable total life cycle costs.

4.3.1 Basic Reliability Activities.

Reliability growth management is part of the system engineering process. Formal design guidelines that will “build in” reliability early on needs to be documented in the program plan. Performing an analysis that shows that the designer’s proposed paper design, when built, will meet contractual requirements, provides design assurance. Emphasis should be placed to “build in” reliability up front rather than on extensive test validation. A program manager can better assess reliability efforts by using “best practices” design guidelines (design-in reliability, low indenture level verification, failure modes identification) and test-in reliability. A sample of these areas follows:

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a. Apportionment – a process of allocating the overall system requirement to values for each of the subsystems or components.

b. Failure mode, effects and criticality analysis (FMECA) – a tool used to identify potential failure modes and their impact on the system. These modes are candidates for elimination or reduction. The analysis lists the effects of component failures, provides a basis for eliminating mission-critical, single-point failures, identifies areas requiring special in-process inspection and controls, and the need for BIT or testability.

c. Stress Margins - parts should be designed to operate at the extremes of the parameter range. Items should be designed such that part drift with time, temperature, humidity, etc are within their rated tolerances.

d. Stress analysis – designed to precipitate in a short time latent defects that would be likely over a much longer period of time in actual use. These failures may be due to design or manufacturing process defects. Stress limits may exceed design or material limits in some cases.

e. Highly Accelerated Life Testing (HALT) – is an enhanced step-stress test that simultaneously subjects a product to highly accelerated levels of thermal cycling and vibration. Often employed in conjunction with Highly Accelerated Stress Screening (HASS).

f. Highly Accelerated Stress Screening (HASS) – is based on the specified operational and destruct stress limits that are typically established during HALT. HASS is then performed on 100% of the production items to accelerate the removal of infant mortality failures.

g. Environmental Stress Screening (ESS) – typically consists of a series of tests designed to remove latent part and manufacturing process defects through application of environmental stimuli (e.g., random vibration, thermal cycling) prior to fielding equipment.

h. Physics of Failure (PoF) – a technique used for identifying and understanding the physical processes and mechanisms of failure. The purpose of using PoF tools is to design out failures prior to testing and fielding. Electronic applications can be conducted at the board and device level employing vibration, thermal, and fatigue analysis tools. Mechanical component applications include solid modeling, dynamics simulation, and finite element analysis tools used in support of determining component fatigue failure mechanisms. PoF techniques may be applied both in design and manufacturing. Also known as Predictive Technology, Predictive Engineering and Physics of Reliability.

i. Critical Items List/Analysis – items requiring special attention due to complexity, application of state-of-the-art technology, high cost, single source, or single failure point components.

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j. Software reliability assessment – a software assessment should be performed. This should identify tools (metrics) to be used to measure the "goodness" of the software development process.

k. Failure Reporting and Corrective Action System (FRACAS) and Failure Review Board (FRB) – process by which failures of an item are tracked; analysis conducted to determine root cause; and corrective actions identified and implemented to reduce failure occurrence. The FRB is a board that coordinates and reviews the progress of failure data collected throughout the FRACAS cycle. The FRACAS data and communication flow is illustrated below. FRACAS provides visibility on problem areas and effectiveness of corrective actions. As a minimum, the following should be collected:

i. Test environment in which the failure occurred
ii. Failure report number (Test Incident Report (TIR))
iii. Serial number of failed assembly
iv. Failure description
v. Run time at failure
vi. Failure mode and cause
vii. Reliability scoring
viii. Corrective action

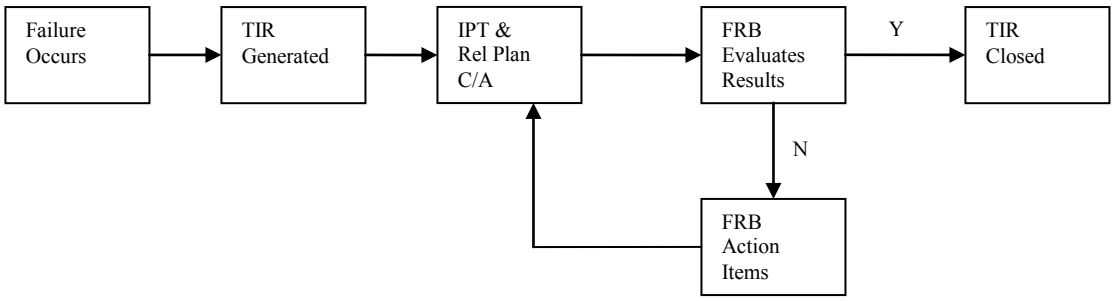


FIGURE 4-1. *RACAS Data and Communication Flow*

l. Fault Tree Analysis (FTA) – a top down model that graphically depicts all known events or combinations of events that can occur that can lead to a specific undesirable event. The FTA and FMECA are supportive techniques, FTA aimed at catastrophic events, FMECA looking at all potential failure modes regardless of severity.

m. Data collection and test monitoring – data generated from testing for use by the various agencies for assessment of reliability should be in a format and have sufficient audit trails so that it can be appropriately used for reliability scoring and analysis purposes.

n. Scoring and Assessment of RAM data – should be scored according to the system's Failure Definition/Scoring Criteria document. For reliability projections based on effectiveness factors, the assessment conferences should develop these factors based on the corrective action information developed by the Corrective Action Review Team (CART).

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4.4 Reliability Growth Process.

4.4.1 Basic Process.

Reliability growth is the result of failure mode discovery and correction, which is an iterative design process. As the design matures, it is investigated to identify actual or potential sources of failures. Further design effort is then spent on these problem areas. The design effort can be applied to either product design or manufacturing process design. The iterative process can be visualized as a simple feedback loop as in FIGURE 4-2. This illustrates that there are four essential elements involved in achieving reliability growth:

a. Failure mode discovery;
b. Feedback of problems identified;
c. Failure mode root cause analysis and proposed corrective action;
d. Approval and implementation of proposed fix.

4.4.2 Additional elements.

Furthermore, for fix implementation, a fifth element may be necessary: Fabrication of hardware and/or software/firmware correction, FIGURE 4-3. Following redesign, follow-on testing at system and/or lower indenture levels serves as: Verification of redesign effort.

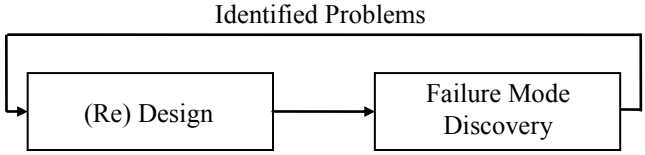


FIGURE 4-2. *Reliability Growth Feedback Model*

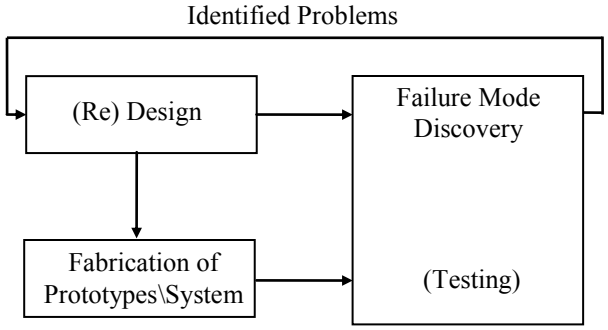


FIGURE 4-3. *Reliability Growth Feedback Model with Hardware*

4.4.3 Type A and Type B Failure Modes.

When a system is tested and failure modes observed, management can make one of two decisions, either not fix the failure mode or fix the failure mode. Therefore, the management

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strategy (MS) places failure modes into two categories called Type A and Type B modes (hence, referred to as A or B modes. A-modes are all failure modes such that when seen during test no corrective action will be taken. This accounts for all modes for which management determines that it is not economically or otherwise justified to take corrective action. Examples of such modes might be failure modes associated with commercial off the shelf (COTS) or legacy systems. B-modes are all failure modes such that when seen during test a corrective action or fix will be attempted. The MS, therefore, partitions the system into an A part and a B part. B-modes may be further classified as BC and BD where BC represents a B-mode which is corrected and BD is one for which the corrective action is delayed. It should be noted that while a mode may be initially classified as an A mode, subsequent conditions may change and it may be reclassified as a B-mode with a fix developed. This failure analysis and corrective action development process is integral to growing system reliability.

4.4.4 Achieving Growth.

Growth is achieved by decreasing the failure rate. The failure rate for A-modes would not change, thus only the B-mode fixes can accomplish growth. It is important to note that while a fix may be developed and implemented for a B-mode, it rarely would totally eliminate the mode's failure rate. Rather a certain percent of the failure rate would be removed still leaving a certain percent of the failure rate of the B-mode. A fix effective factor (FEF) is the fraction decrease in a problem mode failure rate after the corrective action has been made. As stated in a study by the Army Materiel Systems Analysis Activity (AMSAA), an overall fix effectiveness of 0.70 has been observed on average. Some average FEFs vary according to the commodity or technical area. Note that if a FEF was 0.70, on average, the failure rate remaining would be 0.30 or 1 – 0.70, of the initial mode failure rate.

4.4.5 Attaining the Requirement.

An important question is: Can the requirement ever be attained with the planned management strategy and fix effectiveness? In part, this can be looked at by addressing the growth potential (GP). Growth potential is the maximum reliability that can be attained with the system design, management strategy, and FEF. The growth potential will have been attained when all B failure modes have been found and associated fixes incorporated into the system. This is in effect an upper limit on reliability. This growth potential may never actually be achieved in practice.

4.4.6 Growth Rate.

The rate at which reliability grows depends on how rapidly failure mode discovery, failure analysis, fabrication of systems, and retesting/verification can be accomplished. That is, the rate at which a system's reliability is improved is a function of:

a. The rate at which failure modes are surfaced during testing;
b. The turnaround time associated with analyzing/implementing fixes:
    i. Time associated with performing root cause analysis;
    ii. Time associated with the corrective action review and approval process;
    iii. Time associated with physical implementation of approved fixes.
c. Fraction of initial failure rate addressed by fixes - management strategy;
d. Fraction by which failure rate of fixed modes is reduced - fix effectiveness.

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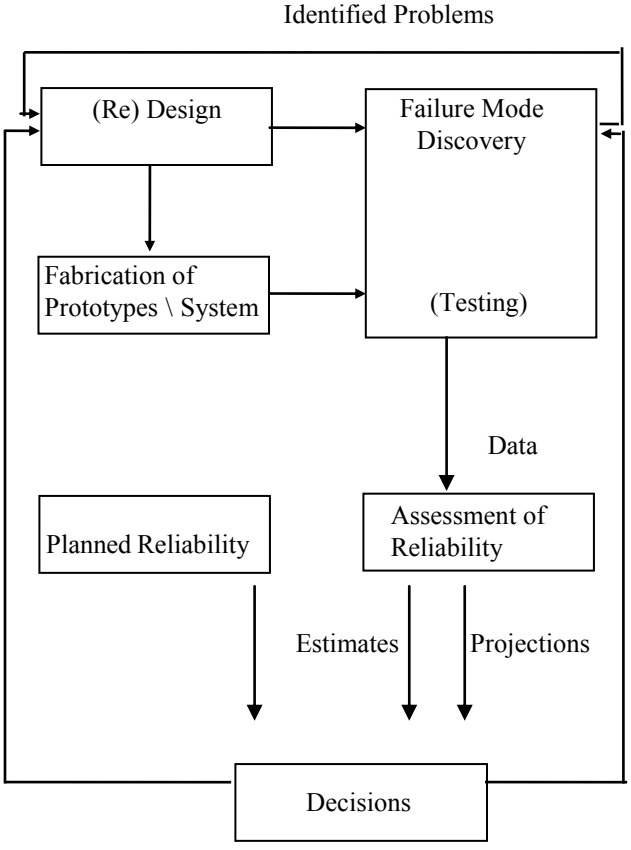
Any of these activities may act as a bottleneck. The cause and degree of the bottleneck may vary from one development program to the next, and even within a single program the causes may vary from one stage of development to the next.

4.4.7 Reliability Growth Management Control Processes.

FIGURE 4-4 illustrates the growth process and associated management processes in a skeleton form. This type of illustration is used so that the universal features of these processes may be addressed. The representation of an actual program or program phase may be considerably more detailed. This detailing may include specific inputs to, and outputs from, the growth process, additional activity blocks, and more explicit decision logic blocks.

4.4.8 Basic Methods.

There are two basic ways that the manager evaluates the reliability growth process. The first method is to utilize assessments (quantitative evaluations of the current reliability status) that are based on information from the detection of failure sources. The second method is to monitor the various activities in the process to assure that the activities are being accomplished in a timely manner and that the level of effort and quality of work are in compliance with the program plan. Each of these methods complements the other in controlling the growth process.



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FIGURE 4-4. *Reliability Growth Management Model (Assessment)*

4.4.9 Comparison of Methods.
The assessment approach is results oriented; however, the monitoring approach, which is activities oriented, is used to supplement the assessments and may have to be relied on entirely early in a program. This is often necessary because of the lack of sufficient objective information in the early program stages.

4.4.10 Assessment.
FIGURE 4-5 illustrates how assessments may be used in controlling the growth process. Reliability growth management differs from conventional reliability program management in two major ways. First, there is a more objectively developed growth standard against which assessments are compared. Second, the assessment methods used can provide more accurate evaluations of the reliability of the present equipment configuration. A comparison between the assessment and the planned value will suggest whether the program is progressing as planned, better than planned, or not as well as planned. If the progress is falling short, new strategies should be developed. These strategies may involve the reassignment of resources to work on identified problem areas or may result in adjustment of the timeframe or a re-examination of the validity of the requirement. FIGURE 4-5 illustrates an example of both the planned reliability growth and assessments.

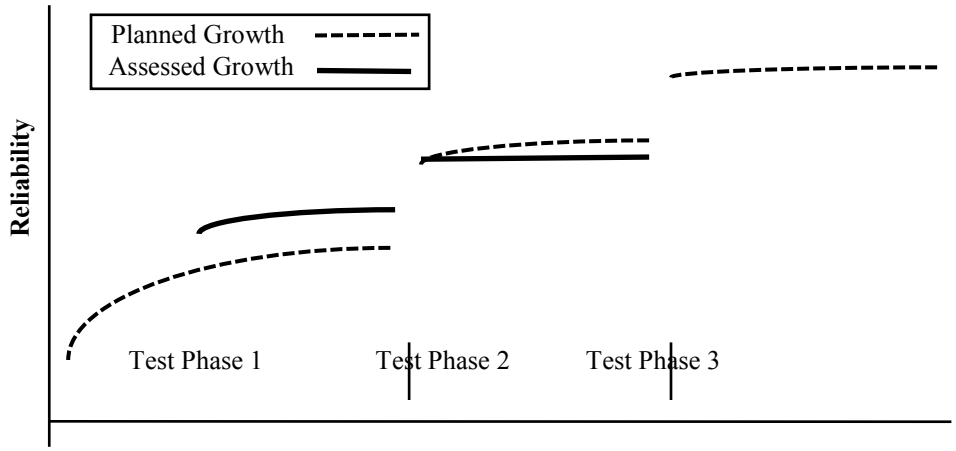


FIGURE 4-5. *Example of Planned Growth and Assessments*

4.4.11 Monitoring.
FIGURE 4-6 illustrates control of the growth process by monitoring the growth activities. Since there is no simple way to evaluate the performance of the activities involved, management based on monitoring is less definitive than management based on assessments. Nevertheless, this activity is a valuable complement to reliability assessments for a comprehensive approach to reliability growth management. But standards for level of effort and quality of work accomplishment should, of necessity, rely heavily on the technical judgment of the evaluator.

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Monitoring is intended to assure that the activities have been performed within schedule and meet appropriate standards of engineering practice. It is not intended to second-guess the designer, e.g., redo his stress calculations. One of the better examples of a monitoring activity is the design review. The design review is planned monitoring of a product design to assure that it will meet the performance requirements during operational use. Such reviews of the design effort serve to determine the progress being made in achieving the design objectives. Perhaps the most significant aspect of the design review is its emphasis on technical judgment, in addition to quantitative assessments of progress.

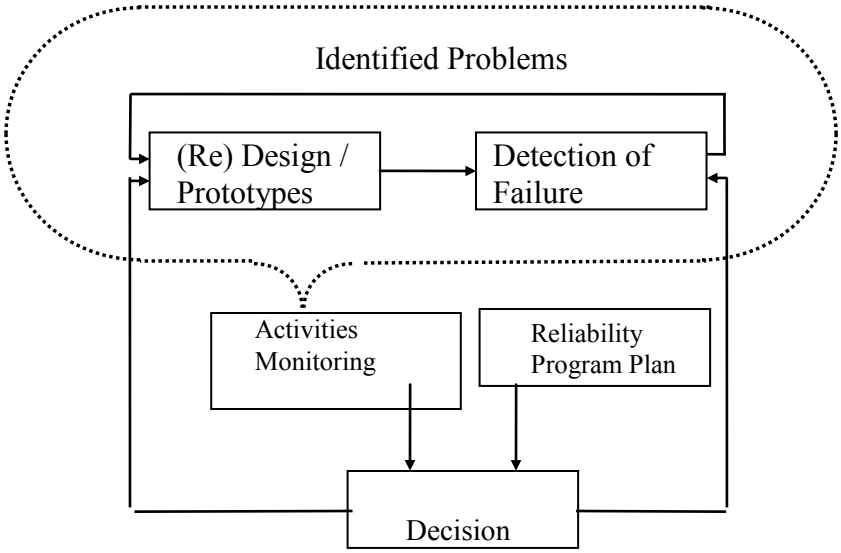


FIGURE 4-6. *Reliability Growth Management Model (Monitoring)*

4.5 Factors Influencing Growth Curve Shape.

This section introduces factors that affect the shape of the growth curve. Such things as the current stage of the development program, the current test phase, the system configuration under test, the timing of design change insertion, and the units of measure for test duration all influence the growth curve's shape.

4.5.1 Stages of the Development Program.

Generally, any system development program is divided into stages having different objectives for each stage. The names and objectives for each stage in a given development program need not be the ones given here. These stages are given as representative of a typical development:

a. Proposal. There is no hardware at this stage. This is the engineering and accounting paper analysis of differing proposed solutions and designs. In this stage the concern is over what are the requirements, can they be met, and if so, how and at what estimated cost?

b. Conceptual. Experimental prototypes are built at this stage. These may bear little resemblance to the actual system. They are for proof-of-principle.

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c. Validation. Prototypes much like the final system are built and tested. This stage tries to achieve the performance and reliability objectives for the system.
d. Full Scale Development. Systems built as though they were in production are tested to work out final design details and manufacturing procedures.

Quantitative reliability growth management can be used during the validation and full-scale development stages of the program. It could be argued that the different nature of the testing going on in these stages is different enough to cause different rates of growth to occur. How much different the types of testing are determines how they will be treated in creating the planning growth curve.

4.5.2 Test Phases.

Within a development stage it is likely that testing will be divided into alternating time periods of testing followed by no testing. Each period of active testing can be viewed as a testing phase. Also, within a development stage it is likely that more than one type of testing will take place (e.g., RAM testing, performance testing). If these other tests that are not specifically for reliability follow the intended operating environment and the intended use stresses well enough, and if design changes are made on the basis of these tests, then the information gathered may be incorporated into the reliability growth test data base. These would also be called reliability growth testing phases. It is to be expected that the reliability will grow from one phase to the next. The reliability growth planning curve should reflect this.

4.5.3 System Configurations.

In an absolute sense, any change to the design of a system constitutes a new configuration. For our purposes, we will term a specific design a new configuration if there has been one significant design change, or enough little design changes, that cause an obviously different failure rate for the system. It is possible that two or more testing phases could be grouped together for analysis based on the configuration tested in these phases being substantially unchanged. It is also possible that one design change is so effective at increasing reliability that a new configuration could occur within a test phase. System configuration decisions can also be made on the basis of engineering judgment. Obviously, the configuration under test has great influence on the growth curve.

4.5.4 Timing of Fixes.

The replacement of a part with another part identical to the first is termed a repair. Replacing, or eliminating, a part due to a design change is termed a fix. Fixes are intended to reduce the rate at which the system fails. Repairs make no change in the failure rate of the system. The time of insertion of a fix affects the pattern of reliability growth.

4.5.5 Test-Fix-Test.

In an absolutely pure test-fix-test program, when a failure is observed, testing stops until a design change is implemented on the system under test. When the testing resumes, it is with a system that has incrementally better reliability. The graph of reliability for this testing strategy is a series of small increasing steps, with each step stretching out longer to represent a longer time between failures. Such a graph can be approximated by a smooth curve. See FIGURE 4-7.

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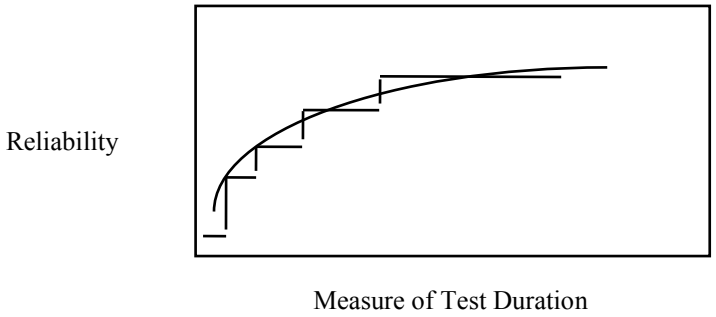


FIGURE 4-7. *Graph of Reliability in a Test-Fix-Test Program*

4.5.6 Pure Test-Fix-Test.

A pure test-fix-test program is impractical in most situations. Testing is likely to continue with a repair, and the fix will be implemented later. Nevertheless, if fixes are inserted as soon as possible and while testing is still proceeding, the stair step like reliability increases and the shape of the approximating curve will be similar, but rise at a slower rate. This is due to the reliability remaining at the same level that it was when the failure happened until the fix is inserted. Thus the steps will all be of longer length, but the same height. Continuing to test after the fix is inserted will serve to verify the goodness of the design change.

4.5.7 Test-Find-Test.

During a test-find-test program the system is also tested to determine problem failure modes. However, unlike the test-fix-test program, fixes are not incorporated into the system during the test. Rather, the fixes are all inserted into the system at the end of the test phase and before the next testing period. Since a large number of fixes will generally be incorporated into the system at the same time, there is usually a significant jump in system reliability at the end of the test phase. The fixes incorporated into the system between test phases are called delayed fixes. See FIGURE 4-8.

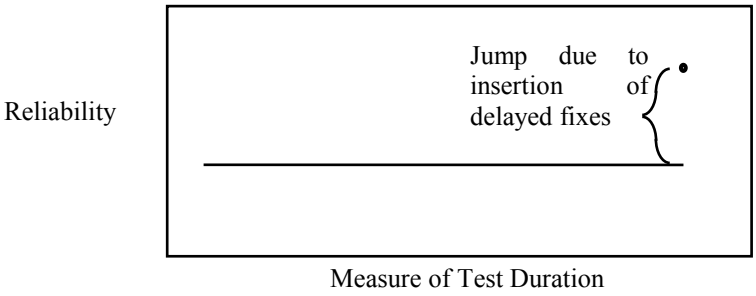


FIGURE 4-8. *Graph of Reliability in a Test-Find-Test Program*

4.5.8 Test-Fix-Test with Delayed Fixes.

The test program commonly used in development testing employs a combination of the two types of fix insertions discussed above. In this case, some fixes are incorporated into the system

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during the test while other fixes are delayed until the end of the test phase. Consequently, the system reliability will generally be seen as a smooth process during the test phase and then jump due to the insertion of the delayed fixes. See FIGURE 4-9.

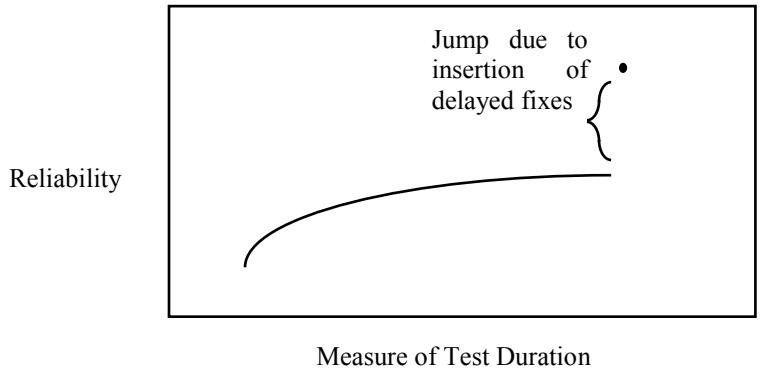


FIGURE 4-9. *Graph of Reliability in a Test-Fix-Test Program with Delayed Fixes*

4.6 Combined Influences of Factors on Reliability Growth Curve Shape.
In order to reach the goal reliability, the development-testing program will usually consist of several major test phases. Within each test phase the fix insertion may be carried out in any one of the three ways discussed above. As an example, suppose that testing were conducted during the validation and full-scale development stages of the program. Each stage would have at least one major test phase, implying a minimum of two major test phases for the program. In this case, there would be $3^2 = 9$ general ways the reliability may grow during the development test.

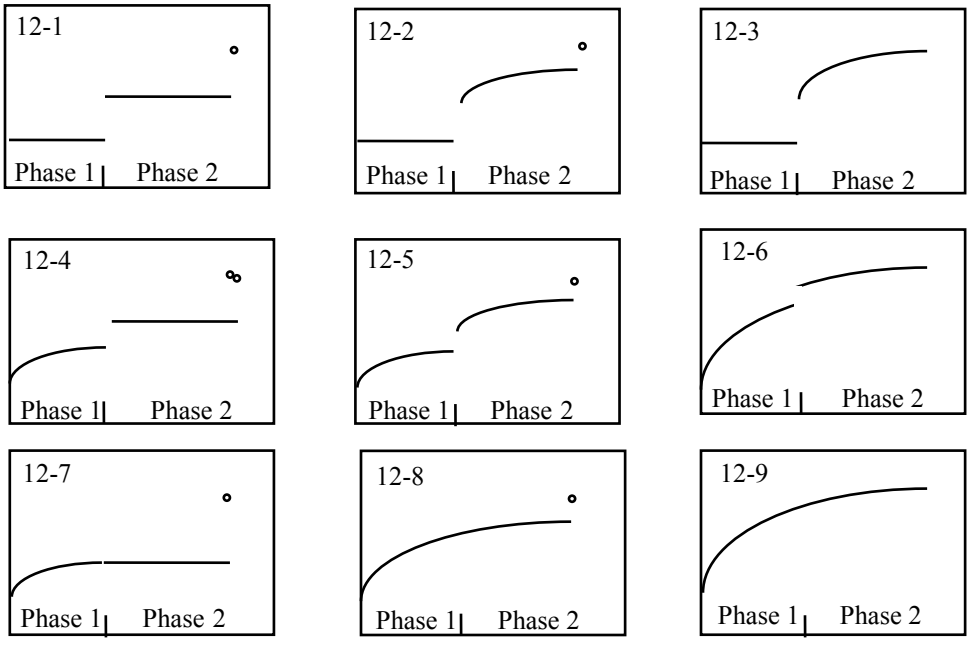


FIGURE 4-10. *The Nine Possible General Growth Patterns for Two Test Phases*

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Row 1 shows Phase 1 as having all fixes delayed until the end of the testing phase. Row 2 shows Phase 1 as having some fixes inserted during test and some delayed. Row 3 shows Phase 1 as having all fixes inserted during test, with none delayed. Column 1 shows Phase 2 as having all fixes delayed until the end of the testing phase. Column 2 shows Phase 2 as having some fixes inserted during test and some delayed. Column 3 shows Phase 2 as having all fixes inserted during test, with none delayed. FIGURE 4-10, charts 12-1 and 12-9 represent the two extremes in possible growth test patterns.

4.6.1 Statistical Advantages of Test-Fix-Test.

There are some distinct statistical advantages to following a complete test-fix-test program:

a. The estimated value of reliability at any point along the smooth growth curve is an instantaneous value. That is, it is not dragged down by averaging with the failures that accrued due to earlier (and hopefully) less reliable configurations;
b. Confidence limits about the true value of reliability can be established.
c. While the impact of the jumps in reliability can be assessed using a mix of some engineering judgment (this will be discussed in the section on Reliability Growth Projection) and direct calculation, the estimate of reliability in a test-fix-test program is based solely on data.
d. In a test-fix-test program, the goodness of the design changes is continuously being assessed in the estimate of reliability.

A development stage may consist of more than one distinct test phase. For example, suppose that testing is stopped part way through the full-scale development stage, and delayed fixes are incorporated into the system. The testing in this case may be considered as two major test phases during this stage, giving three phases for the whole program. If a program had three major test phases then there would be $3^3 = 27$ patterns of reliability growth. Obviously this manner of determining the possible number of growth patterns can be extended to any number of phases.

4.6.2 Growth Curve Re-Initialization.

The differences in the growth curves between phases shown in FIGURE 4-10, charts 12-5 and 12-6 illustrate how changes in factors beyond merely incorporating planned delayed fixes such as testing, replacing prototype subsystems with production ready subsystems, or the correction of a particularly troublesome failure mode can alter growth rates. Underlying chart 12-6 is the assumption that the testing environment and engineering efforts are the same across test phases, thus the continuation of the same growth curve into the succeeding phase, after the jump for delayed fixes. In FIGURE 4-10, chart 12-5 some factor influencing the rate of growth has substantially changed between the phases and is reflected in a new growth curve for the succeeding phase. This is called reinitializing the growth curve. It should be emphasized that re-initialization of a growth curve is only justified if the testing environment is so different as to introduce a new set of failure modes, or the engineering effort is so different as to be best represented as a totally new program.

4.6.3 Shape Changes Due to Calendar Time.

Reliability growth is often depicted as a function of test time for evaluation purposes. For management and presentation purposes it may be desirable to portray reliability growth as a

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function of calendar time. This can be accomplished by determining the number of units of test duration that will have been completed at each measure point in calendar time and then plotting the reliability value that corresponds to the completed test duration above that calendar point. This is a direct function of the program schedule. FIGURE 4-11 shows the reliability growth of a system as a function of test time and calendar time.

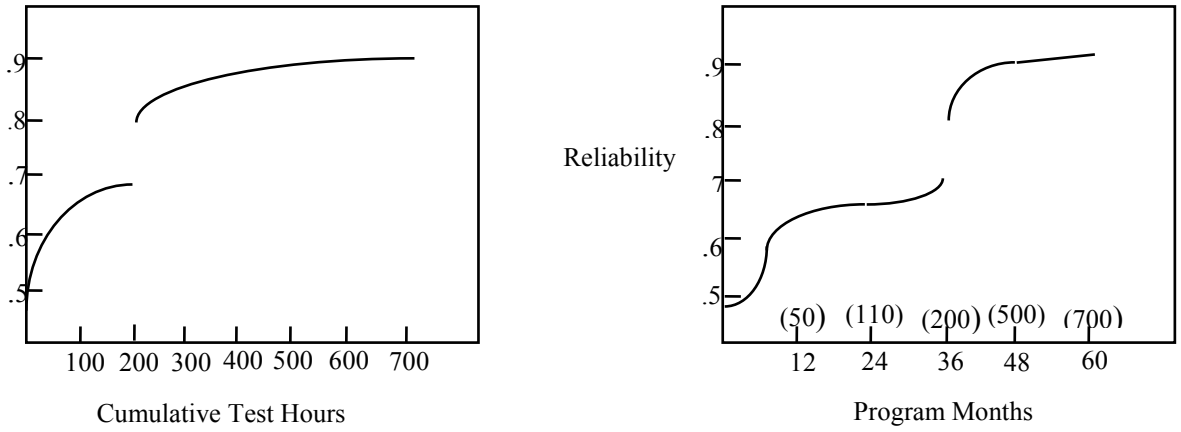


FIGURE 4-11. *Comparison of Growth Curves Based on Test Duration Vs Calendar Time*

4.6.4 Reliability Growth Programmatic Concepts.
There are two levels of consideration for planning and controlling reliability growth: the treatment of growth over the development program (globally over the entire program or on a phase-by-phase basis); and the other considers at what level or levels of indenture the system is tested. The associated management activities concerned include analysis of previous programs (where appropriate), constructing planned curves, and determining or estimating demonstrated and projected reliability values.

4.6.5 Levels of Consideration for Growth.
Planning and controlling reliability growth can be divided as to levels of consideration along both a program basis and an item under test basis.

a. Program considerations:
   i. Global: This approach treats reliability growth on a total basis over the entire development program.
   ii. Local: The other approach treats reliability growth on a phase-by-phase basis.

b. Item Under Test considerations:
   i. System Level: The entire system as it is intended to be fielded is tested.
   ii. Subsystem Level: The obvious meaning is the testing of a major and reasonably complex component of the whole system (e.g., an engine for a vehicle). Sometimes, the subsystem would seem to be an autonomous unit, but because the requirement is for this unit to operate in conjunction with other units to achieve an overall functional goal it is really only part of "the system" (e.g., radar for an air defense system). An additional consideration would be where a system consists of, say, two subsystems, one that is a new development where growth would be expected and planned for, and one that has been developed and used in a similar

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system and thus no growth would be expected. The appropriate level of consideration can be different at different times within the development.

In addition to program and item under test considerations another consideration is that reliability models are classified according to the usage of the system. They fall into two groups -- continuous and discrete models -- and are defined by the type of outcome that the usage provides. Continuous models are those that apply to systems for which usage is measured on a continuous scale, such as time in hours or distance in miles. For continuous models, outcomes are usually measured in terms of an interval or range; for example, mean time/miles between failures. Discrete models are those that apply to systems for which usage is measured on an enumerative or classificatory basis, such as pass/fail or go/no-go. For discrete models, outcomes are recorded in terms of distinct, countable events that give rise to probability estimates.

4.6.6 Analysis of Previous Programs.

Analysis of previous similar programs is used to develop guidelines for predicting the growth during future programs. Such analysis may be performed on either overall programs or individual program phases, or both. Of particular interest are the patterns of growth observed and the effect of program characteristics on initial values and growth rates. Reference [3] provides a useful guide in choosing appropriate growth rates for various system types.

4.7 Reliability Growth Planning, Tracking and Projection.

4.7.1 Planning.

As noted previously, reliability planning addresses program schedules, amount of testing, resources available, and the realism of the test program in achieving its requirements. Planning is quantified and reflected through a reliability growth program plan curve(s). This curve(s) establishes interim reliability goals throughout the test program. The two key planning growth curves – Idealized and Planned – are discussed in the following two sections.

4.7.2 Idealized Growth Curve.

An Idealized Growth Curve is a planned growth curve that consists of a single smooth curve based on initial conditions, an assumed growth rate, and/or planned management strategy. This curve is a strict mathematical function of the input parameters across the measure of test duration (e.g., time, distance, trials), thus the name “Idealized.” No program can be expected to assume this exact mathematical ideal shape, but it is useful in setting interim goals. See FIGURE 4-12.

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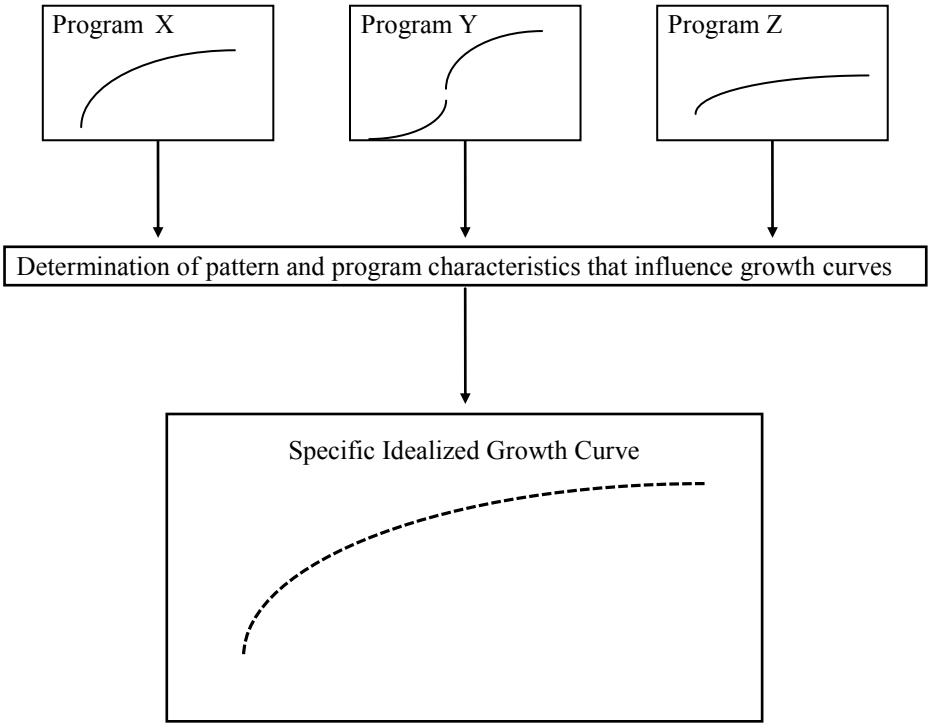
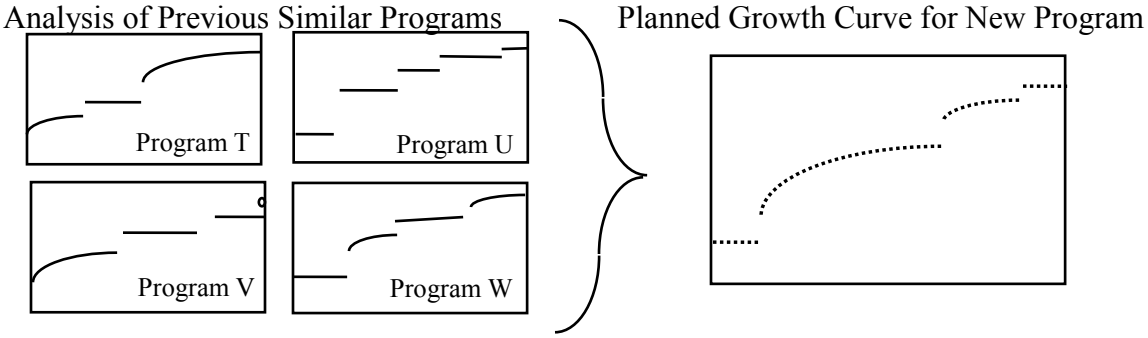


FIGURE 4-12. *Global Analysis Determination of Planned Growth Curve*

4.7.3 Planned Growth Curve.

The planned growth curve is a picture of the anticipated reliability growth for the entire program. It is an essential part of the reliability growth management methodology and is important to any reliability program. This curve is constructed early in the development program generally before hard reliability data are obtained and is typically a joint effort between the program manager and contractor. Its primary purpose is to provide management with guidelines as to what reliability can be expected at any stage of the program and to provide a basis for evaluating the actual progress of the reliability program based upon generated reliability data. The planned growth curve can be constructed on a phase-by-phase basis. See FIGURE 4-13 below.

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Determination of pattern and phase characteristics that influence growth curves.

FIGURE 4-13. *Development of Planned Growth Curve on a Phase by Phase Basis*

4.7.4 Other Planning Considerations.

It is important that sufficient testing be planned and that this testing should be reflective of the Operational Mode Summary/Mission Profile (OMS/MP). In reliability demonstration testing, the concept of operating characteristics curves has been used in planning test time and allowable failures. This concept can be usefully extended to developing reliability growth planning curves where the growth curve follows the Duane failure pattern, i.e., power law expected number of failures. In particular, a system planning curve and associated test duration can be constructed such that if growth occurs in accordance to the planning curve for the planned test duration then, with a prescribed probability, growth test data will be generated that provide a γ statistical lower confidence bound that will meet or exceed the technical requirement (TR). In the above, γ denotes the specified confidence level at which the TR is to be demonstrated. Note the generated test data not only consist of the number of growth test failures but also the cumulative test durations associated with each failure. Also note that the prescribed probability mentioned above will be termed the probability of acceptance since it plays the same role as the probability of acceptance for a fixed configuration demonstration test. For a fixed configuration demonstration test, the discrimination ratio – the reliability associated with the producers risk, α, over the reliability associated with the consumers risk, β – has often been used as a guide in determining test time. As a general rule of thumb, the discrimination ratio of the producer risk (contractor design to MTBF) to the consumer risk (government requirement MTBF (with confidence)) is generally around 2-3. For the system, subsystem and threshold plans of this handbook, the ratio of interest is M_G, the contractor's goal MTBF to the reliability technical requirement, TR (which is to be demonstrated with confidence). Now because we are growing reliability over a test program rather than having a fixed configuration for a demonstration test, the ratios expected would typically be smaller. Tables of probability of acceptance for ratios of *M(T)/TR* from 1.00 to 3.00 and the expected number of failures up to 100 are provided for demonstrating *TR* with 70%, 80% and 90% confidence. FIGURE 4-14 presents the three confidences of demonstrating *TR* with at least 0.50 probability as a function of the *M(T)/TR* ratio and the expected number of failures. The test duration corresponding to a point *(x, y)* on a confidence curve in FIGURE 4-14, can be shown to satisfy the following:

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$$T = (1 - x)\,(TR)\,xy \qquad 4.7\text{-}1$$

Where $x = E(N)$, the expected number of failures over test duration T, and $y = M(T)/TR$. Note also that

$$M(T) = \left(\frac{M_I}{(1-\alpha)}\right)(T/t_I)^{\alpha} \qquad 4.7\text{-}2$$

In the above equation, α denotes the growth rate parameter for the planning curve.

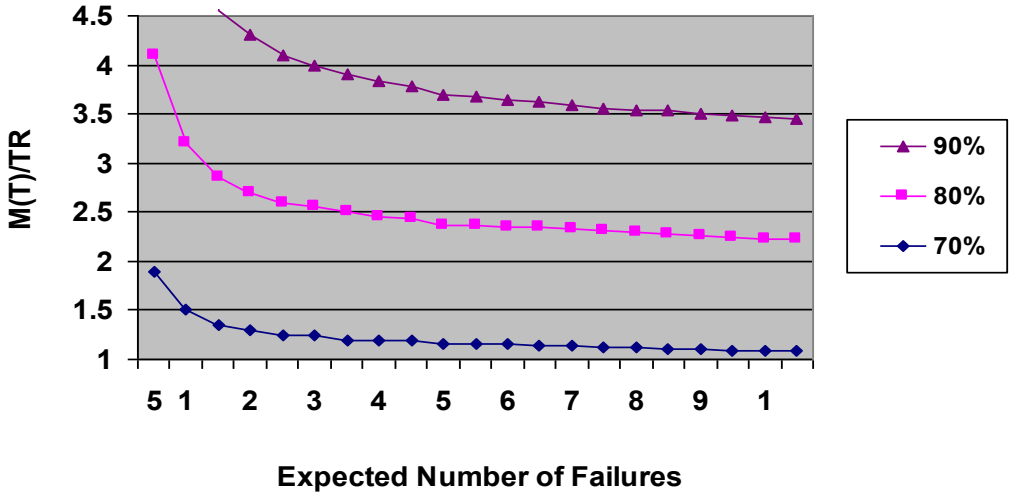


FIGURE 4-14. *Probability of demonstrating TR w/% Confidence as a Function of M(T)/TR and Expected Number of Failures*

4.7.5 Reliability Growth Tracking.

Reliability growth tracking provides management the opportunity to gauge the progress of the reliability effort for the system based on test results. This is done by determining if system reliability is increasing with time (i.e., growth is occurring) and to what degree (i.e., growth rate) and estimating the demonstrated reliability which is an estimate based on test data for the system configuration under test at the end of the test phase. This latter estimate is based on the actual performance of the system tested and not on some future configuration.

Reliability growth tracking offers a viable method for combining test data from several configurations to obtain a demonstrated reliability estimate for the current system configuration, provided the reliability growth tracking model adequately represents the combined test data. The fit of the data to the model should be ascertained by visual plots and statistically tested to determine the adequacy of it as a model for tracking growth.

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4.7.6 Demonstrated Reliability.

A demonstrated reliability value is based on actual test data and is an estimate of the current attained reliability. The assessment is made on the system configuration currently undergoing test, not on an anticipated configuration, nor a prior configuration. This number allows for the effects of even recently introduced fixes into the system as its calculation incorporates the trend of growth established over the history, to date, of the development program.

4.7.7 Reliability Growth Tracking Curve.

The reliability growth tracking curve is the curve that best fits the data being analyzed. It can be based on data solely within one phase or data from several phases. Whatever period of testing is used to form a database, this curve is the statistical best representation from a family of growth curves of the overall reliability growth of the system. It depicts the trend of growth that has been established over the database. Thus, if the database covers the entire program to date, the right end point of this curve is the current demonstrated reliability. See FIGURE 4-15 below.

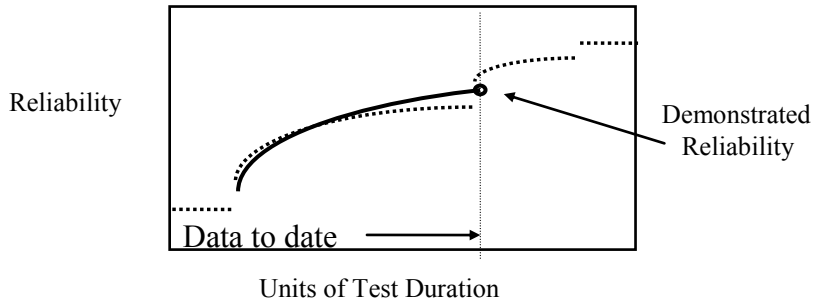


FIGURE 4-15. *Reliability Growth Tracking Curve*

4.8 Reliability Growth Projection.

4.8.1 Projected Reliability.

A reliability projection is an assessment of reliability that can be anticipated at some future point in the development program given corrective action, be it at the start of the next phase (e.g., AMSAA/Crow Projection Model) or some future point in time (AMSAA Maturity Projection Model (AMPM)). The projection is based on the achievement to date and engineering assessments of future program characteristics. Projection is a particularly valuable analysis tool when a program is experiencing difficulties because it enables investigation of program alternatives. One can determine reliability "potential" by sensitizing on the fix effectiveness factors for a proposed management strategy. Projections can be used as a system or subsystem "maturity" metric such as the initial failure rate surfaced.

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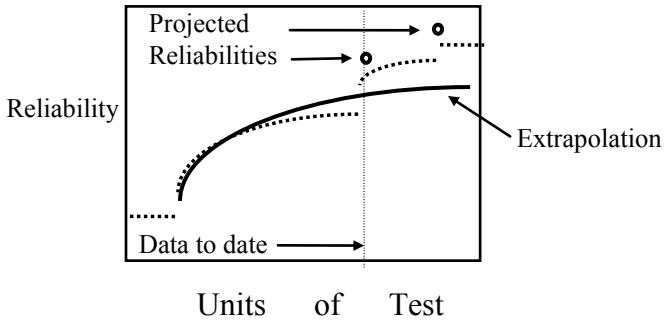


FIGURE 4-16. *Extrapolated and Projected Reliabilities*

4.8.2 Extrapolated Reliability.
Extrapolating a growth curve beyond the currently available data shows what reliability a program might be expected to achieve, as a function of additional test duration, provided the conditions of test and the engineering effort to improve reliability are maintained at their present levels (i.e., the established trend continues). The farther reliability is extrapolated the more problematic the result.

4.9 Threshold.
A threshold is a value, in the rejection region of a statistical test of hypothesis, which indicates that an achieved or demonstrated reliability below the value is not in conformance with the idealized growth curve. A threshold value is not a lower confidence bound on the true reliability; it is used simply to conduct a test of hypothesis. Threshold values are computed at particular points in time, referred to as milestones, which are major decision points.

4.10 Threshold Program.
The threshold program is a tool for determining, at selected program milestones, whether the reliability threshold of a system is failing to progress according to the idealized growth curve established prior to the start of the growth test. The program can be used to compare a reliability point estimate, which is based on actual failure data from a growth test, against a theoretical threshold value. The test statistic in this procedure is the point estimate calculated from the test data. If this estimate falls at or below the threshold value this would raise a red flag and indicate that the achieved reliability is statistically not in conformance with the idealized growth curve. At that point management might want to take some kind of action to restore reliability to a higher level, perhaps through restructuring the program, more intensive corrective action process, change of vendors, additional lower level testing, etc.

4.11 Reliability Growth Models.
Reliability software is available from several vendors. Please see Appendix C Section 7.

4.12 References.

[3] Ellner, Paul M. and Trapnell, Bruce, AMSAA Interim Note IN-R-184, AMSAA Reliability Growth Data Study, January 1990

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5 RELIABILITY GROWTH

5.1 Reliability Growth Planning.

5.1.1 Background.

The goal of reliability growth planning is to optimize testing resources, quantify potential risks, and plan for successful achievement of reliability objectives. A well thought out reliability growth plan can serve as a significant management tool in scoping out the required resources to enhance system reliability and improve the likelihood of demonstrating the system reliability requirement. The principal goal of the growth test is to enhance reliability by the iterative process of surfacing failure modes, analyzing them, implementing corrective actions (fixes), and testing the "improved" configuration to verify fixes and continue the growth process by surfacing remaining failure modes. A critical aspect underlying this process is ensuring that there are adequate resources available to support the desired growth path. This includes addressing program schedules, amount of testing, resources available, and the realism of the test program in achieving its requirements. Planning activities include establishing test schedules, determining resource availability in terms of facilities and test equipment, and identifying test personnel, data collectors, analysts and engineers. Another factor necessary for a successful growth program is allowing for sufficient calendar time during the program to analyze, gain approval and implement corrective actions. Planning is quantified and reflected through a reliability growth program plan curve. This curve may be used to establish interim reliability goals throughout the test program. Two significant benefits of reliability growth planning are:

a. Can perform trade-offs with test time, initial reliability, final reliability, confidence levels, requirements, etc to develop a viable test program.
b. Can assess the feasibility of achieving a requirement given schedule and resource constraints by using historical values for parameters (e.g., growth rate).

5.1.2 Planning Model Limitations.

All reliability growth planning models have limitations. The foremost limitation is that the testing utilized for reliability growth planning should be reflective of the Operation Mode Summary/Mission Profile (OMS/MP). The OMS/MP provides how and under what conditions the system will be used when fielded. These might include hours of operation in the possible modes of usage, environmental conditions expected to operate in, their frequencies etc. If, then, the growth test environment during development reasonably simulates the mission environment stresses then it may be feasible to use the growth test data to statistically estimate the demonstrated technical reliability,(i.e., engineering), requirement (denoted by TR) for system reliability. Such use of the growth test data could eliminate the need to conduct a follow-on reliability demonstration test. The classical demonstration test requires that the system configuration be held constant throughout the test. This type of test is principally conducted to assess and demonstrate the reliability of the system configuration under test, and has often been used for compliance testing. The adaptation of Operating Characteristic (OC) methodology for the reliability growth situation further has allowed development of growth curves so that one can demonstrate requirements with stated confidence.

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5.1.3 Demonstrate reliability requirement with statistical confidence.

Typically a reliability growth plan attempts to lay out a feasible growth path from a current estimate of reliability to a value sufficiently high at the end of testing. In most cases it is of interest to demonstrate the requirement with statistical confidence, e.g. 80%. For many systems with high reliability requirements a statistical demonstration that relies only on data from the Initial Operational Test and Evaluation (IOT&E) or other fixed configuration testing may not be feasible. In order to accomplish this and utilize all applicable data (e.g. under OMS/MP conditions), the concepts of an OC analysis have been extended to the reliability growth setting. Government (consumer) and contractor (producer) statistical risks have been expressed in terms of the underlying growth curve parameters, test duration, and reliability requirement. In particular, a system planning curve and associated test duration can be constructed such that if growth occurs in accordance to the planning curve for the planned test duration then, with a prescribed probability, growth test data will be generated that provide a γ statistical lower confidence bound that will meet or exceed the technical requirement (TR). In the above, γ denotes the specified confidence level at which the TR is to be demonstrated. These risks have been shown to depend solely on the expected number of failures during the growth test and the ratio of the MTBF to be achieved at the end of the growth program to the MTBF technical requirement to be demonstrated with confidence. Formulas have been developed for computing these risks as a function of the test duration and growth curve planning parameters. With the addition of OC curve methodology, planning now can address demonstrating the technical requirement with some stated confidence, as opposed to demonstrating the requirement as a point estimate as was the case in the original handbook.

5.1.4 Management.

Of significant interest to management, the general guidelines for developing planning curves have evolved so that actions under control of management, historical information, and engineering judgment, have been incorporated into the planning process. Additionally, management now plays a role in what failure modes get corrective actions (A-modes and B-modes), what resources are put into corrective actions, how long it takes to develop and implement corrective actions, and when growth might be expected to begin. Crow's Extended Model for projecting reliability sub-divides the B-modes into two categories - BC modes and BD modes. The BC-modes are incorporated in test while the BD-modes are delayed until after test (test-fix-find-test).

5.1.5 Threshold Methodology.

In addition, as cited in the Section 4, threshold methodology has been developed that can be used to statistically test the hypothesis that an achieved or demonstrated reliability is in conformance with the planned idealized growth curve. These threshold values can be computed for one or more major decision points or milestones. FIGURE 5-1 illustrates a planned idealized growth curve along with threshold values.

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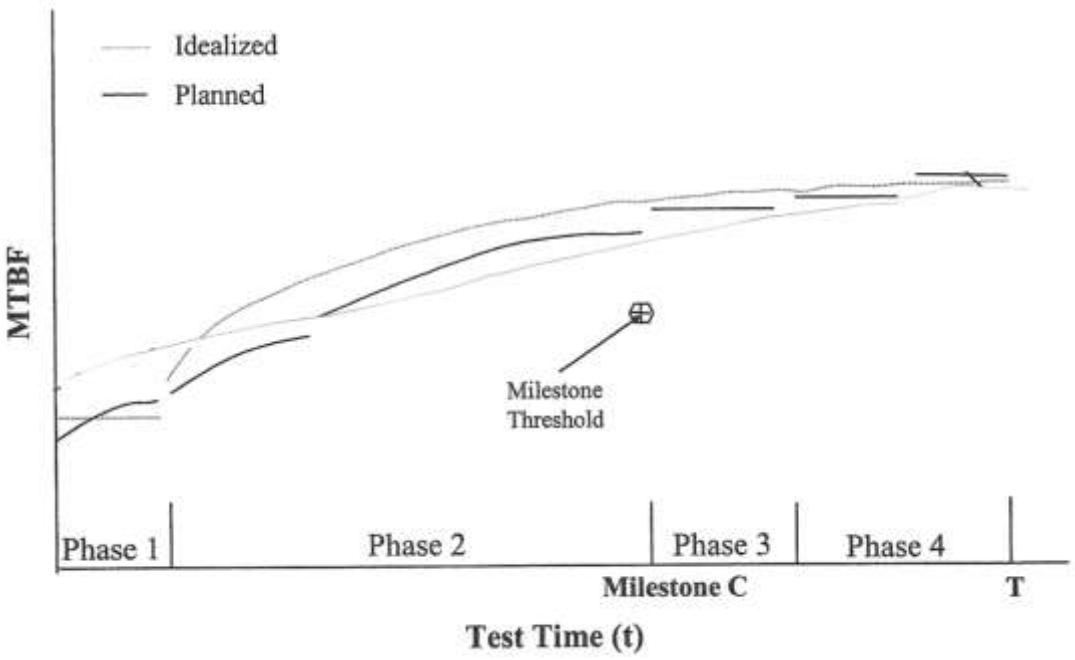


FIGURE 5-1. *Planned Growth Curves with Milestone Threshold*

There are two basic approaches for constructing planned growth curves:

a. Determine the idealized pattern that is expected or desirable and use this as a guide for the detailed planned curve.
b. Proposed plan curve first which satisfies the requirement and interim milestones, the idealized curve is then constructed and evaluated to determine if this pattern is realistic when compared to historical experience.

5.1.6 Planning Areas.

There are two key planning areas: elements under management control and potential risk elements during the planning phase. Elements under management control include:

a. Management Strategy (MS): fraction of system initial failure rate addressed by fixes.
b. Rate at which failure modes are surfaced.
c. Turnaround time for analyzing and implementing corrective action or fixes.
d. Fix effectiveness: percent reduction in rate of occurrence of fixed modes.

The potential risk elements during the planning phase include:

a. Initial MTBF (M_I) and initial time period (t_I);
b. Ratio of initial MTBF, M_I, to final MTBF, M_F;
c. Growth Rate, α;
d. Total Test Time, T.

Under the first risk element, the initial period t_I needs to be large enough so that growth can begin by t_I, i.e., failure mode(s) need to be surfaced and associated corrective action(s) implemented by t_I. For this we need t_I such that there is a high probability of observing at least one failure (when no *MS* assigned) or at least one correctable failure mode. For the latter case

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we need *3*(M_I/MS)* to ensure a 0.95 probability of surfacing at least one correctable failure mode by t_I, where *MS* is the management strategy. To aid in the assessment for the next two risk elements (M_I/M_F and α), (previously cited in Section 4, reference 3), on experiences of tracking reliability for a wide range of systems under development by the Army gives rise to the results in TABLE I below. Presented are means, medians, and ranges of values of systems studied for growth rate (discrete and continuous), ratios for initial to final or mature MTBF's, and fix effectiveness factors.

TABLE I. AMSAA Reliability Growth Data Study Summary: Historical Growth Parameter Estimates

PARAMETER	MEAN/MEDIAN	RANGE
Growth Rate α		
One Shot (Missiles)	0.46/0.47	0.27-0.64
Time or Distance Based	0.34/0.32	0.23-0.53
Initial to Mature Ratio M_I/M_F	0.30/0.27	0.15-0.47
Fix Effectiveness Factors[1]	0.70/0.71	0.55-0.85

The information in the above table may be used as a guide in determining the reasonableness of the ratio of the initial MTBF to the final or required MTBF, growth rates needed to attain requirements, and the range of fix effectiveness factors based on this historical data. For planning purposes, the Management Strategy, *MS*, during early or prototype testing could exceed 0.95; during subsequent testing or Engineering and Manufacturing Development (EMD) it is recommended that the *MS* should be at most 0.95.

The following provides a check list for reviewing reliability growth planning curves:

a. Goal reliability needs to be sufficiently high to have adequate probability of passing Initial Operational Test and Evaluation (IOT&E).
b. The expected initial reliability, M_I or R_I, should be based on expected maturity and prior information, e.g., from previous or similar systems, technology development, or such information as available.
c. The ratio of the initial MTBF M_I to the goal MTBF M_F should not be too low, e.g., less than 0.15.
    i. Desirable to have the ratio above usual historical range of 0.20 to 0.35 the given trend towards reduced test durations.
    ii. Need to achieve sufficient growth in design phase prior to EMD test phase to increase the ratio of M_I to M_F beyond the historical range.
d. The initial test period, t_I, should not be chosen too small relative to M_I or R_I

[1] Personal anecdotal experiences by several AMSAA and ATEC/AEC analysts suggest that software fixes have somewhat higher FEFs, on the order of 0.90 – 0.95 or better.

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i. The initial test period should be large enough so that, with high probability, at least one correctable failure mode occurs during the initial test period.
ii. Corrective action(s) should be implemented by the end of period to commence growth process by t_I as depicted in planning curve.

e. For a realistic initial period t_I and initial reliability (M_I or R_I), the test duration, which includes t_I, should be large enough so that the goal reliability can be attained by IOT&E with a historically achievable growth rate, α.
f. The expected number of failures associated with the planning curve and test duration should be sufficiently large to allow enough corrective action opportunities to grow from the initial to goal reliability.
g. There needs to be sufficient calendar time, facility assets, and engineering personnel to ensure timely implementation of effective corrective actions to surface failure modes prior to IOT&E.
h. If corrective actions are to be implemented at only a few designated points during the development program, then the depicted expected growth pattern should reflect this.

5.1.6.1 Planning Models.
The following five models can be utilized to aid in planning a reliability growth program for a complex system. Which model to be employed depends on the particular circumstances of the program and what viable assumptions that can be made. The possible models include:

a. Planning Model based on the Power Law (AMSAA/Crow, Duane Postulate),
b. System Level Planning Model (SPLAN);
c. Subsystem Level Planning Model (SSPLAN);
d. Planning Model based on Projection Methodology (PM2), and
e. Threshold Program (TP).

The last model is not a growth model per se but rather a program or methodology to develop interim goals to ascertain whether the program is proceeding in accordance with the planned growth curve. The reliability point estimate is compared to a reliability value that is used to mark off a rejection region for the purpose of conducting a test of hypothesis to determine if the achieved reliability of a system is growing according to plan.
Models a., b. and c. are based on the power law; the key difference between a. versus b. and c. is the latter models incorporate OC analysis into determining the planned model, and thus a is restricted to demonstrating a technical requirement (TR) as a point estimate as opposed to demonstrating the TR with confidence. PM2 is derived from a fundamental relationship between the expected number of failure modes surfaced and the cumulative test duration. Exact expressions for the expected number of surfaced failure modes and system failure intensity as functions of test time are presented under the assumption that the surfaced modes are mitigated through corrective actions. We obtain a non-empirical relationship between the mean test duration between system failures and cumulative test duration that can be utilized for reliability growth planning. A significant advantage to the PM2 approach is that it does not rely on an empirically derived relationship such as the Duane based approach. One needs to determine the number of hours, or miles, per unit per month over the test period. Finally, fix implementation periods need to be specified within the planned test schedule. The following sections provide short overviews for each of the five models after which each will be developed in detail.

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5.1.6.2 Planning Model based on AMSAA/Crow Model (Duane Postulate) Overview.

5.1.6.1.1 Purpose.

The purpose of the AMSAA/Crow model is to construct idealized system reliability growth curves, identify test time and growth rate required to improve system reliability, and aid in demonstrating the system reliability requirement as a point estimate.

5.1.6.1.2 Assumptions.

This approach assumes that within a phase, reliability growth can be modeled as a Non-Homogeneous Poisson Process (NHPP) with power law Mean Value Function (MVF), $\theta(t) = \lambda t^{\beta}$. It also assumes that based on the failures and test time within a phase, the cumulative failure rate is linear on a log-log scale. This is a local, within test phase pattern for reliability growth comparable to the global pattern noted by Duane. While applicable to systems measured on a continuous scale, it also may be used for high reliability and a large number of trials for go/no go systems such as missiles.

5.1.6.1.3 Limitations.

Limitations of the model include:

a. sufficient opportunities for fix implementation are required to allow growth to be portrayed as a smooth curve;
b. the expected number of failures needs to be sufficiently large;
c. the portion of testing utilized for reliability growth planning should be reflective of the OMS/MP.

5.1.6.1.4 Benefits.

The model allows for generation of a target idealized growth curve; can be utilized for discrete data when there are a large number of trials and low probability of failure.

5.1.6.1.5 Planning Factors.

The idealized curve has a baseline value M_I over the initial test phase which ends at time t_1. M_I is the average MTBF over the first phase. From t_1 to the end of testing at time T, the idealized curve increases steadily according to a learning curve pattern till it reaches the final reliability requirement M_F. T and the growth rate α are iterated to develop the plan satisfying the constraints. Subsequent to publishing the 1981 version of MIL-HDBK-189 and prior to the development of SPLAN, a function, *Prob*, was developed that assured a designated probability of observing at least one failure in the initial time t_I (subsequently the management strategy (*MS*) was also included). In this case the function is:

$$\boldsymbol{Prob = 1 - e^{-\left(\frac{t_I}{M_I}\right)}} \qquad 5.1\text{-}1$$

which for Prob=0.95 results in t_1 approximately equal to 3 times M_I. After development of *MS*, $3*(M_I/MS)$ was used to satisfy the Prob of 0.95.

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5.1.6.3 SPLAN Model Overview.

5.1.6.3.1 Purpose.

The purpose of the SPLAN model is to construct idealized system reliability growth curves, identify test time required to improve system reliability, and aid in demonstrating the system reliability requirement with confidence.

5.1.6.3.2 Assumptions.

There are two assumptions associated with the SPLAN model:

a. test duration is continuous and;
b. the number of failures during test follows from a NHPP with power law Mean Value Function (MVF).

5.1.6.3.3 Limitations.

Model limitations of SPLAN include:

a. sufficient opportunities for fix implementation are required to allow growth to be portrayed as a smooth curve;
b. the expected number of failures needs to be sufficiently large;
c. the portion of testing utilized for reliability growth planning should be reflective of the OMS/MP;
d. the initial test length should be reasonably small to allow for reliability growth and;
e. the initial MTBF cannot be specified independent of the length of the initial test phase.

5.1.6.3.4 Benefits.

SPLAN also has a number of benefits. The model:

a. allows for generation of a target idealized growth curve;
b. can specify desired probability of achieving goal requirement with confidence using Operating Characteristics (OC) Curves this will be referenced in section detailing SPLAN and;
c. can be utilized for discrete data when there are a large number of trials and low probability of failure.

5.1.6.3.5 Planning Factors.

The initial conditions or planning factors include the test time over the initial test phase before implementation of corrective actions t_I, the initial average MTBF M_I, the final MTBF M_F, the management strategy *MS,* and the specified probability of observing at least 1 correctable or B-mode failure, *Prob.* Four of the conditions or factors are chosen and the fifth is determined. The probability of observing at least one correctable or B-mode failure, *Prob,* is given by the following equation:

$$\mathbf{Prob = 1 - e^{-\left(\frac{t_I * MS}{M_I}\right)}} \qquad 5.1\text{-}2$$

As a general rule, the initial time period should be at least approximately three times greater than the ratio of the initial MTBF to the management strategy to ensure a high probability, say 0.95,

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of surfacing at least one failure by the end of $t_{I.}$. This initial test time is sufficient such that after $t_{I.}$ one can assume the MIL-HDBK-189 growth pattern applies. This pattern does not apply over the first test phase since it would imply an MTBF of zero at the origin.
Note that the planning factors include an area not part of the original MIL-HDBK-189, which management has control – management strategy or the fraction of the initial failure intensity that is expected to be addressed through corrective actions.

5.1.6.4 SSPLAN Model Overview.

5.1.6.4.1 Purpose.
The purpose of SSPLAN is to develop subsystem reliability growth planning curves that achieve a system MTBF objective with a specified confidence, and determine the subsystem test times required to meet an MTBF objective with specified confidence. Both growth and non-growth systems may be included.

5.1.6.4.2 Assumptions.
SSPLAN assumptions include:

a. test duration is continuous;
b. the system may be represented as a series of independent subsystems;
c. for growth subsystems, the number of failures is in accordance with a NHPP with power law MVF.

5.1.6.4.3 Limitations.
The limitation of the SSPLAN model include:

a. sufficient opportunities need to exist to insert fixes for each growth subsystem in order to portray subsystem growth via smooth curves;
b. the expected number of failures needs to be sufficiently large for each growth subsystem;
c. the portion of subsystem testing utilized for reliability growth planning should preferably be reflective of the OMS/MP;
d. problems with subsystem interfaces may not be captured;
e. the initial test length must be reasonably small to allow for reliability growth and;
f. the length of the initial test phase for each growth subsystem should be specified with an associated average initial MTBF.

5.1.6.4.4 Benefits.
The benefits of SSPLAN are that the model:

a. allows generation of a target idealized growth curve, based on subsystem growth testing;
b. can specify desired probability of achieving a goal requirement with confidence and;
c. can aggregate test duration from common subsystems on system variants under test.

5.1.6.4.5 Planning Factors.
The factors include both system level and subsystem level. For the system level they include the system objective M_F or technical requirement TR. The subsystem planning factors for developing the system planning curve include the subsystem initial test time t_I, the subsystem

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initial MTBF M_I, the management strategy *MS*, and the probability of observing at least one B-mode failure for the subsystem Prob. The three strategies or options are:

a. t_I, M_I, *MS*
b. t_I, *MS*, *Prob*,
c. M_I, *MS*, *Prob*.

5.1.6.5 PM2 Model Overview.

5.1.6.5.1 Purpose.
The U.S. Army Materiel Systems Analysis Activity (AMSAA) has developed a new reliability growth planning model, referred to as Planning Model Based on Projection Methodology or PM2 Reference to be cited in section detailing PM2. The new model can:

a. aid in constructing a reliability growth planning curve over a developmental test program useful to program management;
b. serve as a baseline against which reliability assessments can be compared and;
c. highlight the need to management when reallocation of resources is necessary. PM2 does not have a growth rate parameter, nor is there a comparable quantity.

5.1.6.5.2 Assumptions.
There are a number of reasonable assumptions associated with PM2. The first assumption is that there are a large number of potential failure modes. As a rule of thumb, the potential number of failure modes should be at least five times the number of failure modes that are expected to be surfaced during the planned test period. Second, each failure mode time to first occurrence is assumed exponential. Finally, it is assumed that each failure mode occurs independently and causes system failure.

5.1.6.5.3 Limitations.
All reliability growth planning models have limitations. The first limitation associated with PM2 is that the portion of testing utilized for reliability growth planning should be reflective of the Operation Mode Summary/Mission Profile (OMS/MP). This limitation is not unique to PM2 – it is a limitation associated with all reliability growth planning models. Second, one needs to postulate a baseline test schedule. That is, one should determine the number of hours, or miles, per unit per month over the test period. Finally, fix implementation periods need to be specified within the planned test schedule.

5.1.6.5.4 Benefits.
There are a number of benefits associated with the new planning model. PM2 is unique in comparison to other reliability growth planning models in that it utilizes planning parameters that are directly influenced by program management. Some of these parameters include:

a. the initial system MTBF;
b. the fraction of the initial failure rate addressable via corrective action (referred to as management strategy);
c. the goal system MTBF;
d. the average fix effectiveness of corrective actions;
e. the duration of developmental testing and;
f. the average delay associated with fix implementation.

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A second benefit of PM2 is that the model can determine the impact of changes to the planned test schedule, and associated fix implementation periods. Third, PM2's measures of programmatic risk are not sensitive to the length of the initial test phase (which is a limitation of the original MIL-HDBK-189 planning model). Finally, PM2 can be applied to programs with limited opportunities for implementation of corrective actions.

5.1.6.6 PM2 Discrete Model Overview.

5.1.6.6.1 Purpose.
Based on research conducted by J. B. Hall in his PhD Dissertation and documented in ATEC TN, "Reliability Growth Planning for Discrete Systems," a new model of the PM2 for discrete systems, PM2 Discrete Model (PM2-Discrete), was developed. The purpose of PM2-Discrete, is not just a planning model, but also a planning methodology that possesses concomitant measures of programmatic risk and system maturity.

5.1.6.6.2 Assumptions.

a. Initial failure mode probabilities of occurrence $p_1,\ldots,p_k$ constitute a realization of an independent and identically distributed (iid) random sample $P_1,\ldots,P_k$ such that $P_i \sim Beta(n,x)$ for each $i=1,\ldots,k$.
b. The number of trials $t_1,\ldots,t_k$ until failure mode first occurrence constitutes a realization of a random sample $T_1,\ldots,T_k$ such that $T_i \sim Geometric(p_i)$ for each $i=1,\ldots,k$.
c. Potential failure modes occur independently of one another and their occurrence is considered to constitute a system failure.

When failures are observed during testing their corresponding failure modes are identified, and management may (or may not) address them via corrective action. If a given failure mode (e.g., failure mode *i*) is addressed, it is assumed that either: (1) an FEF is assigned by expert judgment with very detailed knowledge regarding the proposed engineering design modification or; (2) a demonstrated FEF, $\hat{d}_i$, is used.

5.1.6.6.3 Limitations.
Same limitations as PM2-continuous except for the usage domain.

5.1.6.6.4 Benefits.
First methodology specifically developed for discrete systems. First quantitative method available for formulating detailed plans in the discrete usage domain.

5.1.6.7 Threshold Program Model Overview.

5.1.6.7.1 Purpose.
The purpose of the threshold program is to determine, at selected program milestones, whether the reliability of a system is failing to progress according to the idealized growth curve

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established prior to the start of the growth test. Compare a reliability point estimate (based on actual failure data) against a theoretical threshold value.

5.1.6.7.2 Assumptions.
Test duration is continuous. Reliability growth is governed by a NHPP with Power Law MVF. The threshold program embodies a methodology that is best suited for application during a reliability growth test referred to as the Test-Analyze-Fix-Test (TAFT) program. Under a TAFT program, when a failure is observed, testing stops until the failure is analyzed and a corrective action is incorporated on the system. Testing then resumes with a system that has (presumably) a better reliability. The graph of the reliability for this testing strategy is a series of small increasing steps that can be approximated by a smooth idealized curve.

5.1.6.7.3 Limitations.
All limitations for SPLAN apply.

5.1.6.7.4 Benefits.
For multiple thresholds, it examines conditional distributions of MTBF estimate given previous threshold(s) not breached. It provides a hypothesis test of whether growth is occurring along a specified idealized curve. It can be utilized for system or subsystem growth curves.

5.1.6.7.5 Planning Factors.
Same as SPLAN and AMSAA/Crow. Only three inputs – the total test time T, the final MTBF M_F and the growth rate alpha – are necessary to define the idealized curve which the program uses to build a distribution of MTBF values. The program does, however, require four additional inputs – the initial MTBF M_I, initial time t_I, Milestone time, and the milestone MTBF distribution percentile point.

5.1.6.8 AMSAA/Crow or Original MIL-HDBK-189.

5.1.6.8.1 Introduction.
While Appendix B of the original 1981 Handbook lists 8 discrete and 17 continuous growth models, the Handbook is based on Duane's work and L.H. Crow's more generalized work. James T. Duane, an engineer with General Electric's Motor and Generator Department, published a paper titled "Learning Curve Approach to Reliability Monitoring" in IEEE Transactions on Aerospace, Vol. 2, No. 2, 1964. Duane analyzed data for several systems developed by General Electric in an effort to determine if any systematic changes in reliability improvement occurred during development for these systems. The paper recorded his observation that if changes to improve reliability (which are now termed fixes) are incorporated into the design of a system under development, then on a log-log plot, the graph of cumulative failure rate vs. cumulative test time is linear. This observation has become known as the "Duane Postulate." This empirically derived statement is the key to one of the most commonly accepted growth model in use today. A graph given in Duane's paper is shown as straight lines and these lines are based on least squares fit of the data. The negative slope of each line is defined to be the growth rate, α, for that line. The material in Sections 5.1.12.1.2, 5.1.12.1.3 and 5.2.7 is as stated in Appendix A, AMSAA TR-652, AMSAA Reliability Growth Guide, September 2000 [1].

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5.1.6.8.2 Duane's Growth Model.
Let *N(t)* = the total number of failures by time *t*. Then the average failure rate, also called cumulative failure rate *C(t)*, can be found by dividing *N(t)* by *t*.

$$C(t) = \frac{N(t)}{t} \qquad 5.1\text{-}3$$

Let δ be the *y*-intercept on a log-log plot of the straight line that Duane postulated. The slope-intercept formula for this line then becomes:

$$Log\ C(t) = \delta - \alpha\ Log\ t \qquad 5.1\text{-}4$$

where log denotes the natural (base e) logarithm (although any base could be used) and α is the slope of the line.

The Duane Postulate:

$Log\ C(t) = \delta - \alpha\ Log\ t$

If we let $\delta = ln\ \lambda$ and take anti-logs, then

$$C(t) = \lambda t^{1-\alpha} \qquad 5.1\text{-}5$$

Multiplying *C(t)* by *t* gives *N(t)*, and multiplying $t^{-\alpha}$ by *t* adds 1 to the exponent, $t^{1-\alpha}$.

So

$$N(t) = \lambda t^{1-\alpha} \qquad 5.1\text{-}6$$

$$r(t) = \frac{d\,N(t)}{d\,t} = \lambda(1-\alpha)t^{-\alpha} \qquad 5.1\text{-}7$$

Duane's model thus has two parameters, α and λ. The first, α is the shape parameter and determines the shape of the growth curve. The second, λ, is the scale or size parameter for the curve. With these two parameters, the cumulative number of failures *N(t)*, the average failure rate *C(t),* and the instantaneous failure rate *r(t)* can be calculated for any time *t* within the test. Further, given α and λ, it is possible to solve for *t*, the amount of testing time it will take to

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achieve a specific reliability. This assumes that the factors affecting reliability growth remain unchanged across the development.

5.1.6.8.3 Drawbacks to Duane's Method.
Duane stated that α could be universally treated as being 0.5, as that seemed to be the modal value within his database. This has since been shown to be unrealistic, as per Table I. First, it does not allow for different test environments causing failures to be surfaced at different rates. Secondly, it does not account for differences in types of systems under development, and it does not account for different levels of engineering efforts causing different rates of fix insertion. The reliability values calculated using his method is treated as being deterministic. That is, there is no allowance for the variation that is typically observed about an estimated value, and there is no way of judging whether the observed value, which rarely matches the estimated value, is close enough. Further, there is no way to check whether the model is valid for the current test situation. All Duane growth curves pass through the origin of the graph. That is, the item under test is imputed to have zero reliability at the start of test.

5.1.6.9 The AMSAA/Crow Growth Model.
Dr. Crow, while at the U.S. Army Materiel Systems Analysis Activity's Reliability and Maintainability Division, and based on his PhD dissertation, published Reliability Analysis for Complex, Repairable Systems, Technical Handbook No. 138, December 1975, AMSAA, Aberdeen Proving Ground, Maryland. In this report, Dr. Crow explored the advantages of using a Non-homogeneous Poisson Process (NHPP) with a Weibull intensity function to model several phenomena, including reliability growth. If system failure times follow the Duane Postulate, then they can be modeled as a NHPP with Weibull intensity function. To make the transition from Duane's formulae to the Weibull intensity functional forms, β has to be substituted for $1-\alpha$. Thus the parameters in the Crow model are α and β, where β determines the shape of the curve. The physical interpretation of β (called the growth parameter) is the ratio of the current (instantaneous) MTBF to average (cumulative) MTBF at time t.

This stochastic interpretation immediately brings the benefits of statistics to the formulae that Duane had derived. The parameters λ and β can be determined using maximum likelihood estimators (MLE's) rather than β being assumed to be fixed. Further, hypothesis tests and confidence limits can be determined for the parameters, and Goodness-of-Fit tests can be performed on the model. This eliminates the first two drawbacks of Duane's model. We will discuss later how Crow handles the problem of imputed zero reliability at the start of test.

One should take note that even though the growth rate estimate $\hat{\alpha}$ can be calculated from Crow's growth parameter estimate, $\hat{\beta}$, and it is still interpreted as the estimate of the negative slope of a straight line on a log-log plot, Crow's estimates of λ and β are somewhat different from the ones derived using Duane's procedures. This follows from the fact that the estimation procedure is MLE, not least squares, thus each model's parameters correspond to different straight lines, respectively.

The December 2005, System Reliability Toolkit, a Practical Guide for Understanding and Implementing a Program for System Reliability, by RIAC and DACS, points out that the 1981 MIL-HDBK-189 suggests the Duane model for planning and the AMSAA/Crow model for

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assessment and tracking. Actually, with the planning improvement additions (see discussion below) the AMSAA/Crow model also may be used for planning purposes.

5.1.6.9.1 Background.
As noted in Section 4.8, the reliability planning curve may extend over all the test phases or just over one test phase. Typically a smooth growth curve is portrayed which represents the overall expected pattern growth over the test phases. For ease of discussion, we will measure reliability by the MTBF metric and test duration by time. The smooth curve is termed the idealized growth curve and is usually specified by a simple mathematical formula that utilizes several parameters. This Handbook utilizes the following form based on the power law expression for the expected number of failures as a function of cumulative test time

$$E(N(t)) = \lambda t^{\beta}. \qquad 5.1\text{-}8$$

Then the idealized reliability growth pattern with failure intensity function $\rho(t)$ is given by,

$$\boldsymbol{\rho(t) = \lambda\beta t^{\beta-1}} \qquad 5.1\text{-}9$$

For growth one has $0<\beta<1$. Thus the above has a singularity at $t=0$. Although all the statistical procedures developed for the power law are based on assuming $N(t)$ is a Poisson process with failure intensity $\rho(t)$ for $t > 0$ this singularity causes difficulties with respect to planning. In particular, using $\rho(t)$ to portray growth for all $t > 0$ suggest that the initial MTBF is zero.
Letting $N(t)$ denote the number of failures by t, the corresponding expected number of failures by t is given by,

$$E(N(t)) = \lambda t^{\beta} \qquad 5.1\text{-}10$$

For planning purposes, the allocated test time T is divided into p test phases that end at the cumulative test times $t_1 < t_2 < \ldots < t_p = T$. Reliability growth may occur within a test phase. However, this original approach does not assume the growth pattern governed by Equation 5.1-9 holds within each test phase. For example, no corrective actions may be applied within some test phases. The test phases typically are separated by blocks of calendar time during which a significant number of corrective actions are implemented to failure modes surfaced in the preceding test phase. Thus, jumps in reliability are typically expected from test phase to test phase. For planning, the handbook only assumes that Equation 5.1-10 ends of the test phases, i.e., for $t = t_i$. In particular, the points with coordinates $\left(t_i, \frac{E(N(t_i))}{t_i}\right)$ are assumed to lie on a straight line on a log-log plot of $\frac{E(N(t_i))}{t_i}$ versus t_i with slope equal to α, called the growth rate. The parameter β that appears in Equation 5.1-10 is equal to $1-\alpha$. The assumption is motivated by Duane's empirical relationship (Duane, 1964). Duane's observations were based on fitting straight lines to test data using a log-log scale and thus presumably applied to the observed pattern of growth during a test phase as opposed to across test phases although this is not

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addressed in (Duane, 1964). The methodology does not assume Equation 5.1-10 holds within all the test phases but it does utilize a power law relationship for estimating reliability in a test phase, i.e., for tracking reliability growth within a test phase. The use of Equation 5.1-10 for this purpose seems more tied to Duane's observations.

This approach utilizes average MTBF values over each test phase to depict the planned reliability growth path. For test phases during which no corrective actions are expected to be implemented this average MTBF would also be the instantaneous MTBF. The methodology to obtain these test phase average MTBF values is based on first specifying an idealized curve that satisfies Equation 5.1-10 at the end of each test phase. The idealized curve is frequently determined by specifying a planned or assessed average MTBF, M_I, over the initial test period, the total test time over all the test phases, T, and the goal or required MTBF, M_F, to be attained at T. The growth rate α can then be solved for and the constants $\beta = 1 - \alpha$ and λ that appear in Equation 5.1-10 can be obtained. For test phase i, the average failure rate $\emptyset_i$ is defined by,

$$\emptyset_i = \frac{E\left(N(t_i)\right) - E\left(N(t_{i-1})\right)}{t_i - t_{i-1}} \qquad 5.1\text{-}11$$

The corresponding average *MTBF* for phase i is taken to be the reciprocal of $\emptyset_i$. The pattern of test phase *MTBF* averages so obtained do not explicitly take into account parameters that can be directly influenced by program management. Such parameters include the management strategy (MS), the average fix effectiveness factor (μ_d), the corrective action lag time, and the scheduled monthly RAM test hours and corrective action periods. If one explicitly takes into account these factors, the growth pattern exhibited by the resulting test phase MTBFs, even when displayed on a test time basis, will often look more irregular than the pattern of average test phase MTBFs obtained from the original MIL-HDBK-189 approach. In particular, this growth pattern may not be well represented by a pattern consistent with Equation 5.1-9.

The reciprocal of the idealized failure intensity given in Equation 5.1-8 is considered to be a representation of the overall MTBF growth trend over the test program after the first test phase. Note for growth one has α greater than zero. Thus as t approaches zero the MTBF implied by Equation 5.1-9 approaches zero. Therefore, for planning purposes, the handbook utilizes Equation 5.1-9 to represent the overall growth trend only for $t > t_1$. The handbook simply utilizes a constant or average failure rate, $\emptyset_1 = M_1^{-1}$, over the first test phase. The constant $\emptyset_1$ is chosen such that Equation 5.1-10 is satisfied for $t = t_1$. Doing so, it follows that the *MTBF* growth trend consistent with Equation 5.1-9 for $t > t_1$ and $\emptyset_1$ is given by,

$$MTBF\,(t) = \begin{cases} M_1 & 0 \le t \le t_1 \\ M_1 \left(\frac{t}{t_1}\right)^{\alpha} (1-\alpha)^{-1} & t > t_1 \end{cases} \qquad 5.1\text{-}12$$

Although all the statistical procedures developed for the power law are based on assuming $N(t)$ is a Poisson process with failure intensity $\rho(t)$ for $t > 0$ this singularity causes difficulties with respect to planning. Using 5.1-9 for $t > 0$ suggests that the initial *MTBF* is zero. Thus for planning purposes one uses the power law to represent the idealized overall growth pattern only

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for values of test time t beyond an initial test phase of length t_I. For t in the initial test phase one either assumes that no growth is taking place and the constant *MTBF* over this period is M_I or that M_I represents an average *MTBF* over the initial test phase. By an average *MTBF* we mean that M_I equals the length of the initial test phase divided by the expected number of failures over the initial test phase. Thus for planning, assuming a constant *MTBF* in the initial test phase,

$$MTBF(t) = M_I \quad \text{for } 0 \le t \le t_I \text{ and } MTBF(t) = 1/\lambda\beta t^{\beta-1} \text{ for } t > t_I \qquad 5.1\text{-}13$$

In terms of the expected number of failures one has

$$E(N(t)) = \lambda_I t \text{ for } 0 \le t \le t_I \text{ and } E(N(t)) = \lambda t^{\beta} \text{ for } t > t_I \text{ where } \lambda_I = M_I^{-1}. \qquad 5.1\text{-}14$$

To make the expected number of failures a continuous function of test time one needs to have λ_I and t_I satisfy the equation

$$\lambda_I t_i = \lambda t_i^{\beta} \qquad 5.1\text{-}15$$

This yields

$$\lambda = \frac{t_I^{\alpha}}{M_I} \qquad 5.1\text{-}16$$

Finally, replacing λ in 5.1−13 one obtains

$$MTBF(t) = M_I \text{ for } 0 \le t \le t_I \text{ and } MTBF(t) = \{M_I/(1-\alpha)\}(t/t_I)^{\alpha} \text{ for } t > t_I \qquad 5.1\text{-}17$$

The expressions in 5.1-17 are used for planning.

5.1.7 Planning Model Issues.
In using Equation 5.1-12 one needs to be careful not to automatically equate M_I to the planning parameter, M_I, defined as the initial MTBF. In general $M_I \le M_I$. The two *MTBF*s should be equated only if no growth is planned over the first test phase, since M_I is the planned average *MTBF* over the initial test phase. The growth rate α is used as a measure of programmatic risk with respect to being able to grow from M_I to $M_F = MTBF(T)$ in test time T. The higher α is relative to past experience, the greater the risk of attaining M_F. From Equation 5.1-12 we can see that $MTBF(T)$ is a strictly increasing function of the ratio T/t_I and can be made as large as desired by making t_I sufficiently small. Thus for any given T, M_I, and growth rate α one can always find a small enough t_I such that $MTBF(T)$ will equal the desired value. This implies that α as a measure of programmatic risk is only as meaningful as the choice of t_I. In particular, one should guard against artificially lowering α by selecting t_I so small that no significant amount of fix implementation is expected to occur until a corrective action period that is beyond t_I. The

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strong dependence of the global parameter α on the length of the initial test phase is not a desirable attribute for planning purposes.

A reliability projection concept, growth potential, is useful in considering the reasonableness of the idealized curve. The growth potential $MTBF_{GP}$ is the theoretical value that would be reached if all B-modes were surfaced and corrected with the assumed or assessed FEFs. Assuming an average *FEF,* μ_d, management strategy *MS*, and initial MTBF, M_I, one can express the growth potential *MTBF* as follows:

$$MTBF_{GP} = M_I /(1 - (MS)\mu_d) \qquad 5.1\text{-}18$$

If the final *MTBF* on the idealized growth curve is not below the *MTBF* growth potential for reasonable planning values of *MS* and μ_d then even if the growth rate α appears modest it might not be sustainable over the entire period over which the idealized power law model has be applied. In such a case, one might consider applying separate power law idealized curves the major test phases. However, applying the power law with the new origin located at the beginning of the subsequent test phase implies – even with a lower growth rate – that the MTBF is initially growing much more rapidly than it grows towards the conclusion of the previous test phase. This is due to the singularity of the power at the origin. Thus portraying growth by using separate idealized curves governed by the power law over adjacent test phases implicitly assumes that a new set of potential "vital few" failures modes have been introduced in the subsequent test phase. This could be the case if new functionality has been added or if different test conditions prevail.

Finally, we note that Equation 5.1-12 implies that, even with a reasonable choice for t_I, any value of M_F can eventually be obtained since there is no upper limit implied by Equation 5.1-12. This is true even using a growth rate that appears to be reasonable based on past experience with similar types of systems. However, one should keep in mind that if the planning curve extends over many thousands of hours, the planned growth rate might not be sustainable due to resource constraints besides test time and due to technological constraints. Past comparable growth rates may have been estimated from test data over one test phase of a much shorter duration and the system may also have been relatively immature.

5.1.8 Examples.
Suppose we have a goal or required MTBF of M_F at a milestone of T = xxx test hours where $T > t_I$. Using equations given by 5.1-17 one can use the associated growth rate as a programmatic risk factor. That is, using T, M_F, t_I, and M_I we can derive a growth rate α that satisfies the equations in 5.1-17. We can see from Equation 5.1-17 that the growth rate depends on the ratio T/t_I. The larger the ratio the smaller the growth rate needed to satisfy Equation 5.1-17. We can derive a reasonable growth rate by choosing t_I sufficiently small, regardless the value of T. This is due to the singularity of the failure intensity function at $t = 0$. To avoid choosing t_I too small and arriving at a high-risk growth rate (see TABLE I for historical growth rates), one should consider when the growth process or test, find, analyze, and fix process might reasonably begin. That is, in order for growth to begin one needs to observe when failures might reasonably begin, not only any failure but also B-mode failures. As noted in the summary of the

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Power Law or AMSAA/Crow Model, we wanted a high chance of observing a B-mode failure, e.g., 0.95. This may be expressed as the ratio of the B-mode failure rate to the initial failure rate - λ_B/λ_I. Equate this ratio to the management strategy, *MS*. Then the value of t_I is chosen so that the probability of observing a B-mode is sufficiently high, say, 0.95. This relationship is given by

$$Prob = 1 - e^{-(MS)\lambda_I t_I} \qquad 5.1\text{-}19$$

and note that on choosing Prob = 0.95 yields a value of t_I of approximately $3*(M_I/MS)$. (See L. H. Crow – "On the Initial System Reliability, Annual Reliability and Maintainability Symposium Proceedings, 1986.)

5.2 Detailed Statements on Planned Growth Curve Development MIL-HDBK-189 (1981). [2]

5.2.1 Planned Growth Curves.

Development of the planned growth curve is an application of the "lessons learned" from previous program experiences to predict the growth that can be expected in a future program. The importance of this curve needs to be understood. When hard reliability data have begun to be generated, the results will be compared with the predicted values given by the planned curve to determine if the reliability growth is progressing satisfactorily.

5.2.2 General Development of the Planned Growth Curve.

The detailed planned growth curve provides, as precisely as possible, the phase-by-phase development of reliability improvement that is expected. Each test phase should be carefully considered to determine the type of testing that will be conducted and the impact of the fixes that can be anticipated. The role of the idealized growth curve is to substantiate that the planned growth follows a learning curve which, based on previous experience, is reasonable and can be expected to be achieved. The following paragraphs describe how planned growth curves may be developed for specific programs. Every program can, however, be expected to require some modification of the suggested procedures.

5.2.3 General Concepts for Construction.

In general, there are two basic approaches for constructing planned growth curves. The first method is to determine the idealized growth pattern that is expected or desirable, and to use this as a guide for the detailed planned curve. The second method is just the reverse. In this case, a proposed planned curve is first developed which satisfies the requirement and interim milestones. The idealized curve is then constructed and evaluated to determine if this learning curve is reasonable when compared to historical experience. If not acceptable, a new detailed curve would need to be developed.

5.2.4 Understanding the Development Program.

Development of planned growth curves requires a fairly complete understanding of the proposed development program, particularly the reliability program and all other program activities and constraints that will affect reliability. In the case of mechanical equipment, an understanding of

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the hardware is useful in evaluating the delays that should be associated with design changes. For complex test programs, a logic diagram should be used to show the relationships between those phases in which failure modes will be found and those phases that will have the resultant "fixes" in the hardware. The expected policy for incorporating fixes needs to be understood. For systems with high reliability, the expected number of failures during the test program should be determined to give an indication of the number of fixes that can be anticipated. For initial estimating purposes this may be based on the starting MTBF.

5.2.5 Portraying the Program in Total Test Units.
Although the planned growth curve is usually portrayed in final form as a function of calendar time for management use, the analytical development of the curve is done as a function of test units. Test units may be hours, miles, rounds, or similar units; and in some cases, the use of multiple units (e.g., both miles and rounds) may be appropriate. FIGURE 5-2 shows an example of a development program portrayed in calendar time, and FIGURE 5-3 shows the same program portrayed in cumulative miles.

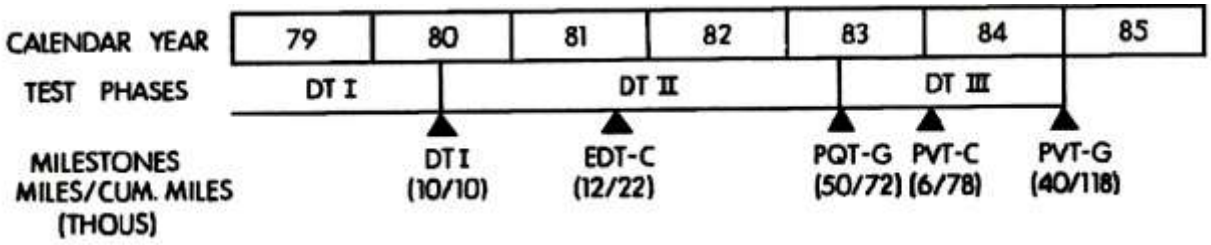
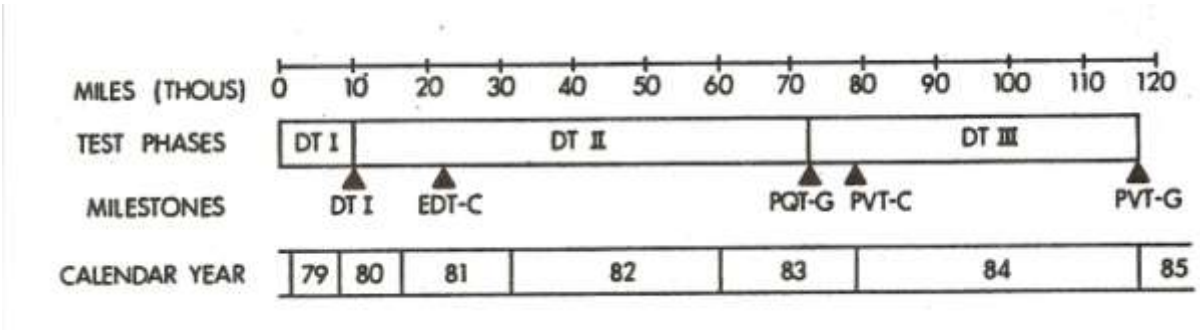


FIGURE 5-2. *Development Program Portrayed in Calendar Time*



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FIGURE 5-3. *Development Program Portrayed in Test Units*

5.2.6 Determining the Starting Point.
The initial reliability for a system under development will typically not be known at the time when the planned curve is developed. A starting point for the planned growth curve may however, be determined from (1) using information from previous programs on similar systems, (2) specifying a minimum level of reliability that management requires to be demonstrated early in order to have assurance that the reliability goals will be met, and (3) conducting an engineering assessment of the design together with any previous test data that may exist e.g., bench test, prototype test. The practice of arbitrarily choosing a starting point, such as 10% of the requirement, is not recommended. Every effort to obtain information even remotely relevant to a realistic starting point should have been exhausted before an arbitrary figure can be used.

5.2.7 Example of Determining a Starting Point.
A planned growth curve is to be developed for a ground vehicle development program. One of the first steps in this process is to determine a starting point for this curve. To establish a starting point, the reliability growth experience of a predecessor system is analyzed. It is found that an initial mean miles between failures (MMBF) of 183 miles was demonstrated during early engineering development. The predicted *MMBF* was 580 miles. Therefore, at this point in development, the achievement was 183/580 = 0.32 of predicted. The system under development has about the same degree of design maturity as did its predecessor; but since the reliability program emphasis is somewhat greater, it is expected that perhaps 0.35 of the prediction, rather than 0.32, will be achieved. With a prediction of 410 miles for the current system, 0.35 (410) = 143 would be expected as a starting point. To further rationalize this estimate, some pre-development testing of the proposed system resulted in 5 failures in 493 miles. No significant design changes were incorporated during test, so the *MMBF* may be estimated as 493/5 = 99 miles. Some design change is planned prior to engineering development testing. Using engineering analysis methods similar to those described in Appendix A, it is estimated that 2 of these failures will be affected by design change. It is also estimated that the design changes will be 70% effective. The *MMBF* expected on entering engineering design testing is then 493/ (5 - 0.7(2)) = 137 miles. This value gives additional support to the estimate of 143 miles.

5.2.8 Development of the Idealized Growth Curve.
During development, management should expect that certain levels of reliability be attained at various points in the program 'in order to have assurance that reliability growth is progressing at a sufficient rate to meet the requirement. The idealized curve portrays an overall characteristic pattern that is used to determine and evaluate intermediate levels of reliability and construct the program planned growth curve. Growth profiles on previously developed, similar type systems provide significant insight into the reliability growth process and are valuable in the construction of idealized growth curves. Reliability growth information on previous programs should be used whenever possible to develop the idealized curve directly or as input into a model for development of the idealized curve.

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5.2.9 Idealized Growth Model Based on Learning Curve Concept.
If documented reliability histories for similar type systems are not available to provide a basis for the idealized curve of the system under consideration, then a general method based on the learning curve concept is an alternative. Appendix C provides a survey of various reliability growth models. If the learning curve pattern for reliability growth assumes that the cumulative failure rate versus cumulative test time is linear on a log-log scale, then the following method is appropriate for construction of the idealized growth curve. This method is based on the test phase structure of a development program for reliability growth, as discussed in Section 4. This approach gives a realistic method for placing the initial MTBF at the proper point in time and portrays a growth pattern, which has a meaningful interpretation in terms of test phase reliability growth.

5.2.10 Summary of Method.
The idealized growth curve *M(t)* discussed in this section has the form shown in Figure 4 and portrays a general profile for reliability growth throughout system testing. The idealized curve has the baseline value M_I over the initial test phase, which ends at time t_1. The value M_I is the average MTBF over the first test phase. From time t_1 to the end of testing at time T, the idealized curve *M(t)* increases steadily according to a learning curve pattern till it reaches the final reliability requirement M_F. The slope of this curve on the log-log plot in FIGURE 5-4 is the growth parameter α. The parametric equation for *M(t)* on this portion of the curve is

$$M(t) = M_I \left(\frac{t}{t_1}\right)^{\alpha} (1-\alpha)^{-1}. \qquad 5.2\text{-}1$$

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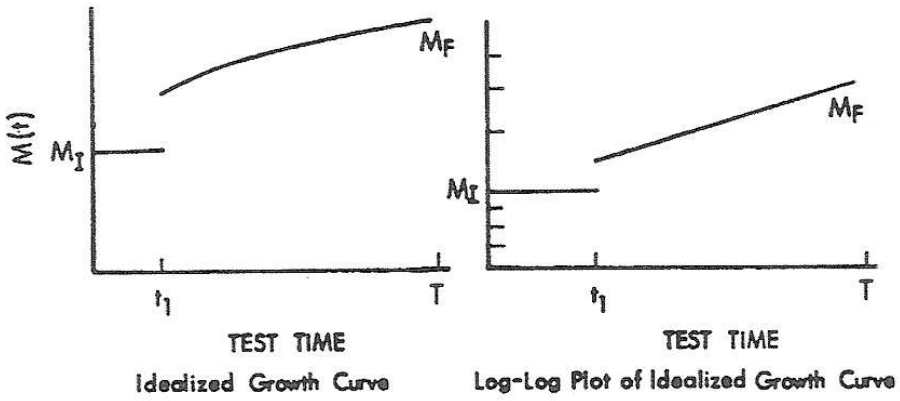


FIGURE 5-4. *Idealized Growth Curve*

5.2.11 Basis of Model.
This model assumes that the cumulative failure rate versus cumulative test time is linear on a log-log scale when plotted at the ends of test phases or reporting periods. See FIGURE 5-5. It is not assumed that the cumulative failure rates follow the same pattern within test phases. In fact, if delayed fixes are incorporated into the system at the end of a test phase, or the reliability is held constant during a test phase, then this linear pattern within test phases would not hold. To illustrate this approach let $t_1, t_2, \cdots, t_k$ denote the cumulative test times which correspond to the ends of test phases. It is assumed that $N(t_i)/t_i$ versus t_i , $i = 1, 2, \ldots, k$, are linear on a 1og-log scale, where $N(t_i)$ is the cumulative number of failures by time t_i. That is, log $N(t_i)/t_i$ is linear with respect to log t_i. This implies that log $N(t_i)/t_i$ can be expressed as log $N(t_i)/t_i = \sigma - \alpha \log t_i$, where σ and α are, respectively, intercept and slope parameters. Let λ_I denote the initial average failure rate for the first test phase, i.e., $\lambda_I = N(t_1)/t_1$. Since log $\lambda_I = \sigma - \alpha \log t_1$, it follows that $\sigma = \log \lambda_I + \alpha \log t_1$.

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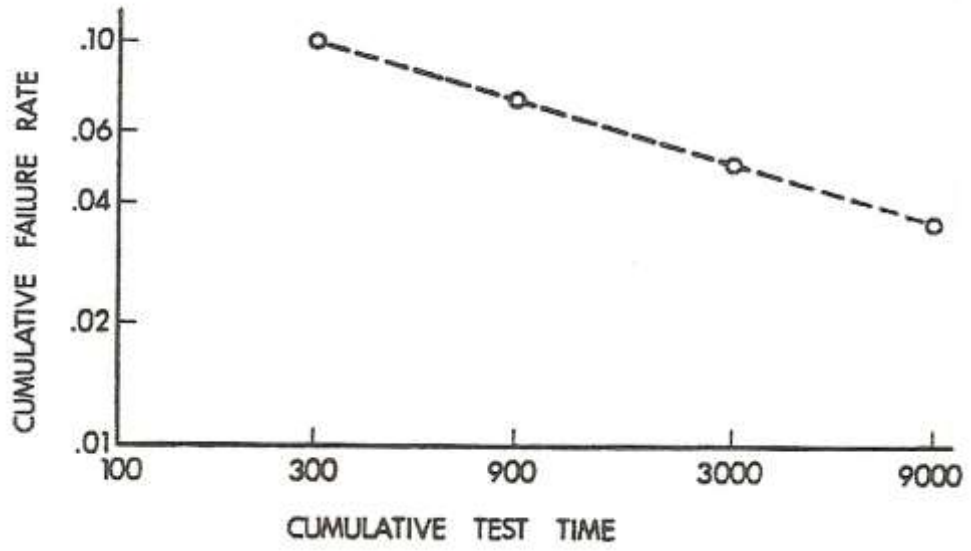


FIGURE 5-5. *Example of Log-Log Plot at Ends of Test Phases*

Therefore, log $N(t_i) / t_i$ = log λ_I - α log(t_i / t_1). Consequently, the cumulative failure rate can be expressed as

$$N(t_1)/t_i = \lambda_I (t_i/t_1)^{-\alpha}. \qquad 5.2\text{-}2$$

This gives

$$N(t_i) = \lambda_I t_i (t_i /t_1)^{-\alpha}, \text{ or equivalently, } N(t_i) = \lambda_I t_1 (t_i / t_1)^{1-\alpha}. \qquad 5.2\text{-}3$$

The average failure rate over the test interval t_{i-1} to t_i (the i-th test phase) is the total number of failures during this period divided by the length of the interval t_i - t_{i-1}. Therefore, the linearity of the cumulative failure rates at ends of test phases implies that the average failure rate λ_i for the i-th test phase is

$$\lambda_i = (N(t_i)- N(t_{i-1}))/(t_i-t_{i-1}) \qquad 5.2\text{-}4$$

where

$$N(t_i) = \lambda_I t_1 (t_i / t_1)^{1-\alpha} \qquad 5.2\text{-}5$$

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See FIGURE 5-6.

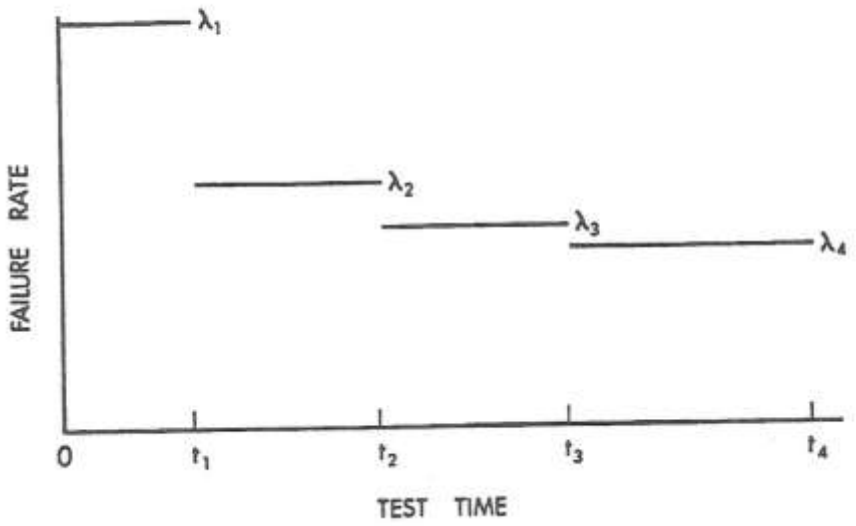
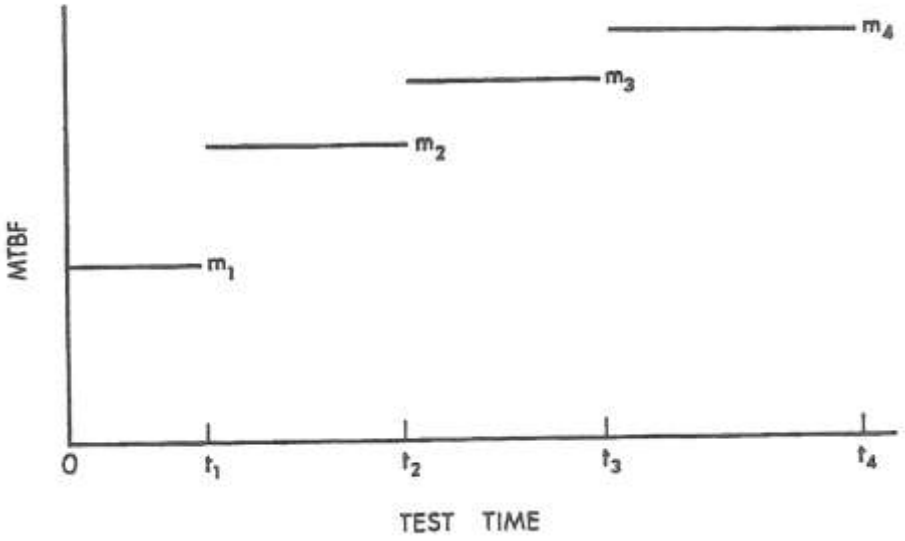


FIGURE 5-6. *Average Failure Rates over Test Phases*

In terms of failure rate, this result for the average failure rates over the test phases is all that can be concluded from the linearity on a log-log scale of the cumulative failure rates at ends of test phase. The reliability growth of the system in terms of MTBF is reflected by the increase in the average MTBF's $M_i = 1/\lambda_i$ over the test program. See FIGURE 5-7.



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FIGURE 5-7. *Average MTBF's over Test Phases*

Now, the curve defined by $r(t) = d/dt\ [N(t)] = \lambda_I(1-\alpha)(t/t_1)^{-\alpha}$ crosses the average failure rates λ_i for each test phase. See FIGURE 5-8.

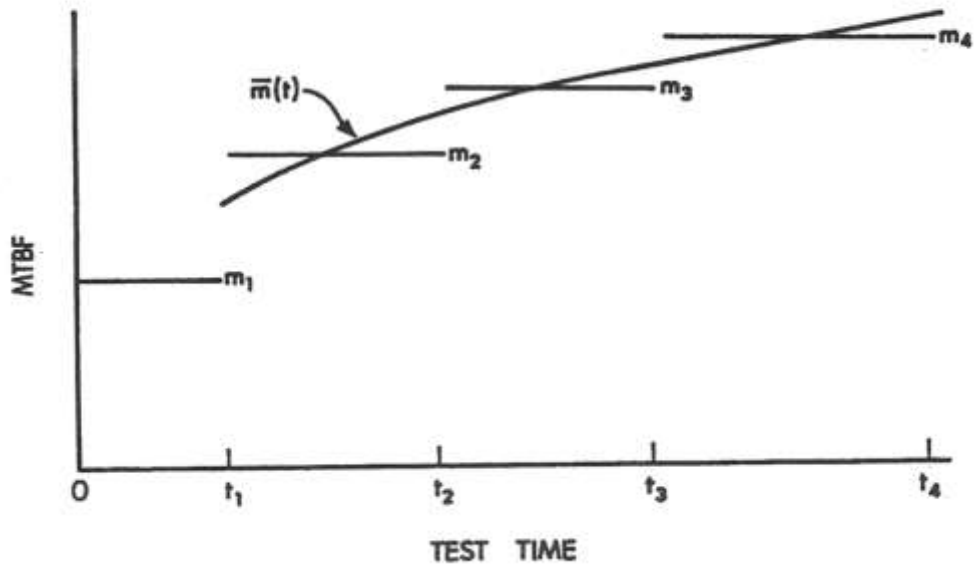


FIGURE 5-8. *Average MTBF's and Modified $\underline{\bar{m}}(t)$ Curve*

For any test phase the area under the curve $r(t)$ is equal to the area under the average failure rate. Therefore, for any test phase the average failure rate can be determined from $r(t)$. The reciprocal of the curve, $r(t)$

$$\bar{m}(t) = \{r(t)\}^{-1} = M_I\,(t/t_1)^{\alpha}(1-\alpha)^{-1} \qquad 5.2\text{-}6$$

also crosses the average MTBF M_I for each test phase. See FIGURE 5-9.

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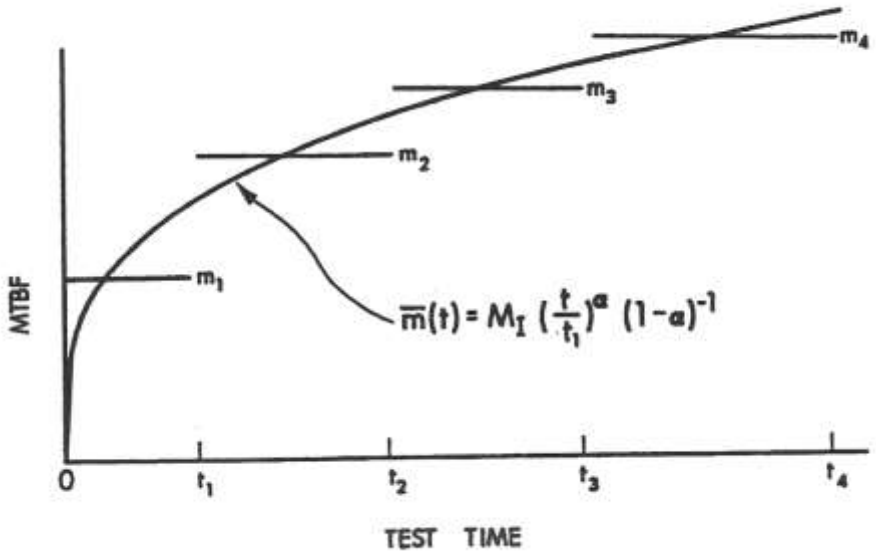


FIGURE 5-9. *Average MBF's and $\overline{m}(t)$ Curve*

The actual underlying pattern of reliability growth is represented by the increase in the test phase average MTBF's. The growth in the individual test phase does not follow the smooth line $\overline{m}(t)$. In particular, note that the curve $\overline{m}(t)$ gives a value of 0 at test time 0, which is, of course, not a realistic value for the actual system MTBF at the beginning of development testing. However, the curve, $\overline{m}(t)$ can generally be viewed as reflecting a meaningful trend for the average MTBF's after the first test phase. See FIGURE 5-10.

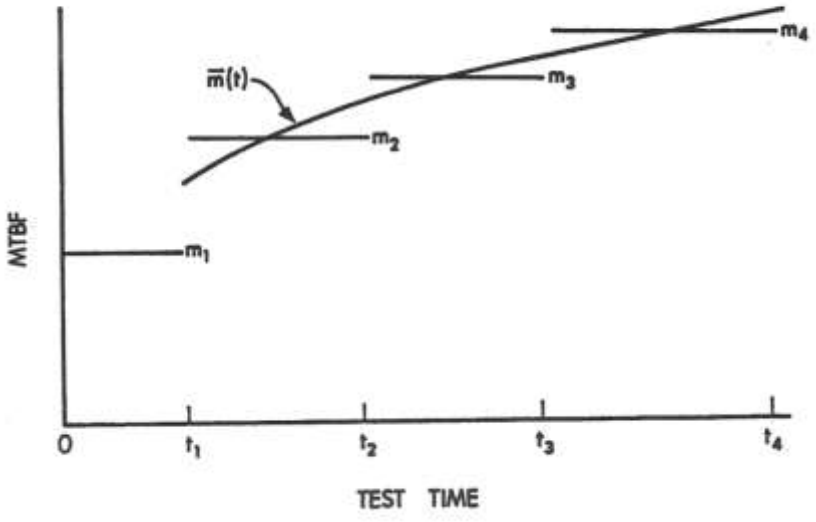


FIGURE 5-10. *Average MTBF's and Modified $\overline{m}(t)$ Curve*

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It is further noted that the baseline for reliability growth in terms of average MTBF's is the initial average MTBF $M_I = 1/\lambda_I$. Therefore, a practical and meaningful idealized growth curve is one that equals *M(t)* over the first test phase and equals the curve $\overline{m}(t)$ over the remaining test time. This curve is denoted by *M(t)*. See FIGURE 5-11 and FIGURE 5-12.

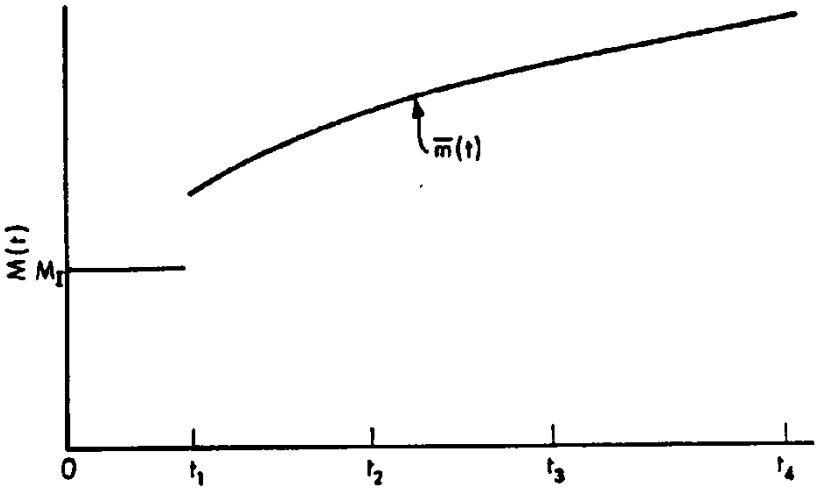


FIGURE 5-11. *Idealized Growth Curve*

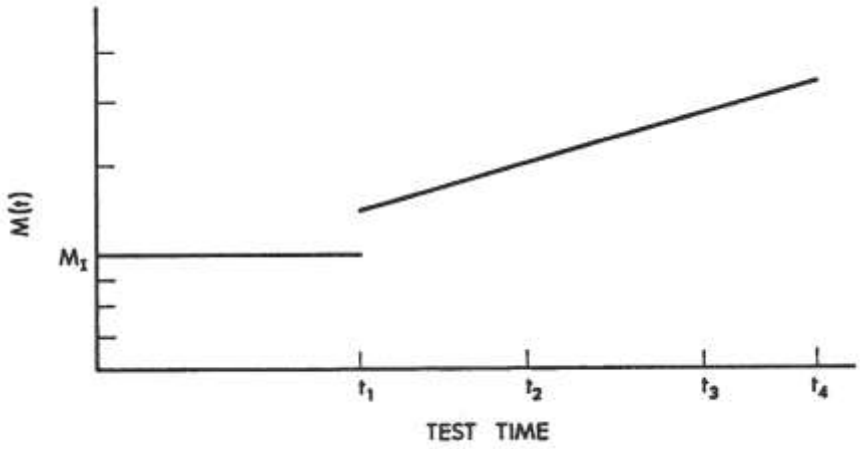


FIGURE 5-12. *Log-Log Plot of Idealized Growth Curve M(t)*

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The idealized growth curve shows that the initial average MTBF over the first test phase is M_I, and that reliability growth from this average begins at t_1. This jump is indicative of delayed fixes incorporated into the system at the end of the first test phase. The idealized curve $M(t)$ is a guide for the average MTBF over each test phase. Further given that

$$M(t) = M_I (t / t_1)^{\alpha} (1 - \alpha)^{-1} \quad \text{for } t > t_1, \qquad 5.2\text{-}7$$

then the average failure rate and the average MTBF for the i-th test phase can be determined by $\lambda_I = (N(t_i) - N(t_{i-1})) / (t_i - t_{i-1})$, and $m_i = 1/\lambda_i$, where $N(t_i) = \lambda_I t_1 (t_i / t_1)^{1-\alpha}$. See FIGURE 5-13.

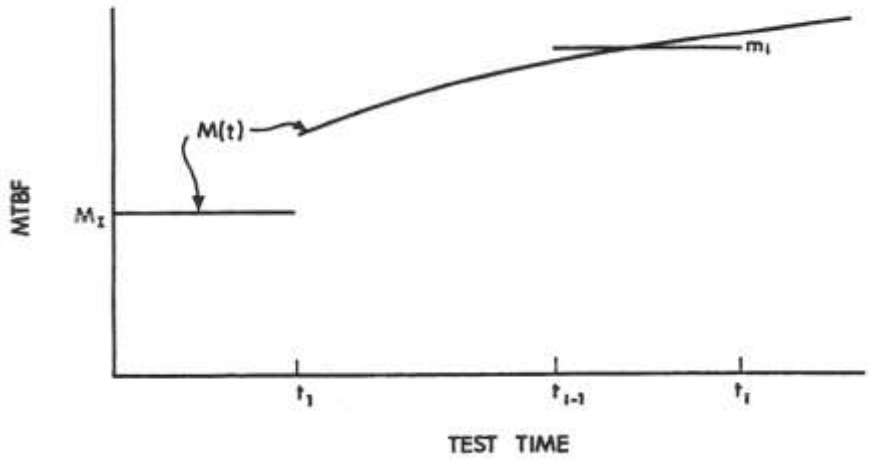


FIGURE 5-13. *Average MTBF over i-th Test Phase*

In the application of the idealized growth curve model, the final MTBF value M_F to be attained at time T is set equal to $M(T)$, i.e.,

$$M_I (T / t_1)^{\alpha} (1 - \alpha)^{-1} = M_F. \qquad 5.2\text{-}8$$

Also, the parameters M_I and t_1 of this model have the physical interpretations that M_I is the initial average MTBF for the system and t_1 is the length of the first test phase in the program. The parameter α is a growth parameter.

5.2.12 Procedures for Using Idealized Growth Curve Model.
This section contains problems, solutions, and numerical examples which illustrate the application of the idealized growth model discussed previously. The following notation and formulas are given for completeness.

List of Notations.

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T	the cumulative test time over the test program.
$t_1, t_2, \bullet\bullet\bullet, t_k$	the cumulative test times corresponding to the ends of test phases ($t_k = T$).
$N(t_i)$	the cumulative number of failures by time t_i.
$H_i = N(t_i)-N(t_{i-1})$	the number of failures during the i-th test phase.
$\lambda_i = H_i/(t_i - t_{i-1})$	the average failure rate over the i-th test phase.
M_F	the final MTBF at time T.
$M_i = 1/\lambda_i$	the average MTBF over the i-th test phase.
$\lambda_1 = \lambda_1$	the subscript I denotes initial average failure rate.
$M_1 = 1 / \lambda_1$	the initial average MTBF.
α	growth parameter.

5.2.13 Idealized Growth Model.

The idealized growth model $M(t)$ is given by

$$M(t) = \begin{cases} M_I \text{ for } 0 < t < t_1 \\ M_I\,(t / t_1)^{\alpha}(1 - \alpha)^{-1} \text{ for } t > t_1. \end{cases} \qquad 5.2\text{-}9$$

where t_1 is the end of the first test phase.

Under this model

$$M_F = M_I\,(T / t_1)\,(1 - \alpha)^{-1} \qquad 5.2\text{-}10$$

and

$$N(t_i) = \lambda_I t_1\,(t_i / t_1)^{1-\alpha}. \qquad 5.2\text{-}11$$

5.2.13.1 Case 1. How to Determine the Idealized Growth Curve.

5.2.13.1.1 Objective.
Determine the idealized growth curve.

5.2.13.1.2 Given Conditions.
T - the cumulative test time over the program
t_1 - the test time for the first test phase.
M_I - the average MTBF over the first test phase.
M_F - the final MTBF at time T.

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5.2.13.1.3 Solution for Case 1.

Set

$$M_F = M_I (T/ t_1)^{\alpha} (1 - \alpha)^{-1} . \quad 5.2\text{-}12$$

Find α such that

$$M_F / M_I = (T/t_1)^{\alpha}(1 - \alpha)^{-1} \quad 5.2\text{-}13$$

That is, find α such that:

$$Log (M_F / M_I) = \alpha \log (T / t_1) - \log (1- \alpha). \quad 5.2\text{-}14$$

Then the idealized curve is given by

$$M(t) = \begin{cases} M_I & 0 < t < t_1 \\ M_I (t/t_1)^{\alpha}(1 - \alpha)^{-1} \end{cases} \quad 5.2\text{-}15$$

See FIGURE 5-14.

The following expression for α is a good second order approximation that is sufficient whenever α is less than 0.5:

$$\alpha = -\log (T / t_1) - 1 + [(1 + \log(T / t_1))^2 + 2 \log (M_F / M_I)]^{1/2}. \quad 5.2\text{-}16$$

The logarithms in this expression are natural logarithms.

5.2.13.1.4 Example of Case 1.

Suppose that the initial MTBF for the system is estimated to be 45 hours and a final MTBF of 110 hours is desired after 10,000 hours of testing. For this program, the first test phase is 1,000 hours. This is the point where delayed fixes will first be introduced into the system. Further, some reliability growth is planned during the first test phase so that an average MTBF of $M_I = 50$ hours is anticipated during the first phase. Determine the idealized growth curve. The parameter $\alpha = .23$ is found as the solution to:

$$\text{Log } (110 /50) = \alpha \log (10{,}000 / 1{,}000) - \log (1 - \alpha). \quad 5.2\text{-}17$$

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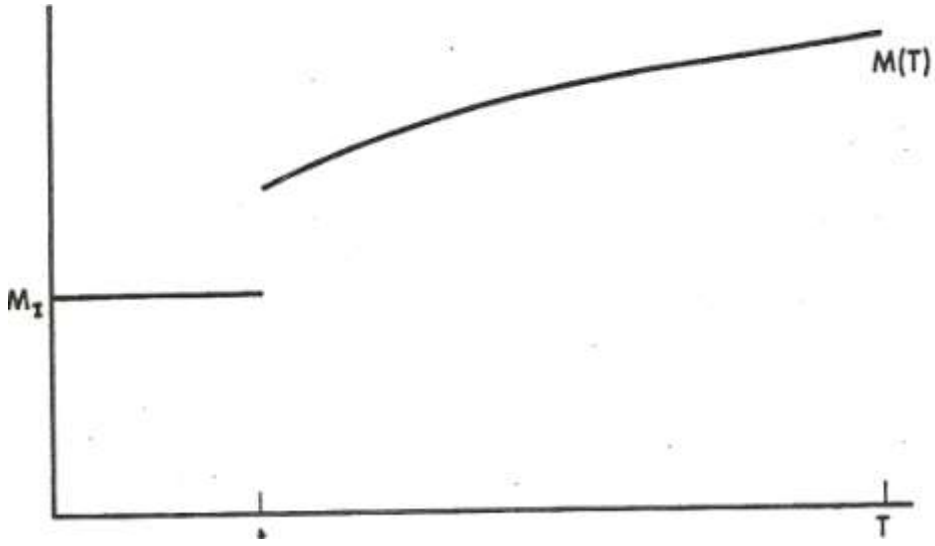
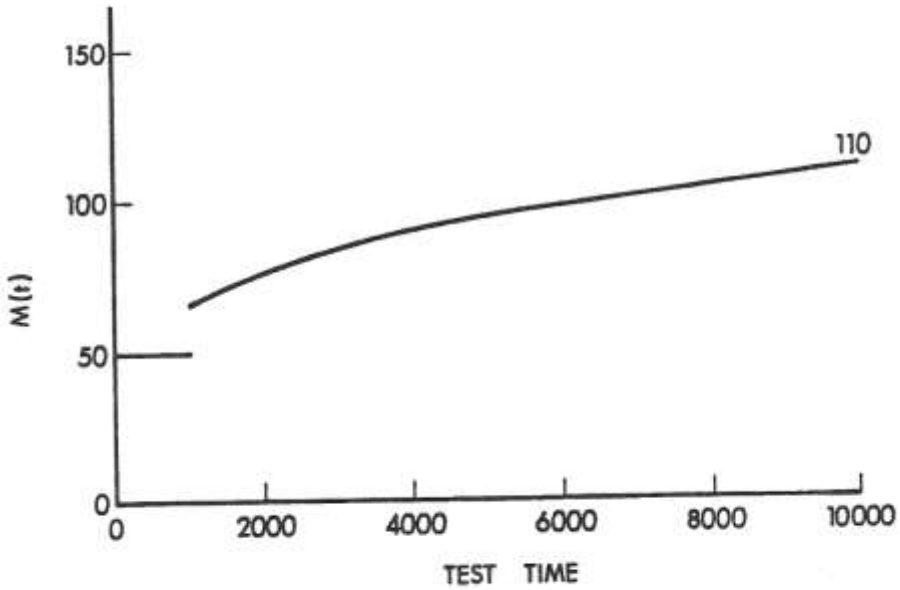


FIGURE 5-14. *Idealized Growth Curve*

Therefore, if $\alpha = 0.23$ is acceptable the idealized growth curve is given by

$$M(t) = \begin{cases} 50 & 0 < t < 1{,}000 \\ (50/0.77)\,(t/1{,}000)^{0.23} & t > 1{,}000 \end{cases} \qquad 5.2\text{-}18$$



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FIGURE 5-15. *Example of Idealized Growth Curve.*

5.2.13.2 Case 2. How to Determine the MTBF for a Test Phase.

5.2.13.2.1 Objective.
Determine the average MTBF M_i for the i-th test phase.

5.2.13.2.2 Given Conditions.
The idealized growth curve

$$M(t) = \begin{cases} M_I, & 0 < t < t_1 \\ M_I (t/t_1)^{\alpha} (1-\alpha)^{-1}, & t > t_1 \end{cases} \quad \text{5.2-19}$$

Is given and the ends of test phases $t_1, t_2, \ldots, t_k$ are known.

5.2.13.2.3 Solution for Case 2.
The average number of failures for the i-th test phase is determined by

$$H_i = N(t_i) - N(t_{i-1}) \quad \text{5.2-20}$$

where

$$N(t_i) = \lambda_I t_1 (t_i / t_1)^{1-\alpha} \text{ and } \lambda_I = 1/M_I. \quad \text{5.2-21}$$

The average MTBF for the i-th test phase is given by

$$M_i = (t_i - t_{i-1})/H_i \quad \text{5.2-22}$$

5.2.13.2.4 Example of Case 2.
In the example in Section 0(how to determine the idealized growth curve) the first test phase was identified from 0 to 1,000 hours. Suppose the program consists of four additional test phases at 1,000-2,500; 2,500-5,000; 5,000-7,000; and 7,000-10,000 hours. To determine the average MTBF's to be expected over these periods if reliability growth follows the idealized curve use

$$M(t) = \begin{cases} 50 & 0 < t < 1{,}000 \\ (50/0.77)(t/1{,}000)^{0.23} & t > 1{,}000 \end{cases} \quad \text{5.2-23}$$

From the example in Section 5.2.1.11.1.

From the idealized growth curve the parameters are $\lambda_I = 0.02$ and $\alpha = 0.23$. Therefore, the average number of failures for the i-th test phase is $H_i = N(t_i) - N(t_{i-1})$ where

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$$N(t_i) = 0.02\ (1{,}000)\ (t_i / 1{,}000)^{0.77}, \qquad 5.2\text{-}24$$

For t_1 = 1,000; t_2 = 2,500; t_3 = 5,000; t_4 =7,000; and t_5 = 10,000.

The average number of failures H_i and the average MTBF M_i for each test phase are presented in . AMSAA Reliability Growth Data Study Summary: Historical Growth Parameter Estimates, in TABLE II below. The average MTBF's are plotted in FIGURE 5-16.

TABLE II. Average number of Failures for H_i, M_i

Phase *i*	H_i	$T_i - t_{i-1}$	M_i
1	20.0	1000	50
2	20.5	1500	73
3	28.6	2500	87
4	20.4	2000	98
5	28.3	3000	106

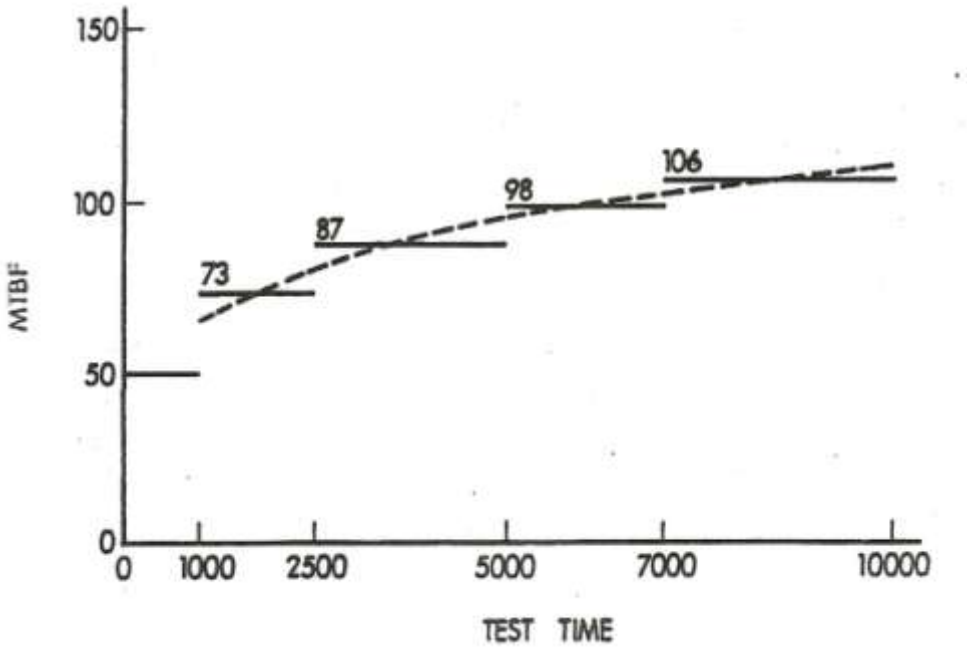


FIGURE 5-16. *Example of Average MTBF's*

5.2.13.3 Case 3. How to Determine how much Test Time is Needed.

5.2.13.3.1 Objective.

Determine how much test time, T, is needed to attain a final MTBF of M_F.

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5.2.13.3.2 Given Conditions.

a. The first test phase is from 0 to t.
b. The initial average MTBF is M_I.
c. The growth parameter is α.

5.2.13.3.3 Solution for Case 3.

The idealized growth curve at time t is

$$M(t) = M_I (t/t_1)^{\alpha} (1-\alpha)^{-1} \qquad 5.2\text{-}25$$

Find T such that $M(T) = M_F$. That is, find T such that

$$\log T = \log t_I + 1/\alpha \, [\log (M_F / M_I) + \log (1-\alpha)]. \qquad 5.2\text{-}26$$

5.2.13.3.4 Example of Case 3.

The average MTBF over the first test phase of $t_I = 700$ hours is estimated to be 1 hour. With a growth parameter of $\alpha = 0.4$, how many test hours are needed to attain a goal of 3 hours MTBF?

From the above, the cumulative test time T necessary to grow from 1 hour MTBF to 3 hours MTBF needs to satisfy

$$\log T = \log 700 + (1/0.4)[\log 3 + \log 0.6] = 8.02. \qquad 5.2\text{-}27$$

That is, $T = 3{,}043$ hours.

5.2.13.4 Test Phase Reliability Growth.

Based on the activities and objectives of the program, the reliability growth plan should indicate for each test phase the levels of reliability that are expected to be achieved. Specifically, for each test phase where an assessment will be made, the following points should be clearly expressed by the reliability program plan:

a. Whether the reliability will be held constant over the test phase or reliability growth is planned during the test, i.e., fixes will be introduced into the system during the test phase.
b. If it is planned to hold the reliability constant, then the level of reliability expected during the phase should be specified.
c. If reliability growth is planned during the test phase, then the reliability objective for the system on test at the end of the test phase should be specified.
d. If delayed fixes are planned at the end of the test phase, then the reliability objective for the beginning of the next test phase should be given.

In addressing the test phase reliability objectives it is useful to consider the effectiveness of the test and redesign efforts. A test phase of a given length can be expected to identify a certain

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number of failure modes. There are three responses that can be made to each identified failure mode:

a. Incorporate a design change during the test phase.
b. Incorporate a design change after the test phase.
c. Incorporate no design change.

5.2.13.4.1 Design Changes During the Test Phases (Test-Fix-Test).
The planned growth curve should reflect the extent of design changes planned during each test phase; and, of course, implicit in this determination is the extent to which design changes are not planned. Historical information may be useful as well as engineering analysis methods described in Appendix A. The rate of growth during test phases is, of course, primarily dependent upon the extent of design changes that are planned.

5.2.13.4.2 Design Changes After the Test Phase (Test-Find-Test).
The growth that takes place between test phases is the result of action taken on failure modes discovered during a previous test phase that is not incorporated until the end of the test phase. This growth cannot, however, be verified until some of the next phase of testing is accomplished. FIGURE 5-17 illustrates the effect of deferring redesign from the test phase to a separate redesign phase.

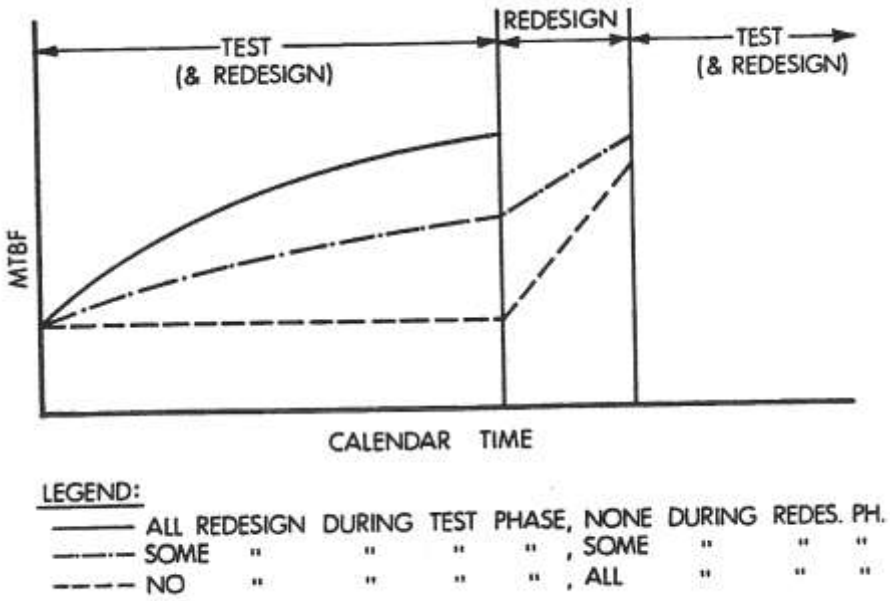


FIGURE 5-17. *Effect of Deferring Redesign*

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As more redesign is deferred, the effectiveness is reduced, because of the inability to detect ineffective design changes and newly introduced failure modes. Analytically, then, the redesign phase can be viewed as a delay of design changes that are identified during test, and some allowance should be made for the lesser effectiveness of delayed redesign. When working in terms of test time, a distinct redesign effort will be shown as a vertical jump. It should be recognized, however, that a certain amount of calendar time is required to achieve the jump. This calendar time may be completely distinct from the calendar time used for testing, as illustrated in FIGURE 5-18, but more commonly, time constraints require that at least some of the time is concurrent with the previous test phase, as illustrated in FIGURE 5-19. Overlapping redesign and test in this fashion will tend to yield a less effective redesign, since it is started somewhat prematurely. A guide to quantifying the growth between test phases is the computation of the percentage jumps that have been historically observed on similar systems or equipments.

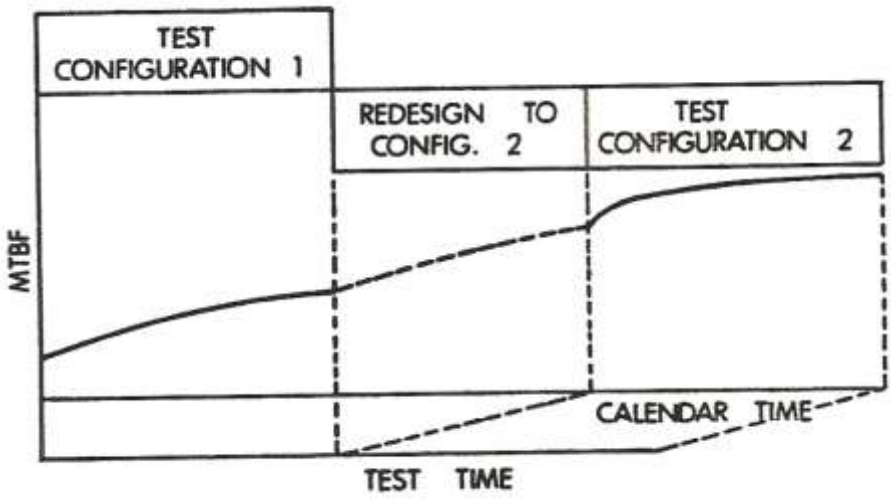


FIGURE 5-18. *Accounting for Calendar Time Required for Redesign*

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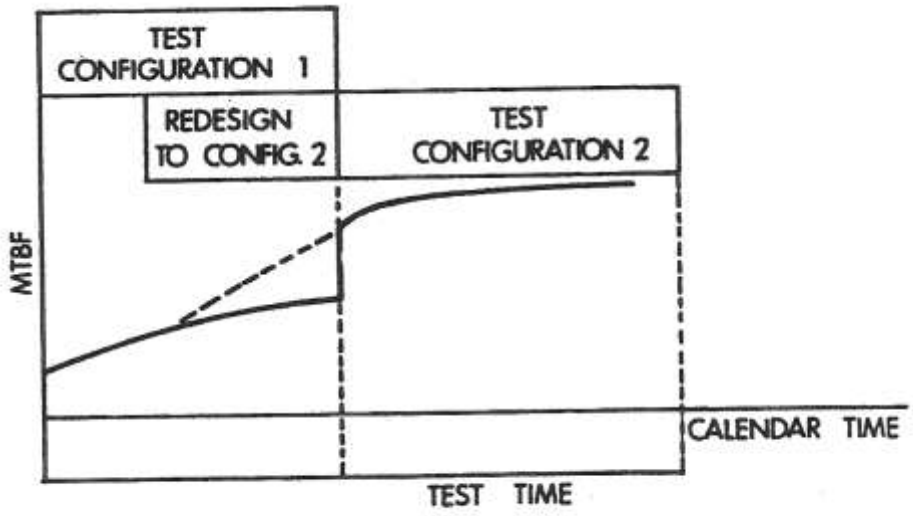


FIGURE 5-19. *Accounting for Calendar Time Required for Redesign*

Incorporate no Design Change.
There will be a certain percentage of failures for which no design changes will be made. There may be an inability to identify appropriate changes, or the identified changes may not be cost effective or may be too time-consuming to pursue.

5.2.14 Examples of Growth Curve Development.
The following examples illustrate the development of planned growth curves for two systems.

5.2.14.1 Example of Growth Curve Development for a Fire Control System.
The project manager for a fire control system wished to construct a planned growth curve while this system was still in the early stages of an accelerated, competitive development program. The growth curve was needed to assist in scheduling test phases for the program, to use as a reference for evaluating planned growth curves submitted by the competing contractors, and to serve as a baseline for tracking demonstrated reliability during development testing.

5.2.14.2 Given Conditions.
Mission reliability requirements spelled out in the Decision Coordinating Papers called for 80 hours *MTBF* during Development Testing/Operational Testing (DT/OT), 110 hours *MTBF* during the Follow-on Evaluation (FOE), and 140 hours *MTBF* during the Initial Production Test (IPT). These reliability requirements were to be demonstrated by fixed configuration testing during the respective test phases. Each test phase was planned to last for 1,100 hours. Preceding and following these formal test phases, the contractors were to perform an undetermined amount of in-house testing and attempt design fixes of any failure modes that were discovered. A mission reliability of 150 hours *MTBF* was required by the end of the first year of production. Unfortunately, some reliability growth had to be planned during the early phases of production. In this instance, however, some failure mode fixes on production items were considered necessary because of the accelerated nature of the program and the 10 months lead-time required to implement a fix from the time of discovery of the problem failure mode. Two further

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conditions on the development program were the limited number of fire control units available for testing and the limited amount of calendar time available for testing. These limitations necessitated a total test time for the formal test phases and contractor in-house testing of no more than 14,000 hours.

5.2.14.3 Problem.
The basic problem of constructing a planned reliability growth curve for the fire control system required decisions about several parameters of the overall test program. The first decision to be made was how much total test time should be planned in order to achieve the final reliability requirement of 150 hours *MTBF*. Then it had to be decided when the test periods should be scheduled for the three test phases DT/OT, FOE, and IPT. The primary tool for making these decisions was to be the idealized reliability growth curve.

5.2.14.4 Construction of Idealized Curve.
From the results of the initial development testing, it was projected that approximately 34 failures would occur during the first 1,700 hours of testing. Since there was not enough calendar time to find, evaluate and fix any failure mode during this initial testing, the *MTBF* over this period was projected to be a constant equal to 1,700/34 = 50.0 hours. Furthermore, it was known that 150 hours *MTBF* should eventually be achieved and that no more than 14,000 hours of test time was available. It was, therefore, of interest to know what kind of idealized curve would result if the maximum possible test time of 14,000 hours was utilized. The conditions of this example: with T= 14,000, t_I = 1,700, M_I = 50.0, and M_F = 150.0. The growth parameter α is obtained by

$$\alpha = - \log(14{,}000/1{,}700) - 1 + ((1 + \log(14{,}00/1{,}700))^2 + 2 \log(150/50))^{1/2} = 0.34. \qquad 5.2\text{-}28$$

An α value of 0.34 is only moderately high, but it is indicative of a relatively aggressive development program that would require management emphasis on the analysis and fixing of problem failure modes. Using a test time of less than 14,000 hours would result in a projected a greater than 0.34 and would therefore require an even more dynamic reliability growth program. Because such a shortened program would have an increased risk of not achieving the required reliability, the program planners for this fire control system decided to schedule the full 14,000 hours of test time for reliability growth effort. The idealized growth curve for this development program is shown in FIGURE 5-20 and FIGURE 5-21.

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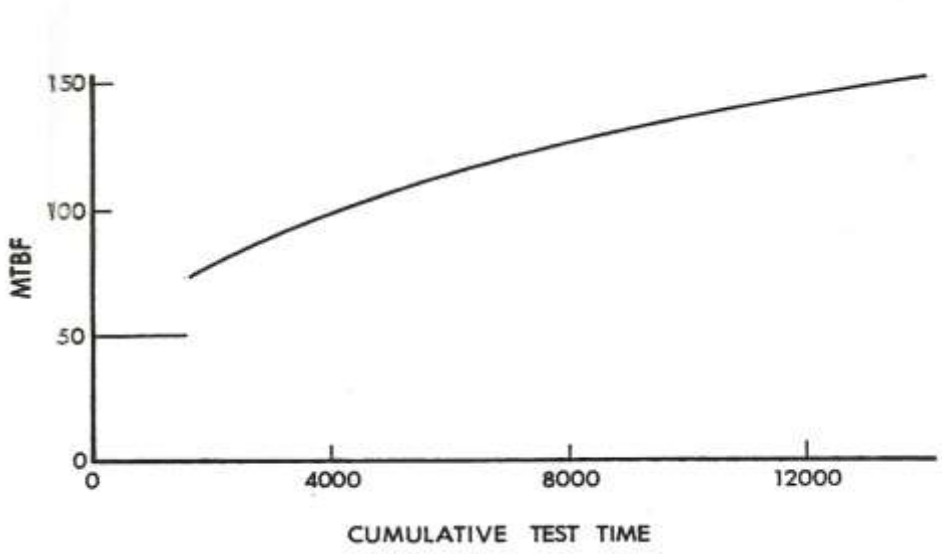


FIGURE 5-20. *Idealized Growth Curve*

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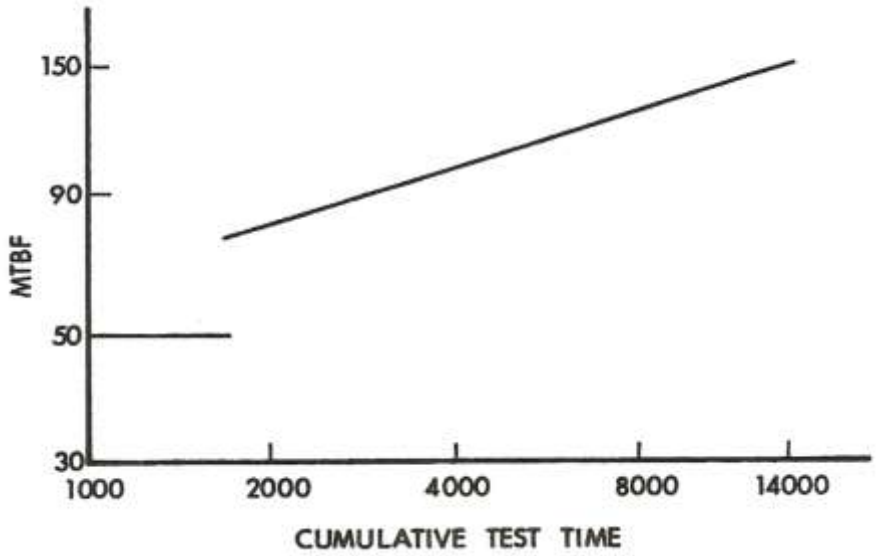


FIGURE 5-21. *Idealized Growth Curve on Log-Log Scale*

5.2.14.5 Construction of Planned Curve.
Once the idealized curve had been constructed, it was used as a basis for developing a planned growth curve. The three test phases were to be scheduled in the testing program during periods when the corresponding reliability requirements could reasonably be expected to be achieved. An appropriate way of judging what average reliability could be demonstrated during a given test period was to utilize the information contained in the idealized growth curve. In FIGURE 5-22, the curve reaches 80 hours *MTBF* at 2,100 hours of testing. It is clear, then, that over any test phase, which begins at 2,100 hours of cumulative test time, the average *MTBF* should equal or exceed 80 hours. Consequently, DT/OT was scheduled to begin at 2100 hours of cumulative test time.

By the same argument, the FOE was scheduled to begin at 5,500 hours of cumulative test time, because the idealized curve in Figure 22 showed that the FOE requirement of 110 hours *MTBF* could be achieved in 5,500 hours of testing. The beginning of IPT was scheduled in a similar manner. As stated in the given conditions, these three test phases were to last for 1,100 hours each, and the fire control systems undergoing test were to remain in a fixed configuration throughout each test phase. This latter condition implied that the reliability during each test phase should be constant, and the planned growth curve should therefore show a constant reliability during these periods of testing.

After each test phase, the reliability was expected to be increased sharply by the incorporation of delayed fixes. In addition, testing was to be halted after 1,700 hours of test time in order to incorporate design fixes into new system prototypes. The planned growth curve had to indicate jumps in reliability at each of these points in the test program. During the test time outside the formal test phases, steady reliability growth was planned because of continual fixing of problem failure modes. The resulting planned growth curve is shown in FIGURE 5-22.

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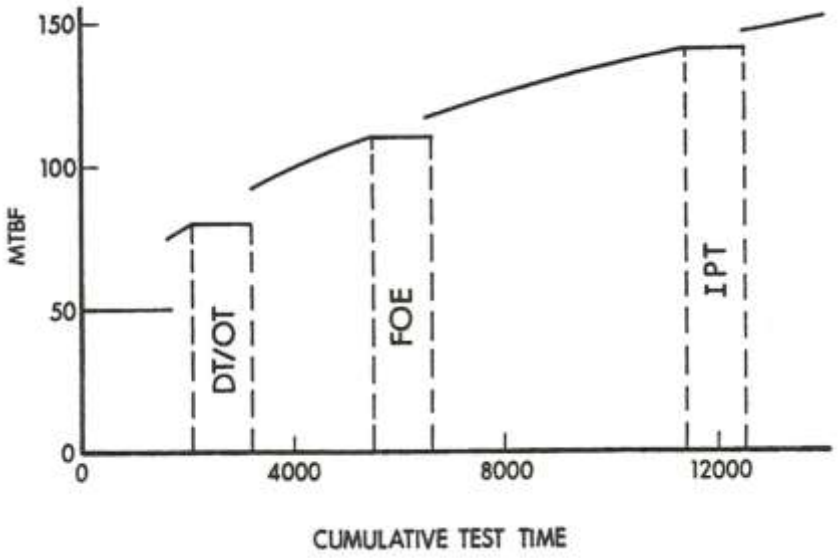


FIGURE 5-22. *Planned Growth Curve*

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5.3 System Level Planning Model (SPLAN).

The material in this section is based on [3] Operating Characteristic Analysis for Reliability Growth Programs, AMSAA TR-524, August 1992.

5.3.1 Introduction.

A well thought out reliability growth plan can serve as a significant management tool in scoping out the required resources to enhance system reliability and demonstrate the system reliability requirement. The principal goal of the growth test is to enhance reliability by the iterative process of surfacing failure modes, analyzing them, implementing corrective actions (fixes), and testing the "improved" configuration to verify fixes and continue the growth process by surfacing remaining failure modes. If the growth test environment during EMD reasonably simulates the mission environment stresses then it may be feasible to use the growth test data to statistically demonstrate the technical, (i.e., engineering), requirement (denoted by TR) for system reliability. Such use of the growth test data could eliminate the need to conduct a follow-on reliability demonstration test. The classical demonstration test requires that the system configuration be held constant throughout the test. This type of test is principally conducted to assess and demonstrate the reliability of the configuration under test. Associated with the demonstration test are statistical consumer and producer risks. In our context, they are frequently termed the Government and contractor risks, respectively. In broad terms, the Government risk is the probability of accepting a system when the true technical reliability is below the *TR* and the contractor risk is the probability of rejecting a system when the true technical reliability is at least the contractor's target value (set above the *TR*). An extensive amount of test time may be required for the reliability demonstration test to suitably limit these statistical risks. Moreover, this allotted test time would be principally devoted to demonstrating the system *TR* associated with the configuration under test instead of to enhancing the system reliability through the reliability growth process of sequential configuration improvement. In today's austere budgetary environment, it is especially important to make maximum use of test resources. With proper planning, a reliability growth program can be an efficient procedure for demonstrating the system reliability requirement while reliability improvements are being achieved via the growth process.

5.3.2 Background.

During a reliability growth test phase, the system configuration is changing due to the activity of surfacing failure modes, analyzing the modes, and implementing fixes to the surfaced modes. It is often reasonable to portray this reliability growth in an idealized manner, i.e., by a smooth rising curve that captures the overall pattern of growth. The curve relates a measure of system reliability, e.g., mean-time-between-failures (MTBF), to test duration (e.g., hours). The functional form used to express this relationship in is given by,

$$\boldsymbol{M(t)} = \left(\frac{\boldsymbol{M_I}}{(\mathbf{1}-\alpha)}\right)\left(\frac{\boldsymbol{t}}{\boldsymbol{t_I}}\right)^{\alpha} \qquad 5.3\text{-}1$$

In this equation, *M(t)* typically denotes the MTBF achieved after t test hours. The exponent α is termed the growth rate and represents the slope of the assumed linear relationship between *ln{M(t)}* and *ln(t)*, where *ln* denotes the base e logarithm function. The parameters t_I, M_I may be thought of as defining the initial conditions. Note that t_I from a previous section is equivalent to

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t_I as defined above. In particular, M_I may be interpreted as the MTBF associated with the initial configuration entering the reliability growth test. In this interpretation, t_I would be the planned cumulative test time until one or more fixes are incorporated, i.e., equal to *Prob* defined in Equation 5.1-1. An alternate and more general interpretation of M_I and t_I would be to regard M_I as the anticipated average MTBF over an initial test period, t_I.

In the above discussion, we have referred to *M(t)* as the MTBF and have measured test duration by time units, e.g., *t* hours. We will continue to refer to *M(t)* and test duration *t* in this fashion; however, more generally, *M(t)* may denote mean-miles-to-failure or mean-rounds-to-failure (for a large number of rounds). The corresponding measures of test duration would be test mileage or rounds expended, respectively.

As indicated in Section 5.2.3, we should consider using the data generated during the reliability growth test phase to demonstrate the system reliability technical requirement (TR) at a specified confidence level **γ.** This section addresses the case where the data consists of individual failure times $0 < t_1 < t_2 < \ldots < t_n \leq T$ for *n* observed mission reliability failures during test time *T*, where Equation 5.3-1 is assumed to hold for $0 < t \leq T$. Since the 1981 MIL-HDBK-189 growth model governed by Equation 5.3-1 is being assumed in this section, we will also require that the observed number of failures by test duration *t*, denoted by *N(t)*, be a non-homogeneous Poisson process with intensity function $\rho(t) = \{M(t)\}^{-1}$.

The growth curve planning parameters α, t_I, M_I, and the test time *T* should be chosen to reasonably limit the consumer (Government) and producer (contractor) statistical risks referred to in Section 5.2.3. Prior to presenting the relationship between these risks and the parameters mentioned above, it is instructive to review the determination of these risks for a reliability demonstration test based on a constant configuration.

The parameters defining the reliability demonstration test consist of the test duration T_{DEM}., and the allowable number of failures *c*. Define the random variable F_{obs} to be the number of failures that occur during the test time T_{DEM}. Denote the observed value of F_{obs} by f_{obs}. Then the "acceptance" or "passing" criterion is simply $f_{obs} \leq c$.

Let *M* denote the MTBF associated with the constant configuration under test. Then F_{obs} has the Poisson probability distribution given by,

$$Prob\,(F_{obs} = i\,) = \left(e^{\left(\frac{-T_{DEM}}{M}\right)}\right)\left(\frac{\left(\frac{T_{DEM}}{M}\right)^{i}}{i!}\right) \qquad 5.3\text{-}2$$

Thus the probability of acceptance, denoted by Prob(*A; M, c,* T_{DEM}), as a function of *M*, *c*, and T_{DEM} is given by,

$$Prob(A; M, c, T_{DEM}\,) = Prob\,(F_{obs} \leq c) =$$
$$\sum_{i=0}^{c} Prob\,(F_{obs} = i) \sum_{i=0}^{c}\left(e^{\left(\frac{-T_{DEM}}{M}\right)}\right)\left(\frac{\left(\frac{T_{DEM}}{M}\right)^{i}}{i!}\right) \qquad 5.3\text{-}3$$

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To ensure "passing the demonstration test" is equivalent to demonstrating the TR at confidence level γ (e.g., γ = 0.80 or γ = 0.90), we must choose *c* such that,

$$f_{obs} \leq c \iff TR \leq \lambda_{\gamma}(f_{obs}) \qquad 5.3\text{-}4$$

where $TR > 0$ and $\lambda_{\gamma}(f_{obs})$ denotes the value of the 100 γ percent lower confidence bound when (f_{obs}) failures occur in the demonstration test of length T_{DEM}. Note that $\lambda_{\gamma}(f_{obs})$ is a lower confidence bound on the true (but unknown) *MTBF* of the configuration under test. It is well known (see Proposition 1 in Appendix on OC Derivations) that the following choice of *c* satisfies 5.3-4.

Choose *c* to be the largest non-negative integer *k* that satisfies the inequality

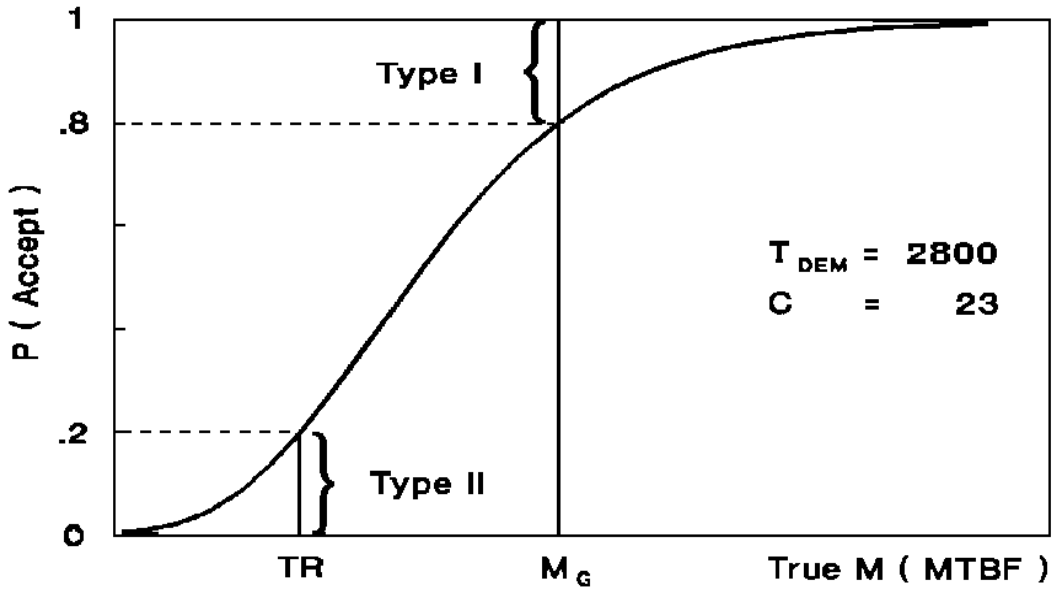
$$\sum_{i=0}^{k} e^{-T_{Dem}/TR} \frac{(T_{Dem}/TR)^{i}}{i!} \leq 1-\gamma \qquad 5.3\text{-}5$$

Note *c* is well-defined provided

$$\exp(-T_{Dem}/TR) \leq 1-\gamma \qquad 5.3\text{-}6$$

Throughout this section, we assume 5.3-6 holds and that *c* is defined as above.

Recall that the operating characteristic (OC) curve associated with a reliability demonstration test is the graph of the probability of acceptance, i.e., Prob (A;M,c,T_{DEM}) given in Equation 5.3-3, as a function of the true but unknown constant MTBF, *M* as depicted on FIGURE 5-23.



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FIGURE 5-23. *Example OC Curve for Reliability Demonstration Test*

The Government (or consumer risk) associated with this curve, called the Type II risk, is defined by

$$\text{Type II} \quad \underline{\Delta} \quad \text{Prob}\,(A; TR, c, T_{Dem}) \qquad 5.3\text{-}7$$

Thus, by the choice of *c*,

$$\text{Type II} \quad \leq \quad 1 - \gamma \qquad 5.3\text{-}8$$

For the contractor (producer) to have a reasonable chance of demonstrating the TR with confidence γ, the system configuration entering the reliability demonstration test must often have a MTBF value, say M_G (the contractor's goal MTBF) that is considerably higher than the TR. The probability that the producer fails the demonstration test given the system under test has a true MTBF value of M_G is termed the producer (contractor) or Type I risk. Thus

$$\text{Type I} \quad = \quad 1 \quad - \quad \text{Prob}\,(A; M_G, c, T_{Dem}) \qquad 5.3\text{-}9$$

If the Type I risk is higher than desired, then either a higher value of M_G should be attained prior to entering the reliability demonstration test or T_{DEM} should be increased. If T_{DEM} is increased then *c* may have to be readjusted for the new value of T_{DEM} to remain the largest non-negative integer that satisfies inequality 5.3-5.

The above numbered equations and inequalities express the relationships between the reliability demonstration test parameters *c* and T_{DEM}, the requirement parameters TR and γ, and the associated risk parameters (the consumer and producer risks). These relationships are fundamental in conducting tradeoff analyses involving these parameters for planning reliability demonstration tests. In the next section we will present relationships between the defining parameters for a reliability growth curve (M_I, t_I, α, and *T*), the requirement parameters (*TR* and γ), and the associated statistical risk parameters (the consumer and producer risks). Once these relationships are in hand, tradeoffs between these parameters may be utilized to consider demonstrating the *TR* at confidence level γ by utilizing reliability growth test data.

5.3.3 Reliability Growth Operating Characteristic (OC) Analysis.

In the previous section, it was noted that for a reliability demonstration test, passing the test could be stated in terms of the allowable number of failures, *c*. It was noted that if c is properly chosen, then passing the test is equivalent to demonstrating the *TR* at confidence level γ, i.e.,

$f_{obs} \leq c \quad \Leftrightarrow \quad TR \leq \ell_\gamma (f_{obs})$.

In the presence of reliability growth, observing *c* or fewer failures is not equivalent to demonstrating the TR at a given confidence level. The cumulative times to failure as well as the number of failures must be considered when using reliability growth test data to demonstrate the *TR* at a specified confidence level γ. Thus, the "acceptance" or "passing" criterion must be stated

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directly in terms of the γ lower confidence bound on *M(T)* calculated from the reliability growth data. These data will be denoted by (*n*, *s*) where n is the number of failures occurring in the growth test of duration *T* and $s = (t_1, t_2, \ldots, t_n)$ is the vector of cumulative failure times. In particular, t_i denotes the cumulative test time to the ith failure and $0 < t_1 < t_2 \ldots\ldots < t_n \le T$ for $n \ge 1$. We also refer to the random vector (*N*, *S*) which takes on values (*n*, *s*) for $n \ge 1$. Unless otherwise stated, throughout the remainder of this section, (*N*, *S*) will be conditioned on $N \ge 1$. Using the lower confidence bound methodology developed for reliability growth data as stated by Crow in [4], we would define our acceptance criterion by the inequality

$$TR \le \ell_\gamma (n, s) \qquad 5.3\text{-}10$$

where $\ell_\gamma(n,s)$ is the γ statistical lower confidence bound on *M(T),* calculated for $n \ge 1$. Thus, the probability of acceptance is given by,

$$\text{Prob}\,(TR \le L_\gamma(N,S)) \qquad 5.3\text{-}11$$

where the random variable $L_\gamma(N,S)$ takes on the value $\ell_\gamma(n,s)$ when (*N*, *S*) takes on the value (*n*, *s*). For $n \ge 1$, we define,

$$\ell_\gamma (n, s) \;\triangleq\; \left(\frac{2n}{z_\gamma (n)} \right)^2 \hat{M}_n (T) \qquad 5.3\text{-}12$$

where $z_\gamma(n)$ is the unique positive value of *z* such that

$$(1/I_1(z)) \sum_{j=1}^{n} \frac{(z/2)^{2j-1}}{j!(j-1)!} \;=\; 1-\gamma \qquad 5.3\text{-}13$$

In the above, the function I_1 denotes the modified Bessel function of order one defined as follows:

$$I_1(z) \;\triangleq\; \sum_{j=1}^{\infty} \frac{(z/2)^{2j-1}}{j!(j-1)!} \qquad 5.3\text{-}14$$

In Equation (5.3-12), $\hat{M}_n(T)$ denotes the maximum likelihood estimate (MLE) for *M(T)* when *n* failures are observed.

$$\hat{M}_n(T) \;=\; T / \left(n \hat{\beta}_n \right) \qquad 5.3\text{-}15$$

where

$$\hat{\beta}_n \;=\; n / \left(\sum_{i=1}^{n} \ln (T/t_i) \right) \qquad 5.3\text{-}16$$

The distribution of (*N, S*) and hence that of L_γ (*N, S*) is completely determined by the test duration *T* together with any set of parameters that define a unique reliability growth curve of the form given by Equation 5.3-1 in Section 5.3.2. Thus, the value of a probability expression such

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as given in 5.3-11 also depends on T and the assumed underlying growth curve parameters. One such set of parameters, as seen directly from Equation 5.3-1, is t_1, M_1, α together with T. In this growth curve representation, t_1 may be arbitrarily chosen subject to $0 < t_1 < T$. Alternately, scale parameter $\lambda > 0$ and growth rate α, together with T, can be used to define the growth curve by the equation

$$M(t) = 1/(\lambda \beta t^{\beta-1}), \quad 0 < t \leq T \qquad 5.3\text{-}17$$

where $\beta = 1 - \alpha$. Note by Equation 5.3-17,

$$1/\lambda = (M(T))\beta T^{\beta-1} \qquad 5.3\text{-}18$$

Thus, the growth curve can also be expressed as,

$$M(t) = (M(T))(t/T)^{\alpha}, \quad 0 < t \leq T \qquad 5.3\text{-}19$$

By Equation 5.3-19 we see that the distribution of (N, S) and hence that of L_γ (N, S) is determined by (α, T, $M(T)$). Unless otherwise stated, throughout the remainder of this section, the distributions for (N, S) and for random variables defined in terms of (N, S) will be with respect to a fixed but unspecified set of values for α, T, $M(T)$ subject only to $\alpha<1$, $T>0$, and $M(T)>0$. The same considerations apply to any associated probability expressions. In particular, the probability of acceptance, i.e., *Prob* ($TR \leq L_\gamma(N, S)$), is a function of (α, T, $M(T)$).

To further consider the probability of acceptance, we must first consider several properties of the system of lower confidence bounds generated by L_γ (N, S) as specified via Equations 5.3-12 through 5.3-16. The statistical properties of this system of bounds directly follow from the properties of a set of conditional bounds derived as specified in [4]. These latter bounds are conditioned on a sufficient statistic W that takes on the value

$$w = \sum_{i=1}^{n} \ln(T/t_i) \qquad 5.3\text{-}20$$

when (N, S) takes on the value (n, s).

Let L_γ (N, S; w) denote the random variable L_γ (N, S) conditioned on $W = w>0$. In accordance with [4] it is shown that L_γ (N, S; w) generates a system of γ lower confidence bounds on $M(T)$, i.e.,

$$\text{Prob}\left(L_\gamma(N, S; w) \leq M(T)\right) \geq \gamma \qquad 5.3\text{-}21$$

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for each set of values $(\alpha, T, M(T))$ subject to $\alpha<1$, $T>0$, and $M(T)>0$. Note that the value of w is not known prior to conducting the reliability growth test. Thus, to calculate an OC curve for test planning, i.e., a priori, we wish to base our acceptance criterion on $L_\gamma(N, S)$ as in 5.3-11 and not on the conditional random variable $L_\gamma(N, S; w)$. We can utilize Equation 5.3-21 to show (see Propositions 2, 3, and 4 in Appendix for OC Derivations) that the Type II or consumer risk for $M(T)$=TR is at most 1-γ (for any α<1 and T>0),

$$\text{Type II} \quad = \quad \text{Prob}\,(\text{TR} \le L_\gamma(\text{N},\text{S})) \quad \le \quad 1-\gamma \qquad \text{5.3-22}$$

for any α<1 and $T>0$, provided $M(T) = TR$.

To emphasize the functional dependence of the probability of acceptance on the underlying true growth curve parameters $(\alpha, T, M(T))$, this probability is denoted by Prob $(A; \alpha, T, M(T))$. Thus,

$$\text{Prob}\left(\text{A};\alpha,\text{T},\text{M}(\text{T})\right) \quad \underline{\Delta} \quad \text{Prob}\,(\text{TR} \le L_\gamma(\text{N},\text{S})) \qquad \text{5.3-23}$$

where the distribution of (N, S) and hence that of $L_\gamma(N, S)$ is determined by $(\alpha, T, M(T))$. It can be shown that Prob $(A; \alpha, T, M(T))$ only depends on the values of $M(T)/TR$ (or equivalently $M(T)$ for known TR) and $E(N)$. The ratio $M(T)/TR$ is analogous to the discrimination ratio for a constant configuration reliability demonstration test of the type considered in Section 5.5.2. Note $E(N)$ denotes the expected number of failures associated with the growth curve determined by $(\alpha, T, M(T))$. More explicitly, the following equations can be derived (see Propositions 5 and 6 in Appendix on OC Derivations):

$$\text{E}(\text{N}) \quad = \quad \text{T}/\{(1-\alpha)\text{M}(\text{T})\} \qquad \text{5.3-24}$$

And

$$\text{Prob}\,(\text{A};\alpha,\text{T},\text{M}(\text{T})) = \quad (1-\text{e}^{-\mu})^{-1} \sum_{\text{n}=1}^{\infty}\left[\text{Prob}\left(\frac{\chi^2_{2\text{n}}}{z^2_\gamma(\text{n})} \ge \frac{1}{2\mu\text{d}}\right)\right]\text{e}^{-\mu}\left(\frac{\mu^n}{\text{n!}}\right) \qquad \text{5.3-25}$$

where $\mu \underline{\Delta} E(N)$ and $d \underline{\Delta} M(T)/TR$. Note 5.3-25 shows that the probability of acceptance only depends on μ and d. Thus, we will subsequently denote the probability of acceptance by Prob (A;μ,d). By 5.3-22,

$$\text{Type II} \quad = \quad \text{Prob}\,(\text{A};\mu,1) \quad \le \quad 1-\gamma \qquad \text{5.3-26}$$

Thus, the actual value of the Government or consumer risk solely depends on μ and is at most $1-\gamma$. To consider the producer or contractor risk, Type I, let α_G denote the contractor's target or goal growth rate. This growth rate should be a value the contractor feels he can achieve for the growth test. Let M_G denote the contractor's MTBF goal. This is the MTBF value the contractor plans to achieve at the conclusion of the growth test of duration T. Thus, if the true growth curve

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has the parameters α_G and M_G, then the corresponding contractor risk of not demonstrating the TR at confidence level γ (utilizing the generated reliability growth test data) is given by,

$$\text{Type I} = 1 - \text{Prob}(A; \mu_G, d_G) \qquad 5.3\text{-}27$$

where

$$d_G = M_G / TR \quad \text{and} \quad \mu_G = T/\{(1-\alpha_G)M_G\} \qquad 5.3\text{-}28$$

If the Type I risk is higher than desired, there are several ways to consider reducing this risk while maintaining the Type II risk at or below $1-\gamma$. Since Prob $(A; \mu_G, d_G)$ is an increasing function of μ_G and d_G, the Type I risk can be reduced by increasing one or both of these quantities, e.g., by increasing T. To further consider how the Type I statistical risk can be influenced, we express d_G and μ_G in terms of TR, T, α_G, and the initial conditions (M_I, t_I). Using Equations 5.3-1 and 5.3-19 with $\alpha = \alpha_G$ and $M(T) = M_G$, by 5.3-28 we can show

$$M_G / TR = d_G = \left\{ \frac{M_I}{(1-\alpha_G) t_I^{\alpha_G} TR} \right\} T^{\alpha_G} \qquad 5.3\text{-}29$$

and

$$E(N) = \mu_G = (t_I^{\alpha_G} / M_I) T^{1-\alpha_G} \qquad 5.3\text{-}30$$

Note for a given requirement TR, initial conditions (M_I, t_I), and an assumed positive growth rate α_G, the contractor risk is a decreasing function of T via Equations 5.3-27, 5.3-29, and 5.3-30. These equations can be used to solve for a test time T such that the contractor risk is a specified value. The corresponding Government risk will be at most $1-\gamma$ and is given by Equation 5.3-26.

Section 5.3.3 contains two examples of an OC analysis for planning a reliability growth program. The first example illustrates the construction of an OC curve for given initial conditions (M_I, t_I) and requirement TR. The second example illustrates the iterative solution for the amount of test time T necessary to achieve a specified contractor (producer) risk, given initial conditions (M_I, t_I) and requirement TR. These examples use Equations 5.3-29 and 5.3-30 rewritten as in Equations 5.3-1 and 5.3-24, respectively, i.e.,

$$M(T) = \left(\frac{M_I}{1-\alpha} \right) \left(\frac{T}{t_I} \right)^{\alpha} \quad \text{and} \quad E(N) = \frac{T}{(1-\alpha) M(T)} \qquad 5.3\text{-}31$$

The quantities $d = M(T)/TR$ and $\mu = E(N)$ are then used to obtain an approximation to Prob $(A; \mu, d)$. Approximate values are provided in the Appendix F on the Probability of Demonstrating the TR with Confidence for a range of values for μ and d. The nature of this approximation is also discussed in the Appendix F.

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5.3.4 Application.

5.3.4.1 Example 1.

Suppose we have a system under development that has a technical requirement (TR) MTBF of 100 hours to be demonstrated with 80 percent confidence. For the developmental program, a total of 2800 hours test time (T) at the system level has been predetermined for reliability growth purposes. Based on historical data for similar type systems and on lower level testing for the system under development, the initial MTBF (M_I) averaged over the first 500 hours (t_I) of system-level testing was expected to be 68 hours. Using these data, an idealized reliability growth curve was constructed such that if the tracking curve followed along the idealized growth curve, the TR, MTBF of 100 hours would be demonstrated with 80 percent confidence. The growth rate (α) and the final MTBF ($M(T)$) for the idealized growth curve were 0.23 and 130 hours, respectively. The idealized growth curve for this program is depicted on FIGURE 5-24.

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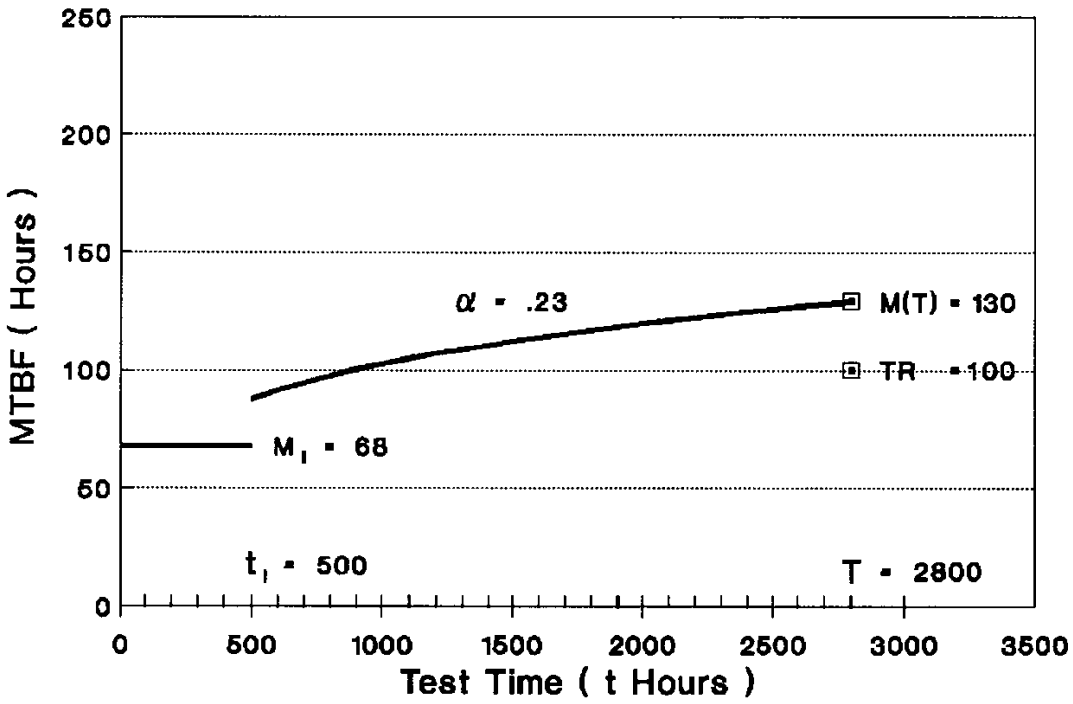


FIGURE 5-24. *Idealized Reliability Growth Curve*

For this example, suppose we want to determine the operating characteristic (OC) curve for the program. For this, we need to consider alternate idealized growth curves where the *M(T)* vary but the M_I and t_I remain the same values as those for the program idealized growth curve; i.e., M_I = 68 hours and t_I = 500 hours. In varying the *M(T)*, this is analogous to considering alternate values of the true MTBF for a reliability demonstration test of a fixed configuration system. For this program, one alternate idealized growth curve was determined where *M(T)* equals the TR whereas the remaining alternate idealized growth curves were determined for different values of the growth rate. These alternate idealized growth curves along with the program idealized growth curve are depicted on FIGURE 5-25.

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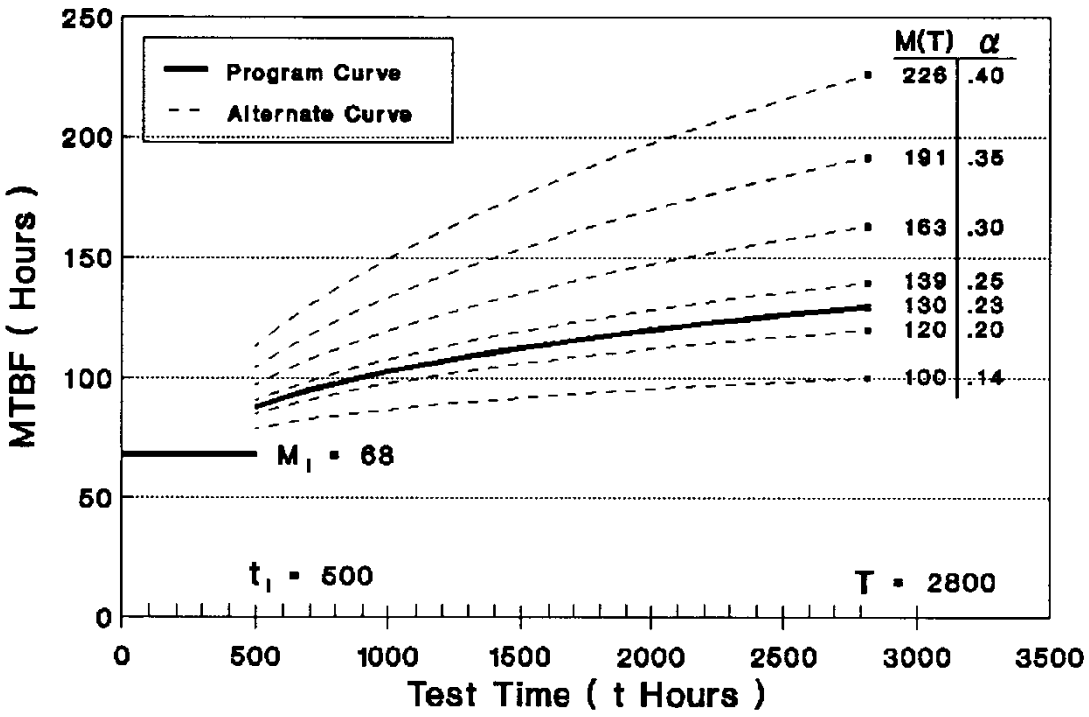


FIGURE 5-25. *Program and Alternate Idealized Growth Curves*

Now, for each idealized growth curve we find *M(T)* and the expected number of failures *E(N)* from 5.3-31. Using the ratio *M(T)/TR* and *E(N)* as entries in the tables contained in Appendix F, we determine, by double linear interpolation, the probability of demonstrating the *TR* with 80 percent confidence. This probability is actually the probability that the 80 percent lower confidence bound (80 percent LCB) for *M(T)* will be greater than or equal to the *TR*. These probabilities represent the probability of acceptance *(P(A))* points on the OC curve for this program which is depicted below in FIGURE 5-26. The *M(T)*, α, *E(N)*, and *P(A)* for these idealized growth curves are summarized in TABLE III.

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TABLE III. Example Planning Data

M (T)	α	E (N)	P (A)
100	0.14	32.6	0.15
120	0.20	29.2	0.37
130	0.23	28.0	0.48
139	0.25	26.9	0.58
163	0.30	24.5	0.77
191	0.35	22.6	0.90
226	0.40	20.6	0.96

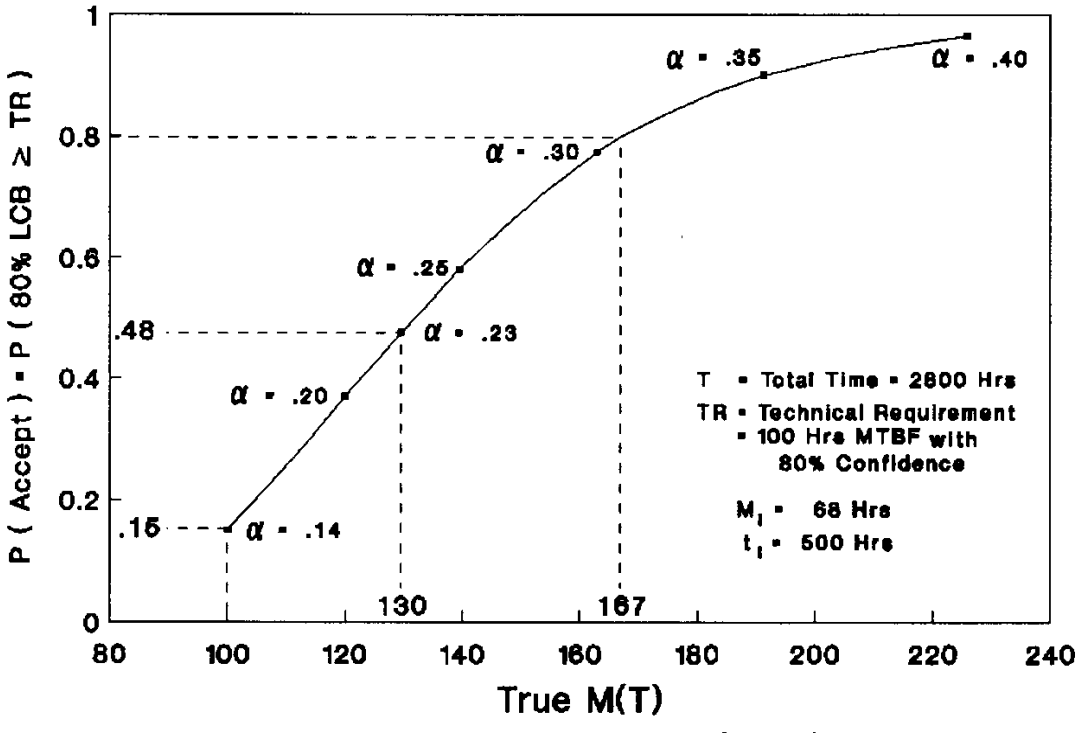


FIGURE 5-26. *Operating Characteristic (OC) Curve*

From the OC curve, the Type I or producer risk is 0.52 (1-0.48) which is based on the program idealized growth curve where *M(T)* = 130. Note that if the true growth curve were the program idealized growth curve, there is still a 0.52 probability of not demonstrating the *TR* with 80 percent confidence. This occurs even though the true reliability would grow to *M(T)* = 130 which is considerably higher than the *TR* value of 100. The Type II or consumer risk, which is based on the alternate idealized growth curve where *M(T)* = *TR* = 100, is 0.15. As indicated on the OC curve, it should be noted that for this developmental program to have a producer risk of 0.20, the contractor would have to plan on an idealized growth curve with *M(T)* = 167.

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5.3.4.2 Example 2.
Consider a system under development that has a technical requirement (*TR*) MTBF of 100 hours to be demonstrated with 80 percent confidence, as in Example 1. The initial MTBF (M_I) over the first 500 hours (t_I) of system level testing for this system was estimated to be 48 hours which, again as in Example 1, was based on historical data for similar type systems and on lower level testing for the system under development. For this developmental program, it was assumed that a growth rate (α) of 0.30 would be appropriate for reliability growth purposes. Now, for this example, suppose we want to determine the total amount of system level test time (*T*) such that the Type I or producer risk for the program idealized reliability growth curve is 0.20; i.e., the probability of not demonstrating the *TR* of 100 hours with 80 percent confidence is 0.20 for the final MTBF value (*M(T)*) obtained from the program idealized growth curve. This probability corresponds to the probability of acceptance (*P(A)*) point of 0.80 (1-0.20) on the operating characteristic (OC) curve for this program.

Now, to determine the test time *T* which will satisfy the Type I or producer risk of 0.20, we first select an initial value of *T* and, as in Example 1, find *M(T)* and the expected number of failures (*E(N)*) from 5.3-31. Then, again, using the ratio *M(T)/TR* and *E(N)* as entries in the tables contained in Appendix F, we determine, by double linear interpolation, the probability of demonstrating the TR with 80 percent confidence. An iterative procedure is then applied until the *P(A)* obtained from the table equals the desired 0.80 within some reasonable accuracy. For this example, suppose we selected 3000 hours as our initial estimate of *T* and obtained the following iterative results:

TABLE IV. Example Planning Data

T	M(T)	E(N)	P(A)
3000	117.4	36.5	<0.412
4000	128.0	44.6	<0.610
5000	136.8	52.2	<0.793
5500	140.8	55.8	0.815
5400	140.0	55.1	0.804
5300	139.2	54.4	0.790
5350	139.6	54.7	0.796
5375	139.8	54.9	0.800

Based on these results, we determine *T* = 5375 hours to be the required amount of system level test time such that the Type I or producer risk for the program idealized growth curve is 0.20.

5.3.5 Summary.
The concepts of an operating characteristic (OC) analysis have been extended to the reliability growth setting. Government (consumer) and contractor (producer) statistical risks have been expressed in terms of the underlying growth curve parameters, test duration, and reliability requirement. In particular, for a given confidence level, these risks have been shown to depend solely on the expected number of failures during the growth test and the ratio of the MTBF to be achieved at the end of the growth program to the MTBF technical requirement to be

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demonstrated with confidence. Formulas have been developed for computing these risks as a function of the test duration and growth curve planning parameters.

The methodology developed and illustrated in this section should be of interest to RAM analysts responsible for structuring realistic reliability growth programs to achieve and demonstrate program objectives with reasonable statistical risks. In particular, this methodology allows the RAM analysts to construct a reliability growth curve that considers both the Government and contractor risks prior to agreeing to a reliability growth program.

5.4 Subsystem Level Planning Model (SSPLAN).

5.4.1 Subsystem Reliability Growth.

This section is based on material as stated in [5] Developing a Subsystem Reliability Growth Program Using the Subsystem Reliability Growth Planning Model (SSPLAN), AMSAA TR-555, September 1994.

5.4.1.1 Benefits and Special Considerations.

Conducting a subsystem reliability growth program prior to the start of system level testing can:

a. reduce the amount of system level testing,
b. reduce or eliminate many failure mechanisms (problem failure modes) early in the development cycle where they may be easier to locate and correct,
c. allow for the use of subsystem test data to monitor reliability improvement,
d. increase product quality by placing more emphasis on lower level testing and
e. provide management with a strategy for conducting an overall reliability growth program.

Thus, subsystem reliability growth offers the potential for significant savings in testing cost. To be an effective management tool for planning and assessing system reliability in the presence of reliability growth, it is important for the subsystem reliability growth process to adhere as closely as possible to the following considerations:

a. Potential high-risk interfaces need to be identified and addressed through joint subsystem testing,
b. Subsystem usage/test conditions need to be in conformance with the proposed system level operational environment as envisioned in the Operational Mode Summary/Mission Profile (OMS/MP),
c. Failure Definitions/Scoring Criteria (FD/SC) formulated for each subsystem need to be consistent with the FD/SC used for system level test evaluation.

5.4.1.2 Overview of SSPLAN Approach.

The subsystem reliability growth planning model, SSPLAN, provides the user with a means to develop subsystem testing plans for demonstrating a system mean time between failure (MTBF) goal prior to system level testing. (The MTBF goal is also referred to as the MTBF objective ($MTBF_{obj}$).) In particular, the model is used to develop subsystem reliability growth planning curves that, with a specified probability, achieve a system MTBF objective with a specified

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confidence level. More precisely, associated with the subsystem MTBFs growing along a set of planned growth curves for given subsystem test durations is a probability; this is termed the probability of acceptance (P_A), the probability that the system MTBF objective will be demonstrated at the specified confidence level. The complement of P_A, $1\text{-}P_A$, is termed the producer's (or contractor's) risk: the risk of not demonstrating the system MTBF objective at the specified confidence level when the subsystems are growing along their target growth curves for the prescribed test durations. Note that P_A also depends on the fixed MTBF of any non-growth subsystem and on the lengths of the demonstration tests on which the non-growth subsystem MTBF estimates are based.

SSPLAN estimates P_A for a given value of the final combined growth subsystem MTBF ($MTBF_{G,sys}$) by simulating the reliability growth of each subsystem and calculating a statistical lower confidence bound (LCB) for the final system MTBF based on the growth and non-growth subsystem simulated failure data. If the system LCB, at the specified confidence level, meets or exceeds the specified MTBF goal, then the trial is labeled a success. SSPLAN runs as many as 5000 trials, and estimates P_A as the number of successes divided by the number of trials.

One of the model's primary outputs is the growth subsystem test times. If the growth subsystems were to grow along the planning curves for these test times then the probability would be P_A that the subsystem test data demonstrate the system MTBF objective, $MTBF_{obj}$, at the specified confidence level. The model determines the subsystem test times by using a specified fixed allocation of the combined final failure intensity to each of the individual growth subsystems.

As a reliability management tool, the model can serve as a means for prime contractors to coordinate/integrate the reliability growth activities of their subcontractors as part of their overall strategy in implementing a subsystem reliability test program for their developmental systems.

5.4.1.3 List of Notations.
There are some variant terms in the following parameter list to show that the form of some parameters depends on the context in which they are used. For example, T, $T_{D,I}$ and $T_{G,i}$ indicate, respectively, that time may be used generically, specifically for non-growth subsystem i and specifically for growth subsystem i. Also, for notational convenience, several parameters that can vary by subsystem are sometimes written without a subsystem subscript. However, subscripts are used where required for clarity.

t	subsystem test time
T	total subsystem test time $\left(0 < t \le T\right)$
$F(t)$	total number of subsystem failures by time t
$E\,[F\,(t)]$	expected number of subsystem failures by time t
λ	AMSAA model scale parameter for growth subsystem
β	AMSAA model shape (or growth) parameter for growth subsystem
α	growth rate
t_I	initial time period for subsystem growth test
$MTBF$	Mean Time Between Failure
M_I	initial average MTBF over interval

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λ_I	initial average failure intensity over interval
MS	management strategy, ($0 < MS \leq 1$)
$\rho(t)$	instantaneous failure intensity at time t, $[\rho(t) > 0]$
$M(t)$	instantaneous MTBF at time t
$MTBF_{obj}$	system MTBF objective to be demonstrated with confidence γ
P_{obi}	probability of acceptance associated with demonstrating $MTBF_{obj}$
LCB	lower confidence bound
D	demonstration (non-growth) test data or estimator subscript
G	growth test data or estimator subscript
i	subsystem index number
$T_{D,i}$	total amount of demonstration or "equivalent demonstration" (non-growth) test time for subsystem i
$T_{G,i}$	total amount of growth test time for subsystem i
$T_{MAX,i}$	specified maximum allowable growth test time for subsystem i. Thus $T_{G,i} \leq T_{MAX,i}$
$n_{D,i}$	number of failures during a demonstration test of length $T_{D,i}$ for a non-growth *subsystem* i. Also, number of "equivalent demonstration" failures for growth subsystem i during growth test.
$n_{G,i}$	number of failures during a test time $T_{G,i}$ for a growth subsystem i
$M_{D,i}$	demonstration (constant) MTBF for non-growth subsystem i
$\rho_{D,i}$	equals $M_{D,i}^{-1}$
$M_{G,i}$	Final *MTBF* for growth subsystem i
$\rho_{G,i}(T_{G,i})$	equals $M_{G,i}^{-1}$
^	denotes an estimate when placed over a parameter
$\hat{\rho}_{D,i}$	estimate of $\rho_{D,i}$
$\hat{\rho}_{G,i}(T_{G,i})$	estimate of $\rho_{G,i}(T_{G,i})$
χ^2_{df}	chi-squared random variable with "df" degrees of freedom
ρ_{SYS}	final system failure intensity
$\rho_{G,SYS}$	total failure intensity contribution of growth subsystems to ρ_{SYS}
a_i	fraction of $\rho_{G,SYS}$ allocated to growth subsystem i
M_{SYS}	final system MTBF
$M_{G,SYS}$	final MTBF of combined growth subsystems, i.e., $M_{G,sys} = \rho_{G,sys}^{-1}$
$N_{D,SYS}$	system demonstration "equivalent" number of failures
$T_{D,SYS}$	system demonstration "equivalent" test time
MLE	maximum likelihood estimate
~	symbol for "distributed as" a specified random variable
$\hat{M}_{D,i}$	subsystem i MTBF estimate of demonstration or "equivalent demonstration" MTBF
$\hat{M}_{G,i}$	subsystem i MLE for final MTBF of growth subsystem
$\hat{M}_{SYS}$	estimate of final system MTBF
γ	specified confidence level for demonstrating $MTBF_{obj}$
$\chi^2_{df,\gamma}$	chi-squared 100γ percentile point for df degrees of freedom

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$\hat{\rho}_i$	estimate of final subsystem *i* failure intensity
$\hat{\rho}_{SYS}$	estimate of final system failure intensity
K	number of subsystems
$LCB_{D,i,\gamma}$	subsystem *i* LCB at confidence level from demonstration data
$LCB_{G,i,\gamma}$	subsystem *i* LCB at confidence level from growth data
$(C_F)_i$	cost per failure for subsystem *i*
$(C_T)_i$	cost per hour for subsystem *i*
C_{Total}	total testing cost
$C_i[T_{D,i}]$	cost contribution of non-growth subsystem *i* to C_{Total} as a function of $T_{D,i}$
$C_i[\rho_{G,i}(T_{G,i})]$	cost contribution of growth subsystem *i* to C_{Total} as a function of $\Delta_{G,i}(T_{G,i})$
$(M_{G,sys})_{NEW}$	new value of $M_{G,sys}$ to use in search routine
$(M_{G,sys})_{LB}$	lower bound for $M_{G,sys}$
$(M_{G,sys})_{UB}$	upper bound for $M_{G,sys}$
$(P_A)_{LB}$	estimated associated with $(M_{G,sys})_{LB}$
$(P_A)_{UB}$	estimated associated with $(M_{G,sys})_{UB}$
$(P_A)_{GOAL}$	desired P_A

5.4.2 SSPLAN Methodology.

5.4.2.1 Model Assumptions.

The SSPLAN methodology assumes that a system may be represented as a series of $K \geq 1$ independent subsystems. (The theory allows for $K = 1$ but the current computer implementation requires $K \geq 2$.)

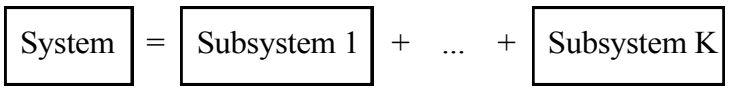


FIGURE 5-27. *System Architecture*

This means that a failure of any single subsystem results in a system level failure and that a failure of a subsystem does not influence (either induce or prevent) the failure of any other subsystem. SSPLAN allows for a mixture of test data from growth and non-growth subsystems, but in its current implementation, at least one growth subsystem is required to run the model. For growth subsystems, the model assumes that the number of failures occurring over a period of test time follows a non-homogeneous Poisson process (NHPP) with mean value function

$$E[F(t)] = \lambda t^{\beta} \qquad (\lambda, \beta, t > 0) \qquad 5.4\text{-}1$$

E[F(t)] is the expected number of failures by time *t,* λ is the scale parameter and β is the growth (or shape) parameter. The parameters λ and β may vary from subsystem to subsystem and will be subscripted by a subsystem index number when required for clarity. Non-growth subsystems are assumed to have constant failure rates.

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5.4.2.2 Mathematical Basis for Growth Subsystems.

5.4.2.2.1 Initial Conditions.

The power function shown in 5.4-1 together with the initial conditions described in this section provides a framework for a discussion of the way SSPLAN develops reliability growth curves. Together they provide a starting point for describing each growth subsystem's MTBF as a function of the parameters, β and t. Since it is not convenient to directly work with for planning purposes, we relate λ to an initial or average subsystem MTBF over an initial period of test time. First, we note that the growth parameter, β, is related to the growth rate, α, by the following:

$$\beta = 1 - \alpha \qquad (\beta > 0) \tag{5.4-2}$$

For planned growth situations, α must be in the interval (0, 1). Additional guidance on choosing α may be gained from the AMSAA Reliability Growth Data Study, January 1990 previously cited in Section 3, Reference [3]. The initial conditions for the model consist of:

a. an initial time period, t_I (for example, the amount of planned test time prior to the implementation of any corrective actions), and
b. the initial MTBF, M_I, representing the average MTBF over the interval $(0, t_I]$.

From this, note that

$$\lambda_I = \frac{1}{M_I} \qquad (M_I > 0) \tag{5.4-3}$$

is the average failure intensity over the interval . The fact that 5.4-1 must be consistent with the initial conditions allows the scale parameter, λ, to be expressed in terms of planning parameters t_I, M_I, and α. To do so, note the expected number of failures by time t_I is:

$$E[F(t_I)] = \lambda_I t_I \qquad (t_I > 0) \tag{5.4-4}$$

Using 5.4-1, we see that the expected number of failures by time t_I is also given by

$$E[F(t_I)] = \lambda t_I^{\beta} \tag{5.4-5}$$

By equating 5.4-4 and 5.4-5 and by using the relationship from 5.4-2, an expression for λ may be developed:

$$\lambda = \lambda_I t_I^{\alpha} = \frac{t_I^{\alpha}}{M_I} \tag{5.4-6}$$

In addition to using both M_I and t_I as initial growth subsystem input parameters, the model allows a third possible input parameter, termed the planned management strategy, *MS*, which represents the fraction of the initial subsystem failure intensity that is expected to be addressed through corrective actions. The relationship among these three parameters is addressed in the following discussion.

Since reliability growth occurs when correctable failure modes are surfaced and (successful) fixes are incorporated, it is desired to have a high probability of observing at least one correctable failure by time t_I. In what follows we will utilize a probability of 0.95. From our

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assumptions, the number of failures that occur over the initial time period t_I is Poisson distributed with expected value $\lambda_I t_I$. Thus

$$0.95 \;=\; 1-e^{-(\,ms\times\lambda_I\times t_I\,)} \;=\; 1-e^{-\left(\frac{ms\times t_I}{M_I}\right)} \quad (0<MS<1) \qquad 5.4\text{-}7$$

From 5.4-7 it is evident that specifying any two of the parameters is sufficient to determine the third parameter. Thus, there are three options for the user when entering the initial conditions for growth subsystems.

5.4.2.2.2 Failure Intensity and Mean Time Between Failures – MTBF.
The derivative with respect to time of the expected number of failures function 5.4-1 is:

$$\rho(t) \;=\; \lambda\beta t^{\beta-1} \qquad 5.4\text{-}8$$

The function $\rho(t)$ represents the instantaneous failure intensity at time t. The reciprocal of $\rho(t)$ is the instantaneous MTBF at time t:

$$M(t) \;=\; \frac{1}{\rho(t)} \qquad 5.4\text{-}9$$

Equations 5.4-8 and 5.4-9 provide much of the foundation for a discussion of how SSPLAN develops reliability growth curves for growth subsystems. FIGURE 5-28 shows a graphical representation of subsystem reliability growth.

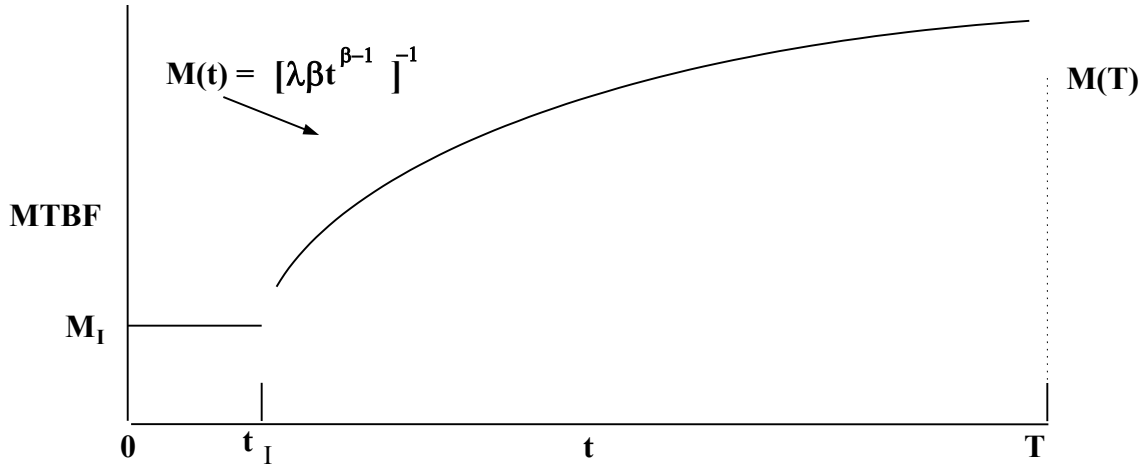


FIGURE 5-28. *Reliability Growth based on AMSAA Continuous Tracking Model*

5.4.2.3 Mathematical Basis for Non-growth Subsystems.
Based on the constant failure rate assumption, the input parameters that characterize a non-growth subsystem are its fixed reliability estimate, M, and the length of the demonstration test, T, upon which the constant MTBF estimate is based.

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5.4.2.4 Algorithm for Estimating Probability of Acceptance P_A.
Rather than use purely analytical methods, SSPLAN uses simulation techniques to estimate the probability of achieving a system MTBF objective with a specified confidence level. This estimate of P_A is calculated by running the simulation with a large number of trials.

Using the parameters that have been inputted and calculated at the subsystem level, the model generates "test data" for each subsystem for each simulation trial, thereby developing the data required to produce an estimate for the failure intensity for each subsystem. The test intervals and estimated failure intensities corresponding to the set of subsystems that comprise the system provide the necessary data for each trial of the simulation.

The model then uses a method developed for discrete data (the Lindström-Madden Method) to "roll up" the subsystem test data to arrive at an estimate for the final system reliability at a specified confidence level, namely, a statistical lower confidence bound (LCB) for the final system MTBF. In order for the Lindström-Madden method to be able to handle a mix of test data from both growth and non-growth subsystems, the model first converts all growth (G) subsystem test data to an "equivalent" amount of demonstration (D) test time and "equivalent" number of demonstration failures. This conversion process is done so that all subsystem results are expressed in a common format, namely, in terms of fixed configuration (non-growth) test data. (The equivalent demonstration test time and the equivalent demonstration number of failures are, respectively, the length of time and the number of failures a non-growth test would have to achieve to produce an {MTBF point estimate, MTBF LCB} pair that is equivalent to the respective estimates from a growth test.) By treating growth subsystem test data in this way, a standard lower confidence bound formula for time-truncated demonstration testing may be used to compute the system reliability LCB for the combination of "converted" growth and non-growth test data.

SSPLAN can run as many as 5000 trials. For each simulation trial, if the LCB for the final system MTBF meets or exceeds the specified system MTBF objective, then the trial is termed a success. An estimate for the probability of acceptance is the ratio of the number of successes to the number of trials.

5.4.2.4.1 Algorithm Topics.
The algorithm for estimating the probability of acceptance is described in greater detail by expanding upon the following four topics:

a. generating "test data" estimates for growth subsystems
b. generating "test data" estimates for non-growth subsystems
c. converting growth subsystem data to "equivalent" demonstration data
d. using the Lindström-Madden method for computing system level statistics

5.4.2.4.2 Generating Estimates for Growth Subsystems.
There are two quantities of interest for each growth subsystem for each trial of the simulation -

a. the total amount of test time, $T_{G,i}$, and
b. the estimated failure intensity at that time, $\hat{\rho}_{G,i}(T_{G,i})$.

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To calculate $T_{G,i}$, note that from the initial input conditions we have values for the growth parameter, β (using 5.4-2), and the scale parameter, λ (using 5.4-3 and 5.4-6). Also, note that the final growth subsystem MTBF, $M_{G,SYS}$, can be calculated by dividing the final MTBF, $M_{G,i}$, of the combined growth subsystems, $M_{G,SYS}$, by the subsystem failure intensity allocation a_i. Equations 5.4-8 and 5.4-9 can then be combined and rearranged to solve for $T_{G,i}$:

$$T_{G,i} = \frac{1}{\left[\lambda \beta M_{G,i}\right]^{\left[\frac{1}{(\beta-1)}\right]}} \qquad (\beta \neq 1) \qquad 5.4\text{-}10$$

To generate the estimated failure intensity, $\hat{\rho}_{G,i}(T_{G,i})$, the model uses , λ, β, $T_{G,t}$ and 5.4-10 with $t = T_{G,i}$ to calculate a Poisson distributed random number, $n_{G,i}$, which serves as an outcome for the number of growth failures during a simulation trial. The model then generates a chi-squared random number with $2n_{G,i}$ degrees of freedom and uses relation 5.4-11 below as specified in [6] for obtaining a random value from the distribution for the estimated growth parameter, conditioned on the number of growth failures, $n_{G,i}$, during the trial:

$$\hat{\beta} \sim \frac{\left(2\beta n_{G,i}\right)}{\chi^2_{2n_{G,i}}} \qquad 5.4\text{-}11$$

Note: β is obtained from the initial input and 5.4-2. One can show, $n_{G,i}$ and the maximum likelihood estimates (MLE's) for λ and β, satisfy the following:

$$n_{G,i} = \hat{\lambda} T_{G,i}^{\hat{\beta}} \qquad \left(\hat{\lambda},\hat{\beta},T_{G,i} > 0\right) \qquad 5.4\text{-}12$$

In light of equation 5.4-1, this result is not surprising. Using MLE's for the parameters in 5.4-8 yields:

$$\hat{\rho}_{G,i}\left(T_{G,i}\right) = \hat{\lambda}\hat{\beta}T_{G,i}^{\hat{\beta}-1} \qquad 5.4\text{-}13$$

Rearranging terms in (5.4-13) we obtain:

$$\hat{\rho}_{G,i}\left(T_{G,i}\right) = \frac{\hat{\lambda} T_{G,i}^{\hat{\beta}} \hat{\beta}}{T_{G,i}} \qquad 5.4\text{-}14$$

Substituting (5.4-13) into (5.4-14) we conclude:

$$\hat{\rho}_{G,i}\left(T_{G,i}\right) = \frac{n_{G,i}\hat{\beta}}{T_{G,i}} \qquad 5.4\text{-}15$$

Thus using $n_{G,i}$ and the corresponding conditional estimate for β generated from 5.4-11, an estimate for the failure intensity, $\hat{\rho}_{G,i}(T_{G,i})$, can be obtained for each growth subsystem for each trial of the simulation. Note the same value for $T_{G,i}$ is used on all the simulation trials.

5.4.2.4.3 Generating Estimates for Non-growth Subsystems.

There are two quantities of interest for each non-growth subsystem for each trial of the simulation:

a. the total amount of test time, $T_{n,i}$, and
b. the estimated failure intensity, $\hat{\rho}_{D,i}\left(T_{D,i}\right)$.

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The total amount of test time, $T_{n,i}$, is an input planning parameter that represents the length of the demonstration test on which the non-growth subsystem MTBF estimate is based. To generate the estimated failure intensity, $\hat{\rho}_{D,i}(T_{D,i})$, the model first calculates (this is done only once for each non-growth subsystem in SSPLAN) the expected number of failures:

$$E\left[F\left(T_{D,i}\right)\right] = \frac{T_{D,i}}{M_{D,i}} \qquad 5.4\text{-}16$$

where $M_{n,i}$ is an input planning parameter representing the constant MTBF for the non-growth subsystem. The expected number of failures from 5.4-16 is then used as an input parameter (representing the mean of a Poisson distribution) to a routine that calculates a Poisson distributed random number, $n_{D,i}$, which is an outcome for the number of failures during a simulation trial. An estimate for the failure intensity follows:

$$\hat{\rho}_{D,i}\left(T_{D,i}\right) = \frac{n_{D,i}}{T_{D,i}} \qquad 5.4\text{-}17$$

5.4.2.4.4 Calculating Lower Confidence Bound for System MTBF.
After all subsystem estimates have been calculated for a particular trial, SSPLAN uses a two-step approach to calculate the system reliability lower confidence bound by:

a. Converting all growth subsystem data to "equivalent" demonstration data, that is, data from a fixed configuration. These data consist of:
   i. $T_{D,i}$ - subsystem *i* equivalent demonstration test time and
   ii. $n_{D,i}$ - subsystem *i* equivalent demonstration number of failures
b. Using the Lindström-Madden method to obtain system level statistics for calculating the LCB for the system MTBF.

5.4.2.4.4.1 Converting Growth Subsystem Data to "Equivalent" Demonstration Data.
There are two equivalency relationships that must be maintained for the approach to be valid, namely, the demonstration data and the growth data must yield:

a. The same subsystem MTBF point estimate:

$$\hat{M}_{D,i} = \hat{M}_{G,i} \qquad 5.4\text{-}18$$

b. And the same subsystem MTBF lower bound at a specified confidence level γ:

$$LCB_{D,i,\gamma} = LCB_{G,i,\gamma} \qquad 5.4\text{-}19$$

Starting with the left side of the second equivalency relationship, 5.4-19, note that the lower confidence bound formula for time-truncated demonstration testing is:

$$LCB_{D,i,\gamma} = \frac{2T_{D,i}}{\chi^2_{2n_{D,i}+2,\gamma}} \qquad 5.4\text{-}20$$

Where $T_{D,i}$ is the demonstration test time, $n_{D,i}$ is the demonstration number of failures, γ is the specified confidence level and $\chi^2_{2n_{D,i}+2,\gamma}$ is a chi-squared 100 γ percentile point with $2n_{D,i}+2$

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degrees of freedom. Using an approximation equation developed by Crow, the lower confidence bound formula for growth testing (the right side of 5.4-19) is:

$$LCB_{G,i,\gamma} \approx \frac{n_{G,i}\,\hat{M}_{G,i}}{\chi^2_{n_{G,i}+2,\gamma}} \qquad 5.4\text{-}21$$

where $n_{G,i}$ is the number of growth failures during the growth test, $\hat{M}_{G,i}$ is the MLE for the MTBF and $\chi^2_{n_{G,i}+2,\gamma}$ is a chi-squared 100γ percentile point with $n_{G,i}+2$ degrees of freedom.

Since we want 5.4-20 and 5.4-21 to yield the same estimate, we begin by equating their denominators:

$$2n_{D,i}+2 = n_{G,i}+2 \quad \Rightarrow \quad n_{D,i} = \frac{n_{G,i}}{2} \qquad 5.4\text{-}22$$

Equating numerators from 5.3-20 and 5.3-21 yields:

$$2T_{D,i} = n_{G,i}\,\hat{M}_{G,i} \quad \Rightarrow \quad T_{D,i} = \frac{n_{G,i}\,\hat{M}_{G,i}}{2} \qquad 5.4\text{-}23$$

Thus 5.4-19 holds for $n_{D,i}$ and $T_{D,i}$ given by 5.4-22 and 5.4-23 respectively in terms of the simulated growth test data. Dividing 5.4-23 by 5.4-22:

$$\hat{M}_{D,i} = \frac{T_{D,i}}{n_{D,i}} = \hat{M}_{G,i} \qquad 5.4\text{-}24$$

Thus 5.4-18 is also satisfied. By 5.4-23 we obtain:

$$T_{D,i} = \frac{n_{G,i}}{2\,\hat{\rho}_{G,i}(T_{G,i})} \qquad 5.4\text{-}25$$

From 5.4-13 we have:

$$T_{D,i} = \frac{n_{G,i}}{2\hat{\lambda}\,\hat{\beta}\,T_{G,i}^{\hat{\beta}-1}} \qquad 5.4\text{-}26$$

Multiplying both numerator and denominator of 5.4-26 by $T_{G,i}$, replacing the estimate of the expected number of failures (in the denominator) by the observed number of growth failures and canceling the term $n_{G,i}$ in the numerator and denominator yields:

$$T_{D,i} = \frac{T_{G,i}}{2\hat{\beta}} \qquad 5.4\text{-}27$$

SSPLAN uses 5.4-22 and 5.4-27 in converting growth subsystem data to equivalent demonstration data.

5.4.2.4.4.2 Using the Lindström-Madden Method for Computing System Level Statistics.

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A continuous version of the Lindström-Madden method for discrete subsystems is used to compute an approximate lower confidence bound (LCB) for the final system MTBF from subsystem demonstration (non-growth) and "equivalent" demonstration (converted growth) data. The Lindström-Madden method typically generates a conservative LCB, which is to say the actual confidence level of the LCB is at least the specified level. It computes the following four estimates in order:

a. The equivalent amount of system level demonstration test time. (Since this estimate is the minimum demonstration test time of all the subsystems, it is constrained by the least tested subsystem.)
b. The estimate of the final system failure intensity, which is the sum of the estimated final growth subsystem failure intensities and non-growth subsystem failure rates
c. The equivalent number of system level demonstration failures, which is the product of the previous two estimates.
d. The approximate LCB for the final system MTBF at a given confidence level, which is a function of the equivalent amount of system level demonstration test time and the equivalent number of system level demonstration failures.

In equation form, these system level estimates are, respectively:

$$T_{D,SYS} = \min T_{D,i} \quad \text{for } i = 1 \ldots K \qquad 5.4\text{-}28$$

$$\hat{\rho}_{sys} = \sum_{i=1}^{K} \hat{\rho}_i \qquad 5.4\text{-}29$$

where $\hat{\rho}_i = \frac{1}{\hat{M}_{D,i}}$ and $\hat{M}_{D,i} =$ the demonstration or equivalent demonstration MTBF estimate for subsystem i.

$$N_{D,Sys} = T_{D,SYS} \times \hat{\rho}_{SYS} \qquad 5.4\text{-}30$$

$$LCB_{\gamma} = \frac{2T_{D,SYS}}{\chi^2_{2n_{D,SYS}+2,\gamma}} \qquad 5.4\text{-}31$$

5.4.2.5 Calculation of Testing Costs.
SSPLAN can be used to calculate the cost of carrying out a subsystem reliability growth plan for any given solution. The model does not address the initial start-up, or fixed costs since they are the same for any solution. The model does address all costs that are a function of the number of failures and all costs that are a function of time, as shown respectively in the following formula:

$$C_{Total} = \sum_{i \in \{all\,subsystems\}} \left\{ E\left[F_i(T_i)\right] \times \left(C_F\right)_i + T_i \times \left(C_T\right)_i \right\} \qquad 5.4\text{-}32$$

In 5.4-32, for each subsystem i, T_i denotes the amount of test time, $E\left[F_i(T_i)\right]$ is the expected number of failures by time T_i, $(C_F)_i$ is the cost per failure, and $(C_T)_i$ is the cost per unit of time (usually per hour). Therefore, the total testing cost, C_{Total}, is the sum, over all subsystems, of the costs associated with testing each subsystem. Once again, it is useful to treat growth and non-growth subsystems separately.

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5.4.2.5.1 Calculating Cost for Growth Subsystems.

For a given solution, we can calculate the cost contribution to C_{Total} of a growth subsystem i in terms of $T_{G,i}$ and growth parameters λ_i, β_i by directly using 5.4-32 with $T_i = T_{G,i}$. Note by 5.4-1 $E[F_i(T_{G,i})] = \lambda_i T_{G,i}^{\beta_i}$. Alternately, we can express this cost in terms of the achieved subsystem failure intensity, $\rho_{G,i}(T_{G,i})$, and λ_i, β_i. To write the cost equation in terms of the subsystem failure intensity, we begin by obtaining an expression for $T_{G,i}$ from 5.4-8:

$$\rho_{G,i}(T_{G,i}) = \lambda_i \beta_i T_{G,i}^{\beta_i - 1} \qquad (\lambda_i, \beta_i, T_{G,i} > 0) \tag{5.4-33}$$

Isolating the $T_{G,i}$ term on one side of 5.4-33 yields:

$$T_{G,i}^{\beta_i - 1} = \frac{\rho_{G,i}(T_{G,i})}{\lambda_i \beta_i} \tag{5.4-34}$$

Raising both sides of 5.4-34 to the $1/(\beta_i - 1)$ power:

$$T_{G,i} = \frac{[\rho_{G,i}(T_{G,i})]^{\left[\frac{1}{(\beta_i - 1)}\right]}}{[\lambda_i \beta_i]^{\left[\frac{1}{(\beta_i - 1)}\right]}} \tag{5.4-35}$$

(Note $\beta_i \neq 1$ since subsystem i is a growth subsystem.). Substituting from 5.4-2 yields the following intermediate result:

$$T_{G,i} = [\rho_{G,i}(T_{G,i})]^{-\left(\frac{1}{\alpha_i}\right)} [\lambda_i \beta_i]^{\left(\frac{1}{\alpha_i}\right)} \qquad (0 < \alpha_i < 1) \tag{5.4-36}$$

Now, to obtain an expression for $E[F_i(T_{G,i})]$, we begin with 5.4-1:

$$E[F_i(T_{G,i})] = \lambda_i T_{G,i}^{\beta_i} \tag{5.4-37}$$

Substituting for $T_{G,i}$ from 5.4-36 yields:

$$E[F_i(T_{G,i})] = \lambda_i [\rho_{G,i}(T_{G,i})]^{-\left(\frac{\beta_i}{\alpha_i}\right)} [\lambda_i \beta_i]^{\left(\frac{\beta_i}{\alpha_i}\right)} \tag{5.4-38}$$

Rearranging terms in 5.4-38 yields:

$$E[F_i(T_{G,i})] = \lambda_i^{\left(1+\frac{\beta_i}{\alpha_i}\right)} [\rho_{G,i}(T_{G,i})]^{-\left(\frac{\beta_i}{\alpha_i}\right)} \beta_i^{\left(\frac{\beta_i}{\alpha_i}\right)} \tag{5.4-39}$$

Finally, the cost contribution in 5.4-32 of growth subsystem i can be expressed in terms of its failure intensity using 5.4-36 and 5.4-39:

$$C_i[\rho_{G,i}(T_{G,i})] = [\rho_{G,i}(T_{G,i})]^{-\left(\frac{\beta_i}{\alpha_i}\right)} \lambda_i^{\left(1+\frac{\beta_i}{\alpha_i}\right)} \beta_i^{\left(\frac{\beta_i}{\alpha_i}\right)} (C_F)_i + [\rho_{G,i}(T_{G,i})]^{-\left(\frac{1}{\alpha_i}\right)} [\lambda_i \beta_i]^{\left(\frac{1}{\alpha_i}\right)} (C_T)_i \tag{5.4-40}$$

5.4.2.5.2 Calculating Cost for Non-growth Subsystems.

To obtain the cost contribution of a non-growth subsystem, we use 5.4-16 to express $E[F_i(T_{D,i})]$ in terms of $T_{D,i}$ and $M_{D,i}$:

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$$C_i[T_{D,i}] = \left(\frac{T_{D,i}}{M_{D,i}}\right)(C_F)_i + T_{D,i}(C_T)_i \qquad 5.4\text{-}41$$

where $M_{D,i}^{-1} = \rho_{D,i}$.

5.4.2.6 Methodology for a Fixed Allocation of Subsystem Failure Intensities.
The methodology utilizes a fixed allocation, a_i, of $\rho_{G,SYS}$ to each growth subsystem i. Thus $\rho_{G,i}(T_{G,i}) = \alpha_i \rho_{G,SYS}$. For this allocation, SSPLAN first determines if a solution exists that satisfies the criteria given by the user during the input phase. Specifically, SSPLAN checks to see if the desired probability of acceptance can be achieved with the given failure intensity allocations and maximum subsystem test times. If a solution does exist, SSPLAN will proceed to find the solution that meets the desired probability of acceptance within a small positive number epsilon.

5.4.2.6.1 Determining the Existence of a Solution.
To determine if a solution is possible, SSPLAN uses 5.4-8 and 5.4-9 for each subsystem, with T set to the subsystem's maximum test time, to calculate the maximum possible MTBF for each subsystem. The maximum subsystem MTBF is multiplied by its failure intensity allocation to determine its influence on the system MTBF. For example, if a subsystem can grow to a maximum MTBF of 1000 hours and it has a failure intensity allocation of 0.5 (that is, its final failure intensity accounts for half of the total final failure intensity due to all of the growth subsystems), then that particular subsystem will limit the combined growth subsystem maximum MTBF to 500 hours. In other words, the maximum MTBF to which the growth portion of the system can grow, $MTBF_{G,sys}$, is the minimum of the products (subsystem final MTBF multiplied by the subsystem failure intensity allocation) from among all the growth subsystems. The probability of acceptance, P_A, is then estimated using $MTBF_{G,sys}$. If the estimated P_A is less than the desired P_A, then no solution is possible within the limits of estimation precision for P_A, and SSPLAN will stop with a message to that effect.

5.4.2.6.2 Finding the Solution.
On the other hand, if the estimated P_A is greater than or equal to the desired P_A, then a solution exists. If, by chance, the desired P_A has been met (within a small number epsilon) then SSPLAN will use $MTBF_{G,sys}$ as its solution. It is more likely, however, that the estimate corresponding to $MTBF_{G,sys}$ exceeds the requirement, meaning that the program resulting in $MTBF_{G,sys}$ contains more testing than is necessary to achieve the desired. SSPLAN proceeds, then, to find a value for $MTBF_{G,sys}$ that meets the desired within epsilon.

To save time, P_A is initially estimated using a reduced number of iterations equal to one tenth of the requested number. As soon as the estimated P_A approaches the desired P_A, the full number of iterations is used.

For a given fixed failure intensity allocation, P_A increases as $MTBF_{G,sys}$ increases. Every value of $MTBF_{G,sys}$ determines a unique set of reliability growth curves, and thus a unique P_A. To find the set of growth curve test times that give rise to the desired P_A, SSPLAN first finds the upper and lower bounds for $MTBF_{G,sys}$. The initial upper bound for $MTBF_{G,sys}$ is the value

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found in verifying the existence of a solution; this value is the maximum possible value for $MTBF_{G,sys}$ (based on the maximum test times inputted by the user). The initial lower bound for $MTBF_{G,sys}$ is chosen arbitrarily; if the value chosen results in a P_A that is higher than the desired P_A, then the lower bound for $MTBF_{G,sys}$ is successively decreased until the resulting P_A is less than the desired P_A. At that point, upper and lower bounds for $MTBF_{G,sys}$ have been established, and SSPLAN uses a linear interpolation to find the value of $MTBF_{G,sys}$ that gives rise to an estimated P_A that meets the desired P_A. At each step of the search, $MTBF_{G,sys}$ is updated using the following equation (actually, the algorithm does all comparisons in terms of failure intensities, but the equation below shows the comparisons in terms of MTBFs to be consistent with those stated in [4] :

$$\frac{\left(MTBF_{G,sys}\right)_{NEW} - \left(MTBF_{G,sys}\right)_{LB}}{\left(MTBF_{G,sys}\right)_{UB} - \left(MTBF_{G,sys}\right)_{LB}} = \frac{\left(P_A\right)_{GOAL} - \left(P_A\right)_{LB}}{\left(P_A\right)_{UB} - \left(P_A\right)_{LB}} \qquad 5.4\text{-}42$$

where $\left(MTBF_{G,sys}\right)_{UB}$ and $\left(MTBF_{G,sys}\right)_{LB}$ refer to the upper and lower bounds, respectively, for $MTBF_{G,sys}$; $(P_A)_{UB}$ and $(P_A)_{LB}$ refer to the estimated P_A values associated with each of the preceding $MTBF_{G,sys}$ values, respectively; and $\left(MTBF_{G,sys}\right)_{NEW}$ is the new value of $MTBF_{G,sys}$ to be used in the search algorithm.

The bounds are systematically updated during the search as follows. If the estimated value of P_A associated with $\left(MTBF_{G,sys}\right)_{NEW}$ is less than the desired probability of acceptance, $(P_A)_{GOAL}$, then $\left(MTBF_{G,sys}\right)_{NEW}$ becomes the new lower bound for the next search. If the estimated P_A is greater than the desired P_A, then $\left(MTBF_{G,sys}\right)_{NEW}$ becomes the new upper bound. The solution is found when the estimated P_A is within epsilon of the desired P_A or when the lower and upper bounds on $MTBF_{G,sys}$ are within epsilon of each other.

5.5 Planning Model based on Projection Methodology (PM2)

5.5.1 PM2 Overview of Approach.

The following material is as stated in [7] Planning Model Based on Projection Methodology (PM2), AMSAA TR-2006-09, March 2006.

In the following sections, exact expressions for the expected number of surfaced failure modes and system failure intensity as functions of test time are presented under the assumption that the surfaced modes are mitigated through corrective actions. These exact expressions depend on a large number of parameters. Functional forms are derived to approximate these quantities that depend on only a few parameters. Such parsimonious approximations are suitable for developing reliability growth plans and portraying the associated planned growth path. Simulation results indicate that the functional form of the derived parsimonious approximations can adequately represent the expected reliability growth associated with a variety of patterns for the failure mode initial rates of occurrence. A sequence of increasing MTBF target values can be constructed from the parsimonious MTBF projection approximation based on:

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a. planning parameters that determine the parsimonious approximation;

b. corrective action mean lag time with respect to implementation and;

c. the test schedule that gives the number of planned Reliability, Availability, and Maintainability (RAM) test hours per month and specifies corrective action implementation periods.

5.5.2 Background and Outline of PM2 Topics.

To mature the reliability of a complex system under development it is important to formulate a detailed reliability growth plan. One aspect of this plan is a depiction of how the system's reliability is expected to increase over the developmental test period. The depicted growth path serves as a baseline against which reliability assessments can be compared. Such baseline planning curves for Department of Defense (DoD) systems have frequently been developed in the past utilizing the assumed reliability growth pattern specified in Military Handbook 189. This growth relationship is between the reliability, expressed as the mean test duration between system failures and a continuous measure of test duration such as time or mileage. The equation governing this growth pattern was motivated by the empirically derived linear relationship observed for a number of data sets by Duane (1964), between the developmental system cumulative failure rate and the cumulative test time when plotted on a log-log scale. In the following sections, we obtain a non-empirical relationship between the mean test duration between system failures and cumulative test duration that can be utilized for reliability growth planning. This relationship is derived from a fundamental relationship between the expected number of failure modes surfaced and the cumulative test duration. For convenience, we will refer to the test duration as test time and measure the reliability as the mean time between system failures (MTBF). The functional form of this fundamental relationship is well known and is easily established without recourse to empiricism in accordance with An Improved Methodology for Reliability Growth Projections, AMSAA TR-357, June 1982. We obtain an approximation to this relationship that is suitable for reliability growth planning. One significant advantage to the PM2 approach is that it does not rely on an empirically derived relationship such as the Duane based approach. We will show how the cumulative relationship between the expected number of discovered failure modes and the test time naturally gives rise to a reliability growth relationship between the expected system failure intensity and the cumulative test time. The presented approximation for the resulting growth pattern avoids a number of deficiencies associated with the Duane/MIL-HDBK-189 approach to reliability growth planning.

Section 5.5.3 develops the exact expected system failure intensity and parsimonious approximations suitable for reliability growth planning. These functions of test time are derived from the exact and planning approximation relationships between the expected number of surfaced failure modes and the cumulative test time. The exact relationship is expressed in terms of the number of potential failure modes, k, and the individual initial failure mode rates of occurrence. Parsimonious approximations to this relationship are obtained. The first approximation utilizes k and several additional parameters. The second approximation discussed is the limiting form of the first approximation as k increases. This approximation is suitable for complex systems or subsystems. The approximations are derived through consideration of an

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MTBF projection equation. This equation arises from considering the problem of estimating the system MTBF at the start of a new test phase after implementing corrective actions to failure modes surfaced in a preceding test phase. This MTBF projection has been documented in The AMSAA Maturity Projection Model based on Stein Estimation, AMSAA TR-751, July 2004 and is described in Section 5.5.3.

Section 5.5.4 contains simulation results. The simulations are conducted to obtain actual patterns for the cumulative number of surfaced failure modes versus test time for random draws of initial mode failure rates from several parent populations, and for a geometric sequence of initial mode failure rates. The resulting stochastic realizations are compared to the theoretical expected number of potential surfaced failures modes and to the parsimonious approximations. Random draws for mode fix effectiveness factors (FEFs) (fraction reductions in initial failure mode rates of occurrence due to mitigation) are used to simulate corrective actions to surfaced failure modes. Using the simulated corrective actions, the relationship between the expected system failure intensity and cumulative test time is simulated for various sets of mode initial failure rates. This relationship is obtained under the assumption that the system failure intensity associated with a cumulative test time t reflects implementation of corrective actions to the modes surfaced by t with the associated randomly drawn FEFs. The resulting system MTBF versus test time relationship is compared to the corresponding relationship established for planning purposes.

Section 5.5.5 derives expressions for a reliability projection scale parameter that is utilized in the parsimonious approximations. The projection parameter is expressed in terms of basic planning parameters. The resulting MTBF approximations are compared to the reciprocals of the exact expected system failure intensity and stochastic realizations of the system failure intensity, and to MIL-HDBK-189 MTBF approximations based on planning parameters. The comparisons are done for several reliability growth patterns.

Section 5.5.6 addresses the relationship between the theoretical upper bound on the achievable system MTBF, termed the growth potential, and the planning parameters. The projection scale parameter considered in Section 5.5.5 is then expressed in terms of planning parameters and the MTBF growth potential. It is shown that the scale parameter becomes unrealistically large if the goal MTBF is chosen too close to the growth potential or if the allocated test time to grow from the initial to goal MTBF is inadequate.

Section 5.5.7 indicates how to construct a sequence of MTBF target values that start at an expected or measured initial MTBF and end at the goal MTBF. It is shown that the parsimonious approximation to the reciprocal of the expected system failure intensity can be used for this purpose in conjunction with a test schedule that specifies the expected monthly RAM hours to be accumulated on the units under test and the planned corrective action periods.

5.5.3 Derived Reliability Growth Patterns.

5.5.3.1 Assumptions.

The system has a large number of potential failure modes with initial rates of occurrence of $\lambda_1, \ldots, \lambda_k$. The modes are candidates for corrective action if they are surfaced during test. All

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failure modes independently generate failures according to the exponential distribution and the system fails whenever a failure mode occurs. It is also assumed that corrective actions do not create new failure modes.

5.5.3.2 Background Information.
The first step in obtaining a functional form for the expected failure intensity as a function of test time and planning parameters that is based on non-empirical considerations involves the relationship between the expected number of failure modes surfaced and test duration. This relationship was considered by Crow (1982) for the case where test duration is continuous. In this paper, we are measuring test duration in a continuous fashion. Test time will be used as a generic measure of test duration for this continuous case. The relationship is easily obtained by expressing the number of surfaced modes by test time t as a sum of mode indicator functions. In particular, let $I_i(t)$ denote the indicator function for mode i. The indicator function takes on the value one if mode i occurs by t and equals zero otherwise. The number of modes surfaced by t is given by,

$$M(t) = \sum_{i=1}^{k} I_i(t) \qquad 5.5\text{-}1$$

The expected value of $M(t)$ is equal to,

$$\mu(t) = \sum_{i=1}^{k} E(I_i(t)) = \sum_{i=1}^{k} (1 - e^{-\lambda_i t}) = k - \sum_{i=1}^{k} e^{-\lambda_i t} \qquad 5.5\text{-}2$$

This expected value function implies a functional form for the expected failure intensity and corresponding MTBF as a function of test time t given that corrective actions have been incorporated to all the failure modes surfaced by t. One component of the expected failure intensity is due to the failure modes not yet surfaced by t. This component is simply given by the derivative of $\mu(t)$. Note,

$$\frac{d\mu(t)}{dt} = \sum_{i=1}^{k} \lambda_i e^{-\lambda_i t} \qquad 5.5\text{-}3$$

In (Ellner et al., 2000) it is shown that the expression in 5.5-3 is the expected failure intensity due to all the modes not surfaced by t. To show this observe that the failure intensity due to these modes can be expressed as the random variable $\Lambda_U(t)$ where

$$\Lambda_U(t) = \sum_{i=1}^{k} \lambda_i (1 - I_i(t)) \qquad 5.5\text{-}4$$

The expected value of $\Lambda_U(t)$ is given by,

$$E(\Lambda_U(t)) = \sum_{i=1}^{k} \lambda_i \{1 - E(I_i(t))\} = \sum_{i=1}^{k} \lambda_i e^{-\lambda_i t} = \frac{d\mu(t)}{dt} \qquad 5.5\text{-}5$$

Before considering the other components of the system failure intensity, we will address obtaining parsimonious approximations to the expected number of failure modes surfaced by t and the corresponding failure intensity due to the unsurfaced failure modes. The exact expressions for these quantities are given by 5.5-2 and 5.5-5. Note that these expressions depend

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on k +1 parameters, namely the number of potential failure modes k and the initial failure mode rates of occurrence λ_i for $i=1,\ldots,k$.

5.5.3.3 Parsimonious Approximations.

5.5.3.3.1 Expected Number of Modes and its Derivative.
To obtain parsimonious approximations to the expected number of modes surfaced by t and its derivative, we consider an optimization problem under the assumption that all corrective actions are delayed until t. Let N_i denote the number of failures that occur by t due to mode i. Then $\hat{\lambda}_i = \frac{N_i}{t}$ denotes the standard Maximum Likelihood Estimate (MLE) of λ_i. Consider the estimator for λ_i given by,

$$\tilde{\lambda}_i = \theta \cdot \hat{\lambda}_i + (1-\theta) \cdot avg(\hat{\lambda}_i) \qquad 5.5\text{-}6$$

where $avg(\hat{\lambda}_i)$ denotes the arithmetic average of the k, $\hat{\lambda}_i$ and $\theta \in [0,1]$ is chosen to minimize the expected sum of squared errors between $\tilde{\lambda}_i$ and λ_i, i.e. $E\left[\sum_{i=1}^{k}(\tilde{\lambda}_i - \lambda_i)^2\right]$. The value of θ that solves this optimization problem can be shown to be θ_S (Ellner et al., 2004) where:

$$\theta_S = \frac{Var[\lambda_i]}{\frac{\lambda}{k \cdot t}\left(1-\frac{1}{k}\right) + Var[\lambda_i]} \qquad 5.5\text{-}7$$

for $\lambda = \sum_{i=1}^{k}\lambda_i$, $\bar{\lambda} = \frac{\lambda}{k}$, and $Var[\lambda_i] = \frac{\sum_{i=1}^{k}(\lambda_i - \bar{\lambda})^2}{k}$. The estimate of λ_i given by (5.5-6) with θ equal to θ_S has been called the Stein estimate of λ_i (Ellner et al., 2004). Note this is a theoretical estimate in the sense that it cannot be computed from the data since it involves the unknown values of k, λ, and $Var[\lambda_i]$. The quantity $Var[\lambda_i]$ can also be expressed as follows:

$$Var[\lambda_i] = \left(\frac{1}{k}\right)\left(\sum_{i=1}^{k}\lambda_i^2 - \frac{\lambda^2}{k}\right) \qquad 5.5\text{-}8$$

From the definition of $\tilde{\lambda}_i$ and the fact that N_i equals zero for a failure mode unobserved by t we have that the Stein assessment for the failure rate contribution of a failure mode not observed by t is given by,

$$\tilde{\lambda}_i = (1-\theta_S) \cdot \left(\frac{N}{k \cdot t}\right) \qquad 5.5\text{-}9$$

Thus, the Stein assessment of the failure intensity due to all the failure modes not surfaced by t equals,

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$$\sum_{i\in\overline{obs}}\tilde{\lambda}_i=(k-m)\cdot(1-\theta_S)\cdot\left(\frac{N}{k\cdot t}\right)=\left(1-\frac{m}{k}\right)\cdot(1-\theta_S)\cdot\left(\frac{N}{t}\right) \qquad 5.5\text{-}10$$

where m denotes the number of surfaced modes by t and $\overline{obs}$ denotes the index set for the failure modes not surfaced by t. From 5.5.-7and 5.5-8 one can show,

$$1-\theta_S=\frac{\left(\frac{\lambda}{k\cdot t}\right)\cdot\left(1-\frac{1}{k}\right)}{\left(\frac{1}{k}\right)\cdot\left(\sum_{i=1}^{k}\lambda_i^2-\frac{\lambda^2}{k}\right)+\left(\frac{\lambda}{k\cdot t}\right)\cdot\left(1-\frac{1}{k}\right)} \qquad 5.5\text{-}11$$

Replacing $1-\theta_S$ in 5.5-10 by the final expression in 5.5-11 and simplifying yields,

$$\sum_{i\in\overline{obs}}\tilde{\lambda}_i=\frac{\left(1-\frac{1}{k}\right)\cdot\left(1-\frac{m}{k}\right)\cdot\left(\frac{N}{t}\right)}{\left(1-\frac{1}{k}\right)+\left(\frac{\sum_{i=1}^{k}\lambda_i^2}{\lambda}-\frac{\lambda}{k}\right)\cdot t} \qquad 5.5\text{-}12$$

Equation 5.5-12 gives the Stein assessment for the failure intensity due to all the failure modes not surfaced by t. Note that m can be regarded as an estimate of the expected number of modes surfaced by t, i.e. $\mu(t)$. Additionally, in light of Equation 5.5-5, the left hand side of 5.5-12 can be viewed as an estimate of $\frac{d\mu(t)}{dt}$. From 5.5-3, it follows that the derivative of $\mu(t)$ at $t=0$ equals λ. Finally, observe that $\frac{N}{t}$ in 5.5-12 is the maximum likelihood estimate of the initial failure rate λ under the assumption that all corrective actions are delayed to t. Let $h(t)$ denote $\frac{d\mu(t)}{dt}$. Simulation results for a number of cases (where k and λ_i are known) conducted in support of (Ellner et al., 2004) have indicated that the Stein assessment given in 5.5-12 yields good estimates of $h(t)$ when all the corrective actions are delayed. The value $h(t)$ that is being estimated does not depend on the corrective action process. Only the estimate of λ given by $\frac{N}{t}$ depends on the assumption that all corrective actions are delayed until t. Thus the right hand side of 5.5-12 with m and $\frac{N}{t}$ replaced by good approximations of $\mu(t)$ and $\lambda=\left.\frac{d\mu(t)}{dt}\right|_{t=0}$ respectively should yield a good approximation for $h(t)$ regardless of the corrective action process for the cases where the Stein estimate of $h(t)$ given by 5.5-12 are accurate. The above discussion regarding equation 5.5-12 suggests that a reasonable choice for our parsimonious approximation to $\mu_k(t)$ should satisfy the following differential equation and associated initial conditions:

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$$\frac{d\mu_k(t)}{dt} = \frac{\left(1-\frac{1}{k}\right)\left(1-\frac{\mu_k(t)}{k}\right)\lambda}{\left(1-\frac{1}{k}\right)+\left(\frac{\sum_{i=1}^{k}\lambda_i^2}{\lambda}-\frac{\lambda}{k}\right)t} \qquad 5.5\text{-}13$$

where $\mu_k(0)=0$ and $\left.\frac{d\mu_k(t)}{dt}\right|_{t=0} = \lambda$.

For the case where all the λ_i are equal, one can show that $\mu_k(t)=\mu(t)$ for all $t \geq 0$. Thus, in what follows we only consider the case where not all λ_i are equal. The solution to the resulting differential equation, for this case, with the specified initial conditions is,

$$\mu_k(t) = k\cdot\left[1-\left(1+\beta_k\cdot t\right)^{-p_k}\right] \qquad 5.5\text{-}14$$

where

$$\beta_k = \left(1-\frac{1}{k}\right)^{-1}\left[\left(\frac{1}{\lambda}\right)\left(\sum_{i=1}^{k}\lambda_i^2\right)-\frac{\lambda}{k}\right] \qquad 5.5\text{-}15$$

and

$$p_k = \frac{\lambda}{k\cdot\beta_k} \qquad 5.5\text{-}16$$

The solution was obtained by the method of integrating factors (Boyce et al., 1965). The solution can be verified by directly substituting 5.5-14 and its derivative into the differential equation for $\mu_k(t)$ and noting that $\mu_k(t)$ satisfies the specified initial conditions. Observe that $\mu_k(t)$ can be expressed in terms of *t* and three constants, namely *k*, λ and β_k. The corresponding parsimonious approximation for *h(t)* is $\frac{d\mu_k(t)}{dt}$, which will be denote by $h_k(t)$.

It is interesting to note that $\mu_k(t)$ given by 5.4-14 is the same expression that one can obtain for the expected number of software bugs surfaced in execution time *t* given by the doubly stochastic exponential order model presented in (Miller, 1985) for the case where the initial bug occurrence rates $\lambda_1,\ldots,\lambda_k$ constitute a realization of a random sample of size *k* from a gamma random variable. The density function of this random variable is given by,

$$f(x) = \begin{cases} \dfrac{x^{\alpha_k}\cdot e^{-\left(\frac{x}{\beta_k}\right)}}{\Gamma(\alpha_k+1)\cdot\beta_k^{(\alpha_k+1)}} & \text{for } x>0 \\ 0 & \text{otherwise} \end{cases} \qquad 5.5\text{-}17$$

In this density function, *Γ* denotes the gamma function, β_k is defined by 5.5-15, and α_k+1 equals p_k. This result is shown in (Ellner et. al., 2000) where $\mu_k(t)$ denotes the expected number of surfaced modes by time *t* that will be mitigated by a corrective action.

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5.5.3.3.2 Expected System Failure Intensity and MTBF.
Next, we will consider the expected system failure intensity after *t* test hours and a corresponding parsimonious approximation, given that corrective actions are implemented to all the surfaced failure modes. Let d_i denote the fraction reduction in the rate of occurrence of mode *i* due to the corrective action (termed a fix). The reduction factor is termed the fix effectiveness factor (FEF) for failure mode *i*. Let $\Lambda(t)$ denote the failure intensity of the system given that fixes have been applied to all the failure modes surfaced by *t*. Then,

$$\Lambda(t) = \sum_{i=1}^{k} (1 - d_i I_i(t))\lambda_i \qquad 5.5\text{-}18$$

The corresponding expected failure intensity is $\rho(t)$ where,

$$\rho(t) = E(\Lambda(t)) = \sum_{i=1}^{k} (1 - d_i E(I_i(t)))\lambda_i = \sum_{i=1}^{k} (1 - d_i)\lambda_i + \sum_{i=1}^{k} d_i \lambda_i e^{-\lambda_i t} \qquad 5.5\text{-}19$$

This expression for the expected failure intensity was presented in (Crow, 1982).

For reliability growth planning purposes, assessments of individual failure mode FEFs will not be available. Thus, in place of $\rho(t)$, we will use a parsimonious approximation, denoted by $\rho_k(t)$, that utilizes an average fix effectiveness factor. It follows from 5.5-5 that λ-$h(t)$ is the expected failure intensity due to the failure modes surfaced by *t* prior to mitigation. Assume these modes are mitigated with an average FEF of μ_d. Then the expected failure intensity due to the surfaced failure modes after mitigation can be approximated by $(1 - \mu_d) \cdot (\lambda - h(t))$. Thus the parsimonious approximation for $\rho(t)$ will be defined as follows:

$$\rho_k(t) = (1 - \mu_d) \cdot (\lambda - h_k(t)) + h_k(t) \qquad 5.5\text{-}20$$

We also define the parsimonious MTBF approximation of $MTBF(t) = (\rho(t))^{-1}$ for reliability growth planning by $MTBF_k(t) = (\rho_k(t))^{-1}$.

For planning, it can be useful to add a term, λ_A, to the expressions for $\rho(t)$ and $\rho_k(t)$ given by 5.5-19 and 5.5-20 respectively. This term represents the failure rate due to all the failure modes that will not be corrected, even if surfaced … referred to as A-modes (Crow, 1982). This term for planning purposes would be given by the quantity $(1 - MS) \cdot \lambda$. However, since this term does not contribute to the difference between $\rho(t)$ and $\rho_k(t)$ we will not consider it further in this section or Section 5.5.4.

It may be difficult to select a value of *k* for planning purposes. For complex systems or subsystems it is reasonable to use the limiting forms of $\mu_k(t)$, $h_k(t)$, and $\rho_k(t)$ as $k \to \infty$. Consider the limit as $k \to \infty$ of these functions. In taking the limit, we hold λ fixed and assume the limit of β_k is positive as *k* increases, say $\beta_\infty \in (0, \infty)$. Under these conditions one can show the three functions converge to limiting functions which will be denoted by $\mu_\infty(t)$, $h_\infty(t)$, and $\rho_\infty(t)$, respectfully. One can show,

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$$\mu_\infty(t) = \left(\frac{\lambda}{\beta_\infty}\right)\ln(1+\beta_\infty \cdot t) \qquad 5.5\text{-}21$$

and

$$h_\infty(t) = \frac{d\mu_\infty(t)}{dt} = \lambda(1+\beta_\infty \cdot t)^{-1} \qquad 5.5\text{-}22$$

Also, $\rho_\infty(t)$ is given by 5.5-20 with $h_k(t)$ replaced by $h_\infty(t)$.

5.5.4 Simulation.

5.5.4.1 Simulation Overview.

We wish to compare the parsimonious approximations to realized and expected reliability growth patterns with respect to a number of quantities. To do so we will generate a number of realized reliability growth patterns via simulation in Mathematica. We will consider cases where the failure mode initial rates of occurrence are realizations of a specified parent population for several choices of the parent distribution. We will also generate reliability growth patterns for a deterministically specified sequence of failure mode initial rates of occurrence that have been found to be useful in representing initial bug rates of occurrence in software programs under development (Miller, 1985). The simulation consists of the following steps:

a. Specify inputs. This includes items such as,
   i. test duration,
   ii. the number of failure modes, and
   iii. the sequence or parent population governing the initial mode failure rates.
b. Produce mode initial failure rates. Failure rates are either stochastically generated, or deterministically calculated. In the stochastic case, failure rates are generated by drawing realizations of a random sample from a specified gamma, Weibull, lognormal or log-logistic as suggested by Meeker and Escobar in Statistical Methods for Reliability Data, John Wiley & Sons, Inc, New York, 1998 (Meeker et al., 1998) parent population. In the deterministic case, failure rates are calculated in accordance with a specified geometric sequence.
c. Generate mode failure times. The mode failure times are generated via a function of randomly generated uniform numbers, and the mode initial failure rates.
d. Generate mode fix effectiveness factors. The FEFs are generated by drawing realizations of a random sample from a beta distribution with mean 0.80, and coefficient of variation 0.10.
e. Examine quantities and plots of interest.

5.5.4.1 Simulation Results.

Results below display plots of the expected and realized number of surfaced failure modes for stochastic λ_i generated from a log-logistic (FIGURE 5-29), and deterministic λ_i calculated from a geometric sequence (FIGURE 5-31). Also shown are plots of the reciprocals (i.e. MTBFs) of the expected and realized system failure intensities for log-logistic λ_i (FIGURE 5-30), and geometric λ_i (FIGURE 5-32). The geometric initial mode failure rates are given by

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$$\lambda_i = a \cdot b^i \qquad 5.5\text{-}23$$

for $i = 1,\ldots,k$ where $0 < a$ and $0 < b < 1$. All the displayed quantities have been averaged over ten replications of simulation steps b. through d. above.

The intent of the plots is to see whether the functional form of the parsimonious approximations are reasonably compatible with respect to (1) the expected number of surfaced failure modes as a function of test time, and (2) the reciprocal of the expected system failure intensity as a function of test time. Corrective actions are assumed to be implemented to all the failure modes surfaced by *t* with the simulated mode fix effectiveness factors. The value of μ_d in Equation 5.5-20 is set equal to $\frac{1}{k}\sum_{i=1}^{k} d_i$ to generate the parsimonious approximations to the exact expected failure intensity and corresponding MTBF. Additionally, for the results displayed below, $k = 1{,}500$ and $\lambda = 10^{-1}$. The value of the scale parameter obtained from Equation 5.5-15 does not provide adequate parsimonious approximations except when the parent population is gamma or the scale parameter is sufficiently small. Thus for the specified *k*, μ_d, and λ, the scale parameters β_k and β_∞ of the parsimonious approximations were fitted to the exact expected number of surfaced failure modes function by using maximum likelihood estimates. These estimates were obtained from the simulated mode first occurrence times. This was accomplished by assuming the generated initial mode failure rates $\lambda_1,\ldots,\lambda_k$ represented a realization of a random sample of size *k* from a gamma distribution with scale parameter β_k and mean λ/k. This procedure provided a "best statistical fit" of the parsimonious functional approximations for $\mu(t)$ and $\rho(t)$, with respect to the scale parameter, over the entire planning period of interest, i.e. 10,000 hours.

The parsimonious approximations for $\mu(t)$ and $\rho(t)$ based on the limiting forms for $\mu_k(t)$ and $\rho_k(t)$ as *k* increases will tend to be too large for values of *t* when $\mu(t)$ is too close to *k*. We have observed that the limiting approximations are adequate for $\mu(t)$ and $\rho(t)$ over the range of *t* for which $\mu(t) \le k/5$. Thus for complex systems, or subsystems, the limiting approximation functional forms should be adequate representations of $\mu(t)$ and $\rho(t)$ over most test periods of interest.

The red curves in the figures below represent the exact expected number of surfaced modes (FIGURE 5-29 and FIGURE 5-31) or the reciprocal of the exact expected system failure intensity (FIGURE 5-30 and FIGURE 5-32). The dots in each figure represent a corresponding stochastic realization. The green curves display the finite *k* approximations while the blue curves display the corresponding limiting approximations. The displayed curves and stochastic realizations are averages over ten replications of simulation steps b. through d. Similar results were obtained for the cases where the λ_i were generated from gamma, lognormal, and Weibull parent populations.

For comparison purposes, the MIL-HDBK-189 system MTBF based on Equation 5.1-12 was fitted to the reciprocal of the expected system failure intensity (the red curves). The MIL-HDBK-189 curves are displayed in yellow and were fitted utilizing all the observed simulated cumulative times of failure. The use of all cumulative failure times requires that fixes be

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implemented when failure modes are observed. The simulation was carried out in this manner to allow the parameters of the MIL-HDBK-189 curves to be statistically fitted via the maximum likelihood estimation procedure in (Department of Defense, 1981). As for the other displayed quantities, the averages of 10 replicated MIL-HDBK-189 MTBF curves are shown.

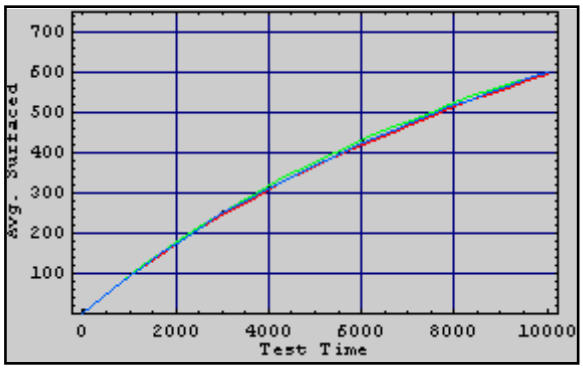


FIGURE 5-29. *Average Number of Surfaced Modes (Loglogistic)*

Notice the high degree of accuracy displayed in FIGURE 5-29 for the finite and infinite *k* approximations despite violating the gamma assumption used to statistically fit the parsimonious approximations.

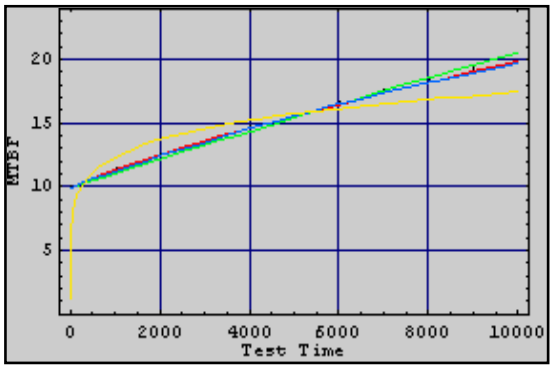


FIGURE 5-30. *Reciprocal of the Failure Intensity (Loglogistic)*

FIGURE 5-30 displays a high degree of accuracy for the statistically fitted PM2 MTBF approximations despite violating the MLE assumption that the initial mode failure rates are gamma distributed. In addition, the PM2 approximations of the MTBF appear favorable to that of the MIL-HDBK-189 model.

FIGURE 5-31 and FIGURE 5-32 below are analogous to FIGURE 5-29 and FIGURE 5-30, respectively. The only difference is the generation procedure associated with the initial mode failure rates utilized in the analysis. In this case, failure rates are deterministically calculated in accordance with a geometric sequence. The results are similar.

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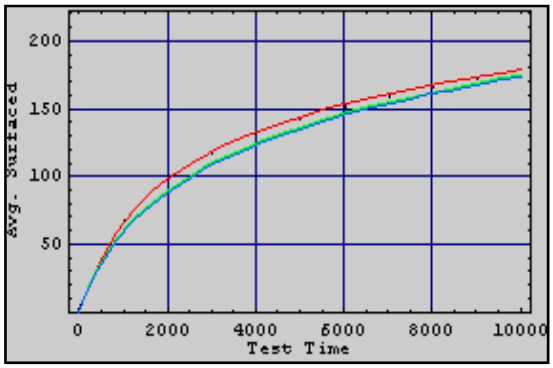


FIGURE 5-31. *Average Number of Surfaced Modes (Geometric)*

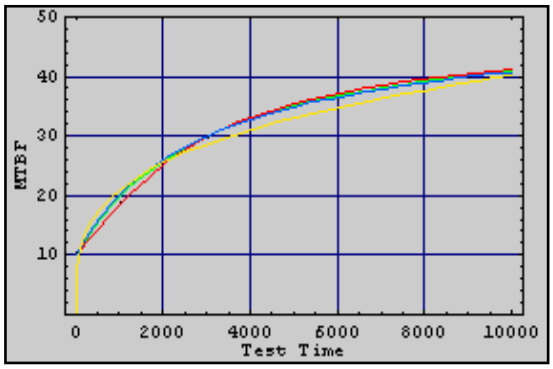


FIGURE 5-32. *Reciprocal of the Failure Intensity (Geometric)*

5.5.5 Using Planning Parameters to Construct the Parsimonious
5.5.6 MTBF Growth Curve.

5.5.6.1 Methodology.

5.5.6.1.1 Planning Formulae not Using Failure Mode Classification.

In the previous sections a functional form for the planned MTBF growth curve was developed. It was indicated that this functional form was compatible with a number of potential growth patterns. In Section 5.5.4, the simulation produced failure mode first occurrence times from a set of initial mode failure rates. For each simulation replication, the parsimonious MTBF growth pattern was derived from a statistically fitted parsimonious expression for the expected number of failure modes function. This was accomplished by utilizing the mode first occurrence times to obtain a MLE of the scale parameter β subject to the initial failure intensity λ held fixed to a specified value (e.g. $\lambda = 0.10$ in Section 5.5.4). In practice, the initial mode rates of occurrence will not be available to obtain the planning curve parameter β.

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In this section, we develop formulas for β in terms of the planning parameters T, M_I, M_G, and average FEF μ_d (and k for the finite case). We will also address the question of how well the parsimonious MTBF planning curves based on the resulting values of β captures several potential reliability growth patterns that depend on realized values of $\lambda_1,\ldots,\lambda_k$ and $d_1,\ldots,d_k$.
As indicated in Section 5.5.3.3.2, the form of the parsimonious expected system failure intensity is,

$$\rho_{PL}(t)=(1-\mu_d)(\lambda-h(t))+h(t) \qquad 5.5\text{-}24$$

For complex systems,

$$h(t)=\frac{\lambda}{1+\beta\cdot t} \qquad 5.5\text{-}25$$

where $0\le t\le T$. For the finite k case, the equation for $\rho_{PL}(t)$ remains the same with $h(t)$ replaced by $h_k(t)=\frac{\lambda}{(1+\beta_k\cdot t)^{p_k+1}}$ where $p_k=\lambda/k\cdot\beta_k$ and β_k denotes the planning value of β for finite k.
To develop formulas for β in terms of planning parameters, let $\theta(t)$ denote the expected fraction of λ attributed to the failure modes surfaced by t. Thus,

$$\theta(t)=\frac{\lambda-h(t)}{\lambda}=1-\frac{h(t)}{\lambda} \qquad 5.5\text{-}26$$

This yields,

$$h(t)=\lambda\cdot(1-\theta(t)) \qquad 5.5\text{-}27$$

It follows that,

$$\rho_{PL}(t)=(1-\mu_d)\{\lambda\cdot\theta(t)\}+\lambda\cdot\{1-\theta(t)\}=\{1-\mu_d\cdot\theta(t)\}\cdot\lambda \qquad 5.5\text{-}28$$

Let M_G denote the goal MTBF at $t=T$ and $\lambda_G=M_G^{-1}$. Then we set

$$\lambda_G=\rho_{PL}(T)=\{1-\mu_d\cdot\theta(T)\}\cdot\lambda \qquad 5.5\text{-}29$$

Thus,

$$\theta(T)=\left(\frac{1}{\mu_d}\right)\left(1-\frac{M_I}{M_G}\right) \qquad 5.5\text{-}30$$

For finite k let $\theta(t)=\theta_k(t)$ where $\theta_k(t)=1-\frac{h_k(t)}{\lambda}=1-(1+\beta_k\cdot t)^{-(p_k+1)}$. In the above β_k is the solution to the equation 5.5-30 with $\theta(T)=\theta_k(T)$ where $p_k=\lambda/k\cdot\beta_k$.
Note for the complex system case,

$$\theta(t)=1-\frac{h(t)}{\lambda}=\frac{\beta\cdot t}{1+\beta\cdot t} \qquad 5.5\text{-}31$$

Therefore, for this case

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$$\frac{\beta \cdot T}{1+\beta \cdot T} = \left(\frac{1}{\mu_d}\right)\left(1-\frac{M_I}{M_G}\right) \qquad 5.5\text{-}32$$

Solving for β yields,

$$\beta = \left(\frac{1}{T}\right)\left(\frac{1-\frac{M_I}{M_G}}{\mu_d - \left(1-\frac{M_I}{M_G}\right)}\right) \qquad 5.5\text{-}33$$

5.5.6.1.2 Planning Formulae Using Failure Mode Classifications.
In some cases, the set of failure modes can be split into two categories termed A-modes and B-modes (Crow, 1982). The B-modes are failure modes that will be mitigated if surfaced during test. The A-modes are those that will not receive a corrective action even if observed during test. For this case, the parsimonious expected failure intensity would be,

$$\rho_{PL}(t) = \lambda_A + (1-\mu_d')(\lambda_B - h_B(t)) + h_B(t) \qquad 5.5\text{-}34$$

where λ_A is the failure intensity due to A-modes, λ_B is the initial failure intensity due to B-modes (thus $\lambda = \lambda_A + \lambda_B$), $h_B(t)$ is the expected failure intensity due to the set of B-modes not surfaced by t, and μ_d' is the average FEF that would be realized for the B-modes if all were surfaced during test. For complex systems, $h_B(t)$ is given by $\frac{\lambda_B}{1+\beta \cdot t}$. It can be shown that for this case planning, formula 5.5-33 becomes,

$$\beta = \left(\frac{1}{T}\right)\left(\frac{1-\frac{M_I}{M_G}}{MS \cdot \mu_d' - \left(1-\frac{M_I}{M_G}\right)}\right) \qquad 5.5\text{-}35$$

where $MS = \lambda_B / \lambda$. The planning parameter MS is termed the management strategy. This represents the fraction of λ that is due to the initial B-mode failure intensity. For the finite k case, $h_B(t)$ is given by $\frac{\lambda_B}{(1+\beta_k \cdot t)^{p_k'+1}}$ where $p_k' = \frac{\lambda_B}{k \cdot \beta_k}$. The value β_k solves Equation 5.5-30 with $\theta(T)$ replaced by $\theta_{k,B}(T) = 1-(1+\beta_k \cdot T)^{-(p_k'+1)}$ and μ_d replaced by $MS \cdot \mu_d'$.

5.5.6.2 Comparisons of MTBF Approximations Using Planning Parameters.
In what follows, we will not use failure mode categories. Unlike the planned MTBF growth curve, $MTBF_{PL}(t) = \rho_{PL}^{-1}(t)$, the average MTBF growth path generated from the simulation replications depends on the particular parent population of the λ_i or deterministic formula used to generate the λ_i, together with the generated mode FEFs drawn from a beta distribution. Thus, this average MTBF growth path over $t \in [0, T]$ depends on far more than just k, T, M_I, M_G, and μ_d. Hence one cannot expect that the planned growth path from M_I to M_G, based solely on the planning parameters, will always closely match the averaged reciprocals of the exact expected system failure intensity. However, as indicated in the preceding sections, the functional form of

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the parsimonious MTBF planning curve is more compatible with respect to the realized MTBF growth pattern than the MIL-HDBK-189 power law MTBF growth pattern. Additionally, the planning parameters are easier to interpret and directly influence than those utilized in the MIL-HDBK-189 approach.

In a number of instances of practical interest the parsimonious MTBF model based on the planning parameters closely approximates the averaged exact MTBF growth patterns. To consider this, we compare the parsimonious MTBF planning curve to the reciprocals of the realized stochastic system failure intensity and expected system failure intensity. For a given simulation replication we will stochastically generate from a given parent population or deterministically calculate $\lambda_1,\ldots,\lambda_k$, together with corresponding mode FEFs $d_1,\ldots,d_k$. The FEFs are generated on each replication from a beta distribution with a mean of 0.80 and coefficient of variation of 0.10.

To calculate the planning value of β on each simulation replication, set $M_I = \lambda^{-1}$ where $\lambda = \sum_{i=1}^{k} \lambda_i$ and choose $\mu_d = \frac{1}{k}\sum_{i=1}^{k} d_i$ (one could alternately choose μ_d to be the expected value of the beta distribution). The value of M_G is set equal to the reciprocal of the realized value of the stochastic system failure intensity at $t = T$. Then Equation 5.5-30 with the appropriate form of $\theta(t)$ is used to obtain the planning β for the finite k and complex system cases. The corresponding finite k and complex system MTBF planning curves for the replication are given by $MTBF_{PL}(t) = \rho_{PL}^{-1}(t)$ where $\rho_{PL}(t)$ is specified in Equation 5.5-24.

The plots below in FIGURE 5-33, FIGURE 5-34, FIGURE 5-36 and FIGURE 5-38 compare the average of ten replicated MTBF finite and infinite k planning curves (green and blue curves, respectively) to the corresponding averages of the reciprocals of the following failure intensities: (1) stochastic realizations of the system failure intensity (black dots); (2) the expected system failure intensity (red curves) and; (3) MIL-HDBK-189 planning curve failure intensities (yellow curves). For our examples the test period is T = 10,000 hours and k = 1,500.

One problem encountered in utilizing planning parameters to generate the MIL-HDBK-189 curves is that, as noted in Section 5.4.3, these curves employ an average MTBF over a selected initial test phase (since the curves interpolate back to a zero MTBF at $t = 0$). To generate a MIL-HDBK-189 planning curve on each simulation replication using M_I, M_G, and T we have used the initial system MTBF metric given in (Crow, 2004) for power law growth curves. The expression for M_I given in (Crow, 2004), denoted by $M_{I,CE}$, is given by

$$M_{I,CE} = \frac{\Gamma\left(1+\frac{1}{\beta_M}\right)}{(\lambda_M)^{1/\beta_M}} \qquad 5.5\text{-}36$$

where λ_M and β_M are the MIL-HDBK-189 power law parameters utilized in Equations 5.1-9 and 5.1-10. The rationale for the M_I metric given by 5.5-36 is not discussed in (Crow, 2004). However, it may have been motivated by the following fact: for a non-homogeneous Poisson

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process of the number of failures experienced by test time t with mean value function $\lambda_M \cdot t^{\beta_M}$, the time to first failure is Weibull distributed. Moreover, the mean of this Weibull random variable is equal to the right-hand side of Equation 5.5-36.

Averages of ten replicated power law MTBF approximations shown in yellow in the figures below were obtained using the M_1 metric presented in (Crow, 2004) as follows:

1. Set $M_1 = \lambda^{-1}$ where $\lambda = \sum_{i=1}^{k} \lambda_i$ and $\lambda_1, \ldots, \lambda_k$ are the realized failure mode initial rates of occurrence for the replication;
2. On each simulation replication of the realized stochastic system failure intensity, set M_G equal to the reciprocal of the realized failure intensity at $t = T$;
3. Set $M_1 = M_{1,CE}$ and $M_G = \left\{\lambda_M \cdot \beta_M \cdot T^{\beta_M - 1}\right\}^{-1}$. Solve for λ_M and β_M, and;
4. Set $MTBF(t) = \{\rho_M(t)\}^{-1}$ for $0 \le t \le T$ where $\rho_M(t) = \lambda_M \cdot \beta_M \cdot t^{\beta_M - 1}$. The values for λ_M and β_M that satisfy the two equations in step 3 can readily be shown to satisfy the equations $M_1 = \dfrac{T\left\{\Gamma\left(1 + \dfrac{1}{\beta_M}\right)\right\}}{\left(\dfrac{T}{\beta_M \cdot M_G}\right)^{\frac{1}{\beta_M}}}$, and $\lambda_M = \dfrac{T^{1-\beta_M}}{\beta_M \cdot M_G}$.

First, we consider several cases where the λ_i are considered a realization of a random sample of size k from a specified parent population. The reference (Miller, 1985) considers a class of software reliability models where the initial bug rates of occurrence are assumed to represent such realizations.

5.5.6.2.1 Gamma Parent Population.

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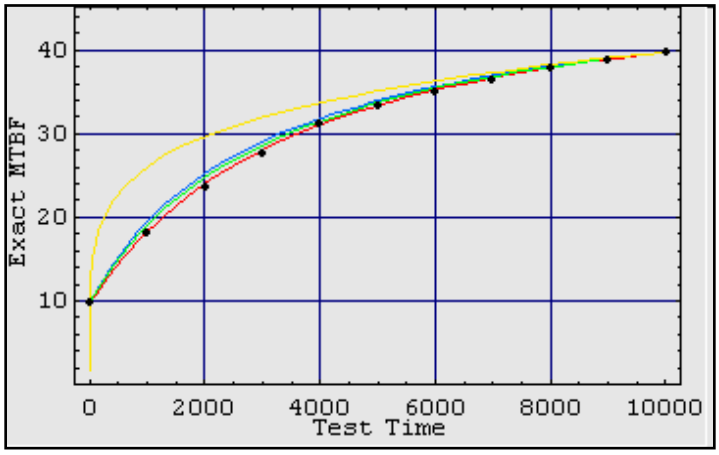


FIGURE 5-33. *Reciprocal of the Failure Intensity (Gamma)*

In FIGURE 5-33 above, the parent population is a gamma distribution. This distribution has been utilized as a parent population for initial bug rates of occurrence in the software reliability literature (Fakhre-Zakeri et. al., 1992). For this case, the averages of the MTBF planning curves for the finite and infinite cases are quite close to the averages of the reciprocals of the realized stochastic and expected system failure intensities. The close approximation is a consequence of the relation mentioned in Section 5.5.4.2 between the solution of the differential equation that defines the parsimonious function $\mu_k(t)$ for the expected number of failure modes surfaced by t and the gamma distribution.

5.5.6.2.2 Lognormal Parent Population.

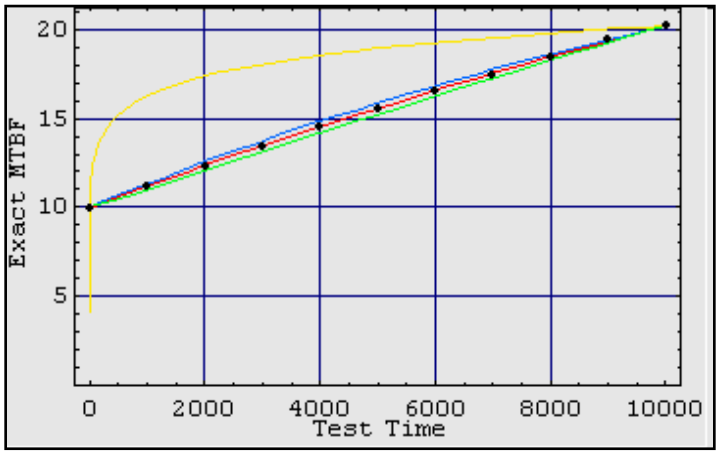


FIGURE 5-34. *Reciprocal of the Failure Intensity (Log Normal)*

In FIGURE 5-34 above, a lognormal parent population is utilized. For this population, the averaged parsimonious MTBF approximations based on the finite and infinite planning values of β are again close to the averages of the reciprocals of the realized stochastic and expected system

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failure intensities. However, this close agreement does not always occur, as illustrated in FIGURE 5-36. To consider this further, look at the portion of the initial failure intensity that the top *w* failure modes comprise as a function of *w*. The top *w* failure modes refer to a set of failure modes of size *w* whose initial failure rates are at least as large as the initial failure rates of the remaining *k*-*w* modes. More formally, for $i = 1, \ldots, k$ let $\lambda_{(i)}$ denote the ordered mode initial failure rates such that $\lambda_{(1)} \geq \cdots \geq \lambda_{(k)}$. Also let $\eta(w) \equiv \left(\frac{1}{\lambda}\right)\sum_{i=1}^{w} \lambda_{(i)}$ for $w = 1, \ldots, k$ where $\lambda = \sum_{i=1}^{k} \lambda_i$.

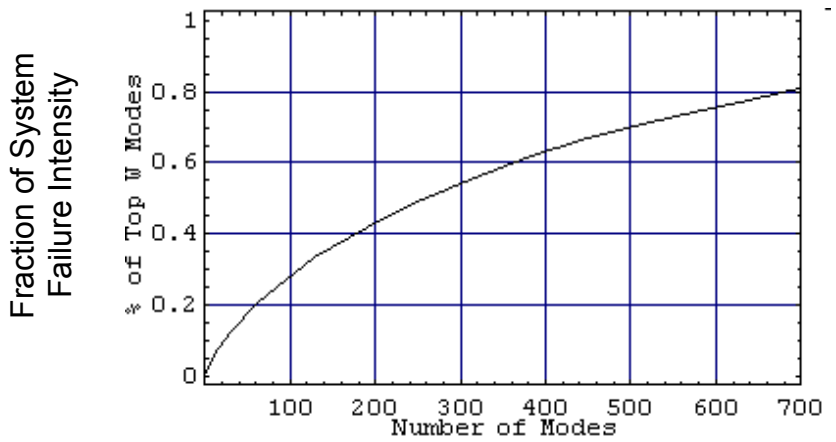
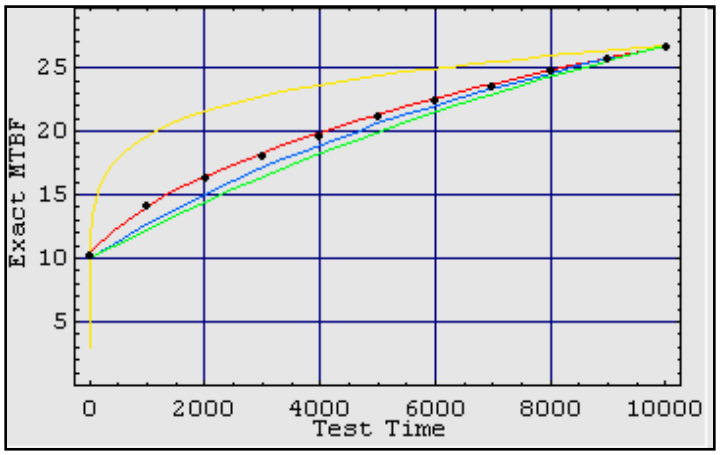


FIGURE 5-35. *Top W Modes (Log Normal)*

The graphs of the average, $\bar{\eta}(w)$, of the ten replicated functions $\eta(w)$ is displayed in FIGURE 5-35 for the case where on each replication, $\lambda_1, \ldots, \lambda_k$ was drawn from a lognormal distribution with mean $0.10/k$. This lognormal distribution was also used to generate the ten replicated graphs of the realized and expected system failure intensities on which FIGURE 5-34 is based. Note in FIGURE 5-35 that $\bar{\eta}(100)$ is less than 0.40.



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FIGURE 5-36. *Reciprocal of the Failure Intensity (Log Normal)*

FIGURE 5-37 is the corresponding graph of $\bar{\eta}(w)$ versus w for the MTBF average curves displayed in FIGURE 5-36. The ten replicated system failure intensity curves utilized in FIGURE 5-36 are based on the same ten sets of initial mode failure rates used to construct the graph in FIGURE 5-37. As for FIGURE 5-34 and FIGURE 5-35, the lognormal parent population used had a mean equal to $0.10/k$. However the variance was larger than that of the lognormal utilized for FIGURE 5-34 and FIGURE 5-35. This resulted in $\bar{\eta}(100) > 0.50$.

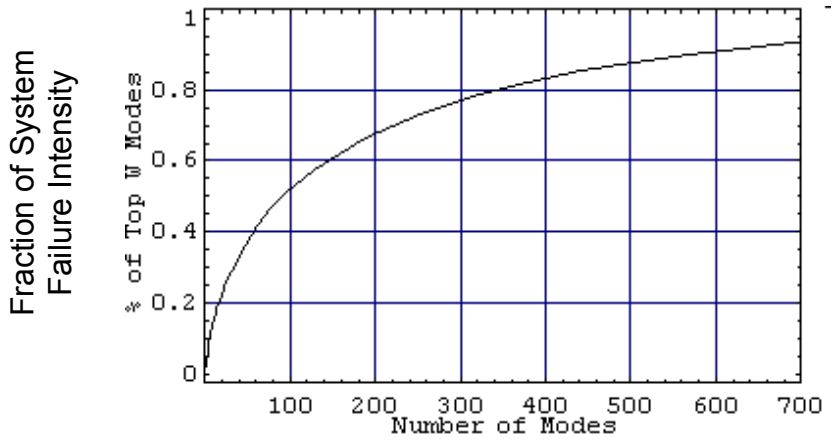


FIGURE 5-37. *Top W Modes (Log Normal)*

In general, the closer the graph of $\bar{\eta}(w)$ is to the line through the origin with slope equal to $1/k$, the closer the averaged MTBF planning curve will be to the averaged reciprocals of the realized stochastic and expected system failure intensities. This follows from the fact that the solution to the differential equation, $\mu_k(t)$, converges to $\mu(t)$ as $\lambda_1, \ldots, \lambda_k$ approach a common value $\lambda_0 > 0$. For the case where $\lambda_i = \lambda_0$ for $i = 1, \ldots, k$ on each replication, the graph of $\bar{\eta}(w)$ versus w for $w = 1, \ldots, k$ lies on the line through the origin with slope equal to $1/k$. Note the graph of $\bar{\eta}(w)$ in FIGURE 5-35, although not close to the line through the origin with slope $1/k$, is closer to this line than is the graph of $\bar{\eta}(w)$ displayed in FIGURE 5-37. This is consistent with the planning approximation in FIGURE 5-34 being more accurate than in FIGURE 5-36. The lognormal distribution for the λ_i on which FIGURE 5-36 and FIGURE 5-37 are based would not be a realistic candidate in many instances for the parent population.

5.5.6.2.3 Geometric Initial Mode Failure Rates.
Finally, we consider geometric failure rates, a case where the initial mode failure rates are specified deterministically. For this case recall that $\lambda_i = a \cdot b^i$ for $i = 1, \ldots, k$ where $a > 0$ and $0 < b < 1$. According to Miller (1985), such bug initial failure rates have been estimated during replicated – run software debugging experiments (Nagel, 1984, 1982). For such a deterministic case, only the set of associated FEFs $d_1, \ldots, d_k$ are regenerated from the beta distribution on each replication. Note that one can show,

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$$\lambda = \sum_{i=1}^{k} \lambda_i = a \cdot \sum_{i=1}^{k} b^i = a \cdot b \cdot \left(\frac{1-b^k}{1-b} \right) \qquad 5.5\text{-}37$$

Thus,

$$M_I = \left(\frac{1-b}{a \cdot b} \right) \left(1 - b^k \right)^{-1} \qquad 5.5\text{-}38$$

Also, as before on each simulation replication, λ_G is set equal to the realized value of the stochastic system failure intensity at $t = T$ and $M_G = \lambda_G^{-1}$. To calculate the MTBF planning curve on the replication, in addition to T, M_I, and M_G, the average FEF value μ_d must be specified. We set $\mu_d = \frac{1}{k} \sum_{i=1}^{k} d_i$ (an alternate possibility would be to set μ_d equal to the specified mean of the beta distribution). FIGURE 5-38 displays the averages of the ten MTBF planning curves for the finite and infinite cases over the replications (green and blue curves, respectively). These average curves are compared to the averaged reciprocals of the realized stochastic system failure intensities (black dots) and the expected system failure intensities (red curve). Also displayed, is the average of the ten MIL-HDBK-189 planning curves (yellow curve). These are based on T, M_G, and $M_I = M_{I,CE}$.

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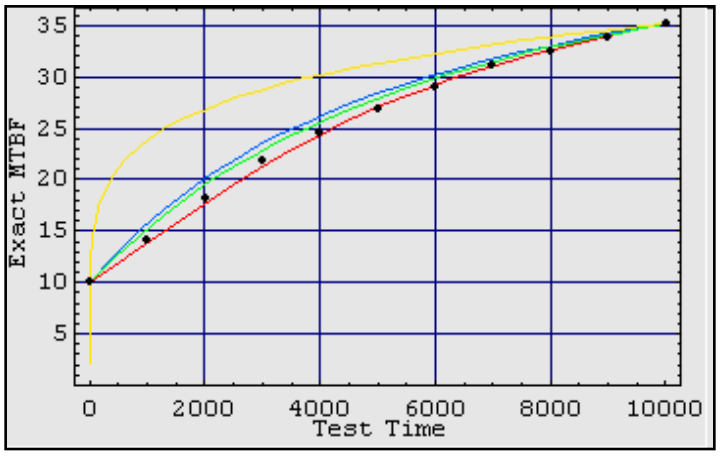


FIGURE 5-38. *Reciprocal of the Failure Intensity (Geometric)*

FIGURE 5-39 below, displays the graph of $\bar{\eta}(w) = \eta(w)$ versus w.

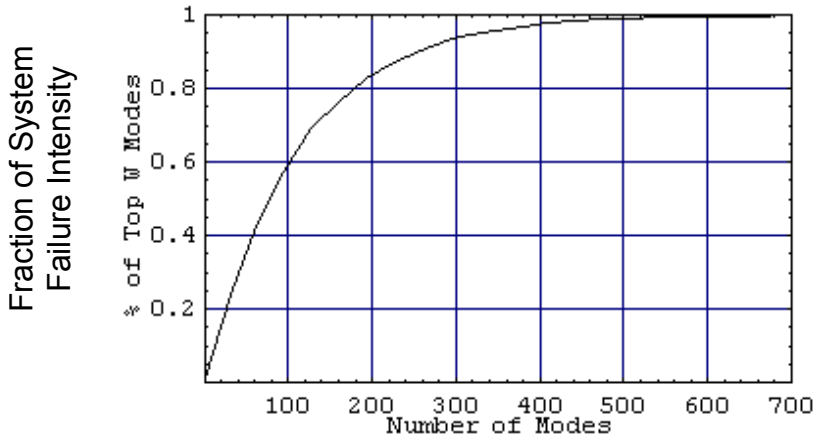


FIGURE 5-39. *Top W Modes (Geometric)*

Even though the top 100 modes account for 60% of the initial failure intensity, the finite and infinite averages of the 10 replicated PM2 planning curves are reasonably compatible with the averages of the reciprocals of the 10 corresponding realized stochastic system failure intensities and expected system failure intensity graphs. The parameter b governs the percent of the initial failure intensity contributed by $\lambda_{(1)}, \ldots, \lambda_{(k)}$. One can show $\eta(w) = \frac{1-b^w}{1-b^k}$ for $w = 1, \ldots, k$. Also, holding λ and b fixed we obtain $\eta_\infty(w) = \lim_{k\to\infty} \eta(w) = 1 - b^w$ for $w = 1, 2, \ldots$.

As for the lognormal case, if $\eta(100)$ is lowered from its current value, then this will generally lead to the finite and infinite average of the PM2 MTBF planning curves being closer to the averages of the reciprocals of the realized and expected system failure intensities.

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5.5.7 Reliability Growth Potential.

5.5.7.1 Growth Potential in Terms of Planning Parameters.

In contrast to the MIL-HDBK-189 planning model, the PM2 planning approximation recognizes a ceiling for the MTBF. To obtain this limiting value we note that $\lim_{t\to\infty} h(t) = 0$. From Equation 5.5-24 we then obtain

$$\lim_{t\to\infty} \rho_{PL}(t) = (1-\mu_d)\cdot\lambda \qquad 5.5\text{-}39$$

This limiting value of $\rho_{PL}(t)$ is termed the growth potential failure intensity and is denoted by ρ_{GP}. The growth potential MTBF is defined to be $\rho_{GP}^{-1}(t)$ and is denoted by M_{GP}. Note,

$$M_{GP} = \frac{M_I}{1-\mu_d} \qquad 5.5\text{-}40$$

From Equation 5.5-28 it is clear that $\rho_{PL}(t)$ is a strictly decreasing function of t whose limit as $t\to\infty$ is ρ_{GP}. Equivalently, $MTBF_{PL}(t) = \rho_{PL}^{-1}(t)$ is a strictly increasing function of t whose limit as $t\to\infty$ is M_{GP}.

The above comments with respect to $\rho_{PL}(t)$ and $MTBF_{PL}(t)$ also apply to the case where failure modes are classified into A-modes and B-modes. However, for this case,

$$M_{GP} = \frac{M_I}{1-(MS)\cdot\mu_d'} \qquad 5.5\text{-}41$$

where μ_d' now denotes the average FEF with respect to the B-modes.

5.5.7.2 Planning Parameter β in Terms of Growth Potential.

The complex system planning formula for β, given by Equation 5.5-33, can be rewritten in terms of the MTBF growth potential. One can show,

$$\beta = \left(\frac{1}{T}\right)\left(\frac{\frac{M_G}{M_I}-1}{1-\frac{M_G}{M_{GP}}}\right) \qquad 5.5\text{-}42$$

This formula applies whether one or two failure mode categories are utilized, as long as the appropriate expression for M_{GP} is applied. For a logically consistent set of reliability growth planning parameters, one must have $M_I < M_G < M_{GP}$. Note this ensures that $\beta > 0$.

5.5.7.3 Plausibility Metrics for Planning Parameters.

Observe that Equation 5.5-42 shows that if T is chosen to be unrealistically small for growing from M_I to M_G then the resulting value of β will be unduly large. This would be reflected in the function,

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$$\theta(t) = \frac{\beta \cdot t}{1 + \beta \cdot t} \qquad 5.5\text{-}43$$

rising towards one at an unrealistic rate. For example, a large β could imply that $\theta(t_0) = 0.80$ for an initial time segment $[0, t_0]$ for which past experience indicates it would not be feasible to surface a set of failure modes that accounted for 80% of the initial failure intensity. An unrealistically large β, and corresponding $\theta(t)$ function, could also arise by choosing M_G to be an excessively high percentage of M_{GP}.

This discussion also pertains to the case where two failure mode categories are utilized. For this case $\theta(t)$ is replaced by $\theta_B(t)$, or the left-hand side of Equation 5.5-43. The value $\theta_B(t)$ denotes the expected fraction of λ_B attributed to the B-modes surfaced by t. The scale parameter β utilized in obtaining $\theta_B(t)$ is still given by Equation 5.5-42. However, to obtain $\theta_B(t)$, M_{GP} is computed via Equation 5.5-41.

A second potentially useful metric for judging whether the planning parameters give rise to a reasonable value for β is the implied average mode failure rate for the expected set of surfaced modes over a selected initial reference test period $[0, t_0]$. Denoting this average mode failure rate by $\rho_{avg}(t_0)$ we have,

$$\rho_{avg}(t_0) = \frac{\lambda - h(t_0)}{\mu(t_0)} \qquad 5.5\text{-}44$$

Classifying failure modes into A and B modes we have,

$$\rho_{avg,B}(t_0) = \frac{\lambda_B - h_B(t_0)}{\mu_B(t_0)} \qquad 5.5\text{-}45$$

Where $\rho_{avg,B}(t_0)$ denotes the average B-mode failure rate for the set of B-modes expected to be surfaced during $[0, t_0]$. In Equation 5.5-45, $\mu_B(t_0)$ is the expected number of surfaced B-modes over $[0, t_0]$. For complex systems, Equation 5.5-44 yields,

$$\rho_{avg}(t_0) = \frac{(\beta \cdot t_0)^2}{t_0 \cdot (1 + \beta \cdot t_0) \cdot \ln(1 + \beta \cdot t_0)} \qquad 5.5\text{-}46$$

where β is given by Equation 5.5-42 with M_{GP} expressed by Equation 5.5-40. Likewise, for complex systems, Equation 5.5-45 implies,

$$\rho_{avg,B}(t_0) = \frac{(\beta \cdot t_0)^2}{t_0 \cdot (1 + \beta \cdot t_0) \cdot \ln(1 + \beta \cdot t_0)} \qquad 5.5\text{-}47$$

where β is given by Equation 5.5-45 with M_{GP} expressed by Equation 2.6-42. For a given M_I (and MS for $\rho_{avg,B}(t_0)$), $\rho_{avg}(t_0)$ and $\rho_{avg,B}(t_0)$ can be expressed in terms of T and M_G for $M_I \le M_G < M_{GP}$. Denote these expressions for $\rho_{avg}(t_0)$ and $\rho_{avg,B}(t_0)$ by $\rho_{avg}(T, M_G; t_0)$ and $\rho_{avg}(T, M_G; t_0)$, respectively. The following properties of these functions can be useful in judging whether T and M_G are reasonable planning values:

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a. $\rho_{avg}(T, M_G; t_0)$ and $\rho_{avg,B}(T, M_G; t_0)$ are continuous positive strictly decreasing functions of T and strictly increasing functions of M_G for $M_I \le M_G < M_{GP}$;
b. $\rho_{avg}(T, M_G; t_0)$ and $\rho_{avg,B}(T, M_G; t_0)$ approach infinity as T approaches zero or M_G approaches M_{GP} and;
c. $\rho_{avg}(T, M_G; t_0)$ and $\rho_{avg,B}(T, M_G; t_0)$ approach zero as T approaches infinity or M_G approaches M_I.

A third potentially useful metric to judge the reasonableness of the planning parameters is the expected number of unique failure modes or unique B-modes surfaced over an initial test interval $[0, t_0]$ they imply. Prior experience with similar developmental programs or initial data from the current program can serve as benchmarks.

5.5.8 Generating a Planned Reliability Growth Path.

Once the planning parameters are chosen, the parsimonious approximation for the expected failure intensity can be used to generate a detailed reliability growth planning curve. For example, suppose a test schedule is laid out that gives the planned number of RAM miles accumulated on the units under test per month. Also suppose the test schedule specifies blocks of calendar time for implementing corrective actions. Finally, for planning purposes let us assume that in order for a failure mode to be addressed in an upcoming corrective action period, it must occur four months prior to the start of the period. For this situation, the MTBF could be represented by a constant value between the ends of corrective action periods and between the start of testing and the end of the first scheduled corrective action period (CAP). For such a test plan, jumps in MTBF would be portrayed at the conclusion of each CAP. The increased MTBF after the jump is given by $MTBF(t_i) = \{\rho(t_i)\}^{-1}$ where t_i denotes the accumulated test time, as determined from the monthly schedule, by the calendar date that occurs four months prior to the start of the ith CAP. Since $MTBF(t_i)$ depends on a large number of parameters it would be approximated by the parsimonious approximation $\{\rho_k(t_i)\}^{-1}$ or $\{\rho_\infty(t_i)\}^{-1}$. In such a manner a sequence of target MTBF steps would be generated that grow from the initial MTBF value to a goal MTBF value.

FIGURE 5-40 below, depicts a detailed reliability growth planning curve for a complex system for the case where A and B failure mode categories are utilized.

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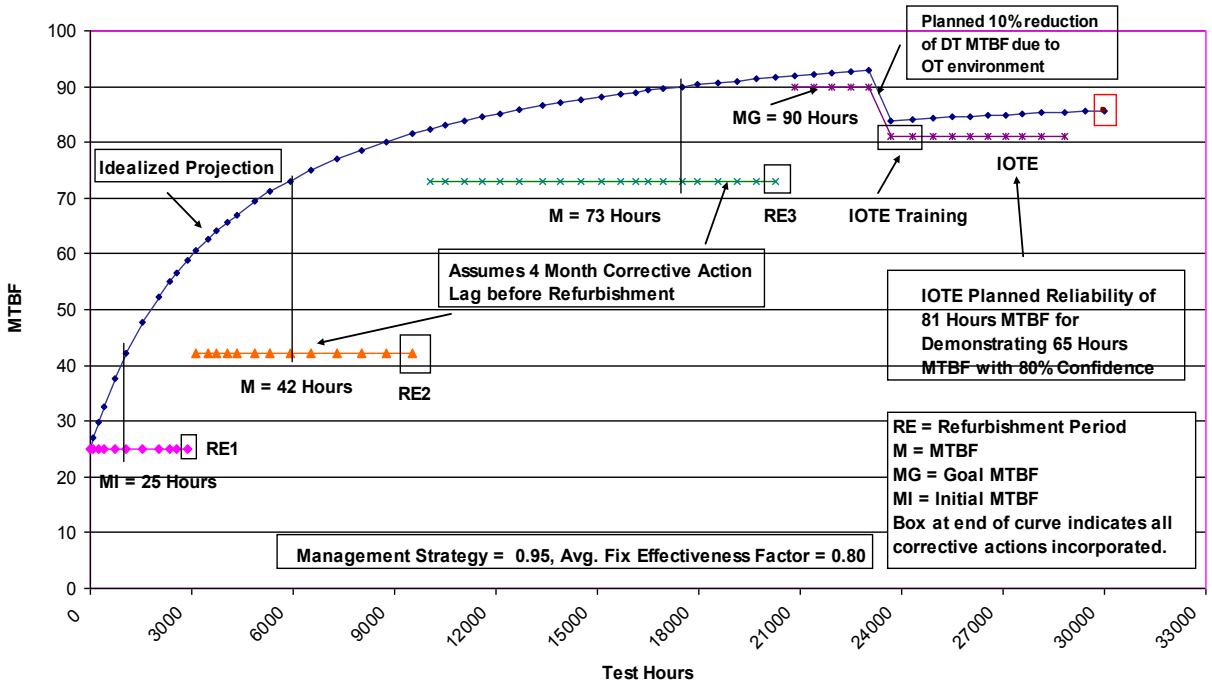


FIGURE 5-40. *PM2 Reliability Growth Planning Curve*

The blue curve in FIGURE 5-40 represents $MTBF(t) = \{\rho_{PL}(t)\}^{-1}$ where $\rho_{PL}(t)$ is given by Equation 5.5-34 and β is obtained from the planning parameters by Equation 5.5-35. Note the value $MTBF(t)$ is the system MTBF one expects to obtain once all corrective actions to B-modes surfaced during test period $[0,t]$ are implemented. The MTBF steps are constructed from the blue curve, the schedule of corrective action periods (CAPs), and the assumed average corrective action implementation lag. In FIGURE 5-40, note that the goal MTBF, M_G, was chosen to be larger than $M_R = 65$ hours, the MTBF to be demonstrated during a follow-on IOT&E. This test is an operational demonstration test of the system's suitability for fielding. Such a test is mandated by public law for major DoD developmental systems. In such a demonstration test it may be required to demonstrate M_R with a measure of assurance. In the figure we have utilized as our measure of assurance a demonstration of M_R at the 80% statistical confidence level. To have a reasonable probability of achieving such a demonstration, the system must enter the IOT&E with an MTBF value of M_R^+ which is greater than M_R. The needed value of M_R^+ can be determined by a well-known statistical procedure from the IOT&E test length, the desired confidence level of the statistical demonstration, and the specified probability of being able to achieve the statistical demonstration. After determining M_R^+ one can consider what the goal MTBF, M_G, should be at the conclusion of the development test. The value of M_G should be the goal MTBF to be attained just prior to the IOT&E training period that proceeds the IOT&E. The goal MTBF associated with the development test environment must be chosen sufficiently above M_R^+ so that the operational test environment does not cause the reliability of the test units to fall

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below M_R^+ during the IOT&E. The significant drop in MTBF often seen could be attributable to operational failure modes that were not encountered during the developmental test. In Figure 40, a derating factor of 10% was used to obtain M_G from M_R^+, i.e., in the figure $M_G = \frac{M_R^+}{0.90}$.

5.6 PM2-Discrete.

The material in this section is as specified in [7].

5.6.1 Purpose.

This report outlines a new reliability growth planning methodology that may be used to construct detailed reliability growth programs and associated planning curves for *discrete systems*[2]. The purpose and utility of a reliability growth planning curve is to:

a. Portray the planned reliability achievement of a system as a function of test exposure, as well as other important programmatic resources.
b. Serve as a baseline against which demonstrated reliability values may be compared throughout the test program (for tracking purposes).
c. Illustrate and quantify the feasibility in a potential test program in achieving interim and final reliability goals. An example of an interim reliability goal is the AAE test threshold (DA 2007).

5.6.2 Impact.

The mathematical developments presented herein constitute the first reliability growth planning methodology ever developed specifically for discrete systems. Thus, it represents the first on only existing quantitative method available that reliability practitioners and program managers may use for formulating detailed reliability growth plans (in the discrete usage domain). Note also that the methodology herein, hereafter referred to as PM2-Discrete, is not just a reliability growth planning model. It is a robust reliability growth planning methodology that possesses concomitant measures of programmatic risk and system maturity. For instance, PM2-Discrete offers several reliability growth management metrics of fundamental interest that practitioners may use when assessing the efficacy of a proposed T&E plan. These metrics include:

a. Expected number of failures observed by trial t.
b. Expected number of failure modes observed by trial t.
c. Expected reliability on trail t under failure mode mitigation.
d. Expected *reliability growth potential*[3].
e. Expected probability of failure on trial t due to a new failure mode.
f. Expected fraction surfaced of the system probability of failure on trial t.

The model equations associated with these metrics, as well as the required inputs are briefly outlined in the following section.

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[2] A ***discrete system*** is a system whose test exposure is measured in terms of discrete trials, shots, or demands, e.g., guns, rockets, missile systems, torpedoes etc.

[3] The ***reliability growth potential*** is the theoretical upper-limit on reliability achieved by finding and fixing all B-modes with a specified level of fix effectiveness. A B-mode is a failure mode that will be addressed via corrective action, if observed during testing.

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5.6.3 List of Notations.

k	Total potential number of failure modes.
m	Number of observed failure modes.
x	Shape parameter of the beta distribution, e.g., represents pseudo failures.
n	Shape parameter of the beta distribution, e.g., represents pseudo time.
c	Max allowable number of failures.
μ	Average fix effectiveness factor.
γ	Derating factor due to transition from a DT to an OT environment.
δ	Lag-time to corrective action implementation.
T	Total test number of trials.
T_L	Total test number of trials to the lag-time before the last CAP. This is potentially where the development effort stops.
t_I	Length, i.e., number of trials in the initial test phase.
t_i	First occurrence trial of failure mode i.
p_i	Initial failure probability for failure mode i.
R_I	Initial system reliability.
R_A	Portion of R_I comprised of A-modes.
R_B	Portion of R_I comprised of B-modes.
R_R	Reliability requirement for the system.
R_G	Reliability goal for the system before derating.
R_F	Final reliability target on the growth curve after derating.
R_{GP}	Reliability growth potential.
$R(t)$	System reliability on trial t.
$f(t)$	Expected number of failures on trial t.
$\mu(t)$	Expected number of failure modes observed on or prior to trial t.
$h(t)$	Expected probability of failure on trial t due to a new failure mode.
$\phi(t)$	Expected fraction surfaced of the B-mode probability of failure on trial t.
$\Gamma(t)$	Euler gamma function evaluated at t.
$\psi(t)$	Psi gamma function evaluated at t, defined as $\psi(t) \equiv \Gamma'(t)/\Gamma(t)$.
$\ell_{1-\alpha}(c,T)$	$(1-\alpha)\cdot 100$ percent LCB on system reliability based on c failures and T trials.

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5.6.4 Model Assumptions.

a. Initial failure mode probabilities of occurrence $p_1,\ldots,p_k$ constitute a realization of an independent and identically distributed (*iid*) random sample $P_1,\ldots,P_k$ such that $P_i \sim Beta(n,x)$ for each $i=1,\ldots,k$. We utilize the following Probability Density Function (PDF) parameterization,

$$f(p_i) \equiv \begin{cases} \dfrac{\Gamma(n)}{\Gamma(x)\cdot\Gamma(n-x)} \cdot p_i^{x-1} \cdot (1-p_i)^{n-x-1} & p_i \in [0,1] \\ 0 & \text{otherwise} \end{cases} \qquad 5.6\text{-}1$$

where the shape parameters n and x represent pseudo trials and failures, respectively, and $\Gamma(x) \equiv \int_0^\infty t^{x-1} \cdot e^{-t} dt$ is the Euler gamma function. The associated mean, and variance of the P_i are given respectively by,

$$E(P_i) = \frac{x}{n} \qquad 5.6\text{-}2$$

and

$$Var(P_i) = \frac{x \cdot (n-x)}{n^2 \cdot (n+1)} \qquad 5.6\text{-}3$$

b. The number of trials $t_1,\ldots,t_k$ until failure mode first occurrence constitutes a realization of a random sample $T_1,\ldots,T_k$ such that $T_i \sim Geometric(p_i)$ for each $i=1,\ldots,k$.

c. Potential failure modes occur independently of one another and their occurrence is considered to constitute a *system failure*.

d. When failures are observed during testing their corresponding failure modes are identified, and management may (or may not) address them via corrective action. If a given failure mode (e.g., failure mode i) is addressed, it is assumed that either: (1) an FEF is assigned by expert judgment with very detailed knowledge regarding the proposed engineering design modification or; (2) a demonstrated FEF, $\hat{d}_i$, is used.

5.6.5 Management Metrics & Model Equations.

5.6.5.1 Overview.

The methodology presented herein consists of deriving several model equations of immediate interest. These model equations constitute the analytical framework from which a number of different reliability growth management metrics may be estimated. These metrics (outlined below) give managers and practitioners the means to gauge the development effort of discrete systems, and build off the methodology advanced in (Hall 2008b). In this section, we now address the more complicated case where corrective actions may be installed on system

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prototypes anytime after failure modes are first discovered. These equations are extensions of the earlier ones in the sense that they are unconditional expectations of their counterparts (i.e., unconditioned on the P_i for $i = 1,\ldots,k$). The resulting expressions in the following sections are found to be functions of the two beta shape parameters, rather than the vector of unknown failure probabilities inherent to the system. Equation numbers from our earlier publication (Hall 2008b) are given for cross-reference.

5.6.5.2 Expected Reliability.
Previously we discussed the notion of the expected initial system reliability, or the reliability of the system in its current configuration (i.e., before corrective actions are applied). We now consider the expected reliability of the system on trial t that would be achieved if observed failure modes are mitigated via a specified level of fix effectiveness. Per Equation 8 in (Hall 2008b), the expected reliability of a discrete system on trial t conditioned on the vector of unknown failure probabilities $\vec{P} \equiv (P_1,\ldots,P_k)$ is given as,

$$R_k\left(t \mid \vec{P}\right) \equiv \prod_{i=1}^{k}\left[1-\left(1-\left[1-\left(1-P_i\right)^{t-1}\right]\cdot d_i\right)\cdot P_i\right] \qquad 5.6\text{-}4$$

The unconditional expectation of $R_k\left(t \mid \vec{P}\right)$ w.r.t. the P_i for $i = 1,\ldots,k$ is,

$$R_k(t) \equiv E\left[R_k\left(t \mid \vec{P}\right)\right] \approx \left[1-\left(1-\left[1-\frac{\Gamma(n-x+t-1)\cdot\Gamma(n+1)}{\Gamma(n-x)\cdot\Gamma(n+t)}\right]\cdot\bar{d}\right)\cdot\frac{x}{n}\right]^k \qquad 5.6\text{-}5$$

where $\bar{d} \equiv \sum_{i\in obs} d_i \Big/ m$ is an average FEF. This expression models the true but unknown expected reliability of a discrete system on trial t, where corrective actions may be implemented at any time after their associated failure modes are first discovered. The parameters n and x in 5.6-5 are estimated via the MLE procedure. Note that our model is independent of the *A-mode* / *B-mode* classification scheme, as A-modes need only be distinguished from B-modes via a zero FEF (i.e., $d_i = 0$ if failure mode i is not observed, or not corrected). Notice that 5.6-5 is a function of the parameters n and x, and the average FEF $\bar{d}$, rather than the individual failure probabilities $p_1,\ldots,p_k$ and the individual FEF $d_1,\ldots,d_k$. As a result, 5.6-5 is an approximation of 5.6-4. The accuracy of this approximation depends on the number of observed failure modes that are corrected, as well as the variance of the assessed FEF. The approximation is best when all observed failure modes are corrected, and when the variance of the assessed FEF is small. Under the additional assumption that the assigned FEF are independent of the magnitudes of $p_1,\ldots,p_k$, one can show the expected value as $k \to \infty$ only depends on n, x and average FEF for all failure modes. When taking the limit as $k \to \infty$, the average FEF is treated as a constant equal to $\sum_{\forall i} d_i \Big/ k$. In practice, this average FEF is assessed as $\bar{d}$, e.g., ibid. Finally, notice that the initial condition of 5.6-5 equates to the unconditional expected initial reliability of the system as required,

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$$R_k\left(t=1\right)=\left(1-\frac{x}{n}\right)^k \qquad 5.6\text{-}6$$

It is also desirable to study the limiting behavior of 5.6-5 as $k\to\infty$, since the total potential number of failure modes inherent to a complex system is typically large, and since k is unknown. After reparameterizing 5.6-5 via 5.6-6, our limiting approximation simplifies too,

$$\hat{R}_\infty\left(t\right)\equiv\lim_{k\to\infty}\hat{R}_k\left(t\right)=\hat{R}_{\infty,I}^{\,1-\bar{d}\left(\frac{t-1}{\hat{n}_\infty+t-1}\right)}=\left(\hat{R}_{\infty,I}\right)^{\wedge}\left[1-\bar{d}\cdot\left(\frac{t-1}{\hat{n}_\infty+t-1}\right)\right] \qquad 5.6\text{-}7$$

where $\hat{R}_{\infty,I}$ and $\hat{n}_\infty$ are obtained via the MLE procedure outlined in (Hall 2008b).

Equation 5.6-7 has rather significant applications to reliability growth planning. By expressing 5.6-7 in terms of reliability growth planning parameters, one my generate the idealized planning curve for a discrete system as,

$$R\left(t\right)\equiv R_A\cdot R_B^{\,1-\mu\cdot\left(\frac{t-1}{n+t-1}\right)}=R_A\cdot R_B{}^{\wedge}\left[1-\mu\cdot\left(\frac{t-1}{n+t-1}\right)\right] \qquad 5.6\text{-}8$$

where

a. $R_A\in\left(0,1\right)$ is the portion of system reliability comprised of failure modes that will not be addressed via corrective action.
b. $R_B\in\left(0,1\right)$ is the portion of system reliability comprised of failure modes that will be addressed via corrective action.
c. $\mu\in\left(0,1\right)$ is the planned average fix effectiveness.
d. $n\in\square^+$ is the shape parameter of the beta distribution that represents pseudo trials.

Clearly, formulae are required for the parameters R_A, R_B, and n before 5.6-8 may be utilized in a reliability growth planning context. In the next few sections, these formulae are derived and found to be functions of only a small number of planning parameters, e.g., thereby yielding a parsimonious approximation. Most importantly, these planning parameters can be directly controlled by program management, and easily quantified throughout the developmental test program for tracking purposes. Before these formulae may be derived, the notions of Management Strategy and growth potential must first be defined in the present context, i.e., in the context of the discrete usage domain.

5.6.5.3 Management Strategy.
In the continuous time domain, Management Strategy (MS) is defined as the fraction of the initial system failure intensity addressed via the corrective action effort,

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$$\text{(continuous time domain)}\quad MS \equiv \frac{\lambda_B}{\lambda_I} = \frac{\lambda_B}{\lambda_A + \lambda_B} \qquad 5.6\text{-}9$$

In 5.6-9, λ_B and λ_A denote the portion of the system failure intensity associated comprised of failure modes that are, and are not, addressed by corrective action, respectively. λ_I represents the total initial system failure intensity. Note that in the continuous time domain, MS may be defined via 5.6-9 since the failure intensity is mathematical modeled as an additive function, i.e., the sum of failure mode rates of occurrences. Recall via 5.6-4, however, system reliability is mathematically modeled as a multiplicative function in the discrete usage domain, i.e., the product of failure mode success probabilities. Thus for discrete systems, the initial reliability may be expressed as,

$$R_I \equiv R_A \cdot R_B \qquad 5.6\text{-}10$$

or equivalently, $\ln R_I = \ln R_A + \ln R_B$. In light of this relationship, one may define the *MS* for discrete systems in an analogous (but not equivalent) fashion, in comparison to that of the continuous time domain. The *MS* for discrete systems is given by,

$$\text{(discrete usage domain)}\ MS \equiv \frac{\ln R_B}{\ln R_I} = \frac{\ln R_B}{\ln R_A + \ln R_B} \qquad 5.6\text{-}11$$

5.6.5.4 Formulae for R_A, and R_B.

Using 5.6-11, one may now derive the required formulae for R_B. The desired expression, derived directly from 5.6-11 is,

$$R_B = R_I^{MS} \qquad 5.6\text{-}12$$

From 5.6-11 and 5.6-12, the desired expression for R_A is,

$$R_A = \frac{R_I}{R_B} = \frac{R_I}{R_I^{MS}} = R_I^{1-MS} \qquad 5.6\text{-}13$$

Notice from 5.6-12 and 5.6-13 that $R_A \cdot R_B = R_I^{1-MS} \cdot R_I^{MS} = R_I$, as desired.

5.6.5.5 Reliability Growth Potential.

The earliest notion of the reliability growth potential of a system was first expressed in a paper written by Virene (1968). The concept was advanced further by other researchers, and is a characteristic of a number of reliability growth models such as Crow (1984), Ellner & Wald (1995), Crow (2003, 2004), Ellner & Hall (2004, 2006), and Hall (2008a-c). The growth potential represents the theoretical upper-limit on reliability achieved by finding and effectively correcting all B-modes in a system with a specified fix effectiveness. Using 5.6-8 and 5.6-11, one may derive an expression for the growth potential with applications in reliability growth planning. The growth potential is defined as,

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$$\begin{aligned} R_{GP} \equiv \lim_{t\to\infty} R(t) &= \lim_{t\to\infty} R_A \cdot R_B^{1-\mu\cdot\left(\frac{t-1}{n+t-1}\right)} \\ &= R_A \cdot R_B^{1-\mu} \\ &= R_I^{1-MS} \cdot R_I^{MS\cdot(1-\mu)} \\ &= R_I^{1-MS\cdot\mu} \end{aligned} \qquad 5.6\text{-}14$$

The importance of this expression cannot be overemphasized. Specifically, it states that the theoretical upper-limit on reliability that can be achieved for a discrete system depends on only three quantities: (1) the initial reliability of the system; (2) the magnitude of the problem that is addressed (e.g., MS) and; (3) the average level of fix effectiveness achieved, μ. Thus, the growth potential represents an asymptote on idealized planning curve 5.6-8. To appreciate this point, one must realize that some reliability growth planning models, e.g., Military Handbook 189 model (DoD 1981) do not possess a growth potential. Thus, it is very easy for practitioners to develop growth curves whose final reliability goal is higher than the growth potential. This means that it is easy to develop impossible reliability growth plans with models that do not possess a growth potential. Thus, it is important to be able to estimate the growth to assess the feasibility of proposed reliability growth plans for discrete systems.

5.6.5.6 Formula for *n*.

Let T_L denote the trial number at the lag-time before the last corrective action period. Then $R(T_L)$ is interpreted as the potentially representing the final reliability of the system before production. Thus, the formula for the parameter n is found *s.t.* $R(T_L) = R_G$, where R_G is the reliability goal for the system. After some detailed calculation, one will find that $R(T_L) = R_G$ implies,

$$n = (T_L - 1)\cdot\frac{\ln(R_{GP}/R_G)}{\ln(R_G/R_I)} \qquad 5.6\text{-}15$$

Recall that the condition $n \in \square^+$ must hold for the idealized curve 5.6-8 to be meaningful. Notice from 5.6-15, if the reliability goal is chosen *s.t.* $R_G \geq R_{GP}$, then $n \leq 0$. This emphasizes the importance for the practitioner to be mindful of the growth potential when specifying a final reliability goal for the system to achieve.

5.6.5.7 Expected Number of Failures.

Let r denote the number of test phases corresponding to the fixed configurations of the system w.r.t. reliability. Let $T_1,\ldots,T_r$ denote the total number of trials conducted in each of the r test phases. Since failure modes may be addressed via corrective action during scheduled corrective action periods between test phases, the initial failure probabilities $P_1,\ldots,P_k$ may be reduced throughout the total number of trials, $T = \sum_{j=1}^{r} T_j$, conducted in the test program. Thus, let $P_{i,j}$

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represent the failure probability for failure mode $i = 1,\ldots,k$, in test phase $j = 1,\ldots,r$. Then, the conditional expected number of failures in T trials is given by,

$$f_k\left(T \mid P_i\right) \equiv \sum_{j=1}^{r}\sum_{i=1}^{k} T_j \cdot P_{i,j} \qquad 5.6\text{-}16$$

The unconditional expectation of 5.6-16 w.r.t. the $P_{i,j}$ for $i = 1,\ldots,k$ is,

$$f_k\left(T\right) \equiv \sum_{j=1}^{r}\sum_{i=1}^{k} \frac{T_j \cdot x_j}{n_j} = \sum_{j=1}^{r} \frac{k \cdot T_j \cdot x_j}{n_j} \qquad 5.6\text{-}17$$

where x_j and n_j are the shape parameters of the beta parent population of failure mode probabilities of occurrence in test phase $j = 1,\ldots,r$. Using the reparameterization 5.6-6, the limiting approximation of 5.6-17 as $k \to \infty$ is,

$$\begin{aligned}
\hat{f}_\infty\left(T\right) \equiv \lim_{k\to\infty} \hat{f}_k\left(T\right) &= \lim_{k\to\infty} \sum_{j=1}^{r} \frac{k \cdot T_j \cdot \hat{x}_{k,j}}{\hat{n}_{k,j}} \\
&= \lim_{k\to\infty} \sum_{j=1}^{r} k \cdot T_j \cdot \left(1 - \hat{R}_{k,j}^{1/k}\right) \\
&= \sum_{j=1}^{r} \ln \hat{R}_{\infty,j}^{-T_j} \\
&= \ln \prod_{j=1}^{r} \hat{R}_{\infty,j}^{-T_j}
\end{aligned} \qquad 5.6\text{-}18$$

In the context of reliability growth planning, the total expected number of failures associated with a given T&E planned is expressed as,

$$f\left(T\right) \equiv \sum_{j=1}^{r} \ln R_j^{-T_j} = \ln R_1^{-T_1} + \cdots + \ln R_r^{-T_r} \qquad 5.6\text{-}19$$

where the terms $\ln R_j^{-T_j}$ are interpreted as the expected number of failures in test phase $j = 1,\ldots,r$.

5.6.5.8 Expected Number of Failure Modes.
The conditional expected number of unique failure modes observed on or before trial t is given by,

$$\mu_k\left(t \mid \vec{P}\right) \equiv k - \sum_{i=1}^{k}\left(1 - P_i\right)^t \qquad 5.6\text{-}20$$

The resulting unconditional expectation of 5.6-20 w.r.t. the P_i for $i = 1,\ldots,k$ is,

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$$\mu_k(t) \equiv E\left[\mu_k\left(t \mid \vec{P}\right)\right] = k - \sum_{i=1}^{k} E\left[\left(1-P_i\right)^t\right] = k - k \cdot \left[\frac{\Gamma(n)\cdot\Gamma(n-x+t)}{\Gamma(n-x)\cdot\Gamma(n+t)}\right] \quad 5.6\text{-}21$$

These expressions have the following convenient interpretation: the expected number of unique failure modes observed in t trials is equivalent to the total potential number of failure modes in the system minus the expected number of failure modes that will not be observed in t trials. The initial condition of 5.6-21 implies that the expected number of failure modes observed on trial $t=0$ (i.e., before testing begins) is $\mu_k(t=0)=0$, as required. An estimate of 5.6-21 is obtained by using the finite k MLE for the beta shape parameters n and x.

To derive the limiting behavior of 5.6-21, we have used the reparameterization 5.6-6 and taken the limit as $k \to \infty$. After some detailed calculation we find,

$$\hat{\mu}_\infty(t) \equiv \lim_{k\to\infty} \hat{\mu}_k(t) = \hat{n}_\infty \cdot \ln \hat{R}_{\infty,I} \cdot \left[\psi(\hat{n}_\infty) - \psi(\hat{n}_\infty + t)\right] \quad 5.6\text{-}22$$

where $\hat{n}_\infty$ is an MLE. Recall via the well-known recurrence formula for the psi-gamma functions that $\sum_{j=0}^{t-1} \frac{1}{n_\infty + j} = \psi(n+t) - \psi(n)$. Using this recurrence formula with 5.6-22, the expected number of failure modes observed on or before trial t in a reliability growth planning context may be calculated by.

$$\begin{aligned} \mu(t) &\equiv n \cdot \ln R_I \cdot \left[\psi(n) - \psi(n+t)\right] \\ &= \sum_{j=0}^{t-1} \frac{\ln R_I^{-n}}{n+j} \end{aligned} \quad 5.6\text{-}23$$

5.6.5.9 Expected Probability of Failure due to a New Mode.
The conditional expected probability of discovering a new failure mode on trial t is given as,

$$h_k\left(t \mid \vec{P}\right) \equiv 1 - \prod_{i=1}^{k}\left[1 - \left(1-P_i\right)^{t-1} \cdot P_i\right] \quad 5.6\text{-}24$$

The unconditional expectation of 5.6-24 w.r.t. the P_i for $i = 1,\ldots,k$ is,

$$h_k(t) \equiv E\left[h_k\left(t \mid \vec{P}\right)\right] = 1 - \left[1 - \frac{\Gamma(n-x+t-1)\cdot\Gamma(x+1)}{\mathrm{B}(x,n-x)\cdot\Gamma(n+t)}\right]^k \quad 5.6\text{-}25$$

Equation 5.6-25 is estimated by using the finite k MLE for n and x obtained via the MLE procedure in (Hall 2008b). Notice that the initial condition of 5.6-25 equates to,

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$$h_k(t=1) = 1 - \left[1 - \frac{\Gamma(n-x)\cdot\Gamma(x+1)}{\mathrm{B}(x,n-x)\cdot\Gamma(n+1)}\right]^k = 1 - \left(1 - \frac{x}{n}\right)^k = 1 - R_k(t=1) \qquad 5.6\text{-}26$$

This means that the expected probability of discovering a new failure mode on the first trial is equivalent to the initial system probability of failure, as desired.

After reparameterizing via 5.6-6, the limiting approximation of 5.6-25 as $k \to \infty$ simplifies to,

$$\hat{h}_\infty(t) \equiv \lim_{k\to\infty} \hat{h}_k(t) = 1 - \hat{R}_{\infty,I}^{\frac{\hat{n}_\infty}{\hat{n}_\infty + t - 1}} = 1 - \hat{R}_{\infty,I} \wedge \left(\frac{\hat{n}_\infty}{\hat{n}_\infty + t - 1}\right) \qquad 5.6\text{-}27$$

where $\hat{R}_{\infty,I}$ and $\hat{n}_\infty$ are MLE. Using 5.6-12, 5.6-13, and 5.6-27, the reliability growth planning application of this metric may be assessed via,

$$h(t) \equiv 1 - R_A \cdot R_B^{\frac{n}{n+t-1}} \qquad 5.6\text{-}28$$

where n is given by 5.6-15. The expressions above estimate the expected probability of discovering a new failure mode on trial t, and can be utilized as a measure of programmatic risk. For example, as the development effort (e.g., TAFT process) continues, we would like the estimate of $h_k(t) \to 0$. This condition indicates that program management has observed the dominant failure modes in the system. Conversely, large values of $h_k(t)$ indicate higher programmatic risk w.r.t. additional unseen failure modes inherent to the current system design. Effective management and goal setting of $h_k(t)$ would be a good practice to reduce the likelihood of the customer encountering unknown failure modes during fielding and deployment.

5.6.5.10 Expected Fraction Surfaced of System Probability of Failure.
The portion of system unreliability on trial t associated with failure modes that have already been observed during testing (e.g., the probability of observing repeat failure modes with continued testing) is given in (Hall 2008b) as,

$$\varphi(t \mid \vec{P}) \equiv 1 - \prod_{i=1}^{k} \left(1 - \left[1 - (1 - P_i)^{t-1}\right] \cdot P_i\right) \qquad 5.6\text{-}29$$

The unconditional expectation of (5.6-29) w.r.t. the P_i for $i = 1, \ldots, k$ is,

$$\varphi_k(t) \equiv E\left[\varphi(t \mid \vec{P})\right] = 1 - \left(1 - \left[\frac{x}{n} - \frac{\Gamma(n-x+t-1)\cdot\Gamma(x+1)}{\mathrm{B}(x,n-x)\cdot\Gamma(n+t)}\right]\right)^k \qquad 5.6\text{-}30$$

Using 5.6-26 and 5.6-30 we express the expected probability of failure on trial t due to a repeat failure mode as a fraction of initial system unreliability. This fraction is given by,

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$$\phi_k(t) \equiv \frac{\varphi_k(t)}{h_k(t=1)} = \frac{1-\left(1-\left[\frac{x}{n}-\frac{\Gamma(n-x+t-1)\cdot\Gamma(x+1)}{\mathrm{B}(x,n-x)\cdot\Gamma(n+t)}\right]\right)^k}{1-R_k(t=1)} \qquad 5.6\text{-}31$$

An estimate of 5.6-31 is obtained by substituting the true beta parameters by their corresponding MLE. The initial condition of 5.6-31 is $\phi_k(t=1)=0$, which means that the expected probability of failure on the first trial due to a repeat failure mode is zero, as required.
To take the limit of 5.6-31 as $k\to\infty$, we proceed in a similar fashion as above by using the reparameterization (5.6-6). After simplification we obtain,

$$\hat{\phi}_\infty(t) \equiv \lim_{k\to\infty}\hat{\phi}_k(t) = \frac{1-\hat{R}_{\infty,I}^{\frac{t-1}{\hat{n}_\infty+t-1}}}{1-\hat{R}_{\infty,I}} = \frac{1-\hat{R}_{\infty,I}\wedge\left(\frac{t-1}{\hat{n}_\infty+t-1}\right)}{1-\hat{R}_{\infty,I}} \qquad 5.6\text{-}32$$

where $\hat{R}_{\infty,I}$ and $\hat{n}_\infty$ are MLE. Using 5.6-12, and 5.6-13, Equation 5.6-32 may be expressed for application in reliability growth planning by,

$$\phi(t) \equiv \frac{1-R_A\cdot R_B^{\frac{t-1}{n+t-1}}}{1-R_I} \qquad 5.6\text{-}33$$

where n is given by 5.6-15. The value of these expressions is that they may be used as a system maturity metric. For instance, a good management practice would be to specify goals for $\phi_\infty(t)$ at important program milestones in order to track the progress of the development effort w.r.t. the maturing design of the system (from a reliability standpoint). Small values of $\phi_\infty(t)$ indicate that further testing is required to find and effectively correct additional failure modes. Conversely, large values of $\phi_\infty(t)$ indicate that further pursuit of the development effort to increase system reliability may not be economically justifiable (i.e., the cost may not be worth the gains that could be achieved). Finally, note that program management can eliminate at most the portion $\phi_\infty(t)$ from the initial system unreliability prior to trial t regardless of when fixes are installed or how effective they are (i.e., since this metric is independent of the corrective action process).

5.7 Threshold Program.

5.7.1 Introduction.

A threshold value is not a statistical lower confidence bound on the true reliability. Rather, it is a reliability value that is used simply to mark off a rejection region for the purpose of conducting a test of hypothesis to determine if the achieved reliability of a system is not growing according to plan. The threshold derivations for a single threshold are presented in Appendix E.

The threshold program is a tool for determining, at selected program milestones, whether the reliability of a system is failing to progress according to the idealized growth curve established prior to the start of the growth test. The threshold program embodies a methodology that is best

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suited for application during a reliability growth test referred to as the test-fix-test program. Under this program, when a failure is observed, testing stops until the failure is analyzed and a corrective action is incorporated on the system. Testing then resumes with a system that has (presumably) a better reliability. In some references, this is also referred to as a test-analyze-fix-test or TAFT. The graph of the reliability for this testing strategy is a series of small increasing steps that can be approximated by a smooth idealized curve.

The test statistic in this procedure is the reliability point estimate that is computed from test failure data. If the reliability point estimate falls at or below the threshold value (in the rejection region), this would indicate that the achieved reliability is statistically not in conformance with the idealized growth curve and without some remedial action to restore the system reliability to a higher level such as a program restructuring effort, a more intensive corrective action process, a change of vendors, additional lower level testing, etc. requirements may not be achieved.

Recall that the initial time T_I marks off a period of time in which the initial reliability of the system is essentially held constant while early failures are being surfaced. Corrective actions are then implemented at the end of this initial phase, and this gives rise to improvement in the reliability. Therefore, to make threshold assessments during the period of growth, milestones should be established at points in time that are sufficiently beyond T_I.

Note also that reliability increases during test until it reaches its maximum value of M_F by the end of the test at T_F. Growth usually occurs rapidly early on and then tapers off toward the end of the test phase. Therefore, in order to have sufficient time to verify that remedial adjustments (if needed) to the system are going to have the desired effect of getting the reliability back on track, milestones must be established well before T_F.

In actual practice, it is possible that the actual milestone test time may differ, for a variety of reasons, from the planned milestone time. In that case, one would simply recalculate the threshold based on the actual milestone time.

5.7.2 Background.

There are only three inputs – the total test time T_F, the final MTBF M_F and the growth rate α – necessary to define the idealized curve to build a distribution of MTBF values. The initial MTBF M_I and the initial time period T_I are not required because this implementation assumes that the curve goes through the origin. In general, this is not a reasonable assumption to make for planning purposes, but for the purposes of this program, the impact is negligible, especially since milestones are established sufficiently beyond T_I. If more than one milestone is needed the subsequent milestones are conditional in the sense that milestone k cannot be reached unless the system gets through the previous *k-1* milestones.

A program would develop a distribution of MTBF values by generating a large number of failure histories from the parent curve defined by T_F, M_F and α. Typically, the number of failure histories may range from 1000 to 5000, where each failure history corresponds to a simulation run. The threshold value is that reliability value corresponding to a particular percentile point of an ordered distribution of reliability values. A percentile point is typically chosen at the 10th or 20th percentile when establishing the rejection region – a small area in the tail of the distribution

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that allows for a test of hypothesis to be conducted to determine whether the reliability of the system is "off" track.

The test statistic in this procedure is the reliability point estimate that is computed from test failure data, which is compared to the threshold reliability value.

5.7.3 Application

5.7.4 Example.

The process begins with a previously constructed idealized growth curve with a growth rate of 0.25 and reliability growing to a final MTBF (requirement) of 70 hours by the end of 1875 hours of test time. These parameters - α, M_F, and T along with a milestone, selected at 1000 hours, and a threshold percentile value of 20% was selected. The failure history number was set at 2500 histories. The resulting reliability threshold of approximately 46 hours was computed. Now, suppose that a growth test is subsequently conducted for T = 1875 hours. Using the AMSAA/Crow tracking model, an MTBF point estimate is computed based on the first 1000 hours of growth test data. If the resulting MTBF point estimate at the selected milestone is above the threshold value, there is not strong statistical evidence to reject the null hypothesis that the system is growing according to plan. If the resulting MTBF point estimate at the 1000 hour milestone is at or below the threshold value, then there is strong statistical evidence to reject the null hypothesis and a red flag would be raised. This red flag is a warning that the achieved reliability, as computed with the AMSAA model, is statistically not in conformance with the pre-established idealized growth curve, and that the information collected to date indicates that the system may be at risk of failing to meet its requirement. This situation should be brought to the attention of management, testers and reliability personnel for possible remedial action to get the system reliability back on track.

5.8 References.

1. William J. Broemm, Paul M. Ellner, W. John Woodworth, AMSAA TR-652, AMSAA Reliability Growth Guide, September 2000

2. MIL HDBK-189, Reliability Growth Management, 13 February 1981

3. Ellner, Paul and Mioduski, Robert, AMSAA TR-524, Operating Characteristic Analysis for Reliability Growth Programs, August 1992

4. Crow, Larry H., AMSAA TR-197, Confidence Interval Procedures for Reliability Growth Analysis, June 1977

5. McCarthy, Michael and Mortin, David, and Ellner, Paul, and Querido, Donna, AMSAA TR-555, Developing a Subsystem Reliability Growth Program Using the Subsystem Reliability Growth Planning Model (SSPLAN), September 1994

6. Crow, Larry H., AMSAA TR-138, Reliability Analysis for Complex Repairable Systems, 1974

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Check the source to verify that this is the current version before use.

7. Ellner, Paul M., and Hall, Brian J., AMSAA TR2006-09, Planning Model Based on Projection Methodology (PM2), March 2006

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6 RELIABILITY GROWTH TRACKING.

6.1 Introduction.

Reliability growth tracking is an area of reliability growth that provides management the opportunity to gauge the progress of the reliability effort for a system. The choice of the correct model to use is dependent on the management strategy for incorporating corrective actions in the system. However, it is important to note that the AMSAA/Crow test-fix-test model does not assume that all failures in the data set receive a corrective action. Based on the management strategy some failures may receive a corrective action and some may not. This section contains material on the AMSAA Continuous Tracking Model (RGTMC), the AMSAA Discrete Tracking Model (RGTMD) developed in [2].

Reliability growth tracking has many significant benefits, many of which make the process not subject to opinion or bias, but are rather statistically based and therefore estimation is placed on a sound and consistent basis. The following is a partial list of these tracking methodology benefits.

a. Uses all failure data (no purging). Purging failures that had a fix for the problems had been a particular problem. These failures often were completely purged from the database after a fix was proposed. With Crow's work, the power model eliminated the need to purge since the methodology does that analytically without need for user intervention or bias. This may be seen from the estimate of MTBF - $T/\hat{\beta}$n - where $0<\hat{\beta}<1$ $\left(\hat{\beta} = 1 - \hat{\alpha}\right)$ for growth so that the denominator reduces the number of failures in accordance with a growth situation.
b. Statistically estimates the current reliability (demonstrated value) and may be used to determine if the requirement has been demonstrated as a point estimate or with confidence.
c. Provides a framework such that statistical confidence bounds on reliability and the parameters of the model may be estimated.
d. Allows for a statistical test of the model applicability through goodness-of-fit tests.
e. Determines the direction of reliability growth from the test data.
    i Positive growth ($\alpha>0$)
    ii. No growth ($\alpha=0$)
    iii. Negative growth ($\alpha<0$)
f. Highlights to management shortfalls in achieved reliability compared to planned reliability.
g. Provides a metric for tracking progress that may provide a path for early transition into next program phase.

Important elements of reliability growth tracking analysis include proper failure classification, test type, configuration control, and data requirements. Many of these elements are spelled out in the FRACAS. Typical data requirements for tracking analysis include cumulative time/miles to failure (continuous systems), cumulative trials/rounds to failure (discrete systems), and total test time or total trials/rounds. Again, the FRACAS should be used as a guide.

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6.1.1 Definition and Objectives of Reliability Growth Tracking.

Reliability growth tracking is a process that allows management the opportunity to gauge the progress of the reliability effort for a system by obtaining a demonstrated numerical measure of the system reliability during a development program based on test data. Some objectives of reliability growth tracking include:

a. determining if system reliability is increasing with time (i.e., growth is occurring) and to what degree (i.e., growth rate), and
b. estimating the demonstrated reliability (i.e., a reliability estimate based on test data for the system configuration under test at the end of the test phase). This estimate is based on the actual performance of the system tested and is not based on some future configuration.

Reliability growth tracking allows for the situation where the configuration of the system may be changing as a result of the incorporation of corrective actions to problem failure modes. In the presence of reliability growth, the data from earlier configurations may not be representative of the current configuration of the system. On the other hand, the most recent test data, which would best represent the current system configuration, may be limited so that an estimate based upon the recent data would not, in itself, be sufficient for a valid determination of reliability. Because of this situation, reliability growth tracking may offer a viable method for combining test data from several configurations to obtain a demonstrated reliability estimate for the current system configuration, provided the reliability growth tracking model adequately represents the combined test data.

6.1.2 Managerial Role.

The role of management in the reliability growth tracking process is twofold:

a. to systematically plan and assess reliability achievement as a function of time and other program resources (such as personnel, money, available prototypes, etc.,) and,
b. to control the ongoing rate of reliability achievement by the addition to or reallocation of these program resources based on comparisons between the planned and demonstrated reliability values.

To achieve reliability goals, it is important that the program manager be aware of reliability problems during the conduct of the development program so that effective system design changes can be funded and implemented. It is essential, therefore, that periodic assessments (tracking) of reliability be made during the test program (usually at the end of a test phase) and compared to the planned reliability goals. A comparison between the assessed and planned values will suggest whether the development program is progressing as planned, better than planned, or not as well as planned. Thus, tracking the improvement in system reliability through quantitative assessments of progress is an important management function.

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6.1.3 Types of Reliability Growth Tracking Models.

Reliability growth tracking models are distinguished according to the level at which testing is conducted and failure data are collected. They fall into two categories: system level and subsystem level. For system level reliability growth tracking models, testing is conducted in a full-up integrated manner, failure data are collected on an overall system basis, and an assessment is made regarding the system reliability. For subsystem level reliability growth tracking models, the subsystems are tested and the failure data are collected on an individual subsystem basis -- the subsystem data are then "rolled up" to arrive at an estimate for the demonstrated system reliability.

System level reliability growth tracking models are further classified according to the usage of the system. They fall into two groups -- continuous and discrete models -- and are defined by the type of outcome that the usage provides. Continuous models are those that apply to systems for which usage is measured on a continuous scale, such as time in hours or distance in miles. For continuous models, outcomes are usually measured in terms of an interval or range; for example, mean time/miles between failures. Discrete models are those that apply to systems for which usage is measured on an enumerative or classificatory basis, such as pass/fail or go/no-go. For discrete models, outcomes are recorded in terms of distinct, countable events that give rise to probability estimates.

6.1.4 Model Substitution.

In general, continuous models are designed for continuous data, and discrete models are designed for discrete data. In the event a designated model is unavailable for use, it may be possible to use a continuous model for discrete data or a discrete model for continuous data. The latter case is generally not a practical option, though. (The AMSAA Subsystem Tracking Model, for example, is a continuous model that may be used with discrete data, subject to the conditions mentioned at the end of this paragraph.) In cases involving model substitution, the "substitute" model is used as an approximation for the intended model, and the original data appropriate for the intended model must be converted to a format appropriate for the substitute model. Note that in applying a continuous model to discrete data, the results of the approximation improve as the number of trials increases and the probability of failure decreases.

6.1.5 List of Notations.

Discrete Parameters:

N	number of trials = sample size
S	success
F	failure
NS	number of successes
NF	number of failures
U	unreliability
R	reliability

Continuous Parameters:

MTTF mean time/trials to failure
MTBF mean time/trials between failures

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By way of an example, we show a method for converting discrete data to a continuous format and vice versa. Suppose that from a sample size of N = 5 trials the following outcomes are observed, where *S* denotes a success and F denotes a failure:

$$S \quad S \quad S \quad S \quad F$$

The number of successes, *NS*, is four; the number of failures, *NF*, is one; and $N = NS + NF$.

To begin, note that in discrete terms:

$$U = probability(failure) = \frac{NF}{N} \qquad 6.1\text{-}1$$

The reciprocal of *U*, namely *N/NF*, may be viewed as a measure of the number of trials to the number of failures, *MTTF*, thus allowing a continuous measure to be related to a discrete measure:

$$MTTF = \frac{1}{U} \qquad 6.1\text{-}2$$

In the example, MTTF = 5 and MTBF = 4, so that:

$$MTBF = MTTF - 1 \qquad 6.1\text{-}3$$

Substituting (6.1-2) into (6.1-3) and noting that $R = 1 - U$ results in:

$$MTBF = \frac{1}{U} - 1 = \frac{R}{1 - R} \qquad 6.1\text{-}4$$

Equation (6.1-4) is used to convert a discrete measure to a continuous measure. To convert a continuous measure to a discrete measure, rearrange (6.1-4) and solve for *R*:

$$MTBF + 1 = \frac{1}{1 - R} \qquad 6.1\text{-}5$$

$$R = 1 - \frac{1}{MTBF + 1} \qquad 6.1\text{-}6$$

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6.1.6 Some Practical Data Analysis Considerations.

While the above sections provide important benefits and aspects of growth tracking, there are many tasks that should be performed either before running the models or in conjunction with running the models. A number of these have evolved from practical applications by analysts using the methodologies over the past twenty-some years, and are not necessarily dependent on the growth methodology but rather practical statistical and graphical analyses of data. For example, a thorough review and analysis of the data should be performed in order to more completely understand the data, where there are shortcomings, identify whether data from different tests might be aggregated, how the data might be aggregated or broken up, question whether there are outliers which would affect results, do plots of data or analyses of the failure modes suggest anything, etc. In some cases the last analysis performed might be running the final model for estimation of point and interval values and looking into projections. Progress is gauged by obtaining a demonstrated numerical measure of system, or subsystem, reliability throughout a development program based on test data.

Not to belabor the point of analyzing the data in this reliability growth handbook, but good analysis does play a major role in developing the most reasonable estimates of a system's reliability. The following paragraphs are provided in hopes that they may help in guiding analysis. First, we present a generalized reliability evaluation approach in the following FIGURE 6-1. It is recognized that although this figure goes beyond modeling and tracking (projection), it does however give a flow of general actions and top side analyses that constitute a good approach to tracking growth analysis.

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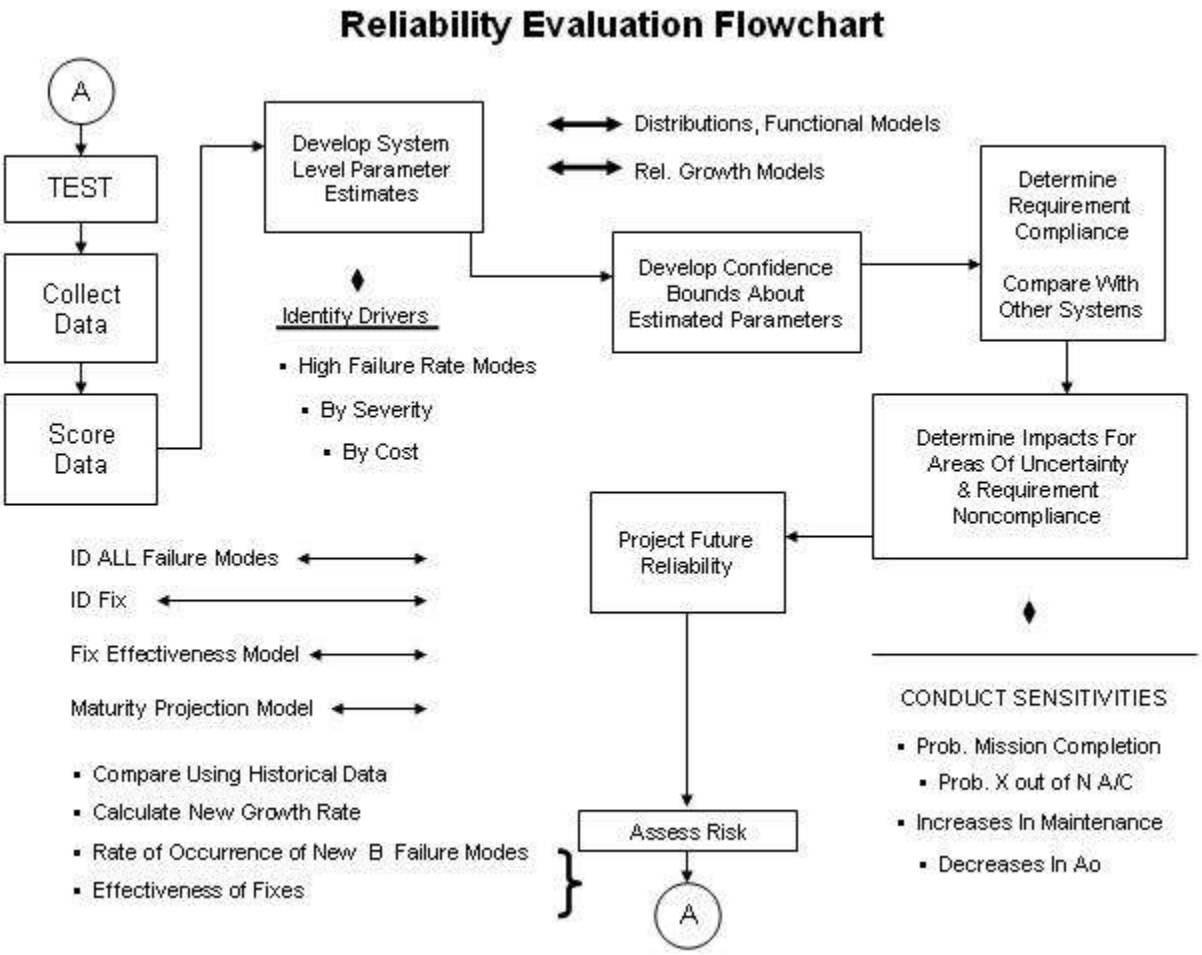


FIGURE 6-1. *Reliability Evaluation Flowchart*

Assuming that the proper ground work has been laid out for the collection and documentation of data so that detailed analyses can be performed, the following are suggestions for initial actions that might be taken, a number of which may not seem to directly impact running the tracking models, but which often lead to identifying problems and leading to a more informed analysis. For example, they may not lead to specific statistics or methods but rather lead to a better understanding of the data, the underlying processes and subsequently lead to a more informed and unbiased estimation. The following paragraphs contain some suggestions for preparation and analysis of test data.

a. Review the data for consistency, omissions, etc. Group data by configuration or other logical break out, order data by time, and plot data. This allows you to see trends, identify suspect outliers.
b. Develop functional reliability models, e.g. series versus parallel.
c. Can data be aggregated? Look for reasons why data may be different, e.g., different test conditions, configurations, random differences, other. Get estimates

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and use different methodologies. And remember, you have variability in both individual test items and from item to item.

d. Compare the data with previous test phase/testing/predecessor systems. Have there been improvements? Are they reflected in an improvement in reliability?

e. Identify failure modes and stratification. Identify driver failure modes for possible corrective action. Where do the failure modes occur by major subsystems? A generalized Pareto Chart of failure modes is illustrated below in FIGURE 6-2 In addition; an example break out of system failures by major subsystems is illustrated. Where appropriate, the following question might be asked. Were frequently occurring failure modes fixed? Better to have 5 failures in 100 hours of test than 1 failure in 20 hours of test even though they have the same MTBF.

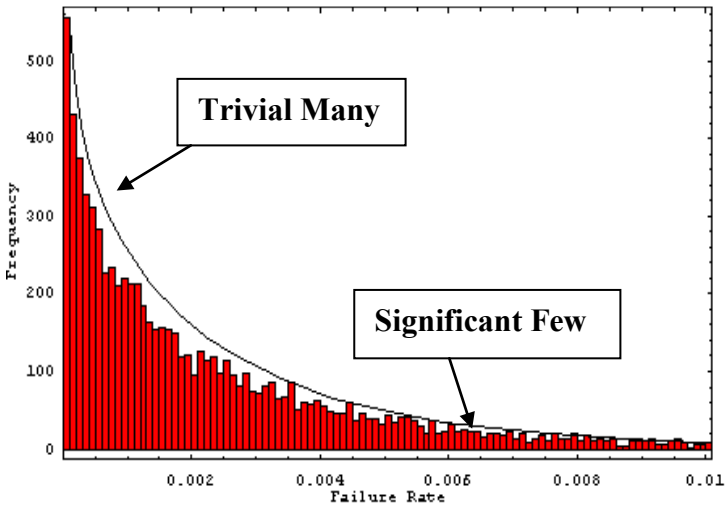


FIGURE 6-2. *Pareto Chart of Failure Modes*

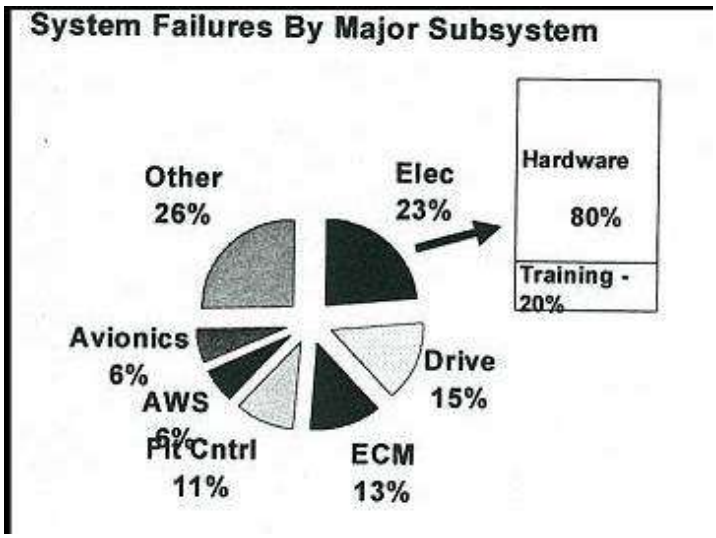


FIGURE 6-3. *System Failures by Major Subsystem*

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f. Determine what conditions may have impacted data. Determine impacts of data analysis on program.
g. Determine applicability of the growth model: before using a statistical model, such as the power law model, one should decide whether the model is in reasonable agreement with the failure pattern exhibited by the data.
h. There are at least two ways to look for trends in the system failure data after chronologically ordering the times to failure – plot or graphical techniques and statistical tests.

(1) Regarding graphical techniques, plot cumulative failure rate versus cumulative time on a log-log scale as for the Duane log-log plot or graphically plot cumulative failure (y-axis) vs. cumulative operating time (x-axis) (FIGURE 6-4). Both provide simple, effective means to visually assess whether or not a trend exists and whether to model using a HPP (times between failure are independent identically exponentially distributed) or NHPP (times between failure tend to increase or decrease with time or age). TABLE V will be used to illustrate the latter plot for two systems under test.

TABLE V. System Arrival Times

Failure Order Number (i)	System A Arrival Times ($TTSF_i$)	System B Arrival Times ($TTSF_i$)
1	15	177
2	42	242
3	74	293
4	117	336
5	168	368
6	233	395
7	410	410

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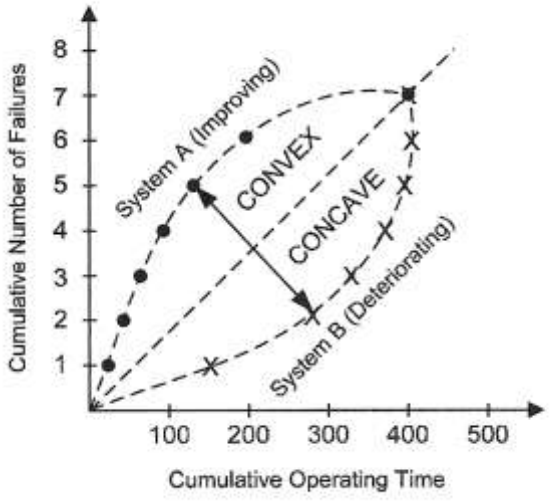


FIGURE 6-4. *Cumulative Failures Vs Cumulative Operating Time*

A Convex Curve implies the system is improving (time between failures increasing) while a Concave Curve implies the system is deteriorating (time between failures decreasing). Plots that generally fall along a straight line indicate no trend.

(2)Perform trend test such as the LaPlace Trend Test. The Laplace test statistic can be used to determine if the times between failures are tending to increase, decrease or remain the same. The underlying probability model for the Laplace test is a NHPP having a log-linear intensity function. This test is better at finding significance when the choice is between no trend and a NHPP model. In other words, if the data come from a system following the exponential law, this test will generally do better than any test in terms of finding significance.

If we have *r-1* chronologically ordered gap times at $t_1, t_2, \cdots t_{r-1}$ with the observation period ending at time t_r. The LaPlace Trend Test Statistic, *Z*, tests the hypothesis

H_0: HPP – Homogeneous Poisson Process
H_1: NHPP – i.e., Monotone Time Trend

$$Z = \frac{\sum_{i=1}^{r-1} t_i - \frac{1}{2(r-1)t_r}}{t_r\sqrt{\frac{r-1}{12}}} \sim N(0,1) \qquad 6.1\text{-}7$$

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Compare Z to percentiles of the standard normal distribution. For the data for systems A and B, the Z values are calculated to be -2.01 and +2.00, respectively, with Z_A inferring times between failure increasing (growth) and Z_B inferring times between failure are tending to decrease (degradation).

i. Apply growth methodology, conduct goodness-of-fit, and calculate MTBF point and interval estimates.
   1. Determine which tracking model is appropriate based on objectives and model assumptions satisfied.
   2. Use Lindstrom-Madden Method for determination of confidence bounds as appropriate. This methodology is applicable for combining subsystem data for which each subsystem may or may not operate for the entire mission length.
   3. Perform Goodness-of-Fit Test to determine if the tracking model adequately represents the data. There are two tests, one for the continuous model when individual times of failure are known (Cramer-von-Mises Statistic), the other when all failure times are not known but known to an interval of test time(Chi-square statistic). If the goodness-of-fit test does not provide strong evidence against the model and there are no non-statistical considerations that argue against using the model to represent the growth pattern exhibited by the data, then one can make the non-statistical decision to analyze the data based on the model representation and associated statistical techniques.

      It is noted that in using the Chi-square goodness-of-fit test, reference often is made only to the number of categories or cells and not to the total number of observations. However, in order that the approximation of the distribution to that in Chi-square tables should be close, the sample size must be sufficiently large so that none of the cell frequencies is less than 1 and not more than 20 per cent of the cells are less than 5. It is noted that there exist other criteria for judging whether that the approximation is close. The above is at least a guide for judging the adequacy of the sample size for cells. Note also that the Chi-Square goodness-of-fit test is not sensitive to trend or order of testing, but only to deviations from the expected frequencies without regard to order.

   4. Assess risk – If tracking growth is below the idealized planned growth, what is the growth rate required to get back on track? Are the fixes effective? Is there enough time for fix implementation and test verification?
      i. Perform sensitivity analyses.
      ii. Calculate a new growth rate α required to get back on track.
      ii. Is the new growth rate too aggressive?

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iii. Does the AMPM model show a high percentage of failure rate revealed?
iv. Are fix effectiveness factors required in excess of historical values?
v. Is there sufficient time for fix implementation and test verification?

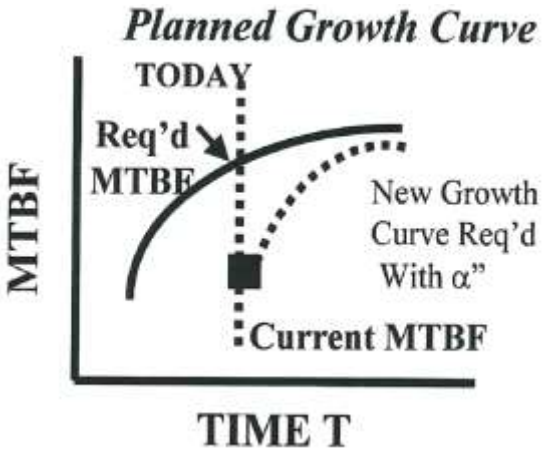


FIGURE 6-5. *Planned Growth Curve*

6.2 Tracking Models Overview.

There are three models that can be utilized in tracking reliability through test:

a. The Reliability Growth Tracking Model – Continuous (RGTMC);
b. The Reliability Growth Tracking Model – Discrete (RGTMD) and;
c. The Subsystem Level Tracking Model (SSTRACK). The following sections provide an overview of each tracking model.

6.2.1 Reliability Growth Tracking Model – Continuous (RGTMC) Overview.

6.2.1.1 RGTMC Purpose.

The purpose of the RGTMC is to assess the improvement in the reliability, within a single test phase, of a system during development for which usage is measured on a continuous scale. The model may utilize both if failure times are known and if failure times are only known to an interval (grouped data).

6.2.1.2 RGTMC Assumptions.

The assumptions associated with the RGTMC are:

a. test time is continuous and;
b. (2) failures, within a test phase, are occurring according to a NHPP with power law MVF.

6.2.1.3 RGTMC Limitations.

The limitation of the RGTMC include:

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a. the model will not fit the test data if large jumps in reliability occur as a result of the applied fix implementation strategy;
b. the model will be inaccurate if the testing does not adequately reflect the OMS/MP;
c. if a significant number of non-tactical fixes are implemented, the growth rate and associated system reliability will be correspondingly inflated as a result and;
d. with respect to contributing to the reliability growth of the system, the model does not take into account reliability improvements due to delayed corrective actions.

6.2.1.4 RGTMC Benefits.
There are also a number of benefits of the RGTMC:

a. the model can gauge demonstrated reliability versus planned reliability;
b. the model can provide statistical point estimates and confidence intervals for MTBF and growth rate and;
c. the model allows for statistical goodness-of-fit testing.

6.2.2 Reliability Growth Tracking Model – Discrete (RGTMD) Overview.

6.2.2.1 RGTMD Purpose.
The purpose of the RGTMD is to track reliability of one-shot systems during development for which usage is measured on a discrete basis, such as trials or rounds.

6.2.2.2 RGTMD Assumptions.
The assumptions of the RGTMD are:

a. test duration is discrete (i.e. trials, or rounds);
b. trials are statistically independent;
c. the number of failures for a given system configuration is distributed according to a binomial random variable and;
d. the cumulative expected number of failures through any initial sequence of configuration is given by the power law.

6.2.2.3 RGTMD Limitations.
The MLE solution may occur on the boundary of the constraint region of reliability, which can give an unrealistic estimate of zero for the initial reliability. Also, for the RGTMD one cannot perform goodness-of-fit tests if there are a limited number of failures.

6.2.2.4 RGTMD Benefits.
The benefits of the RGTMD include the following:

a. can gauge demonstrated reliability versus planned reliability; and
b. provides approximate lower confidence bounds for system reliability (when the MLE solution does not lie on the boundary), and
c. it is the only AMSAA model for discrete reliability growth tracking.

6.2.3 Subsystem Level Tracking Model (SSTRACK) Overview.

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6.2.3.1 SSTRACK Purpose.

The purpose of the SSTRACK model is to assess system level reliability from the use of component, or subsystem, test data.

6.2.3.2 SSTRACK Assumptions.

The assumptions associated with the SSTRACK model include:

a. subsystem test duration is continuous;
b. the system can be represented as a series of independent subsystems and;
c. for each growth subsystem, the reliability improvement is in accordance with a NHPP with power law MVF.

6.2.3.3 SSTRACK Limitations.

All of the limitations associated with the RGTMC apply to the SSTRACK model – for each subsystem. Also, the SSTRACK model does not address reliability problems associated with subsystem interfaces.

6.2.3.4 SSTRACK Benefits.

The benefit of the SSTRACK model consists of:

a. can provide statistical point estimates and approximate confidence intervals on system reliability based on subsystem test data;
b. can accommodate a mixture of growth and non-growth subsystem test data and;
c. can perform goodness-of-fit test for the NHPP subsystem assumptions.

6.3 Tracking Models.

6.3.1 Reliability Growth Tracking Model – Continuous.

The AMSAA/Crow Continuous Reliability Growth Tracking Model may be used to track the reliability improvement of a system during a development test phase for which usage is measured on a continuous scale. The model may also be used for tracking the reliability of one-shot (discrete) systems if there are a large number of trials and the system demonstrates high reliability during test.

6.3.1.1 Basis for the Model.

List of Notations.

t_i	cumulative test time when design modification *i* is made
K	final entry in a sequence of test times; point where the last design modification is made
λ_i	constant failure rate during *i*-th time interval
F_i	number of failures during *i*-th time interval
θ_i	mean value function for F_i
f	a particular realization of F_i

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e	exponential function
t	cumulative test time
F(t)	total number of system failures by time *t*
$\theta(t)$	mean value function for *F(t)*
$\rho(y)$	failure rate for configuration *i* where
$\rho(t)$	instantaneous system failure rate at time *t*; also referred to as the failure intensity function
λ	scale parameter of parametric function $\rho(t)$;
β	shape parameter of parametric function $\rho(t)$;
m(t)	instantaneous mean time between failures at time *t*
T	total test time
F	total observed number of failures by time *T*
X_i	cumulative time to *i*-th failure
^	denotes an estimate when placed over a parameter
L	lower confidence coefficient
U	upper confidence coefficient
γ	desired confidence level
-	denotes an unbiased estimate when placed over a parameter
α	significance level

The model is designed for tracking system reliability within a test phase and not across test phases. Accordingly, the basis of the model is described in the following way. Let the start of a test phase be initialized at time zero, and let $0 = t_0 < t_1 < t_2 < \ldots < t_k$ denote the cumulative test times on the system when design modifications are made. Assume the system failure rate is constant between successive t_i's, and let λ_i denote the constant failure rate during the *i*-th time interval $[t_{i-1}, t_i)$. The time intervals do not have to be equal in length. Based on the constant failure rate assumption, the number of failures F_i during the *i*-th time interval is Poisson distributed with mean $\theta_i = \lambda_i (t_i - t_{i-1})$. That is,

$$\text{Prob}(F_i = f) = \frac{(\theta_i)^f e^{-\theta_i}}{f!} \qquad (f = 0,1,2,\ldots) \qquad 6.3\text{-}1$$

During developmental testing programs, if more than one system prototype is tested and if the prototypes have the same basic configuration between modifications, then under the constant failure rate assumption, the following are true:

a. the time t_i may be considered as the cumulative test time to the *i*-th modification, and;

b. F_i may be considered as the cumulative total number of failures experienced by all system prototypes during the *i*-th time interval $[t_{i-1}, t_i)$.

The previous discussion is summarized graphically:

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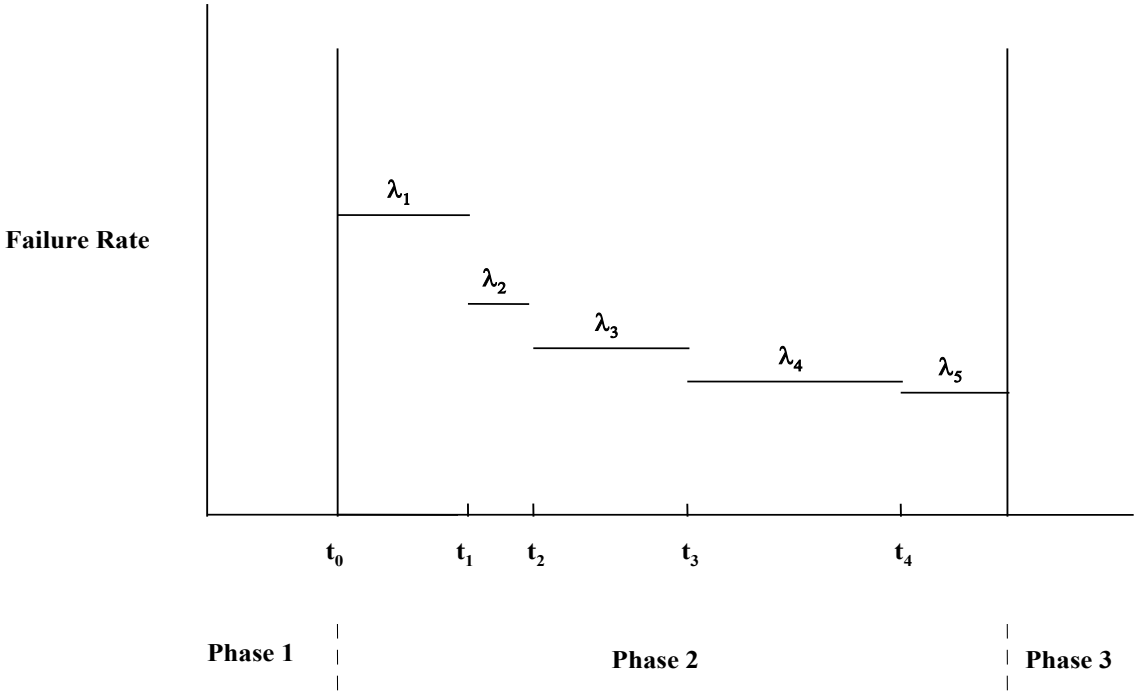


FIGURE 6-6. *Failure Rates Between Modifications*

Let *t* denote the cumulative test time, and let *F(t)* be the total number of system failures by time *t*. If t is in the first time interval:

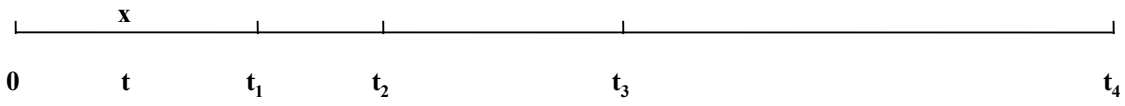


FIGURE 6-7. *Timeline for Phase 2 (t in first time interval)*

then *F(t)* has the Poisson distribution with mean $\lambda_1 t$. Now if *t* is in the second time interval:

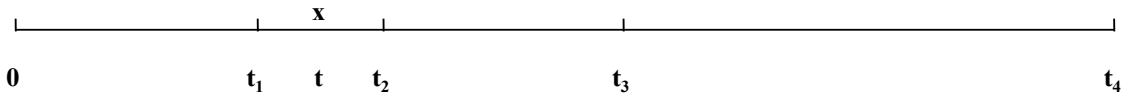


FIGURE 6-8. *Timeline for Phase 2 (t in second time interval)*

then *F(t)* is the number of system failures F_1 in the first time interval plus the number of system failures in the second time interval between t_1 and *t*. The failure rate for the first time interval is λ_1, and the failure rate for the second time interval is λ_2. Therefore, the mean of *F(t)* is the sum of the mean of $F_1 = \lambda_1 t_1$ plus the mean number of failures from t_1 to *t*, which is $\lambda_2(t - t_1)$. That is, *F(t)* has mean $\theta(t) = \lambda_1 t_1 + \lambda_2(t - t_1)$.

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When the failure rate is constant (homogeneous) over a test interval, then *F(t)* is said to follow a homogeneous Poisson process with mean number of failures of the form λt. When the failure rates change with time, e.g., from interval 1 to interval 2, then under certain conditions, *F(t)* is said to follow a non-homogeneous Poisson process (NHPP). In the presence of reliability growth, *F(t*) would follow a NHPP with mean value function:

$$\theta(t) = \int_o^t \rho(y)dy \qquad 6.3\text{-}2$$

Where $(\gamma) = \lambda_i, \gamma \in [t_{i-1}, t_i)$. From 6.3-2, for any $t > 0$,

$$\text{Prob}[F(t) = f] = \frac{[\theta(t)]^f e^{-\theta(t)}}{f!} \qquad (f = 0,1,2,...) \qquad 6.3\text{-}3$$

The integer-valued process $\{F(t), t>0\}$ may be regarded as a NHPP with intensity function $\rho(t)$. The physical interpretation of $\rho(t)$ is that for infinitesimally small Δt, $\rho(t)\,\Delta t$ is approximately the probability of a system failure in the time interval $(t, t+\Delta t)$; that is, it is approximately the instantaneous system failure rate. If $\rho(t) = \lambda$, a constant failure rate for all t, then a system is experiencing no growth over time, corresponding to the exponential case. If $\rho(t)$ is decreasing with time, $(\lambda_1 > \lambda_2 > \lambda_3 ...)$, then a system is experiencing reliability growth. Finally, $\rho(t)$ increasing over time indicates deterioration in system reliability.

Based on the learning curve approach, which is outlined in detail in the section on the AMSAA/Crow Discrete Reliability Growth Tracking Model, the AMSAA/Crow Continuous Reliability Growth Tracking Model assumes that $\rho(t)$ may be approximated by a continuous, parametric function. Using a result established for the Discrete Model:

$$E[F(t)] = \lambda t^{\beta} \qquad 6.3\text{-}4$$

and the instantaneous system failure rate $\rho(t)$ is the change per unit time of E[F(t)]:

$$\rho(t) = \frac{d}{dt} E[F(t)] = \lambda\beta t^{\beta-1} \qquad (\lambda, \beta, t > 0) \qquad 6.3\text{-}5$$

With a failure rate $\rho(t)$ that may change with test time, the NHPP provides a basis for describing the reliability growth process within a test phase.

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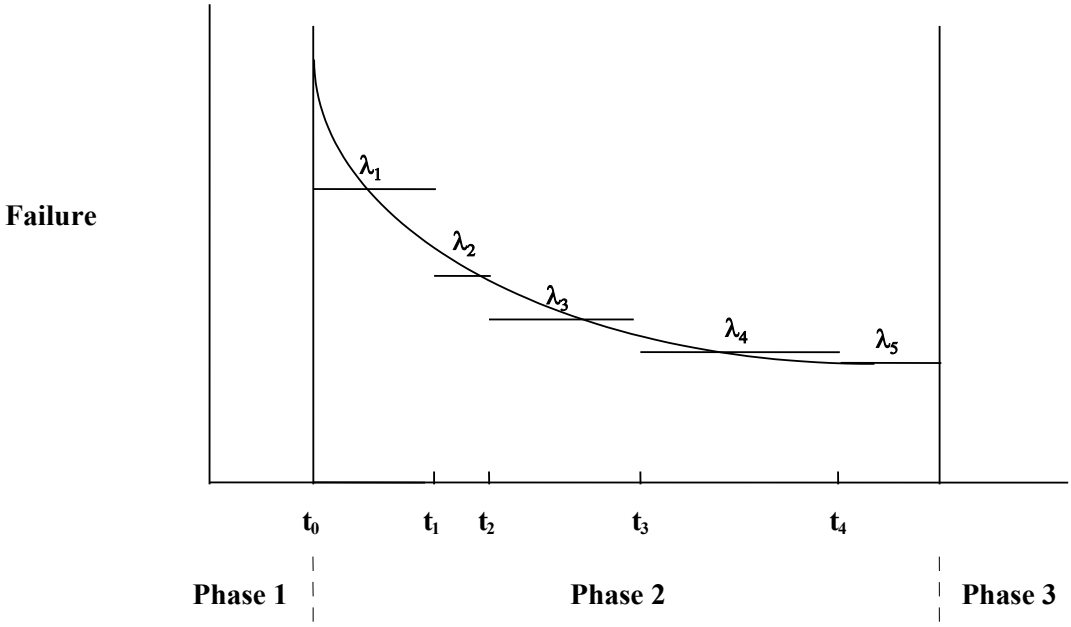


FIGURE 6-9. *Parametric Approximation to Failure Rates Between Modifications*

6.3.1.2 Methodology.
The AMSAA Continuous Reliability Growth Tracking Model assumes that within a test phase failures are occurring according to a non-homogeneous Poisson process with failure rate (intensity of failures) represented by the parametric function:

$$\rho(t) = \lambda\beta t^{\beta-1} \qquad (\lambda, \beta, t > 0) \qquad 6.3\text{-}6$$

where the parameter λ is referred to as the scale parameter because it depends upon the unit of measurement chosen for t, the parameter β is referred to as the growth or shape parameter because it characterizes the shape of the graph of the intensity function (Equation (6.3.-6) and FIGURE 6-9), and t is the cumulative test time. Under this model the function:

$$m(t) = \frac{1}{\rho(t)} = \left(\lambda\beta t^{\beta-1}\right)^{-1} \qquad 6.3\text{-}7$$

is interpreted as the instantaneous mean time between failures (MTBF) of the system at time t. When t corresponds to the total cumulative time for the system; that is, $t=T$, then $m(T)$ is the demonstrated MTBF of the system in its present configuration at the end of test.

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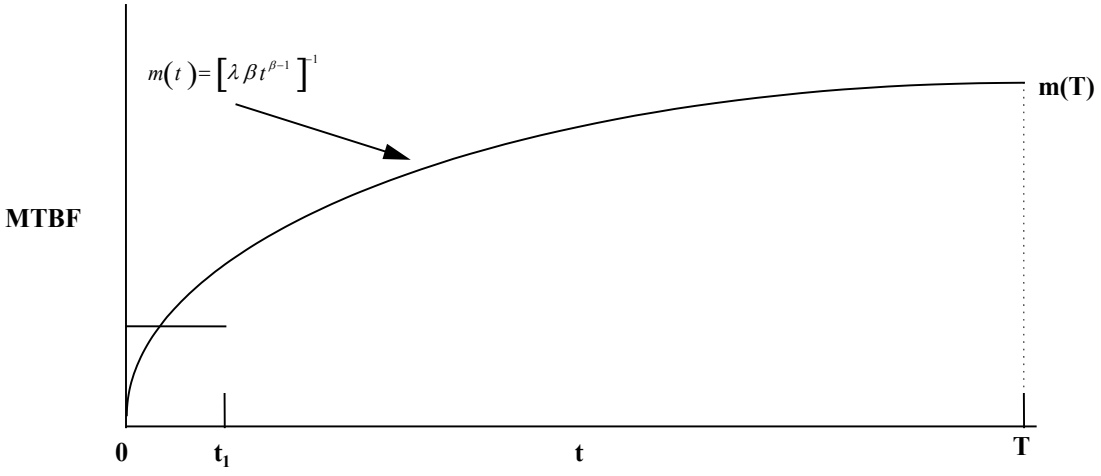


FIGURE 6-10. *Test Phase Reliability Growth based on AMSAA/Crow Continuous Tracking Model*

Note that the theoretical curve is undefined at the origin. Typically the MTBF during the initial test interval $[0,t_1]$ is characterized by a constant reliability with growth occurring beyond t_1.

6.3.1.3 Cumulative Number of Failures.
The total number of failures *F(t)* accumulated on all test items in cumulative test time *t* is a Poisson random variable, and the probability that exactly *f* failures occur between the initiation of testing and the cumulative test time *t* is:

$$\text{Prob}[\text{F(t)} = \text{f}] \quad = \quad \frac{[\theta\,(\text{t})]^{\text{f}}\ \text{e}^{-\theta(t)}}{\text{f!}} \qquad 6.3\text{-}8$$

in which $\theta(t)$ is the mean value function; that is, the expected number of failures expressed as a function of test time. To describe the reliability growth process, the cumulative number of failures is a function of the form $\theta(t)=\lambda t^{\beta}$, where λ and β are positive parameters.

6.3.1.4 Number of Failures in an Interval.
The number of failures occurring in the interval from test time t_1 until test time t_2, where $t_2 > t_1$ is a Poisson random variable with mean:

$$\theta(t_2)-\theta(t_1) \quad = \quad \lambda\left(t_2^{\beta}-t_1^{\beta}\right) \qquad 6.3\text{-}9$$

According to the model assumption, the number of failures that occur in any time interval is statistically independent of the number of failures that occur in any interval which does not overlap the first interval, and only one failure can occur at any instance of time.

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6.3.1.5 Intensity Function.
The intensity function in equation (6.3-6) is sometimes referred to as a failure rate; it is not the failure rate of a life distribution, rather it is the failure rate of a process, namely a NHPP.

6.3.1.6 Estimation Procedures for Individual Failure Time Data Model.
Modeling reliability growth as a non-homogeneous Poisson process permits an assessment of the demonstrated reliability by statistical procedures. The method of maximum likelihood provides estimates for the scale parameter λ and the shape parameter β, which are used in the estimation of the intensity function $\rho(t)$ in (6.3-6). In accordance with (6.3-7), the reciprocal of the current value of the intensity function is the instantaneous mean time between failures (MTBF) for the system. Procedures for point estimation and interval estimation for the system MTBF are described in more detail. A goodness-of-fit test to determine model suitability is also described.

The procedures outlined in this section are used to analyze data for which (a) the exact times of failure are known and (b) testing is conducted on a time terminated basis or the tests are in progress with data available through some time. The required data consist of the cumulative test time on all systems at the occurrence of each failure as well as the accumulated total test time *T*. To calculate the cumulative test time of a failure occurrence, it is necessary to sum the test time on every system at the point of failure. The data then consist of the *F* successive failure times $X_1 < X_2 < X_3 <\ldots< X_F$ that occur prior to *T*. This case is referred to as the Option for Individual Failure Time Data.

6.3.1.6.1 Point Estimation.
The method of maximum likelihood provides point estimates for the parameters of the failure intensity function (6.3.-6). The maximum likelihood estimate (MLE) for the shape parameter β is:

$$\hat{\beta} = \frac{F}{F \ln T - \sum_{i=1}^{F} \ln X_i} \qquad 6.3\text{-}10$$

By equating the observed number of failures by time *T* (namely *F*) with the expected number of failures by time *T* (namely *E[F(T)]*) and by substituting MLE's in place of the true, but unknown, parameters in (6.3-10) we obtain:

$$\mathrm{F} = \hat{\lambda}\,\mathrm{T}^{\hat{\beta}} \qquad 6.3\text{-}11$$

from which we obtain an estimate for the scale parameter λ:

$$\hat{\lambda} = \frac{\mathrm{F}}{\mathrm{T}^{\hat{\beta}}} \qquad 6.3\text{-}12$$

For any time $t > 0$, the failure intensity function is estimated by:

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$$\hat{\rho}(t) = \hat{\lambda}\hat{\beta}\, t^{\hat{\beta}-1} \qquad 6.3\text{-}13$$

In particular, (6.3-13) holds for the total test time *T*. By substitution from (6.3-11), the estimator can be written as:

$$\hat{\rho}(T) = \hat{\lambda}\,\hat{\beta}\, T^{\hat{\beta}-1} = \hat{\beta}\left(\frac{\hat{\lambda}\, T^{\hat{\beta}}}{T}\right) = \hat{\beta}\left(\frac{F}{T}\right) \qquad 6.3\text{-}14$$

where *F/T* is the estimate of the intensity function for a homogeneous Poisson process. Hence the fraction $(1-\hat{\beta})$ of the initial failure intensity is effectively removed by time *T,* resulting in (6.3-14).

Finally, the reciprocal of $\hat{\rho}(T)$ provides an estimate of the mean time between failures of the system at the time T and represents the system reliability growth under the model:

$$\hat{m}(T) = \frac{1}{\hat{\rho}(T)} = \left(\hat{\lambda}\hat{\beta}\, T^{\hat{\beta}-1}\right)^{-1} \qquad 6.3\text{-}15$$

6.3.1.6.2 Interval Estimation.
Interval estimates provide a measure of the uncertainty regarding a parameter. For the reliability growth process, the parameter of primary interest is the system mean time between failures at the end of test, *m(T).* The probability distribution of the point estimate for the intensity function at *T,* $\hat{\rho}(T)$, is the basis for the interval estimate for the true (but unknown) value of the intensity function at *T,* $\rho(T)$.

These interval estimates are referred to as confidence intervals and may be computed for selected confidence levels. The values in TABLE VI facilitate computation of two-sided confidence intervals for *m(T)* by providing confidence coefficients L and U corresponding to the lower bound and upper bound, respectively. These coefficients are indexed by the total number of observed failures F and the desired confidence level γ. The two-sided confidence interval for *m(T)* is thus:

$$L_{F,\gamma}\,\hat{m}(T) \le m(T) \le U_{F,\gamma}\,\hat{m}(T) \qquad 6.3\text{-}16$$

TABLE VIII may be used to compute one-sided interval estimates (lower confidence bounds) for *m(T)* such that:

$$L_{F,\gamma}\,\hat{m}(T) \le m(T) \qquad 6.3\text{-}17$$

Note that both tables are to be used only for time terminated growth tests. Also, since the number of failures has a discrete probability distribution, the interval estimates in (6.3-16) and (6.3-17) are conservative; that is, the actual confidence level is slightly larger than the desired confidence level γ.

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TABLE VI. Lower (L) And Upper (U) Coefficients for Confidence Intervals for MTBF from Time Terminated Reliability Growth Test

γ	0.8		0.9		0.95		0.98	
F	L	U	L	U	L	U	L	U
2	0.261	18.660	0.200	38.660	0.159	78.660	0.124	198.700
3	0.333	6.326	0.263	9.736	0.217	14.550	0.174	24.100
4	0.385	4.243	0.312	5.947	0.262	8.093	0.215	11.810
5	0.426	3.386	0.352	4.517	0.300	5.862	0.250	8.043
6	0.459	2.915	0.385	3.764	0.331	4.738	0.280	6.254
7	0.487	2.616	0.412	3.298	0.358	4.061	0.305	5.216
8	0.511	2.407	0.436	2.981	0.382	3.609	0.328	4.539
9	0.531	2.254	0.457	2.750	0.403	3.285	0.349	4.064
10	0.549	2.136	0.476	2.575	0.421	3.042	0.367	3.712
11	0.565	2.041	0.492	2.436	0.438	2.852	0.384	3.441
12	0.579	1.965	0.507	2.324	0.453	2.699	0.399	3.226
13	0.592	1.901	0.521	2.232	0.467	2.574	0.413	3.050
14	0.604	1.846	0.533	2.153	0.480	2.469	0.426	2.904
15	0.614	1.800	0.545	2.087	0.492	2.379	0.438	2.781
16	0.624	1.759	0.556	2.029	0.503	2.302	0.449	2.675
17	0.633	1.723	0.565	1.978	0.513	2.235	0.460	2.584
18	0.642	1.692	0.575	1.933	0.523	2.176	0.470	2.503
19	0.650	1.663	0.583	1.893	0.532	2.123	0.479	2.432
20	0.657	1.638	0.591	1.858	0.540	2.076	0.488	2.369
21	0.664	1.615	0.599	1.825	0.548	2.034	0.496	2.313
22	0.670	1.594	0.606	1.796	0.556	1.996	0.504	2.261
23	0.676	1.574	0.613	1.769	0.563	1.961	0.511	2.215
24	0.682	1.557	0.619	1.745	0.570	1.929	0.518	2.173
25	0.687	1.540	0.625	1.722	0.576	1.900	0.525	2.134
26	0.692	1.525	0.631	1.701	0.582	1.873	0.531	2.098
27	0.697	1.511	0.636	1.682	0.588	1.848	0.537	2.068
28	0.702	1.498	0.641	1.664	0.594	1.825	0.543	2.035
29	0.706	1.486	0.646	1.647	0.599	1.803	0.549	2.006
30	0.711	1.475	0.651	1.631	0.604	1.783	0.554	1.980
35	0.729	1.427	0.672	1.565	0.627	1.699	0.579	1.870
40	0.745	1.390	0.690	1.515	0.646	1.635	0.599	1.788
45	0.758	1.361	0.705	1.476	0.662	1.585	0.617	1.723
50	0.769	1.337	0.718	1.443	0.676	1.544	0.632	1.671
60	0.787	1.300	0.739	1.393	0.700	1.481	0.657	1.591
70	0.801	1.272	0.756	1.356	0.718	1.435	0.678	1.533
80	0.813	1.251	0.769	1.328	0.734	1.399	0.695	1.488
100	0.831	1.219	0.791	1.286	0.758	1.347	0.722	1.423

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For $F > 100$, $L \approx \left(1 + z_{.5+\gamma/2} \Big/ \sqrt{2F}\right)^{-2}$ *and* $U \approx \left(1 - z_{.5+\gamma/2} \Big/ \sqrt{2F}\right)^{-2}$

in which $z_{.5+\gamma/2}$ is the $100 \times \left(.5 + \gamma/2\right) - th$ percentile of the standard normal distribution.

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TABLE VII. Lower Confidence Interval Coefficients for MTBF From Time Terminated Reliability Growth Test

TIME TERMINATED RELIABILITY GROWTH TEST															
	Confidence Level γ								Confidence Level γ						
F	.50	.60	.70	.80	.90	.95	.99	F	.50	.60	.70	.80	.90	.95	.99
2	.761	.606	.480	.369	.261	.200	.124	51	.987	.939	.891	.838	.771	.720	
3	.823	.680	.559	.447	.333	.263	.174	52	.987	.940	.892	.840	.773	.722	.637
4	.860	.727	.611	.501	.385	.312	.215	53	.988	.941	.893	.841	.775	.724	.640
5	.884	.760	.649	.542	.426	.352	.250	54	.988	.941	.894	.843	.777	.727	.643
6	.901	.784	.678	.574	.459	.385	.280	55	.988	.942	.895	.844	.778	.729	.645
7	.914	.803	.701	.600	.487	.412	.305	56	.988	.942	.896	.845	.780	.731	.648
8	.924	.818	.720	.622	.511	.436	.328	57	.988	.943	.897	.847	.782	.733	.650
9	.932	.830	.736	.640	.531	.457	.349	58	.989	.944	.898	.848	.784	.735	.653
10	.938	.841	.749	.656	.549	.476	.367	59	.989	.944	.899	.849	.785	.737	.655
11	.943	.849	.761	.670	.565	.492	.384	60	.989	.945	.900	.850	.787	.739	.657
12	.948	.857	.771	.683	.579	.507	.399	61	.989	.945	.901	.852	.788	.741	.659
13	.951	.864	.780	.694	.592	.521	.413	62	.989	.946	.901	.853	.790	.742	.662
14	.955	.870	.788	.704	.604	.533	.426	63	.990	.946	.902	.854	.792	.744	.664
15	.958	.875	.795	.713	.614	.545	.438	64	.990	.947	.903	.855	.793	.746	.666
16	.960	.880	.802	.721	.624	.556	.449	65	.990	.947	.904	.856	.794	.748	.668
17	.962	.884	.808	.729	.633	.565	.460	66	.990	.948	.905	.857	.796	.749	.670
18	.964	.888	.814	.736	.642	.575	.470	67	.990	.948	.905	.858	.797	.751	.672
19	.966	.891	.819	.742	.650	.583	.479	68	.990	.948	.906	.859	.799	.752	.674
20	.968	.895	.823	.748	.657	.591	.488	69	.990	.949	.907	.860	.800	.754	.676
21	.969	.898	.828	.754	.664	.599	.496	70	.991	.949	.907	.861	.801	.756	.678
22	.971	.900	.832	.759	.670	.606	.504	71	.991	.950	.908	.862	.803	.757	.680
23	.972	.903	.836	.764	.676	.613	.511	72	.991	.950	.909	.863	.804	.759	.681
24	.973	.905	.839	.769	.682	.619	.518	73	.991	.951	.909	.864	.805	.760	.683
25	.974	.908	.842	.773	.687	.625	.525	74	.991	.951	.910	.865	.806	.761	.685
26	.975	.910	.846	.777	.692	.631	.531	75	.991	.951	.911	.866	.807	.763	.687
27	.976	.912	.849	.781	.697	.636	.537	76	.991	.952	.911	.866	.809	.764	.688
28	.977	.914	.851	.785	.702	.641	.543	77	.991	.952	.912	.867	.810	.766	.690
29	.978	.915	.854	.788	.706	.646	.549	78	.992	.952	.912	.868	.811	.767	.692
30	.978	.917	.857	.792	.711	.651	.554	79	.992	.953	.913	.869	.812	.768	.693
31	.979	.919	.859	.795	.715	.656	.560	80	.992	.953	.914	.870	.813	.769	.695
32	.980	.920	.861	.798	.719	.660	.565	81	.992	.953	.914	.871	.814	.771	.696
33	.980	.922	.863	.801	.722	.664	.570	82	.992	.954	.915	.871	.815	.772	.698
34	.981	.923	.866	.804	.726	.668	.574	83	.992	.954	.915	.872	.816	.773	.699
35	.981	.924	.868	.806	.729	.672	.579	84	.992	.854	.916	.873	.817	.774	.701
36	.982	.926	.869	.809	.733	.676	.583	85	.992	.955	.916	.874	.818	.776	.702
37	.982	.927	.871	.811	.736	.680	.587	86	.992	.955	.917	.874	.819	.777	.704
38	.983	.928	.873	.814	.739	.683	.591	87	.992	.955	.917	.875	.820	.778	.705
39	.983	.929	.875	.816	.742	.687	.595	88	.992	.956	.918	.876	.821	.779	.707
40	.984	.930	.876	.818	.745	.690	.599	89	.993	.956	.918	.876	.822	.780	.708
41	.984	.931	.878	.820	.747	.693	.603	90	.993	.956	.919	.877	.823	.781	.709
42	.984	.932	.879	.822	.750	.696	.606	91	.993	.956	.919	.878	.824	.782	.711
43	.985	.933	.881	.824	.753	.699	.610	92	.993	.957	.920	.878	.825	.783	.712
44	.985	.934	.882	.826	.755	.702	.613	93	.993	.957	.920	.879	.826	.784	.713
45	.985	.935	.884	.828	.758	.705	.617	94	.993	.957	.920	.880	.826	.785	.714
46	.986	.935	.885	.830	.760	.707	.620	95	.993	.957	.921	.880	.827	.786	.716
47	.986	.936	.886	.832	.762	.710	.623	96	.993	.958	.921	.881	.828	.787	.717
48	.986	.937	.888	.833	.764	.713	.626	97	.993	.958	.922	.881	.829	.788	.718
49	.987	.938	.889	.835	.767	.715	.629	98	.993	.958	.922	.882	.830	.789	.719
50	.987	.939	.890	.837	.769	.718	.632	99	.993	.958	.923	.883	.831	.790	.721

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6.3.1.6.3 Goodness-of-Fit (GOF).
For the case where the individual failure times are known, a Cramér-von Mises statistic is used to test the null hypothesis that a non-homogeneous Poisson process with failure intensity function (6.3-6) properly describes the reliability growth of a system. To calculate the statistic, an unbiased estimate of the shape parameter β is used:

$$\bar{\beta} = \frac{F-1}{F}\hat{\beta} \qquad 6.3\text{-}18$$

This unbiased estimate of β is for a time terminated reliability growth test with F observed failures. The goodness-of-fit statistic is:

$$C_F = \frac{1}{12F} + \sum_{i=1}^{F}\left[\left(\frac{X_i}{T}\right)^{\bar{\beta}} - \frac{2i-1}{2F}\right]^2 \qquad 6.3\text{-}19$$

where the failure times X_i must be ordered so that $0 < X_1 \le X_2 \le \ldots < X_F$. The null hypothesis that the model represents the observed data is rejected if the statistic C_F exceeds the critical value for a chosen significance level α. Critical values of C_F for $\alpha = .20, .15, .10, .05, .01$ are shown in TABLE VIII where the table is indexed by F, the total number of observed failures.

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TABLE VIII. Critical Values for Cramer-Von Mises Goodness-Of-Fit Test

For Individual Failure Time Data

F \ α	.20	.15	.10	.05	.01
2	.138	.149	.162	.175	.186
3	.121	.135	.154	.184	.23
4	.121	.134	.155	.191	.28
5	.121	.137	.160	.199	.30
6	.123	.139	.162	.204	.31
7	.124	.140	.165	.208	.32
8	.124	.141	.165	.210	.32
9	.125	.142	.167	.212	.32
10	.125	.142	.167	.212	.32
11	.126	.143	.169	.214	.32
12	.126	.144	.169	.214	.32
13	.126	.144	.169	.214	.33
14	.126	.144	.169	.214	.33
15	.126	.144	.169	.215	.33
16	.127	.145	.171	.216	.33
17	.127	.145	.171	.217	.33
18	.127	.146	.171	.217	.33
19	.127	.146	.171	.217	.33
20	.128	.146	.172	.217	.33
30	.128	.146	.172	.218	.33
60	.128	.147	.173	.220	.33
100	.129	.147	.173	.220	.34

For F > 100 use values for F = 100; α = significance level.

Besides using statistical methods for assessing model goodness-of-fit, one should also construct an average failure rate plot or a superimposed expected failure rate plot (as shown in FIGURE 6-11). These plots, derived from the failure data, provide a graphic description of test results and should always be part of the reliability analysis.

6.3.1.6.4 Example.

The following example demonstrates the option for individual failure time data in which two prototypes of a system are tested concurrently with the incorporation of design changes. (The data in this example are used subsequently for one of the growth subsystems in the example for the Subsystem Tracking Model - SSTRACK.) The first prototype is tested for 132.4 hours, and the second is tested for 167.6 hours for a total of T = 300 cumulative test hours. TABLE V shows the time on each prototype and the cumulative test time at each failure occurrence. An

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asterisk denotes the failed system. There are a total of F = 27 failures. Although the occurrence of two failures at exactly 16.5 hours is not possible under the assumption of the model, such data can result from rounding and are computationally tractable using the statistical estimation procedures described previously for the model. Note that the data are from a time terminated test.

TABLE IX. Test Data for Individual Failure Time Option

(An asterisk denotes the failed system.)

Failure Number	Prot. #1 Hours	Prot. #2 Hours	Cum Hours	Failure Number	Prot. #1 Hours	Prot. #2 Hours	Cum Hours
1	2.6*	.0	2.6	15	60.5	37.6*	98.1
2	16.5*	.0	16.5	16	61.9*	39.1	101.1
3	16.5*	.0	16.5	17	76.6*	55.4	132.0
4	17.0*	.0	17.0	18	81.1	61.1*	142.2
5	20.5	.9*	21.4	19	84.1*	63.6	147.7
6	25.3	3.8*	29.1	20	84.7*	64.3	149.0
7	28.7	4.6*	33.3	21	94.6*	72.6	167.2
8	41.8*	14.7	56.5	22	104.8	85.9*	190.7
9	45.5*	17.6	63.1	23	105.9	87.1*	193.0
10	48.6	22.0*	70.6	24	108.8*	89.9	198.7
11	49.6	23.4*	73.0	25	132.4	119.5*	251.9
12	51.4*	26.3	77.7	26	132.4	150.1*	282.5
13	58.2*	35.7	93.9	27	132.4	153.7*	286.1
14	59.0	36.5*	95.5	End	132.4	167.6	300.0

By using the 27 failure times listed under the columns labeled “Cumulative Hours” in TABLE IX and by applying equations (6.3.-10), (6.3.-12), (6.3.-13) and (6.3.-15), we obtain the following estimates. The point estimate for the shape parameter is $\hat{\beta} = 0.716$; the point estimate for the scale parameter is $\hat{\lambda} = 0.454$; the estimated failure intensity at the end of the test is $\hat{\rho}(T) = 0.0645$ failures per hour; the estimated MTBF at the end of the 300-hour test is $\hat{m}(T) = 15.5$ hours. As shown in FIGURE 6-11 superimposing a graph of the estimated intensity function (6.3-14) atop a plot of the average failure rate (using six 50-hour intervals) reveals a decreasing failure intensity indicative of reliability growth.

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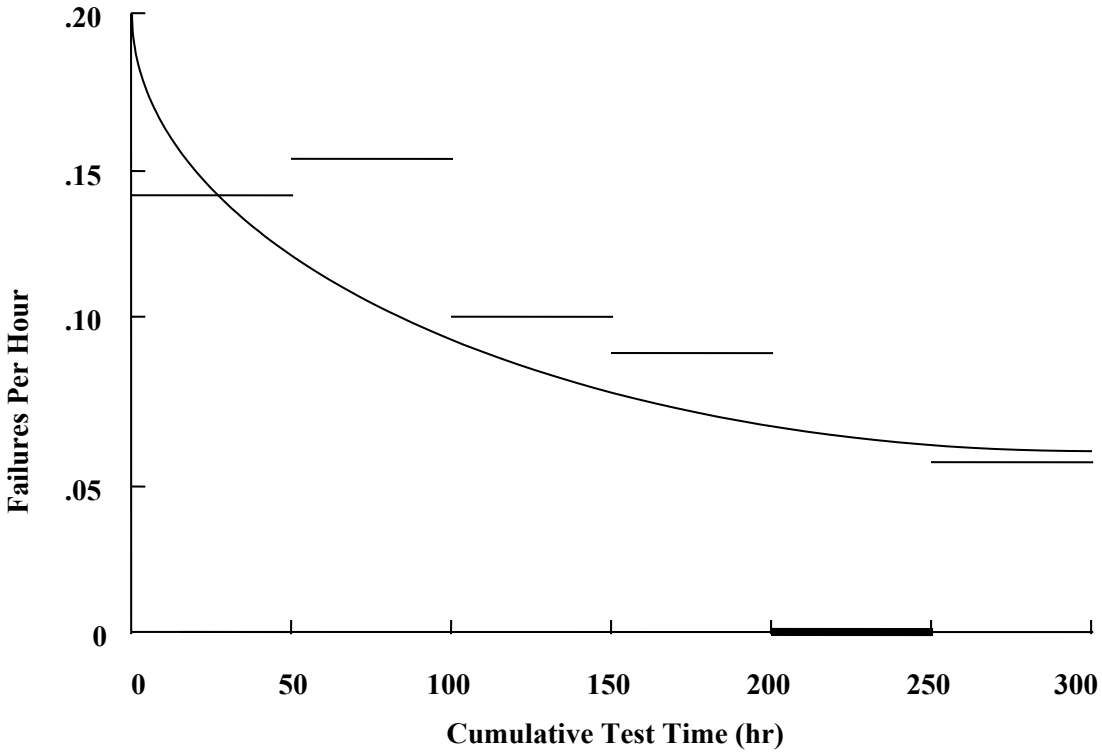


FIGURE 6-11. *Estimated Intensity Function*

Using (6.3-16), TABLE VII and a confidence level of 90 percent, the two-sided interval estimate for the MTBF at the end of the test is [9.9, 26.1]. These results and the estimated MTBF tracking growth curve (substituting t for T in (6.3-15)) are shown in Figure 6-11.

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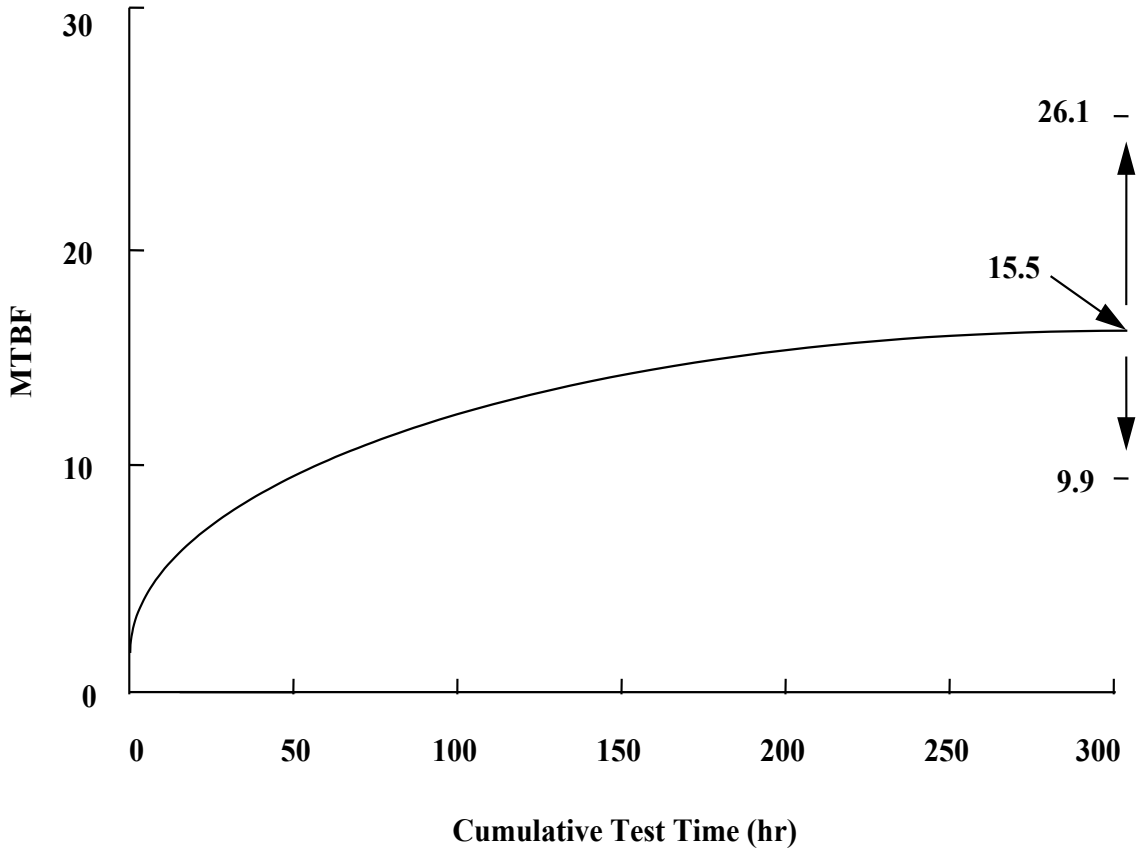


FIGURE 6-12. *Estimated MTBF Function with 90% Interval Estimate at T=300 Hours*

Finally, to test the model goodness-of-fit, a Cramér-von Mises statistic is compared to the critical value from TABLE VIII corresponding to a chosen significance level $\alpha = 0.05$ and total observed number of failures $F = 27$. Linear interpolation is used to arrive at the critical value. Since the statistic, 0.091, is less than the critical value, 0.218, we accept the hypothesis that the AMSAA/Crow Continuous Reliability Growth Tracking Model is appropriate for this data set.

6.3.1.7 Option for Grouped Data.

6.3.1.7.1 List of Notations.

K	number of intervals (or groups) or the last group
i	Interval number
t_i	time at beginning (or end) of interval
F_i	observed number of failures in interval $[t_{i-1}, t_i)$
t_K	total test time
^	denotes an estimate when placed over a parameter
β	shape parameter $(\beta > 0)$
λ	scale parameter $(\lambda > 0)$

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$\rho(t)$	instantaneous failure intensity at time *t*
$m(t)$	instantaneous MTBF at time *t*
M_K	MTBF for the last group
E_K	expected number of failures in the last group
ρ_K	failure intensity for the last group
F	total observed number of failures
L	lower confidence coefficient
U	upper confidence coefficient
γ	specified confidence level
E_i	expected number of failures in interval *i*
K_R	number of intervals after recombination of intervals
O_i	observed number of failures in interval *i*
χ^2	chi-squared value

Reliability growth parameters can be estimated in accordance with the AMSAA/Crow Continuous Tracking Model even if the exact times of failure are unknown and all that is known is the number of failures that occurred in each interval of time provided there are at least three intervals and at least two intervals have failures. This case is referred to as the Option for Grouped Data. This section describes the estimation procedures and goodness-of-fit procedures for analyzing such data and provides an example of model usage. In the following discussion, the words "group" and "interval" are interchangeable.

6.3.1.7.2 Point Estimation.
The required data consist of the total number of failures in each of K intervals of test time. The first interval always starts at test time zero so that $t_0 = 0$. The groups do not have to be of equal length. The observed number of failures in the interval from t_{i-1} to t_i is denoted by *F*.

The method of maximum likelihood provides point estimates for the parameters of the model. The maximum likelihood estimate for the shape parameter β is the value that satisfies the following nonlinear equation:

$$\sum_{i=1}^{K} F_i \left[\frac{t_i^{\hat{\beta}} \ln t_i - t_{i-1}^{\hat{\beta}} \ln t_{i-1}}{t_i^{\hat{\beta}} - t_{i-1}^{\hat{\beta}}} - \ln t_K \right] = 0 \qquad 6.3\text{-}20$$

in which $t_0 \ln t_0$ is defined as zero. By equating the total expected number of failures to the total observed number of failures:

$$\hat{\lambda}\, t_K^{\hat{\beta}} = \sum_{i=1}^{K} F_i \qquad 6.3\text{-}21$$

and solving for $\hat{\lambda}$, we obtain an estimate for the scale parameter:

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$$\hat{\lambda} = \frac{\sum_{i=1}^{K} F_i}{t_K^{\hat{\beta}}} \qquad 6.3\text{-}22$$

Point estimates for the intensity function $\rho(t)$ and the mean time between failures function $m(t)$ are calculated as in the previous section that describes the Option for Individual Failure Time Data; that is,

$$\hat{\rho}(t) = \hat{\lambda}\hat{\beta}\, t^{\hat{\beta}-1} \qquad \left(\hat{\lambda}, \hat{\beta}, t > 0\right) \qquad 6.3\text{-}23$$

$$\hat{m}(t) = [\hat{\rho}(t)]^{-1} \qquad \left(\hat{\lambda}, \hat{\beta}, t > 0\right) \qquad 6.3\text{-}24$$

The functions in (6.3-23) and (6.3-24) provide instantaneous estimates that give rise to smooth continuous curves, but these functions do not describe the reliability growth that occurs on a configuration basis representative of grouped data. Under the model option for grouped data, the estimate for the MTBF for the last group, $\hat{M}_K$, is the amount of test time in the last group divided by the estimated expected number of failures in the last group:

$$\hat{M}_K = \frac{t_K - t_{K-1}}{\hat{E}_K} \qquad 6.3\text{-}25$$

where the estimated expected number of failures in the last group $\hat{E}_K$ is:

$$\hat{E}_K = \hat{\lambda}\left(t_K^{\hat{\beta}} - t_{K-1}^{\hat{\beta}}\right) \qquad 6.3\text{-}26$$

From (6.3-25) we obtain an estimate for the failure intensity for the last group:

$$\hat{\rho}_K = \frac{1}{\hat{M}_K} \qquad 6.3\text{-}27$$

6.3.1.7.3 Interval Estimation.
Approximate lower confidence bounds and two-sided confidence intervals may be computed for the MTBF for the last group. Using (6.3.25) and TABLE II, a two-sided approximate confidence interval for M_K may be calculated from:

$$L_{F,\gamma}\,\hat{M}_K \leq M_K \leq U_{F,\gamma}\,\hat{M}_K \qquad 6.3\text{-}28$$

and using (6.3-25) and TABLE III, a one-sided approximate interval estimate for M_K may be calculated from:

$$L_{F,\gamma}\,\hat{M}_K \leq M_K \qquad 6.3\text{-}29$$

where F is the total observed number of failures and γ is the desired confidence level.

6.3.1.7.4 Goodness-of-Fit.
A chi-squared goodness-of-fit test is used to test the null hypothesis that the AMSAA/Crow Continuous Reliability Growth Tracking Model adequately represents a set of grouped data. The expected number of failures in the interval from t_{i-1} to t_i is approximated by:

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$$\hat{E}_i = \hat{\lambda}\left(t_i^{\hat{\beta}} - t_{i-1}^{\hat{\beta}}\right) \qquad 6.3\text{-}30$$

Adjacent intervals may have to be combined so that the estimated expected number of failures in any combined interval is at least five. Let the number of intervals after this recombination be K_R, and let the observed number of failures in the *i*-th new interval be O_i and the estimated expected number of failures in the *i*-th new interval be $\hat{E}_i$. Then the statistic:

$$\chi^2 = \sum_{i=1}^{K_R} \frac{\left(O_i - \hat{E}_i\right)^2}{\hat{E}_i} \qquad 6.3\text{-}31$$

is approximately distributed as a chi-squared random variable with $K_R - 2$ degrees of freedom. The null hypothesis is rejected if the χ^2 statistic exceeds the critical value for a chosen significance level. Critical values for this statistic can be found in tables of the chi-squared distribution.

Besides using statistical methods for assessing model goodness-of-fit, one should also construct an average failure rate plot or a superimposed expected failure rate plot (as shown in FIGURE 6-11). Derived from the failure data, these plots provide a graphic description of test results and should always be part of the reliability analysis.

6.3.1.7.5 Example.
The following example uses aircraft data to demonstrate the option for grouped data. (The data in this example are used subsequently for one of the growth subsystems in the example for the AMSAA/Crow Subsystem Tracking Model - SSTRACK.) In this example, an aircraft has scheduled inspections at intervals of twenty flight hours. For the first 100 hours of flight testing the results are:

TABLE X. Test Data for Grouped Option

Start Time	End Time	Observed Number of Failures
0	20	13
20	40	16
40	60	5
60	80	8
80	100	7

There are a total of $F = 49$ observed failures from $K = 5$ intervals. The solution of equation (6.3.-20) for $\hat{\beta}$ yields an estimate of 0.753 for the shape parameter. From (6.3.-22) the scale parameter estimate is 1.53. For the last group, the intensity function estimate is 0.379 failures per flight hour and the MTBF estimate is 2.6 flight hours. TABLE XI shows that those adjacent intervals do not have to be combined after applying (6.3-30) to the original intervals. Therefore, $K_R = 5$.

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TABLE XI. Observed Versus Expected Number of Failures

For Test Data for Grouped Data Option

Start Time	End Time	Observed Number of Failures	Estimated Expected Number of Failures
0	20	13	14.59
20	40	16	9.99
40	60	5	8.77
60	80	8	8.07
80	100	7	7.58

To test the model goodness-of-fit, a chi-squared statistic of 5.5 is compared to the critical value of 7.8 corresponding to 3 degrees of freedom and a 0.05 significance level. Since the statistic is less than the critical value, the applicability of the model is accepted.

6.3.2 Reliability Growth Tracking Model – Discrete (RGTMD).

6.3.2.1 Background.

The material in this section is as presented in [2]. Reliability growth tracking methodology may also be applied to discrete data in a manner that is consistent with the learning curve property observed by J.T. Duane for continuous data. Growth takes place on a configuration by configuration basis. Accordingly, this section describes model development and maximum likelihood estimation procedures for assessing system reliability for one-shot systems during development.

6.3.2.2 Basis for Model.

The motivation for the AMSAA/Crow Discrete version of the AMSAA/Crow Reliability Growth Tracking Model comes from the learning curve approach for continuous data.

6.3.2.3 List of Notations.

t	cumulative test time
K(t)	cumulative number of failures by time *t*
c(t)	cumulative failure rate by time *t*
ln	natural logarithm function (base e)
δ	constant term representing the y-intercept of a linear equation
α	constant term representing the slope of a linear equation
λ	scale parameter ($\lambda > 0$ $\lambda > 0$) of power function
β	shape parameter ($\beta > 0$) of power function; $\beta = 1 - \alpha$
i	configuration number
T_i	cumulative number of trials through configuration *i*
Σ	summation of
N_i	number of trials in configuration *i*

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K_i	cumulative number of failures through configuration *i*
M_i	number of failures in configuration *i*
$E[K_i]$	expected value of K_i
f_i	probability of failure for configuration *i*
g_i	probability of failure for trial *i*
R_i	reliability for configuration *i* (or trial *i*)
^	denotes an estimate when placed over a parameter

Let *t* denote the cumulative test time, and let *K(t)* denote the cumulative number of failures by time *t*. The cumulative failure rate, *c(t)*, is the ratio:

$$c(t) = \frac{K(t)}{t} \qquad 6.3\text{-}32$$

While plotting test data from generators, hydro-mechanical devices and aircraft jet engines, Duane observed that the logarithm of the cumulative failure rate was linear when plotted against the logarithm of the cumulative test time:

$$\ln c(t) = \delta - \alpha \ln t \qquad 6.3\text{-}33$$

By letting $\delta = \ln \lambda$ for the y-intercept and by exponentiating both sides of (6.3-33), the cumulative failure rate becomes:

$$c(t) = \lambda t^{-\alpha} \qquad 6.3\text{-}34$$

By substitution from (6.3-32),

$$\frac{K(t)}{t} = \lambda t^{-\alpha} \qquad 6.3\text{-}35$$

Multiplying both sides of (6.3-35) by *t* and letting $\beta = 1 - \alpha$, the cumulative number of failures by *t* becomes:

$$K(t) = \lambda t^{\beta} \qquad 6.3\text{-}36$$

This power function of *t* is the learning curve property for *K(t)*, where $\lambda, \beta > 0$.

6.3.2.4 Model Development.
To construct the AMSAA/Crow Discrete Reliability Growth Tracking Model, we use the power function developed from the learning curve property for *K(t)* to derive an equation for the probability of failure on a configuration basis. We refer to this situation where growth takes place on a configuration basis (and the number of trials in at least one of the configurations is greater than one) as the grouped data option. In the presence of reliability growth, the failure probability trend for the grouped data option appears graphically as a sequence of decreasing, horizontal steps.

We then note the special case where the configuration size is one for all configurations, develop an equation for the probability of failure, and refer to this special case as the option for trial by

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trial data. In a growth situation, the failure probability trend for this option is described graphically as a decreasing, smooth curve.

Model development proceeds as follows. Suppose system development is represented by *i* configurations. (This corresponds to $i-1$ configuration changes, unless fixes are applied at the end of the test phase, in which case there would be *i* configuration changes.) Let N_i be the number of trials during configuration *i*, and let M_i be the number of failures during configuration *i*. Then the cumulative number of trials through configuration *i*, namely T_i, is the sum of the N_i for all *i*:

$$T_i = \sum N_i \qquad 6.3\text{-}37$$

and the cumulative number of failures through configuration i, namely K_i, is the sum of the M_i for all *i*:

$$K_i = \sum M_i \qquad 6.3\text{-}38$$

We express the expected value of K_i as $E[K_i]$ and define it as the expected number of failures by the end of configuration *i*. Applying the learning curve property to $E[K_i]$ implies:

$$E[K_i] = \lambda T_i^{\beta} \qquad 6.3\text{-}39$$

We introduce a term for the probability of failure for configuration one, namely f_1, and use it to develop a generalized equation for f_i in terms of the T_i and N_i. From (6.3-39), the expected number of failures by the end of configuration one is:

$$E[K_1] = \lambda T_1^{\beta} = f_1 N_1 \Rightarrow f_1 = \frac{\lambda T_1^{\beta}}{N_1} \qquad 6.3\text{-}40$$

Applying (6.3-39) again and noting that the expected number of failures by the end of configuration two is the sum of the expected number of failures in configuration one and the expected number of failures in configuration two, we obtain:

$$E[K_2] = \lambda T_2^{\beta} = f_1 N_1 + f_2 N_2 = \lambda T_1^{\beta} + f_2 N_2 \Rightarrow f_2 = \frac{\lambda T_2^{\beta} - \lambda T_1^{\beta}}{N_2} \qquad 6.3\text{-}41$$

By this method of inductive reasoning we obtain a generalized equation for the failure probability, f_i, on a configuration basis:

$$f_i = \frac{\lambda T_i^{\beta} - \lambda T_{i-1}^{\beta}}{N_i} \qquad 6.3\text{-}42$$

and use (6.3-42) for the grouped data option.

For the special case where $N_i = 1$ for all *i*, (6.3-42) becomes a smooth curve, g_i, that represents the probability of failure for the option for trial by trial data:

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$$g_i = \lambda i^{\beta} - \lambda(i-1)^{\beta} \quad 6.3\text{-}43$$

In (6.3-43), *i* represents the trial number. Note that $T_0 = 0$, so that (6.3-42) reduces to (6.3-40) when $i = 1$. Also, for $i = 1$ in (6.3-43), $g_1 = \lambda$. Using (6.3-42) we obtain an equation for the reliability (probability of success) for the *i*-th configuration:

$$R_i = 1 - f_i \quad 6.3\text{-}44$$

and using (6.3-6.) we obtain an equation for the reliability for the *i*-th trial:

$$R_i = 1 - g_i \quad 6.3\text{-}45$$

Equations (6.3- 42), (6.3- 43), (6.3- 44) and (6.3- 45) are the exact model equations for tracking the reliability growth of discrete data using the AMSAA/Crow Discrete Reliability Growth Tracking Model.

6.3.2.5 Estimation Procedures.
This section describes procedures for estimating the parameters of the AMSAA/Crow Discrete Reliability Growth Tracking Model. It also includes an approximation equation for calculating reliability lower confidence bounds and an example illustrating these concepts.

The estimation procedures described below provide maximum likelihood estimates (MLE's) for the model's two parameters, λ and β, where λ is the scale parameter and β is the shape (or growth) parameter. The MLE's for λ and β allow for point estimates for the probability of failure:

$$\hat{f}_i = \frac{\hat{\lambda}T_i^{\hat{\beta}} - \hat{\lambda}T_{i-1}^{\hat{\beta}}}{N_i} = \frac{\hat{\lambda}\left(T_i^{\hat{\beta}} - T_{i-1}^{\hat{\beta}}\right)}{N_i} \quad 6.3\text{-}46$$

and the probability of success (reliability):

$$\hat{R}_i = 1 - \hat{f}_i \quad 6.3\text{-}47$$

for each configuration *i*.

6.3.2.6 Point Estimation.
Let $\hat{\lambda}$ and $\hat{\beta}$ be the MLE's for λ and β respectively, i.e. let $(\hat{\lambda}, \hat{\beta}) = (\lambda, \beta)$ such that (λ, β) maximizes the discrete model likelihood function over the region $0 \le R_i \le 1$ for $i = 1, \ldots, K$. Let $\hat{R}_i$ denote the corresponding estimate of R_i. If $0 < \hat{R}_i < 1$ for $i = 1, \ldots, K$ then the point $(\lambda, \beta) = (\hat{\lambda}, \hat{\beta})$ satisfies the following likelihood equations:

$$\sum_{i=1}^{K}\left[\lambda T_i^{\beta}\ln T_i - \lambda T_{i-1}^{\beta}\ln T_{i-1}\right]\left\{\frac{M_i}{\left[\lambda T_i^{\beta} - \lambda T_{i-1}^{\beta}\right]} - \frac{N_i - M_i}{\left[N_i - \lambda T_i^{\beta} + \lambda T_{i-1}^{\beta}\right]}\right\} = 0 \quad 6.3\text{-}48$$

and

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$$\sum_{i=1}^{K}\left[T_i^{\beta}-T_{i-1}^{\beta}\right]\left\{\frac{M_i}{\left[\lambda T_i^{\beta}-\lambda T_{i-1}^{\beta}\right]}-\frac{N_i-M_i}{\left[N_i-\lambda T_i^{\beta}+\lambda T_{i-1}^{\beta}\right]}\right\} \quad = \quad 0 \qquad 6.3\text{-}49$$

We recommend using the model MLE's only for this case. Situations can occur when the likelihood is maximized at a point $(\hat{\lambda},\hat{\beta})$ such that $\hat{R}_1 = 0$ and $(\hat{\lambda},\hat{\beta})$ does not satisfy Equations (6.3-48) and (6.3-49). One such case occurs for the trial-by-trial model when a failure occurs on the first trial. If one wishes to use the model in such an instance we suggest either (i) initializing the model so that at least the first trial is a success or (ii) using the grouped version and initializing with a group that contains at least one success. This should typically produce maximizing values $\hat{\lambda},\hat{\beta}$ that satisfy Equations (6.3-48) and (6.3-49) with $0<\hat{R}_i<1$ for $i = 1, \ldots K$. Procedure (*i*) is especially appropriate if performance problems associated with an early design cause the initial failure(s). Since the assessment of the achieved reliability will depend on the model initialization and groupings, the basis for the utilized data and groupings should be considered part of the assessment. A goodness-of-fit test should be used to explore whether the model provides a reasonable fit to the data and groupings. If there is insufficient failure data to perform such a test, a binomial point estimate and lower confidence bound based on the total number of successes and trials would provide a conservative assessment of the achieved reliability R_K under the assumption that $R_K \geq R_i$ for $i = 1, \ldots, K$.

From (6.3-48) and (6.3-49) we note the following data requirements for using the model:

K	number of configurations (or the final configuration)
M_i	number of observed failures for configuration *i*
N_i	number of trials for configuration *i*
T_i	cumulative number of trials through configuration *i*

6.3.2.7 Interval Estimation.
A one-sided interval estimate (lower confidence bound) for the reliability of the final (last) configuration may be obtained from the approximation equation:

$$LCB_{\gamma} \quad \approx \quad 1-\left(1-\hat{R}_K\right)\left(\frac{\chi^2_{\gamma,n+2}}{n}\right) \qquad 6.3\text{-}50$$

where

LCB_{γ}	an approximate lower confidence bound at the gamma (γ) confidence level for the reliability of the last configuration, where γ is a decimal number in the interval (0,1)
$\hat{R}_K$	a maximum likelihood estimate for the reliability of the last configuration
n	the total number of observed failures (summed) over all configurations *i, (I = 1..K)*
$\chi^2_{\gamma,n+2}$	the gamma percentile point of the chi-squared distribution with *n+2* degrees of freedom

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6.3.2.8 Goodness-of-Fit.
Provided there is sufficient data to obtain at least five expected number of failures per group, a chi-squared goodness-of-fit test may be used to test the null hypothesis that the AMSAA/Crow Discrete Reliability Growth Tracking Model adequately represents a set of grouped discrete data or a set of trial by trial data. If these conditions are met, then one may use the chi-squared goodness-of-fit procedures outlined previously for the Continuous Reliability Growth Tracking Model.

Besides using statistical methods for assessing model goodness-of-fit, one should also construct an average failure rate plot or a superimposed expected failure rate plot (as shown in FIGURE 6-11). Derived from the failure data, these plots provide a graphic description of test results and should always be part of the reliability analysis.

6.3.2.9 Example.
The following example is an application of the grouped data option of the AMSAA/Crow Discrete Reliability Growth Tracking Model for a system having four configurations of development test data:

TABLE XII. Test Data for Grouped Option

Configuration Number, i $K = 4$	Observed Number of Failures in Configuration i M_i	Number of Trials in Configuration i N_i	Cumulative Number of Trials Through Configuration i T_i
1	5	14	14
2	3	19	33
3	4	15	48
4	4	20	68

This is represented graphically as:

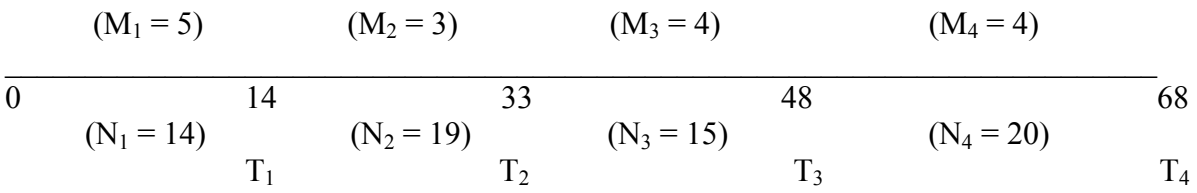


FIGURE 6-13. *Test Data for Grouped Data Option*

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The solution of (6.3-48) and (6.3-49) provides MLE's for λ and β corresponding to 0.595 and 0.780, respectively. Using (6.3-46) and (6.3-47) results in the following table:

TABLE XII. Estimated Failure Rate and Estimated Reliability By Configuration

Configuration Number, i $K = 4$	Estimated Failure Probability for Configuration i $\hat{f}_i$	Estimated Reliability for Configuration i $\hat{R}_i$
1	.333	.667
2	.234	.766
3	.206	.794
4	.190	.810

A plot of the estimated failure rate by configuration is:

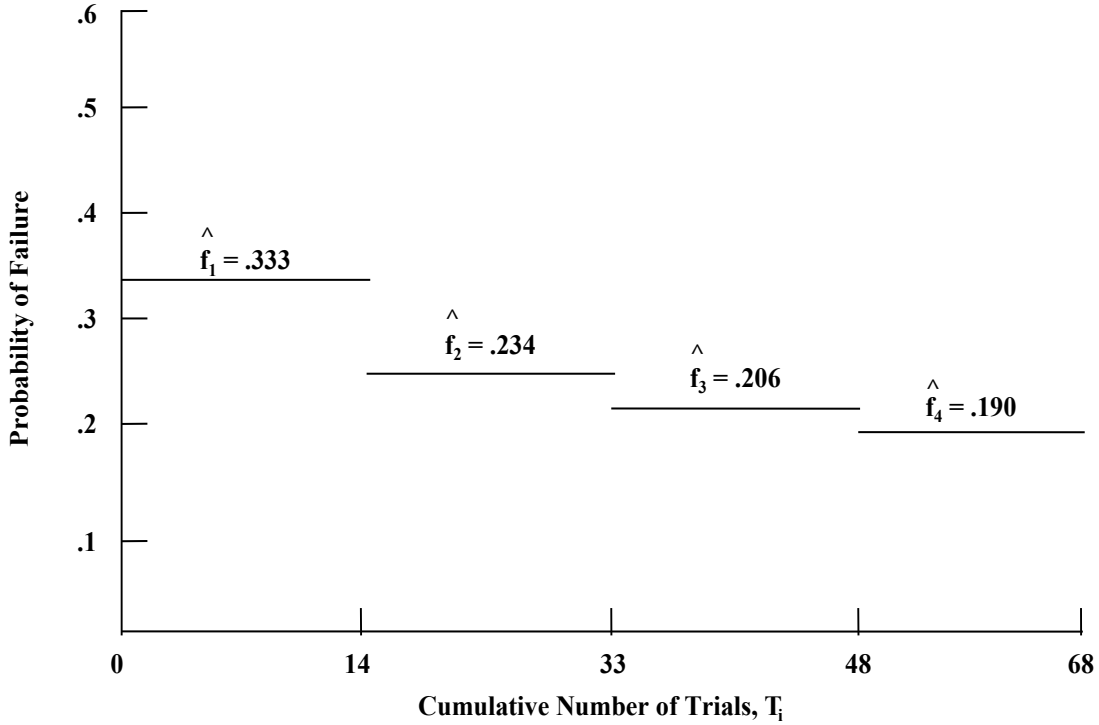


FIGURE 6-14. *Estimated Failure Rate by Configuration*

and a plot of the estimated reliability by configuration is:

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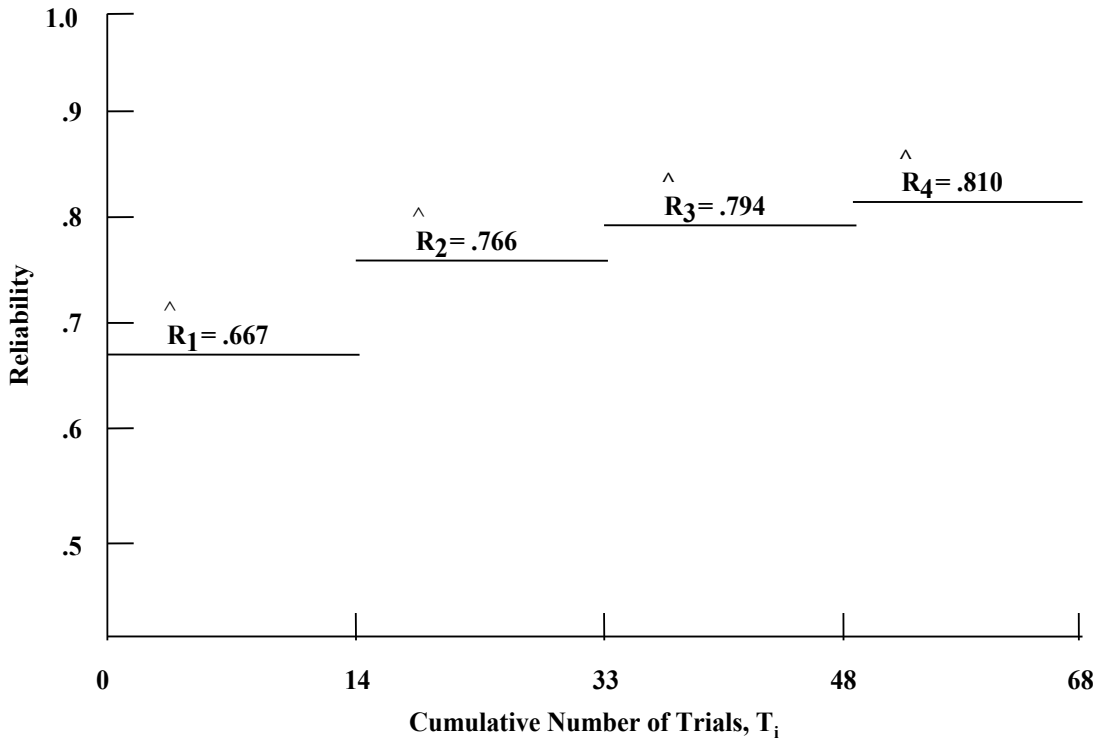


FIGURE 6-15. *Estimated Reliability by Configuration*

Finally, (6.3-50) is used to generate the following table (TABLE XIII of approximate LCB's for the reliability of the last configuration:

TABLE XIII. Table of Approximate Lower Confidence Bounds (LCB's) For Final Configuration

Confidence Level	LCB
0.50	0.806
0.75	0.783
0.80	0.777
0.90	0.761
0.95	0.747

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6.3.3 Subsystem Level Tracking Model (SSTRACK).

6.3.3.1 Background and Conditions for Usage.

The AMSAA Subsystem Tracking Model (SSTRACK) is a tool for assessing system level reliability from lower level test results. The methodology was developed to make greater use of component or subsystem test data in estimating system reliability. By representing the system as a series of independent subsystems, the methodology permits an assessment of the system level demonstrated reliability at a given confidence level from the subsystem test data. This system level assessment is permissible provided the:

a. Subsystem test conditions/usage are in conformance with the proposed system level operational environment (as embodied in the Operational Mode Summary/Mission Profile [OMS/MP]);
b. Failure Definitions/Scoring Criteria (FD/SC) formulated for each subsystem are consistent with the FD/SC used for system level test evaluation;
c. Subsystem configuration changes are well documented and;
d. High risk interfaces are identified and addressed through joint subsystem testing.

The SSTRACK methodology supports a mix of test data from growth and non-growth subsystems. Statistical goodness-of-fit procedures are used for assessing model applicability for growth subsystem test data. For non-growth subsystems, the model uses fixed configuration test data in the form of the total test time and the total number of failures. The model applies the Lindström-Madden method as specified in [3] for combining the test data from the individual subsystems. Twenty-five subsystems can be represented by the current implementation of the model. SSTRACK is a continuous model, but it may be used with discrete data if the number of trials is large and the probability of failure is small.

A potential benefit of this methodology is that it may allow for reduced system level testing by combining lower level subsystem test results in such a manner that system reliability may be demonstrated with confidence. Another potential benefit is that it may allow for an assessment of the degree of subsystem test contribution toward demonstrating a system reliability requirement. Finally, as mentioned, it may serve as an effective means of combining test data from dissimilar sources, namely growth and non-growth subsystems.

Besides the two provisos stated in the opening paragraph regarding OMS/MP conformance and FD/SC consistency, a caveat in using the methodology is that high-risk subsystem interfaces should be identified and addressed through joint subsystem testing. Also, as in any reliability growth test program, growth subsystem configuration changes must be properly documented for the methodology to provide meaningful results.

The primary output from the SSTRACK computer implementation is a table of approximate lower confidence bounds for the system reliability (MTBF) for a range of confidence levels.

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List of Notations.

$\wedge$	denotes an estimate when placed over a parameter
M	Mean Time Between Failures (MTBF)
D	demonstration
G	growth
LCB	Lower Confidence Bound
γ	confidence level
T	(total) test time
N	(total) number of failures
$\chi^2_{df,\gamma}$	chi-squared percentile point for df degrees of freedom and γ confidence
β	growth parameter from reliability growth tracking model

To be able to handle a mix of test data from growth and non-growth subsystems, the methodology converts all growth subsystem test data to its "equivalent" amount of demonstration test time and "equivalent" number of demonstration failures so that all subsystem results are expressed in a common format; namely, in terms of fixed configuration (non-growth) test data. By treating growth subsystem test data in this way, a standard lower confidence bound formula for fixed configuration test data may be used to compute an approximate system reliability lower confidence bound for the combination of growth and non-growth data. The net effect of this conversion process is that it reduces all growth subsystem test data to "equivalent" demonstration test data while preserving the following two important equivalency properties:

The "equivalent" demonstration data estimators and the growth data estimators must yield:

(1) the same subsystem MTBF point estimate and;

(2) the same subsystem MTBF lower confidence bound.

In other words, the methodology maintains the following relationships, respectively:

$$\hat{M}_D = \hat{M}_G \tag{6.3-51}$$

$$LCB_\gamma(D) = LCB_\gamma(G) \tag{6.3-52}$$

where

$$\hat{M}_D = \frac{T_D}{N_D} \tag{6.3-53}$$

$$LCB_\gamma(D) = \frac{2T_D}{\chi^2_{2N_D+2,\gamma}} \tag{6.3-54}$$

Reducing growth subsystem test data to "equivalent" demonstration test data using the following equations closely satisfies the relationships cited above:

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$$N_D = \frac{N_G}{2} \qquad 6.3\text{-}55$$

$$T_D = \hat{M}_G \times \frac{N_G}{2} = \frac{T_G}{2\hat{\beta}} \qquad 6.3\text{-}56$$

The growth estimate for the MTBF, $\hat{M}_G$, and the estimate for the growth parameter, $\hat{\beta}$, are described in the sections on point estimation for system level Continuous Reliability Growth Tracking Models.

The model then uses the above equations to compute an approximate lower confidence bound for the serial system reliability (MTBF) from non-growth subsystem demonstration data and growth subsystem "equivalent" demonstration data as described in the following section on the Lindström-Madden method.

6.3.3.2 Lindström-Madden Method.
In addition to using the notation defined in the previous section on Methodology, subsequent equations use the following notation:

List of Notations.

sys	system level
min	minimum of
K	number of subsystems in serial system
ρ	failure rate
i	subscript for subsystem number
Σ	summation of

To compute an approximate lower confidence bound (LCB) for the system MTBF from subsystem demonstration and "equivalent" demonstration data, the AMSAA SSTRACK model uses an adaptation of the Lindström-Madden method by computing the following four estimates:

1. the equivalent amount of system level demonstration test time. (This estimate is a reflection of the least tested subsystem because it is the minimum demonstration test time of all the subsystems.);
2. the current system failure rate, which is the sum of the estimated failure rate from each subsystem *i, i* = 1..*K*;
3. the "equivalent" number of system level demonstration failures, which is the product of the previous two estimates and;
4. the approximate LCB for the system MTBF at a given confidence level, which is a function of the equivalent amount of system level demonstration test time and the equivalent number of system level demonstration failures.

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In equation form, these system level estimates are, respectively:

$$T_{D,sys} = \min T_{D,i} \quad for\ i = 1..K \tag{6.3-57}$$

$$\hat{\rho}_{sys} = \sum_{i=1}^{K} \hat{\rho}_i \tag{6.3-58}$$

where

$$\hat{\rho}_i = \frac{1}{\hat{M}_{D,i}} \tag{6.3-59}$$

$$\hat{M}_{D,i} = the\ current\ MTBF\ estimate\ for\ subsystem\ i$$

$$N_{D,sys} = \hat{\rho}_{sys} \times T_{D,sys} \tag{6.3-60}$$

and

$$LCB_{\gamma} = \frac{2T_{D,sys}}{\chi^2_{2N_{D,sys}+2,\gamma}} \tag{6.3-61}$$

6.3.3.3 Example.
The following example is an application of the AMSAA Subsystem Level Reliability Growth Tracking Model to a system composed of three subsystems: one non-growth and two growth subsystems. Besides showing that SSTRACK can be used for test data gathered from dissimilar sources (namely, non-growth and growth subsystems), this particular example was chosen to show that system level reliability estimates are influenced by:

a. the least tested subsystem and;
b. the least reliable subsystem, that is, the subsystem with the largest failure rate.

Subsystem 1 in this example is a non-growth subsystem consisting of fixed configuration data of 8,000 hours of test time and 2 observed failures.

Subsystem 2 is a growth subsystem with individual failure time data. In 900 hours of test time there were 27 observed failures occurring at the following cumulative times: 7.8, 49.5, 49.5, 51.0, 64.2, 87.3, 99.9, 169.5, 189.3, 211.8, 219.0, 233.1, 281.7, 286.5, 294.3, 303.3, 396.0, 426.6, 443.1, 447.0, 501.6, 572.1, 579.0, 596.1, 755.7, 847.5, and 858.3.

Subsystem 3 is also a growth subsystem with individual failure time data. In 400 hours of test time there were 16 observed failures occurring at the following cumulative times: 15.04, 25.26, 47.46, 53.96, 56.42, 99.57, 100.31, 111.99, 125.48, 133.43, 192.66, 249.15, 285.01, 379.43, 388.97, and 395.25.

The table below shows the pertinent statistics for each subsystem *i*. It is here that all growth (*G*) subsystem test data are reduced to equivalent demonstration (*D*) test data.

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TABLE XIV. Subsystem Statistics

Statistics (i = 1,2,3)	Subsystem 1 (Non-growth)	Subsystem 2 (Growth)	Subsystem 3 (Growth)
$T_{G,i}$	N/A	900	400
$N_{G,i}$	N/A	27	16
$\hat{M}_{G,i}$	N/A	46.53	31.37
$N_{D,i} = \frac{N_{G,i}}{2}$	2	13.5	8
$T_{D,i} = \hat{M}_{G,i} \times N_{D,i}$	8000	628.19	250.95
$\hat{M}_{D,i} = \hat{M}_{G,i} = \frac{T_{D,i}}{N_{D,i}}$	4000	46.53	31.37
$\hat{\rho}_i = \frac{1}{\hat{M}_{D,i}}$	2.50×10^{-4}	2.149×10^{-2}	3.188×10^{-2}

System level statistics are computed by applying the Lindström-Madden method to the equivalent demonstration data from each subsystem.

$$T_{D,sys} = \min T_{D,i(i=1,2,3)} = 251.0 \qquad 6.3\text{-}62$$

$$\hat{\rho}_{sys} = \sum_{i=1}^{3} \hat{\rho}_i = 5.362 \times 10^{-2} \qquad 6.3\text{-}63$$

$$\hat{M}_{D,sys} = \frac{1}{\hat{\rho}_{sys}} = 18.7 \qquad 6.3\text{-}64$$

$$N_{D,sys} = T_{D,sys} \times \hat{\rho}_{sys} = 13.5 \qquad 6.3\text{-}65$$

$$LCB_{.80} = \frac{(2 \times T_{D,sys})}{\chi^2_{2N_{D,sys}+2,.80}} = 14.32 \quad (confidence\ level = 80\%) \qquad 6.3\text{-}66$$

Finally, a table of approximate lower confidence bounds (TABLE XV) is shown for the system reliability (MTBF) for a range of confidence levels.

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TABLE XV. System Approximate Lower Confidence Bounds

Confidence Level (in percent)	LCB for System MTBF
50	**17.77**
55	**17.19**
60	16.62
65	16.07
70	15.51
75	14.93
80	14.32
85	13.66
90	12.87
95	11.82
98	10.78
99	10.15

There may be cases when not all subsystems function throughout the mission. That is, the OMS/MP may specify some subsystems, or all, may function only a specific percent of the system operating time. In this situation we introduce the term w_i, subsystem i utilization factor (weight) where $0<w_i\leq 1$. The following adjustments are made to the Lindström-Madden method:

$$T_{D,SYS} = min\frac{T_{D,i}}{w_i}\ (i = 1,\cdots,\mathrm{K}) \qquad 6.3\text{-}67$$

$$\hat{\rho}_{SYS} = \sum_{i=1}^{K} w_i\,\hat{\rho}_i. \qquad 6.3\text{-}68$$

For our example, all w_i=1, then the results above apply. If, however, w_1=w_2=1and w_3= 0.6 then

$T_{D,SYS}$= min $\frac{T_{D,i}}{w_i}$ $(i = 1,2,3)$ = min(8000, 628.19, 418.25) = 418.25

$\hat{\rho}_{SYS} = 0.040868$

$$\hat{M}_{D,SYS} = 24.6$$

$N_{D,SYS} = T_{D,SYS} \times \hat{\rho}_{SYS} = 17.9$

$LCB_{.80} = \frac{2\times T_{D,SYS}}{\chi^2_{2N_{D,SYS}+2,0.80}} = \frac{836.5}{37.8} = 22.13$

So, the reduction of operating time for subsystem 3 results in an increase in MTBF from18.7 to 24.6 (not unexpected), but equivalent system test time $T_{D,SYS}$ increases from 251 to 418

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increasing the degrees of freedom from 29 to 37.8. Subsystem 3 previously had the least reliability, now subsystem 2 has the larger contribution to system unreliability. Note that the increase in degrees of freedom results in a more narrow spread between $\widehat{M}_{SYS}$ and $LCB_{0.80}$ - from 4.4 to 2.5 - a significant reduction.

One lesson to learn from the above examples is to understand how the system works, system requirements, what percentages of time subsystems operate relative to system operation, and how this impacts system reliability and how one needs to test items and subsystems in order to maximize utility of information and resources. Merely looking at system level testing and overall system numbers is not enough; one needs to comprehend the entire process.

6.4 References.

1. MIL-HDBK-189, Reliability Growth Management, 13 February 1981

2. Crow, Larry, Methodology Office Note 1-83, AMSAA Discrete Reliability Growth Model, March 1983

3. Lloyd, D. K., and M. Lipow; Reliability: Management, Methods, and Mathematics, Prentice Hall, NJ; 1962; p. 227

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7 RELIAIBLITY GROWTH PROJECTION.

7.1 Reliability Projection Background.

The reliability growth process applied to a complex system undergoing development involves surfacing failure modes, analyzing the modes, and implementing corrective actions (termed fixes) to the surfaced modes. In such a manner, the system configuration is matured with respect to reliability. The rate of improvement in reliability is determined by (1) the on-going rate at which new problem modes are being surfaced, (2) the effectiveness and timeliness of the fixes, and (3) the set of failure modes that are addressed by fixes.

At the end of a test phase, program management usually desires an assessment of the system's reliability associated with the current configuration. Often, the amount of data generated from testing the current system configuration is severely limited. In such circumstances, if the failure data generated over a number of system configurations is consistent with a reliability growth model, we can pool the data over the tested configurations to estimate the parameters of the growth model. This in turn will yield a reliability tracking curve that gives estimates of the configuration reliabilities. The resulting assessment of the system's current reliability is called a demonstrated estimate since it is based solely on test data.

If the current configuration is the result of applying a group of fixes to the previous configuration, there could be a statistical lack of fit in tracking reliability growth between the previous and current configurations. In such a situation, it may not be valid to use a reliability growth tracking model to pool configuration data to assess the reliability of the current configuration. We always have the option of estimating the current configuration reliability based only on failure data generated for this configuration. However, such an estimate may be poor if little test time has been accumulated since the group of fixes was implemented. In this situation, program management may wish to use a reliability projection method. Such methods are typically based on assessments of the effectiveness of corrective actions and failure data generated from the current and previous configurations.

A second situation in which a reliability projection is often utilized is when a group of fixes are scheduled for implementation at the end of the current test phase, prior to commencing a follow-on test phase. Program management often desires a projection of the reliability that will be achieved by implementing the delayed fixes. This type of projection can be based solely on the current test phase failure data and engineering assessments of the effectiveness of the planned fixes. The current test phase could consist of several system configurations if not all the fixes to surfaced problem modes are delayed. In this instance, we can still obtain a projection of the reliability with one of the methodologies of this section. In addition, if there are at least three such configurations and a tracking model fits the growth pattern over these configurations, and then an approach utilized in the Crow-Extended Model may also be applicable.

Another situation in which a projection can be useful is in assessing the plausibility of meeting future reliability milestones, i.e., milestones beyond the commencement of the follow-on test. The model utilizing the first occurrences of B-mode failures and an average FEF can provide such projections based on failure data generated to date and fix effectiveness assessments for all

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implemented and planned fixes to surfaced problem modes, assuming the rate of occurrence of problem failure modes remains consistent with the experienced problem failure mode occurrence pattern.

Section 7.2 presents several basic concepts used in connection with the reliability projection models, and establishes notation and presents assumptions that are used throughout this section. Notation and assumptions directed toward a particular method are introduced in the corresponding section. Section 7.3 presents short summaries for each of the models noting their purpose, assumptions, limitations, and benefits.

Sections 7.4 and 7.5 present two reliability projection models and associated statistical procedures. In Section 7.4, the AMSAA/Crow Projection Model is discussed. This model is used to estimate the system failure intensity at the beginning of a follow-on test phase based on information from the previous test phase. This information consists of problem mode first occurrence times, the number of failures associated with each problem mode, and the total number of failures due to modes that will not be addressed by fixes. Additionally, the projection uses engineering assessments of the planned corrective actions to problem modes surfaced during the test phase. The associated statistical estimation procedure assumes that all the corrective actions are implemented at the end of the current test phase but prior to commencing the follow-on test phase. This model addresses the continuous case, i.e., where test duration is measured in a continuous fashion such as in hours or miles.

Section 7.5 presents Crow's Extended Reliability Projection Model, which is applicable to the test-fix-find-test situation in which fixes may be incorporated in testing or as delayed fixes at the end of the test phase.

In Section 7.6 a reliability projection model is presented that addresses, for the continuous case, the situation where one wishes to utilize test data generated over one or more test phases to project the impact of fixes to surfaced problem failure modes. This model is called the AMSAA Maturity Projection Model – Continuous (AMPM-Continuous). The model does not require that the fixes be all delayed to the end of the current test phase. It only assumes the fixes are implemented prior to the time at which a projection is desired. Also, projections may be made for milestones beyond the start of the next test phase.

Section 7.7 presents the AMPM based on Stein Estimation. This approach does not require one to distinguish between A-modes and B-modes other than through the assignment of a zero, or positive FEF, respectively, to surfaced modes. A significant difference between the Stein approach and the other methods is that the Stein projection is a direct assessment of the realized system failure rate after failure mode mitigation.

Section 7.8 presents the Discrete Projection Model (DPM), a reliability growth projection model for one-shot systems whose approach in many aspects serves as a discrete analogue to the continuous AMPM-Stein projection model.

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7.2 Basic Concepts, Notation and Assumptions.

In addition to utilizing a statistical tracking model over the test phase, one may wish to use a reliability growth projection model. The basic objective is to obtain an estimate of the reliability at a current or future milestone based on management strategy, planned and/or implemented fixes, assessed fix effectiveness, and the statistical estimate of problem mode rates of occurrence. One would then analyze the sensitivity of reliability projection to program planning parameters (e.g., fix effectiveness, etc) and determine the maturity of the system or subsystem based on maturity metrics such as MTBF, rate of occurrence of problem modes or percent of problem mode initial failure rate surfaced. The benefits of reliability growth projection are:

a. Assesses reliability improvements due to surfaced problems and corrective actions
b. Provides important maturity metrics
    i. Projections of MTBF
    ii. Rate of occurrence of new problem modes
    iii. Percent surfaced of problem mode initial failure rate
c. Projects expected number of new problem modes during additional testing
d. Assesses system reliability growth potential
e. Quantifies impact of successful fixes and overall engineering and management strategy

Database considerations for projection methodology and data requirements for projection analysis are:

a. Database Considerations
    i. Failure mode classification
    ii. Test exposure (i.e. land, water, etc…)
    iii. Configuration control
    iv. Engineering assessments of fix effectiveness
b. Data Requirements
    i. First occurrence time for each distinct correctable failure mode
    ii. Occurrence time for each repeat of a distinct correctable failure mode
    iii. Number of non-correctable failures
    iv. Fix effectiveness factor for each correctable failure mode or the average over all
    v. Total test duration
    vi. Corrective action times for each mode for Crow Extended.

There are two basic projection models, one based on the Crow Power Law Model – AMSAA/Crow Projection Model (ACPM)-which assumes fixes are delayed but implemented prior to the next test phase. Subsequently, the Crow Extended Reliability Projection Model was developed wherein both delayed and non-delayed fixes are permitted. The method accomplishes this by suitably combining the Crow Projection Model with the MIL-HDBK 189 Tracking Model.

The other basic projection model -AMPM- is based on the approach of viewing subsystem failure rates $(\lambda_1, \cdots, \lambda_K)$ as a realization of a random sample $\underline{\Lambda} = (\Lambda_1, \cdots, \Lambda_K)$ from the

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gamma distribution, $\Gamma(\alpha, \beta)$. This allows one to utilize all the B-mode times to first occurrence observed during Test Phase I to estimate the gamma parameters - α, β. This type of projection is based on the Phase I B-mode first occurrence times, whether the associated B-mode fix is implemented within the current test phase or delayed (but implemented prior to the projection time). In addition to the B-mode first occurrence times, the projections are based on an average fix effectiveness factor (FEF). This average is with respect to all the potential B-modes, whether surfaced or not. However, as in the ACPM, this average FEF is assessed based on the surfaced B-modes. For the AMPM model and AMPM-Stein, the set of surfaced B-modes would typically be a mixture of B-modes addressed with fixes during the current test phase as well as those addressed beyond the current test phase. AMPM-Stein only provides projection at the end of the test phase, where all corrective actions must be delayed. It utilizes individual mode FEF's for surfaced modes only. It does not need to consider FEF's for un-surfaced modes.

Throughout this section, we will regard a potential failure mode as consisting of one or more potential failure sites with associated failure mechanisms. Fixes are often applied to failure modes surfaced through testing. In accordance with [1], if a B-mode is defined to be a failure mode then we would apply a fix to it, if the mode were surfaced. All other failure modes will be referred to as A-modes. A surfaced mode might be regarded as an A-mode if (1) a fix is not economically justifiable, or (2) the underlying failure mechanisms associated with the mode are not sufficiently understood to attempt a fix. Thus the rate of failure due to the set of A-modes is constant as long as the failure modes are not reclassified.

As stated in [1], an approximation is proposed to the expected value of a random value consisting of: (1) a constant failure rate for the A-mode failure rate, (2) a failure rate for the seen B-modes reduced by a fix-effectiveness factor, and (3) and a bias term *h(T)* that represents the rate of occurrence of new B-modes at the end of the test phase.

For the ACPM, a projection applies to *test-find-test management strategy* where all fixes are implemented at the end of test. The Crow Extended Reliability Projection Model applies to the *test-fix-find-test management strategy* wherein some fixes may be implemented in testing, the remaining at the conclusion of testing. In order to provide the assessment and management metric structure for corrective actions during and after a test, two types of B modes are defined. BC failure modes are corrected during test. BD failure modes are delayed to the end of the test. A failure mode, as before, is those failure modes that will not receive a corrective action. These classifications define the management strategy and can be changed. It is noted that with Crow's Extended Model system failure rate consists of four terms: (1) AMSAA/Crow tracking model failure rate, (2) less the failure rate of the BD modes, (3) plus the failure rate of the BC modes reduced by fix-effectiveness factors for these modes, and (4) plus the rate of occurrence of the BD failure modes. Crow presents many metrics that might be used in assessment and feasibility of meeting requirements. Might this be moved to the section on Crow's extended model? Then maybe cite the extended model reference to that section.

It has been suggested to use ACPM to project reliability at the start of the next phase of testing and to use AMPM to project reliability at some future point in time assuming the B-mode failure rate of occurrence pattern continues. With the development of the Crow Extended Model, this may also be used to project reliability at the start of the next phase of testing. AMPM is

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particularly useful when the assumptions of Crow Extended may not hold, e.g., the tracking model assumed does not fit. In such a case, the tracking model should not be utilized if the fit is not good. AMPM is particularly useful if non-tactical fixes are being applied during the test phase. It is also noted that these projection models may be used to determine reliability "potential" by sensitizing on FEFs. Additionally, they may be used as a system or subsystem "maturity" metric to reveal the percent of initial failure rate surfaced:

a. The more failure rate surfaced, the more opportunity for growth
b. The less failure rate surfaced, the more testing required (This also presents potential to look at problems beyond testing).

For the Test-Find-Test strategy, these delayed corrective actions are often incorporated as a group at the end of the test phase and the result is generally a distinct jump in the system reliability. A projection model estimates this jump in reliability due to the delayed fixes. This is called a "projection." These models do not simply extrapolate the tracking curve beyond the current test phase, although such an extrapolation is frequently referred to as a reliability projection. Reliability projection through extrapolation implicitly assumes that the conditions of test do not change and that the level of activities that promote growth essentially remain constant (i.e., the growth rate, α, remains the same). In the past, this was sometimes done for the test-fix-test strategy. However, the situation in which such an extrapolation is inappropriate is when a significant group of fixes to failure modes that occurred in the test phase is to be implemented at the conclusion of the test phase.

Here we will simply point out several things to keep in mind when applying a model. First note that for some models, the estimation procedure of the system failure rate is only valid when all the fixes are delayed to the end of the test phase. This ensures that the failure rate is constant over the test phase. If this is not the case, alternate projection models and/or estimation procedures must be utilized, such as the Crow Extended Model. Thus, one should graphically and statistically investigate whether all fixes have been delayed. This would imply that $\rho(t)$ is constant during the test phase. Occasionally, a developer will assert that all the fixes will be implemented at the end of the test phase. At times, such a statement merely implies that the long-range fixes will not be implemented until the test's conclusion. However, even in such cases it is not unusual that expedient short-term fixes are applied during the test period to allow completion of the test without undue interference from known problems. As mentioned earlier, sometimes the "fix" is simply to attempt to avoid exercising portions of the system functionality with known problems. In such instances projection methodology that depends on the $\rho(t)$ remaining constant during the test phase should not be used.

Also, although there are no hard and fast rules, one needs to surface enough distinct B-modes to allow the rate of occurrence of new B-mode failures, $h(T)$, to be statistically estimated. This implies that there must be enough B-modes so that the graph of the cumulative number of B-modes versus the test time appears regular enough and in conformance with the projection model's assumed mean value function that parameters of this function can be statistically estimated. In fact, one should visually compare the plot of the cumulative number of observed B-modes verses test time to the statistically fitted curve of the estimated expected number of B-modes verses test time. Such a visual comparison can help determine if the assumed mean value function for the expected number of B-modes as a function of test time captures the observed

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trend. There are also statistical tests for the null hypothesis that *E(M(T))* is the mean value function based on the fact that for any time truncated Poisson process, conditioned on the number of observed B-modes over the time period [0, *T*], the cumulative times of B-mode first occurrences are order statistics of a random sample drawn from the distribution given by

$$F(t) = E(M(t))/E\big(M(T)\big) \quad \text{for } 0 \leq t \leq T. \qquad 7.2\text{-}1$$

7.2.1 List of Notation

K	Number of potential B-modes that reside in the system
λ_i	Initial rate of occurrence of B-mode *i* $(i = 1, \cdots, K)$
λ_A	Contribution of A-modes to system failure intensity
λ_B	B-mode contribution to initial system failure intensity
T	Total duration of conducted test, typically measured in hours or miles.
N_A	Number of A-mode failures that occur over [0,*T*]
N_B	Number of B-mode failures that occur over [0,*T*]
m	Number of distinct B-modes surfaced over [0,*T*]
$M(t)$	Random variable of number of distinct B-modes surfaced by test duration *t*
$\mu(t)$	The expected value of $M(t)$
t_i	Time of first occurrence of B-mode *i* $(i = 1, \cdots, K)$
$\underset{\sim}{t}$	Vector of B-mode first occurrence times $(t_1, \cdots, t_m)$
N_i	Number of failures associated with B-mode *i* that occurs during test
d_i	Fix effectiveness factor (FEF) for B-mode *i*. The factor d_i is the fraction of λ_i removed by the fix.
obs	The index set associated with the m B-modes that are surfaced during test
E	Expectation operator
V	Variance operator
MLE	Maximum likelihood estimator
^	When placed over a parameter, it denotes an estimate
~	"Distributed as"
$\approx$	"Approximated by"
$\cong$	"Approximately equal to"

7.2.2 Assumptions

a. At the start of test, there is a large unknown constant number, denoted by *K*, of potential B-modes that reside in the system (which could be a complex subsystem);
b. Failure modes (both types A and B) occur independently;
c. Each occurrence of a failure mode results in a system failure;
d. No new modes are introduced by attempted fixes.

Additional notation and assumptions germane to a particular model will be introduced in the section dealing with the model.

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7.3 Projection Models.

Reliability growth projection is an area of reliability growth that provides assessments of system reliability which take into account the impact of either delayed or non-delayed corrective actions. Five reliability growth projection models will be considered: (1) the AMSAA/Crow Projection Model (ACPM); (2) the Crow Extended Reliability Projection Model, (3) the AMSAA Maturity Projection Model (AMPM) and; (4) the AMSAA Maturity Projection Model based on Stein estimation (AMPM-Stein), and (5) the Discrete Projection Model (DPM). The following sections provide short overviews for each of the five models after which each will be developed in detail.

7.3.1 AMSAA/Crow Projection Model (ACPM).

7.3.1.1 Purpose.

The purpose of ACPM is to estimate the system reliability at the beginning of a follow-on test phase by taking into consideration the reliability improvement from delayed fixes.

7.3.1.2 Assumptions.

The Assumptions of the model are:

a. test duration is continuous,
b. corrective actions are implemented as delayed fixes at the end of the test phase,
c. failure modes can be categorized between A-Modes and B-Modes,
d. failure modes occur independently and cause system failure,
e. there are a large number of potential B-Modes, relative to the number of surfaced modes,
f. the number of B-Modes surfaced can be approximated by a NHPP with Power law MVF.

7.3.1.3 Limitations.

The limitations of the ACPM include:

a. all corrective actions must be delayed,
b. FEFs are often a subjective input,
c. and projection accuracy can be degraded via reclassification of A-Modes to B-Modes.

7.3.1.4 Benefits.

The benefits of the ACPM are: (1) can project the impact of delayed corrective actions on system reliability and (2) projection takes into account the contribution to the system failure rate due to unobserved problem failure modes.

7.3.2 Crow Extended Reliability Projection Model

7.3.2.1 Purpose.

The purpose of the Crow Extended Model is to estimate the system reliability at the beginning of a follow-on test phase by taking into consideration the reliability improvement from fixes incorporated during testing and those delayed fixes incorporated at the conclusion of the test phase.

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7.3.2.2 Assumptions.
Same as ACPM except that B-modes are subdivided into BC (fixes incorporated) and BD (fixes delayed).

7.3.2.3 Limitations.
Does not explicitly use BC-mode FEFs.

7.3.2.4 Benefits.
Same as ACPM plus includes implemented fixes but does not explicitly use BC-mode FEFs. Added capability includes pre-emptive fixes at time *T* for failure modes that have not experienced a failure. It is assumed that for these failure modes an estimate – by analysis, analogy, or other test – of the failure rate is available.

7.3.3 AMSAA Maturity Projection Model (AMPM).

7.3.3.1 Purpose.
The purpose of AMPM is to provide estimates of the following taking into consideration delayed and non-delayed fixes:

a. the B-Mode initial failure intensity,
b. the expected number of B-Modes surfaced,
c. the percent surfaced of the B-Mode initial failure intensity,
d. the rate of occurrence of new B-Modes, and
e. the projected reliability.

7.3.3.2 Assumptions.
Model assumptions include:

a. test duration is continuous,
b. corrective actions are implemented prior to the time at which projections are made,
c. failure modes independently occur and cause system failure,
d. failure rates can be modeled as a realization of a random sample from a gamma distribution, and
e. modes can be classified as A-Modes or B-Modes.

7.3.3.3 Limitations.
All limitations of ACPM apply.

7.3.3.4 Benefits.
All benefits from ACPM apply. Corrective actions can be implemented during test, or be delayed. Reliability can be projected for future milestones. Additionally, in situations where there is an apparent steepness of cumulative number of B-modes versus cumulative test time over an early portion of testing after which this rate of occurrence slows, there is methodology to partition the early modes from the remaining modes. These early B-modes must be aggressively and effectively corrected. Additionally methodology exists to handle cases where there is an early "gap" or if there appears to be a difference in the average FEFs in early or start-up testing versus the remainder of testing (an apparent or visual difference in failure rate in the initial testing).

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7.3.4 AMPM based on Stein Estimation (AMPM-Stein).

7.3.4.1 Purpose.
Same as AMPM.

7.3.4.2 Assumptions.
All corrective actions must be delayed, otherwise same as AMPM.

7.3.4.3 Limitations.
FEFs are often a subjective input and there must be at least one repeat failure mode.

7.3.4.4 Benefits.

a. reliability projection has been shown (via simulations conducted to date) to provide greater accuracy than that of the ACPM (given model assumptions hold),
b. allows for natural trade-off analysis between reliability improvement and incremental cost,
c. no need to group identified failure modes into A-Modes and B-Mode classes, and
d. it requires less data than AMSAA/Crow and AMPM and does not require mode first occurrence times. Further, it provides the developer a way to incorporate major refurbishment and schedule events.

7.3.5 Discrete Projection Model (DPM).

7.3.5.1 Purpose.
The model, developed by J. Brian Hall and Ali Mosleh in "A Reliability Growth Projection Model for One-Shot Systems," AMSAA, Aberdeen Proving Ground, MD, Technical Report No. TR2006-140, will not be suitable for application to all one-shot development programs, but it is useful in cases where one or more failure modes are, or can be, discovered in a single trial; and catastrophic failure modes have been previously discovered, and corrected. The model is unique in the area of reliability growth projection, and offers an alternative to the popular competing risks approach.

7.3.5.2 Assumptions.

a. A trial results in a dichotomous occurrence/non-occurrence of B-mode i such that N_{ij} ~ Bernoulli (p_i) for each $i = 1,\ldots, k$, and $j = 1,\ldots,T$.
b. Initial failure probabilities $p_1 \ldots p_k$ constitute a realization of an s-random sample $P_1,\ldots, P_k$ such that $P_i \sim$ Beta $(n,x)$for each $i = 1,\ldots, k$.
c. Corrective actions are delayed until the end of the current test phase, where a test phase is considered to consist of a sequence of T s-independent Bernoulli trials.
d. Failures associated with different failure modes arise s-independently of one another on each trial. As a result, the system must be at a stage in development where catastrophic failure modes have been previously discovered & corrected, and are therefore not preventing the occurrence of other failure modes.

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e. There is at least one repeat failure mode. If there is not at least one repeat failure mode, the moment estimators, and the likelihood estimators of the beta parameters do not exist.

7.3.5.3 Limitations.
FEFs are often a subjective input and there must be at least one repeat failure mode.

7.3.5.4 Benefits.
The model provides a method for approximating the vector of failure probabilities associated with a complex one-shot system, which is based on our derived shrinkage factor given by (7.7-5). The benefit of this procedure is that it not only reduces error, but also reduces the number of unknowns requiring estimation from k +1 to only three. Also, estimates of mode failure probabilities, whether observed or unobserved during testing, will be positive.

7.4 The AMSAA/Crow Projection Model (ACPM).

7.4.1 Background.
The material in this section is as presented in [1]. In this section, we consider the case where all fixes to surfaced B-modes are implemented at the end of the current test phase prior to commencing a follow-on test phase. Thus, all fixes are delayed fixes. The current test phase will be referred to as Phase I and the follow-on test phase as Phase II.

The AMSAA/Crow reliability projection model and associated parameter estimation procedure was developed to assess the reliability impact of a group of delayed fixes. In particular, the model and estimation procedure allow assessment of what the system failure intensity will be at the start of Phase II after implementation of the delayed fixes. Denoting this failure intensity by *r(T)*, where *T* denotes the duration of Test Phase I, the AMSAA/Crow assessment of *r(T)* is based on: (1) the A and B mode failure data generated during Phase I test duration *T*; and (2) assessments of the fix effectiveness factors (FEFs) for the B-modes surfaced during Phase I. Since the assessments of the FEFs are often largely based on engineering judgment, the resulting assessment, $\hat{r}(T)$, of the system failure intensity after fix implementations is called a reliability projection as opposed to a demonstrated assessment (which would be based solely on test data).

The AMSAA/Crow projection model and estimation procedure was motivated by the desire to replace the widely used "adjustment procedure." The adjustment procedure assesses *r(T)* based on reducing the number of failures N_i due to B-mode *i* during Phase I to $\left(1-d_i^*\right)N_i$, where d_i^* is the assessment of d_i. Note $\left(1-d_i^*\right)N_i$ is an assessment of the expected number of failures due to B-mode *i* that would occur in a follow-on test of the same duration as Phase I. The adjustment procedure assesses *r(T)* by $\hat{r}_{adj}(T)$ where

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7.4-1

$$\hat{r}_{adj}(T) = \frac{N_A}{T} + \frac{\sum_{i \in obs}\left(1 - d_i^*\right)N_i}{T}$$

Crow[1] shows that even if the assessed FEFs are equal to the actual d_i, the adjustment procedure systematically underestimates *r(T)*. This bias, i.e.,

7.4-2

$$B(T) = E\{r(T) - \hat{r}_{adj}(T)\} > 0$$

is calculated in [1] by considering the random set of B-modes surfaced during Phase I. In particular, the adjustment procedure is shown to be biased since it fails to take into account that, in general, not all the B-modes will be surfaced by the end of Phase I. Before discussing how the AMSAA/Crow methodology addresses this bias we will list some additional notation and assumptions associated with the AMSAA/Crow model.

7.4.2 AMSAA/Crow Model Notation and Additional Assumptions.

7.4.3 List of Notations.

D_i	The conditional random variable for B-mode *i* (*i* = 1, …, *K*) whose realization is the fix effectiveness factor d_i if mode *i* occurs during test Phase I.
μ_d	Expected value of D_i
T	Length of Test Phase I
r(T)	System failure intensity at beginning of Test Phase II after implementation of delayed B-mode fixes. Viewed as a random variable whose value is determined by the set of B-modes surfaced during Test Phase I and the associated fix effectiveness factors.
$\rho(T)$	Expected value of *r(T)* with respect to random set of B-modes surfaced in Test Phase I, conditioned on the fix effectiveness factor values. We write $\rho(T) = E(r(T))$.
$\hat{r}_{adj}(T)$	Adjustment procedure assessment of the value taken on by *r(T)*
B(T)	Bias incurred by assessing the value of *r(T)* by $\hat{r}_{adj}(T)$. Thus, $B(T) = E\{r(T) - \hat{r}_{adj}(T)\}$
ρ_{GP}	Growth potential system failure intensity
M_{GP}	Growth potential system MTBF, i.e., $M_{GP} = (\rho_{GP})^{-1}$
$h(t)$	Expected rate of occurrence of new B-modes at test duration *t*. Note: $h(t) = \frac{d\,\mu(t)}{d\,t}$
$h_c(t), r_c(t), \rho_c(t)$	Crow/AMSAA model approximations to $h(t)$, *r(t)*, $\rho(t)$ respectively
M(T), M_c(T)	Denote $(\rho(T))^{-1}$ and $(\rho_c(T))^{-1}$ respectively

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7.4.4 Additional Assumptions for AMSAA/Crow.

a. The time to first occurrence is exponentially distributed for each failure mode.
b. No fixes to B-modes are implemented during Test Phase I. Fixes to all B-modes surfaced during Phase I are implemented prior to Phase II.
c. The fix effectiveness factors (FEFs) d_i associated with the B-modes surfaced during Phase I are realized values of the random variables D_i ($i = 1, \ldots, K$) where
   i. the D_i are independent;
   ii. the D_i have common mean value μ_d; and
   iii. the D_i are independent of $M(T)$.
d. The random process for the number of distinct B-modes that occur over test interval $[0,t]$, i.e. $M(t)$, is well approximated by a non-homogeneous Poisson process with mean value function $\mu_c(t) = \lambda t^{\beta}$ for some $\lambda, \beta > 0$.

7.4.5 Methodology.

The AMSAA/Crow model assesses the value of the system failure intensity, $r(T)$, after implementation of the Phase I delayed fixes. This assessment is taken to be an estimate of the expected value of $r(T)$, i.e., an estimate of $\rho(T) = E(r(T))$. In [1] (and in Section 7.3.2) it is shown that:

$$\rho(T) = \lambda_A + \sum_{i=1}^{K}(1-d_i)\lambda_i + \sum_{i=1}^{K} d_i\,\lambda_i e^{-\lambda_i T} \qquad 7.4\text{-}3$$

The traditional adjustment procedure assessment for the value of $r(T)$ is actually an estimate of $\lambda_A + \sum_{i=1}^{K}(1-d_i)\lambda_i$ since (as shown later in this subsection)

$$E(\hat{r}_{adj}(T)) = \lambda_A + \sum_{i=1}^{K}(1-d_i^*)\lambda_i \qquad 7.4\text{-}4$$

where d_i^* is an assessment of d_i. Thus, by (7.4-3) and (7.4-4), the adjustment procedure has the bias $B(T)$ where

$$\begin{aligned} B(T) &= E\{r(T) - \hat{r}_{adj}(T)\} \\ &= \rho(T) - E(\hat{r}_{adj}(T)) \end{aligned}$$

$$= \sum_{i=1}^{K}(d_i^* - d_i)\lambda_i + \sum_{i=1}^{K} d_i\,\lambda_i\,e^{-\lambda_i T} \qquad 7.4\text{-}5$$

It follows that for $d_i^* = d_i \quad (i = 1, \cdots, K)$

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$$B(T) \;=\; \sum_{i=1}^{K} d_i\, \lambda_i\, e^{-\lambda_i T}$$

This shows that even with perfect knowledge of the d_i (i.e., when $d_i^* = d_i$), the adjustment procedure provides a biased underestimate of the value of *r(T)*. The AMSAA/Crow procedure attempts to reduce this bias by estimating *B(T)* given by (7.4-5).

To estimate B(T), the AMSAA/Crow Model uses an approximation to B(T). This approximation is obtained in two steps. The first step is to regard the d_i in (7.4-5) as realizations of random variables $D_i \;\; (i = 1, \cdots, K)$ that satisfy assumption number 3 in the "Additional Assumptions for AMSAA/Crow." Then B(T) is approximated by the expected value (with respect to the D_i) of $\sum_{i=1}^{K} D_i\, \lambda_i\, e^{-\lambda_i T}$. Thus the initial approximation arrived at for *B(T)* in (7.4-5) is

$$B(T) \;\approx\; E\left(\sum_{i=1}^{K} D_i\, \lambda_i\, e^{-\lambda_i T}\right) \;=\; \mu_d \sum_{i=1}^{K} \lambda_i\, e^{-\lambda_i T} \qquad 7.4\text{-}6$$

where $\mu_d = E(D_i) \;\; (i = 1, \cdots, K)$. The final step to obtain the AMSAA/Crow approximation of *B(T)* is to replace the sum $\sum_{i=1}^{K} \lambda_i\, e^{-\lambda_i T}$ in (7.4-6) by a two parameter function of *T*. The AMSAA/Crow Model replaces this sum by the power function

$$h_c(T) \;=\; \lambda \beta T^{\beta - 1} \qquad for \quad \lambda, \beta > 0 \qquad 7.4\text{-}7$$

The form in equation 7.4-7 is chosen based on the desire for a mathematically tractable estimation problem and an empirical observation. Based on an empirical study, Crow [1] states that the number of distinct B-modes surfaced over a test period $[0, t]$ can often be approximated by a power function of the form

$$\mu_c(t) \;=\; \lambda t^{\beta} \qquad for \quad \lambda, \beta > 0 \qquad 7.4\text{-}8$$

In equation 7.4-8, Crow [1] interprets $\mu_c(t)$ as the expected number of distinct B-modes surfaced during the test interval $[0, t]$. More specifically, [1] assumes the number of distinct B-modes occurring over $[0, t]$ is governed by a non-homogeneous Poisson process with $\mu_c(t)$ as the mean value function. Thus

$$h_c(t) \;=\; \frac{d\, \mu_c(t)}{d\, t} \;=\; \lambda \beta t^{\beta - 1} \qquad 7.4\text{-}9$$

represents the expected rate at which new B-modes are occurring at test time *t*.

In Annex 1 of Appendix G, under the previously stated assumptions, it is shown that the expected number of distinct B-modes surfaced over $[0, t]$ is given by

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$$\mu(t) = \sum_{i=1}^{K}\left(1-e^{-\lambda_i t}\right) \qquad 7.4\text{-}10$$

Thus the expected rate of occurrence of new B-modes at test time t is

$$h(t) = \frac{d\,\mu(t)}{d\,t} = \sum_{i=1}^{K}\lambda_i e^{-\lambda_i t} \qquad 7.4\text{-}11$$

Equation 7.4-11 shows that the initial approximation to the bias *B(T)*, given in (7.4-6) can be expressed as

$$B(T) \approx \mu_d\, h(T) \qquad 7.4\text{-}12$$

By replacing *h(T)* in equation (7.4-12) by $h_c(T)$ given in (7.4-9), we arrive at the final AMSAA/Crow Model approximation to *B(T)*, namely

$$B_c(T) = \mu_d\, h_c(T) = \mu_d\,\lambda\,\beta\,T^{\beta-1} \qquad 7.4\text{-}13$$

Returning to our expression in (7.4-3) for the expected value of the system failure intensity after incorporation of the Phase I delayed fixes, i.e., $\rho(T) = E(r(T))$, we can now write down the AMSAA/Crow Model approximation for $\rho(T)$. This approximation, by (7.4-13), is given by:

$$\begin{aligned}\rho_c(T) &= \lambda_A + \sum_{i=1}^{K}(1-d_i)\lambda_i + B_c(T)\\ &= \lambda_A + \sum_{i=1}^{K}(1-d_i)\lambda_i + \mu_d\left(\lambda\,\beta\,T^{\beta-1}\right)\end{aligned}$$

Next, consider the AMSAA/Crow procedure for estimating $\rho_c(T)$. This estimate is taken as the assessment of the system failure intensity after incorporation of the delayed fixes.

Consider the first term in the expression for $\rho_c(T)$ given in equation (7.4-14), i.e., λ_A. Since the A-modes are not fixed, the A-mode failure rate λ_A is constant over [0,*T*]. Thus, we simply estimate λ_A by

$$\hat{\lambda}_A = \frac{N_A}{T} \qquad 7.4\text{-}14$$

where N_A is the number of A-mode failures over [0,*T*]. Note

$$E(\hat{\lambda}_A) = \frac{E(N_A)}{T} = \frac{\lambda_A T}{T} = \lambda_A \qquad 7.4\text{-}15$$

Next consider estimation of the summation $\sum_{i=1}^{K}(1-d_i)\lambda_i$ in the expression for $\rho_c(T)$. By the second assumption in the "Additional Assumptions for AMSAA/Crow," all fixes are delayed until Test Phase I has been completed. This implies the failure rate for B-mode i $(i=1,\cdots,K)$ remains constant over [0,*T*]. Thus, we simply estimate λ_i by

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$$\hat{\lambda}_i \;=\; \frac{N_i}{T} \qquad (i = 1, \cdots, K)$$

7.4-16

where N_i denotes the number of failures during [0,*T*] attributable to B-mode *i*. Note

$$E\left(\hat{\lambda}_i\right) \;=\; \frac{E\left(N_i\right)}{T} \;=\; \frac{\lambda_i T}{T} \;=\; \lambda_i$$

7.4-17

equations (7.4-16) and (7.4-18) suggest we assess $\lambda_A + \sum_{i=1}^{K}\left(1 - d_i\right)\lambda_i$ by

$$\hat{r}_{adj}(T) \;=\; \hat{\lambda}_A + \sum_{i=1}^{K}\left(1 - d_i^*\right)\hat{\lambda}_i \;=\; \frac{N_A}{T} + \sum_{i=1}^{K}\left(1 - d_i^*\right)\left(\frac{N_i}{T}\right) \qquad 7.4\text{-}18$$

Observe $N_i = 0$ if B-mode *i* does not occur during [0,*T*]. Thus

$$\hat{r}_{adj}(T) \;=\; \frac{N_A}{T} + \sum_{i \in obs}\left(1 - d_i^*\right)\left(\frac{N_i}{T}\right) \qquad 7.4\text{-}19$$

where obs = {*i* | B-mode *i* occurs during [0,*T*]}. Note the adjustment procedure estimate has the form

$$\hat{r}_{adj}(T) \;=\; \frac{N^*}{T} \qquad 7.4\text{-}20$$

where

$$N^* \;=\; N_A + \sum_{i \in obs}\left(1 - d_i^*\right)N_i \qquad 7.4\text{-}21$$

is the "adjusted" number of failures. For given fix effectiveness factor (FEF) assessments, d_i^*, note that

$$\begin{aligned} E\left(\hat{r}_{adj}(T)\right) \;&=\; T^{-1}\left\{E\left(N_A\right) + \sum_{i=1}^{K}\left(1 - d_i^*\right)E\left(N_i\right)\right\} \\ &=\; T^{-1}\left\{\lambda_A T + \sum_{i=1}^{K}\left(1 - d_i^*\right)\left(\lambda_i T\right)\right\} \\ &=\; \lambda_A + \sum_{i=1}^{K}\left(1 - d_i^*\right)\lambda_i \end{aligned} \qquad 7.4\text{-}22$$

Thus, as stated earlier, we see that the adjustment procedure estimate only provides an assessment for a portion of the expected system failure intensity, namely

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$$\lambda_A + \sum_{i=1}^{K} (1 - d_i)\lambda_i$$

Returning to the fundamental equation for the AMSAA/Crow Model approximation to the expected system failure intensity, i.e. equation 7.3-14,

$$\rho_c(T) \;=\; \lambda_A + \sum_{i=1}^{K} (1 - d_i)\lambda_i + \mu_d \left(\lambda \beta T^{\beta-1}\right)$$

Next consider the assessment of the fix effectiveness factors d_i. The assessment d_i^* will often be based largely on engineering judgment. The value chosen for d_i^* should reflect several considerations:

a. How certain we are that the root cause for B-mode i has been correctly identified;

b. the nature of the fix, e.g., its complexity;

c. past FEF experience; and

d. any germane testing (including assembly level testing).

Note that equation 7.4-20 shows that we need only assess FEFs for those B-modes that occur during [0,*T*] to make an assessment of $\lambda_A + \sum_{i=1}^{K} (1 - d_i)\lambda_i$.

To assess the mean FEF, $\mu_d = E(D_i)$, we utilize our assessments d_i^* for $i \in obs$. Let m be the number of distinct B-modes surfaced over [0,*T*]. Then we assess μ_d by

$$\mu_d^* \;=\; \frac{1}{m} \sum_{i \in obs} d_i^* \qquad \text{7.4-23}$$

To complete our assessment of the expected system failure intensity after incorporation of delayed fixes, we will now address the assessment of $h_c(T) \;=\; \lambda \beta T^{\beta-1}$. To develop a statistical estimation procedure for λ and β, the AMSAA/Crow Model regards the number of distinct B-modes occurring in an interval [0,*t*], denoted by $M(t)$, as a random process. The model assumes that this random process can be well approximated, for large K, by a non-homogeneous Poisson process with mean value function $\mu_c(t) \;=\; E(M(t)) \;=\; \lambda t^{\beta}$, where λ, β, t > 0. As noted earlier in (7.4-9), $h_c(t) \;=\; \frac{d\,\mu_c(t)}{d\,t}$. The data required to estimate λ and β are (1) the number of distinct B-modes, *m*, that occur during [0,*T*] and (2) the B-mode first occurrence times $0 < t_1 \le t_2 \le \cdots \le t_m \le T$. Crow [1] states that the maximum likelihood estimates of λ and β, denoted by $\hat{\lambda}$ and $\hat{\beta}$ respectively, satisfy the following equations:

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$$\hat{\lambda} T^{\hat{\beta}} = m \qquad 7.4\text{-}24$$

$$\hat{\beta} = \frac{m}{\sum_{i=1}^{m} \ln\left(\frac{T}{t_i}\right)} \qquad 7.4\text{-}25$$

Note equation 7.4-25 merely says that the estimated number of distinct B-modes that occur during $[0,T]$ should equal the observed number of distinct B-modes over this period. Solving equation 7.4-25 for $\hat{\lambda}$ we can write our estimate for $h_c(T)$ in terms of m and $\hat{\beta}$ as follows:

$$\hat{h}_c(T) = \hat{\lambda}\hat{\beta}T^{\hat{\beta}-1} = \left(\frac{m}{T^{\hat{\beta}}}\right)\hat{\beta}T^{\hat{\beta}-1} = \frac{m\hat{\beta}}{T} \qquad 7.4\text{-}26$$

Crow [1] notes that conditioned on the observed number of distinct B-modes m, i.e. $M(T) = m$, the estimator

$$\bar{\beta}_m = \left(\frac{m-1}{m}\right)\hat{\beta} \qquad m \geq 2 \qquad 7.4\text{-}27$$

is an unbiased estimator of β, i.e.,

$$E(\bar{\beta}_m) = \beta \qquad 7.4\text{-}28$$

Thus, we will also consider estimating $h_c(T) = \lambda\beta T^{\beta-1}$ by using $\bar{\beta}_m$. This leads to the estimate

$$\bar{h}_c(T) = \frac{m\bar{\beta}_m}{T} \qquad 7.4\text{-}29$$

Finally, to complete our assessment of the system failure intensity, we need to assess the AMSAA/Crow Model expected system failure intensity $\rho_c(T)$. Recall, by equation (7.3-14)

$$\rho_c(T) = \lambda_A + \sum_{i=1}^{K}(1-d_i)\lambda_i + \mu_d h_c(T) \qquad 7.4\text{-}30$$

Piecing together our assessments for the individual terms in (7.4-31) we arrive at the following assessment for $\rho_c(T)$ based on $\hat{\beta}$:

$$\hat{\rho}_c(T) = \frac{N_A}{T} + \sum_{i=1}^{K}(1-d_i^*)\left(\frac{N_i}{T}\right) + \left(\frac{1}{m}\sum_{i\in obs} d_i^*\right)\left(\frac{m\hat{\beta}}{T}\right)$$

$$= \frac{1}{T}\left\{N_A + \sum_{i=1}^{K}(1-d_i^*)N_i + \hat{\beta}\sum_{i\in obs} d_i^*\right\}$$

Since $N_i = 0$ for $i \notin obs$, we finally obtain

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$$\hat{\rho}_c(T) = \frac{1}{T}\left\{N_A + \sum_{i\in obs}\left(1-d_i^*\right)N_i + \hat{\beta}\sum_{i\in obs} d_i^*\right\} \quad 7.4\text{-}31$$

Likewise, we arrive at the following alternate assessment for $\rho_c(T)$ based on $\bar{\beta}_m$ (provided $m \geq 2$):

$$\bar{\rho}_c(T) = \frac{1}{T}\left\{N_A + \sum_{i\in obs}\left(1-d_i^*\right)N_i + \bar{\beta}_m\sum_{i\in obs} d_i^*\right\} \quad 7.4\text{-}32$$

Note both estimates of $\rho_c(T)$ are of the form

$$Estimate\ \rho_c(T) = \frac{1}{T}\left\{N^* + (estimate\ \beta)\sum_{i\in obs} d_i^*\right\} \quad 7.4\text{-}33$$

where N^* is the "adjusted" number of failures over [0,*T*]. Recall the historically used adjustment procedure assessment for the system failure intensity, after incorporation of delayed fixes, is given by $\hat{r}_{adj}(T) = \frac{N^*}{T}$. Also recall $\bar{\beta}_m = \left(\frac{m-1}{m}\right)\hat{\beta} < \hat{\beta}$. Thus we see by equations 7.4-32 and 7.4-33

$$\hat{r}_{adj}(T) < \bar{\rho}_c(T) < \hat{\rho}_c(T) \quad 7.4\text{-}34$$

Also of interest is an assessment of the reciprocal of $\rho_c(T)$, i.e. $\mathrm{M}_c(T) = \{\rho_c(T)\}^{-1}$. The assessment for the system mean time between failures after incorporation of the delayed fixes, denoted by *M(T),* is taken to be the AMSAA/Crow Model assessment of $\mathrm{M}_c(T)$. The assessments of $M_c(T)$ based on $\hat{\rho}_c(T)$ and $\bar{\rho}_c(T)$ are denoted by $\hat{\mathrm{M}}_c(T)$ and $\overline{\mathrm{M}}_c(T)$ respectively. Thus

$$\hat{\mathrm{M}}_c(T) = \{\hat{\rho}_c(T)\}^{-1} \quad 7.4\text{-}35$$

and

$$\overline{\mathrm{M}}_c(T) = \{\bar{\rho}_c(T)\}^{-1} \quad 7.4\text{-}36$$

By equation 7.4-35 we have

$$\hat{\mathrm{M}}_c(T) < \overline{\mathrm{M}}_c(T) < \{\hat{r}_{adj}(T)\}^{-1} \quad 7.4\text{-}37$$

In Section 7.4.5 we will argue that $\bar{\rho}_c(T)$ generally provides a more accurate assessment of $\rho_c(T)$ than does $\hat{\rho}_c(T)$. However, somewhat surprisingly at first thought, in Section 7.4.5 we identify conditions under which $\hat{\mathrm{M}}_c(T)$ generally provides a more accurate assessment of $\mathrm{M}_c(T)$ than does $\overline{\mathrm{M}}_c(T)$.

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7.4.6 Reliability Growth Potential.

Consider the expression in (7.4-3) for $\rho(T)$, the expected system failure intensity after incorporation of the delayed fixes. If we let $T \to \infty$ and denote the resulting limit of $\rho(T)$ by ρ_{GP} :

$$\rho_{GP} \quad = \quad \lim_{T\to\infty} \rho(T) \quad = \quad \lambda_A + \sum_{i=1}^{K} (1-d_i)\lambda_i \qquad 7.4\text{-}38$$

The expression ρ_{GP} is called the growth potential failure intensity. Its reciprocal is referred to as the growth potential MTBF. The growth potential MTBF represents a theoretical upper limit on the system MTBF. This limit corresponds to the MTBF that would result if all B-modes were surfaced and corrected with specified fix effectiveness factors. Note ρ_{GP} is estimated by

$$\hat{\rho}_{GP} \quad = \quad \frac{1}{T}\left(N_A + \sum_{i\in obs} (1-d_i^*)N_i \right) \qquad 7.4\text{-}39$$

If the reciprocal $(\hat{\rho}_{GP})^{-1}$ lies below the goal MTBF then this may indicate that achieving the goal is high risk.

7.4.7 Maximum Likelihood Estimator versus the Unbiased Estimator for β.

Recall that the estimator $\bar{\beta}_m \quad = \quad \left(\frac{m-1}{m}\right)\hat{\beta}$ conditioned on $M(T)=m$, with $m \geq 2$, is unbiased for β, i.e. $E(\bar{\beta}_m) \quad = \quad \beta$. Furthermore the variances of $\bar{\beta}_m$ and $\hat{\beta}$, denoted by $V(\bar{\beta}_m)$ and $V(\hat{\beta})$ respectively, satisfy the following:

$$\begin{aligned} V(\bar{\beta}_m) \quad &= \quad V\left(\left(\frac{m-1}{m}\right)\hat{\beta}\right) \\ &= \quad \left(\frac{m-1}{m}\right)^2 V(\hat{\beta}) \quad < \quad V(\hat{\beta}) \end{aligned} \qquad 7.4\text{-}40$$

for $m \geq 2$. Equation 7.4-41 together with the unbiased property of $\bar{\beta}_m$, suggest that $\bar{\beta}_m$ provides a more accurate assessment of β than does $\hat{\beta}$.

Next consider the assessments of $h_c(T)$ based on $\hat{\beta}$ and $\bar{\beta}_m$. Recall the AMSAA/Crow Model assumes that $M(t)$, $t > 0$, is a non-homogeneous Poisson process with mean value function $\mu_c(t) \quad = \quad E(M(t)) \quad = \quad \lambda t^{\beta} \quad for \quad \lambda, \beta > 0$. Thus, in particular, $M(T)$ is Poisson distributed with mean $E(M(T)) \quad = \quad \lambda T^{\beta}$.

Using this fact, it can be shown that $\bar{h}_c(T)$ is an approximately unbiased estimator of $h_c(T)$ under most conditions of practical interest, where it is understood that $\bar{h}_c(T)$ denotes a conditional estimator, conditioned on $M(T) \geq 2$. To be more explicit, $\bar{h}_c(T)$, when viewed as an estimator (as opposed to an estimated value), is a random variable which is a function of $M(T)$

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and the random vector of B-mode first occurrence times $(T_1,\cdots,T_{M(T)})$. When $M(T)=m$ and $(T_1,\cdots,T_{M(T)})=(t_1,\cdots,t_m)$, the estimator $\bar{h}_c(T)$ takes on the value $\frac{m\,\bar{\beta}_m}{T}$ where

$$\bar{\beta}_m \;=\; \left(\frac{m-1}{m}\right)\hat{\beta} \;=\; \left(\frac{m-1}{m}\right)\left(\frac{m}{\sum_{i=1}^{m}\ln\left(\frac{T}{t_i}\right)}\right) \;=\; \frac{m-1}{\sum_{i=1}^{m}\ln\left(\frac{T}{t_i}\right)}$$

The estimator $\bar{h}_c(T)$ can be shown to satisfy the following:

$$E\left(\bar{h}_c(T)\right) \;\cong\; h_c(T) \qquad 7.4\text{-}41$$

provided $\Pr(M(T)=0) \;\cong\; 0$, where Pr denotes the probability function for $M(T)$.
Consider the variances of the estimators $\bar{h}_c(T)$ and $\hat{h}_c(T)$ conditioned on $M(T)=m$. For $m\geq 2$,

$$\begin{aligned} V\left(\bar{h}_c(T)\,|\,M(T)=m\right) \;&=\; V\left(\frac{m\,\bar{\beta}_m}{T}\right) \\ &=\; \left(\frac{m}{T}\right)^2 V\left(\bar{\beta}_m\right) \;=\; \left(\frac{m}{T}\right)^2 V\left(\left(\frac{m-1}{m}\right)\hat{\beta}\right) \\ &=\; \left(\frac{m}{T}\right)^2\left(\frac{m-1}{m}\right)^2 V\left(\hat{\beta}\right) \;=\; \left(\frac{m-1}{T}\right)^2 V\left(\hat{\beta}\right) \\ &<\; \left(\frac{m}{T}\right)^2 V\left(\hat{\beta}\right) \;=\; V\left(\frac{m\hat{\beta}}{T}\right) \\ &=\; V\left(\hat{h}_c(T)\,|\,M(T)=m\right) \end{aligned} \qquad 7.4\text{-}42$$

Now consider the variances of $\bar{h}_c(T)$ and $\hat{h}_c(T)$ conditioned on $M(T)\geq 2$. Since equation 7.4-43 holds for each $m\geq 2$, we have

$$V\left(\bar{h}_c(T)\,|\,M(T)\geq 2\right) \;<\; V\left(\hat{h}_c(T)\,|\,M(T)\geq 2\right) \qquad 7.4\text{-}43$$

Equations 7.4-42 and 7.4-44 suggest that the estimator $\bar{h}_c(T)$ provides a more accurate estimate of $h_c(T)$ than does the estimator $\hat{h}_c(T)$ when two or more distinct B-modes occur during [0,T]. We now investigate the bias of the estimators $\hat{\rho}_c(T)$ and $\bar{\rho}_c(T)$. To do so, let

$$\tilde{\rho}_c(T) \;=\; \frac{1}{T}\left\{N_A + \sum_{i\in obs}\left(1-d_i^*\right)N_i + \tilde{\beta}\sum_{i\in obs}d_i^*\right\}$$

where $\tilde{\beta}\in\left\{\hat{\beta},\bar{\beta}_m\right\}$. Also let $\tilde{h}_c(T) \;=\; \frac{m\,\tilde{\beta}}{T}$. By equation 7.4-27 and equation 7.4-30 we have

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$$\tilde{\rho}_c(T) = \left\{\frac{N_A}{T} + \sum_{i\in obs}(1-d_i^*)\frac{N_i}{T} + \left(\frac{1}{m}\sum_{i\in obs} d_i^*\right)\left(\frac{m\tilde{\beta}}{T}\right)\right\}$$
$$= \frac{N_A}{T} + \sum_{i=1}^{K}(1-d_i^*)\frac{N_i}{T} + \mu_d^*\,\tilde{h}_c(T)$$

Thus the expected value of $\tilde{\rho}_c(T)$ is

$$E(\tilde{\rho}_c(T)) = \lambda_A + \sum_{i=1}^{K}(1-d_i^*)\lambda_i + E\left(\mu_d^*\,\tilde{h}_c(T)\right) \qquad 7.4\text{-}44$$

Recall by equation 7.4-31,

$$\rho_c(T) = \lambda_A + \sum_{i=1}^{K}(1-d_i)\,\lambda_i + \mu_d\,h_c(T).$$

Thus by Equation 7.4-45, we have

$$E(\tilde{\rho}_c(T)) - \rho_c(T) =$$
$$\sum_{i=1}^{K}(d_i - d_i^*)\,\lambda_i + E\left\{(\mu_d^* - \mu_d)\,\tilde{h}_c(T)\right\} + \mu_d\left\{E(\tilde{h}_c(T)) - h_c(T)\right\} \qquad 7.4\text{-}45$$

By equation 7.4-46 we see that even if our assessments of μ_d and the d_i are perfect, the estimator $\tilde{\rho}_c(T)$ will have a residual bias of,

$$\mu_d\,\{E(\tilde{h}_c(T)) - h_c(T)\}.$$

To reduce this residual bias as much as possible, we wish to make the bias $E(\tilde{h}_c(T)) - h_c(T)$ as small as possible. Since $\bar{h}_c(T)$ is almost an unbiased estimator for $h_c(T)$, this suggests we use

$$\bar{\rho}_c(T) = \frac{1}{T}\left\{N_A + \sum_{i\in obs}(1-d_i^*)N_i + \bar{\beta}_m\sum_{i\in obs} d_i^*\right\}$$

to assess $\rho_c(T)$.

Next, we discuss the assessment of $M_c(T) = \{\rho_c(T)\}^{-1}$. To do so let $\tilde{M}_c(T) = \{\tilde{\rho}_c(T)\}^{-1}$. Thus

$$\tilde{M}_c(T) = \left[\frac{1}{T}\left\{N_A + \sum_{i\in obs}(1-d_i^*)N_i + \tilde{\beta}\sum_{i\in obs} d_i^*\right\}\right]^{-1}$$
$$= \left[\frac{N_A}{T} + \sum_{i\in obs}(1-d_i^*)\frac{N_i}{T} + \left(\frac{1}{m}\sum_{i\in obs} d_i^*\right)\tilde{h}_c(T)\right]^{-1} \qquad 7.4\text{-}46$$

Also

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$$\mathrm{M}_c(T) = \left[\lambda_A + \sum_{i=1}^{K}(1-d_i)\lambda_i + \mu_d h_c(T)\right]^{-1} \qquad 7.4\text{-}47$$

We have shown that to minimize the bias $E\{\tilde{\rho}_c(T)\} - \rho_c(T)$. we should use $\bar{\rho}_c(T)$ instead of $\hat{\rho}_c(T)$ to estimate $\rho_c(T)$. However, we wish to demonstrate that one should not infer from this that $\overline{\mathrm{M}}_c(T)$ must have a smaller bias than $\hat{\mathrm{M}}_c(T)$ as an estimator of $\mathrm{M}_c(T)$. To demonstrate this we will consider a simple case, the instance when the bias of $\tilde{\mathrm{M}}_c(T)$ is approximately equal to the bias of $\{\tilde{h}_c(T)\}^{-1}$. Thus in the following assume

$$E\left[\tilde{M}_c(T)\middle|M(T) \geq 2\right] - M_c(T)$$

$$\cong \mathrm{E}\left[\{\tilde{h}_c(T)\}^{-1}\middle| M(T) \geq 2\right] - \{h_c(T)\}^{-1} \qquad 7.4\text{-}48$$

One instance where equation 7.4-49 would be expected to hold is when $\lambda_A \cong 0$ and $d_i \cong 1$ for i = 1, …, K. For such conditions, we have by equation 7.3-48 that $M_c(T) \cong \{h_c(T)\}^{-1}$. Also, in such a case, it is reasonable that $d_i^* \cong 1$ for $i \in obs$ and, with high probability, $\frac{N_A}{T} \cong 0$. By equation 7.4-47, we see that such conditions would imply

$$E\left[\tilde{M}_c(T)\middle|M(T) \geq 2\right] \cong E\left[\{\tilde{h}_c(T)\}^{-1}\middle| M(T) \geq 2\right].$$

The above expectations and all subsequent expectations in this section are with respect to all the random quantities for given d_i^*, conditioned on $M(T) \geq 2$. These random quantities are the number of A-mode failures and the number of distinct B-modes experienced over [0, T], and the random vector of B-mode first occurrence times
(T_1, …, $T_{M(T)}$).

Now consider the expected values of $\{\hat{h}(T)\}^{-1}$ and $\{\bar{h}_c(T)\}^{-1}$conditioned on $M(T) \geq 2$. From the fact that the number of distinct B-modes occurring over $[0,T]$ is Poisson with mean λT^{β} it can be shown

$$\{h_c(T)\}^{-1} < E\left[\{\hat{h}_c(T)\}^{-1}\middle| M(T) \geq 2\right]$$
$$< E\left[\{\bar{h}_c(T)\}^{-1}\middle| M(T) \geq 2\right] \qquad 7.4\text{-}49$$

for $\mu(T) \geq 3.2$. Thus, when equation 7.4-49 holds, equation 7.4-50 implies,

$$E[\overline{\mathrm{M}}_c(T)|M(T) \geq 2] - M_c(T)$$

$$\cong E\left[\{\bar{h}_c(T)\}^{-1}\middle| M(T) \geq 2\right] - \{h_c(T)\}^{-1}$$

$$> E\left[\{\hat{h}_c(T)\}^{-1}\middle| M(T) \geq 2\right] - \{h_c(T)\}^{-1}$$

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$$\cong E\left[\hat{\mathrm{M}}_c(T)\middle|M(T)\geq 2\right]-M_c(T) \qquad 7.4\text{-}50$$

and

$$E\left[\hat{\mathrm{M}}_c(T)\middle|M(T)\geq 2\right]-M_c(T)$$
$$\cong E\left[\left\{\hat{h}_c(T)\right\}^{-1}\middle|M(T)\geq 2\right]-\{h_c(T)\}^{-1}>0 \qquad 7.4\text{-}51$$

Equations 7.4-51 and 7.4-52 show that for the case considered we should anticipate that $\hat{M}_c(T)$ and $\bar{M}_c(T)$ will have positive biases with the bias of $\bar{M}_c(T)$ larger than that of $\hat{M}_c(T)$. It has not been established whether this holds more generally. If there is concern that $\hat{M}_c(T)$ will have a positive bias and that the bias of $\bar{M}_c(T)$ will exceed that of $\hat{M}_c(T)$, then one may wish to assess $\mathrm{M}_c(T)$ by the more conservative estimator $\hat{M}_c(T)$ (recall $\hat{M}_c(T) < \bar{M}_c(T)$ for $M(T) \geq 2$).

7.4.8 Example.

The following example is taken from [1] and illustrates application of the AMSAA/Crow projection model. Data were generated by a computer simulation with $\lambda_A = 0.02$, $\lambda_B = 0.1$, $K = 100$ and the d_i's distributed according to a beta distribution with mean 0.7. The simulation portrayed a system tested for $T = 400$ hours. The simulation generated $N = 42$ failures with $N_A = 10$ and $N_B = 32$. The thirty-two B-mode failures were due to M=16 distinct B-modes. The B-modes are labeled by the index i where the first occurrence time for mode i is t_i and $0 < t_1 < t_2 < \cdots < t_{16} < T = 400$. These same data plus 12 additional failure modes fixed during testing (Crow's BC-modes) will be used in a subsequent example.

TABLE XVI lists, for each B-mode i, the time of first occurrence followed by the times of subsequent occurrences (if any). Column 3 of the table lists N_i, the total number of occurrences of B-mode i during the test period. Column 4 contains the assessed fix effectiveness factors for each of the observed B-modes. Column 5 has the assessed expected number of type i B-modes that would occur in T=400 hours after implementation of the fix. Finally, the last column contains the base e logarithms of the B-mode first occurrence times. These are used to calculate $\hat{\beta}$.

TABLE XVI. ACPM Projection Example Data

B-mode	Failure Times (hrs)	N_i	d_i^*	$\left(1-d_i^*\right)N_i$	$\ln t_i$
1	15.04, 254.99	2	.67	.66	2.7107
2	25.26, 120.89, 366.27	3	.72	.84	3.2292
3	47.46, 350.2	2	.77	.46	3.8599
4	53.96, 315.42	2	.77	.46	3.9882
5	56.42, 72.09, 339.97	3	.87	.39	4.0328
6	99.57, 274.71	2	.92	.16	4.6009
7	100.31	1	.50	.50	4.6083
8	111.99, 263.47, 373.03	3	.85	.45	4.7184
9	125.48, 164.66, 303.98	3	.89	.33	4.8321
10	133.43, 177.38, 324.95, 364.63	4	.74	1.04	4.8936

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11	192.66	1	.70	.30	5.2609
12	249.15, 324.47	2	.63	.74	5.5181
13	285.01	1	.64	.36	5.6525
14	379.43	1	.72	.28	5.9387
15	388.97	1	.69	.31	5.9635
16	395.25	1	.46	.54	5.9795
Totals		32	11.54	7.82	75.7873

From equation 7.4-1 and TABLE XVI, the adjustment procedure estimate of $r(T) = r(400)$ is

$$\hat{r}_{adj}(400) = \left(\frac{1}{400}\right)\left(N_A + \sum_{i=1}^{16}\left(1 - d_i^*\right)N_i\right)$$

$$= \frac{10 + 7.82}{400} = 0.04455$$

Thus the adjustment procedure estimate of the system MTBF is

$$\left\{\hat{r}_{adj}(400)\right\}^{-1} = \frac{400}{17.82} = 22.45$$

Looking at Equation 7.4-40, we can see that the adjustment procedure estimate of system failure intensity after implementation of the fixes is simply $\hat{\rho}_{GP}$, the estimated growth potential failure intensity. Thus

$$\hat{\rho}_{GP} = \hat{r}_{adj}(400) = 0.04455$$

Also, the estimate of the system growth potential MTBF is

$$\hat{\rho}_{GP}^{-1} = \left\{\hat{r}_{adj}(400)\right\}^{-1} = 22.45$$

To obtain an estimate with less bias of the system's failure intensity and corresponding MTBF at T=400 hours, after incorporation of fixes to the sixteen surfaced B-modes, we use the AMSAA/Crow model estimation equation 7.4-32. This projection is given by

$$\hat{\rho}_c(400) = \hat{\rho}_{GP} + \left(\frac{\hat{\beta}}{400}\right)\sum_{i \in obs} d_i^*$$

$$= 0.04455 + \left(\frac{\hat{\beta}}{400}\right)(11.54) \qquad 7.4\text{-}53$$

The MLE $\hat{\beta}$ is obtained from Equation 7.4-26, i.e.,

$$\hat{\beta} = \frac{m}{\sum_{i=1}^{m}\ln\left(\frac{T}{t_i}\right)} = \frac{m}{m\ln T - \sum_{i=1}^{m}\ln t_i}$$

$$= \frac{16}{16\ln 400 - 75.7873} = 0.7970$$

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Thus, by equation 7.4-53, the AMSAA/Crow projection for the system failure intensity, based on $\hat{\beta}$, is

$$\begin{aligned}\hat{\rho}_c(400) &= 0.04455 + \left(\frac{0.7970}{400}\right)(11.54) \\ &= 0.06754\end{aligned}$$

The corresponding MTBF projection is

$$\{\hat{\rho}_c(400)\}^{-1} = 14.81$$

A nearly unbiased assessment of the system failure intensity, for $d_i^* = d_i$, can be obtained by using $\overline{\beta}_m$ instead of $\hat{\beta}$. Recall by equation 7.4-28,

$$\overline{\beta}_m = \left(\frac{m-1}{m}\right)\hat{\beta} = \left(\frac{15}{16}\right)(0.7970) = 0.7472$$

By equation 7.4-33, the projected system failure intensity based on $\overline{\beta}_m$ is

$$\begin{aligned}\overline{\rho}_c(400) &= \hat{\rho}_{GP} + \frac{\overline{\beta}_m}{T}\sum_{i\in obs} d_i^* \\ &= 0.04455 + \left(\frac{0.7472}{400}\right)(11.54) \\ &= 0.06611\end{aligned}$$

The corresponding MTBF projection is $\{\overline{\rho}_c(400)\}^{-1} = 15.13$

As discussed in Section 7.4.5, we recommend basing the projected system failure intensity on $\overline{\rho}_c(T)$ which uses $\overline{\beta}_m$, but assess the projected system MTBF by using $\hat{\beta}$. Thus in this example we would recommend assessing the projected system failure intensity by

$$\overline{\rho}_c(400) = 0.06611,$$

and the projected system MTBF by

$$\{\hat{\rho}_c(400)\}^{-1} = 14.81.$$

7.5 The Crow Extended Reliability Projection Model.

7.5.1 Background.

The Extended Reliability Growth Projection Model for test-fix-find-test was developed by Crow and presented at RAMS in 2004 [10] to address the common and practical case where some corrective actions are incorporated during test and some corrective actions are delayed and incorporated at the end of the test. This model extends the AMSAA/Crow Model for test-fix-test and the ACPM for test-find-test data. That is, these other two AMSAA/Crow models are special cases of the Crow Extended Model.

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In order to provide the assessment and management metric structure for corrective actions during and after a test, two types of B-modes are defined. BC failure modes are corrected during test. BD failure modes are delayed to the end of the test. A-mode failures are those failure modes that will not receive a corrective action. These classifications define the management strategy and can be changed. The AMSAA/Crow basic test-fix-test model does not utilize the failure mode designation. The ACPM model for test -find-test data utilizes failure mode designation for the situation of A modes and BD modes only. The BC and BD failure mode designation is an important practical aspect of the Extended Model.

Note that in the failure mode designation BC modes are entirely different than BD modes. For example, mode BC-1 would be an entirely different failure mode from failure BD-1 although both have a similar sub designation "1." The test-fix-find-test strategy will fix more failure modes than with the test-fix-test management strategy. During test the A and BD failure modes do not contribute to reliability growth. The corrective actions for the BC failure modes affect the increase in the system reliability during the test. After the incorporation of corrective actions for the BD failure modes at the end of the test, the reliability increases further, typically as a jump. Estimating this increased reliability with test-fix-find-test data is the objective of the Crow Extended Model.

For the Crow Extended Model the achieved MTBF, before delayed fixes due to BC corrective actions should be exactly the same as the achieved failure intensity λ_{CA} for the AMSAA/Crow Model for test-fix-test data. To allow for BC failure modes in the Extended Model, replace λ_S by λ_{CA} and recalling that $\hat{\lambda}_A = \hat{\lambda}_S - \hat{\lambda}_B$, the estimated AMSAA/Crow projected failure intensity is given by the following:

$$\hat{\lambda}_P = \hat{\lambda}_A + \sum_{j=1}^{M}(1-d_j)\frac{N_j}{T} + \bar{d}\hat{h}(T)$$

or

$$\lambda_P = \lambda_{CA} - \lambda_B + \sum_{i=1}^{K}(1-d_i)\lambda_i + dh(T|BD) \qquad 7.5\text{-}1$$

Also, let λ_{BD} be the constant failure intensity for the BD failure modes, and let h(T|BD) be the first occurrence function for the BD failure modes.

The Crow Extended Model projected failure intensity is

$$\lambda_{EM} = \lambda_{CA} - \lambda_{BD} + \sum_{i=1}^{K}(1-d_i)\lambda_i + dh(T|BD) \qquad 7.5\text{-}2$$

The Crow Extended Model projected MTBF is $M_{EM} = 1/\lambda_{EM}$. This is the MTBF after the incorporation of the delayed BD failure modes that we wish to estimate.

Under the Crow Extended Model, the achieved failure intensity, before the incorporation of the delayed BD failure modes, is the first term λ_{CA}. The achieved MTBF at time T before the BD failure modes is $M_{CA} = [\lambda_{CA}]^{-1}$. That is, the achieved MTBF before delayed fixes for the Crow

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Extended Model is exactly the same as the achieved MTBF for the AMSAA/Crow Model for test-fix-test.

The estimate of the projected failure intensity for the model is

$$\hat{\lambda}_{EM} = \hat{\lambda}_{CA} - \hat{\lambda}_{BD} + \sum_{j=1}^{M}(1 - d_j)\frac{N_j}{T} + \bar{d}\hat{h}(T|BD) \qquad 7.5\text{-}3$$

If it is assumed that no corrective actions are incorporated into the system during the test (no BC failure modes), then this is equivalent to assuming that $\beta = 1$ for $\lambda_{CA}$and $\lambda_{CA}$is estimated by $\hat{\lambda}_{CA} = \hat{\lambda}_A + \hat{\lambda}_B$. In general, the assumption of constant failure intensity ($\beta = 1$) can be assessed by a statistical test from the data. For details on estimation and application of the Extended Model refer to [10].

In using the Crow Extended Model, it is important that the classification of a B-mode with respect to the BC and BD categories not be dependent on when the mode occurs during the test phase. In some testing programs, modes that occur in the early portion of the test phase tend to have fixes implemented during the test and are thus classified as BC, while those that occur later are not implemented until after the test phase and are thus classified BD. Under such conditions, the pattern of BD first occurrence times will provide an inaccurate estimate of the failure rate due to the unobserved BD failure modes. This in turn would degrade the accuracy of the MTBF projection.

The test-fix-find-test concept is illustrated in Table XVII and denotes those failure modes that received a corrective action during the test (BC modes) and also those failure modes that will receive a corrective action at the end of the test (BD modes). TABLE XVIII presents the BD Mode first occurrences, frequency of the modes and the effectiveness factors d_i. During test the A and BD failure modes do not contribute to reliability growth. The corrective actions for the BC failure modes affect the increase in the system reliability during the test. After the incorporation of corrective actions for the BD failure modes, the reliability increases. Estimating this increased reliability with test-fix-find-test data is the objective of this model.

For the Extended Model, we assume that the achieved MTBF before delayed fixes, based on Table XVII data should be exactly the same as the achieved MTBF applying the AMSAA/Crow model to all the data. If K is the total number of distinct BD modes then the intensity to be estimated is

$$\lambda_P = \lambda_S - \lambda_B + \sum_{i=1}^{K}(1 - d_i)\lambda_i + dh(T) \qquad 7.5\text{-}4$$

To allow for BC failure modes in the extended model we replace λ_S by λ_{CA} in equation 7.4-4. Also, let λ_{BD} be the constant failure intensity for the BD failure modes, and let $h(T|BD)$ be the first occurrence function for the BD failure modes.

The Extended Model projected failure intensity is

$$\lambda_{EM} = \lambda_{CA} - \lambda_{BD} + \sum_{i=1}^{K}(1 - d_i)\lambda_i + dh(T|BD). \qquad 7.5\text{-}5$$

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The Extended Model projected MTBF is $M_{EM} = 1/\lambda_{EM}$. This is the MTBF after the incorporation of the delayed BD failure modes that we wish to estimate.

Under the Extended Model the achieved failure intensity, before the incorporation of the delayed BD failure modes, is the first term λ_{CA}. The achieved MTBF at time T before the BD failure modes is $M_{CA} = [\lambda_{CA}]^{-1}$. That is, the achieved MTBF before delayed fixes for the data in TABLE XVI is exactly the same as the achieved MTBF for the AMSAA/Crow model.

TABLE XVII. Test-Fix-Find Test Failure Times and Failure Mode Designations

1	0.7	BC1	29	192.7	BD11
2	3.7	BC1	30	213.0	A
3	13.2	BC1	31	244.8	A
4	15	BD1	32	249.0	BD12
5	17.6	BC2	33	250.8	A
6	25.3	BD2	34	260.1	BD1
7	47.5	BD3	35	263.5	BD8
8	54.0	BD4	36	273.1	A
9	54.5	BC3	37	274.7	BD6
10	56.4	BD5	38	282.8	BC11
11	63.6	A	39	285.0	BD13
12	72.2	BD5	40	304.0	BD9
13	99.2	BC4	41	315.4	BD4
14	99.6	BD6	42	317.1	A
15	100.3	BD7	43	320.6	A
16	102.5	A	44	324.5	BD12
17	112.0	BD8	45	324.9	BD10
18	112.2	BC5	46	342.0	BD5
19	120.9	BD2	47	350.2	BD3
20	121.9	BC6	48	355.2	BC12
21	125.5	BD9	49	364.6	BD10
22	133.4	BD10	50	364.9	A
23	151.0	BC7	51	366.3	BD2
24	163.0	BC8	52	373.0	BD8
25	164.7	BD9	53	379.4	BD14
26	174.5	BC9	54	389.0	BD15
27	177.4	BD10	55	394.9	A
28	191.6	BC10	56	395.2	BD16

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TABLE XVIII. BD Failure Mode Data and Effectiveness Factors

BD Mode j	N_j	First Occurrence	FEF d_j
1	2	15.0	0.67
2	3	25.3	0.72
3	2	47.5	0.77
4	2	54.0	0.77
5	3	56.4	0.87
6	2	99.6	0.92
7	1	100.3	0.50
8	3	112.0	0.85
9	3	125.5	0.89
10	4	133.4	0.74
11	1	192.7	0.70
12	2	249.0	0.63
13	1	285.0	0.64
14	1	379.4	0.72
15	1	389.0	0.69
16	1	395.2	0.46
Total	32		11.54

As a note, these same data will be used to analyze a "baseline" using the AMSAA/Crow Tracking Model, then they will be used to illustrate the Crow Extended Model (7.5) (wherein, the B-modes will be divided into fixes implemented in test (BC) and fixes implemented after the test phase (BD)), the ACPM Model (7.4) and lastly the AMPM (7.6).

7.5.2 AMSAA/Crow Tracking Example.

Suppose a development testing program begins at time 0 and is conducted until time T and stopped. Let N be the total number of failures recorded and let $0 < X_1, < X_2, \cdots, X_N < T$ denote the N successive failure times on a cumulative time scale. We assume that the AMSAA/Crow NHPP assumption applies to this set of data. Under the AMSAA/Crow Model the MLEs for λ and β (numerator of MLE for β adjusted from N to N-1 to obtain an unbiased estimate) are

$$\widehat{\lambda} = \frac{N}{T^{\widehat{\beta}}} \quad , \qquad \overline{\beta} = \frac{N-1}{\sum_{i=1}^{N} ln\frac{T}{X_i}}.$$

Applying these equations to the 56 failure times in TABLE XVII we get estimates of

$$\hat{\lambda} = 0.2397 \text{ and } \bar{\beta} = 0.9103 \ (\hat{\beta} = 0.0872).$$

While growth is small, hypothesis testing indicates it being significantly different from 0. Thus growth is occurring and not constant or exponential.

The achieved or demonstrated failure intensity and MTBF at T=400 are estimated by

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$$\hat{\lambda}_{CA} = \hat{\lambda}\hat{\beta}T^{\hat{\beta}-1} \quad \text{or} \quad \hat{\lambda}_{CA} = 0.1274, \text{ and } \hat{M}_{CA} = \left[\hat{\lambda}_{CA}\right]^{-1} = 7.84.$$

And so our "baseline" estimate of reliability at T=400 hours is 7.84 using the AMSAA/Crow tracking model. It is important to note that the AMSAA/Crow test-fix-test model does not assume that all failures in the data set (e.g., above table) receive a corrective action. Based on the management strategy some failures may receive a corrective action and some may not.

For information purposes it is noted that in Crow's 2004 RAMS paper, his example 2 for test-find-test was presented here in Section 7.4.6.

7.5.3 Estimation for the Extended Reliability Growth Model.

The estimate of the projected failure intensity for the Extended Model is

$$\hat{\lambda}_{EM} = \hat{\lambda}_{CA} - \hat{\lambda}_{BD} + \sum_{j=1}^{M}(1 - d_j)\frac{N_j}{T} + \bar{d}\hat{h}(T|BD) \qquad 7.5\text{-}6$$

where the first term is the AMSAA/Crow model estimate applied to all A, BC, and BD data, and the remaining terms are calculated as in the 7.4.6. Example, using only the BD data in Tables XVIII and XIV.

If it is assumed that no corrective actions are incorporated into the system during the test (no BC failure modes), then this is equivalent to assuming that $\beta = 1$ for λ_{CA}, and this term is estimated as in the example in (7.5-6). In general, the assumption of a constant failure intensity *(β=1)* can be assessed by a statistical test from the data.

7.5.4 Test-Fix-Find-Test Example.

In TABLE XVII there are 56 total failures each denoted as a A, BC or BD-mode, and total test time of T=400 hours. For the current example assume that all the failure times X_j are known. There are BC failure modes but assume in this example that only the BD failure modes are designated. Assume the remaining A and BC failure modes are not designated as such.

The first term $\hat{\lambda}_{CA}$, (7.5-6) uses all failure time data in TABLE XVII, as in the AMSAA/Crow Tracking example in 7.5.2. This gives

$$\hat{\lambda}_{CA} = 0.1274. \qquad 7.5\text{-}7$$

For the remaining terms in (7.5-6) the BD data in TABLE XVII and the FEFs given in TABLE XVIII are used. This BD data is the same B data in the 7.4-6 Example so the calculations in (7.5-6) are the same. That is,

M=16, T=400, λ_{BD} = 0.08, $\bar{d}$=0.72.

Also,

$$\sum_{i=1}^{M}(1 - d_j)\frac{N_j}{T} = 0.0196, \text{ and} \qquad 7.5\text{-}8$$

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$\hat{h}(T|BD)$ has parameters $\hat{\lambda}$=0.1820, $\hat{\beta}$=0.7472.

This gives $\bar{d}h(T|BD) = 0.0215$. Therefore,

$$\hat{\lambda}_{EM} = 0.1274 - 0.08 + 0.0196 + 0.0215 \qquad 7.5\text{-}9$$

or

$$\hat{\lambda}_{EM} = 0.0885. \qquad 7.5\text{-}10$$

The Extended Model projected MTBF is $\widehat{M}_{EM} = 11.29$.
The achieved MTBF before the 16 delayed fixes is estimated by $\widehat{M}_{CA} = 7.84$. We therefore have based on the Extended Model estimates that the MTBF grew to 7.84 as a result of corrective actions for BC failure modes during the test, and then jumped to 11.29 as a result of the delayed corrected actions after the test for the BD failure modes.

7.5.5 ACPM Example Using Crow Extended Data.

Now, using the data from Tables XVII and XVIII, we look at applying the AMSAA/Crow Projection Model which projects reliability at the start of the next test phase based on the implementation of the delayed fixes. The ACPM assumes all modes are delayed, which means in this example our BC-modes are assumed to be delayed rather than incorporated in testing, so that system failure rate is estimated by the following:

$$\hat{\rho}_S(T) = \hat{\lambda}_A + \hat{\rho}_{BC,BD}(T)$$

where $\hat{\lambda}_A = \frac{N_A}{T}$ and

$$\hat{\rho}_{BC,BD} = \left[\sum_{i \epsilon obs\,(BC)} (1 - d_i^*)\left(\frac{N_{BC,i}}{T}\right)\right]\left[\sum_{j \epsilon obs\,(BD)} \left(1 - d_j^*\right)\left(\frac{N_{BD,j}}{T}\right)\right] + d_{avg}^*\left(\frac{\hat{\beta}_{BC,BD}(N_{BC}+N_{BD})}{T}\right).$$

The associated projected system MTBF is $\widehat{M}_S(T) = \{\hat{\rho}_S(T)\}^{-1}$.
In order to use this approach all of the BC-modes are assumed to have an FEF of 0.72, the average FEF for the BD-modes.

$\hat{\lambda}_A = 0.025,\ \hat{\rho}_{BC,BD} = 0.068$

Failure intensity $\hat{\beta}$	.093	mtbf 10.745
Failure intensity $\bar{\beta}$	.092	mtbf 10.896
GP intensity	.057	mtbf 17.508

So, at least for this example, it is noted that Crow's Extended Model results in a projected value at the beginning of the next test phase of 11.29 as compared to the ACPM result of 10.90 (using $\bar{\beta}$), a difference of 0.39. The difference in part may be due to the assumption of the BC-mode fixes being delayed has in using the ACPM model.

7.5.6 Extended Reliability Growth Model with Pre-emptive Corrective Actions.

Suppose that in addition to delayed fixes, there are also pre-emptive fixes at time T for failure modes that have not experienced a failure. We assume that these failure modes are in fact B-

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modes in the sense that they would have received delayed corrective actions if they had occurred during the test. Let Q be the number of pre-emptive failure modes receiving a fix at time T, and denote these by $BP_1, BP_2, \cdots .BP_Q$. For each of these failure modes we assume that we have an estimate (by analysis, analogy, or test) of the failure rate l_q before the corrective action, and estimate of the corresponding effectiveness factor, $d_q \cdot q = 1, \ldots, Q$. That is, the failure rate is estimated by l_q before the corrective action and by $(1-d_q)\lambda_q$ after the corrective action. The Extended Model failure intensity estimate, λ^*_{EM}, with the Q pre-emptive corrective actions is the right hand side of 7.5-11 is the modification of $h(t)$ for the BD's due to the pre-emptive mitigation. This is not the corresponding total system failure intensity as implied in equation 7.5-11.

$$\lambda^*_{EM} = h_{BD}(T) - \sum_{i=1}^{Q} \lambda_{i,BD} + \sum_{i=1}^{Q} (1 - d_i)\lambda_{i,BD}$$
$$= h_{BD}(T) - \sum_{i=1}^{Q} d_i \, \lambda_{i,BD} \qquad 7.5\text{-}11$$

Example. Test-Fix-Find-Test with Pre-emptive Corrective Actions.
Consider again the previous example but suppose that at the end of the 400 hour test Q=3 pre-emptive Corrective actions were incorporated into the system in addition to the 16 delayed fixes. Let

$$\lambda_{BP_1} = 0.0010, \lambda_{BP_2} = 0.0005, \lambda_{BP_3} = 0.0007$$

$$d_{BP_1} = 0.60, d_{BP_2} = 0.85, d_{BP_3} = 0.65.$$

Then,

$$\lambda^*_{EM} = 0.0885 - 0.00159 = 0.0869.$$

The corresponding MTBF is $\widehat{M}^*_{EM} = 11.50$.

By incorporating the additional 3 pre-emptive corrective actions the MTBF jumped from 11.29 to 11.50.

7.5.7 Extended Model Management and Maturity Metrics.

In Crow's paper he includes a section of 33 management and maturity metrics, useful for providing management and engineering some practical situations. These are illustrated using data from the examples presented here. The metrics are not presented here; they may be found in the RAMS reference cited earlier.

7.6 The AMSAA Maturity Projection Model (AMPM).

7.6.1 Introduction.

The continuous version of the AMPM assumes the test duration is measured in a continuous scale such as time or miles. Throughout this section AMPM will refer to the continuous version of the model and we will refer to time as the measure of test duration.

The AMPM addresses making reliability projections in several situations of interest. One case corresponds to that addressed by the AMSAA/Crow projection model introduced in [1] and

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discussed in Section 7.4. This is the situation in which all fixes to B-modes are implemented at the end of the current test phase, Phase I, prior to commencing a follow-on test phase, Phase II. The projection problem is to assess the expected system failure intensity at the start of Phase II. Another situation handled by the AMPM estimation procedure is the case where the reliability of the unit under test has been maturing over Test Phase I due to implemented fixes during Phase I. This case includes the situations where

a. all surfaced B-modes in Test Phase I have fixes implemented within this test phase or
b. some of the surfaced B-modes are addressed by fixes within Test Phase I and the remainder are treated as delayed fixes, i.e., are fixed at the conclusion of Test Phase I, prior to commencing Test Phase II.

A third type of projection of interest involves projecting the system failure intensity at a future program milestone. This future milestone may occur beyond the commencement of the follow-on test phase.

All the above type of projections are based on the Phase I B-mode first occurrence times, whether the associated B-mode fix is implemented within the current test phase or delayed (but implemented prior to the projection time). In addition to the B-mode first occurrence times, the projections are based on an average fix effectiveness factor (FEF). This average is with respect to all the potential B-modes, whether surfaced or not. However, as in the AMSAA/Crow model, this average FEF is assessed based on the surfaced B-modes. For the AMPM model, the set of surfaced B-modes would typically be a mixture of B-modes addressed with fixes during the current test phase as well as those addressed beyond the current test phase.

In some instances, a reliability projection for a future milestone can be based on extrapolating a reliability growth tracking curve. Such a curve only utilizes cumulative failure times and does not use B-mode fix effectiveness factors. This is a valid projection approach provided it is reasonable to expect that the observed pattern of reliability growth will continue up through the milestone of interest. However, this pattern could change in a pronounced manner. Reasons for such a change include:

a. a change in the test environment;
b. a different level of future resources to analyze and implement effective corrective actions; and
c. jumps in reliability due to delayed fixes.

If extrapolating the current tracking curve is not deemed suitable due to considerations such as above, the AMPM projection methodology may be useful. Unlike assessments based on the tracking model, the AMPM assessments are independent of the fix discipline, as long as the fixes are implemented prior to the projection milestone date of interest. The AMPM (as well as the Crow/AMSAA projection model) utilizes a non-homogeneous Poisson process with regard to the number of distinct B-modes that occur by test duration t. The associated pattern of B-mode first occurrence times is not dependent on the corrective action strategy, under the assumption that corrective actions are not inducing new B-modes to occur. Thus the AMPM assessment procedure is not upset by jumps in reliability due to delayed groups of fixes. In contrast, reliability growth tracking curve methodology utilizes the pattern of cumulative failure times.

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Such a pattern is sensitive to the corrective action strategy. Thus a reliability growth tracking curve model may not be appropriate for fitting failure data or for extrapolating due to a corrective action strategy that is not compatible with the model.

Note that AMPM reliability projections for a future milestone would be optimistic if corrective actions beyond the current test phase were less effective than the average FEF assessment based on B-modes surfaced through the current test phase. Also, a change in the future testing environment could result in a new set of potential failure modes or affect the rates of occurrence of the original set of failure modes. Either of these circumstances would tend to degrade the accuracy of the AMPM reliability projection.

Another instance in which a reliability projection model would be useful is when the current test phase contains a number of design configurations of the units under test due to incorporation of reliability fixes during the test phase. If there is a lack of fit of the reliability growth tracking model over these configurations then the tracking model should not be used to assess the reliability of the latest configuration or for extrapolation to a future milestone. Such a lack of fit may be due to the corrective action process, i.e., when the fixes are implemented and their effectivity. As pointed out earlier, the AMPM, unlike a tracking model, is insensitive to any nonsmoothness in the expected number of failures versus test time that results from the timing or effectivity of corrective actions. Thus in such a situation, program management may wish to use a projection method such as the AMPM to assess the reliability of the current configuration or to project the expected reliability at a future milestone.

The AMPM can also be used to construct a useful reliability maturity metric. This metric is the fraction of the expected initial system B-mode failure intensity, λ_B, surfaced by test duration t. By this we mean the expected fraction of λ_B due to B-modes surfaced by t. This concept will be expanded upon in a later subsection.

Prior to presenting the model equations and estimation procedures, we will list the associated notation and assumptions.

7.6.2 List of Notations

μ_d	Arithmetic average of the d_i, i.e., $\left(1/K\right)\sum_{i=1}^{K} d_i$
α, β	Parameters for gamma density function, where $\alpha > -1$ and $\beta > 0$ (subscripted by K or ∞ where required for clarity).
$\alpha!$	Denotes the integral $\int_0^\infty x^\alpha e^{-x} dx$ for $\alpha > -1$.
Λ	Gamma random variable.
Ψ	Moment generating function for Λ.
$\Gamma(\alpha, \beta)$	Denotes gamma random variable with parameters $\alpha > -1$, $\beta > 0$.
f_Λ	Denotes density function for $\Lambda \sim \Gamma(\alpha, \beta)$, where

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	$f_{\Lambda}(\lambda) = \frac{\lambda^{\alpha} e^{-\frac{\lambda}{\beta}}}{\alpha! \beta^{\alpha+1}}$ for $\lambda > 0$; = 0 elsewhere
$\underline{\Lambda} = (\Lambda_1, \cdots, \Lambda_K)$	Random sample of size K from $\Gamma(\alpha, \beta)$.
$\underline{\lambda} = (\lambda_1, \cdots, \lambda_K)$	Realization of $\underline{\Lambda}$.
$\lambda_{B,K}$	Expected value of $\sum_{i=1}^{K} \Lambda_i$.
$\lambda_{B,\infty} =$	$\lim_{K \to \infty} \lambda_{B,K}$
$\mu(t; \underline{\lambda})$	Expected number of distinct B-modes conditioned on $\underline{\Lambda} = \underline{\lambda}$
$h(t; \underline{\lambda})$	Expected rate of occurrence of B-modes given $\underline{\Lambda} = \underline{\lambda}$.
$h(t)$	Unconditional expected B-mode rate of occurrence.
$r(t; \underline{\lambda})$	System failure intensity after fixes to B-modes surfaced by t have been implemented, conditioned on $\underline{\Lambda} = \underline{\lambda}$.
$\rho(t; \underline{\lambda})$	Expected value of $r(t; \underline{\lambda})$ with respect to random first occurrence times of B-modes.
$\rho(t)$	Expectation of $\rho(t; \underline{\Lambda})$ with respect to $\underline{\Lambda}$.
$I_i(t)$	Equals 1 if B-mode i occurs by t, equals 0 otherwise.
$u(t; \underline{\lambda})$	Failure intensity at time t due to unsurfaced B-modes, conditioned on $\underline{\Lambda} = \underline{\lambda}$.
$s(t)$	Unconditional expected failure intensity due to set of B-modes surfaced by t, in absence of any fixes.
$\theta(t)$	Expected fraction of $\lambda_{B,K}$ surfaced as a function of t.
t_i	Time of first occurrence of B-mode i.
$\underline{t} = (t_1, \cdots, t_m)$	
$L(m, \underline{t}, \underline{\lambda})$	Likelihood function for the test data $(m, \underline{t})$ given $\underline{\Lambda} = \underline{\lambda}$.
$L(m, \underline{t})$	Expectation of $L(m, \underline{t}, \underline{\Lambda})$.
ln	Natural logarithm (base "e").
Z	$\ln\{L(m, \underline{t})\}$
v_K	(α, β, K)
$\hat{v}_K$	$(\hat{\alpha}_K, \hat{\beta}_K, K)$
obs	Set of indices associated with m observed B-modes.
K_0	Greatest lower bound for set of K-values for which AMPM MLE's are well defined
K_{IBM}	IBM model MLE of K.
$\underline{\underline{\Delta}}$	Defined to be.

7.6.3 Assumptions.

Additional Assumptions for AMPM – Continuous:

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a. The time to first occurrence is exponentially distributed for each failure mode.
b. For $i = 1,2,\cdots,K$, the effectiveness of a fix associated with B-mode *i* is independent of the mode's initial rate of occurrence λ_i.
c. The B-mode initial rates of occurrence $(\lambda_1,\cdots,\lambda_K)$ constitute the realization of a random sample $(\Lambda_1,\cdots,\Lambda_K)$ from a gamma distribution with density f_Λ. This models mode-to-mode variation in the B-mode initial failure rates. That is, we assume the $\Lambda_i\ (i = 1,\cdots,K)$ are independent and identically distributed (IID) random variables, where $\Lambda_i \sim \Gamma(\alpha,\beta)$.

7.6.4 AMPM Development.

The AMPM provides a procedure for assessing the system failure intensity $r(t;\underline{\lambda})$. Recall $r(t;\underline{\lambda})$ denotes the system failure intensity after fixes to all B-modes surfaced by test time t have been implemented.

Note $\underline{\lambda} = (\lambda_1,\cdots,\lambda_K)$ denotes the initial B-mode rates of occurrence. In particular, consider B-mode *i*. If this mode does not occur by *t* then its rate of occurrence at *t* is still λ_i. However, if B-mode *i* occurs by *t* then, by our definition of $r(t;\underline{\lambda})$, the contribution of this mode to $r(t;\underline{\lambda})$ is only $(1-d_i)\lambda_i$ due to the implemented fix (or fixes) to mode *i* by *t*. We may conveniently mathematically express the contribution of B-mode *i* to $r(t;\underline{\lambda})$ by

$$\{1 - d_i I_i(t)\}\lambda_i \qquad 7.6\text{-}1$$

Thus

$$r(t;\underline{\lambda}) = \lambda_A + \sum_{i=1}^{K}\{1 - d_i I_i(t)\}\lambda_i = \lambda_A + \sum_{i=1}^{K}\lambda_i - \sum_{i=1}^{K} d_i\lambda_i I_i(t) \qquad 7.6\text{-}2$$

As in the AMSAA/Crow model, the AMPM assesses the system failure intensity $r(t;\underline{\lambda})$ by an assessment of the expected value of $r(t;\underline{\lambda})$, i.e. $\rho(t;\underline{\lambda}) = E(r(t;\underline{\lambda}))$. Note by (7.6-2) we have

$$\begin{aligned}\rho(t;\underline{\lambda}) &= E(r(t;\underline{\lambda})) \\ &= \lambda_A + \sum_{i=1}^{K}\lambda_i - \sum_{i=1}^{K} d_i\lambda_i E(I_i(t))\end{aligned} \qquad 7.6\text{-}3$$

In Appendix G, Annex 1 we show,

$$E[I_i(t)] = 1 - e^{-\lambda_i t} \qquad 7.6\text{-}4$$

where the expectation is with respect to the time of first occurrence of B-mode *i*. This yields

$$\rho(t;\underline{\lambda}) = \lambda_A + \sum_{i=1}^{K}(1-d_i)\lambda_i + \sum_{i=1}^{K} d_i\lambda_i e^{-\lambda_i t} \qquad 7.6\text{-}5$$

In Section 7.4 (where the argument $\underline{\lambda}$ was suppressed) it was noted that the AMSAA/Crow model approximates $\rho(t;\underline{\lambda})$ by

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$$\rho_c(t;\underline{\lambda}) \;=\; \lambda_A + \sum_{i=1}^{K}(1-d_i)\lambda_i + \mu_d\, h_c(t;\underline{\lambda}) \qquad 7.6\text{-}6$$

with

$$h_c(t;\underline{\lambda}) \;=\; uvt^{v-1} \qquad 7.6\text{-}7$$

for positive constants u, v. This form for the expected rate of occurrence of new B-modes corresponds to approximating the expected number of distinct B-modes occurring over [0, t] by

$$\mu_c(t;\underline{\lambda}) \;=\; ut^{v} \qquad 7.6\text{-}8$$

Recall the AMSAA/Crow procedure estimates the constants u, v by the MLE statistics based on the B-mode first occurrence times observed during Test Phase I, i.e., [0, T]. The summation term in (7.6-6) is assessed as

$$\sum_{i\in obs}\left(1-d_i^*\right)\frac{\mathrm{N_i}}{\mathrm{T}} \qquad 7.6\text{-}9$$

where d_i^* is the assessed fix effectiveness factor for observed B-mode i, and N_i is the number of occurrences of failures during [0,T] attributed to B-mode i. Note in the AMSAA/Crow procedure all fixes are assumed to be delayed to the end of the period [0,T]. Under this assumption $\frac{\mathrm{N_i}}{\mathrm{T}}$ is an unbiased estimate of λ_i. However, if fixes to B-modes are implemented prior to the end of this period (7.6-9) may not be an adequate assessment of $\sum_{i=1}^{K}(1-d_i)\lambda_i$.

The AMPM does not attempt to assess $\rho(t;\underline{\lambda})$ by estimating each λ_i. Instead the AMPM approach is to view $(\lambda_1,\cdots,\lambda_K)$ as a realization of a random sample $\underline{\Lambda} \;=\; (\Lambda_1,\cdots,\Lambda_K)$ from the gamma random variable $\Gamma(\alpha,\beta)$. This allows one to utilize all the B-mode times to first occurrence observed during Test Phase I to estimate the gamma parameters α, β. Thus in place of directly assessing $\rho(t;\underline{\lambda})$, the AMPM uses estimates of α and β to assess the expected value of $\rho(t;\underline{\Lambda})$ where

$$\rho(t;\underline{\Lambda}) \;=\; \lambda_A + \sum_{i=1}^{K}(1-d_i)\Lambda_i + \sum_{i=1}^{K} d_i\Lambda_i e^{-\Lambda_i t} \qquad 7.6\text{-}10$$

This assessed value is then taken as the AMPM assessment of the system failure intensity after fixes to all B-modes surfaced over [0,t] have been implemented. This approach does away with the need to estimate individual λ_i. Trying to adequately estimate individual λ_i could be particularly difficult in the case where many fixes are implemented prior to the end of the period [0,T].

From equation 7.6-10, we see that the expected value of $\rho(t;\underline{\Lambda})$ with respect to the random sample $\underline{\Lambda}$, denoted by $\rho(t)$, is given by

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$$\rho(t) \;=\; \lambda_A + \sum_{i=1}^{K}(1-d_i)E(\Lambda_i) + \sum_{i=1}^{K} d_i\, E\left(\Lambda_i e^{-\Lambda_i t}\right) \qquad 7.6\text{-}11$$

Recall that Λ_i are IID with $\Lambda_i \sim \Lambda$. Thus $E(\Lambda_i) = E(\Lambda)$ and $E\left(\Lambda_i e^{-\Lambda_i t}\right) = E\left(\Lambda e^{-\Lambda t}\right)$ for $i = 1, \cdots, K$. After rearranging terms and replacing $\sum_{i=1}^{K} d_i$ by $K\,\mu_d$, $E(\Lambda_i)$ by $E(\Lambda)$, and $E\left(\Lambda_i e^{-\Lambda_i t}\right)$ by $E\left(\Lambda e^{-\Lambda t}\right)$ we arrive at

$$\rho(t) \;=\; \lambda_A + (1-\mu_d)\{K\,E(\Lambda)\} + \mu_d\left\{K\,E\left(\Lambda\, e^{-\Lambda t}\right)\right\} \qquad 7.6\text{-}12$$

Next note

$$\lambda_{B,K} \;=\; E\left(\sum_{i=1}^{K}\Lambda_i\right) \;=\; K\,E(\Lambda) \qquad 7.6\text{-}13$$

Thus we can express $\rho(t)$ by

$$\rho(t) \;=\; \lambda_A + (1-\mu_d)\lambda_{B,K} + \mu_d\left\{K\,E\left(\Lambda e^{-\Lambda t}\right)\right\} \qquad 7.6\text{-}14$$

To interpret the term $K\,E\left(\Lambda e^{-\Lambda t}\right)$ in (7.6-14) we first note that in Appendix G, Annex 1, it is shown that $\mu(t;\underline{\lambda}) \;=\; \sum_{i=1}^{K}\left(1-e^{-\lambda_i t}\right)$. Thus the expected rate of occurrence of new B-modes at t, given $\underline{\lambda}$, is $h(t;\underline{\lambda}) \;=\; \dfrac{d\,\mu(t;\underline{\lambda})}{dt} \;=\; \sum_{i=1}^{K}\lambda_i e^{-\lambda_i t}$. Consider the average (i.e. expected) value of $h(t;\underline{\Lambda}) = \sum_{i=1}^{K}\Lambda_i e^{-\Lambda_i t}$ over all possible random samples $\underline{\Lambda} = (\Lambda_1, \cdots, \Lambda_K)$, where $\Lambda_i \sim \Lambda$ for $i = 1, \cdots, K$. We obtain,

$$E(h(t;\Lambda)) \;=\; \sum_{i=1}^{K} E\left(\Lambda_i e^{-\Lambda_i t}\right) \;=\; K\,E\left(\Lambda e^{-\Lambda t}\right) \qquad 7.6\text{-}15$$

Let $h(t) = E(h(t;\Lambda))$. Thus $h(t)$ is the unconditional expected rate of occurrence of new B-modes at test time *t* averaged over all possible random samples $\underline{\Lambda}$. By (7.6-14) and (7.6-15) we have

$$\rho(t) \;=\; \lambda_A + (1-\mu_d)\lambda_{B,K} + \mu_d h(t) \qquad 7.6\text{-}16$$

This expression for $\rho(t)$ is similar in form to the Crow/AMSAA approximation to $\rho(t;\underline{\lambda})$ given in Equation (7.6-14): $\rho_c(t;\underline{\lambda}) \;=\; \lambda_A + \sum_{i=1}^{K}(1-d_i)\lambda_i + \mu_d h_c(t)$, where reference to $\underline{\lambda}$ was suppressed in the notation.

The expression in (7.6-16) for the expected system failure intensity after incorporation of B-mode fixes is actually quite appealing to one's intuition if put in a slightly different form. To

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arrive at this form, simply subtract and add the term $h(t)$ on the right hand side of Equation (7.6-16). Doing this we can express $\rho(t)$ by

$$\rho(t) = \lambda_A + (1-\mu_d)\{\lambda_{B,K} - h(t)\} + h(t) \qquad 7.6\text{-}17$$

Now we see that $\rho(t)$ is the sum of three failure intensities. The first is simply the constant failure intensity due to the A-modes. To consider the second failure intensity we will first consider $h(t)$. We have shown that this term is the expected rate of occurrence of new B-modes at test time t averaged over the random samples $\underline{\Lambda}$. Additionally, $h(t)$ is the expected failure intensity contribution to $\rho(t)$ due to the set of B-modes that have not been surfaced by t. To see this, first note that the failure intensity at time t, conditioned on $\underline{\Lambda} = \underline{\lambda}$, due to unsurfaced B-modes is $u(t;\underline{\lambda})$ where

$$u(t;\underline{\lambda}) = \sum_{i=1}^{K} \{1 - I_i(t)\}\lambda_i \qquad 7.6\text{-}18$$

Recall by (7.6-4), $E[I_i(t)] = 1 - e^{-\lambda_i t}$ with respect to the first occurrence of B-mode *i*. Thus by (7.6-18) we have

$$E[u(t;\underline{\lambda})] = \sum_{i=1}^{K} \lambda_i - \sum_{i=1}^{K} \lambda_i E[I_i(t)]$$

$$= \sum_{i=1}^{K} \lambda_i e^{-\lambda_i t} = h(t;\underline{\lambda}) \qquad 7.6\text{-}19$$

It immediately follows from (7.6-19) that $h(t)$ is the unconditional expected failure intensity due to the set of un surfaced B-modes at time t, since $h(t) = E(h(t;\underline{\Lambda}))$.

Finally, we consider the second term of $\rho(t)$ in (7.6-17). In the absence of any fixes, the sum of $h(t)$ and the unconditional expected failure intensity due to the set of B-modes surfaced by *t*, denoted by *s(t),* must equal $\lambda_{B,K}$. Thus $s(t) = \lambda_{B,K} - h(t)$. If we implement fixes to the B-modes surfaced by t with an average FEF equal to μ_d, then the residual expected failure intensity due to the set of surfaced B-modes would be

$$(1-\mu_d)s(t) = (1-\mu_d)\{\lambda_{B,K} - h(t)\} \qquad 7.6\text{-}20$$

In the above equations we can replace $\lambda_{B,K}$ by $h(0)$ since at t=0 all B-modes are unsurfaced. Thus

$$h(0) = \lambda_{B,K} \qquad 7.6\text{-}21$$

As in Section 7.4.4, we call the residual expected failure intensity approached by $\rho(t)$ as t tends towards infinity the growth potential failure intensity, denoted by ρ_{GP}. Since $\lim_{t\to\infty} h(t) = 0$ we have

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$$\rho_{GP} = \lambda_A + (1-\mu_d)\lambda_B \qquad 7.6\text{-}22$$

Note this expression has a form similar to that for the growth potential in the Crow/ AMSAA model. The quantity ρ_{GP}^{-1} is called the growth potential MTBF. The growth potential for the AMPM is used in the same way as indicated in Section 7.4 for the AMSAA/Crow model.

Another useful quantity is the expected fraction of the system expected initial B-mode failure intensity, λ_B, surfaced as a function of test time *t*. Let $\theta(t)$ denote this quantity. Thus, by definition of *s(t),* we have

$$\theta(t) = \frac{s(t)}{\lambda_{B,K}} = \frac{\lambda_{B,K} - h(t)}{\lambda_{B,K}} \qquad 7.6\text{-}23$$

Note that, $\theta(t)$ is independent of the corrective action process. By this it is meant that $\theta(t)$does not depend upon when fixes are implemented or on how effective they are.

The function $\theta(t)$ can usefully serve as a measure of system maturity. Observe that for a test of duration *t*, no matter how effective our fixes are, we can only expect to eliminate at most a fraction equal to $\theta(t)$ of the expected B-mode contribution to the initial system failure intensity. Thus low values of $\theta(t)$ would indicate additional testing is required to surface a set of B-modes that account for a significant part of λ_B. A high value for $\theta(t)$ could indicate that further testing is not cost effective. Resources would be better expended toward formulating and implementing corrective actions for the surfaced B-modes. As part of a reliability growth plan it would be useful to specify goals for $\theta(t)$ at several program milestones.

Next we will express the key AMPM reliability projection quantities in terms of *K* and the gamma parameters α and β. By Appendix G, Annex 2, we have

$$\lambda_{B,K} = K\beta(\alpha+1) \qquad 7.6\text{-}24$$

$$\mu(t) = K\left\{1-(1+\beta t)^{-(\alpha+1)}\right\} \qquad 7.6\text{-}25$$

$$h(t) = \frac{K\beta(\alpha+1)}{(1+\beta t)^{\alpha+2}} = \frac{d\mu(t)}{dt} \qquad 7.6\text{-}26$$

$$\rho(t) = \lambda_A + (1-\mu_d)K\beta(\alpha+1) + \frac{\mu_d K\beta(\alpha+1)}{(1+\beta t)^{\alpha+2}} \qquad 7.6\text{-}27$$

and

$$\theta(t) = 1-(1+\beta t)^{-(\alpha+2)} \qquad 7.6\text{-}28$$

Utilizing Equation (7.6-24) for $\lambda_{B,K}$ we can also express $h(t)$ and $\rho(t)$ as follows:

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$$h(t) = \frac{\lambda_{B,K}}{(1+\beta t)^{\alpha+2}} \qquad 7.6\text{-}29$$

and

$$\rho(t) = \lambda_A + (1-\mu_d)\lambda_{B,K} + \frac{\mu_d \lambda_{B,K}}{(1+\beta t)^{\alpha+2}} \qquad 7.6\text{-}30$$

In the next section, the behavior of the AMPM is considered as K increases. Limiting expressions for the AMPM quantities in (7.6-24) through (7.6-30) will be obtained as $K \to \infty$ under natural assumptions about $\lambda_{B,K}$ and $\beta = \beta_K$. Then parameter estimation procedures will be specified for the finite K AMPM and the limiting parameters as $K \to \infty$.

7.6.5 Limiting Behavior of AMPM.

The limiting behavior of the AMPM as K increases is now considered. To do so, first define step processes

$$\{X_{K,i}(t), 0 \le t < \infty\} \quad for \quad i = 1, \cdots, K \text{ where}$$

$$X_{K,i}(t) = \begin{cases} 1 & if\ B-\text{mode } i \text{ occurs by t} \\ 0 & \text{otherwise} \end{cases}$$

Note

$$\Pr(X_{K,i}(t) \ge 2) = 0 \qquad 7.6\text{-}31$$

and

$$\Pr(X_{K,i}(t) = 1) = 1 - \Pr(X_{K,i}(t) = 0) \qquad 7.6\text{-}32$$

Thus to complete our definition of these processes, we need only specify $\Pr(X_{K,i}(t) = 0)$. To keep the definition of these processes consistent with the AMPM assumptions we define

$$\Pr(X_{K,i}(t) = 0) = \int_0^\infty e^{-\lambda t} f_\Lambda(\lambda) d\lambda \qquad 7.6\text{-}33$$

where $\Lambda \sim \Gamma(\alpha, \beta)$ and f_Λ is the previously defined gamma density function with $\alpha = \alpha_K$ and $\beta = \beta_K$. Note $X_{K,i}(t)$ is the unconditional AMPM indicator function for B-mode i corresponding to the earlier defined conditional indicator function $I_i(t)$ where $\Pr(I_i(t) = 0) = e^{-\lambda_i t}$ and subscript K was suppressed. By (7.6-33) and Appendix G, Annex 2,

$$\Pr(X_{K,i}(t) = 0) = E(e^{-t\Lambda}) = \Psi(-t) = (1+\beta_K t)^{-(\alpha+1)} \qquad 7.6\text{-}34$$

From (7.6-32) and (7.6-34) we obtain

$$\mu_K(t) \triangleq E\left[\sum_{i=1}^{K} X_{K,i}(t)\right]$$

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$$= \sum_{i=1}^{K} \Pr\left(X_{K,i}(t)=1\right) = K - K\left(1+\beta_K t\right)^{-(\alpha+1)} = \mu(t) \qquad 7.6\text{-}35$$

for $(\alpha_K, \beta_K) = (\alpha, \beta)$. Thus the AMPM step processes $\{X_{K,i}(t), 0 \le t < \infty\}$, $1 \le i \le K$, give rise to the previously developed AMPM.

To investigate the behavior of the projection model as K increases, specify the limiting behavior of α_K and β_K. Since β_K is simply a scale factor for test time t it is reasonable to keep β_K fixed, say $\beta_K = \beta_\infty \in (0,\infty)$. Recall by (7.6-24), $\lambda_{B,K} = K\,\beta_K(\alpha_K + 1)$. Regardless of the value of K, $\lambda_{B,K}$ represents the unconditional expected B-mode contribution to the initial system failure intensity. Thus it is natural to let $K\,\beta_K(\alpha_K+1) = \lambda_{B,\infty} \in (0,\infty)$ for all K. Actually, to obtain our results for the limiting behavior of the AMPM we need only insist that

$$\lim_{K\to\infty} \beta_K = \beta_\infty \in (0,\infty) \qquad 7.6\text{-}36$$

and

$$\lim_{K\to\infty} K\,\beta_K(\alpha_K+1) = \lambda_{B,\infty} \in (0,\infty) \qquad 7.6\text{-}37$$

Simply denote β_∞ and $\lambda_{B,\infty}$ by β and λ_B, respectively. Since $\alpha_K + 1 \ge 0$, (7.6-36) and (7.6-37) imply

$$\lim_{K\to\infty} \alpha_K = -1 \qquad 7.6\text{-}38$$

Let $X_K(t)$ be the supposition of the independent step processes $X_{K,i}(t)$, i.e.

$$X_K(t) \triangleq \sum_{i=1}^{K} X_{K,i}(t) \qquad 7.6\text{-}39$$

It is demonstrated that the stochastic process $\{X_K(t), 0 \le t < \infty\}$ converges to a non-homogeneous Poisson process (NHPP) with mean value function $\mu_\infty(t)$ as $K \to \infty$, where

$$\mu_\infty(t) = \left(\frac{\lambda_B}{\beta}\right)\ln(1+\beta t) \qquad 7.6\text{-}40$$

This result suggests that for complex systems or subsystems, we can expect our AMPM process $\{X_K(t), 0 \le t < \infty\}$ to behave like a NHPP $\{X_\infty(t), 0 \le t < \infty\}$ where $X_\infty(t)$ is the number of distinct B-modes that occur by t and $E\{X_\infty(t)\} = \mu_\infty(t)$ given in (7.6-40).

We can now relate the key AMPM reliability projection quantities in (7.6-24) through (7.6-28) which depend on K to the corresponding NHPP quantities. To do so we will subscript the AMPM quantities by K and the NHPP quantities by ∞. Thus, for example, by (7.6-24) and limit condition (7.6-37) we have

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$$\lim_{K\to\infty}\lambda_{B,K} \;=\; \lambda_{B,\infty}\in(0,\infty) \qquad 7.6\text{-}41$$

(where we also denote $\lambda_{B,\infty}$ simply by λ_B). By (7.6-26) we also have

$$\mu_K(t) \;=\; \int_0^t \frac{K\,\beta_K\,(\alpha_K+1)}{(1+\beta_K z)^{\alpha_K+2}}\,dz$$

Thus

$$\lim_{K\to\infty}\mu_K(t) \;=\; \int_0^t\left[\lim_{K\to\infty}\left\{\frac{K\,\beta_K\,(\alpha_K+1)}{(1+\beta_K z)^{\alpha_K+2}}\right\}\right]dz$$

By (7.6-36) through (7.6-38) and (7.6-40) this yields

$$\lim_{K\to\infty}\mu_K(t) \;=\; \int_0^t\frac{\lambda_B dz}{1+\beta z} \;=\; \left(\frac{\lambda_B}{\beta}\right)\ln(1+\beta t) \;=\; \mu_\infty(t) \qquad 7.6\text{-}42$$

Again by (7.6-26), (7.6-36) through (7.6-38) and (7.6-40), we obtain

$$\lim_{K\to\infty}h_K(t) \;=\; \lim_{K\to\infty}\frac{K\,\beta_K(\alpha_K+1)}{(1+\beta_K t)^{\alpha_K+2}}$$

$$=\; \frac{\lambda_B}{1+\beta t} \;=\; \frac{d\mu_\infty(t)}{dt} \;=\; h_\infty(t) \qquad 7.6\text{-}43$$

By (7.6-27), (7.6-36) through (7.6-38) and (7.6-43) we arrive at

$$\lim_{K\to\infty}\rho_K(t) \;=\; \lim_{K\to\infty}\left\{\lambda_A+(1-\mu_d)K\,\beta_K(\alpha_K+1)+\frac{\mu_d\,K\,\beta_K(\alpha_K+1)}{(1+\beta_K t)^{\alpha_K+2}}\right\}$$

$$=\; \lambda_A+(1-\mu_d)\lambda_B+\frac{\mu_d\,\lambda_B}{1+\beta t} \;=\; \lambda_A+(1-\mu_d)\lambda_B+\mu_d\,h_\infty(t)$$

$$=\; \rho_\infty(t) \qquad 7.6\text{-}44$$

Additionally, by (7.6-22) and (7.6-41) we have

$$\lim_{K\to\infty}\rho_{GP,K} \;=\; \lim_{K\to\infty}\{\lambda_A+(1-\mu_d)\lambda_{B,K}\}$$

$$=\; \lambda_A+(1-\mu_d)\lambda_B \;=\; \rho_{GP,\infty} \qquad 7.6\text{-}45$$

Finally, by (7.6-28), (7.6-36) and (7.6-38) we deduce

$$\lim_{K\to\infty}\theta_K(t) \;=\; \lim_{K\to\infty}\left\{1-(1+\beta_K t)^{-(\alpha_K+2)}\right\} \;=\; \frac{\beta t}{1+\beta t}$$

Thus by (7.6-43) we conclude

$$\lim_{K\to\infty}\theta_K(t) \;=\; \frac{\beta t}{1+\beta t} \;=\; \frac{\lambda_B-h_\infty(t)}{\lambda_B} \;=\; \theta_\infty(t) \qquad 7.6\text{-}46$$

7.6.6 Estimation Procedure for AMPM.

In this section we will specify the procedures to estimate key AMPM parameters and reliability measures expressed in terms of these parameters. Estimation equations will be given for the finite K and NHPP variants of the continuous AMPM. The model parameter estimators are

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MLE's. Statistical details and further discussion of the estimation procedures are provided in Appendix G, Annex 3.

Our parameter estimates are written in terms of the following data: m = number of distinct B-modes that occur over a test period of length T, $\underline{t} = (t_1, \cdots, t_m)$ where $0 < t_1 \le t_2 \le \cdots \le t_m \le T$ are the first occurrence times of the m observed B-modes, and n_A = number of A-mode failures that occur over test period T. We will denote an estimate of a model parameter or expression by placing the symbol "^" over the quantity.

The finite K AMPM estimates are based on a specified value of K. If we hold the test data constant and let $K \to \infty$ we obtain AMPM projection estimates that are appropriate for complex subsystems or systems that typically have many potential B-modes. The AMPM limit estimating equations are derived in Appendix E, Annex 3. These equations can also be obtained from MLE equations for the NHPP associated with the AMPM. This process was discussed in Section 7.6.4 and has the mean value function given by equation (7.5-40).

Recall α_K, β_K are the gamma parameters for the AMPM where it is assumed the K initial B-mode failure rates are realized values of a random sample from a gamma random variable $\Gamma(\alpha_K, \beta_K)$.

The MLE for β_K is $\hat{\beta}_K$ where

$$K = \frac{\left(\sum_{i=1}^{m} \ln \frac{1+\hat{\beta}_K T}{1+\hat{\beta}_K t_i}\right)\left(\sum_{i=1}^{m} \frac{1}{1+\hat{\beta}_K t_i}\right) - \left(\frac{m\hat{\beta}_K}{1+\hat{\beta}_K T}\right)\sum_{i=1}^{m} \frac{T - t_i}{1+\hat{\beta}_K t_i}}{\left\{\ln\left(1+\hat{\beta}_K T\right)\right\}\left(\sum_{i=1}^{m} \frac{1}{1+\hat{\beta}_K t_i}\right) - \left(\frac{m\hat{\beta}_K}{1+\hat{\beta} T}\right)T} \qquad 7.6\text{-}47$$

The MLE for α_K is $\hat{\alpha}_K$ where $\hat{\alpha}_K$ can be easily obtained from $\hat{\beta}_K$ and either equation below. These equations are the maximum likelihood equations for α_K and β_K respectively (see Appendix E, Annex 3):

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$$\left(\hat{\alpha}_K + 1\right)^{-1} = m^{-1}\left[K \ln\left(1 + \hat{\beta}_K T\right) - \sum_{i=1}^{m} \ln\left(\frac{1 + \hat{\beta}_K T}{1 + \hat{\beta}_K t_i}\right)\right] \quad 7.6\text{-}48$$

$$\hat{\alpha}_K + 1 = \frac{\frac{m}{\hat{\beta}_K} - \sum_{i=1}^{m} \frac{t_i}{1 + \hat{\beta}_K t_i}}{\frac{(K-m)T}{1 + \hat{\beta}_K T} + \sum_{i=1}^{m} \frac{t_i}{1 + \hat{\beta}_K t_i}} \quad 7.6\text{-}49$$

Using $\left(\hat{\alpha}_K, \hat{\beta}_K, K\right)$ we can estimate all our finite K AMPM quantities where the A-mode failure rate λ_A is estimated by $\hat{\lambda}_A = {}^{n_A}\!/_{T}$ and the average B-mode fix effectiveness factor μ_d is assessed as

$$\mu_d^* = \frac{1}{m} \sum_{i \in obs} d_i^* \quad 7.6\text{-}50$$

In (7.6-50), the assessment d_i^* of the fix effectiveness factor (FEF) for observed B-mode i will often be based largely on engineering judgment. The value of d_i^* should reflect several considerations: (1) How certain we are that the problem has been correctly identified; (2) the nature of the fix, e.g., its complexity; (3) past FEF experience and (4) any germane testing (including assembly level testing).

Note the left-hand side of Equation (7.6-47) requires a value for K before we can numerically solve for $\hat{\beta}_K$. In practice we do not know the value of K. We could attempt to use the data $(m, \underline{t})$ to statistically estimate K. However, graphs presented in the next section illustrate the difficulty in obtaining a reasonable estimate for K even for a large data set that appears to fit the model well. Thus we prefer to take the point of view that we should not attempt to statistically assess K. However, by conducting a standard failure modes and effects criticality analysis (FMECA), we can place a lower bound on K, say K_ℓ. Our experience with the AMPM indicates that if K is substantially higher than m, say, e.g., $K \geq 10m$, then our AMPM projection quantities will be insensitive to the value of K. We believe for a complex system or subsystem it will often be the case that $K_\ell \geq 10m$ or at least the unknown value of K will be $10m$ or higher. The factor 10 may be larger than necessary. We suggest exercising the finite K AMPM with several plausible lower bound values for K and comparing the associated projections with those obtained in the limit as $K \to \infty$. This is illustrated for a data set in the next section.

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To obtain the limiting AMPM projection model estimates consider the sequence of finite K AMPM estimates $\left\langle \hat{\beta}_K \right\rangle_{K > K_0}$ where we assume $\hat{\beta}_K$ satisfies Equation (7.6-47) for each $K > K_0$. In Appendix G, Annex 3 it is shown that

$$\hat{\beta}_\infty = \lim_{K \to \infty} \hat{\beta}_K \in (0, \infty) \qquad 7.6\text{-}51$$

is a finite positive value. Moreover, it is demonstrated that

$$\left\{ \ln\left(1 + \hat{\beta}_\infty T\right) \right\} \sum_{i=1}^{m} \frac{1}{1 + \hat{\beta}_\infty t_i} - \frac{m \hat{\beta}_\infty T}{1 + \hat{\beta}_\infty T} = 0 \qquad 7.6\text{-}52$$

It is also shown that

$$\hat{\alpha}_\infty = \lim_{K \to \infty} \hat{\alpha}_K = -1 \qquad 7.6\text{-}53$$

where for each $\hat{\beta}_K$, $K > K_0$, $\hat{\alpha}_K$ satisfies Equation (7.6-48) or Equation (7.6-49). The limiting AMPM estimates $\hat{\beta}_\infty$ and $\hat{\lambda}_\infty$, given below in Equation (7.6-55), can be shown to be MLE's for parameters β_∞ and $\lambda_{B,\infty}$. Recall these parameters define the NHPP discussed in Section 7.6.4 whose mean value function is given in Equation (7.6-40).

For ease of reference, the finite K AMPM and limiting AMPM estimates for key projection model quantities are listed below and indexed by K and ∞, respectively:

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$$\hat{\lambda}_K = K\hat{\beta}_K\left(\hat{\alpha}_K + 1\right) \quad 7.6\text{-}54$$

$$\hat{\lambda}_\infty = \frac{m\hat{\beta}_\infty}{\ln\left(1+\hat{\beta}_\infty T\right)} \quad 7.6\text{-}55$$

$$\hat{\mu}_K(t) = K\left[1-\left(1+\hat{\beta}_K t\right)^{-\left(\hat{\alpha}_K+1\right)}\right] \quad 7.6\text{-}56$$

$$\hat{\mu}_\infty(t) = \left(\frac{\hat{\lambda}_{B,\infty}}{\hat{\beta}_\infty}\right)\ln\left(1+\hat{\beta}_\infty t\right) \quad 7.6\text{-}57$$

$$\hat{h}_K(t) = \frac{\hat{\lambda}_K}{\left(1+\hat{\beta}_K t\right)^{\hat{\alpha}_K+2}} \quad 7.6\text{-}58$$

$$\hat{h}_\infty(t) = \frac{\hat{\lambda}_{B,\infty}}{1+\hat{\beta}_\infty t} \quad 7.6\text{-}59$$

$$\hat{\rho}_{GP,K} = \hat{\lambda}_A + \left(1-\mu_d^*\right)\hat{\lambda}_{B,K} \quad 7.6\text{-}60$$

$$\hat{\rho}_{GP,\infty} = \hat{\lambda}_A + \left(1-\mu_d^*\right)\hat{\lambda}_{B,\infty} \quad 7.6\text{-}61$$

$$\hat{\rho}_K(t) = \hat{\rho}_{GP,K} + \mu_d^*\hat{h}_K(t) \quad 7.6\text{-}62$$

$$\hat{\rho}_\infty(t) = \hat{\rho}_{GP,\infty} + \mu_d^*\hat{h}_\infty(t) \quad 7.6\text{-}63$$

$$\hat{\theta}_K(t) = 1-\left(1+\hat{\beta}_K t\right)^{-\left(\hat{\alpha}_K+2\right)} \quad 7.6\text{-}64$$

$$\hat{\theta}_\infty(t) = \frac{\hat{\beta}_\infty t}{1+\hat{\beta}_\infty t} \quad 7.6\text{-}65$$

Note (7.6-55) together with (7.6-57) imply

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$$\hat{\mu}_{\infty}(T) \;=\; \left(\frac{\hat{\lambda}_{B,\infty}}{\hat{\beta}_{\infty}}\right)\ln\left(1+\hat{\beta}_{\infty}T\right) \;=\; m \qquad 7.6\text{-}66$$

This agrees with intuition in the sense that $\hat{\mu}_{\infty}(T)$ is an estimate of the expected number of distinct B-modes generated over the test period $[0,T]$ while m is the observed number of distinct B-modes that occur.

Suppose we adopt the view that our "model of reality" for a system or subsystem is the AMPM for a finite K which is large but unknown. Then we can consider the limiting AMPM projection estimates as approximations to the AMPM estimates that correspond to the "true" value of K. Our discussion in this section suggests that over the projection range of $t \geq T$ values of practical interest, the limiting estimates should be good approximations for complex systems or subsystems. In this sense, knowing the "true" value of K is usually unimportant. Note, however, it is useful to have available the computational formulas for the finite K AMPM projection estimators as a function of K. For example, we can compare the graphs of a projection estimator such as $\hat{\rho}_K(t)$ or $\hat{\mu}_K(t)$ over the t range of interest for different values of K to the corresponding limiting estimator. In this fashion we can discern the nature of the convergence, for example, the rapidity of convergence and whether the convergence is strictly increasing or decreasing for t values of interest. This type of graphical analysis is illustrated with an example.

7.6.7 Example.

We will illustrate several key features of the AMPM projection model and associated estimators by applying the model to a data set generated during an Army system development program. Here, we will just focus on the B-modes and let $\hat{\lambda}_A = 0$. This test data set consists of $m = 163$ B-mode first occurrence times generated over $T = 8000$ "equivalent" mission hours.

In Figure 7-1, we display the cumulative number of distinct B-modes versus the mission hours. We also display the graphs of $\hat{\mu}_K(t)$ for several values of K. We can show that the greatest lower bound, K_0, for the set of K-values for which the AMPM estimators are well defined corresponds to a degenerate gamma. This limiting gamma density has zero variance and mean equal to λ, where $\lambda_i = \lambda$ for $i = 1, \cdots, K$. To avoid numerical instability, separate maximum likelihood equations were derived and used for this limiting case. On our graphs we have labeled the curves associated with this case (i.e., $K = K_0$) IBM to indicate that this limiting form for $\mu(t)$ coincides with the IBM model [4]. More explicitly, the IBM model uses this $\mu(t)$ for the expected number of "non-random" failures experienced in t test hours. This limiting form for $\mu(t)$ also is used by Musa in his software reliability basic execution time model [9]. It is interesting to note that the opposite AMPM limiting form, $\mu_{\infty}(t)$, is used by Musa and Okumoto in their Logarithmic Poisson software reliability execution time model [9]. In both of Musa's models, $\mu(t)$ represents the expected number of software failures experienced over test period [0,t], where t denotes execution time.

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Note over the data range, i.e., $0 \le t \le 8000$ hours, the graphs of $\hat{\mu}_K(t)$ are visually indistinguishable for $K_{IBM} \le K \le \infty$. In such circumstances the value of *K* cannot be reasonably assessed from the test data even if one can formally obtain an MLE for *K*. In fact, applying the IBM model (where all λ_i are implicitly assumed to be equal), we can always obtain an MLE for *K* whenever $\frac{1}{m}\sum_{i=1}^{m} t_i < \frac{T}{2}$ (see Musa, Iannino, and Okumoto with respect to the exponential class family [9]). However, it has been our experience that the IBM estimate of *K*, $K_{IBM} = K_0$, is often only marginally higher than m, the observed number of distinct B-modes. Since $\hat{\mu}_K(t)$ approaches *K* as $t \to \infty$, such a low estimate of *K* forces the slope of $\hat{\mu}_K(t)$ to quickly approach zero beyond *T* (Figure 7-2). Note $\hat{h}_K(t)$ is the slope of $\hat{\mu}_K(t)$. Thus we can see that such a low estimate of *K* quickly forces $\hat{h}_K(t)$ close to zero for $t \ge T$. This in turn tends to produce an "optimistic" failure intensity projection, especially when the assessed value of μ_d is high. This follows from the formula

$$\hat{\rho}_K(t) = \hat{\lambda}_A + (1 - \mu_d^*)\hat{\lambda}_{B,K} + \mu_d^* \hat{h}_K(t) \qquad 7.6\text{-}67$$

which applies for $K_{IBM} \le K \le \infty$. Thus a good fit over [*0,T*] is not a sufficient condition to ensure that a projection model will provide reasonable projection estimates for $t \ge T$.

Looking at Figure 7-3, as one might expect, the model with $K = \infty$ appears to provide a more conservative estimate of $\rho_K^{-1}(t)$ for $t \ge T$ than do the finite *K* estimators. However, for $t \ge T$, it is important to note that the $\hat{\mu}_K(t)$, $\hat{\rho}_K^{-1}(t)$ and $\hat{\theta}_K(t)$ graphs, displayed in Figures 7-2, 7-3, and 7-4, respectively, quickly become much closer to the corresponding $K = \infty$ graph than to the $K = K_{IBM}$ graph as K increases above K_{IBM}.

Observe from Figure 7-4, $\hat{\theta}_k(8000) \approx 0.67$ for $K_{IBM} \le K \le \infty$. Thus, whatever the "true" value of *K*, we estimate that the remaining B-modes contribute about $.33\lambda_B$ to the system failure intensity.

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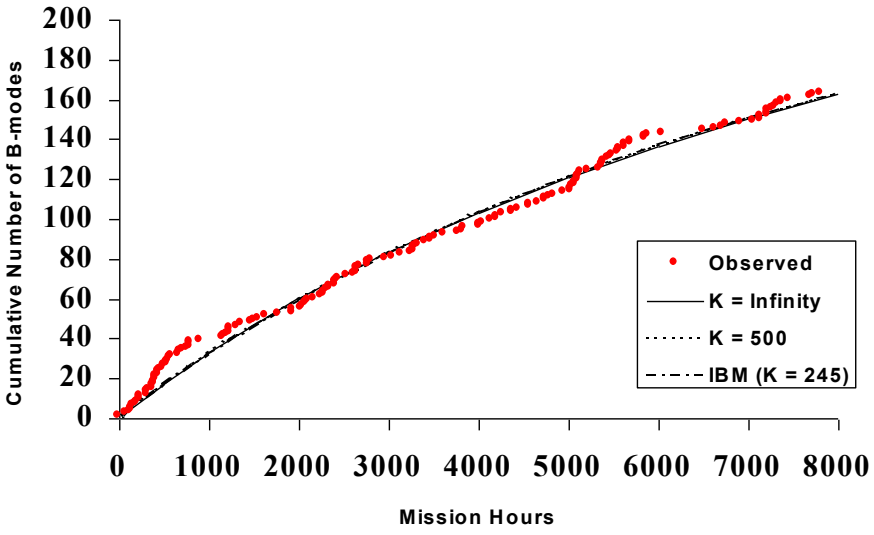


FIGURE 7-1. *Observed Versus Estimate of Expected Number of B-Modes*

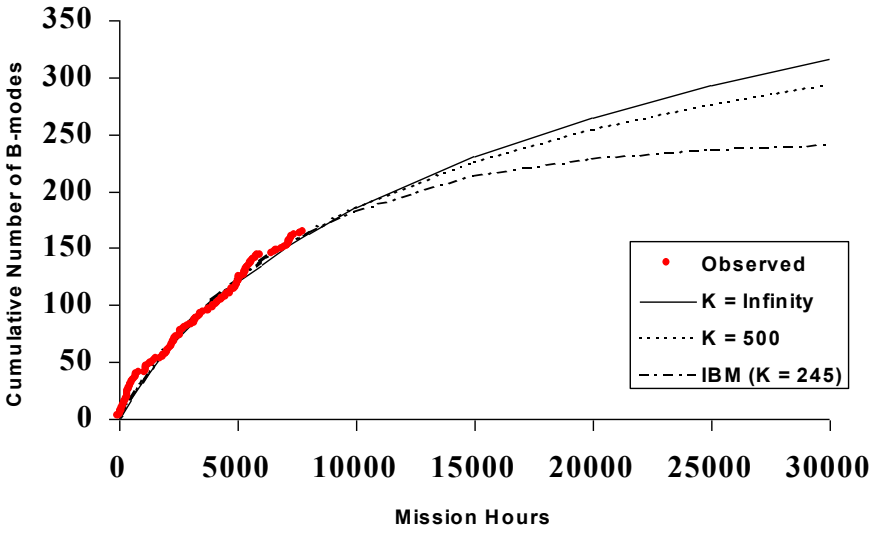


FIGURE 7-2. *Extrapolation of Estimated Expected Number of B-Modes as Function of K. (Data Ends at 8000 Hours)*

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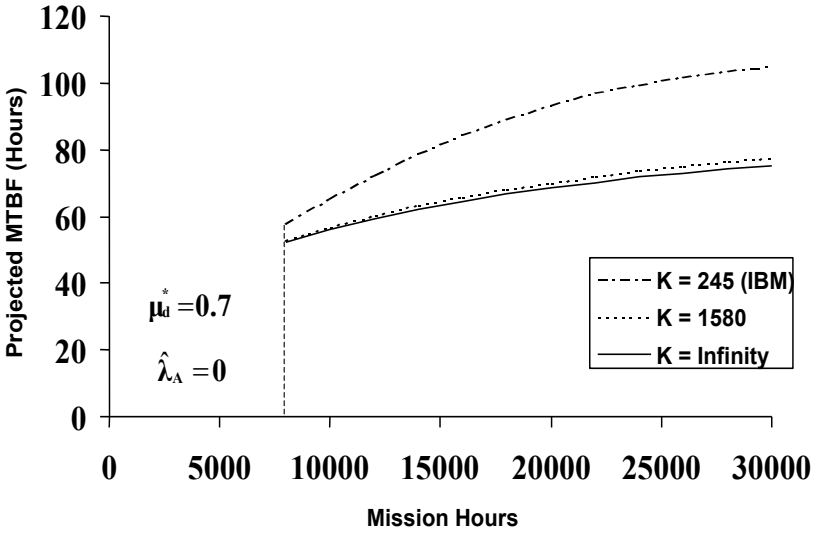


FIGURE 7-3. *Projected MTBF for Different K's.*
(Based on Initial 8000 Hours of Data)

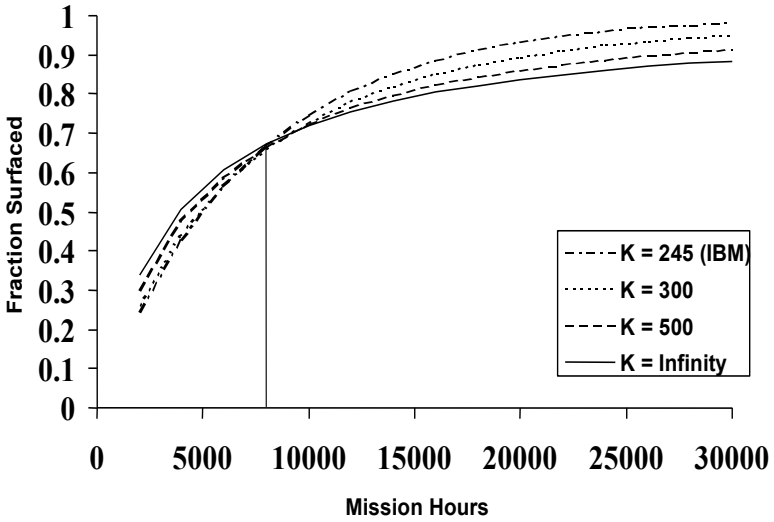


FIGURE 7-4. *Estimated Fraction of Expected Initial B-Mode Failure Intensity Surfaced for Different K's.*
(Based on Initial 8000 Hours of Data)

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7.6.8 AMPM Projection Using Crow Extended Data.

The data in TABLE XVI were analyzed using the AMPM model wherein all of the BD-modes and BC-modes were analyzed using the average FEF of the BD-modes - 0.72. The AMSAA VGS suite was used to analyze the data for the option of individual B-mode time data. When running this program you are asked if the fixes are delayed or not – option 1 is for all fixes delayed, option 2 is for the case where not all fixes were delayed, that is, some (or all) fixes were implemented during the test. In our situation, the BC-modes were implemented in test and the BD-modes were delayed – so option 2 was selected. The results were a projection of reliability at 400 hours of 10.75 hours and a Growth Potential of 15.24 hours. The Cramer Von Mises test for model fit was accepted. The figure below shows a plot of the expected number of B-modes versus the observed first occurrence B-modes by cumulative test time.

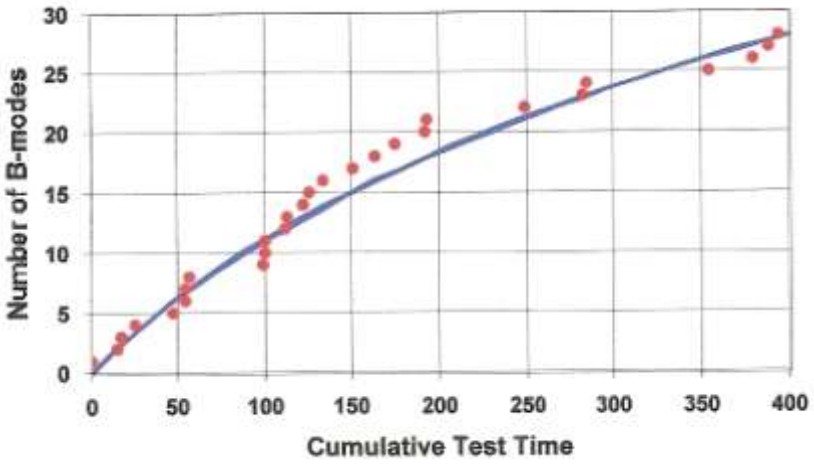


FIGURE 7-5. *Expected (Smooth) vs. Observed (Pts) Number of B-Modes*

The following is an application of an option in the AMPM AMSAA VGS suite that allows for two FEFs partitioned at a point v in the test program. In reviewing the BC-mode failure times it is noted that most of the BC modes occur prior to 175 hours, further that only the BC-1 mode had a repeat of the failure mode. Note that 9 of 12 BC-modes occur prior to 175 test hours. It is recognized that there are BD-modes occurring in this test interval. The option on the VGS AMPM model methodology provides alternative choices for two FEFs with a break or partition at a point v with FEF of d_1 before v and d_2 after v. Because of the lack of repeats of BC-modes, save BC-1, it might be conjectured that the average FEF for the BC-modes might reasonably be greater than the BD-mode average of 0.72. Two alternative FEF values for those modes occurring before 175 hours were investigated – 0.80 and 0.85. In applying this option we are assuming that the BC- and BD-modes occurring before v have an average FEF of d_1. The FEF d_2 of 0.72 was used for both cases. This resulted in projected MTBFs at 400 hours of 11.5 for an initial FEF of 0.80 and 12.0 for an initial FEF of 0.85.

Comparison of Projections for ACPM, Crow Extended and AMPM.

The following table presents a summary of the projected MTBFs for the indicated methodologies based on using the Crow Extended Model data. The AMSAA/Crow Tracking model is presented

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as a baseline. For the baseline AMSAA/Crow Tracking and ACPM cases the unbiased estimate of β was used.

TABLE XIX. Projected MTBFs

Method	Projected MTBF
AMSAA/Crow Tracking[4]	7.84
ACPM[5]	10.90
Crow Extended	11.29
Crow Extended w/Preemptive	11.50
AMPM	10.75
Two FEFs[6]	
.85/.72	12.00
.80/.72	11.5

The results indicate reasonably close results, at least for this example; the correct choice depends on satisfaction of the assumptions and fits of the models. Note that the ACPM case assumes the BC-modes to be delayed rather than incorporated in testing; it is not known how much affect this would have on the result.

It may be noted that for the AMSAA/Crow Tracking result the growth rate was small, with a growth rate α of less than 0.1, but still statistically significantly different from zero. Other examples may result in different projections and orders of projections.

7.6.9 Analysis Considerations for Apparent Failure Mode Rates of Occurrence Changes. There may be times when there appear to be changes in the rate of occurrence of B-modes, especially during initial periods of testing or near the end of testing. This may be seen through plots of the rate of occurrence of B-modes wherein the rate initially is, say, high and after a period of testing the rate of occurrence slows, or the reverse occurs. Such changes may occur due to a number of reasons: initial assembly or infant mortality; quality or weak areas of the design and operator unfamiliarity with equipment; or refurbishing hardware near the end of a test phase. The following paragraphs (7.5.9.1-3) address this situation in three ways, namely using a Gap Method wherein such an initial period is to use only the test data beyond an initial period but retains failure times as measured, a Segmented FEF approach where data are partitioned at a point v chosen such that a relatively high average fix effectiveness factor d_1 is applied to the B-modes occurring on or before v, and a more typical average fix effectiveness factor d_2 is applied

[4] Based on using $\bar{\beta}$.

[5] Based on using $\bar{\beta}$ and assigned FEFs for the BD-modes and the average BD- mode FEF (0.72) for the BC-modes which are assumed to be delayed fixes.

[6] For the AMPM case, two FEFs are an option for the AMSAA Visual Growth Suite program. In the two cases presented, most of the BC modes occur in the first 180 hours (9 of 12), and further only BC-1 has repeats (all 3 within 14 hours of test start). For this reason the two FEFs approach was considered with a break at v at 175 hours with the FEFs indicated before and after the v break or partition. Based on historical data these FEFs (.85 and .80) seemed reasonable.

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to the B-modes surfaced beyond *v*, or lastly re-initializing the data at the point at which the rate appears to change. Each of these approaches and their rationale is discussed in the following sections.

Gap Method

7.6.9.1.1 Rationale for Using the Gap Method.
From previous discussions with regard to the AMSAA/Crow Projection Model and the AMSAA Maturity Projection Model, it was stated that a projection is a reliability estimate that takes into account the contribution of A-mode failures, B-mode failures and the effectiveness of the corrective actions that are implemented to the latter class of failure modes. The statistical reliability projection based on this information is significantly influenced by the initial B-mode rate of occurrence. For some systems under development, the initial rate of occurrence of the correctable failure modes (B-modes) is steep. This can be due to problems associated with infant mortality, initial assembly procedures or operator unfamiliarity with the system. One way of excluding the impact of such start-up problems on the reliability projection is to utilize only the test data beyond an initial time period. This is referred to as "jumping the gap." Another method to address this situation – segmented FEF - is discussed in the next section. For this case where the individual B-mode occurrence times are known, the gap method may be an appropriate strategy to use for a dataset where the initial rate of occurrence of B-modes is very steep due to these start-up problems. Such a situation is described below.

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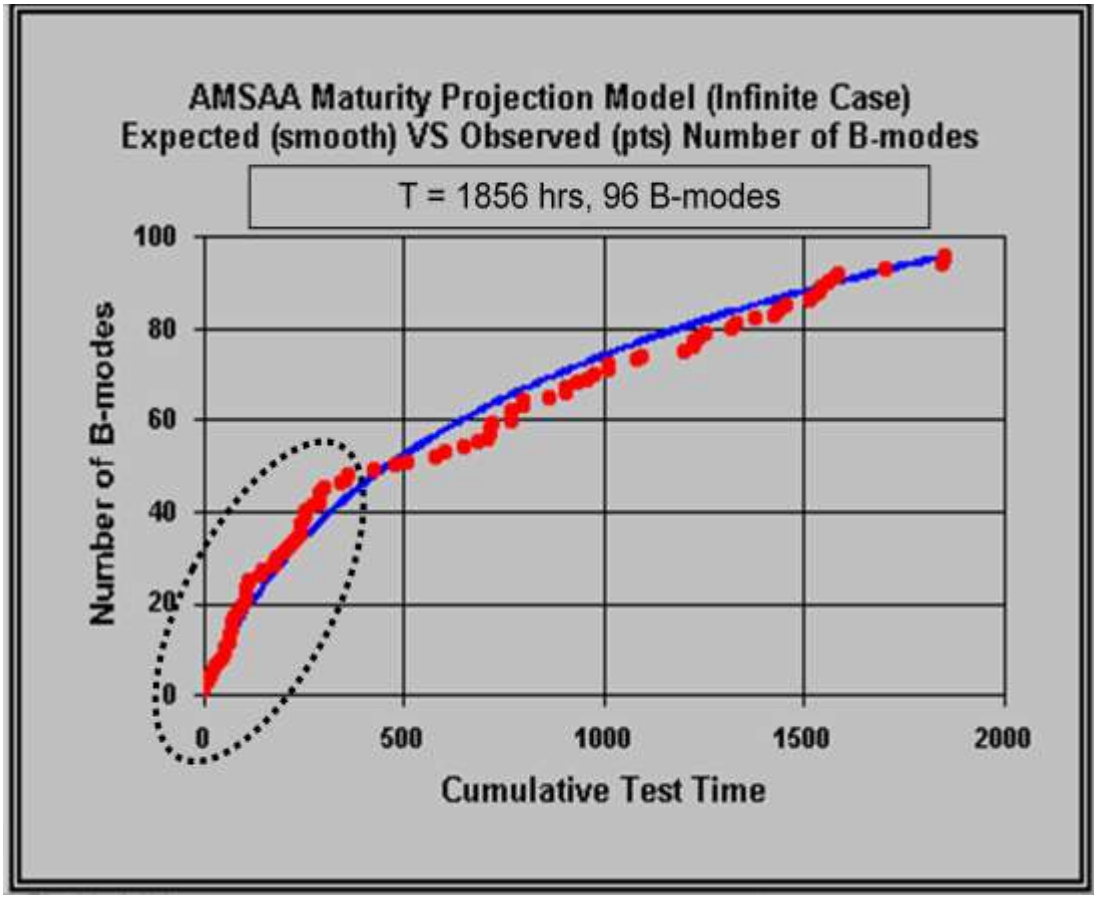


FIGURE 7-6. *Example Curve for Illustrating the Gap Method*

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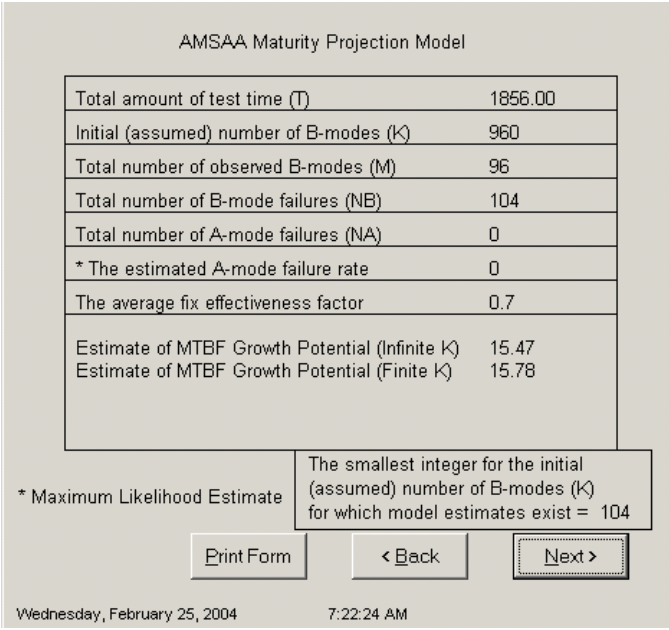


FIGURE 7-7. *Model Results for AMPM*

The above figure is presented for informational and comparison purposes for the Gap Method and the Segmented FEF Method presented in the following section.

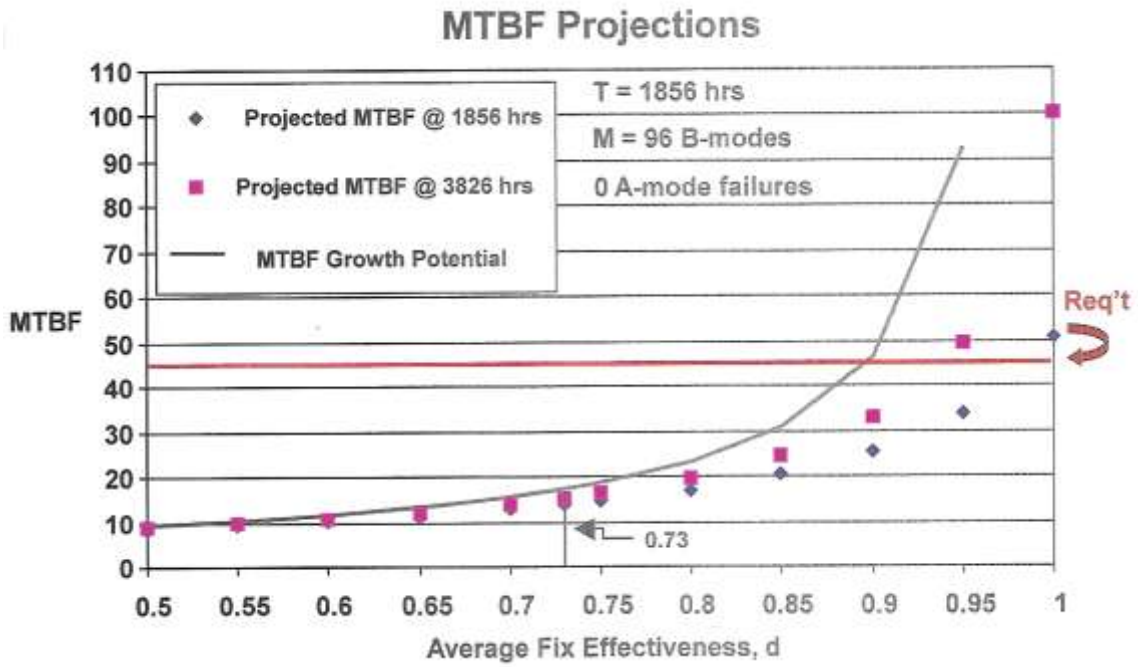


FIGURE 7-8. *MTBF Projection Increases as Fix Effectiveness Improves*

One of the first considerations when doing a reliability analysis is a plot of the data. In this case a projection analysis is a plot of the estimated expected versus the observed number of B-modes surfaced as a function of cumulative test time. The plot will reveal many important aspects of the

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data as well as the model being applied to the data. Besides showing the overall trend of the data and the total number of observed B-modes during the test period, this plot has the following features. First, it provides a visual perspective of the goodness-of-fit of the model and shows (non-statistically) the degree to which the model represents the observed test data. Second, the shape of the curve illustrates whether the rate of occurrence of new B-modes is diminishing with time, as the concavity of this particular curve indicates that the rate is beginning to flatten out. Finally, this curve shows (indirectly) the rate of occurrence of new B-modes at the origin, since for the example curve, nearly a third of the total number of B-modes have occurred within the first 10 percent of the total test duration. The steepness of this initial rate of occurrence indicates there could be start-up problems for which a gap method may be useful. This steepness is further revealed by considering the slope of the above curve, which is the rate of change of the estimated expected number of B-modes with respect to time. This is shown below.

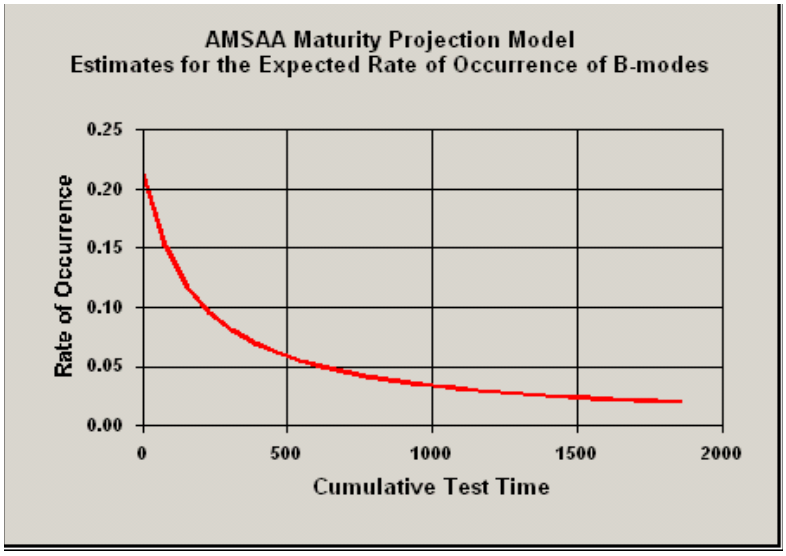


FIGURE 7-9. *Estimated Expected Rate of Occurrence of New B-Modes*

The above curve is referred to as the rate of occurrence, or *h(t)*, curve. Note the downward trend and the steepness at the origin. It can be shown that the initial B-mode failure rate is equal to *h(0)*. For the example curve, the statistical estimate for the initial B-mode failure rate is approximately 0.22, which can also be approximated directly from the *h(t)* curve by reading off the rate of occurrence value at $t = 0$. In addition to providing an estimate for the initial B-mode failure rate, the rate of occurrence function, *h(t)*, plays a key role in projecting the system reliability. Incidentally, for our example, the total test time equals 1856, the number of observed B-modes equals 96, the number of A-mode failures equals 0, the number of projections to make is 1, the total number of B-mode failures (which includes 1st occurrences plus repeats) equals 104, the assumed number (K) of B-modes initially in the system equals 960, and the overall average fix effectiveness equals 0.7. The resulting MTBF projection curve follows.

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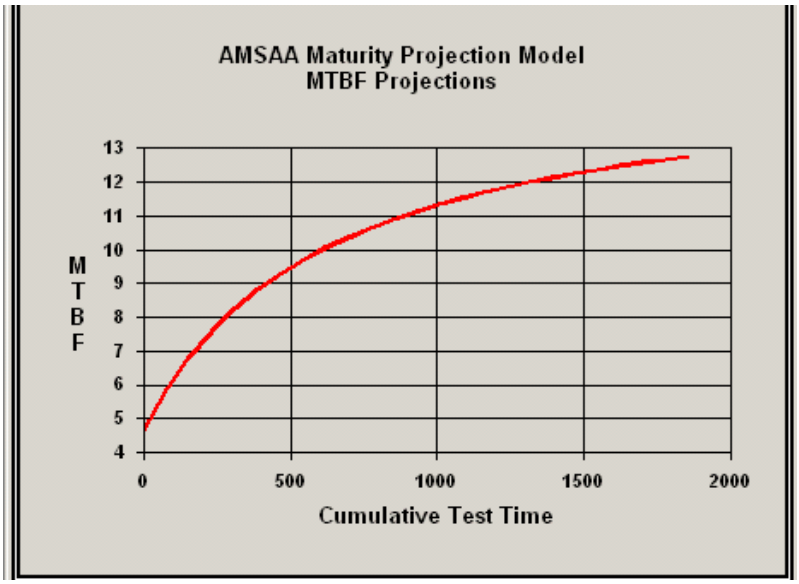


FIGURE 7-10. *MTBF Projection Curve*

Typically, MTBF projections are of most interest at the end of the test period, T, and for milestones beyond T, provided the projections are not made too far beyond the range of the data. The curve above is plotted from the origin and over the entire range of the data simply to capture and display the trend with which the reliability is improving over time. At the end of the test, namely at T = 1856 hours, the projected MTBF is estimated to be approximately 12.8 hours. Given the steep rate of occurrence of B-modes, the gap method may be an appropriate strategy to use. By jumping the gap, this allows us to specify a segment of time at the beginning of the data so that the set of B-modes used in the analysis are those that have occurred after the gap. It is important to note that to use this approach, there ought to be some compelling reason or reasons, something peculiar about the initial test situation, that would justify using the gap method. In choosing a positive gap size, the underlying assumption is that there is a special group of B-modes occurring within the gap whose failure mechanisms are such that they can be assigned very high fix effectiveness factors (perhaps, close to 1), thus excluding them from consideration. It is equivalent to stating that there are two types of B-modes occurring within the gap: those with a collective failure rate λ_1 due to start-up problems and those with a collective failure rate λ_2 which are not due to start-up problems. Things that may contribute to start-up problems within the gap are:

a. Vendor problems
b. Initial assembly problems due to inadequate workmanship
c. Inferior parts selection
d. Excessive variation in the material
e. Operator unfamiliarity with the system

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In choosing potential gap sizes, it is important to use visual as well as statistical methods for assessing model goodness-of-fit. A good tool for visual purposes is a plot of the estimates for the expected versus observed number of B-modes. A good statistical tool for assessing model applicability is the chi-square goodness-of-fit procedure.

7.6.9.1.2 Application of the Gap Method.
Suppose there were start-up problems that gave rise to the steep slope shown at the start of the example curve in Figure 7-6, and that it was determined by engineering analysis that it was reasonable to choose a gap size of $v = 250$ hours (v is a variable that represents the gap size). As noted previously, the total test time is 1856 hours, the number of observed B-modes is 96, the number of A-mode failures is 0, the number of projections to make is 1, the total number of B-mode failures (which includes 1st occurrences plus repeats) is 104, the assumed number (K) of B-modes initially in the system is 960, and the overall average fix effectiveness for the B-modes is 0.7.

The results of the model application are shown below.

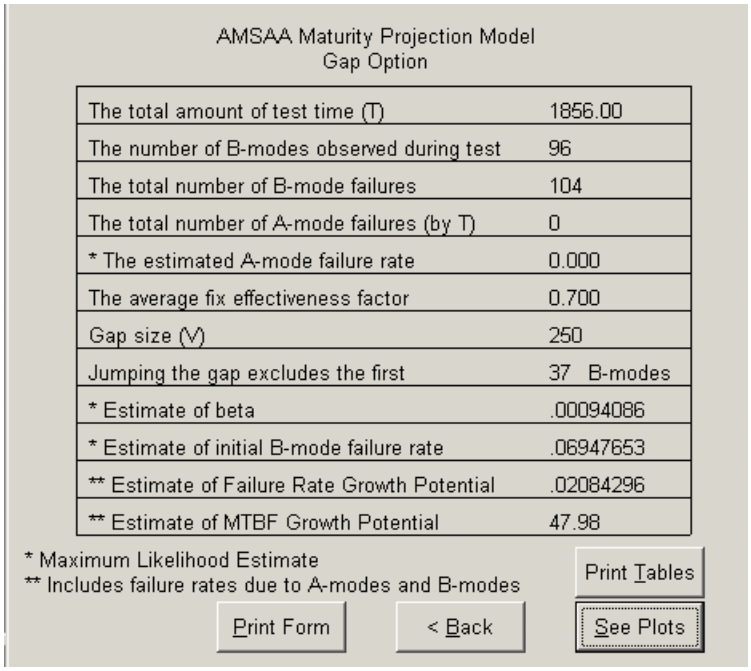


FIGURE 7-11. *AMPM Results Using the Gap Method*

Note that the line "Jumping the gap excludes the first" indicates that the first 37 B-modes are excluded from the analysis. A plot provides a visual perspective of goodness-of-fit. Essentially the model is fitted to the B-modes occurring beyond the gap. The model appears to represent the data very well.

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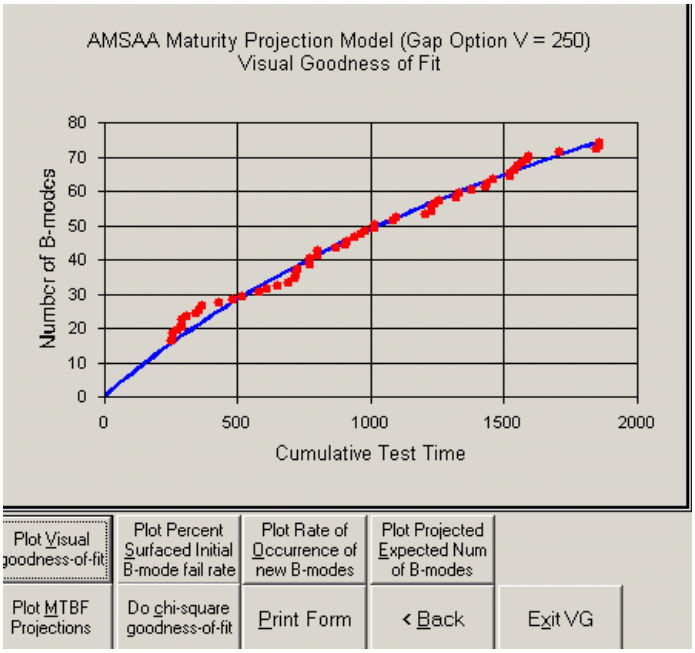


FIGURE 7-12. *Visual Goodness-of-Fit with AMPM (Gap Method, v = 250 Hours)*

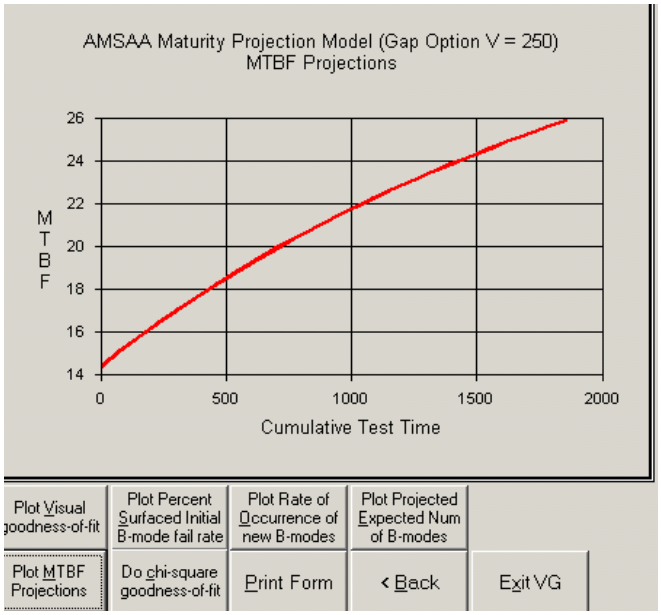


FIGURE 7-13. *Plot of MTBF Projections for AMPM (Gap Option, v = 250 Hours)*

Note that without the gap, the projected MTBF after 1856 hours of test time was approximately 13 hours. With an initial gap size of 250 hours, the projected MTBF after 1856 hours of time is approximately 26 hours.

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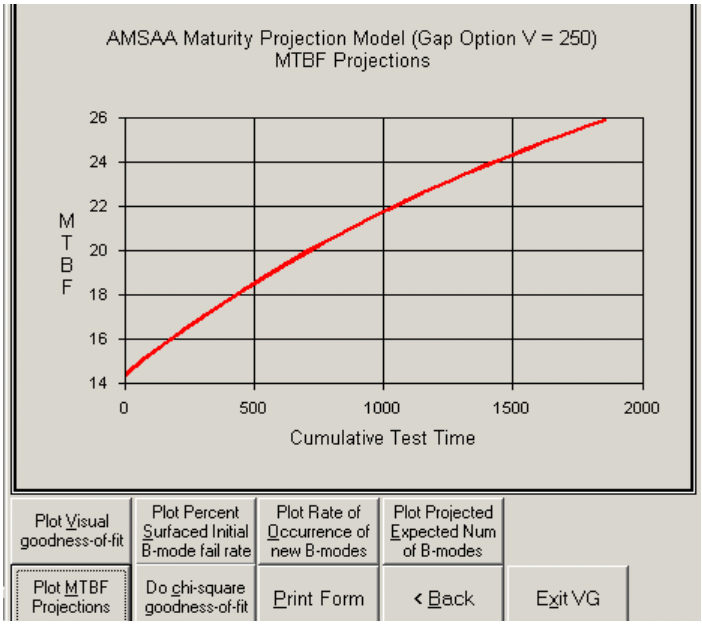


FIGURE 7-14. *Plot of MTBF Projections for AMPM (Gap Option, v = 250 Hours)*

MTBF projections take into account the A-mode failure intensity (zero in our case), the B-mode failure intensity, and fix effectiveness. Recall that fixes may be delayed or incorporated during the test phase and that model estimates are based solely on B-mode first occurrence times. Note that without the gap, the projected MTBF after 1856 hours of test time was approximately 13 hours. With an initial gap size of 250 hours, the projected MTBF after 1856 hours of time is approximately 26 hours.

These same data will be used in the next section to illustrate the segmented FEF approach method and a comparison of the estimates.

Segmented Fix Effectiveness Factor (FEF) Method.

7.6.9.1.3 Rationale for Using the Segmented FEF Method.
The segmented FEF method is another strategy that may be useful for a dataset where the initial rate of occurrence of B-modes is very steep due to start-up problems. With this method, a partition point v is chosen such that a relatively high average fix effectiveness factor d_1 is applied to the B-modes occurring on or before v, and a more typical average fix effectiveness factor d_2 is applied to the B-modes surfaced beyond v. Note that this method is justified only if the early B-modes (those occurring on or before v) are aggressively and effectively corrected. Engineering analysis should be the driving force in choosing the initial segment v, however, statistical analysis may be useful in viewing and/or estimating v. In using this method, the underlying

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assumption is that the B-modes occurring during the early segment are those whose failure mechanisms are so well understood that they can be assigned very high fix effectiveness factors.

7.6.9.1.4 Segmented Fix Effectiveness Factor (FEF) Method.
In situations where there is a steepness of cumulative number of B-modes versus cumulative test time over an early portion of testing after which this rate of occurrence slows as shown in the previous section on Figure 7-6 Example Curve for Illustrating the Gap Method. Again, the same cautions apply to this method as with the gap method, i.e., early B-modes are aggressively and effectively corrected.

Recall that the failure intensity projection equation (for single FEF) is given by

$$\rho(t) = \lambda_A + (1 - d)[\lambda_B - h(t)] + h(t).$$

The failure intensity projection equation for the segmented FEF method is given by

$$\rho(t) = \lambda_A + (1 - d_1)\big(\lambda_B - h(v)\big) + (1 - d_2)(h(v) - h(t) + h(t)$$

where $0 \leq v < T \leq t$. "v" is a partition point such that fix effectiveness factor d_1 is applied to B-modes surfaced on or before v, and fix effectiveness factor d_2 is applied to B-modes surfaced beyond v. Note that when $v=0$ the last equation reduces to the first equation and $h(0) = \lambda_B$.

There are two choices with respect to fix implementation when using the AMPM:

a. All fixes are delayed until the end of the test.
b. Some (or all) fixes are implemented during the test.

(A third option is that no fixes are put in at all, in which case a projection model should not even be used!)

If ALL fixes are implemented at the END of the test phase, then choose the first option. This will provide two additional choices. If there are a number of B-mode repeats, then it is advantageous to use the additional information provided by the repeat data, so select "Case A." If all fixes are delayed and there are no B-mode repeats, then there is no advantage in using repeat information in estimating model parameters, so choose Case B, in which case model estimates will be based solely on B-mode first occurrence times.

On the other hand, if some or all fixes are incorporated during the test, then select "Option 2." There is only one choice for option two and that is to use B-mode first occurrence times only. If there are any doubts regarding fix implementation, use option two. This is essentially the message behind the red button. Click on the red button to see this message.

For the data presented here the following figure shows the results of the software AMPM Segmented Method for d_1=0.95, d_2=0.7, v=250, T=1856 and the total number of B-modes, M=96.

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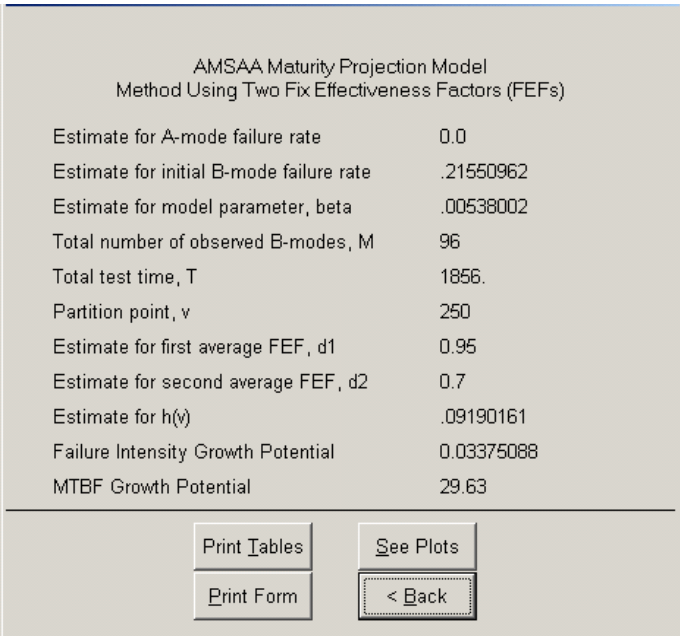
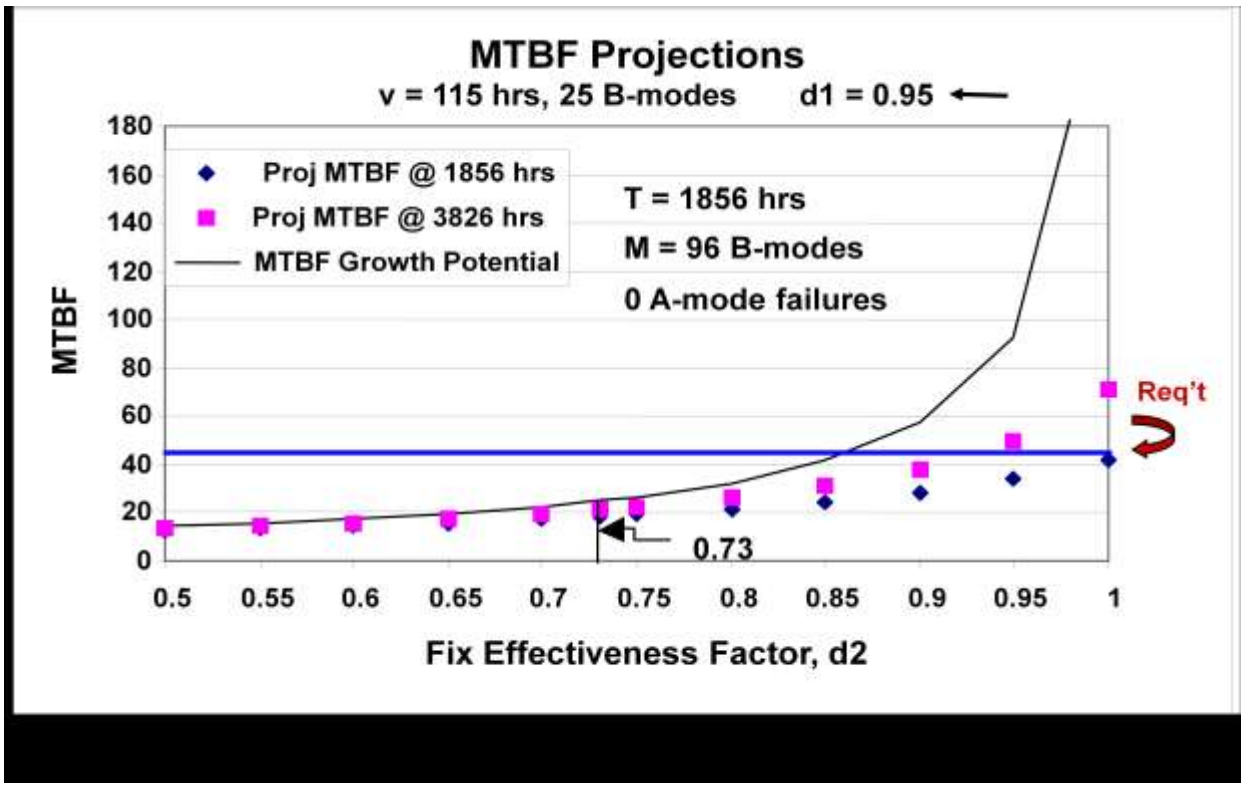


FIGURE 7-15. *AMPM Method Using Two FEFs*

Notice that by using the segmented FEF method, the MTBF growth potential has essentially been doubled from approximately 15 hours to approximately 30 hours.



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FIGURE 7-16. *Moderate Improvement in MTBF Projection Using Segmented FSF Approach*

Note that there is a moderate improvement in MTBF projection using the segmented FEF approach even with higher initial fix effectiveness, however, it is noted that in this example, achieving the requirement is still a challenge due to the high requirement of FEF for d_2 (at least 0.95).

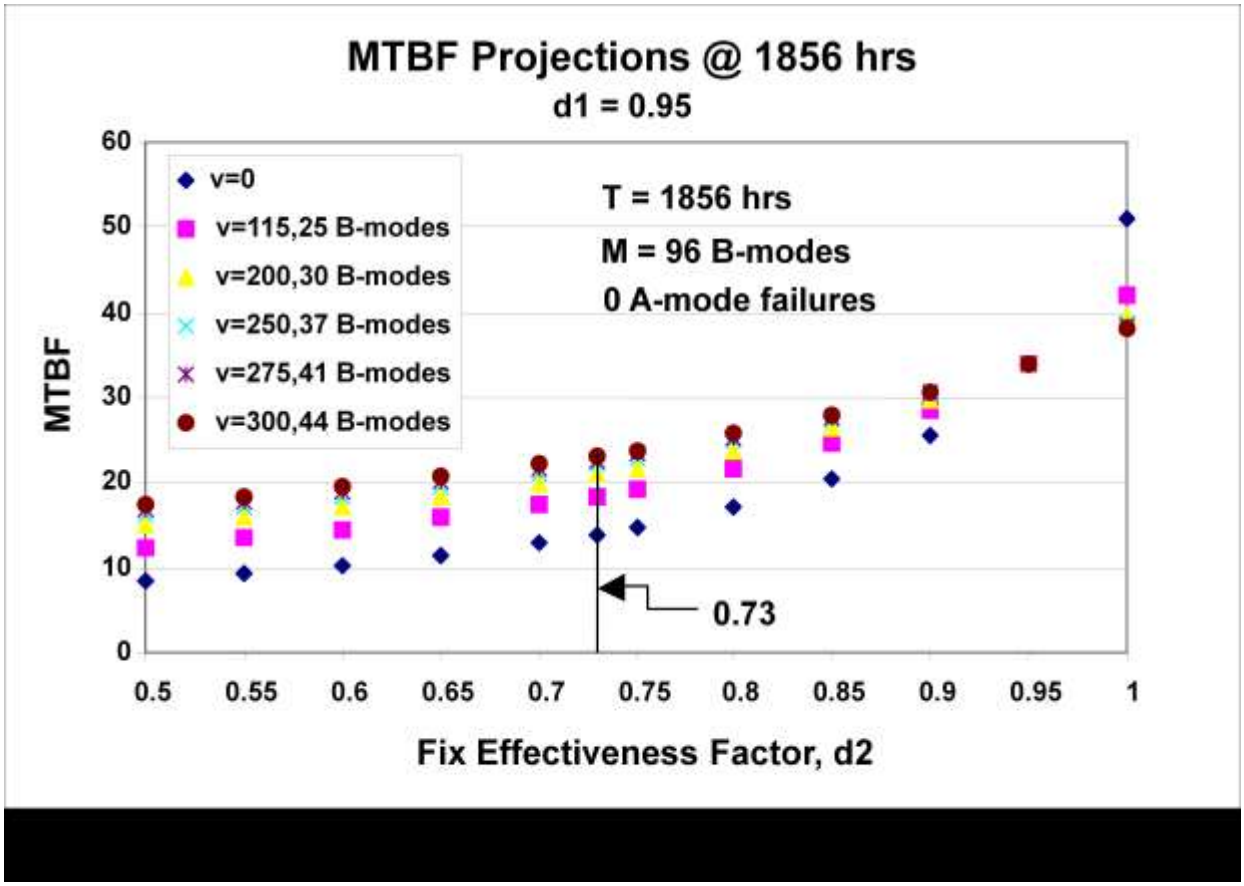


FIGURE 7-17. *"v" Should Be Chosen Based On Engineering Analysis*

The above chart presents MTBF projections for a range of partition points and a range of FEFs. In determining a partition point, "*v*", while statistical analysis and plotting may be useful, it is determination should be driven based on engineering analysis.

Restart Method.
Another approach to analyze the data would be to re-initialize the data beginning at the partition point "*v*." Thus, failure data prior to *v* would not be used in the analysis. Note that a repeat of a failure mode occurring prior to *v* that occurs after *v* may now be the first occurrence of that failure mode and thus included as a "first occurrence."

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7.6.9.1.5 Rationale.

The rationale for implementing this approach versus the Gap Method may be due more for practical analysis or engineering concerns of the data such as significant changes in the systems configuration or possible differences in test conditions.

7.6.9.1.6 Methodology.

The basic difference between the Gap Method and the Restart Method is after determining a "partition" v, the Restart Method reinitializes test time to zero at v. And as noted above, first occurrences of modes prior to v having repeats after v would now, with the next occurrence of that mode, be considered first occurrences. The analysis would then be the same as with the normal AMPM application.

7.7 The AMSAA Maturity Projection Model based on Stein Estimation (AMPM-Stein).

The material in this section is in accordance with [11].

7.7.1 Differences in Technical Approach.

The AMPM-Stein approach does not require one to distinguish between A-modes and B-modes other than through the assignment of a zero, and positive FEF, respectively, to surfaced modes. Also, only FEFs associated with the surfaced modes need be referenced. In particular, unlike the methods in Crow (1982) and Ellner, et al. (1995), no estimate of the arithmetic average of all the FEFs, that would be realized if all the B-modes were surfaced, is required. Another significant difference between the Stein approach and the other methods is that the Stein projection is a direct assessment of the realized system failure rate after failure mode mitigation. The approaches (Crow, 1982), (Corcoran, et al., 1964), and (Ellner, et al., 1995) indirectly attempt to assess the realized system reliability by estimating the expected value of the mitigated system probability of failure or system failure rate, $r(T)$, where $r(T)$ is viewed as a random variable. For example, in Crow (1982) and Ellner, et al. (1995), the realized value of $r(T)$ is assessed as the estimate of a conditional (given λ_i), or unconditional expected value of $r(T)$, respectively, where,

$$r(T) = \lambda_A + \sum_{i=1}^{k} \left(1 - d_i \cdot I_i(T)\right)\lambda_i \qquad 7.7\text{-}1$$

Corcoran, et al. (1964) proceeds in a similar fashion for one-shot systems. In Crow (1982) it is shown that,

$$E\left[r(T)\right] = \lambda_A + \sum_{i \in B}(1 - d_i)\lambda_i + \sum_{i \in B} d_i \cdot \lambda_i \cdot e^{-\lambda_i T} \qquad 7.7\text{-}2$$

To estimate $E\left[r(T)\right]$, the AMSAA/Crow method approximates $\sum_{i \in B} d_i \cdot \lambda_i \cdot e^{-\lambda_i T}$ by $\mu_d \cdot \sum_{i \in B} \lambda_i \cdot e^{-\lambda_i T}$ where

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$$\mu_d = \frac{1}{k_B} \cdot \sum_{i \in B} d_i \qquad 7.7\text{-}3$$

The value μ_d is estimated by

$$\hat{\mu}_d = \frac{1}{m_B} \cdot \sum_{i \in obs(B)} d_i^* \qquad 7.7\text{-}4$$

where d_i^* is an assessment of d_i. The sum $\sum_{i \in B} \lambda_i \cdot e^{-\lambda_i T}$ is estimated (Crow, 1982) by noting that the number of B-modes surfaced by *t* is

$$M(t) = \sum_{i \in B} I_i(t) \qquad 7.7\text{-}5$$

and the expected number of B-modes by *t*, is

$$\mu(t) = E\left[M(t)\right] = \sum_{i \in B} (1 - e^{-\lambda_i t}) \qquad 7.7\text{-}6$$

Thus, the slope of $\mu(t)$ is

$$\frac{d\mu(t)}{dt} = \sum_{i \in B} \lambda_i \cdot e^{-\lambda_i t} \qquad 7.7\text{-}7$$

and represents the expected rate of occurrence of B-modes at *t*. By assuming $\mu(t)$ can be approximated by

$$\mu_c(t) = \lambda_c \cdot T^{\beta_c} \qquad 7.7\text{-}8$$

and that $M(t)$ is a Poisson process with mean value function $\lambda_c \cdot t^{\beta_c}$, Crow (1982) develops an estimation procedure for λ_c and β_c based on the B-mode first occurrence times and number of surfaced B-modes. This yields an estimate of

$$h_c(t) = \frac{d\mu_c(t)}{dt} = \lambda_c \cdot \beta_c \cdot t^{\beta_c - 1} \qquad 7.7\text{-}9$$

which represents the rate of occurrence of new B-modes at time t for the AMSAA/Crow model. The resulting estimate of $h_c(t)$, $\hat{h}_c(t)$, is taken as an assessment of $\sum_{i \in B} \lambda_i \cdot e^{-\lambda_i T}$, and $\hat{\mu}_d \cdot \hat{h}_c(t)$ is utilized as an assessment of $\sum_{i \in B} d_i \cdot \lambda_i \cdot e^{-\lambda_i T}$. The assessment for $E[r(T)]$ (Crow, 1982), and hence the indirect assessment of the realized value of $r(T)$, is then obtained as

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$$\hat{E}[r(T)] = \frac{N_A}{T} + \sum_{i \in obs(B)} (1 - d_i^*) \left(\frac{N_i}{T} \right) + \hat{\mu}_d \cdot \hat{h}_c(T) \qquad 7.7\text{-}10$$

The AMPM approach (Ellner et al., 1995) treats the initial B-mode failure rates λ_i as a realized random sample of size k_B from a gamma random variable, $\Gamma[\alpha, \beta]$, with density

$$f(x) = \begin{cases} \dfrac{x^{\alpha} \cdot e^{-x/\beta}}{\Gamma(\alpha+1) \cdot \beta^{\alpha+1}} & x > 0 \\ 0 & x \le 0 \end{cases} \qquad 7.7\text{-}11$$

where Γ is the gamma function, $\alpha > -1$ and $\beta > 0$. The AMPM approach replaces T with $t \ge T$ and the λ_i in equation (7.7-2) by independent and identically distributed random variables $\Lambda_i \sim \Gamma[\alpha, \beta]$. The expected value of $E[r(t)]$ with respect to $\Lambda_1, \ldots, \Lambda_{k_B}$ is obtained as

$$\rho(t) = \lambda_A + (1 - \mu_d)(\lambda_B - h(t)) + h(t) \qquad 7.7\text{-}12$$

where

$$h(t) = \frac{k_B \cdot \beta \cdot (\alpha+1)}{(1 + \beta \cdot t)^{\alpha+2}} \qquad 7.7\text{-}13$$

and $\lambda_B = h(0)$. Maximum likelihood estimates for λ_B, β and α, denoted by $\hat{\lambda}_{B,k_B}, \hat{\beta}_{k_B}$ and $\hat{\alpha}_{k_B}$ respectively, are obtained based on k_B, the number of surfaced B-modes, and the observed B-mode first occurrence times. The realization of the mitigated system failure rate, $r(t)$, is assessed as the resulting estimate, $\hat{\rho}_{k_B}(t)$, of $\rho(t)$. The limiting values of $\hat{\lambda}_{B,k_B}, \hat{\beta}_{k_B}$, and $\hat{\alpha}_{k_B}$ are obtained and used to derive $\hat{\rho}_{\infty}(t) = \lim_{k_B \to \infty} \hat{\rho}_{k_B}(t)$. This limiting estimate is taken as the assessment of the realized value of $r(t)$ for complex systems (i.e. for large k_B). The use of B-mode first occurrence times to estimate λ_B, β and α allows the AMPM approach (Ellner, et al., 1995) to be used to assess the realized value of $r(t)$ for $t \ge T$. One need only assume that all fixes are incorporated by *t*. In particular, it is not necessary to assume that all fixes are delayed until *t*. Note, however, the AMPM and AMSAA/Crow methods both require an assessment of μ_d, the arithmetic average of the B-mode FEFs that would be realized if all B-modes were surfaced. In addition, both these methods utilize the number of observed B-modes and the B-mode first occurrence times for parameter estimation to obtain an assessment of the realized mitigated system failure rate. Thus, these methods do not solely distinguish between A-modes and surfaced B-modes for estimation purposes by assigning zero FEFs to the former and positive FEFs to the latter.

Finally, we note a connection between the AMPM estimate for $h(t)$ and the Stein projection. It is shown that $h(t)$ is the expected failure rate due to the B-modes not surfaced by *t* (Ellner et al.,

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1995). As suggested by equation (7.7-13) the AMPM estimate based on k_B potential B-modes is

$$\hat{h}_{k_B}(t) = \frac{\hat{\lambda}_{B,k_B}}{(1+\hat{\beta}_{k_B} \cdot t)^{\hat{\alpha}_{k_B}+2}} \qquad 7.7\text{-}14$$

It is shown in Ellner, et al. (1995) that

$$\hat{h}_{\infty}(t) = \lim_{k_B \to \infty} \hat{h}_{k_B}(t) = \frac{\hat{\lambda}_{B,\infty}}{(1+\hat{\beta}_{\infty} \cdot t)} \qquad 7.7\text{-}15$$

where $\hat{\lambda}_{B,\infty}$ and $\hat{\beta}_{\infty}$ are positive constants. For $t = T$ this form will be shown in Section 7.7.4 to be compatible for complex systems with the Stein projection expression for the portion of the mitigated system failure rate attributable to the B-modes not surfaced by *T*. This is interesting to note since the Stein projection approach does not treat the initial B-mode failure rates as a realization of a random sample from some assumed parent population.

7.7.2 Stein Approach to Projection using One Classification of Failure Modes.

Assume the system has $k > 1$ potential failure modes that have initial failure rates $\lambda_1,...,\lambda_k$. It is assumed the modes independently generate failures and that the system fails whenever a failure mode occurs. It is also assumed that corrective actions do not spawn new failure modes and that all fixes are incorporated into the system at the end of a test period of duration *T* hours, or miles.

Let N_i denote the number of failures encountered for mode *i* that occur during the test. The standard Maximum Likelihood Estimate (MLE) of λ_i is

$$\hat{\lambda}_i = \frac{N_i}{T} \qquad 7.7\text{-}16$$

Let $avg(\hat{\lambda}_i)$ denote the arithmetic average of the estimates $\hat{\lambda}_1,...,\hat{\lambda}_k$. The Stein estimators for $\lambda_1,...,\lambda_k$, denoted by $\tilde{\lambda}_1,...,\tilde{\lambda}_k$, are defined by

$$\tilde{\lambda}_i \equiv \theta \cdot \hat{\lambda}_i + (1-\theta) \cdot avg(\hat{\lambda}_i) \qquad 7.7\text{-}17$$

where $\theta \in [0,1]$ is chosen to minimize the expected sum of mean squared errors, $E\left[\sum_{i=1}^{k} (\tilde{\lambda}_i - \lambda_i)^2\right]$. Let θ_S denote the optimal value of θ and refer to θ_S as the Stein shrinkage factor. Vector estimators, such as $(\tilde{\lambda}_1,...,\tilde{\lambda}_k)$ of multidimensional parameters that satisfy such an optimality criterion were considered by Stein (1981). To obtain θ_S note the following:

$$avg(\hat{\lambda}_i) = \frac{1}{k}\sum_{i=1}^{k}\frac{N_i}{T} = \frac{N}{k \cdot T} \qquad 7.7\text{-}18$$

where

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$$N = \sum_{i=1}^{k} N_i \qquad 7.7\text{-}19$$

$$E(N_i) = \lambda_i \cdot T = Var(N_i) \qquad 7.7\text{-}20$$

From (7.7-20) we have

$$E[\hat{\lambda}_i] = \lambda_i \qquad 7.7\text{-}21$$

and

$$Var[\hat{\lambda}_i] = \frac{\lambda_i}{T} \qquad 7.7\text{-}22$$

Finally, from the above, one can show,

$$E\left[\sum_{i=1}^{k} \tilde{\lambda}_i\right] = \lambda \qquad 7.7\text{-}23$$

and

$$Var\left[\sum_{i=1}^{k} \tilde{\lambda}_i\right] = \frac{\lambda}{T} \qquad 7.7\text{-}24$$

Let

$$\lambda \equiv \sum_{i=1}^{k} \lambda_i \qquad 7.7\text{-}25$$

After some detailed calculation, using equations (7.7-14 through 7.7-20) we arrive at the following result

$$E\left[\sum_{i=1}^{k} (\tilde{\lambda}_i - \lambda_i)^2\right] = \theta^2\left(\frac{\lambda}{T}\right) + (1-\theta)^2 \sum_{i=1}^{k} \lambda_i^2 + 2\theta(1-\theta)\left(\frac{\lambda}{k \cdot T}\right) + (1-\theta)^2\left[\frac{\lambda/T - \lambda^2}{k}\right] \qquad 7.7\text{-}26$$

Thus, $E\left[\sum_{i=1}^{k} (\tilde{\lambda}_i - \lambda_i)^2\right]$ is a quadratic polynomial with respect to θ. The polynomial coefficient of θ^2 is equal to

$$\frac{\lambda}{T} + \sum_{i=1}^{k} \lambda_i^2 - \left(\frac{\lambda}{k \cdot T} + \frac{\lambda^2}{k}\right) =$$

$$\frac{\lambda}{T}\left(1 - \frac{1}{k}\right) + \sum_{i=1}^{k} \lambda_i^2 - \frac{\lambda^2}{k} =$$

$$\frac{\lambda}{T}\left(1 - \frac{1}{k}\right) + \sum_{i=1}^{k} (\lambda_i - \bar{\lambda})^2 > 0 \qquad 7.7\text{-}27$$

for $k > 1$, where

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$$\bar{\lambda} \equiv \frac{\lambda}{k} \tag{7.7-28}$$

The quadratic polynomial with respect to θ in equation (7.7-26) has a unique minimum value that can be found by solving the equation

$$\frac{d}{d\theta}\left(E\left[\sum_{i=1}^{k} (\tilde{\lambda}_i - \lambda_i)^2 \right] \right) = 0 \tag{7.7-29}$$

Denoting the unique value of θ that solves equation (7.7-29) by θ_0, we find

$$\theta_0 = \frac{\sum_{i=1}^{k} (\lambda_i - \bar{\lambda})^2}{\frac{\lambda}{T}\left(1 - \frac{1}{k}\right) + \sum_{i=1}^{k} (\lambda_i - \bar{\lambda})^2} \tag{7.7-30}$$

Thus, we have $\theta_0 \in (0,1)$ which shows that θ_S is equal to θ_0. Let

$$Var[\lambda_i] = \frac{\sum_{i=1}^{k} (\lambda_i - \bar{\lambda})^2}{k} \tag{7.7-31}$$

Note by equation (7.7-30),

$$\theta_S = \frac{Var[\lambda_i]}{\left(\frac{\lambda}{k \cdot T}\right)\left(1 - \frac{1}{k}\right) + Var[\lambda_i]} \tag{7.7-32}$$

By equation (7.7-24), we can see that

$$\frac{1}{k}\sum_{i=1}^{k} \frac{\lambda_i}{T} = \frac{1}{k}\sum_{i=1}^{k} Var[\hat{\lambda}_i] \tag{7.7-33}$$

Let $avg\left(Var[\hat{\lambda}_i]\right)$ denote the right side of equation (7.7-33). It is interesting to note by equation (7.7-32) that

$$\theta_S = \frac{Var[\lambda_i]}{avg\left(Var[\hat{\lambda}_i]\right)\left(1 - \frac{1}{k}\right) + Var[\lambda_i]} \tag{7.7-34}$$

This shows that the Stein estimate of λ_i can be expressed as the following weighted combination of $\hat{\lambda}_i$ and $avg\left(\hat{\lambda}_i\right)$;

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$$\tilde{\lambda}_i = \left[\frac{Var[\lambda_i]}{Var[\lambda_i] + avg\left(Var[\hat{\lambda}_i]\right)\left(1-\frac{1}{k}\right)}\right]\hat{\lambda}_i + \left[\frac{avg\left(Var[\hat{\lambda}_i]\right)\left(1-\frac{1}{k}\right)}{Var[\lambda_i] + avg\left(Var[\hat{\lambda}_i]\right)\left(1-\frac{1}{k}\right)}\right]avg\left(\hat{\lambda}_i\right) \qquad 7.7\text{-}35$$

Therefore, the smaller the population variance of the mode failure rates is relative to the average of the variances associated with the individual mode standard estimators, $Var[\hat{\lambda}_i]$, the more $\tilde{\lambda}_i$ is weighted (i.e. "shrunk") towards $avg\left(\hat{\lambda}_i\right)$.

After mitigation of the failure modes surfaced during the test period [0, T], the realized system failure rate is

$$r(t) = \sum_{i\in obs}(1-d_i)\lambda_i + \sum_{i\in\overline{obs}}\lambda_i \qquad 7.7\text{-}36$$

The Stein projection for $r(T)$, denoted by $\rho_S(T)$ is obtained by replacing d_i by an assessed value d_i^* and by estimating λ_i by $\tilde{\lambda}_i$. Thus,

$$\rho_S(T) = \sum_{i\in obs}(1-d_i^*)\tilde{\lambda}_i + \sum_{i\in\overline{obs}}\tilde{\lambda}_i \qquad 7.7\text{-}37$$

Note for mode $i\in\overline{obs}$, by definition $N_i = 0$. Thus, by equation (7.7-17) where $\theta = \theta_S$, we obtain for $i\in\overline{obs}$,

$$\tilde{\lambda}_i = (1-\theta)\cdot\left(\frac{N}{k\cdot T}\right) \qquad 7.7\text{-}38$$

Let *m* denote the number of surfaced modes during [*0, T*]. Then by equations (7.7-37) and (7.7-38),

$$\rho_S(T) = \sum_{i\in obs}(1-d_i^*)\tilde{\lambda}_i + \left(1-\frac{m}{k}\right)(1-\theta_S)\left(\frac{N}{T}\right) \qquad 7.7\text{-}39$$

The Stein projection cannot be directly calculated from the data for a set of d_i^* since k is typically unknown before and after the test and θ_S is a function of $Var[\lambda_i]$, λ, and k (or equivalently, $\sum_{i=1}^{k}\lambda_i^2$, λ, and k). However, approximations to the Stein projection can be obtained that can be calculated from the test data and the assessed FEFs.

7.7.3 Stein Approach to Projection using Two Classifications of Failure Modes.

One can also use the Stein projection approach with two failure mode classifications as is done for the AMSAA/Crow and AMPM models. Strictly speaking, such an application of these models demands that there are a priori ground rules for classifying observed modes into A or B-modes which do not become reclassified. The Stein projection for the two failure mode classification case is given by

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$$\rho_{S,2}(T) = \hat{\lambda}_A + \sum_{i \in obs(B)} (1 - d_i^*)\tilde{\lambda}_i + \sum_{i \in \overline{obs(B)}} \tilde{\lambda}_i \qquad 7.7\text{-}40$$

In equation (7.7-40) λ_A denotes the collective failure rate due to the A-modes and

$$\hat{\lambda}_A = \frac{N_A}{T} \qquad 7.7\text{-}41$$

where N_A is the number of observed A-mode failures. Also, $obs(B)$ denotes the index set of the observed B-modes and $\overline{obs(B)}$ denotes its complement in B. For $i \in B$, $\tilde{\lambda}_{i,2}$ denotes the Stein estimate of λ_i for two classifications. In place of the expression for $\tilde{\lambda}_i$ in equation (7.7-17), $\tilde{\lambda}_{i,2}$ is defined as

$$\tilde{\lambda}_{i,2} \equiv \theta_{S,B} \cdot \hat{\lambda}_i + (1 - \theta_{S,B}) \cdot \frac{\sum_{i \in B} \hat{\lambda}_i}{k_B} \qquad 7.7\text{-}42$$

In equation (7.7-42), k_B is the number of potential B-modes. The shrinkage factor $\theta_{S,B}$ in equation (7.7-42) is the value of $\theta \in [0,1]$ that minimizes $E\left[\sum_{i \in B} (\tilde{\lambda}_{i,2} - \lambda_i)^2\right]$. This optimal value for θ is derived in a manner similar to θ_S. In place of equation (7.7-30),

$$\theta_{S,B} = \frac{\sum_{i \in B} (\lambda_i - \bar{\lambda}_B)^2}{\frac{\lambda_B}{T}\left(1 - \frac{1}{k_B}\right) + \sum_{i \in B} (\lambda_i - \bar{\lambda}_B)^2} \qquad 7.7\text{-}43$$

where

$$\bar{\lambda}_B = \frac{1}{k_B} \sum_{i \in B} \lambda_i \qquad 7.7\text{-}44$$

Finally, we note that the Stein projection for two mode classifications, $\rho_{S,2}(T)$, can be expressed in a manner analogous to equation (7.7-39):

$$\rho_{S,2}(T) = \hat{\lambda}_A + \sum_{i \in obs(B)} (1 - d_i^*)\tilde{\lambda}_{i,2} + \left(1 - \frac{m_B}{k_B}\right)(1 - \theta_{S,B})\left(\frac{N_B}{T}\right) \qquad 7.7\text{-}45$$

In equation (7.7-45), m_B denotes the number of surfaced B-modes and

$$N_B = \sum_{i \in B} N_i = \sum_{i \in obs(B)} N_i \qquad 7.7\text{-}46$$

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7.7.4 Failure Rate due to Unobserved Modes as $k \to \infty$.

The term $\sum_{i \in \overline{obs}} \tilde{\lambda}_i$ in equation (7.7-37) represents the portion of the projected failure rate $\rho_S(T)$ attributed to the failure modes not surfaced by T. Equation (7.7-38) indicates that

$$\sum_{i \in \overline{obs}} \tilde{\lambda}_i = (1-\theta_S)(1-\frac{m}{k})\left(\frac{N}{T}\right) \qquad 7.7\text{-}47$$

Utilizing the expression (7.7-32) for θ_S we obtain

$$1-\theta_S = \frac{\left(\frac{\lambda}{k \cdot T}\right)\left(1-\frac{1}{k}\right)}{Var[\lambda_i]+\left(\frac{\lambda}{k \cdot T}\right)\left(1-\frac{1}{k}\right)} \qquad 7.7\text{-}48$$

Thus,

$$\sum_{i \in \overline{obs}} \tilde{\lambda}_i = \frac{\left(\frac{\lambda}{k \cdot T}\right)\left(1-\frac{1}{k}\right)\left(1-\frac{m}{k}\right)\left(\frac{N}{T}\right)}{Var[\lambda_i]+\left(\frac{\lambda}{k \cdot T}\right)\left(1-\frac{1}{k}\right)} \qquad 7.7\text{-}49$$

Using equation (7.7-31) we obtain,

$$\sum_{i \in \overline{obs}} \tilde{\lambda}_i = \frac{\left(\frac{\lambda}{k \cdot T}\right)\left(1-\frac{1}{k}\right)\left(1-\frac{m}{k}\right)\left(\frac{N}{T}\right)}{\left(\frac{1}{k}\right)\left(\sum_{i=1}^{k}\lambda_i^2-\frac{\lambda^2}{k}\right)+\left(\frac{\lambda}{k \cdot T}\right)\left(1-\frac{1}{k}\right)} \qquad 7.7\text{-}50$$

or equivalently,

$$\sum_{i \in \overline{obs}} \tilde{\lambda}_i = \frac{\left(1-\frac{1}{k}\right)\left(1-\frac{m}{k}\right)\left(\frac{N}{T}\right)}{\left(1-\frac{1}{k}\right)+\left(\frac{\sum_{i=1}^{k}\lambda_i^2}{\lambda}-\frac{\lambda}{k}\right)\cdot T} \qquad 7.7\text{-}51$$

To consider the limiting behavior of the expression in equation (7.7-51) for large k, denote λ_i by $\lambda_{i,k}$ for $i = 1,\ldots,k$ where

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$$\sum_{i=1}^{k} \lambda_{i,k} = \lambda \qquad 7.7\text{-}52$$

and $\lambda \in (0,\infty)$ with $k > 1$. Let $\underset{--k}{\lambda} \equiv (\lambda_{1,k},...,\lambda_{k,k})$. Consider the maximization problem P: $\max_{\underset{--k}{\lambda}} \left(\frac{1}{\lambda} \sum_{i=1}^{k} \lambda_{i,k}^2 \right)$ subject to $\lambda_{i,k} \geq 0$ and (7.6-52). All maximizers $\underset{--k}{\lambda}^{0}$ for problem P are of the form

$$\lambda_{i,k}^{0} = \begin{cases} 0 & i \neq l \\ \lambda & i = l \end{cases} \qquad 7.7\text{-}53$$

Thus, the maximum value for problem P equals

$$\frac{1}{\lambda} \sum_{i=1}^{k} \left(\lambda_{i,k}^{0} \right)^2 = \lambda \qquad 7.7\text{-}54$$

This shows that for any $k > 1$ with at least two non-zero $\lambda_{i,k}$ one has $0 < \frac{\sum_{i=1}^{k} \lambda_{i,k}^2}{\lambda} < \lambda$. This implies that for complex systems or subsystems (i.e. for large k)

$$\sum_{i \in \overline{obs}} \tilde{\lambda}_i \cong \frac{\hat{\lambda}}{1 + \beta_S \cdot T} \qquad 7.7\text{-}55$$

where

$$\hat{\lambda} = \frac{N}{T} \qquad 7.7\text{-}56$$

and $0 < \beta_S < \lambda$ with

$$\beta_S = \frac{1}{\lambda} \sum_{i=1}^{k} \lambda_i^2 \qquad 7.7\text{-}57$$

Using two mode classifications, in a similar fashion one can show that for complex systems or subsystems

$$\sum_{i \in \overline{obs(B)}} \tilde{\lambda}_{i,2} \cong \frac{\hat{\lambda}_B}{1 + \beta_{S,B} \cdot T} \qquad 7.7\text{-}58$$

where $\hat{\lambda}_B = \frac{N_B}{T}$ and $0 < \beta_{S,B} < \lambda_B$, where $\beta_{S,B} = \frac{1}{\lambda_B} \sum_{i \in B} \lambda_i^2$ and $\lambda_B = \sum_{i \in B} \lambda_i$.

The functional form in equation (7.7-58) for approximating $\sum_{i \in \overline{obs(B)}} \tilde{\lambda}_{i,2}$ is the same form utilized by the AMPM model for large k to estimate the failure rate due to the unsurfaced B-modes at the end of the test period. In contrast to the AMPM, the Stein projection approach leads to this form for large k without assuming the initial B-mode failure rates are a realization of a random sample from an assumed parent population. The AMSAA/Crow projection estimates the failure rate at

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the end of test due to the unsurfaced B-modes by $\hat{\lambda}_c \cdot \hat{\beta}_c \cdot T^{\hat{\beta}_c - 1}$, where $\hat{\lambda}_c$ and $\hat{\beta}_c$ are statistical estimates of λ_c and β_c. This functional form arises from the AMSAA/Crow assumption that the number of B-modes that are surfaced by *t* is $M(t)$ where $M(t)$ is a Poisson process with mean value function $\lambda_c \cdot t^{\beta_c}$ for $\lambda_c > 0$ and $\beta_c > 0$.

7.7.5 AMPM-Stein Approximation using MLE.

As shown in the previous section, the Stein projection depends on the unknown constants *k*, $\lambda = \sum_{i=1}^{k} \lambda_i$, and $Var[\lambda_i]$. We now consider an approximation to the Stein projection obtained for a given *k* and for when *k* is unknown but large. To obtain the approximations we assume $\lambda_1,...,\lambda_k$ is a realization of a random sample from a gamma distribution with density function given in equation (7.7-11). We will use the data N_i to obtain MLEs for α and β, denoted by $\hat{\alpha}_k$ and $\hat{\beta}_k$. The method of marginal maximum likelihood will be employed (Martz et al., 1982). We initially use the gamma parameterization used in Martz, et al. (1982) (i.e. $\alpha_0 = \alpha + 1$ and $\beta_0 = \frac{1}{\beta}$, to express the MLE equations. After simplification of, we arrive at equations (7.7-59) and (7.7-60) below,

$$\hat{\alpha}_0 = \frac{\hat{\beta}_0 \sum_{j=1}^{k} \left(\frac{N_j}{T + \hat{\beta}_0} \right)}{k - \hat{\beta}_0 \sum_{j=1}^{k} \left(\frac{1}{T + \hat{\beta}_0} \right)} \qquad 7.7\text{-}59$$

and

$$k \cdot \ln[\hat{\beta}_0] + \sum_{j=1}^{k} \sum_{i=1}^{N_j - 1} \left(\frac{1}{\hat{\alpha}_0 + i} \right) - \sum_{j=1}^{k} \ln[T + \hat{\beta}_0] = 0 \qquad 7.7\text{-}60$$

Let $\hat{\lambda}_k \equiv k \cdot \hat{\beta}_k (1 + \hat{\alpha}_k)$ where $\hat{\alpha}_k$ and $\hat{\beta}_k$ denote the MLEs for α and β, respectively, given the system has *k* potential failure modes. Thus, $\hat{\alpha}_k = \hat{\alpha}_0 - 1$ and $\hat{\beta}_k = \frac{1}{\hat{\beta}_0}$. This yields $\hat{\lambda}_k = \left(\frac{k}{\hat{\beta}_0} \right) \hat{\alpha}_0$. Equations (7.7-59) and (7.7-60) can be rewritten in terms of $\hat{\lambda}_k$ and $\hat{\beta}_k$. Upon simplification we obtain,

$$\hat{\lambda}_k = \frac{N}{T} \qquad 7.7\text{-}61$$

and

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$$\left(\frac{N}{\hat{\beta}_k \cdot T}\right)\ln\left[1+\hat{\beta}_k \cdot T\right] - \sum_{j\in obs}\sum_{i=1}^{N_j-1}\frac{1}{1+\left(\frac{i\cdot\hat{\beta}_k\cdot T\cdot k}{N}\right)} = m \qquad 7.7\text{-}62$$

The sum from $i=1$ to N_j-1 in equation (7.7-62) is defined to be zero if $N_j=1$. Next we consider the limiting values of $\hat{\lambda}_k$ and $\hat{\beta}_k$ as k increases. Let $\hat{\lambda}_\infty \equiv \lim_{k\to\infty}\hat{\lambda}_k$ and $\hat{\beta}_\infty \equiv \lim_{k\to\infty}\hat{\beta}_k$. From equation (7.7-61) and (7.7-62) it follows that

$$\hat{\lambda}_\infty = \frac{N}{T} \qquad 7.7\text{-}63$$

and

$$\left(\frac{N}{\hat{\beta}_\infty \cdot T}\right)\ln[1+\hat{\beta}_\infty \cdot T] = m \qquad 7.7\text{-}64$$

One can show that equation (7.7-64) has a unique positive solution $\hat{\beta}_\infty$ if and only if N > m. This condition is equivalent to saying $N_i>1$ for at least one mode *i*. We will assume this is the case. Consider equation (7.7-62) and let $x_k \equiv \hat{\beta}_k \cdot T$. Then one can show $\exists\, x_k \in (0,\hat{\beta}_\infty \cdot T)$ such that x_k satisfies (7.7-62) provided

$$k > \frac{N^2}{\sum_{j\in obs}(N_j-1)N_j} \qquad 7.7\text{-}65$$

From numerical experience, we conjecture that equation (7.7-65) is a necessary and sufficient condition for a solution $x_k \in (0,\hat{\beta}_\infty \cdot T)$ of equation (7.7-62). However, this has not been established. One can utilize the finite k estimate $\hat{\beta}_k$ to obtain an estimate of the shrinkage factor θ_S. The limiting value $\hat{\beta}_\infty$ will be used to estimate θ_S for complex systems or subsystems. To consider this further, let $\Gamma[\alpha,\beta]$ denote a gamma random variable with density $f(x)$. Also let $\Lambda_1,...,\Lambda_k$ be independent and identically distributed gamma random variables with density $f(x)$, given in equation (7.7-11). Define $\Lambda \equiv \sum_{i=1}^{k}\Lambda_i$. Note $E\left[\Gamma[\alpha,\beta]\right]=\beta\cdot(\alpha+1)$ and $Var\left[\Gamma[\alpha,\beta]\right]=\beta^2\cdot(\alpha+1)$. This implies $E[\Lambda]=k\cdot\beta(\alpha+1)$ and $E[(k-1)\cdot s^2(\Lambda_i)] = (k-1)\cdot\beta^2\cdot(\alpha+1)$ where $s^2(\Lambda_i) = \frac{1}{k-1}\sum_{i=1}^{k}(\Lambda_i-\overline{\Lambda})^2$ and $\overline{\Lambda}=\frac{1}{k}\Lambda$. Thus we will approximate $\lambda=\sum_{i=1}^{k}\lambda_i$ by $k\cdot\hat{\beta}_k(\hat{\alpha}_k+1)=\hat{\lambda}_k$ and $k\cdot Var[\lambda_i] = \sum_{i=1}^{k}(\lambda_i-\overline{\lambda})^2$ by $(k-1)(\hat{\beta}_k)^2(\hat{\alpha}_k+1)=\left(\frac{k-1}{k}\right)\cdot\hat{\beta}_k\cdot\hat{\lambda}_k$. By equation (7.6-32),

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$$\theta_S = \frac{k \cdot Var[\lambda_i]}{\left(\frac{\lambda}{T}\right)\left(1-\frac{1}{k}\right)+k \cdot Var[\lambda_i]} \qquad 7.7\text{-}66$$

Thus we approximate θ_S by

$$\hat{\theta}_{S,k} = \frac{\hat{\beta}_k \cdot T}{1+\hat{\beta}_k \cdot T} \qquad 7.7\text{-}67$$

This suggests that for complex systems or subsystems, (i.e. large *k*) a suitable approximation for θ_S is

$$\hat{\theta}_{S,\infty} = \lim_{k\to\infty}\hat{\theta}_{S,k} = \frac{\hat{\beta}_\infty \cdot T}{1+\hat{\beta}_\infty \cdot T} \qquad 7.7\text{-}68$$

One can now obtain approximations to the Stein projection by utilizing $\hat{\theta}_{S,k}$ and $\hat{\theta}_{S,\infty}$. These approximations will be referred to as the finite *k* and infinite *k* AMPM-Stein projections, respectively. For finite *k*, motivated by equation (7.7-17) we define

$$\tilde{\lambda}_{i,k} \equiv \hat{\theta}_{S,k} \cdot \hat{\lambda}_i + (1-\hat{\theta}_{S,k}) avg(\hat{\lambda}_i) \qquad 7.7\text{-}69$$

for $i = 1,\ldots,k$. The corresponding AMPM-Stein projection for the system failure rate after mitigation of surfaced modes, denoted by $\hat{\rho}_{S,k}(T)$, is given by

$$\hat{\rho}_{S,k}(T) = \sum_{i\in obs}(1-d_i^*)\tilde{\lambda}_{i,k} + \sum_{i\in\overline{obs}}\tilde{\lambda}_{i,k} \qquad 7.7\text{-}70$$

Equation (7.7-70) can be rewritten in a manner analogous to the form of equation (7.6-39) utilized for the Stein projection:

$$\hat{\rho}_{S,k}(T) = \sum_{i\in obs}(1-d_i^*)\tilde{\lambda}_{i,k} + (1-\frac{m}{k})(1-\hat{\theta}_{S,k})\left(\frac{N}{T}\right) \qquad 7.7\text{-}71$$

We also obtain

$$\hat{\rho}_{S,\infty}(T) = \lim_{k\to\infty}\hat{\rho}_{k,\infty}(T) = \sum_{i\in obs}(1-d_i^*)\hat{\theta}_{S,\infty}\hat{\lambda}_i + (1-\hat{\theta}_{S,\infty})\left(\frac{N}{T}\right) \qquad 7.7\text{-}72$$

The corresponding MTBF projection is

$$\hat{M}_{S,\infty} \equiv \left(\hat{\rho}_{S,\infty}(T)\right)^{-1} \qquad 7.7\text{-}73$$

One can also apply the AMPM-Stein projection to the case of two mode classifications based on appropriate a priori mode classification rules. Denoting the two-mode AMPM-Stein system failure rate projections by $\hat{\rho}_{S,2,k_B}(T)$ and $\hat{\rho}_{S,2,\infty}(T)$ for the finite and infinite k_B projection respectively, we let

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$$\hat{\rho}_{S,2,k_B}(T) = \frac{N_A}{T} + \sum_{i \in obs(B)} (1 - d_i^*)\tilde{\lambda}_{i,k_B} + \left(1 - \frac{m_B}{k_B}\right)(1 - \hat{\theta}_{S,k_B,B})\left(\frac{N_B}{T}\right) \quad 7.7\text{-}74$$

and

$$\hat{\rho}_{S,\infty,2}(T) = \frac{N_A}{T} + \sum_{i \in obs(B)} (1 - d_i^*)\tilde{\lambda}_{i,\infty} + (1 - \hat{\theta}_{S,\infty,B})\left(\frac{N_B}{T}\right) \quad 7.7\text{-}75$$

where for equations (7.7-74) and (7.7-75),

$$\tilde{\lambda}_{i,k_B} \equiv \hat{\theta}_{S,B,k_B} \cdot \hat{\lambda}_i + (1 - \hat{\theta}_{S,B,k_B}) \underset{i \in B}{avg}(\hat{\lambda}_i) \quad 7.7\text{-}76$$

and

$$\tilde{\lambda}_{i,\infty} \equiv \lim_{k_B \to \infty} \tilde{\lambda}_{i,k_B} \quad 7.7\text{-}77$$

In equations (7.7-74) and (7.7-75), respectively,

$$\hat{\theta}_{S,B,k_B} = \frac{\hat{\beta}_{B,k_B} \cdot T}{1 + \hat{\beta}_{B,k_B} \cdot T} \quad 7.7\text{-}78$$

and

$$\hat{\theta}_{S,B,\infty} = \frac{\hat{\beta}_{B,\infty} \cdot T}{1 + \hat{\beta}_{B,\infty} \cdot T} \quad 7.7\text{-}79$$

In equation (7.7-78) $\hat{\beta}_{B,k_B} = \frac{y_{k_B}}{T}$ is the MLE solution of the following modification of equation (7.7-62):

$$\left(\frac{N_B}{y_{k_B}}\right)\ln[1 + y_{k_B}] - \sum_{j \in obs(B)} \sum_{i=1}^{N_j - 1} \frac{1}{1 + \left(\frac{i \cdot y_{k_B} \cdot k_B}{N_B}\right)} = m_B \quad 7.7\text{-}80$$

In equation (7.7-80), the sum from $i = 1$ to $i = N_j - 1$ is defined to be zero if $N_j = 1$. In equation (7.7-79), $\hat{\beta}_{B,\infty}$ is the limit of the $\hat{\beta}_{B,k}$, and $y_\infty = \hat{\beta}_{B,\infty} \cdot T$ satisfies the following modification of equation (7.7-64)

$$\left(\frac{N_B}{y_\infty}\right)\ln[1 + y_\infty] = m_B \quad 7.7\text{-}81$$

For equations (7.7-80) and (7.7-81) we assume $N_B > m_B$. This guarantees equation (7.7-81) has a unique positive solution, y_∞. If in addition,

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$$k_B > \frac{(N_B)^2}{\sum_{j \in obs(B)} (N_j - 1) N_j} \qquad 7.7\text{-}82$$

(the counterpart of equation (7.7-65)) then equation (7.7-82) will have a solution $y_{k_B} \in (0, y_\infty)$.

7.7.6 AMPM-Stein Approximation using MME.

Using Method of Moment Estimation (MME) a second estimation procedure was utilized to estimate θ_S and obtain associated approximations to the Stein projection for the mitigated system failure rate. The second procedure is a method of moments presented in Chapter 7 of Martz, et al. (1982). Once again assume that $\lambda_1,...,\lambda_k$ is a realization of a sample of size *k* from $\Gamma(\alpha, \beta)$. It is noted in Chapter 7 of Martz, et al. (1982) that the marginal distribution of N_j is given by the density $g(n_j, \alpha_0, \beta_0)$ where

$$g(n_j, \alpha_0, \beta_0) = \frac{\beta_0^{\alpha_0} \cdot T^{n_j} \cdot \Gamma(n_j + \alpha_0)}{\Gamma(\alpha_0) \cdot n_j! \cdot (T + \beta_0)^{n_j + \alpha_0}} \qquad 7.7\text{-}83$$

for $n_j = 0,1,2,...$ where $\alpha_0 = \alpha + 1$, and $\beta_0 = \frac{1}{\beta}$ and $n_j!$ denotes the factorial of n_j. The marginal mean and variance are

$$E[N_j, \alpha_0, \beta_0] = \frac{\alpha_0 \cdot T}{\beta_0} \qquad 7.7\text{-}84$$

and

$$Var[N_j, \alpha_0, \beta_0] = \frac{\alpha_0 \cdot T \cdot (T + \beta_0)}{\beta_0^2} \qquad 7.7\text{-}805$$

It follows that the marginal mean and variance of $\hat{\lambda}_j$ are

$$E[\hat{\lambda}_j, \alpha_0, \beta_0] = \frac{\alpha_0}{\beta_0} \qquad 7.7\text{-}81$$

and

$$Var[\hat{\lambda}_j, \alpha_0, \beta_0] = \frac{\alpha_0 \cdot (T + \beta_0)}{T \cdot \beta_0^2} \qquad 7.7\text{-}827$$

Let $\overline{\lambda}_u \equiv \sum_{i=1}^{k} \frac{\hat{\lambda}_j}{k}$ and $m_u^2 \equiv \sum_{j=1}^{k} \frac{\hat{\lambda}_j^2}{k}$ (the unweighted sample mean and second sample moment about the origin respectively for $\hat{\lambda}_1,...,\hat{\lambda}_k$. In Martz, et al. (1982) it is shown that

$$E[\overline{\Lambda}_u, \alpha_0, \beta_0] = \frac{\alpha_0}{\beta_0} \qquad 7.7\text{-}838$$

and

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$$E[M_u^2, \alpha_0, \beta_0] = \frac{\alpha_0[T \cdot (1+\alpha_0) + \beta_0]}{T \cdot \beta_0^2} \qquad 7.7\text{-}849$$

where $\overline{\Lambda}_u$ and M_u^2 are random variables that take on the values of $\overline{\lambda}_u$ and m_u^2, respectively. This suggests implicitly defining the unweighted moment estimators for α_0 and β_0, denoted by $\widetilde{\alpha}_0$ and $\widetilde{\beta}_0$, respectively, through the following equations:

$$\overline{\lambda}_u = \frac{\widetilde{\alpha}_0}{\widetilde{\beta}_0} \qquad 7.7\text{-}90$$

and

$$m_u^2 = \frac{\widetilde{\alpha}_0[T \cdot (1+\widetilde{\alpha}_0) + \widetilde{\beta}_0]}{T \cdot \widetilde{\beta}_0^2} \qquad 7.7\text{-}91$$

Let $\widetilde{\alpha}_k$ and $\widetilde{\beta}_k$ be the corresponding method of moments estimators for α and β based on assuming k potential failure modes. Thus, $\widetilde{\alpha}_k = \widetilde{\alpha}_0 - 1$ and $\widetilde{\beta}_k = \frac{1}{\widetilde{\beta}_0}$. Let

$$\widetilde{\lambda}_k \equiv k \cdot \widetilde{\beta}_k \cdot (1+\widetilde{\alpha}_k) \qquad 7.7\text{-}92$$

From equations (7.7-90) and (7.7-91) it can be shown that

$$\widetilde{\beta}_0 = \frac{\widetilde{\alpha}_0}{\overline{\lambda}_u} = \frac{k \cdot \overline{\lambda}_u}{k \cdot (m_u^2 - \overline{\lambda}_u) - H \cdot \overline{\lambda}_u} \qquad 7.7\text{-}93$$

where $H = \frac{k}{T}$. From the first equality in equation (7.7-93) we have

$$\widetilde{\beta}_k = \frac{\overline{\lambda}_u}{\widetilde{\alpha}_0} = \frac{\overline{\lambda}_u}{1+\widetilde{\alpha}_k} \qquad 7.7\text{-}94$$

Thus

$$\widetilde{\beta}_k \cdot (1+\widetilde{\alpha}_k) = \overline{\lambda}_u = \frac{1}{k}\sum_{j=1}^{k} \hat{\lambda}_j \qquad 7.7\text{-}95$$

This yields

$$\widetilde{\lambda}_k = k \cdot \widetilde{\beta}_k \cdot (1+\widetilde{\alpha}_k) = \sum_{j=1}^{k} \frac{N_j}{T} = \frac{N}{T} \qquad 7.7\text{-}96$$

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$$\tilde{\beta}_k = \frac{k \cdot (m_u^2 - \lambda_u^2) - H \cdot \lambda_u}{k \cdot \bar{\lambda}_u} \qquad 7.7\text{-}97$$

from the second equality in equation (7.7-93). Note, $k \cdot m_u^2 = \sum_{j=1}^{k} \hat{\lambda}_j^2 = \sum_{j=1}^{k} \left(\frac{N_j}{T} \right)^2 = \frac{1}{T^2} \sum_{j \in obs} N_j^2$.

Also,

$$k \cdot \bar{\lambda}_u^2 = \frac{1}{k \cdot T^2} \left(\sum_{j \in obs} N_j \right)^2 \qquad 7.7\text{-}98$$

$$H \cdot \bar{\lambda}_u = \frac{1}{T^2} \left(\sum_{j \in obs} N_j \right) \qquad 7.7\text{-}99$$

and

$$k \cdot \bar{\lambda}_u = \sum_{j \in obs} \frac{N_j}{T} \qquad 7.7\text{-}100$$

Thus by equation (7.7-97)

$$\tilde{\beta}_k = \frac{\frac{1}{T^2} \sum_{j \in obs} N_j^2 - \frac{1}{k \cdot T^2} \left(\sum_{j \in obs} N_j \right)^2 - \frac{1}{T^2} \sum_{j \in obs} N_j}{\sum_{j \in obs} \frac{N_j}{T}} \qquad 7.7\text{-}101$$

This yields,

$$\tilde{\beta}_k = \frac{\sum_{j \in obs} N_j^2 - \frac{N^2}{k} - N}{T \cdot N} \qquad 7.7\text{-}102$$

One can now obtain the method of moments limit estimators $\tilde{\lambda}_\infty$ and $\tilde{\beta}_\infty$ as k increases. These are $\tilde{\lambda}_\infty = \lim_{k \to \infty} \tilde{\lambda}_k$ and $\tilde{\beta}_\infty = \lim_{k \to \infty} \tilde{\beta}_k$. From equation (7.7-96),

$$\tilde{\lambda}_\infty = \frac{N}{T} \qquad 7.7\text{-}103$$

Also, from equation (7.7-102),

$$\tilde{\beta}_\infty = \frac{1}{T} \left(\frac{\sum_{j \in obs} N_j^2}{N} - 1 \right) \qquad 7.7\text{-}104$$

The moment estimates $\tilde{\beta}_k$ and $\tilde{\beta}_\infty$ of β provide the respective estimates $\tilde{\theta}_{S,k}$ and $\tilde{\theta}_{S,\infty}$ of the Stein shrinkage factor θ_S, where

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$$\widetilde{\theta}_{S,k} = \frac{\widetilde{\beta}_k \cdot T}{1 + \widetilde{\beta}_k \cdot T} \qquad 7.7\text{-}105$$

and

$$\widetilde{\theta}_{S,\infty} = \frac{\widetilde{\beta}_\infty \cdot T}{1 + \widetilde{\beta}_\infty \cdot T} \qquad 7.7\text{-}106$$

The moment estimators of θ_S in equations (7.7-105) and (7.7-106) provide corresponding approximations to the Stein system failure rate projection. Let $\widetilde{\rho}_{S,k}(T)$ and $\widetilde{\rho}_{S,\infty}(T)$ denote these approximations based on $\widetilde{\theta}_{S,k}$ and $\widetilde{\theta}_{S,\infty}$, respectively. In place of equation (7.7-71) for the MLE based approximation of $\rho_S(T)$ we define

$$\widetilde{\rho}_{S,k} = \sum_{i \in obs} (1 - d_i^*)\widetilde{\lambda}_{i,k} + \left(1 - \frac{m}{k}\right)(1 - \widetilde{\theta}_{S,k})\left(\frac{N}{T}\right) \qquad 7.7\text{-}85$$

where, for equation (7.7-107),

$$\widetilde{\lambda}_{i,k} \equiv \widetilde{\theta}_{S,k}\hat{\lambda}_i + (1 - \widetilde{\theta}_{S,k})avg(\hat{\lambda}_i) \qquad 7.7\text{-}86$$

Let,

$$\widetilde{M}_{S,k}(T) \equiv \left(\widetilde{\rho}_{S,k}(T)\right)^{-1} \qquad 7.7\text{-}87$$

denote the corresponding AMPM-Stein MTBF projection based on the finite k moment estimators for θ_S. Next, let

$$\widetilde{\rho}_{S,\infty}(T) \equiv \lim_{k \to \infty} \widetilde{\rho}_{S,k}(T) \qquad 7.7\text{-}88$$

Note, $\lim_{k \to \infty} avg(\hat{\lambda}_i) = \lim_{k \to \infty} \frac{1}{k} \sum_{i \in obs} \hat{\lambda}_i = 0$. Also $\widetilde{\theta}_{S,\infty} \equiv \lim_{k \to \infty} \widetilde{\theta}_{S,k}$ by equations (7.7-105) and (7.7-106), respectively. Thus by equation (7.7-108) $\lim_{k \to \infty} \widetilde{\lambda}_{i,k} = \widetilde{\theta}_{S,\infty}\left(\frac{N_i}{T}\right)$. By equation (7.7-107) this yields

$$\widetilde{\rho}_{S,\infty} = \sum_{i \in obs} (1 - d_i^*)\widetilde{\theta}_{S,\infty}\left(\frac{N_i}{T}\right) + (1 - \widetilde{\theta}_{S,\infty})\left(\frac{N}{T}\right) \qquad 7.7\text{-}89$$

The corresponding MTBF projection is

$$\widetilde{M}_{S,\infty}(T) \equiv \left(\widetilde{\rho}_{S,\infty}(T)\right)^{-1} \qquad 7.7\text{-}90$$

AMPM-Stein projections based on moment estimators can also be developed for the case where failure modes are partitioned into A-modes and B-modes by a priori classification rules. The shrinkage factor, finite k_B approximation to $\theta_{S,B}$ is given by

$$\widetilde{\theta}_{S,B,k_B} = \frac{\widetilde{\beta}_{B,k_B} \cdot T}{1 + \widetilde{\beta}_{B,k_B} \cdot T} \qquad 7.7\text{-}91$$

The estimate $\widetilde{\beta}_{B,k_B}$ in equation (7.7-112) is

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$$\widetilde{\beta}_{B,k_B} = \frac{\sum_{j\in obs(B)} N_j^2 - \left(\frac{1}{k_B}\right) N_B^2 - N_B}{T \cdot N_B} \qquad 7.7\text{-}92$$

Thus,

$$\widetilde{\beta}_{B,\infty} \equiv \lim_{k\to\infty} \widetilde{\beta}_{B,k_B} = \left(\frac{1}{T}\right)\left(\frac{\sum_{j\in obs(B)} N_j^2}{N_B} - 1\right) \qquad 7.7\text{-}93$$

The corresponding large k_B estimate of $\theta_{S,B}$, based on the limit of the method of moments estimator $\widetilde{\beta}_{B,k_B}$, is

$$\widetilde{\theta}_{S,B,\infty} = \frac{\widetilde{\beta}_{B,\infty} \cdot T}{1 + \widetilde{\beta}_{B,\infty} \cdot T} \qquad 7.7\text{-}94$$

The AMPM-Stein system failure rate projections, based on $\widetilde{\theta}_{S,B,k_B}$ and $\widetilde{\theta}_{S,B,\infty}$ when utilizing two mode classifications are denoted by $\widetilde{\rho}_{S,2,k_B}(T)$ and $\widetilde{\rho}_{S,2,\infty}(T)$, respectively. In place of equation (7.7-107) for $\widetilde{\rho}_{S,k}(T)$ we have

$$\widetilde{\rho}_{S,2,k_B}(T) = \frac{N_A}{T} + \sum_{i\in obs}(1-d_i^*)\widetilde{\lambda}_{i,k_B} + \left(1 - \frac{m_B}{k_B}\right)(1-\widetilde{\theta}_{S,B,k_B})\left(\frac{N_B}{T}\right) \qquad 7.7\text{-}95$$

For equation (7.7-113),

$$\widetilde{\lambda}_{i,k_B} \equiv \widetilde{\theta}_{S,B,k_B}\hat{\lambda}_i + (1-\widetilde{\theta}_{S,B,k_B})\underset{i\in B}{avg}(\hat{\lambda}_i) \qquad 7.7\text{-}96$$

The MTBF projection is

$$\widetilde{M}_{S,2,k_B}(T) = \left(\widetilde{\rho}_{S,2,k_B}(T)\right)^{-1} \qquad 7.7\text{-}97$$

Let $\widetilde{\rho}_{S,2,\infty}(T) \equiv \lim_{k_B\to\infty} \widetilde{\rho}_{S,2,k_B}(T)$. Then we can show that

$$\widetilde{\rho}_{S,2,\infty}(T) = \frac{N_A}{T} + \sum_{i\in obs(B)} (1-d_i^*)\widetilde{\theta}_{S,B,\infty}\left(\frac{N_i}{T}\right) + (1-\widetilde{\theta}_{S,B,\infty})\left(\frac{N_B}{T}\right) \qquad 7.7\text{-}98$$

The associated MTBF projection is

$$\widetilde{M}_{S,2,\infty}(T) = \left(\widetilde{\rho}_{S,2,\infty}(T)\right)^{-1} \qquad 7.7\text{-}99$$

7.7.7 Cost versus Reliability Tradeoff Analysis.

At the end of a test phase one might wish to conduct a cost versus reliability tradeoff analysis to assist in selecting a set of surfaced failure modes to address with fixes. For any selected set of surfaced modes, say $Z \subseteq obs$, one could study the underlying root causes of failure to determine

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potential fixes. Based on such a study, a set of positive FEFs, d_i^* for $i \in Z$ could be assessed for the proposed fixes. Actual implementation of these fixes would lower the system failure rate from the initial value, say $\rho(T;\phi) = \sum_{i=1}^{k} \lambda_i$, to a lower failure rate

$$\rho(T;Z) = \sum_{i \in Z} (1 - d_i)\lambda_i + \sum_{i \in (obs - Z)} \lambda_i + \sum_{i \in \overline{obs}} \lambda_i \qquad 7.7\text{-}100$$

with corresponding MTBF $M(T;Z) = (\rho(T;Z))^{-1}$. Note that $\rho(T;Z)$ changes from $\rho(T;\phi)$ as a function of the selected mode set Z only through the FEFs being raised from zero to positive values d_i for $i \in Z$. The assessments for $\rho(T;Z)$ provided by the AMPM-Stein approach have the same property. For example, this can be seen for the AMPM-Stein system failure rate assessment for large k based on MLEs by recalling equation (7.7-72). Since $d_i^* = 0$ for $i \in (obs - Z)$ we have

$$\hat{\rho}_{S,\infty}(T) = \sum_{i \in Z} (1 - d_i^*)\hat{\theta}_{S,\infty}\hat{\lambda}_i + \sum_{i \in (obs - Z)} \hat{\theta}_{S,\infty}\hat{\lambda}_i + (1 - \hat{\theta}_{S,\infty})\left(\frac{N}{T}\right) \qquad 7.7\text{-}101$$

where $\hat{\lambda}_i = \frac{N_i}{T}$. Note by equations (7.7-68) and (7.7-64), $\hat{\theta}_{S,\infty}$ only depends on N and the number of surfaced modes, m. Thus by equation (7.7-123), $\hat{\rho}_{S,\infty}(T)$ is an assessment of $\rho(T;Z)$ that only changes for a given set of test results as Z changes through the resulting change in the FEFs. However, assessments of $\rho(T;Z)$ based on the AMSAA/Crow (Crow, 1982) or AMPM (Ellner et al., 1995) would not change solely due to the change in FEFs brought about by a change in Z. In these methods the modes are partitioned into A-modes and B-modes. If one identifies the modes in Z as the surfaced B-modes (since $d_i^* > 0$ for $i \in Z$) then these assessments would depend on the number of modes in Z and the pattern of first occurrence times for these modes, in addition to the assessed FEFs for $i \in Z$. The dependence of the assessment of $\rho(T;Z)$ on the B-mode first occurrence times indicates that the AMSAA/Crow and the version of the AMPM based on B-mode first occurrence times are not appropriate for this selection problem.

Associated with each selection of Z, one could also assess the cost of implementing all the fixes for the failure modes $i \in Z$. Let $c^*(Z)$ denote this assessed cost. A plot of the points $(\hat{M}_{S,\infty}(T;Z), c^*(Z))$ where $\hat{M}_{S,\infty}(T;Z) = (\hat{\rho}_{S,\infty}(T;Z))^{-1}$ for a number of potential selected sets Z would be useful in identifying the least cost solution Z to meet a reliability goal. Alternately, one could replace $\hat{M}_{S,\infty}(T;Z)$ by the AMPM-Stein assessments of MTBF based on the method of moments estimators.

7.8 Discrete Projection Model.

7.8.1 Introduction.

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7.8.1.1 Background and Motivation.
In this section we present a reliability growth projection model for one-shot systems. The model will not be suitable for application to all one-shot development programs. But it is useful in cases where one or more failure modes are, or can be, discovered in a single trial; and catastrophic failure modes have been previously discovered, and corrected. The model is unique in the area of reliability growth projection, and offers an alternative to the popular competing risks approach.

A survey of discrete reliability growth models indicated limitations when applied to one-shot systems where more than one failure mode is discovered in a given trial. This phenomenon has been encountered on a number of different DoD systems over the years, particularly with smart munitions. This is the primary motivational factor for developing the method in the case considered. A second motivational factor is associated with statistical estimation. Stein [6] developed a statistical estimator based on an optimality criterion; that is, based on minimizing the s-expected sum of squared error. After deriving the required shrinkage factor, this estimator provided good results when utilized in the development of a continuous reliability growth model, known as AMPM-Stein. Simulations conducted by AMSAA indicate that the accuracy in the reliability projections of AMPM-Stein is greater than that of the international standard reliability growth projection model adopted by the International Electrotechnical Commission.
To apply the Stein estimator in the proposed discrete setting, we derived the required shrinkage factor, which is discussed and provided below. In many respects, the presented approach serves as a discrete analogue to the continuous reliability growth projection model AMPM-Stein.

7.8.1.2 Overview.
The methodology of this approach is presented in 7.7.2 which includes: 1) a list of model assumptions; 2) a discussion of the data required; 3) a new method for approximating the vector of failure probabilities inherent to a complex, one-shot system; 4) exact expression for system reliability growth; 5) development of multiple estimation procedures for our model equations; and 6) a graphical method for studying GOF. To highlight model accuracy (e.g., s-bias, and s-variability), Monte-Carlo simulation results are presented in 7.7.3. Concluding remarks are given in 7.7.4.

7.8.1.3 List of Notations.

k	total number of potential failure modes.
m	total number of observed failure modes.
N_{ij}	number of failures for mode i in trial j – zero or unity.
N_i	total number of failures for mode i in T trials.
p_i	true but unknown probability of failure for mode i.
$\hat{p}_i$	MLE of p_i .
$\tilde{p}_I$	theoretical shrinkage factor estimator for p_i .
θ	true but unknown shrinkage factor.
n	beta parameter; pseudo number of trials.
x	beta parameter; pseudo number of failures.
d_i	true but unknown FEF for mode i.
$r(T)$	true but unknown system reliability after mitigation of known failure modes.
$\tilde{r}(T)$	theoretical approximation of $r(T)$ using $\tilde{p}_i$.

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T	total number of trials.

7.8.1.4 Model Assumptions.

a. A trial results in a dichotomous success/failure outcome such that $N_{ij} \sim$ Bernoulli (p_i) for each $i = 1,\ldots, k$, and $j = 1,\ldots,T$.
b. The distribution of the number of failures in T trials for each failure mode is binomial. That is, $N_i \sim$ Binomial (T, p_i) for each $i = 1,\ldots,k$.
c. Initial failure probabilities $p_1 \ldots p_k$ constitute a realization of a s-random sample $P_1,\ldots, P_k$ such that $P_i \sim$ Beta (n,x) for each $i = 1,\ldots, k$.
d. Corrective actions are delayed until the end of the current test phase, where a test phase is considered to consist of a sequence of T s-independent Bernoulli trials.
e. One or more potential failure modes can occur in a given trial, where the occurrence of any one of which causes failure.
f. Failures associated with different failure modes arise s-independently of one another on each trial. As a result, the system must be at a stage in development where catastrophic failure modes have been previously discovered & corrected, and are therefore not preventing the occurrence of other failure modes.
g. There is at least one repeat failure mode. If there is not at least one repeat failure mode, the moment estimators, and the likelihood estimators of the beta parameters do not exist.

7.8.1.5 Data Required.

There are two classes of projection models, each requiring a unique type of data. The first class of models address the case where all fixes are delayed, and the approach presented herein. The second class of projection models address the case where fixes can be either delayed, or non-delayed. In this case, fixes can be implemented during or following the current test phase; hence, the system configuration need not be constant. The data required for reliability growth projection consists of either: count data (i.e., the number of failures for individual failure modes), FOT data (i.e., the times or trials at which failure modes were first discovered), or a mixture of the two.

While estimation procedures have been developed for both classes of projection models, we will only present the case where all fixes are delayed. This requires T, N_i, and d_i for $i = 1,\ldots, m$. The number of trials T, and the count data N_i for observed failure modes are obtained directly from testing. The d_i can be estimated from test data, or assessed via engineering judgment. For many DoD weapon system development programs, FEF are assessed via expert engineering judgment, and assigned in failure prevention review board meetings.

7.8.1.6 Estimation of Failure Probabilities.

The well-known, widely used MLE of a failure probability is given by

$$\hat{p}_i = \frac{N_i}{T} \qquad 7.8\text{-}1$$

The problem with this estimator is that, if there are no observed failures for failure mode j, then $N_j = 0$. Hence, our corresponding estimate of the failure probability is $\hat{p}_j = 0$, which results in an overly optimistic assessment. Therefore, a finite & positive estimate for each failure mode

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probability of occurrence is desired, whether observed during testing or not observed during testing. One way to do this is to utilize a shrinkage factor estimator given by

$$\tilde{\boldsymbol{p}}_{\,i} \equiv \theta \cdot \hat{\boldsymbol{p}}_i + (1-\theta)\cdot \frac{\sum_{i=1}^{k} \hat{\boldsymbol{p}}_i}{\boldsymbol{k}} \qquad 7.8\text{-}2$$

where θ (unknown) is referred to as the shrinkage factor, and k denotes the total potential number of failure modes inherent to the system. The optimal value of $\theta \in (0,1)$ can be chosen to minimize the s-expected sum of squared error, but it must be derived consistently with the specific case considered, and $r.v.$ in question. The associated optimality criterion can be mathematically expressed as

$$\frac{d}{d\theta} \mathrm{E}\left[\sum_{i=1}^{k} (\tilde{p}_i - p_i)^2\right] = 0 \qquad 7.8\text{-}3$$

To derive θ uniquely for our application, we have first expressed the mathematical expectation in (7.8-3) as a quadratic polynomial with respect to θ by assuming that the distribution of the number of failures in T trials conditioned on a given failure mode is binomial, which gives

$$\mathrm{E}\left[\sum_{i=1}^{k} (\tilde{p}_i - p_i)^2\right] = \theta^2 \cdot \left(\frac{p}{T} - \frac{\sum_{i=1}^{k} p_i^2}{T}\right) + 2\theta(1-\theta)\cdot\left(\frac{p}{k\cdot T} - \frac{\sum_{i=1}^{k} p_i^2}{k\cdot T}\right)$$
$$+ (1-\theta)^2 \cdot \left(\frac{p}{k\cdot T} - \frac{\sum_{j=1}^{k} p_j^2}{k\cdot T} + \sum_{i=1}^{k} p_i^2 - \frac{p^2}{k}\right) \qquad 7.8\text{-}4$$

where $p \equiv \sum_{i=1}^{K} p_i$. Using (7.8-4), we have derived the solution to (7.8-3), which we conveniently express as

$$\theta = \qquad 7.8\text{-}5$$

This result is significant for a number of reasons. First, we have expressed the shrinkage factor in terms of quantities that can be easily estimated; namely, the s-mean, and s-variance of the p_i . Second, we have reduced the number of unknowns requiring estimation from (k +1) to only three. The (k +1) unknowns to which we refer include the unknown failure probabilities $p_1,\ldots,p_k$ and the unknown value of k. Finally, estimating (or providing appropriate treatment to) these unknowns yields an approximation of the vector of failure probabilities associated with a complex, one-shot system, where each failure probability (observed or unobserved) is finite, and positive.

7.8.1.7 Reliability Growth Projection.
Let $obs \equiv \{i: N_i > 0 \text{ for } i=1,\ldots,k\}$ represent the index of failure modes observed during testing, and let $obs' \equiv \{j : N_j = 0 \text{ for } j=1, \ldots,k\}$ denote its compliment. After mitigation to (all or a portion of) failure modes observed during testing, we define the true, but unknown system reliability growth as

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$$r(T) \equiv \prod_{i \epsilon obs}[1-(1-d_i)\cdot p_i] \prod_{j \epsilon obs'}(1-p_j) \qquad 7.8\text{-}6$$

where $d_i \epsilon$ [0,1] represents the FEF of failure mode *i,* the true but unknown fraction reduction in initial mode failure probability *i* due to implementation of a unique corrective action. In our model, $(1\text{-}d_i)\cdot p_i$ represents the true reduction ii in failure probability *i* due to correction as originally developed by Corcoran et al. [5]. It will typically be the case that $d_i \epsilon$ (0,1), as d_i =0 models the condition where a given failure mode is not addressed (e.g., an A-mode), and d_i =1 corresponds to complete elimination of the failure mode's probability of occurrence. We would only expect to completely eliminate a failure mode's probability of occurrence when the corrective action consists of the total removal of all components associated with the mode. Notice that our model does not require utilization of the A-mode/B-mode classification scheme proposed in [10], as A-modes need only be distinguished from B-modes via a zero FEF.
The theoretical assessment of (7.8-6) is given by

$$\tilde{r}\,(T) \equiv \prod_{i \epsilon obs}[1-(1-d_i)\tilde{p}_i] \cdot \prod_{j \epsilon obs'}(1-\tilde{p}_j) \qquad 7.8\text{-}7$$

where $\tilde{p}_i$ is expressed via (7.8-2). Note that (7.8-7) is theoretical because *k* is unknown, the d_i for i =*1,.. k* are unknown, and the p_i for i =*1,..., k* (upon which the shrinkage factor is based) are unknown. In the following section, we present several approximations to (7.7-7), which are derived from our estimation method for the vector of the p_i in combination with classical moment-based and likelihood-based procedures for the beta parameters. We also derive unique limiting approximations to (7.8-7).

7.8.2 Estimation Procedures.

7.8.2.1 Parametric Approach.

Assume that the initial mode probabilities of failure $p_{1,},\ldots\ p_k$ constitute a realization of a *s*-random sample $P_{1,}\ldots\ P_k$ from a beta distribution with the parameterization

$$f(p_i) \equiv \frac{\Gamma(n)}{\Gamma(x)\cdot\Gamma(n-x)} \cdot p_i^{x-1} \cdot (1\text{-}\ p_i)^{\text{n-x-1}} \qquad 7.8\text{-}8$$

for *pi* ϵ [0,1], and 0 otherwise; where n represents pseudo trials, x represents pseudo failures, and

$$\Gamma(x) \equiv \int_0^\infty t^{x-1}\cdot e^{-t}\,dt$$

is the gamma function. The above beta assumption not only facilitates convenient estimation of (7.8-5), but models mode-to mode *s*-variability in the initial failure probabilities of occurrence. The source of such *s*-variability could result from many different factors including, but not limited to, variation in environmental conditions, manufacturing processes, operating procedures, maintenance philosophies, or a combination of the above. As indicated by Ellner & Wald [12], the approach of modeling *s*-variability in complex systems is not new Based on our beta assumption with parameterization given by (7.8-8), the associated s-mean, and s-variance are given respectively by

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$$E(P_i) = \frac{x}{n}\ , \qquad 7.8\text{-}9$$

and

$$\text{Var}\ (P_i) = \frac{x \cdot (n-x)}{n^2 \cdot (n-1)} \qquad 7.8\text{-}10$$

Notice that (7.8-5) is in terms of only three unknowns; namely, the population *s*-mean of the failure probabilities, the population *s*-variance of the failure probabilities, and *k*. The first two unknowns are approximated by (7.8-9), and (7.8-10), respectively, which are in terms of the two unknown beta shape parameters. MME and MLE procedures are utilized to approximate these parameters. The third, final unknown, *k*, is treated in two ways. First, we assume a value of *k*, which can be done in applications where the system is well understood. Second, we allow *k* to grow without bound to study the limiting behavior of our model equations. This is suitable in cases where the number of failure modes is unknown, and the system is complex.

7.8.2.2 Moment-based Estimation Procedure.
Moment estimators for the beta shape parameters, per the special case we consider (i.e., where all failure probabilities are estimated via the same number of trials), are given by

$$\breve{n}_k = \frac{k \cdot (\bar{p}_u - m_u^2)}{k \cdot m_u^2 - \frac{k \cdot \bar{p}_u}{n} - \left(k - \frac{k}{n}\right) \cdot \bar{p}_u^2} \qquad 7.8\text{-}11$$

and

$$\breve{x}_k = \breve{n}_k \cdot \bar{p}_u\ , \qquad 7.8\text{-}12$$

where $\bar{p}_u \equiv \frac{\sum_{i=1}^{k} \hat{p}_i}{k}$, and $m_u^2 \equiv \frac{\sum_{i=1}^{k} \hat{p}_i^2}{k}$ are the un-weighted first, and second sample moments, respectively. Using the above MME for the beta parameters with (7.8-5), our approximation of θ can be expressed as

$$\breve{\theta}_k = \frac{1}{\left(\frac{\breve{n}_k}{T}\right)\left(1 - \frac{1}{k}\right) + 1} \qquad 7.8\text{-}13$$

Using (7.8-13), the moment-based shrinkage factor estimate of p_i for finite *k* is then given by

$$\breve{\tilde{p}}_{k,i} = \breve{\theta}_k \cdot \hat{p}_i + (1 + \breve{\theta}_k)\left(\frac{N}{k \cdot T}\right) \qquad 7.8\text{-}14$$

where $N \equiv \sum_{i=1}^{k} N_i$ is the total number of failures observed in *T* trials. Let the total number of observed failure modes be denoted by $m = |obs|$, which implies that there are $|obs'| = k\text{-}m$ unobserved failure modes. Then by (7.8-7), (7.8-13), and (7.8-14), the MME-based reliability growth projection for an assumed number of failure modes is given by

$$\breve{\tilde{p}}_{k,i} = \breve{\theta}_k \cdot \hat{p}_i + (1 - \breve{\theta}_k)\left(\frac{N}{k \cdot T}\right) \qquad 7.8\text{-}15$$

where d_i^* estimates d_i .

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Because the total potential number of failure modes associated with a complex system is typically large & unknown, it is desirable to study the limiting behavior of (7.8-15) as $k \rightarrow \infty$. The reliability projection under these conditions simplifies to

$$\breve{\breve{r}}_{\infty}(\mathrm{T}) = \lim_{k\to\infty} \breve{\breve{r}}_{k}(T) =$$

$$\prod_{i\epsilon obs}\left[1 - (1 - d_i^*) \cdot \breve{\breve{p}}_{\infty,i}\right] \cdot \exp\left[-\left(1 - \breve{\theta}_{\infty}\right) \cdot \left(\frac{N}{T}\right)\right] \qquad 7.8\text{-}16$$

where

$$\breve{\breve{p}}_{\infty,i} = \breve{\theta}_{\infty} \cdot \hat{p}_i\,, \qquad 7.8\text{-}17$$

$$\breve{\theta}_{\infty} = \frac{T}{\breve{n}_{\infty} + T} \qquad 7.8\text{-}18$$

$$\breve{n}_{\infty} \equiv \lim_{k\to\infty} \breve{n}_{k} = \frac{\sum_{i=1}^{m} \hat{p}_i - \sum_{i=1}^{m} \hat{p}_{i=1}^{2}}{\sum_{i=1}^{m} \hat{p}_{i=1}^{2} - \sum_{i=1}^{m} \frac{\hat{p}_i}{T}} \qquad 7.8\text{-}19$$

all of which are in terms of failure data that are readily available. From (7.8-12), we can see that $\breve{x}_{\infty} \equiv \lim_{k\to\infty} \breve{x}_{k} = 0$, which implies that the *s*-mean, and *s*-variance of the beta distribution both converge to zero as $k \rightarrow \infty$. Hence, the distribution becomes degenerate in the limit.

7.8.2.3 Likelihood-based Estimation Procedure.
The method of marginal maximum likelihood provides estimates of the beta parameters n, and x that maximize the beta marginal likelihood function. For an assumed number of total potential failure modes, the estimates denoted by $\hat{n}_k$, and $\hat{x}_k$, respectively are obtained by solving the following two likelihood equations simultaneously:

$$\sum_{i=1}^{k} \sum_{j=0}^{N_i - 1} \left(\frac{1}{\hat{x}_k + j}\right) - \sum_{i=1}^{k} \sum_{j=0}^{T - N_j - 1} \left(\frac{1}{\hat{n}_k - \hat{x}_k + j}\right) = 0 \qquad 7.8\text{-}20$$

and

$$\sum_{i=1}^{k} \sum_{j=0}^{T - N_i - 1} \left(\frac{1}{\hat{n}_k - \hat{x}_k + f}\right) - \sum_{i=1}^{k} \sum_{j=0}^{T-1} \left(\frac{1}{\hat{n}_k + j}\right) = 0 \qquad 7.8\text{-}21$$

which are defined to be zero if $N_i = 0$. The starting values for the associated numerical routine to obtain such estimates can be chosen to be the un-weighted moment estimators given by (7.8-11), and (7.8-12). Without loss of generality, the finite *k* likelihood-based estimates $\hat{\theta}_k$, and $\hat{\breve{p}}_{k,i}$ are obtained analogously to that of (7.8-13), and (7.8-14) with appropriate substitution of the MLE in place of the MME. This provides the likelihood-based estimate of system reliability growth

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$$\hat{\hat{r}}_k(\mathrm{T}) = \prod_{i\epsilon obs}\left[1 - (1 - d_i^*)\cdot \hat{\hat{p}}_{k,i}\right]\cdot \left[1 - \left(1 - \hat{\theta}_k\right)\cdot \left(\frac{N}{k\cdot T}\right)\right]^{k-m} \qquad 7.8\text{-}22$$

To estimate the limiting behavior of (7.8-22), we will re-parameterize (7.8-20) and (7.8-21), and take limits of these equations as $k \to \infty$. The true but unknown reliability of the system at the beginning of the current test phase is a realization of the product $\prod_{i=1}^{k}(1 - P_i)$), where P_i is interpreted as a *s*-independent beta *r.v.* The mathematical expectation of this quantity with respect to the P_i for i =1,…, k is $\hat{R}_k = \left(1 - \frac{\hat{x}_k}{\hat{n}_k}\right)^k$, which yields the useful parameterization

$$\hat{x}_k = \hat{n}_k \cdot \left(1 - \hat{R}_k^{1/k}\right)$$

where $\hat{R}_k$ denotes an MLE of the unconditional *s*-expected initial system reliability. Notice that $\hat{x}_k \to 0$ as $k \to \infty$. This does not come as much of a surprise because we would expect the likelihood-based estimate of the beta parameter x to exhibit the same behavior as that of the moment based estimate, which also converges to zero as k grows without bound. By substituting this parameterization into (7.8-20), and taking the limit, we derive the following MLE-based approximation for the *s*-expected initial system reliability of a complex one-shot system for $k \to \infty$:

$$\hat{R}_\infty = \exp\left[\frac{1}{\sum_{j=0}^{T-1}\left(\frac{-m}{\hat{n}_\infty + j}\right)}\right] \qquad 7.8\text{-}23$$

where $\hat{n}_\infty$ denotes the limit of the MLE for the beta parameter n (i.e., pseudo trials). This result is significant for a number of reasons. First, we derived a new estimate for the *s*-expected initial reliability of a one-shot system, which is a basic quantity of interest to program managers, and reliability practitioners. This quantity also serves as an estimate of the current demonstrated reliability of a one-shot system. This offers an alternative to the typical reliability point estimate calculated as the ratio of the number of successful trials to the total number of trials. Second, we expressed this quantity in terms of only one unknown, which has reduced the estimation procedure to solving one equation for $n \to \infty$. To derive this equation, we proceed in a similar fashion as above. Let $x_k = \frac{n_k \cdot k \cdot \left(1 - R_k^{1/k}\right)}{k} = \frac{\mathrm{n_k}\cdot\gamma_\mathrm{k}}{\mathrm{k}}$, where $\gamma_k = k \cdot \left(1 - R_k^{1/k}\right)$. Note that $\gamma_k$is finite and positive as $k \to \infty$. By substituting this parameterization into (7.8-21), and taking the limit, the estimate $\hat{n}_\infty$ for the beta parameter n is found such that

$$\sum_{i=1}^{m}\sum_{j=0}^{T-N_i-1}\left(\frac{1}{\hat{n}_\infty}\right) =$$

$$= m \cdot \sum_{j=0}^{T-1}\left(\frac{1}{(\hat{n}_\infty + j)} - \frac{1}{(\hat{n}_\infty)^2 \cdot \sum_{i=0}^{T-1}\left(\frac{1}{\hat{n}_\infty + i}\right)}\right) \qquad 7.8\text{-}24$$

Hence, the resulting limiting behavior of the likelihood-based estimate for one-shot system reliability growth is given by

$$\hat{\hat{r}}_\infty \equiv \lim_{k\to\infty} \hat{\hat{r}}_k\,(T) =$$

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$$\prod_{i\epsilon obs}\left[1-(1-d_i^*)\cdot\hat{\hat{p}}_{\infty,i}\right]\cdot exp\left[-(1-\hat{\theta}_\infty)\cdot\left(\frac{N}{T}\right)\right] \quad 7.8\text{-}25$$

where

$$\hat{\hat{p}}_{\infty,i}=\hat{\theta}_\infty\cdot\hat{p}_i\ , \quad 7.8\text{-}26$$

$$\hat{\theta}_\infty=\frac{T}{\hat{n}_\infty+T} \quad 7.8\text{-}27$$

and $\hat{n}_\infty$ is found as the solution of (7.8-24).

7.8.2.4 Goodness-of-Fit.
The GOF of the model can be graphically studied by plotting the cumulative number of observed failure modes versus trials against the estimate of the cumulative *s*-expected number of observed failure modes on trial *t* given by

$$\hat{\mu}(t)=\left(\frac{m}{\sum_{j=1}^{T-1}\left(\frac{1}{\hat{n}_\infty+j}\right)}\right)\cdot\left(\frac{\Gamma'(\hat{n}_\infty+t)}{\Gamma(\hat{n}_\infty+t)}-\frac{\Gamma(\hat{n}_\infty)}{\Gamma(\hat{n}_\infty)}\right) \quad 7.8\text{-}28$$

where $\hat{n}_\infty$ is found as the solution to (7.8-24), and $\Gamma'(x)/\Gamma(x)$ is the digamma function.

7.8.3 Monte-Carlo Simulation Study

7.8.3.1 Overview.
In previous sections, we have introduced a new model that will be helpful in estimating the demonstrated reliability, and reliability growth of one-shot systems. In light of this new model, a natural concern in its application is the accuracy associated with the resulting reliability estimates. To study model accuracy, we have developed a Monte-Carlo simulation, which consists of the following steps:

a. Specification of simulation inputs such as the total potential number of failure modes, and trials.
b. *S*-random generation of failure probabilities via a beta *r.v.*
c. *S*-random generation of failure histories via a Bernoulli *r.v.*
d. *S*-random generation of fix effectiveness factors via a beta *r.v.*
e. Estimation of the model parameters, and equations presented above.
f. Error estimation between the true, and estimated reliability growth.

Steps a. through f. can be viewed as simulating data analogous to that captured during a single developmental test consisting of *T* trials for a one-shot system comprised of *k* failure modes. These steps are replicated, which corresponds to simulating a sequence of developmental tests. Simulation inputs remain constant during each replication of the simulation. Failure probabilities, and fix effectiveness factors, however, are stochastically generated anew during each replication. After the simulation is replicated, all failure data, parameter estimates,

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reliability projections, and error terms are saved, and analyzed. In the next section, we present simulation results based on a given set of inputs. Simulation output consists of summary statistics, and associated relative error probability densities.

7.8.2.2 Simulation Results.

7.8.2.2.1 Summary.
Via heuristics, stable simulation results are obtained at 100 replications of the simulation. The presented results are based on 300 replications with T =350 trials, k =50 failure modes, $\mu = 2 \cdot 10^{-3}$ for the population s-mean of the failure probabilities, and $\sigma^2 = 2 \cdot 10^{-4}$ for the population *s*-variance of the failure probabilities. The values of these inputs greatly reduce the volume of failures, and failure modes observed during simulation, as a conservative scenario with respect to the volume of failure data available for estimation purposes is desired. For example, only 4 of 50 failure modes were observed on average in the simulated developmental tests. In addition, only a total of 39 failures were observed on average. This is indicative of the high initial reliability of the system, as specified via the inputs above. We wish to emphasize two points. First, it is important not to confuse the difference between the number of replications, and the number of trials, *T*. Clearly, as $T \to \infty$, all failure modes will eventually be observed. However, we are simulating 350 trials per replication of a highly reliable system, and therefore we only observe about 4 of the 50 failure modes consistently on average per replication (i.e., each replication simulates 350 trials). The simulation results are stable in that a small volume of failure data is available for estimation purposes per replication, and there is not much *s*-variability in the reliability growth estimates after 100 replications. Second, a large number of trials does not imply a large volume of failure data. For example, a large number of trials is relative to the initial reliability of the system. In the presented case, 350 trials did not yield a large volume of failure data, as the true unconditional s-expected initial system reliability was 0.9047. The arithmetic average (over all replications) of our corresponding estimate given by (7.8-23) was 0.9029. The table below shows arithmetic averages of the true and estimated reliability projections based on our approach.

TABLE XX. Reliability Projections

THEORETICAL		ESTIMATED			
True	Stein	MME k	MME ∞	MLE k	MLE ∞
0.9763	0.9740	0.9738	0.9786	0.9756	0.9784

The column titled 'True' is computed via the arithmetic average of (7.8-6) over all replications. Similarly, the second column titled Stein is calculated by the arithmetic average of (7.8-7) over all replications. Both of these quantities are theoretical, as they are in terms of the true, but unknown $p_1,\ldots,p_k$, and k. The remaining four columns in TABLE XX are estimates of the true reliability growth based on the arithmetic averages of (7.8-15), (7.8-16), (7.8-22), and (7.8-25), respectively, over all replications. The true value of k =50 was utilized in (7.8-15), and (7.8-22), which are shown in the third, and fifth columns, respectively. The sensitivity of not knowing k is given by (7.8-16), and (7.8-25), which are shown in the fourth, and sixth columns, respectively.

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By addressing four of the 50 failure modes on average (over all replications) with a *s*-mean FEF of 0.80, the system reliability was improved from 0.9047 to 0.9763. By inspection Table XXI, the reliability projections appear quite accurate. There is, however, an element of uncertainty in studying aggregate results, as deviations in model accuracy do occur from one replication to the next. In some cases, reliability projections are conservative, whereas others are optimistic. By computing the arithmetic averages of the projections (over all replications), a portion of the error associated with the conservative estimates is canceled with that of the optimistic, thereby muting deviations in projection error that would otherwise be encountered via a single application of the model in one test phase. To address these concerns, the relative error terms obtained in each replication of the simulation are computed, and analyzed. The error analyses associated with the moment-based and likelihood-based reliability growth estimates are presented in the following two sub-sub-sections, respectively.

7.8.2.2.2 Accuracy of Moment-based Projections.
FIGURE 7-18 displays relative error plots for the moment-based reliability growth projections using a finite and infinite number of modes, respectively. Using (7.8-6), (7.8-15), and (7.8-16), the relative error for these projections is given respectively by

$$\check{E}_{k,r} \equiv \frac{r(T)-\check{r}_k(T)}{r(T)}, \qquad 7.8\text{-}29$$

and

$$\check{E}_{\infty,r} \equiv \lim_{k\to\infty} \check{E}_{k,r} = \frac{r(T)-\check{r}_\infty}{r(T)}. \qquad 7.8\text{-}30$$

FIGURE 7-18 displays the histograms for the relative error terms obtained from the simulation. MLE is utilized to approximate the parameters of an *s*-normal distribution, which is shown to accurately portray the probability densities of the relative error. The error densities for both the finite and infinite *k* reliability growth projections are similar. All error terms are within ±2.5% of the true reliability. Both projections
possess *s*-bias with the finite *k* approach providing a slight underestimate, and the infinite *k* approach providing a slight overestimate.

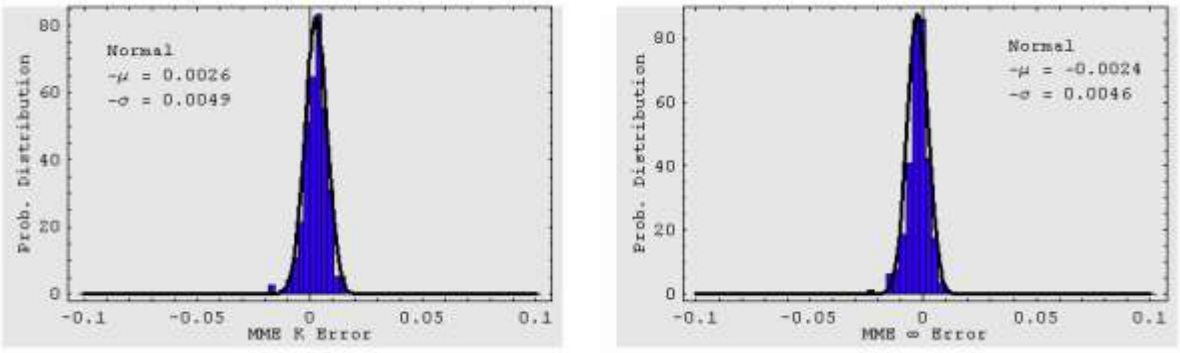


FIGURE 7-18. *Relative Error of Moment-based Projection*

Based on the estimated *s*-normal distribution for the finite *k* moment-based reliability growth projection $Pr\{\check{E}_{k,r} < |x|\} = 0.90 \Rightarrow |x| = 0.0091$. In other words, the projection error in (7.8-

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15) is within ±0.0091. 90% of the time for the simulated conditions specified above. Likewise, error in the infinite k moment-based reliability growth projection (7.8-16) is within ±0.0085, 90% of the time. Without loss of generality, the error results for these projections are very similar to that of the moment-based projections. The only notable difference is that the accuracy is slightly greater using an MLE procedure. Overall, the projection error in (7.8-22), and (7.8-25) is less than ±0.0076, and ±0.0081, respectively, 90% of the time.

7.8.2.2.3 Accuracy of Likelihood-based Projections.
Using (7.8-22), and (7.8-25), the relative error in the likelihood-based projections are obtained analogously to that shown in the previous section. Without loss of generality, the error results for these projections are very similar to that of the moment-based projections. The only notable difference is that the accuracy is slightly greater using an MLE procedure. Overall, the projection error in (7.8-22) and (7.8-25) is less than ±0.0076, and ±0.0081, respectively, 90% of the time.

7.8.2.2.4 General Observations.
The results shown in the previous sections highlight model accuracy for one set of simulation inputs. Clearly, there are infinitely many combinations of inputs under which model accuracy could be studied. Several different combinations of inputs in conjunction with their simulation output have been analyzed in an effort to generalize the conditions for which model accuracy is high (e.g., $\check{E}_{k,r} \leq 0.10$). Based on these analyses, it may be noted that model accuracy is not simply a function of using (for estimation purposes) a large volume of failure data, or observing a proportional majority of failure modes in the system. Rather, model accuracy is found to be a function of obtaining good estimates for the dominant failure modes of the system. In the presented simulation results, only 4 of the 50 failure modes were observed on average, but these failure modes represented about 90% of the system unreliability. In addition, 10 failures were observed on average for each of the modes, which provided good estimates for their associated probabilities of occurrence.

Finally, with respect to the accuracy of the limiting behavior of the model, empirical evidence obtained via simulation suggests that if k is sufficiently greater than m, the projections given by (7.8-16), and (7.8-25) will be insensitive to the value of k. Experience with the model suggests that the condition $k \geq 5 \cdot m$ is a good rule-of-thumb for the convergence of these estimators for complex systems.

7.8.4 Concluding Remarks.
This model offers an alternative to the popular competing risks approach. It is suitable for application when one or more failure modes can be discovered in a single trial, and when catastrophic failures modes have been previously discovered, and corrected. Equation (7.8-6) is the logically derived model. The theoretical estimate of (7.8-6) is given by (7.8-7). The practical estimates of (7.8-7) are given by (7.8-15), (7.8-16), (7.8-22), and (7.8-25).

The model provides a method for approximating the vector of failure probabilities associated with a complex one-shot system, which is based on our derived shrinkage factor given by (7.8-5). The benefit of this procedure is that it not only reduces error, but reduces the number of

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unknowns requiring estimation from $k+1$ to only three. Also, estimates of mode failure probabilities, whether observed or unobserved during testing, will be finite, and positive.

Unique limits of the model equations are derived, which have yielded interesting simplifications. The limiting approximations of the model equations include (7.8-16)-(7.8-19), and (7.8-23)-(7.8-27). In particular, a mathematically-convenient functional form is derived for the *s*-expected initial system reliability of a one-shot system (7.8-23). This quantity serves as an estimate of the current demonstrated reliability of a one-shot system, and offers an alternative to the typical reliability point estimate calculated as the ratio of the number of successful trials to the total number of trials.

Finally, Monte-Carlo simulation results are shown to highlight model accuracy with respect to resulting estimates of reliability growth. While all error terms were within ±2.5% of their reliability estimates, the approximated s-normal distributions above indicate that the projection error is within ±0.9% (i.e., ±0.0091), with a probability of 0.90.

7.9 References.

1. Crow, Larry, AMSAA TR-357, An Improved Methodology for Reliability Growth Projections, June 1982

4. Rosner, N., Proceedings National Symposium on Reliability and Quality Control, System Analysis – Nonlinear Estimation Techniques, New York, NY: IRE, 1961, pp 203-207

5. Ellner, Paul M. and Wald, Lindalee C., 1995 Proceedings Annual Reliability and Maintainability Symposium, AMSAA Maturity Projection Model, January 1995

6. MIL-HDBK-189, Reliability Growth Management, 13 February 1981

7. Ellner, Paul M. and Wald, L. and Woodworth J., Proceedings of Workshop on Reliability Growth Modeling: Objectives, Expectations and Approaches, A Parametric Empirical Bayes Approach to Reliability Projection, The Center for Reliability Engineering, 1998

9. Musa, J. and Okumoto K., Proceedings of 7th International Conference on Software Engineering, A Logarithmic Poisson Execution Time Model for Software Reliability Measurement, 1984, pp. 230-238

10. Crow, Larry H., Proceedings of RAMS 2004 Symposium, An Extended Reliability Growth Model for Managing And Assessing Corrective Actions, pp 73-80

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8 Notes

8.1 Intended use.

This handbook provides guidance to help in the management of reliability growth through the acquisition process.

8.2 Superseding information.

This handbook is in lieu of the materials in MIL-HDBK-189, 1981.

8.3 Subject term (Keyword listing).

AMSAA Crow
Fix Effectiveness Factor
Poisson Process
PM2

8.3.1 Reliability.

Reliability is the probability that an item will perform its intended function for a specified time and under stated conditions, which are consistent with that of the Operations Mode Summary/Mission Profile (OMS/MP).

8.3.2 Operational Mode Summary/Mission Profile.

Defines the concept of deployment, mission profile or details as to how equipment utilized, per cent operating time/mileage in various operating modes and percent of operating time/mileage, etc in operational environment or conditions (temperature, vibration, percent miles on terrain/road types, etc) under which equipment is utilized.

8.3.3 Reliability Growth.

Reliability growth is the positive improvement in a reliability parameter over a period of time due to implementation of corrective actions to system design, operation and maintenance procedures, or the associated manufacturing process.

8.3.4 Reliability Growth Management.

Reliability growth management is the management process associated with planning for reliability achievement as a function of time and other resources, and controlling the ongoing rate of achievement by reallocation of resources based on comparisons between planned and assessed reliability values.

8.3.5 Repair.

A repair is the repair of a failed part or replacement of a failed item with an identical unit in order to restore the system to be fully mission capable.

8.3.6 Fix.

A fix is a corrective action that results in a change to the design, operation and maintenance procedures, or to the manufacturing process of the item for the purpose of improving its reliability.

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8.3.7 Failure Mode.
A failure mode is an individual failure for which a failure mechanism is determined. Individual failure modes may exhibit a given failure rate until a change is made in the design, operation and maintenance, or manufacturing process.

8.3.8 A-Mode.
An A-mode is a failure mode that will not be addressed via corrective action.

8.3.9 B-Mode.
A B-mode is a failure mode that will be addressed via corrective action, if exposed during testing. One caution with regard to B-mode failure correction action is during the test program, fixes may be developed that address the failure mode but are not fully compliant with the planned production model. While such fixes may appear to improve the reliability in test, the final production fix would need to be tested to assure adequacy of the corrective action.

8.3.10 Fix Effectiveness Factor (FEF).
A FEF is a fraction representing the fraction reduction in an individual initial mode failure rate due to implementation of a corrective action.

8.3.11 Growth Potential (GP).
Growth potential is a theoretical upper limit on reliability which corresponds to the reliability that would result if all B-modes were surfaced and fixed with an assessed FEF.

8.3.12 Management Strategy (MS).
MS is the fraction of the initial system failure intensity due to failure modes that would receive corrective action if surfaced during the developmental test program.

8.3.13 Growth rate.
A growth rate is the negative of the slope of the cumulative failure rate for an individual system plotted on log-log scale. This quantity is representative of the rate at which the system's reliability is improving as a result of implementation of corrective actions. A growth rate between (0,1) implies improvement in reliability, a growth rate of 1 implies no growth, and a growth rate greater than 1 implies reliability decay.

8.3.14 Poisson Process.
A Poisson process is a counting process for the number of events, $N(t)$, that occur during test interval $[0,t]$, where t is a measure of test duration. The counting process is required to have the following properties: (1) the number of events in non-overlapping intervals are stochastically independent; (2) the probability that exactly one event occurs in the interval $[t.\ t+\Delta t]$ equals $\lambda_t * \Delta t + o(\Delta t)$ where λ_t is a positive constant, which may depend on t, and $o(\Delta t)$ denotes an expression of $\Delta t \succ 0$ that becomes negligible in size compared to Δt as Δt approaches zero; and (3) the probability that more than one event occurs in an interval of length Δt equals $o(\Delta t)$. The above three properties can be shown to imply that $N(t)$ has a Poisson distribution with mean equal to $\int_0^t \lambda_s\ ds$, provided λ_s is an integrable function of s.

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8.3.15 Homogeneous Poisson Process (HPP).
A HPP is a Poisson process such that the rate of occurrence of events is a constant with respect to test duration *t*.

8.3.16 Non-Homogeneous Poisson Process (NHPP).
A NHPP is a Poisson process with a non-constant recurrence rate with respect to test duration t.

8.3.17 Idealized Growth Curve (IGC).
An IGC is a planned growth curve that consists of a single smooth curve portraying the expected overall reliability growth pattern across test phases and is based on initial conditions, assumed growth rate, and/or planned management strategy.

8.3.18 Planned Growth Curve (PGC).
A PGC is a plot of the anticipated system reliability versus test duration during the development program. The PGC is constructed on a phase-by-phase basis and as such may consist of more than one growth curve.

8.3.19 Reliability Growth Tracking Curve.
A reliability growth tracking curve is a plot of the best statistical representation of system reliability to demonstrated reliability data versus total test duration. This curve is the best statistical representation in comparison to the family of growth curves assumed for the overall reliability growth of the system.

8.3.20 Reliability Growth Projection.
Reliability growth projection is an assessment of reliability that can be anticipated at some future point in the development program. The rate of improvement in reliability is determined by (1) the on-going rate at which new problem modes are being surfaced, (2) the effectiveness and timeliness of the fixes, and (3) the set of failure modes that are addressed by fixes.

8.3.21 Exit Criterion (Milestone Threshold).
Reliability value that needs be exceeded to enter the next test phase. Threshold values are computed at particular points in time, referred to as milestones, which are major decision points that may be specified in terms of cumulative hours, miles, etc. Specifically, a threshold value is a reliability value that corresponds to a particular percentile point of an order distribution of reliability values. A reliability point estimate based on test failure data that falls at or below a threshold value (in the rejection region) indicates that the achieved reliability is statistically not in conformance with the idealized growth curve.

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Appendix A Engineering Analysis

A.1 Scope

A.1.1 Purpose

The majority of reliability growth data analyses are statistical analyses. Statistical analyses view growth as being the result of a smooth, continuous process. In fact, reliability growth occurs in a series of finite steps corresponding to discrete design changes. Mathematical models describe the smooth expectation of this discrete process. Rather than being concerned about whether specific design changes are effected rapidly or slowly – or whether they are very effective, not effective, or even detrimental—the statistical models work with the overall trend. In most situations, this is a desirable feature as it focuses attention on long term progress rather than on day-to-day problems and fixes. The application of statistical analyses relies on analogy. For example, the growth pattern observed for program A may be used as a planned growth model for program B, because the programs are similar. As another example, the growth pattern observed early in program B may be extrapolated to project the growth expected later in the program because of similarities between the early and later portions of the program. The difficulty that occurs in applying the analogy approach is that perfectly analogous situations rarely exist in practice. The engineering analyses described in this section rely on synthesis. That is, they build up estimates based on a set of specific circumstances. There is still, however, reliance on analogy; but the analogies are applied to the parts of the problem rather than to the whole. Although synthesis may be used to provide a complete buildup of an estimate, it is simpler and more common to use synthesis to account for the differences, or lack of perfect analogy, between the baseline situation and the situation being analyzed.

A.1.2 Application

The general approach to growth planning and long term projection is similar to that used for assessment and short-term projection purposes. The main difference is that for planning and long term projection purposes, attention must be directed to program characteristics and general hardware characteristics, since specific design changes are unknown at the time of program planning. For assessment and short term projection purposes, attention must be directed to the specific hardware changes made or anticipated. For the most part, the program and general hardware characteristics can be ignored, since they have already their role in determining the specific hardware changes. The only difference between assessment and short term projection is whether a change has been incorporated in the hardware or not. The analysis is the same in both cases except that recent test results may be incorporated in the assessment. Is should also be noted that the type of assessment described in this section, because judgment involved in arriving at it, is particularly suitable for use within an organization. For inter-organization use, completely objective demonstrated values, computed by a means acceptable to the organizations concerned, are usually necessary.

A.2 Assessment and Short Term Projection

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A.2.1 Application.
At times, it is desirable to assess or project reliability growth by means of engineering analysis rather than statistical analysis. This detailed look is usually desirable in the following situations:

a. When near the end of a test phase, design changes have been, or will be incorporated without adequate demonstration. It is highly desirable to analyze these unverified "fixes" separately on their unique merits, rather than treating the "fixes" as average ones with a statistical model.
b. When a major design change is made, or will be made, in the future. Such a change often causes a jump in reliability that is unrelated to the growth process prior to the change, since it represents a departure from a pure "find and fix" routine.
b. When there are few distinct test and fix phases. In this case growth projections by statistical extrapolation may not be appropriate.
c. When it is desired to evaluate possible courses of design improvements. By considering the failure modes observed and possible corrective actions available, a desirable course of design improvement can be determined. For example, it can be determined if correction of the single worst problem will bring the system reliability up to an acceptable level.

A.2.2 **Objective.**
When a failure mode is observed on test, it becomes desirable to anticipate the improvement that can be expected in a system if that failure mode is subjected to design improvement. The ultimate improvement possible is to completely remove the failure mode or reduce its rate of occurrence to zero. The practical lover limit on the failure rate is limited by the state of the art, and even this value can be attained only under perfect conditions. The failure rate actually attained will usually be somewhat higher than the state of the art limit because unforeseen minor faults in the design and the failure rates of the parts are involved.

A.2.3 Design Changes.
Although this appendix emphasizes reliability analysis of design changes for reliability improvement, all design changes should be analyzed in this manner, since every design change has a potential for enhancing or degrading system reliability. This requires that the reliability management system be linked to the configuration management system and other pertinent programs such as for maintainability and producibility.

A.2.4 Significant Factors.
Some of the factors affecting the expected effectiveness of design change for reliability are listed below. For convenience in application, these are categorized as factors that create reference values and factors that influence estimates.

a. Factors that create reference values:
   i. What is the failure rate being experienced in similar applications?
   ii. What is the failure rate of components to be left unchanged?
   iii. What is the analytically predicted failure rate?
   iv. What failure rate is suggested by laboratory or bench tests?
   v. How successful has the design group involved been in previous redesign efforts?

b. Factors that influence estimates:

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i. Is the failure cause known?
ii. Is the likelihood of introducing or enhancing other failure modes small?
iii. Are there other failure modes indirect competition with the failure mode under consideration?
iv. Have there been previous unsuccessful design changes for the failure mode under consideration?
v. Is the design change evolutionary, rather than revolutionary?
vi. Does the design group have confidence in the redesign effort?

A.2.5 Explanation of Factors.

A.2.5.1 What is the failure rate being experienced in similar applications?
The failure rate that a component experiences in similar applications serves an objective reference point indicative of what may reasonably be expected of that component.

A.2.5.2 What is the failure rate of components to be left unchanged?
Since it is usually unreasonable to expect one of the worst components in a system to be among the best as the result of a design change, the average failure rate of components to be left unchanged can be used as a rough optimistic limit. Although the guidance provided by this reference value is not very firm and may easily be overridden by other factors, there are three reasons to encourage its use. First, it raises the general question of over-optimism. Second, it is a valid and common approach to reliability improvement to bring problem components into conformance with the other components in the system. Third, this reference value is among the easiest to determine.

A.2.5.3 What is the analytically predicted failure rate?
The failure rate for the failure mode under consideration may, in some cases, be analytically predicted using techniques such as probabilistic design approaches, physics-of-failure modeling and simulation such as multi-body dynamic modeling, finite element analysis, and component life prediction. As an analysis of this type may not consider all unforeseen peculiarities in the design or application, such a value should be viewed as optimistic limited mitigated only by the experience and demonstrated track record of the design group.

A.2.5.4 How successful has the design group involved been in previous redesign efforts?
The success rate of the design group provides another objective point of reference. For example, one organization has found that corrective actions are normally not more than 80 percent effective. Usually, this index is evaluated as the proportion of design changes that result in eliminations (essentially) of the failure mode, or it is evaluated as the average proportion of failure rate reduction. In both of these cases, the range of failure rate values under consideration is between the current value and zero. The effectiveness of the design group may also be determined by the average proportion of the predicted improvement that is attained. In this case, the range of failure rate values under consideration is between the current value and the predicted value. This measure of effectiveness is more precise, but also more cumbersome, to work with. If this measure is used, it must be treated as an influence rather than a reference value.

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A.2.5.5 Is the failure of cause known?
Knowledge of the failure cause relies heavily on the ability to perform a failed part analysis. Only when the failure cause and the precise failure mechanism are known can a design change be expected to be fully effective. At the other end of the spectrum are problems that must be attacked by trial and error because the failure cause is (at least, initially) unknown, In this case, he expected effectiveness will be close to zero. Nevertheless, this type of a change may be used to gain insights that will give higher expectations in future changes.

A.2.5.6 Is the likelihood of introducing or enhancing other failure modes small?
The likelihood of other failure modes being affected by design change can usually be evaluated by use of failure mode and effect analysis. Attention should be directed to components that are adjacent to the affected one in either a functional or physical sense.

A.2.5.7 Are there other failure modes indirect competition with the failure mode under consideration?
It is a special, particularly difficult situation when a component or assembly has other failure modes in direct competition with the failure mode under consideration. These are usually characterized by opposite failure mode descriptions such as tight, loose; or high, low. In a situation like this, there is no single, conservative direction, and avoiding one failure mode often results in backing into another. Seals on rotating shafts are an example of this type of problem. An application may initially have a leakage problem. Going to a tighter seal often results in a wear problem, and changing to multiple seals often causes the outer seals to run dry. The optimism solution in a case like this is usually a less-than-satisfactory compromise. And it is not unheard of to end up eventually with the original design.

A.2.5.8 Have there been previous unsuccessful design changes for the failure mode under consideration?
Each unsuccessful design change for a specific failure mode will, in itself, lead to lower expectations for the effectiveness of further changes. This is caused by selecting the most promising alternative first. However, previous unsuccessful changes may have provided sufficient information on the failure mechanism to outweigh this factor.

A.2.5.9 Is the design change evolutionary rather than revolutionary?
Idealistically, an evolutionary change involves a single, small deviation from previous practice. Increases in either the magnitude or number of deviations make the change more revolutionary. When a design is refined in an evolutionary manner, the expectation is for improvement to occur with each iteration. A revolutionary design change is, however, virtually the same as a new design fresh from the drawing board (for the subsystem and components concerned). Thus, the redesigned part of the system may have an initial MTBF only, say, 10 or 20 percent of the predicted value. The revolutionary change may, however, have a potential inherently higher than the original design.

A.2.5.10 Does the design group have confidence in the redesign effort?
Although subjective and intuitive, the confidence of the design group should reflect all of the factors previously discussed. Because of this, any analysis of reliability growth expectations should be compared against this intuitive feel; and, of course, the two opinions should compare

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well. As with any kind of cross-checking, the objective is to ferret out any errors and oversights. The main point is that an adequate analysis of reliability growth expectations cannot be accomplished without input from the design group.

A.2.6 Methodology.
There are two major steps involved in estimating the effect of a design improvement. The first step involves using any reference values that can be determined to roughly define the range within which the new reliability value is expected to be. The second step involves considering the effect of the various influencing factors to narrow down to a likely point within this range. It must be emphasized that this methodology is a thought-process guide rather than an explicit procedure to be followed blindly. Some of the listed factors may be meaningless or inappropriate for a given design change. Some may be overshadowed by other factors. And some combinations of factors may have a net effect that is not consistent with linearly additive relationship suggested in the example to follow. Special cases, such as component failure with acceptable reliability that is to be modified for other reasons, will require adaption of the basic procedure.

A.2.7 Example

A.2.7.1 Objective
This example is intended to illustrate a general methodology that may be used to predict the effectiveness of design changes. This may be used as a method of assessment for design changes incorporated in the hardware, but not adequately tested. It may also be used to make short term projections. This example considers just a single design change. It must be emphasized that the methodology is intended as a guide to reasoning, and no quantitative precision is implied.

A.2.7.2 Problem statement.
The failure mode under consideration is weld cracking in a travel lock of a howitzer. The design change to be incorporated is an increased weld fillet size.

A.2.7.3 Analysis

A.2.7.3.1 Termination of reference values.
The first step is to determine any reference values that are obtainable as shown is Table A.I.

TABLE A.I. Reference Values

Current Failure Rate	0.0005 Failures per round, as demonstrated by test
Analytical Prediction	None
Test Results	Lab test (accelerated) show about a 4 to 1 improvement, suggesting a failure rate of about 0.00012 is attainable
Failure rate of similar components in similar applications	None sufficiently comparable.
Success ratio of the design group	In general, they have been capable of removing

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	60% of the failure rate, implying 0.0002 as an expected failure rate.
Average failure rate of unchanged components	The system failure rate is 0.004, and there are roughly 300 active, or failure-prone components. 0.004/300=0.000013

A.2.7.3.2 Design change features.
The second step is to determine features of the design change that would influence the failure rate to be attained as shown is Table A.II.

TABLE A.II. Design Change Features

Is the failure cause known?	Moderately well. Analysis of broken welds showed no significant flaws; thus ruling out a quality problem. The level of forces encountered is not well known, and there is a question about the stress concentration in the vicinity of the weld.
Is there a likelihood of introducing other failure modes?	No other related failure modes are foreseen.
Are there competing failure modes?	No.
Is the design change evolutionary?	Yes. This is a single, relatively minor change.
Have there been previous unsuccessful design changes for the failure mode under consideration?	Yes. This is the second change. The first change increased the cross-section of the stop. This caused some improvement, but the same type of cracking persists. Further increase in cross-section is impossible without a major change.
Does the design group have confidence in this change?	Their confidence is moderate.

A.2.7.3.3 Defining and refining estimates.
The third and fourth steps in the process involve defining the region of interest in terms of reference values and then refining estimates within (or perhaps slightly beyond) this region by consideration of the influencing factors. This process is shown graphically for illustrative purposes in FIGURE A-1. Point A represents a likely failure value, ignoring the influencing consideration. In this case, the lab test results were felt to be realistic and considerably more concrete than the general expectation, although the two values are in reasonable good agreement. The failure rate of other components does little more in this case than to provide assurance that the failure rate is only being brought into "reasonable conformance" to the rest of the system, rather than surpassing it. Line A-B represents the detrimental influence expected from some lack of knowledge of the failure cause. Since the failure cause is not known exactly, the lab testing may not have adequately reproduced the failure cause. Line B-C represents the influence expected from other failure modes that may be aggravated by change. No influence is expected. Line C-D represents the influence expected from other competing failure modes. No influence is

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expected. Line E-F represents the detrimental influence expected from this being the second design correction attempt. Line F-G takes into consideration the confidence that the design group had in this change. Since their feelings are consistent with the analysis up to this point, no effect is shown. This analysis, then predicts a failure rate of about 0.00025 after the design change. Similar analyses for other design changes may then be combined to estimate the effect at the system level. Finally, it must be emphasized again that this type is estimator is highly subjective.

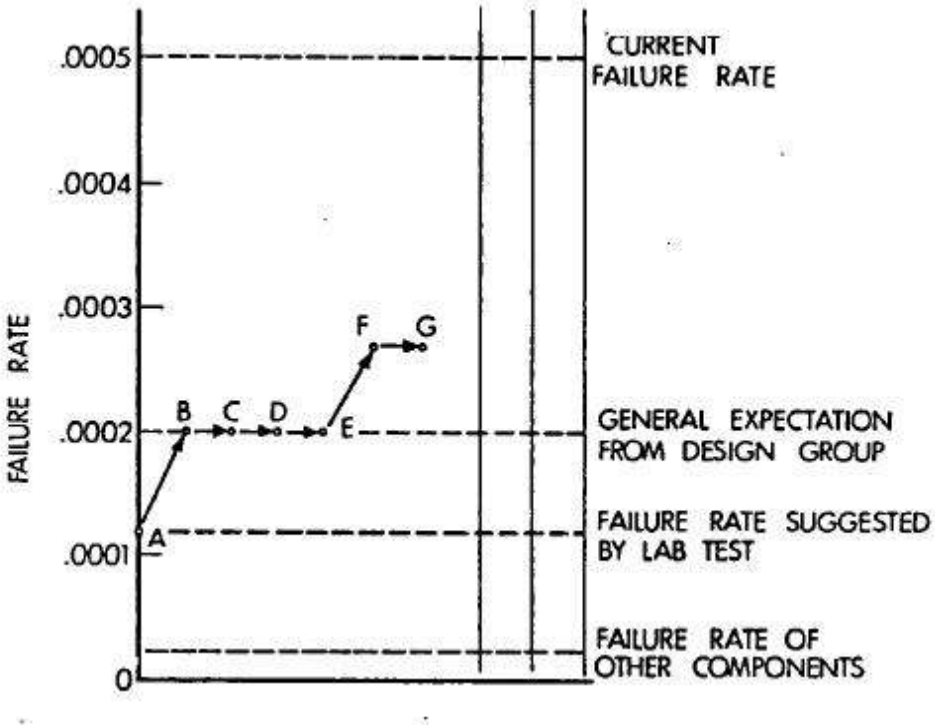


FIGURE A-1. *Defining and Refining Estimates.*

A.3 Planning and Long Term Projection

A.3.1 Purpose.

From an academic standpoint, growth planning and long range projection have as their purpose the determination of the reliability growth that can be expected for a given set of program alternatives. From a more practical standpoint, a set of such analyses enable the program planner to evaluate the benefits and drawbacks of various alternatives.

A.3.2 Approach.

Basically, growth planning and long range projection consider program constraints, activities, and sequencing to judge whether they will encourage or deter growth and to what extent. The three main variable of interest are the number of failure sources identified, the time required to perform the various activities, and the effectiveness of redesign efforts. Particular care must be taken when evaluating these variables to ensure that the sequencing of events is properly accounted for.

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A.3.3 Organization or program characteristics.
The basic reliability feedback model will be used as a means of organizing and assimilating program characteristics. Because of the significance of hardware fabrication time, the fabrication of hardware element is included in the mode as illustrated in FIGURE A-2.

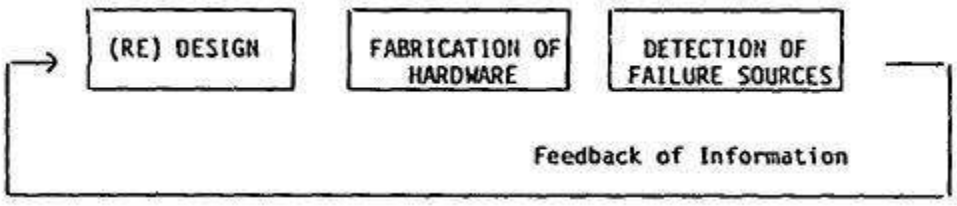


FIGURE A-2. *Feedback Model.*

A.3.4 Program-related questions.
The four major elements of the reliability growth feedback model can be further broken down to a set of specific program-related questions. In the following list of questions, *T* is used to indicate time related questions, # is used to indicate questions related to the number of identified failure modes, and *E* is used to indicate questions related to the effectiveness of corrective actions.

a. Detection of Failure Sources

(1) Are the test durations and the number of systems on test adequate or excessive? (T,#)
If the amount of testing is too small, the number of failure modes identified will be too small to properly guide redesign effort. On the other hand, once the redesign direction is well established, but changes are not incorporated in the test hardware, not all of the newly identified failure modes will be useful. In effect, we are testing "yesterday's" design once it has served its purpose of providing design guidance.

(2) To what extent can and will failed part analysis be performed to determine what failed and why it failed? (T, #)
For most types of equipment, this is a minimal problem, and the time required may be negligible. However, missiles and munitions (as examples) often require special instrumentation to determine what failed, and the determination of what failed may be a time-consuming process.

(3) Will early tests investigate the later life characteristics of the system? (T,#)
Frequently, early tests are relatively short. When longer tests are run later in the development phase, new failure modes associated with wear out may be observed. It is important that they are observed early enough in the program to allow for corrective action and verification.

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b. Feedback of information

(1) Is the feedback system responsive? (T) and

(2) Can information be lost by the feedback system (#)

A well-designed information feedback system should experience no problems in either of these areas, but these questions must be addressed since flaws in the feedback system are as critical as flaws elsewhere in the loop and are more easily corrected.

(3) Can failures find a home in the organization? (T)
A significant amount of time may be expended determining the responsibility for a given failure mode.

c. Redesign Effort based on Problems Identified (and nonreliability reasons).

(1) What general emphasis is to be placed on initiating a corrective action? (T)
In an aggressive reliability program, each failure mode will be analyzed and corrective action at least considered. Less aggressive programs may wait for pattern failures to occur before investigating a failure mode.

(2) How severe are other design constraints? (E)
As other design constraints become more severe, the number of design alternatives becomes more limited. As an example, on one type of equipment approximately 30% of the design changes for reliability have involved some weight increase. This suggests that if a program for equipment of this type is severely weight constrained, approximately 30% of the usual design alternatives must be ruled out.

(3) What design changes for non-reliability reasons can be anticipated? (#)
This is very closely related to the above question, but it is convenient to view the restriction of reliability growth and the (possible) introduction of reliability problems when design changes are made for other reasons.

One approach that has been used it to treat design changes for non-reliability reasons the same as changes for reliability reasons. For example: if 40% of all design changes for reliability reasons were "unsuccessful," in that the failure mode was not essentially removed or another was introduced, we may estimate that 40% of all design changes for non-reliability reasons would cause reliability problems.

(4) Have allowances been made in terms of dollars and time for problems which will surface late in development? (T,E)

If a program has been planned for success at each stage, there is no margin for error; and the unexpected, yet inevitable, problems are difficult to accommodate. In the early program stages, there are usually enough variables in the program to accommodate problems. However, near the

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end of a development program, there may be nothing left to trade off. When planning for reliability growth, it must be recognized that it is possible to approach the end of a development effort with an identified problem an identified "fix", but insufficient time or money to incorporate the fix.

(5) What is the strength of the design team, and what amount of design support will it receive from the reliability function? (T,E)

The main interests are the time required to effect design changes (on paper) and the effectiveness of the changes. These will be affected by the size and competence of the design team and also by the support it is given and the disciplines that are imposed. In general, design principles, such as the use of proven components, or the conduct of a failure mode and effects analysis increase design effectiveness at the expense of time and money.

d. Fabrication of Hardware

What intervals of time can be expected between the time that component design changes are finalized and the time that the components are ready to be tested? (t)

Within a given system, this can easily range from nearly zero in cases where off the shelf components can be used; to many months, in cases where special tooling is required. As a minimum, the longest lead time components should be identified and from these a probable longest lead time determined. This provides a rough estimate of the minimum lead time required before a new design configuration can be place on test. All lead times will have some impact on the practical attainment of reliability growth; but as a first cut, the long lead time components yield the most information. It is also worthwhile noting that identification of a reliability problem in a long lead time component may be a signal of a reliability growth problem that is not otherwise identified.

(1) What provisions are there to replace or repair components that fail on test? (T,E)

Ideally, replacement and repair procedures during test should duplicate those planned for the fielded equipment. However, since there may be no, or few, spare for the prototypes on test, some compromises may be necessary. Testing delays may be necessary while replacement parts are fabricated, or extraordinary repairs may be made to keep the equipment on test. When extraordinary repairs are made, the validity of some subsequently discovered failure modes may be questionable. For example, a casting that is cracked by testing may be repaired by welding, instead of being replaced as it would be in field use. If cracking subsequently occurs in another area of the casting, there may be a question whether the cracking is a result of a design deficiency or a result of residual stresses caused by welding. This doubt effectively reduces the number of identified failure modes.

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A.3.5 Synthesis.
The above questions can be used as a guide to program characteristics that will influence reliability growth. The program characteristics can then be used to synthesize growth expectations for the program.

A.3.6 Analysis

A.3.7 Example

A.3.7.1 Objective.
This example is intended to illustrate the general type of reasoning used to synthesize growth expectations. It does not cover a complete program and it is somewhat simplified, but additional details will vary greatly from one program to the next. It considers a development of a weapon for which the majority of design changes will occur between tests. It must be emphasized that in spite of the apparent mathematical precision, the estimates should be viewed as just ballpark figures.

A.3.7.2 Problem Statement.
The first prototype weapon is to be tested for 10,000 rounds. An MRBF of 200 is anticipated, implying that 50 failures are expected during the test. From experience with similar systems in early stages of development, it is expected that the 50 failures will be in about 20 different modes. The average failure rate in a mode is expected to be:

$$1 \div (200 * 20) = 0.00025$$

A.3.7.3 Analysis of improvement in existing failure modes.
What results can be expected when the second prototype tested? First, of the 20 modes expected, it is anticipated that about 08 will have design corrections attempted, and changes are expected to reduce the failure rates by 60%. Thus, the combined failure rate expected for these modes is (18)*(0.40)*(0.00025)= 0.0018. For the other two failure modes, no design correction will have been made. One is expected to be a long lead time change which won't be reflected until the third prototype, and the other is expected to be impossible to improve without exceeding the weight constraint. Thus, for these two modes, the combined failure rate is expected to be 2*(0.00025)=0.0005. Or, for the entire system, a failure rate of 0.0018 + 0.0005= 0.0023 can be expected, implying MRMF of 1/0.0023-435, provided no new failure modes are introduced.

A.3.7.4 Analysis of new failure modes anticipated.
To take into consideration any new failure modes, a calculation will first be made of the residual failure modes otherwise expected when testing the second prototype. The planned test duration for the second prototype is 15,000 rounds. With an MRBF of 435, about 34 failures are expected which based on previous experience, suggests that about 15 modes will be found. Because some wear out characteristics are expected, it is anticipated that the later life test experience beyond 10,000 rounds will expose 2 =new failure modes. Furthermore, an additional 2 new failure

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modes are expected from the dozen or so design changes motivated by non-reliability considerations. With about 15 +2+ 2+= 19 modes expected, previous experience suggests that about 46 failures can be expected. And the expected MRBF is therefore 15000/46= 326 MRBF.

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Appendix B Reliability Case Plan Outline

B.1 Scope

B.1.1 Purpose

B.1.2 Application.

Note that an overall RAM plan would include Maintainability and Testability. Details for these sections of the Plan are not included.

System and Technical Descriptions
- Historical Information
- System Hardware and Software Elements
- Contractor, Government Furnished Equipment

Reference Documents
- Government
- Contractor
- Standards

Management, Organization and Control
- RAM Organization
- Organizational Ties
- Teaming
- Issues Resolution Process
- Defect Data Management Process
- Relationships with Other Performance Areas
- Responsibilities
- Engineering
- Test/Testing
- Reliability Working Groups
- Technical Interchange Meetings
- Monitor and Control of Suppliers

Reporting
- Case Reports
  - Living Document – Updated Throughout Contract Period of Performance
  - Case Report Content
    - System Description
    - Reliability Requirements
    - Risk Areas
    - Strategy
    - Evidence
    - Metrics
    - Limitations on Use (boundaries and limitations on system use
    - Conclusions and Recommendations
    - Reviews
    - Reliability Program Reviews
    - Design Reviews

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- Internal
- Customer

Reliability Risk Approach
- Risk Identification
- Risk Abatement/Mitigation Approach and Process

Program Plans and Objectives
- Contractual Requirements
  - Schedules
  - Hardware, Software and Test/Testing Requirements

Reliability Requirements
- Management
- Translation of Contract Requirements to Operational Requirements
  - Usage Conditions
  - Warranty Provisions, Contract Incentives
- Reliability Metrics
  - Performance vs. Predictions
  - Analysis
  - MTBSA, MTBEFF, MTBMA, MTBF
  - Software
- Predictions
- Allocations and Changes
- Block Diagrams
- Modeling
- Benchmarking/Comparative Studies
  - Risk Approach
  - Strategy
  - Supportability
  - Compliance Verification

Environment, Usage, and Stress/Loading

Parts Procurement, Control, and Obsolescence
- Parts Program and Standardization
- Diminishing Resources

Design Phase Analysis
- Reliability Assessment Analysis
- Reliability Design Control
  - Design Criteria and Design for Robustness
  - Configuration Control
  - Stress Analysis – HALT, HASS
  - Electronic Stress and De-Rating

- Design Analysis Techniques
  - Physics of Failure, Finite Element Analysis, etc
- Critical Items List
- Failure Mode, Effects and Criticality Analysis
- Fault Tree Analysis
- Fault Tolerance
- Durability, Endurance, Wear-out and Service Life Analysis

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- Integrated Diagnostics and Prognostics
- Development and Test Phase Analysis/Activities
  - Reliability Growth
    - Planned Curves
    - Tracking Reliability
    - Projecting Reliability
  - Growth Methodology for Tracking and Evaluation
  - Reliability Testing
    - HALT, HASS
    - Qualification and Demonstration Testing for contract compliance
  - Environmental Testing
  - Data Collection
    - Failure Reporting, Analysis, and Corrective Action System Plan (FRACAS) and Failure Review Board (FRB)
      - Failure analysis and corrective action requirements and resources
      - Implementation process and timing plans and requirements
    - Reliability Demonstration/Qualification Testing
- Software Reliability Assessment
  - Architecture
  - Software Engineering Institute Capability Maturity Model (as appropriate)
  - Prediction
  - Measurement
- Manufacturing and Production Aspects and Impacts
  - Organizational Responsibilities, Planning for QA, Vendors
  - Processes, Activities to Control Defects
    - Robust Design
    - Six Sigma
    - Statistical Process Control
  - Screens
    - ESS/HASS
- Follow-on Activities
  - Field Data and Data Collection

B.1.3 References

1. RMS Reliability, Maintainability, and Supportability Guidebook, 3rd Edition, SAE International RMS Committee (G-11), 195.
2. A Guide to Preparing and Reviewing A Government/Contractor Reliability Program Plan, AMSAA Technical Report No. 441, Trapnell, Bruce S., April 1988.
3. Reliability Toolkit: Commercial Practices Edition, Reliability Analysis Center.
4. Product Reliability, Maintainability, and Supportability Handbook, ARINC Research Corporation, CRC Press, 1995.

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Appendix C Reliability Growth Models

C.1 Scope

The intent of this appendix is to provide an overview of a number of the various mathematical models for reliability growth that have been developed over the last four-plus decades. Technical references are given where a more complete discussion may be found. The research is courtesy of J. Brian Hall, US ATEC.

C.2 Overview.

Many reliability growth models have been developed over the past several decades. The purpose of these models is to help program managers and reliability practitioners address the formidable tasks of planning, tracking, and projecting the reliability improvement of a system throughout the development process. This literature review briefly covers the majority of the work (not all) done in the field. Planning models, tracking models, and projection models are given in the following three sections, respectively. More comprehensive works, such as, handbooks, surveys, and guides are given in B.5. A synopsis of associated theoretical results, simulation studies, real-world applications, personal-perspectives, and related statistical procedures (i.e., CI construction, parameter estimation, and GOF testing) is given in C.6.

C.3 Reliability Growth Planning.

C.3.1 Duane's Model (1964).

In 1964 J.T. Duane [7], who at the time was an aerospace engineer with General Electric Company in Erie, PA, discovered that if changes to improve reliability are incorporated into the design of a system, then the cumulative failure rate versus cumulative test time plotted on a log-log scale exhibits a linear relationship. This relationship is sometimes referred to as the Duane Postulate. Duane discovered this by developing cumulative failure rate plots for a broad range of aircraft equipment, including complex hydro-mechanical devices, aircraft generators, and jet engines. FIGURE C-1 below shows an example of a typical Duane Plot, where the parameter α represents the overall rate of reliability improvement throughout the course of the development program. The parameter α is commonly referred to as the growth rate and represents the negative of the slope of the logarithm of the cumulative failure rate.

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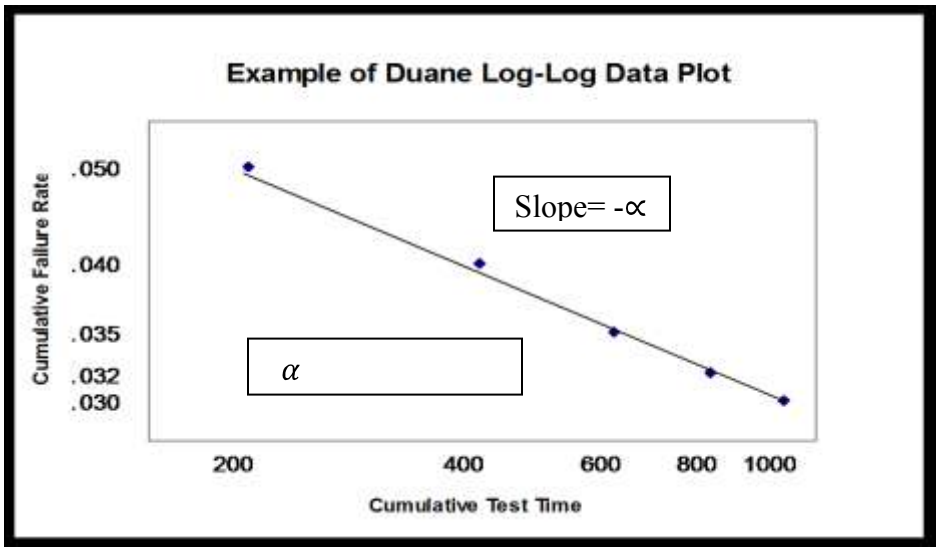


FIGURE C-1. *Duane Reliability Growth Plot.*

Duane's original intent[7] for the methodology was to monitor or track the reliability improvement in major subsystems for various aircrafts. This was done via the above assumed linear relationship, which approximately accounts for the overall change in the sequence of MTBF steps associated with successively redesigned configurations of a system throughout the TAFT process. While Duane's original intent was to monitor reliability improvement, the model has had significant ramifications throughout the field of reliability growth. According to Ebeling [160] the Duane growth model is "the earliest developed and most frequently used reliability growth model." In fact, the *Duane Postulate* is utilized as a fundamental assumption in many other reliability growth models that will be discussed below. An early and detailed application of the Duane model is presented by Selby and Miller [20].

C.3.2 Selby-Miller RPM Model (1970).
Selby and Miller [20] present an approach to reliability planning and management of complex weapon systems, which they refer to as "Reliability Planning Management (RPM)." The basic concept behind the RPM model includes its proposed "patterned reliability growth" approach to planning. This "patterned reliability growth" methodology follows directly from Duane's postulate and states that the cumulative failure rate versus cumulative test duration on a log-log scale is approximately linear with slope or growth rate ?. While this concept is not new, the RPM model appears to be the first application of the Duane postulate for reliability growth planning (as opposed to its original intent of reliability growth monitoring).

[7] Duane's original intent was to monitor reliability of a complex system undergoing design improvements, so his approach is mostly associated with reliability growth tracking. I have presented it in this section, however, since it can also be used as a planning tool. Moreover, the Duane postulate is used as a fundamental assumption in so many other growth models, his method marks a natural starting-point for which to begin this literature review.

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C.3.3 MIL-HDBK-189 Planning Model (1981).

The purpose of the MIL-HDBK-189 model [49] is to construct a reliability growth planning curve over the developmental test program useful to program management. The planning curve serves as a baseline against which reliability assessments can be compared, and it can highlight the need to management when reallocation of resources is necessary. The model is based on the Duane Postulate and consists of an idealized system reliability growth curve[8] that portrays the profile for reliability growth throughout the developmental test period and has a constant MTBF during the initial test phase. The planning parameters that define the idealized growth curve include: (1) the initial MTBF, (2) length of the initial test phase (i.e., reliability demonstration test for the initial MTBF), (3) the final MTBF (e.g., reliability requirement, or goal), (4) the growth rate and (5) the duration of the entire growth program. Some historical data on growth rates for Army systems is discussed by Ellner and Trapnell in [89]. The model also gives a set of expected MTBF steps during each test phase in the growth program. Corrective action periods are scheduled between each of the test phases where fixes are applied to previously observed failure modes. These improvements increase system reliability iteratively and result in an increasing sequence of MTBF steps, as displayed in FIGURE C-2.

2 A ***reliability growth idealized curve*** is a planning curve that consists of a single smooth curve based on initial conditions, assumed growth rate and management strategy [144].

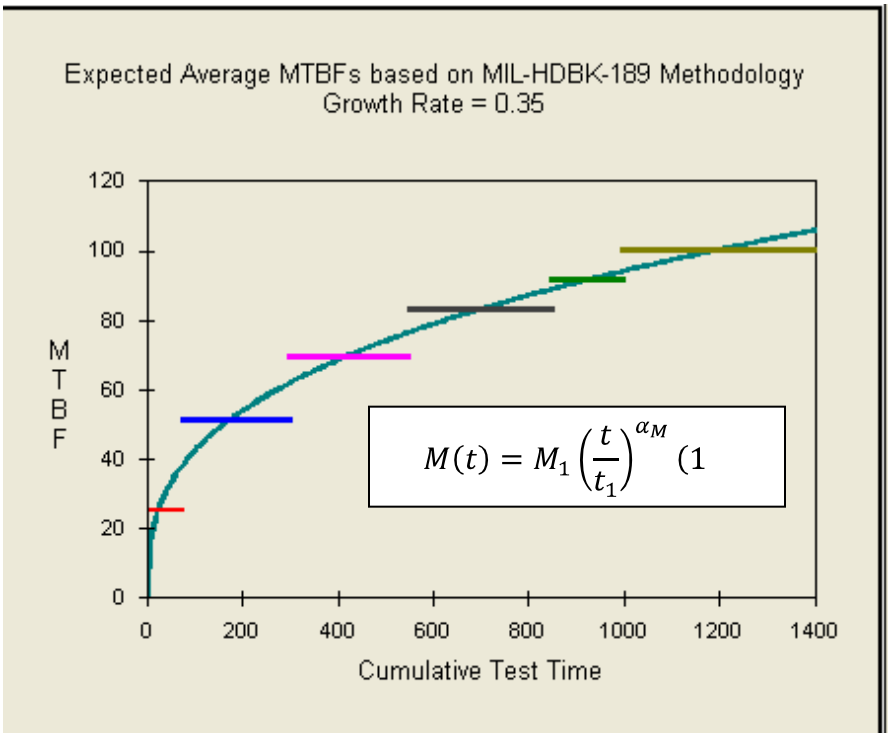


FIGURE C-2. *L-HDBK-189 Planning Curve.*

[8] A ***reliability growth idealized curve*** is a planning curve that consists of a single smooth curve based on initial conditions, assumed growth rate and management strategy [144].

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Ellner and Ziad [76] later studied the statistical precision and robustness[9] of the biased and unbiased estimators of MTBF of MIL-HDBK-189 model. They conclude that the precision of the estimators strongly depend on the expected number of failures. Also robustness between the biased and unbiased estimators is approximately equivalent.

C.3.4 AMSAA System-Level Planning Model (1992).
The SPLAN discussed by Ellner et al. [144] is another variant of the MIL-HDBK-189 model that can be used to construct system reliability growth test plans and associated idealized system reliability growth curves. The model can also prescribe the required test duration to achieve a system reliability requirement as a point estimate. This model gives several new options for determining various planning parameters, which is convenient for conducting sensitivity analyses. For example, given any four of the five planning parameters mentioned above SPLAN will determine the value of the remaining parameter. Most often, the initial MTBF, final MTBF, growth rate, and length of the initial test phase are provided to determine the required test time. FIGURE C-3 shows an example growth curve generated by SPLAN.

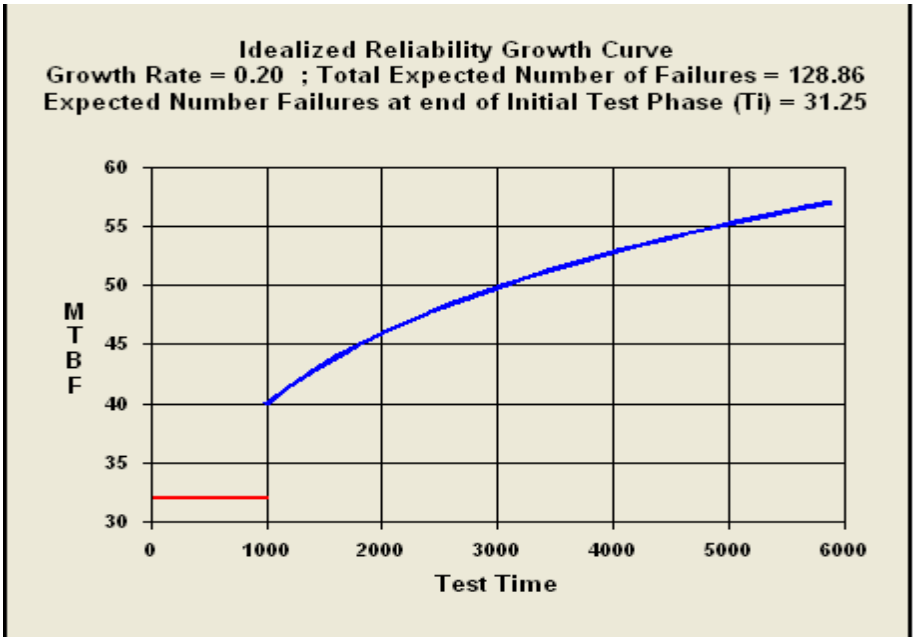


FIGURE C-3. *SPLAN Planning Curve.*

C.3.5 Ellner's Subsystem Planning Model (1992).
The SSPLAN model was developed by Ellner et al. [102] and [117] to develop system or subsystem reliability growth test plans that achieve a given system-level MTBF objective with a specified level of confidence. That is, SSPLAN determines the subsystem test times and subsystem reliabilities required to demonstrate a system MTBF objective at a given level of statistical confidence. Other work on SSPLAN includes Ellner and Mioduski's [100] operating

[9] ***Robustness*** refers to the effect on estimator statistical precision due to discrete configuration changes in system reliability.

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characteristic analysis for the model. Consumer and producer's risks are expressed in terms of the model parameters. For a given confidence level, they show that these risks only depend on the expected number of failures observed during testing, and the ratio of the demonstrated MTBF with confidence over the MTBF requirement. Formulas are developed for computing these risks as a function of the test duration and growth curve planning parameters.

C.3.6 Mioduski's Threshold Program (1992).
The Threshold Program was developed by Mioduski at AMSAA but no publication on the model is known to exist. However, the model is discussed by Broemm in [163]. The program determines at selected program milestones (e.g., thresholds), if the demonstrated reliability of a system is failing to improve as prescribed by the MIL-HDBK-189 idealized curve. It consists of a hypothesis test that compares a reliability point estimate for a system (based on actual failure data) against the theoretical threshold value consistent with the planning curve. Associated threshold values are established early in the acquisition process for program milestones or major decision-points. The test statistic in the procedure is the reliability point estimate (i.e., MTBF) computed from test data for individual system configurations. If the test statistic is inside the rejection region for the test, the program gives statistical evidence at a specified significance level that the system's reliability is not in conformance with the approved reliability growth program plan.

C.3.7 Ellner-Hall PM2 Model (2006).
The purpose of PM2, discussed by Ellner and Hall in [166], is to construct a reliability growth program planning curve for systems under development. Exact expressions are presented for the expected number of surfaced failure modes and system failure intensity as functions of test time. These exact expressions depend on a large number of parameters, but functional forms are derived to approximate these quantities that only depend on a small number of parameters. Simulation results are presented which show that the functional form of the derived parsimonious approximations can adequately represent the expected reliability growth associated with a variety of parent distributions for the initial failure rates of the system. The main difference of this model in comparison to those above is that it is independent of the NHPP assumption and utilizes parameters directly influenced by program management, such as: (1) initial MTBF; (2) MS; (3) goal MTBF; (4) average lag-time associated with fix implementation; (5) total test time; (6) average FEF; (7) the number and placement of CAP and (8) the planned monthly RAM test hours. Another benefit of PM2 is that it is the first planning model to take into consideration the lag-time due to implementation of corrective actions. An example of the type of detailed reliability growth plan that can be constructed using PM2 is shown in FIGURE C-4 below. The vertical lines displayed in the figure correspond to four months prior to the corrective action periods (or refurbishment periods) where fixes are installed to known failure modes. The significance of this is that only failure modes discovered before the four-month lag-time are addressed in the corresponding corrective action periods. The lag-time can be due to many factors but is mainly due to the time associated with root-cause analysis and the corrective action review, approval, and implementation process.

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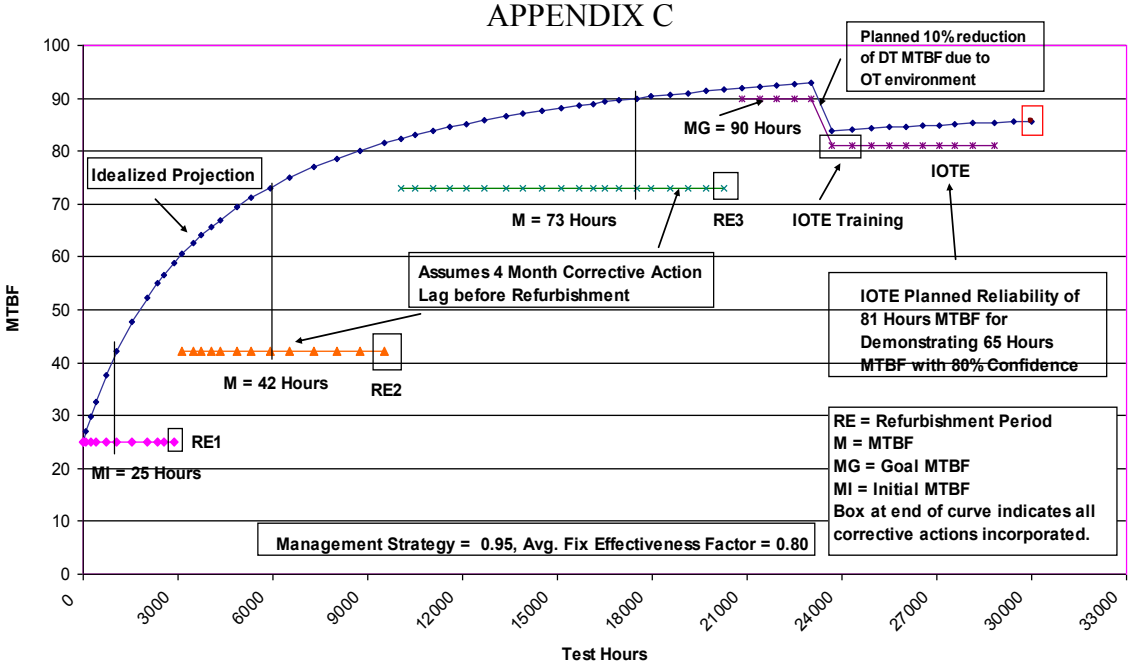


FIGURE C-4. *PM2 Planning Curve.*

C.4 Reliability Growth Tracking.

C.4.1 Weiss' Model (1956).

Weiss [1] developed methods for monitoring and extrapolating reliability growth of guided missile systems with Poisson-type failures. In this approach, the MTTF is believed to change over a sequence of successive trials as a result of finding and fixing failure modes in a system. MLE procedures are utilized to determine if reliability is increasing or decreasing, and identify the uncertainty of the reliability estimate. The model is shown to lead to a logistic-type reliability growth curve. Expressions are given for the estimated MTTF obtained from test data, as well as its variance.

C.4.2 Aroef's Model (1957).

Aroef [2] develops a reliability growth tracking model for continuous systems. He assumes that the rate of reliability improvement of a system is directly proportional to the growth achieved at a given time, and inversely proportional to test duration squared. The resulting differential equation takes-on the form $\frac{df(t)}{dt} = \alpha \frac{f(t)}{t^2}$ The solution is

$$f(t) = \theta \exp\left[\frac{f(t)}{t^2}\right]$$

where α is the growth rate and θ is the upper-limit on reliability (i.e., MTBF) that can be achieved as $t \rightarrow \infty$.

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C.4.3 Rosner's IBM Model (1961).
Rosner [3] develops what has become known as the IBM model, which is an expression for the system intensity function (i.e., rate of occurrence of failure). He assumes that the rate of occurrence of failure at time t is proportional to the number of non-random defects remaining in the system at time t. The resulting differential equation is expressed as dN(t)/ dt = −b N(t), which has the solution *N(t)* = a exp[− bt]. The constants, a and b, are approximated by regression. An interesting feature of the model includes its ability to estimate the required test duration for the system to be at a given "fraction corrected" (i.e., a fraction of the original failures that have been corrected). The model also estimates the number of non-random failures remaining at a given time.

C.4.4 Lloyd-Lipow Model (1962).
Lloyd and Lipow [4] develop a growth model to estimate the reliability of a system comprised of a single failure mode. The test program is assumed to be conducted in a series of trials. If the system fails in a given trial, a corrective action is implemented, and is mathematically modeled with a finite probability of being successful in mitigating the occurrence of the failure mode. The model has a simple exponential form given by
Rn = 1 – A exp$[-C(n-1)]$, where Rn is the reliability of the system in the n-th trial and the model-parameters, A and C , are estimated via test data. They also present a second model,

$R_k = R_\infty - \frac{\propto}{k}$, for estimating the reliability of a system in a given stage, in this case stage *k*. MLE and LS procedures are developed for estimating the model parameters R_∞ and α. A lower confidence limit on R_k is also discussed, in addition to other potential functional forms of reliability growth models.

C.4.5 Chernoff-Woods Model (1962).
Chernoff and Woods [5] present several exponential regression reliability growth models. One model of interest, due to its simplicity, estimates the probability that a system will successfully operate after a given number, r, failures have occurred and been subsequently corrected. The model is given by the simple exponential form Pr = 1 - exp (− (α +βr)), where
α > 0 and β > 0 are parameters estimated by a LS method.

C.4.6 Wolman's Model (1963).
Wolman [6] advanced the idea of assignable cause failure modes (e.g., AssignC failure modes). He assumes all assignable cause failures occur with equal probability in each trial and are completely eliminated upon initial observation. Hence, reliability is improved over a sequence of trials. Wolman assesses the reliability at stage k by Rk = 1 – q0 – (M + 1 – k)q where q0 denotes the probability of a non-AssignC failure mode, M is the initial number of AssignC failure modes and q is the probability of occurrence of a single AssignC failure mode. Probabilistic assessments for the model are provided via Markov chain approach. No estimation procedures reported.

C.4.7 Cox-Lewis Model (1966).
Cox and Lewis [11] proposed, perhaps, one of the first NHPP models, which is sometimes referred to as the exponential-law or the log-linear model. It takes the form m(t) = exp[α t + β],

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where α and β are parameters. The parameters are estimated from test data and GOF test procedures are developed. Clearly, the model reduces to a HPP when $\alpha = 0$.
Also, reliability growth is modeled when $\alpha < 0$.

C.4.8 Barlow-Scheuer Model (1966).
Barlow and Scheuer [12] also propose a k-stage reliability growth model, where the outcome of each stage is utilized to improve the system in remaining stages. In their trinomial framework, exactly one of three outcomes can occur in a given stage: success, inherent failure, or assignable cause failure. The reliability in the i-th stage is given by $r_i = 1 - q_0 - q_i$, where q_0 is the probability of an inherent failure and q_i is the probability of an assignable cause failure. MLE procedures are given for q_0 and q_i under the restriction that they are non-increasing. A conservative LCB on the reliability of the system in its final configuration is also presented. Other work on this model includes the CI procedures developed by Olsen [42].

C.4.9 Virene's Gompertz Model (1968).
Virene [16] considered the utility of the trinomial Gompertz equation for reliability growth modeling. The reliability assessment followed from $R = b^{c^t}$, where $b,c \in (0,1)$ and the parameter a is the upper-limit on reliability as time $t \rightarrow \infty$. He provides estimation procedures for the three model parameters as well as numerical examples.

C.4.10 Pollock's Model (1968).
Pollock [19] developed one of the first (if not the first) Bayesian reliability growth models. He modeled the parameters as random variables with appropriate prior distributions allowing one to project system reliability any time after initiation of the test with, or without, test data. Precision statements on the projection and estimation routines are given.

C.4.11 Crow's Continuous Tracking Model (1974).
In [26] Crow gives the first stochastic interpretation of the Duane Postulate. This is the first time the instantaneous failure rate for reliability growth given by Duane's model was re-parameterized and recognized as being the Weibull hazard rate function for a repairable system. The model is given by $r(t) = \lambda \beta t^{\beta - 1}$, where λ and β are model parameters. This observation allowed the development of statistical estimation and goodness of fit (GOF) procedures for reliability growth, which were also presented for time-truncated data. The same procedures were also developed by Crow [28] shortly thereafter for failure-truncated data. Both failure and time truncated estimation is discussed by Crow in [29]. CI procedures on MTBF are presented in [28]. Associated estimation procedures are based on the method of ML, and the GOF results are based on a Cramer-Von Mises test statistic. Crow gives numerical examples illustrating these procedures and a discussion of Army applications for the methodology.

These results have had a significant impact on reliability growth and repairable systems reliability modeling, as they have served as a methodological foundation for many subsequent approaches. Crow gives more comprehensive treatments to all the normal statistical procedures for the Weibull process in [32], [40], and [52]. This includes MLE procedures, hypothesis tests, and confidence bounds for model parameters (time and failure-truncated testing). Simultaneous confidence bounds on model parameters and a GOF test for the model is also given. Crow

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elaborates upon several applications of the methodology including: reliability growth, mission reliability, maintenance policies, industrial accidents, and applications in the medical field.
Using his stochastic interpretation of the *Duane Postulate*, the resulting model became known as the RGTMC, presented by Crow in [25-27] and [30]. This model is used to assess the improvement in the reliability of a system (within a single test phase) during development for which usage is measured on a continuous scale. Applications to reliability analysis for complex, repairable systems are discussed by Crow in [32]. Four "real-world" examples are given by Crow in a much later paper [87]. FIGURE C-5 below shows a plot of the MVF (i.e., expected number of failures) versus test time against the actual number of failures observed during testing of an Army system.

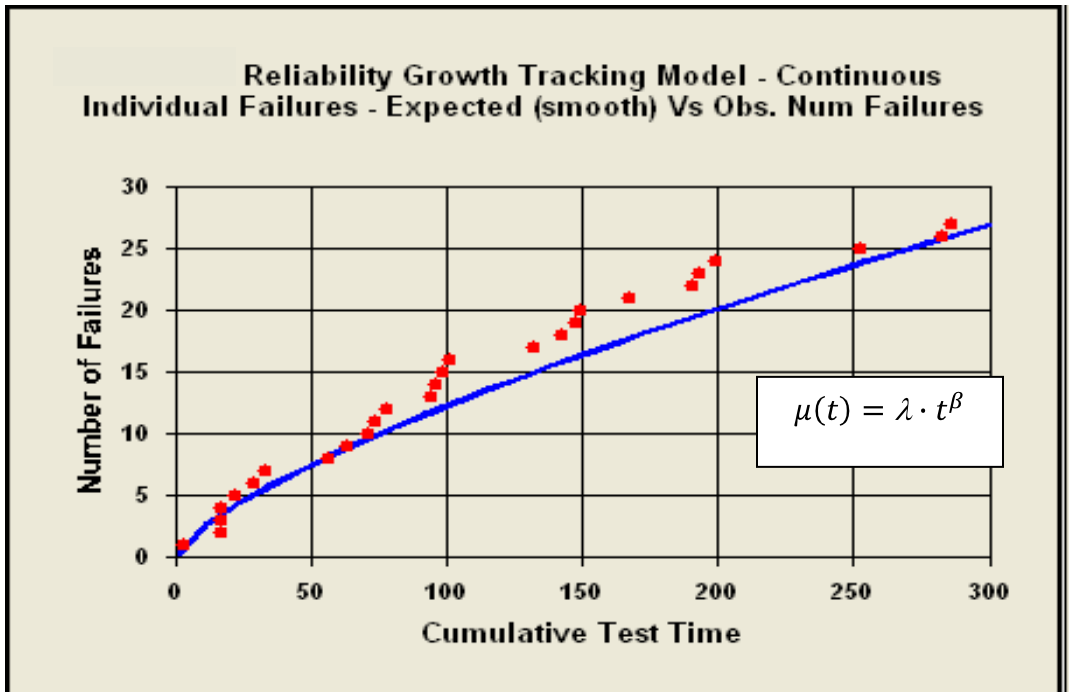


FIGURE C-5. *MVF vs. Test Time against Number of Failures*.

Other work on this model includes Crow's MLE procedure [74] for the parameters of the RGTMC in the case where there is missing data (i.e., incomplete data). This practical reliability growth estimation procedure assumes that the actual failure history over the problem interval is unknown. Such a phenomenon occurs when failure information over a period of testing is determined to be incorrect, which leads to the reporting of either to many, or to few failures.
Based on these techniques the observed number of failures over the problem interval is adjusted to "more realistically reflect the actual growth pattern." Hence, a valid reliability growth curve can then be fitted to the data and used for evaluation purposes. Two GOF procedures are developed (i.e., one based on the Cramer-Von Mises TS for individual failure time data, and the second based on a chi-squared r.v. for grouped failures). These new procedures are illustrated by several numerical applications. Years later, Crow [109] developed ML and CI procedures for failure data generated from multiple systems under test.

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C.4.12 Lewis-Shedler Model (1976).
Lewis and Shedler [33] offer an extension of the Cox-Lewis model by developing estimation procedures for the exponential polynomial model for powers of n = 1,…,10. The extension addresses models of the form m(t) = $[\alpha_0 + \alpha_1 t + \alpha_2 t^2 + \cdots + \alpha_{10} t^{10}]$.

C.4.13 Singpurwalla's Model (1978).
Using time-series[10] methods, Singpurwalla developed a discrete reliability growth model to: determine if the binomial parameter pi (i.e., the probability of success at stage i =1,…,k) is increasing after design modifications are applied in each stage. The model obtains estimates of p_i at the present stage, and also forecasts p_i at future stages (i.e., beyond stage *k*).

C.4.14 Crow's Discrete Tracking Model (1983).
The RGTMD was developed by Crow [55] for tracking the reliability of one-shot systems during development; such as, guns, rockets, missiles, torpedoes, mortars etc. Statistical estimation, CI and GOF procedures are given for both grouped data or for data captured during a trial-by-trial basis. The model is fundamentally based on the NHPP assumption derived from the Duane Postulate. More specifically, the model is constructed by obtaining an equation for the probability of failure on a configuration basis, using the NHPP power-law function (sometimes referred to as the "learning curve"). This equation and a plot of the reliability growth tracking curve is shown in FIGURE C-6 below.

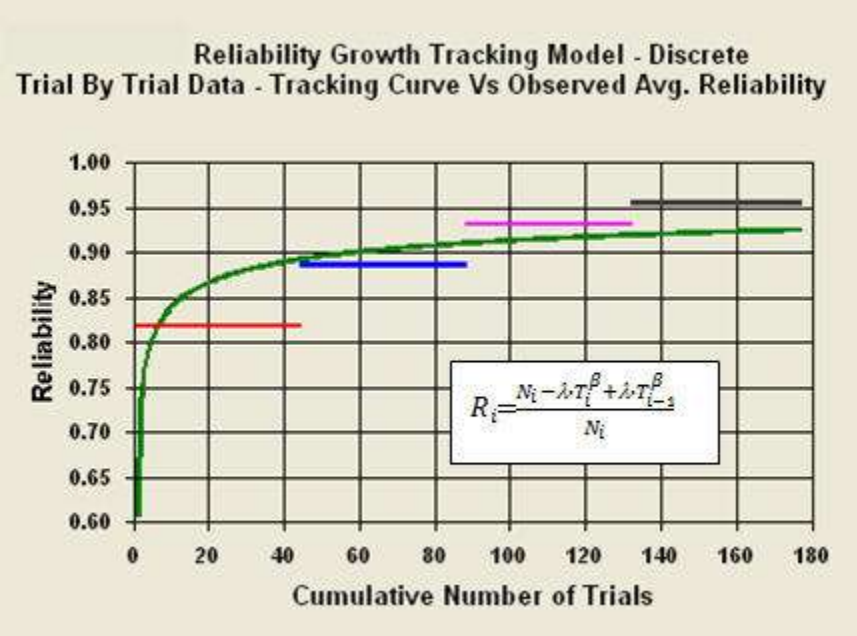


FIGURE C-6. *RGTMD Reliability.*

Other related work on this model was done by Finkelstein [59], and Battacharyya, Fries and

[10] ***Time series*** is defined as a set of observations generated sequentially in time.

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Johnson [84]. Finkelstein developed CAE of the model parameters for the case where only a single trial per configuration is tested. He also performed a simulation study to investigate the behavior of the CAE. He concludes that all attempts to obtain MLE of these parameters were unsuccessful and asserts the consistency of the CAE. Battacharyya, Fries and Johnson generalize the CAE given by Finkelstein in the case where there is a constant pre-specified number of test trials between system configuration changes. Large-sample properties of these estimators to include consistency and normality are developed. Large-sample standard-error formulas and CI procedures are given. Finally, they provide a proof on the consistency of the CAE, which confirms the Finkelstein conjecture. More work was done by Hall and Wessels [145] who formulate an evolutionary programming optimization algorithm to estimate the parameters of the RGTMD [55]. A numerical example is presented, where the standard MLE of the model parameters are compared against the proposed estimates from the optimization algorithm. The estimates are nearly identical. Overall, the algorithm proves to be an effective tool for reliability growth analysis when using the RGTMD.

C.4.15 Robinson-Dietrich Model (1988).
Robinson and Dietrich [78] and [85] develop a reliability growth model for monitoring the progress of the development effort at the system-level while the actual development occurs at the subsystem-level. Using the moments of the subsystem failure rate distributions as they change during testing, they show how the moments of the distribution of the system-level failure rate can be estimated. Using these moments, point-estimates and approximate CI for system reliability growth are derived. A hypothetical example is presented to illustrate some nuances of the methodology. Two additional examples are given on unspecified systems. The first system is comprised of three components, and the second system consists of eleven subsystems in a more complex structure.

C.4.16 Kaplan-Cunha-Dykes-Shaver Model (1990).
Kaplan et al. [90] develop a Bayesian method for assessing reliability during product development. Their "stepwise process" is implemented for analyzing failure data derived from the system and subsystem levels. Bayes' theorem is applied sequentially at each level throughout a number of test stages. The prior distribution and updating procedure at each level utilize engineering judgment to evaluate the significance of failures observed and effectiveness of corrective actions. Overall, the paper includes the development and application of a Bayesian framework for gathering, organizing and incorporating expert knowledge into reliability growth assessment. A notable feature of the approach is that assessments of reliability are derived with a concomitant measure for uncertainty.

C.4.17 Mazzuchi-Soyer Model (1991).
Mazzuchi and Soyer [92], [106], and [110] present a Bayesian approach for assessing reliability growth during system development. At the end of each stage of testing, failures are examined so that modification can be implemented to remove failure modes. They incorporate prior information into an ordered Dirichlet prior distribution for failure probabilities at each stage. The resulting posterior distribution of all relevant quantities is expressed as a mixture of beta, or Dirichlet, distributions. After each stage of testing, the model gives Bayes estimates of system reliability. The method is illustrated by numerical example. Overall, their approach provides a means for incorporating subjective information into reliability assessment, and provides the

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means for analyzing system reliability over successive stages of testing using sequential updating of a Bayesian prior distribution. These results are later extended by Erkanli, Mazzuchi and Soyer [135] who considered both the exponential and Weibull time-to-failure models.

C.4.18 Heimann-Clark PR-NHPP Model (1992).
Heimann and Clark [105] argue that a more accurate reliability assessment of a system can be obtained by explicitly modeling the effect of defects induced during the manufacturing process. To model this phenomenon, they develop a process-related NHPP by replacing the constant scale factor by a process age-dependent function. This function increases asymptotically over process age to a mature process scale factor value. MLE procedures are given for model parameters. The proposed PR-NHPP addresses the questions: "What will the product reliability be after a given age of the manufacturing process?" and "How much reliability growth time will be required to achieve a given product failure intensity goal?" One parameterization of the NHPP is $h(t) = \left(\frac{\beta}{\alpha}\right)\left(\frac{t}{\alpha}\right)^{\beta-1}$. The proposed modification is to replace the constant scale factor α in this equation by the function a(t) = a (1 – exp[-bt]), where *t* is the length of time that the production line has been operational, and *b* is the shape parameter. Numerical examples are given to demonstrate the utility of the proposed PR-NHPP.

C.4.19 Fries' Discrete Learning-Curve Model (1993).
Fries [111] develops a learning-curve approach for discrete reliability growth analysis.
This approach is particularly appropriate for destructive tests of very expensive systems.
Derivations of the new model and of the RGTMD [55] are presented. Approximations of mode parameters are obtained by ML procedures. Extensions of both models are discussed, which account for the distinction between *assignable*[11] and non-assignable cause failure modes. Each model accommodates for the monotonic growth in reliability during system development. The models and estimation procedures are illustrated by two numerical examples. In a later paper [116], Fries gives corrections to the likelihood equations that properly reflect the negative binomial (geometric) behavior of the number of trials until the first observed failure.

C.4.20 Modified-Gompertz Model (1994).
Kececioglu, Jiang, and Vassiliou [115] observe from several datasets that reliability growth data with an S-shaped trend could not be adequately portrayed by the conventional
Gompertz model [16]. They point-out that the reason is due to the model's fixed value of reliability at its inflection point. As a result, only a small fraction of reliability growth data sets following an S-shaped pattern could be fitted. Their proposed solution overcomes this shortcoming by modifying the Gompertz model to include a fourth parameter. This fourth parameter shifts the associated growth curve vertically, thus accommodating for S-shaped growth datasets. The new method is claimed to be more flexible than its predecessor for fitting data with S-shaped trends. The original Gompertz model is given by $R = a\ b^{c^T}$. The modification assumes the form $R = a\ b^{c^T} + d$. Estimation procedures are presented and consist of solving four equations for the four unknown model parameters. A detailed numerical example is given.

[11] An a***ssignable-cause*** failure mode is a failure mode whose root-cause is known and is therefore readily correctable.

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C.4.20 Ellner's Subsystem Tracking Model (1996).
The SSTRACK model was developed by Ellner [144] for assessing system level reliability from lower level subsystem testing. The motivation for this methodology was to make greater use of subsystem test data in estimating system reliability. SSTRACK takes into consideration data from both growth and non-growth subsystems. The model uses the Lindstrom-Madden method [4] for combining test data from individual subsystems. The methodology includes statistical CI and GOF procedures. FIGURE C-7 below shows an example of approximate LCB on system MTBF computed from subsystem data as a function of the desired level of statistical confidence.

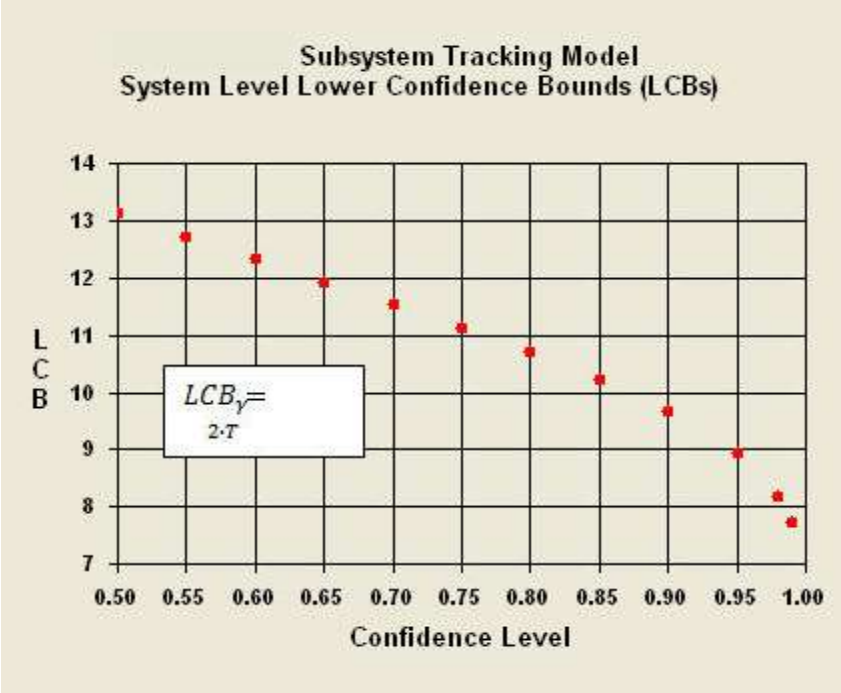


FIGURE C-7. *SSTRACK LCB on MTBF.*

C.4.21 Sen's Alternative to the NHPP (1998).
Sen [136] investigates the statistical inference of current reliability of the Duane model [7]. Exact and large-sample distributional results are derived for the ML and LS estimators of the current failure intensity. The extent of misspecification of the NHPP power-law process [25] to fit failure data of a system experiencing recurrent failures is explored. Simulation results and an illustration is provided to supplement the theoretical findings and demonstrate the presented inference results. Sen concludes that the model is a suitable alternative to Crow's NHPP power law model, in the context of analyzing recurrent failure data from systems undergoing developmental testing. Clarifications on the exact inference procedures are discussed by Sen in [140].

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C.4.22 Donovan-Murphy Model (1999).
Donovan and Murphy [141], [143] and [146] present a new reliability growth model which is claimed to be simpler to plot and provide a better fit to data than the Duane model over the range of slopes normally observed (i.e., $\alpha \leq 0.5$). The model (for MTBF) is derived from variance stabilization transformation theory and takes-on the form $\theta(t) = \alpha\sqrt{t} + \beta$. Simulation results indicate that their model is "more effective" for growth rates less than 0.50 (which is generally the typical range for growth rates). Numerical examples are presented from two published datasets and yield findings consistent with those of the simulation results.

C.4.23 Pulcini's Model (2001).
Pulcini [147] presents an exponential reliability growth model, which incorporates step changes in a system's failure intensity due to engineering design improvements. He gives ML and CI procedures (exact and approximate) for obtaining estimates of the current failure intensity and lifetime expectancy. GOF procedures based on a Cramer-Von Mises TS are also developed. These statistical procedures are based on a scenario where several identical items are put on test and design modifications are introduced to all items at each failure occurrence. A numerical example is given to illustrate the inference, prediction, and test procedures using actual failure data from a single (unspecified) military tank.

C.4.24 Gaver-Jacobs-Glazenbrook-Seglie Model (2003).
Gaver et al. [155] introduce probability models for sequential-stage system reliability growth. These models are appropriate in cases where a system is tested in a series of stages, whereby if a failure occurs in a given stage, later stages are not entered. System success is determined by successful operation in all test stages. At most one defect is assumed to be removed per test. Analytical procedures are developed to calculate the expected probability of field system mission success after completion of a runs-test[12], the distribution of the probability of system field mission success after a successful runs-test, and the expected number of individual system tests required to achieve a run of consecutive system successes, i.e., a runs-test, and hence test termination. Numerical studies indicate that the probability of system field success after a runs test can be quite sensitive to the probabilities that a test activates defects in each of the stages. However, the mean number of tests required to obtain a run of r successful tests appears to be relatively insensitive to these activation probabilities. *Seglie's stopping criterion*[13] [134] is studied quantitatively through a Bayesian model formulation which suggests the criterion provides a simple and effective test stopping-rule for a range of reasonable cost criterion.

C.5 Reliability Growth Projection.

C.5.1 Corcoran-Weingarten-Zehna Model (1964).
Corcoran, Weingarten and Zehna [8] developed the first model for estimating reliability after corrective action. The approach was developed with consideration to estimating reliability in the

[12] A ***runs-test*** is a sequence of tests that is conducted until a specified number of consecutive successful tests is achieved.

[13] ***Seglie's stopping criterion*** consists of stopping all testing after a successful runs-test is achieved.

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final stage of development of an "expensive item." The reliability projection is suitable in cases where corrective actions are installed at the conclusion of a single test phase consisting of N independent trials and where, the number of trial outcomes of interest is a multinomial distributed r.v. with parameters N (total number of trials), p_0 (unknown success probability), and qi (unknown failure probability for failure mode i= 1,...,k). Note that since a multinomial model is used, the equality $p_0 + \sum_{i=1}^{k} q_i = 1$ must be satisfied, which models the condition where at most one failure mode can occur on any given trial. In addition to deriving an exact expression for system reliability under the conditions above, Corcoran, Weingarten and Zehna presented seven different estimators and evaluated them in light of criterion typically adopted for that of point estimation (i.e., bias, consistency, conservatism, and ML). By studying these estimators they showed that an unbiased estimate of the corrected system could not be obtained. They were the first researchers to advance the idea of reducing initial failure probabilities by a fractional amount with consideration to fix effectiveness. They were also the first researchers to take into consideration (and estimate) the portion of system unreliability associated with failure modes that have not yet been observed.

C.5.2 AMSAA/Crow Model (1982).
The ACPM was developed by Crow and discussed in [54] and [56] for estimating system reliability at the beginning of a follow-on test phase. The model takes into consideration the reliability improvement from delayed fixes only, and is suitable for systems whose test duration is continuous. The primary framework for reducing the initial failure rates follows directly from the Corcoran-Weingarten-Zehna model [8]. Two GOF procedures have been developed for the ACPM and are discussed by Ellner in [144]. The first procedure is based on a Cramer von Mises TS for grouped data. The second procedure, which is of the chi-squared type [69], is for individual failure time data. The ACPM is one of the first models to incorporate the important concept of reliability growth potential8 [63]. No CI procedures have been reported. As Crow discusses in [93], this model is also an international standard adopted by IEC and ANSI [120].

C.5.3 Ellner-Wald AMPM Model (1995).
AMPM was developed by Ellner and Wald [121] and was the first projection model to estimate reliability in the case where corrective actions could be either delayed or non-delayed. The benefit of this is that the system's configuration with respect to design and reliability do not have to be constant. The model provides estimates of: (1) the expected number of B-mode surfaced; (2) the percent surfaced of the B-mode initial failure intensity; (3) the rate of occurrence of new B-modes and (4) the projected reliability of the system. FIGURES C-8 through C-11 show each these model equations, comprising the robust reliability growth methodology of AMPM. The model also provides estimates of the reliability growth potential[14]. Estimation procedures are given for both grouped data and individual failure-time data. No CI procedures have been reported. GOF procedures are discussed by Broemm in [163].

[14] ***Reliability growth potential*** is the upper-limit on reliability achieved by finding and correcting all failure modes in a system with a specified fix effectiveness.

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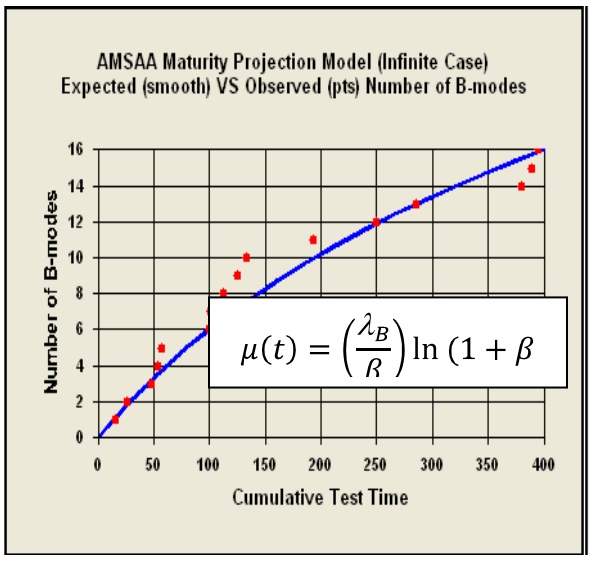


FIGURE C-8. *Expected No. Modes.*

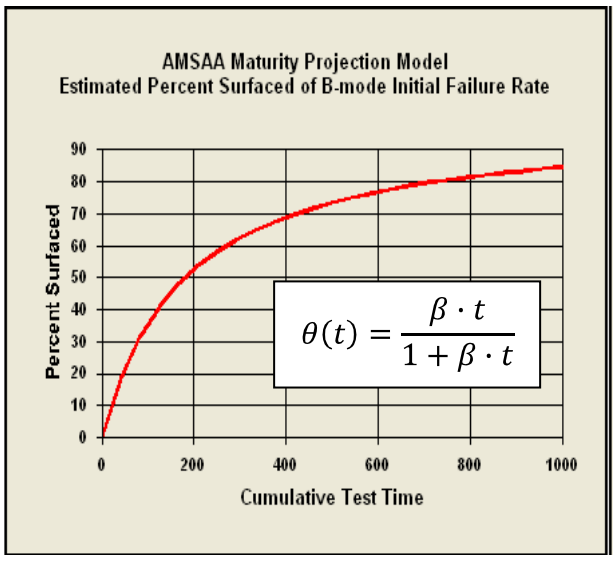


FIGURE C-9. *Percent λB Observed.*

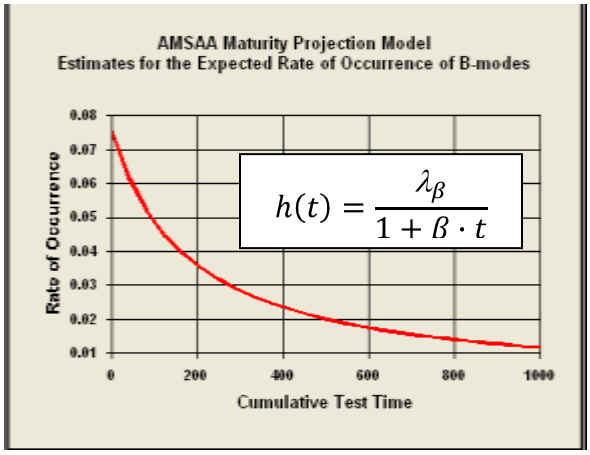


FIGURE C-10. *ROC of New Modes*

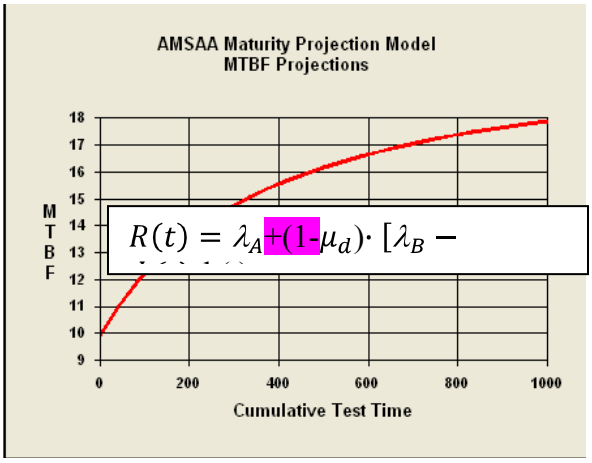


FIGURE C-11. *Reliability Growth*

C.5.4 Clark's Model (1999).

Clark [139] argues that reliability is often overlooked during early system development and many programs experience late growth programs not long before production as a result. He notes that the "popular AMSAA models" are difficult to apply in these cases as they prescribe high test durations even for aggressive growth rates. He formulates a model as an alternative for projecting reliability growth late in development, which is claimed to overcome these shortcomings. The proposed model consists of two main extensions of the ACPM [54]. The first extension includes a technical modification to allow the model to be applied in the case were fixes can be delayed or non-delayed (rather than all delayed). The second extension includes adding a term for the inherent failure rate of the system to determine how close the current reliability is to the maximum that can be achieved and decide when further growth is no longer time or cost effective. The method is illustrated via numerical example on the Airborne Warning

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and Control System Radar System Improvement Program. The results indicate that the model generally projected system reliability well, except when new failure modes introduced into the system by software modifications were not accounted for.

C.5.5 Ellner-Hall AMPM-Stein Model (2004).

The AMPM-Stein model, given by Ellner and Hall [162], is used to estimate the reliability of a system following correction of known failures modes in the case where these fixes are delayed to end of the test. The benefit of this approach is increased accuracy obtained by using a shrinkage factor estimator, given by Stein [50], designed to minimize the expected sum of squared error. The unique feature about this estimation procedure is that all estimates of failure rates are finite and positive (whether they are observed in testing or not observed in testing). Monte-Carlo simulations conducted by AMSAA [156] indicate that the accuracy in the reliability projection associated with AMPM-Stein is greater than that of the international standard adopted by IEC and ANSI [120], namely, the ACPM [54].

C.5.6 Crow-Extended Model (2005)

The purpose of the Crow-Extended model [157] is to estimate reliability in the case where corrective actions can be either delayed or non-delayed (i.e., the same as that for AMPM). This model is a relatively minor extension of the ACPM [54] and RGTMC [30] models and is therefore based on the Duane Postulate. Estimation procedures follow from the two existing models. Crow [157] also provides 33 metrics useful for managing a reliability growth program and introduces the notion of a further failure mode classification scheme (e.g., BD-modes[15] and BC-modes[16]).

C.6 Reliability Growth Surveys and Handbooks.

C.6.1 Crow's Abbreviated Literature Review (1972).

Crow [23] presents an abbreviated literature review of some reliability growth models. A limited number of numerical examples are also presented. The growth models include: Weiss [1], Lloyd-Lipow [4], Wolman [6], Duane [7], Barlow-Scheuer [12], Virene [16], and Pollock [19]. All of these models are discussed herein.

C.6.2 DoD's First Military Handbook on Reliability Growth (1981).

Military Handbook 189 "Reliability Growth Management" [49] is the U.S. DoD's first handbook on reliability growth. The handbook was developed by the U.S. AMSAA with Crow as the principle author of the document. It was first published in February 1981. The handbook offers techniques to enable program managers of DoD weapon systems to plan, evaluate, and control the reliability of their systems during the development process. It also provides procuring activities, and defense contractors with an understanding of the concepts and principles of reliability growth, as well as offer guidelines and procedures to be used in managing a reliability growth program. In the main body of the handbook, two models are briefly introduced including

[15] ***BD-modes*** are "B-Delayed modes" that will not be corrected until the end of the current test phase.

[16] ***BC-modes*** modes are "B-Corrected modes" that will be corrected during testing.

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the MIL-HDBK-189 model [49], and the RGTMC [30], both of which are discussed above. In Appendix B of the handbook, eight discrete and nine continuous reliability growth models are summarized. The discrete models include: two Lloyd-Lipow models [4], Wolman's model [6], the Barlow-Scheuer model [12], Virene's Gompertz model [16], and Singpurwalla's model [45]. The continuous reliability growth models include: Duane [7], RGTMC [30], Cox-Lewis model [11], Lewis-Shedler model [33], Rosner's model [3] and a variant thereof, a continuous Lloyd-Lipow model [4], Aroef's model [2], and an unreferenced exponential model for cumulative MTBF. In Appendix C, the RGTMC [30] and associated statistical procedures are discussed in more detail.

C.6.3 Fries-Sen Survey on Discrete Reliability Growth Models (1996).
Fries and Sen [127] present a comprehensive compilation of model descriptions and characterizations, as well as discuss related statistical methodologies for parameter estimation and CI construction. The interrelationships and assumptions that underlie the various models is also presented. Their survey is extensive covering: single-stage models (e.g., Corcoran et. al [8]), multi-stage models (e.g., Lloyd-Lipow[4]), trinomial models (e.g., Wolman [6], Barlow-Scheuer [12], Weinrich-Gross [43], Mazzuchi-Soyer [110]), Bayes models (e.g., Pollock [19],Kaplan et al. [90], Jewell [65]), exponential-growth models (e.g., Lloyd-Lipow [4]), exponential regression models (e.g., Gross and Kamins [15], Virene [16]), learning curve models (e.g.,Duane [7], RGTMD [55]), and several other model types. This survey is the most comprehensive available on the subject.

C.6.4 DoD's Guide for Achieving RAM (2005).
In August 2005, the OSD published a guide on achieving RAM [163] for DoD systems. Appendix C of the document is devoted solely to reliability growth. A number of associated concepts are discussed including: reliability maturity metrics for failure mode coverage and fix effectiveness, as well as some reliability growth planning, tracking, and projection models. The growth models that are discussed include: RGTMC [30], RGTMD [55], Corcoran-Weingarten-Zehna Model [8], ACPM [54], AMPM [121], Crow-Extended [157], AMPM-Stein [162], and the MIL-HDBK-189 model [49].

C.7 Other Literature (i.e., Theoretical Results, Perspectives and Applications).

C.7.1 Corcoran-Read Simulation Study (1967).
Corcoran and Read [13] present a simulation study of four reliability growth models available at the time. These models include: Chernoff-Woods [5], Barlow-Scheuer [12], Wolman [6] and Lloyd-Lipow [4]. They compare the reliability estimates of these methods with three measures of effectiveness: the popular average squared-error, the average squared-error after applying an inverse sine transformation (used to stabilize the variance of success probability estimates), and a logarithmic transformation applied to the failure probabilities (i.e., the average of the absolute deviation of logarithms of the ratio of error in the failure probabilities). Based on these measures of effectiveness, they conclude that their "general ranking of the preferences" of these methods are 1, 4, 2, and 3, respectively.

C.7.2 Barr's Paper (1970).
Barr [21] considers a class of reliability growth models that accommodate for variations in several important factors including: the interdependencies of assignable-cause failure modes, the

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inclusion of an inherent failure mode, the repair policy and the distribution of initial states of the system. His paper is an exposition of several prediction models appearing in the early literature of reliability growth and identifies their general features. The methods considered include those of Lloyd-Lipow [4], Pollock [19], Weiss [1], and Wolman [6]. The overall problem Barr considered is that of predicting (before testing is undertaken) what the reliability of the system will be after a sequence of trials, and to predict the number of trials required to attain a given reliability. He divides this general class of reliability growth models in three types: single assignable-cause mode models, multiple equally likely assignable-cause mode models, and multiple assignable-cause modes not necessarily equally likely.

C.7.3 Read's Remark on Barlow-Scheuer Estimation Scheme (1971).
Read's remark [22] notes that the Barlow-Scheuer estimation procedure is incomplete. He notes that this is due to not addressing the case where all trials of a stage result in only inherent failures. Read proposes a policy to handle the case which allows estimation of the trial probability of assignable-cause failures.

C.7.4 The AMSAA Reliability Growth Symposium (1972).
Crow [24] provides conference proceedings on a reliability growth symposium sponsored by the U.S. AMSAA. The conference took place on 26-27 September 1972 and was as an outgrowth of recommendations by the Panel on Accelerated Development of Reliability. This panel was chaired by Jack Hope who was then serving on the White House Staff. The purpose of the symposium was to enhance the state of technical and managerial knowledge on reliability growth methodology to benefit the Army's materiel acquisition process. There were over 200 attendees and six papers given. The papers include Selby-Miller [20], Virene [16], Crow [24], Barlow-Proschan-Scheuer [24], Barlow [24], and Corcoran-Read [14].

C.7.5 Langberg-Proschan Theoretical Paper (1979).
Langberg and Proschan [47] present theoretical results on converting reliability growth (or decay) models involving dependent failure times into equivalent models involving only independent random variables. They consider a sequence of such conversions occurring at successive points in time where the independent random variables are becoming stochastically larger (reliability growth). Ultimately, they demonstrate that the limiting distributions in the sequence of dependent models "correctly correspond" to the limiting distributions in the sequence of independent models. No practical reliability growth model is presented, rather, associated results are mainly theoretical and focus on the technical aspects of the aforementioned conversion.

C.7.6 Jewell on Learning-Curve Models (1984).
Jewell [65] constructs a general framework for learning-curve reliability growth models with Bayesian estimation procedures for model parameters. He argues that Bayesian estimation methods must be used to incorporate engineering experience in prior estimates of the parameters of learning-curve models because ML estimators may be very inaccurate and unstable. His main conclusion is that the majority of learning-curve developmental test programs will provide insufficient data to reach the desired precision for manufacturers to make early predictions on reliability when using traditional methods. In particular, he indicates that the use of the Duane

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learning-curve g(t) =k·t$^{\tilde{?}ı}$ leads to technical difficulties in reliability growth application and that an exponential learning-curve g(t) = exp(−ν·t) avoids such problems.

C.7.7 Wong's Letter to the Editor (1988).
Wong [79] discusses the lack of organization in the vast literature on reliability growth and identifies some processes that are not reflected in associated methodologies. These processes include: equipment aging effects, manufacturing learning-curve (i.e., improvement in production processes over time for a single item), industry-wide part improvement (e.g., effects of industry burn-in for electronic components), and methods on the test, analyze, and fix process that only use test duration as an independent variable (e.g., Duane's model [7]). He suggests that authors of reliability growth papers should: specify what kind of reliability growth process they are modeling, and which factors in their model are held constant or randomized to smooth-out effects.

C.7.8 Wronka's Application of the RGTMC (1988).
Wronka [77] shows the benefits that can be obtained by conducting reliability growth tracking early in the development process. He gives an application of the RGTMC [30] for prototypes of a circuit card assembly. Results are presented for the estimation procedure of grouped data and associated GOF test.

C.7.9 Benton and Crow on Integrated Reliability Growth Testing (1989).
Benton and Crow [81] consider the development of reliability growth under integrated reliability growth testing. By integrated testing they refer to a development program consisting of: functional testing, environmental testing, safety testing, performance testing, mobility testing, and dedicated RAM testing. They discuss and apply the concepts of the MIL-HDBK-189 model [49], RGTMC [30] and the ACPM [54] under the framework of these types of integrated tests. They also present results and lessons-learned on some Army programs. Years later, Crow, Franklin and Robbins [118] present a successful application of integrated reliability growth testing in the development of a large switching system. Their claimed benefits include: timely analysis of failed items, accurate problem classification, accurate laboratory failure rates, early identification of failure modes, management metrics for reporting, and reliability growth achievement using all test resources available.

C.7.10 Frank's Corollary of the Duane's Postulate (1989).
In [82] Frank discusses his observation that various types of avionics equipment are found to demonstrate remarkably similar gradual declines in reliability during prolonged service. He proposes a modification of Duane's learning-curve approach by extending its applicability to project a reliability profile over the planned service life for equipment. Frank claims his "revised equations" (not given) can be used to predict changes in equipment reliability, thus providing a capability to more accurately estimate life-cycle support resource requirements and costs.

C.7.11 Gibson-Crow Estimation Method for Fix Effectiveness (1989).
Gibson and Crow [83] develop a "practical and statistically sound" methodology for estimating the average FEF, which is a parameter utilized in some reliability growth models (e.g., Ellner's AMPM [121] and Crow's ACPM [54]). The average FEF, $\bar{?}D$, is basically estimated by using the ACPM in a reverse manner. In this approach, Gibson and Crow estimate the portion of the

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system failure intensity in a follow-on test by the common reliability point estimate, $\lambda = f/T$ (e.g., failures over test time). This value is then equated to the reliability projection equation given by the ACPM. The equation is then algebraically manipulated to solve for the average FEF.

C.7.12 Woods' Study on the Effect of Discounting Failures (1990).
Woods [88] analyzes the effect of failure discounting[17] on the accuracy of two discrete and two continuous reliability growth models. The discrete models include: the Chernoff-Woods exponential regression model [5], and Crow's RGTMD [55]. The continuous models include: Crow's RGTMC [30] and a modification of the same model that only uses data in a given phase (i.e., not cumulative data). Woods concludes that failure discounting has a greater impact on the cumulative growth models than on the non-cumulative (i.e., there is greater bias in the cumulative models, thus yielding more optimistic reliability estimates). He also indicates that the non-cumulative growth models tracked growth patterns better than the cumulative.

C.7.13 Higgins-Constantinides Application (1991).
Higgins and Constantinides [91] present an interesting reliability growth application of the U.S. Navy's EMATT system. EMATT is an open-ocean one-shot expendable target used in simulating combat missions. They faced the dilemma where no one-shot reliability growth model was suitable for their purposes. Additionally, the application of established continuous growth models was deemed inappropriate[18] since the number of trials in each test phase was not relatively large nor was the reliability high. Since none of the classical growth models available at the time could provide suitable approximations, the reliability growth approach adopted for EMATT consisted of fundamentals from the Duane model [7]. Thus, they constructed a reliability growth tracking curve by plotting the cumulative reliability (i.e., cumulative successes over cumulative trials) versus cumulative trials. The results indicate a general reliability improvement trend and they note the difficulty in obtaining precise numerical reliability estimates with limited trials. In their final report [113] published two years later, they apply the RGTMD [55]. The results show that EMATT reliability grew from 0.4 to 0.8 over the several year development program demonstrating its reliability objective requirement of 0.8 (as a point-estimate).

C.7.14 IEC International Standards for Reliability Growth (1991).
The IEC adopts two international standards for reliability growth: IEC Standard 1014 covering "Programs for Reliability Growth," and draft IEC Standard 56 (Central Office) 150 on "Reliability Growth and Estimation Methods." IEC Standard 1014 is issued in 1989 and gives guidelines for improving the reliability and exposing the weaknesses of hardware and software items. This standard also presents basic concepts and descriptions of management, planning, testing, failure analysis, and corrective action techniques. The final draft of IEC Standard 56 (Central Office) 150 becomes IEC Standard 1164 [120] and is issued in 1995. This standard

[17] ***Failure discounting*** is the practice of removing fractions of the previous failures after corrective action has been taken, where no failures for the same cause reoccur in subsequent testing.
[18] Continuous growth models were deemed inappropriate since they can only be utilized as good approximations for tracking the reliability of one-shot systems in the case where the number of trials within each test phase is relatively large and the reliability was relatively high, as stated in MIL-HDBK-189 [49].

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describes Crow's NHPP power-law reliability growth model [30] and related projection model, ACPM [54]. Step-by-step directions on their use are given. All statistical methods for the models are discussed including: MLE, CI, and GOF procedures for failure and time-truncated data. Both standards are discussed by Crow in [93].

C.7.15 Coolas' Application (1991).
Coolas [94] presents a dynamic reliability prediction technique for the DPS 7000, which is a mainframe computer system. Observed measures of field performance, and trends in reliability growth (due to evolving product maturity) are identified. The proposed reliability predictions are based on adjustments to component reliability and reliability growth models following from these observations. As a result of the reliability predictions and associated improvement program, more accurate spare parts provisioning and decrease in maintenance costs are claimed to have been achieved. While not specifically referenced, the Lloyd-Lipow model [4] appears to be used for component level reliability growth assessment.

C.7.16 Bieda's Application (1991).
Bieda [95] presents an analysis addressing product / process design concerns and validation testing issues via reliability growth testing, monitoring, and assessment. The integration of reliability growth test techniques is applied to evaluate the reliability of an unspecified electro-mechanical device. Reliability growth tracking curves are developed using Duane's model [7] and the various relationships between design iterations are identified. Product assurance analyses are performed to help identify design and process-related concerns. Point estimates and one-sided LCB on MTBF are given using the NHPP Weibull process [26]. Results consist of successful demonstration of the relationship between failure detection and corrective action, as well as the achievement of higher reliability through reliability growth testing and use of reliability growth tracking methods.

C.7.17 Ellis' Robustness Study (1992).
Ellis [104] examines the robustness of techniques applied to failure time data to determine if the system failure intensity is changing over time. The techniques include: the Duane model [7], and the RGTMC [30]. Monte-Carlo methods are utilized to simulate failures. MLE procedures based on time-truncated and grouped data are used to approximate model parameters and associated MTBF. Some basic advantages and disadvantages of the models are discussed. The results of the study show that the Duane model indicates reliability growth even in cases when the failure data are generated by an exponential distribution, and that the RGTMC is more suitable for detecting the presence of reliability growth or decay. This is largely due to the more sophisticated statistical procedures developed for the NHPP Weibull process (e.g., point-estimation, CI construction, and GOF testing).

C.7.18 Calabria-Guida-Pulcini Bayes Procedure for the NHPP (1992).
Calabria, Guida and Pulcini [103] develop a Bayesian estimation procedure for the parameters of the NHPP power-law process, originally developed by Crow in 1974 [26]. They provide Bayes estimates of system reliability and the failure intensity for failure-truncated testing. Their Monte-Carlo simulation results show that the procedure is more accurate and efficient than that of ML, even for vague prior information. Years later, they present [128] a nonparametric Bayes-decision framework for complex repairable systems.

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C.7.19 Meth's OSD Perspective on Reliability Growth (1992).
Meth, who at the time was Director of the Weapons Support Improvement Group of the OSD, gives a critical review [101] of reliability growth "myths and methodologies." He asserts that reliability prediction is not a reasonable application of reliability growth and that the various mathematical models may not adequately describe the reliability growth process. He conjectures that understanding of the factors for test planning has not advanced beyond the rules-of-thumb that were initially proposed by Duane [7] in 1964. Meth also challenges the reliability community to "re-examine the reliability growth concept" and how it is being applied.

C.7.20 Demko on Non-Linear Reliability Growth (1993).
Demko [112] identifies a shortcoming to the Duane model [7], namely, that it is insensitive to discontinuities or sudden changes in the reliability growth trend for a system. In other words, Duane's model only considers linear growth on a log-log scale and will not accurately portray non-linear growth (on the same scale). Demko proposes to utilize non-linear, piecewise regression to overcome this shortcoming. Several numerical examples and plots are given to illustrate comparisons of Duane's approach versus that of the proposed. The examples use datasets from programs that demonstrated non-linear growth patterns and show that the proposed method more accurately portrays the growth patterns.

C.7.21 Farquhar and Mosleh on Growth Effectiveness (1995).
Farquhar and Mosleh [124] present an approach for quantifying reliability growth effectiveness. In their approach, they develop a performance parameter, which they present from two perspectives on whether data from reliability growth testing is available or not. If data is not available, a subjective assessment and characterization of attributes that are indicative of the corporate culture is used. When data are available, the parametric variable is quantified by normalizing past performance with reliability growth program goals. Five case studies are utilized to develop the performance parameter. It is then incorporated into an existing reliability growth model, known as the Tracking, Growth and Prediction model. This model was developed by P. F. Verhulst in 1845 and is based on the logistic function characterized by an S-shaped curve. They conclude that their modification to the model provides a conservative estimate of the risk involved in achieving reliability growth goals.

C.7.22 Demko on Reliability Growth Testing (1995).
Demko [123] argues that certain types of testing are not adequate in exposing field-related failure modes. Some of the types of testing mentioned includes: RDGT, EQT, and ESS tests. He claims that these tests yield a high percentage of failure modes that occur only in a chamber-type environment and are not representative of failure modes that would be encountered during field use. Failure modes from over 2 million hours of field data from 13 different types of "283 Avionics Units" are compared against failure modes identified by 5 different companies who performed either RDGT (21K hours), ESS tests (28K hours) or EQT (hours not given). Several plots are given to compare the quantity of failure modes encountered in the field, and in each type of test. The results indicate that the majority of failure modes were found during field use, following ESS testing, RDGT, and EQT.

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C.7.23 Fries-Maillart Stopping Rules (1996).
Fries and Maillart [125] present a method of when to stop testing of a one-shot system when the number of systems to be produced is predetermined and the probabilities of identifying and successfully correcting each failure mode are less than one. The stopping criterion is focused on maximizing the number of systems expected to perform successfully in the field after deployment of the lot. Two rules are presented. The first rule includes stopping the test when the estimated utility[19] (given a failure on the next trial) is less than or equal to the current utility estimate. Motivated by expected value, the second rule is to stop testing when the estimated utility after the next trial (regardless of its outcome) is less than or equal to the current utility estimate. Four discrete reliability growth models are utilized to estimate reliability improvement via Monte-Carlo simulation. The models include: two Lloyd-Lipow models [4], Fries' learning-curve model [111], and Virene's Gompertz model [16]. The results indicate that both stopping rules perform well and can be practically implemented. Specific recommendations are given to implement test-stopping rules in light of several factors, such as, estimation methodology and lot size.

C.7.24 Ebrahimi's MLE for the NHPP (1996).
Ebrahimi [129] develops a general formulation for modeling reliability growth between design modifications. He assumes the model is either a Poisson process or the NHPP power-law process, and the times of design modifications must be known. ML estimates and CI procedures are developed in two cases, depending on the presence of constraints on the system failure intensity. His proposed MLE and CI procedures are illustrated via a limited Monte-Carlo example. Corrections to several typographical and computational errors in Ebrahimi's paper is given years later (i.e., 2002) by Pulcini in [149].

C.7.25 Huang-McBeth-Vardeman One-Shot DT Programs (1996).
Huang, McBeth, and Vardeman [130] develop a method to efficiently conduct developmental testing of one-shot systems that are destroyed in testing upon first use. Dynamic programming is used to identify optimal test-plans that maximize the mean number of effective systems of the final design that can be purchased with the remaining budget. Several suboptimal rules are also considered and their performances are compared to that of the optimal rule.

C.7.26 Xie-Zhao Monitoring Approach (1996).
Xie and Zhao [131] introduce a method called the First-Model-Validation-Then-Parameter-Estimation approach. Their approach, which essentially follows directly from Duane [7], consists of model validation and parameter estimation. The model is validated by plotting the cumulative number of failures versus test duration, using linear regression to derive an equation to fit the data, and calculating the associated correlation coefficient. A subjective assessment of the magnitude of the correlation is the deciding factor on model validation. If the model is reasonable, the equation can be used for prediction purposes. The paper focuses a great deal on the need for more graphical models in reliability growth.

C.7.27 Seglie's OSD Perspective on Reliability Growth (1998).
Written from the perspective of the Chief Scientific Advisor to the Director of Operational Test and Evaluation, OSD, Seglie [134] argues that too many weapon system programs enter

[19] ***Utility*** is defined as the number of systems expected to perform successfully in the field after deployment of a lot.

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operational testing before they are ready (i.e., they have an immature design margin with respect to reliability and ultimately fail their test objectives at great cost). Seglie emphasizes the false predictions that can be reported from the wealth of reliability growth models available, and notes that growth models have demonstrated a poor history of successfully predicting field reliability. He proposes that the role of reliability growth modeling should be focused on prescribing test duration required to reach a level of acceptable reliability before going into operational testing – and not on estimating system or subsystem reliability. He adds that this modest role of reliability growth methodology to develop test plans in determining the amount of test time required for engineers to find the dominant failure modes, analyze them, develop and implement corrective actions, and confirm their fix-effectiveness is still of great importance. In his overall view, the most important role of reliability growth should be to provide information to help programs succeed during test and evaluation. To better do this, he suggests that growth models need to: account for the effects of different environments, be system specific, and be more engineering based.

C.7.28 Hodge-Quigley-James-Marshal Framework (2001).
While no technical details are provided, Hodge et al. [148] discuss a modeling framework that aims to support reliability enhancement decision-making. The main objectives of their approach are to: develop a methodology to support reliability enhancement throughout the design process, and develop a model, referred to as the Reliability Enhancement Methodology Model (REMM), that facilitates the assessment of reliability throughout the product lifecycle. REMM is basically a tracking system to determine how reliability evolves throughout the design process / lifecycle by integrating statistical and engineering understanding of reliability performance. The primary outputs of REMM include point and CI estimates for: product reliability, probability of failure per unit time, and the probability of failure free periods. A list of engineering design concerns (i.e., failure modes, corrective actions) on a given product is also provided. The methodology was implemented by TRW Aeronautical Systems (Lucas Aerospace) in 2001.

C.7.29 Crow's Methods to Reduce LCC (2003).
Crow [150] develops methods for estimating the useful life of a fleet of repairable systems and presents a model for addressing in-service reliability growth. A minimum LCC model and associated MLE procedures are also developed. The methodology is fundamentally based on the NHPP power-law process, originally developed by Crow in 1974 [26]. A numerical example is given for 11 simulated systems to illustrate the LCC methodology. The in-service reliability growth approach follows from the ACPM [54] with a slight modification where an average FEF is utilized (i.e., rather than the individual FEF). All previously reported MLE and GOF procedures apply. A numerical example is given on simulated data for cases with and without the prevalence of wear-out.

C.7.30 Gurunatha-Siegel Six-Sigma Process (2003).
Gurunatha and Siegel [151] formulate 12-step six-sigma quality process for an unspecified complex commercial product developed by Xerox Corporation. As a result of implementing this process, they claim that the company achieved a reliability growth rate that exceeded that of any other program within their corporate history. The 11-step process includes: (1) material selection optimized for reliability and cost, (2) failure mode identification and physics of failure analysis, (3) total LCC calculated for each component, (4) ALT performed with lifetime predictions, (5)

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accuracy measurement on product and process capability, (6) identify critical parameters and their percent contribution to survival and failure, (7) Monte-Carlo simulation on critical parameter uncertainty, (8) design-of-experiments on reliability characteristics, (9) subsystem design modifications to extend product life, (10) statistical process-control for process improvement and, (11) use of lessons-learned for improved reliability growth processes. No quantitative details are provided on the claimed reliability growth achievement of the product.

C.7.31 Yadav-Singh-Goel Approach (2003).
Yadav, Singh, and Goel [152] propose a two-stage model of system reliability growth that they develop with consideration to associated components, functions, and failure modes. The first stage consists of development of a reliability growth plan to achieve program requirements. The second stage of the framework involves a strategy to further improve the system reliability prediction following demonstration of its requirements. The prediction is decomposed by component and a prioritization index is defined to provide a rank order of components based on their potential for improving the accuracy of the system-level reliability prediction. A series system configuration is assumed, and the reliability requirement is allocated equally over all components. A gamma prior distribution is utilized under a Poisson sampling routine, which results in the typical gamma posterior for the distribution of component failure rates. Improvements in the associated system-level reliability prediction are improved by a variance reduction strategy on the component gamma posterior distributions. Methodologies for test cost estimation, and reliability improvement prioritization are given. A numerical example on a hydraulic power rack-and-pinion steering system is presented to demonstrate the proposed two-stage model.

C.7.32 Quigley-Walls CI Procedures (2003).
Quigley and Walls [153] develop inference properties for reliability growth analysis. They assume a Poisson prior distribution for the ultimate number of faults that would be exposed in the system if testing were to continue ad infinitum. Although, they estimate the parameters of the system failure intensity function empirically. Bias and conditions of existence of fixed-point iteration MLE procedures are investigated. The intention of the approach is to support reliability inference in situations where failure data are sparse. Their statistical CI procedures are shown to be suitable for small sample sizes, and are demonstrated by a numerical example.

C.7.33 Smith on Planning (2004).
Smith [158] describes a process for planning and estimating the cost of a reliability growth program under a Performance Based Logistics (PBL) contract. Under this type of contract, a supplier is typically responsible for structuring their reliability and support programs around a defined field availability or reliability goal. This planning process was developed with the intent to minimize cost and performance risks in the execution of a long-term PBL support contract for a complex, repairable system. The process mainly includes a detailed assessment of five areas: expected volume of field usage (e.g., flight hours, mileage), a well-defined field reliability definition, estimates of current field reliability, the goal or future requirement for field reliability, and a schedule for reliability achievement.

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C.7.34 Krasich and Quigley on the Design Phase (2004).
Krasich and Quigley [159] discuss how there has been significantly less attention on reliability growth during the design phase (i.e., most of the literature is developed for growth during the TAFT process). They propose two models that could be utilized to assess reliability growth during design. The first model is a modification of Crow's RGTMC [30], and the second is a modification to Rosner's IBM model [3]. The data required for these models includes: the reliability requirement, a subjective assessment of the initial reliability of the system, an estimate of the number of design modifications, the mean number of faults in the initial design, and an estimate of the effectiveness in mitigating design faults. They indicate that the modified RGTMC is more appropriate when the design activities and modifications are equally spaced and well-planned. Otherwise, in more uncertain situations, the modified IBM model is deemed to be most suitable. The modified RGTMC is illustrated by numerical example to an unspecified industry example.

C.7.35 Mortin-Ellner Paper (2005).
Mortin and Ellner [161] address some of the advancements in reliability growth methodology (offered by AMSAA), and also highlight remaining challenges and areas in the field requiring further development. Some of the reliability growth planning models discussed include: SPLAN [144], PM2 [162], the Threshold Program [163], and SSPLAN [117]. A number of tracking models (e.g., Duane [7], RGTMC [30], RGTMD [55], and SSTRACK [126]) and projection models (e.g., AMPM [121], AMPM-Stein [162], ACPM [54]) are also discussed. One of the remaining challenges in reliability growth that they mention is that, "more projection and tracking methodology needs to be developed for cases where the events are measured on a discrete scale rather than on a continuous basis (e.g., single-shot devices such as missiles).

C.7.36 Acevedo-Jackson-Kotlowitz Application (2006).
Acevedo, Jackson and Kotlowitz [165] discuss how reliability growth achievement can be realized by using a well-educated ALT program. Two product case studies are presented to show how Lucent Technologies performs ALT on critical hardware subsystems used in telecommunication systems. The hardware items studied include: an RF power amplifier module, and radio unit. ALT is used to identify product weaknesses leading to performance degradation over simulated operational lifetimes. Weaknesses are corrected through design changes prior to manufacturing and field deployment. Forecasts for the steady-state product reliability under expected field conditions are given. Crow's NHPP Weibull process [26] is used to estimate steady-state reliability under Type I censoring (time-truncated), which is consistent with the conduct of their ALT. The importance of the paper is how ALT can be incorporated into a reliability growth program, as well as how common reliability growth models can be utilized for assessment purposes when test data is obtained under accelerated conditions.

C.8 Reliability Growth Software.

C.8.1 U.S. AMSAA.
U.S. AMSAA [168] is the first organization to develop a software product for reliability growth analysis. The software program, called the "AMSAA Visual Growth Suite (VGS) can be requested online at AMSAA's website: http://www.amsaa.army.mil/ReliabilityTechnology/VGRequest.htm

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C.8.2 ReliaSoft Corporation.
Founded in 1992, ReliaSoft Corporation [169] performs research and development in the area of reliability engineering and provides a wide array of software products that include the latest theoretical advances within the field. ReliaSoft's major products include: Weibull++ for statistical life data analysis, ALTA for quantitative accelerated life testing data analysis, BlockSim for RBD and FTA, XFMEA for FMEA and FMECA, RCM++ for analysis, data management and reporting of reliability centered maintenance, RENO for probabilistic event and risk analysis, Lambda Predict for reliability prediction analysis, XFRACAS as a failure reporting and corrective action system, MPC 3 for aircraft systems and power-plant analysis, and RGA for reliability growth analysis. The RGA software package includes the following reliability growth models: ACPM [54], Crow-Extended [157], Duane [7], Gompertz [16], Modified Gompertz [62], Logistic [1], Lloyd-Lipow [4] and RGTMC [30].

C.8.3 Relex Corporation.
Relex Corporation [170], founded in 1986, also offers several different kinds of software products for reliability including FTA, FMEA, FMECA, FRACAS, Human Factors Risk Analysis, LCC, Maintainability Prediction, Markov Analysis, Optimization and Simulation, RBD, Reliability Prediction, and Weibull Analysis. While Relex does not have a software product exclusively for reliability growth analysis, some reliability growth tools are offered in the Relex®Weibull software package. The main focus of this product is Weibull analysis but it also offers some reliability growth capabilities; such as, evaluating statistics for reliability growth planning, constructing reliability growth planning curves based on the Duane model [7], and performing reliability growth tracking analyses via the RGTMC model [30].

C.9 Summary.
The preceding sections provide a synopsis of the research that has been done in the field of reliability growth (for hardware). There are three main areas in the field including: planning, tracking, and projection. A wide array of statistical procedures (e.g., MLE and Bayesian point-estimation, CI construction, and GOF tests) is available for many of the proposed methodologies. Models are available for complex systems whose test duration is continuous, as well as for complex systems whose test duration is discrete. This summary covers at least 7 planning models, 25 tracking models, 6 projection models, 4 main comprehensive works on reliability growth, and several dozen related papers covering theoretical results, simulation studies, real-world applications, personal-perspectives, international standards, or related statistical procedures.

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C.10 REFERENCES

[1] H. K. Weiss, "Estimation of Reliability Growth in a Complex System with Poisson-Type Failure," *Operations Research*, vol. 4, num. 5, pp. 532-545,1956.
[2] M. Aroef, "Study of Learning Curves of Industrial Manual Operations," Unpublished Master's Thesis, *Cornell University*, 1957.
[3] N. Rosner, "System Analysis - Nonlinear Estimation Techniques," *Proceeding of the National Symposium on Reliability and Quality Control*, 1961, pp. 203-207.
[4] D. K. Lloyd and M. Lipow, "Reliability Growth Models," *Reliability: Management, Methods, and Mathematics, Prentice-Hall Space Technology Series*, Englewood Cliffs, New Jersey: Prentice-Hall Inc., 1962, pp. 330-348.
[5] H. Chernoff and W. M. Woods, "Reliability growth models-Analysis and applications," *CEIR*, Los Altos, CA, Field Memo, Feb. 1962.
[6] W. Wolman, "Problems in System Reliability Analysis," *Statistical Theory of Reliability*, Marvin Zelen Ed., Madison, Wisconsin: University of Wisconsin Press, 1963, pp. 149- 160.
[7] J. T. Duane, "Learning Curve Approach To Reliability Monitoring," *IEEE Transactions on Aerospace*, vol. 2, pp. 563-566, 1964.
[8] W. J. Corcoran, H. Weingarten and P. W. Zehna, "Estimating Reliability after Corrective Action," *Management Science*, vol. 10, num. 4, pp. 786-798, 1964.
[9] J. E. Bresenham, "Reliability growth models," Dept of Statistics, *Stanford University*, TR 74, Contract Number NONR225 53, AD-605 993, 1964.
[10] NASA, "Reliability growth of US rockets," *Space Technology Labs Inc*, vol. II, Contract NAS 8-11037, 1964. (SECRET).
[11] D. R. Cox and P. A. Lewis, *The Statistical Analysis of Series of Events*. New York: NY: John Wiley and Sons, 1966.
[12] R. E. Barlow and E. M. Scheuer, "Reliability Growth during a Development Testing Program," *Technometrics*, vol. 8, num. 1, pp. 53-60, 1966.
[13] W. J. Corcoran and R. R. Read, "Modeling the Growth of Reliability - Extended Model," *United Technology Center*, Sunnydale, CA, UTC 2140-ITR, 1966.
[14] W.J. Corcoran and R.R. Read, "Comparison of some reliability growth estimation and prediction schemes," *United Technology Center*, Sunnydale, CA, UTC 2140-ITR, 1967.
[15] A. J. Gross, M. Kamins, "Reliability assessment in the presence of reliability growth," *RAND Corporation*, Memo RM-5346-PR, Sep. 1967. Also *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1968, pp. 406-416.
[16] E. P. Virene, "Reliability Growth and the Upper Reliability Limit, How to Use Available Data to Calculate," *Boeing Company*, Seattle, Washington, Report No. D2-24142-1, , AD- 804 150, pp. 33, 1966. Also "Reliability Growth and Its Upper Limit," *IEEE Proceedings of the Annual Symposium on Reliability*, 1968, pp. 265-270.
[17] J. Cozzolino, "Probabilistic Models of Decreasing Failure Rate Processes," *Naval Research Logistics Quarterly*, vol. 15, 1968, pp. 361-374.
[18] E. O. Codier, "Reliability Growth in Real Life," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1968, pp. 458-469.
[19] S. M. Pollock, "A Bayesian reliability growth model," *IEEE Trans. Reliability*, vol. R-17, num. 4, pp. 187-198, Dec. 1968.
[20] J. D. Selby and S. G. Miller, "Reliability Planning and Management (RPM)," *Proceedings ASQC/SRE Seminar*, Milwaukee, WS: ASQC, 1970.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

[21] D. R. Barr, “A class of general reliability growth models,” *Operations Research*, vol. 18, num. 1, pp. 52-65, 1970.
[22] R. R. Read, “A remark on the Barlow-Scheuer reliability growth estimation scheme,” *Technometrics*, vol. 13, num. 1, pp. 199-200, 1971.
[23] L. H. Crow, “Reliability Growth Modeling,” *AMSAA*, Aberdeen Proving Ground, MD, Tech. Rep. TR-55, 1972.
[24] L. H. Crow, “AMSAA Reliability Growth Symposium,” *AMSAA*, Aberdeen Proving Ground, MD, Interim Note 22, 1972.
[25] L. H. Crow, "Reliability Analysis for Complex, Repairable Systems in Reliability and Biometry," *SIAM*, Proschan and R. J. Serfling Eds., Philadelphia, PA, 1974, pp. 379-410.
[26] L. H. Crow, “Reliability Analysis for Complex, Repairable Systems,” *AMSAA*, Aberdeen Proving Ground, MD, Interim Note 26, 1974.
[27] L. H. Crow, “On Reliability Growth Modeling,” *AMSAA*, Aberdeen Proving Ground, MD, Interim Note 27, 1974.
[28] L. H. Crow, “On Reliability Growth Tracking,” *AMSAA*, Aberdeen Proving Ground, MD, Interim Note R-30, 1974.
[29] L. H. Crow, “Reliability Growth Estimation from Failure and Time Truncated Testing,” *AMSAA*, Aberdeen Proving Ground, MD, Interim Note R-31, 1974.
[30] L. H. Crow, "On Tracking Reliability Growth," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1975, pp. 438-443.
[31] Rome Air Development Center, "Reliability Growth Study," *Air Force Systems Command*, Griffiss Air Force Base, NY, RADC-TR-75-253, 1975.
[32] L. H. Crow, “Reliability Analysis of Complex Repairable Systems,” *AMSAA*, Aberdeen Proving Ground, MD, Tech. Rep. 138, 1975.
[33] P. Lewis and G. Shedler, "Statistical Analysis of Non-Stationary Series of Events," *IBM Journal of Research and Development*, vol. 20, pp. 465-482, 1976.
[34] T. Jayachadran and L. R. Moore, “A comparison of reliability growth models,” *IEEE Trans. Reliability*, vol. R-25, pp. 49-51, 1976.
[35] H. Braun and J. M. Paine, “A comparative study of models for reliability growth,” Dept. of Statistics, *Princeton University*, Tech. Rep. 126, ser 2, Jul. 1977.
[36] D. E. Olsen, “Estimating reliability growth,” *IEEE Trans. Reliability*, vol. R-26, pp. 50-53, Apr. 1977.
[37] R. C. Dahiya, “Estimating reliability after corrective action,” *IEEE Trans. Reliability*, vol. R-26, pp. 348-351, Dec. 1977.
[38] A. F. M. Smith, “A Bayesian note on reliability growth during a developmental testing program,” *IEEE Trans. Reliability*, vol. R-26 pp. 346-347, Dec. 1977.
[39] A. J. Bonis, “Reliability growth curves for one-shot devices,” *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1977, pp. 181-185.
[40] L. H. Crow, “Confidence Interval Procedures for Reliability Growth Analysis,” *AMSAA*, Aberdeen Proving Ground, MD, Tech. Rep. 197, 1977.
[41] H. S. Balaban, “Reliability growth models,” *J. Environ. Sciences*, vol. 21, pp. 11-18, 1978.
[42] D.E. Olsen, “A Confidence Interval for the Barlow-Scheuer Reliability Growth Model,” *IEEE Trans. Reliability*, vol. R-27, pp. 308-310, Dec. 1978.
[43] M. C. Weinrich and A. J. Gross, “The Barlow-Scheuer Reliability Growth Models from a Bayesian Viewpoint,” *Technometrics*, vol. 20, num. 3, pp. 249-254, 1978.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

[44] G. L. Sriwastav, “A Reliability Growth Model,” *IEEE Trans. Reliability*, vol. R-27, pp. 306-307, Dec. 1978.
[45] N. D. Singpurwalla, “Estimating reliability growth (or deterioration) using time series analysis,” *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1978, pp. 111-114.
[46] W. M. Woods, “Reliability growth models,” *Avionics Reliability, Its Techniques and Related Disciplines*, num. 261M. C. Jacobsen, ED, NATO, Seine; Advisory Group for Aerospace Research & Development Conf. Proc, 1979.
[47] N. A. Langberg and F. Proschan, "A Reliability Growth Model Involving Dependent Components," *The Annals of Probability*, vol. 7, num. 6, pp. 1082-1087, 1979.
[48] B. S. Dhillon, “Reliability growth: A survey,” *Microelectronics & Reliability*, vol. 20, pp. 743-751, 1980.
[49] U.S. Department of Defense, Reliability Growth Management, “Military Handbook 189,” *Naval Publications and Forms Center*, Philadelphia, 1981.
[50] C. M. Stein, “Estimation of the Mean of a Multivariate Normal Distribution,” *The Annals of Statistics*, vol. 9, pp. 1135-1151, 1981.
[51] B. Littlewood, “Stochastic Reliability-Growth: A Model for Fault-Removal in Computer-Programs and Hardware-Designs,” *IEEE Trans. Reliability*, vol. R-30, num. 4, 1981.
[52] W. Nelson, *Applied Life Data Analysis.* New York, NY: John Wiley & Sons, 1982.
[53] L. H. Crow, "Confidence Interval Procedures for the Weibull Process with Applications to Reliability Growth," *Technometrics*, vol. 24, num.1, pp. 67-72, 1982.
[54] L. H. Crow, “An Improved Methodology for Reliability Growth Projection,” *AMSAA*, Aberdeen Proving Ground, MD, Tech. Rep. TR-357, 1982.
[55] L. H. Crow, "AMSAA Discrete Reliability Growth Model," *AMSAA*, Aberdeen Proving Ground, MD, Office Note 1-83, 1983.
[56] L. H. Crow, "Reliability Growth Projection from Delayed Fixes," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1983, pp. 84-89.
[57] L. H. Crow, "Methods for Reliability Growth Assessment During Development," *In Electronic Systems Effectiveness and Life Cycle Testing*, J. K. Skwirzynski Ed.: Springer-Verlag, 1983.
[58] N. S. Fard and D. L. Dietrich, “Comparison of Attribute Reliability Growth Models,” *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1983, pp. 24-29.
[59] J. M. Finkelstein, “A logarithmic reliability-growth model for single mission systems,” *IEEE Trans. Reliability*, vol. R-32, pp. 508-511, Dec. 1983.
[60] P. B. Trapnell, "Study on Reliability Fix Effectiveness Factors for Army Systems,” *AMSAA*, Aberdeen Proving Ground, MD, Tech. Rep. 388, 1984.
[61] P. B. Trapnell, "Update of Reliability Fix Effectiveness Factors for Army Systems,” *AMSAA*, Aberdeen Proving Ground, MD, Interim Note R-86, 1984.
[62] S. Jiang, D. B. Kececioglu and P. Vassiliou, "Modified Gompertz," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1994.
[63] L. H. Crow, "Methods for Assessing Reliability Growth Potential," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1984, pp. 484-489.
[64] “Reliability growth testing effectiveness,” Rome Air Development Center Report, RADCTR- 84-20, Jan. 1984.
[65] W. S. Jewell, "A General Framework for Learning Curve Reliability Growth Models," *Operations Research*, vol. 32, num. 3, pp. 547-558, 1984.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

[66] D. K. Lloyd, "Forecasting Reliability Growth," *IEEE Quality and Reliability Engineering International*, pp. 19-23, 1986.
[67] D. Miller, “Exponential Order Statistic Models of Software Reliability Growth,” *IEEE Trans. Software Engineering*, vol. SE-12, num. 1, 1986.
[68] R. K. Gates, G. J. Gibson, K.K. McLain, “Reliability growth prediction,” Rome Air Development Center Report, RADC-TR-86-148, Sept. 1986.
[69] R. B. D’Agostino and M. A. Stephens, “Tests of Chi-Squared Type,” in *Goodness-of-Fit Techniques*, New York, NY: Marcel Dekker, 1986, pp. 93-93.
[70] W. P. Clay, "Maximum Likelihood Estimator for the Discrete Application of the AMSAA Growth Model," *AMSAA*, Aberdeen Proving Ground, MD, Tech. Rep. 433, 1987.
[71] Department of Defense, Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production “Military Handbook 781,” *Naval Publications and Forms Center, Philadelphia*, 1987.
[72] D. K. Lloyd, “Forecasting reliability growth,” *Proceedings of the 33rd Annual Technical Meeting*, 1987, pp. 171-175.
[73] N. S. Fard, D. L. Dietrich, “A Bayes reliability growth model for a developmental testing program,” *IEEE Trans. Reliability*, vol. R-36, pp. 568-572, Dec. 1987.
[74] L. H. Crow, "Reliability Growth Estimation With Missing Data - II," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1988, pp. 248-253.
[75] W. M. Woods, J. E. Drake, J.D. Chandler Jr., “Analysis and evaluation of discrete reliability growth models with and without failure discounting,” *U.S. Naval Postgraduate School*, Tech. Rep. NPS55-88-013, Dec. 1988.
[76] P. M. Ellner and T. Ziad, “Statistical Precision and Robustness of the AMSAA Continuous Reliability Growth Estimators,” *AMSAA*, Aberdeen Proving Ground, MD, Tech. Rep. 453, 1988. Also *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1990, pp. 210-214.
[77] J. C. Wronka, "Tracking of Reliability Growth in Early Development,” *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1988, pp. 238-242.
[78] D. Robinson and Duane Dietrich, "A System-Level Reliability Growth Model," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1988, pp. 243-247.
[79] K. L. Wong, "Reliability Growth - Myth or Mess," *IEEE Trans. Reliability*, vol. 37, num. 2, p. 209, 1988.
[80] H. A. Malec, "Reliability Growth from the Customer's Perspective," *IEEE Journal on Selected Areas of Communications*, vol. 6, num. 8, pp. 1287-1293, 1988.
[81] A. W. Benton, "Integrated Reliability Growth Testing," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1989, pp. 160-166.
[82] D. G. Frank, "A corollary to: Duane's postulate on reliability growth," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1989, pp. 167-170.
[83] G. J. Gibson and L. H. Crow, "Reliability Fix Effectiveness Factor Estimation," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1989, pp. 171-177.
[84] G. K. Bhattacharyya, A. Fries, R. A. Johnson, “Properties of continuous analog estimators for a discrete reliability-growth model,” *IEEE Trans. Reliability*, vol. 38, pp. 373-378, Aug. 1989.
[85] D. Robinson and Duane Dietrich, "A Nonparametric-Bayes Reliability Growth Model," *IEEE Trans. Reliability*, vol. 38, num. 5, pp. 591-598, 1989.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

[86] S. E. Rigdon, “Testing goodness-of-fit for the power law process,” *Communications in Statistics: Theory and Methods*, vol. 18, pp. 4665-4676, 1989.
[87] L. H. Crow, "Evaluating the Reliability of Repairable Systems," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1990, pp. 275-279.
[88] W. M. Woods, “The effect of discounting failures and weighting data on the accuracy of some reliability growth models,” *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1990, pp. 200-204.
[89] P. M. Ellner and B. Trapnell, “AMSAA Reliability Growth Data Study,” *AMSAA*, Aberdeen Proving Ground, MD, Interim Note R-184, 1990.
[90] S. Kaplan, G. D. M. Cunha, A. A. Dykes and D. Shaver, "A Bayesian Methodology for Assessing Reliability during Product Development," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1990, pp. 205-209.
[91] P. E. Higgins and A. Constantinides, "Reliability Growth for Expendable Mobile ASW Training Target (EMATT): A 1-Shot Device," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1991, pp. 497-501.
[92] T. A. Mazzuchi and R. Soyer, "A Bayesian Attribute Reliability Growth Model," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1991, pp. 322-325.
[93] L. H. Crow, "New International Standards on Reliability Growth," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1991, pp. 478-480.
[94] G. Coolas, "Dynamic Reliability Prediction: How to Adjust Modeling and Reliability Growth?," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1991, pp. 301-306.
[95] J. Bieda, "Reliability Growth Test Management in a Product Assurance Environment within the Automotive Component Industry," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1991, pp. 317-321.
[96] Kececioglu, Dimitri, B., "Reliability Growth," *Reliability Engineering Handbook*, 4th ed., vol. 2, Englewood Cliffs, New Jersey: Prentice-Hall, 1991, pp. 415-418.
[97] H. F. Martz and R. A. Waller, “Selecting Beta Prior Distributions based on Observed Binomial Sampling Data,” *Bayesian Reliability Analysis*, 2nd ed., Malabar, FL: Krieger, 1991, pp. 312-318.
[98] G. K. Bhattacharyya, J. K. Ghosh, “Estimation parameters in a discrete reliability growth model,” *Journal for Statistics Planning and Inference*, vol. 29, pp. 43-53, 1991.
[99] R. A. Johnson, “Confidence estimation with discrete reliability growth models,” *Journal for Statistics Planning and Inference*, vol. 29, pp. 87-97, 1991.
[100] P. M. Ellner and R. Mioduski, "Operating Characteristic Analysis for Reliability Growth Programs,” *AMSAA*, Aberdeen Proving Ground, MD, Tech. Rep. 524, 1992.
[101] M. A. Meth, “Reliability-growth myths and methodologies: A critical view,” *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1992, pp. 337-342.
[102] P. M. Ellner, D. E. Mortin and D. M. Querido, “Developing a Subsystem Reliability Growth Program using the Subsystem Planning Model,” *AMSAA*, Aberdeen Proving Ground, MD, Division Note R-24, 1992.
[103] R. Calabria, M Guida and G. Pulcini, "A Bayes Procedure for Estimation of Current System Reliability," *IEEE Trans. Reliability*, vol. 41, num. 4, pp. 616-620, 1992.
[104] K. E. Ellis, "Robustness of Reliability Growth Analysis Techniques," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1992, pp. 303-315.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

[105] D. I. Heimann and W. D. Clark, "Process-Related Reliability Growth Modeling - How and Why," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1992, pp. 316-321.
[106] T. A. Mazzuchi and R. Soyer, "Reliability Assessment and Prediction during Product Development," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1992, pp. 468-474.
[107] W. J. Park and Y. G. Kim, "Goodness-of-fit test for the power-law process," *IEEE Trans. Reliability*, vol. 41, pp. 107-111, 1992.
[108] B. Klefsjö and U. Kumar, "Goodness-of-Fit Test for the Power-Law Process based on the TTT-Plot," *IEEE Trans. Reliability*, vol. 41, pp. 593-598, 1992.
[109] L. H. Crow, "Confidence Intervals on the Reliability of Repairable Systems," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1993, pp. 126-134.
[110] T. A. Mazzuchi, R. Soyer, "A Bayes Method for Assessing Product Reliability during Development Testing," *IEEE Trans. Reliability*, vol. 42, pp. 503-510, Sep. 1993.
[111] A. Fries, "Discrete Reliability-Growth Models on Learning-Curve Property," *IEEE Trans. Reliability*, vol. 42, pp. 303-306, Jun. 1993.
[112] E. Demko, "One Non-linear Reliability Growth," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1993, pp. 411-416.
[113] P. E. Higgins, "Final Reliability Growth Results for EMATT, A 1-Shot Device," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1993, pp. 399-403.
[114] M. Xie and M. Zhao, "On Some Reliability Growth Models with Graphical Interpretations," *Microelectronics and Reliability*, vol. 33, pp. 149-167, 1993.
[115] D. Kececioglu, S. Jiang, P. Vassiliou, "The Modified-Gompertz Reliability Growth Model," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1994, pp. 160-165.
[116] A. Fries, "Correction- Discrete Reliability Growth Models based on a Learning-Curve Property," *IEEE Trans. Reliability*, vol. 43, p.475, Sep. 1994.
[117] P. M. Ellner, M. J. McCarthy, D. E. Mortin and D. M. Querido, "Developing a Subsystem Reliability Growth Program using SSPLAN," *AMSAA*, Aberdeen Proving Ground, MD, Tech. Rep. 555, 1994.
[118] L. H. Crow, P. H. Franklin and N. B. Robbins, "Principles of Successful Reliability Growth Applications," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1994, pp. 157-159.
[119] W. J. Park and M. Seoh, "More Goodness-of-Fit Tests for the Power Law Process," *IEEE Trans. Reliability*, vol. 43, pp. 275-278, 1994.
[120] International Electrotechnical Commission "Reliability Growth - Statistical test and estimation methods," *IEC International Standard*, IEC 1164, 1995.
[121] P. M. Ellner and L. C. Wald, "AMSAA Maturity Projection Model," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1995, pp. 174-181.
[122] I. Fakhre-Zakeri and E. Slud, "Mixture Models for Reliability of Software with Imperfect Debugging: Identifiability of Parameters," *IEEE Trans. Reliability*, vol. R-44, num. 1, pp. 104-113, 1995.
[123] E. Demko, "On Reliability Growth Testing," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1995, pp. 162-165.
[124] K. J. Farquhar and A. Mosleh, "An Approach to Quantifying Reliability Growth Effectiveness," *IEEE Proceedings of the Annual Reliability and Maintainability*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

Symposium, 1995, pp. 166-173.
[125] A. Fries and L. M. Maillart, "TAAF (Test-Analyze-And-Fix) Stopping Rules for Maximizing the Operational Utility of One-Shot Systems," *Institute of Defense Analysis*, Arlington, VA, IDA Document D-1880, 1996.
[126] W. J. Broemm, "Subsystem Tracking Model (SSTRACK)," *AMSAA*, Aberdeen Proving Ground, MD, Briefing Chart Rep., 1996.
[127] A. Fries and A. Sen, "A Survey of Discrete Reliability Growth Models," *IEEE Trans. Reliability*, vol. R-45, num. 4, pp. 582-604, 1996.
[128] R. Calabria, M. Guida and G. Pulcini, "A Reliability Growth Model in a Bayes-Decision Framework," *IEEE Trans. Reliability*, vol. 45, num. 3, pp. 505-510, 1996.
[129] N. Ebrahimi, "How to Model Reliability Growth when Times of Design Modifications are Known," *IEEE Trans. Reliability*, vol. 45, num. 1, pp. 54-58, 1996.
[130] M. Y. Huang, D. McBeth and S. B. Vardeman, "Developmental Test Programs for 1 Shot Systems: 2-State Reliability and Binary Development-Test Results," *IEEE Trans. Reliability*, vol. 45, num. 3, pp. 379-385, 1996.
[131] M. Xie and M. Zhao, "Reliability Growth Plot - An Underutilized Tool in Reliability Analysis," *Microelectronics Reliability*, vol. 36, num. 6, pp. 797-805, 1996.
[132] R. D. Baker, "Some new tests of the power law process," *Technometrics*, vol. 38, pp. 256-265, 1996.
[133] D. P. Gaver, and P. A. Jacobs, "Testing or fault-finding for reliability growth: A missile destructive-test example," *Naval Research Logistics*, vol. 44, pp. 623-637, 1997.
[134] E. A. Seglie, "Reliability Growth Programs from the Operational Test and Evaluation Perspective," *Reliability Growth Modeling: Objectives, Expectations, and Approaches*, K. J. Farquar and A. Mosleh Eds., College Park, MD: The Center for Reliability Engineering, University of Maryland, 1998.
[135] A. Erkanli, T. A. Mazzuchi and R. Soyer, "Bayesian Computations for a Class of Reliability Growth Models," *Technometrics*, vol. 40, num. 1, pp. 14-23, 1998.
[136] A. Sen, "Estimation of Current Reliability in a Duane-based Reliability Growth Model," *Technometrics*, vol. 40, num. 4, pp. 334-344, 1998.
[137] E. Crétois and O. Gaudoin, "New Results on Goodness-of-Fit Tests for the Power-Law Process and Application to Software Reliability," *International Journal of Reliability, Quality & Safety Engineering*, vol. 5, pp. 249-267, 1998.
[138] O. Gaudoin, "CPIT Goodness-of-Fit Test for the Power-Law Process," *Communications in Statistics: Theory and Methods*, vol. 27, pp. 165-180, 1998.
[139] J. A. Clark, "Modeling Reliability Growth Late in Development," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 1999, pp. 201-207.
[140] A. Sen, "Correction to: Estimation of Current Reliability in a Duane-based Reliability Growth Model," *Technometrics*, vol. 41, num. 4, p. 385, 1999.
[141] J. Donovan and D. Murphy, "Reliability Growth - A New Graphical Example," *Quality and Reliability Engineering International*, vol. 15, pp. 167-174, 1999.
[142] E. Crétois, M. A. El Aroui and O. Gaudoin, "U-Plot Method for Testing the Goodness-of-Fit of the Power-Law Process," *Communications in Statistics: Theory and Methods*, vol. 28, pp. 1731-1747, 1999.
[143] J. Donovan and D. Murphy, "A New Reliability Growth Model: Its Mathematical Comparison to the Duane Model," *Microelectronics Reliability*, vol. 40, pp. 533-539, 2000.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

[144] P. M. Ellner, W. J. Broemm and W. J. Woodworth, “Reliability Growth Guide,” *AMSAA*, Aberdeen Proving Ground, MD, Tech. Rep. TR-652, 2000.
[145] J. D. Hall and W. R. Wessels, "An Automated Solver to Determine ML Estimates for the AMSAA Discrete Reliability Growth Model," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 2001, pp. 290-295.
[146] J. Donovan and D. Murphy, "Improvements in Reliability Growth Modeling," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 2001, pp. 296-301.
[147] G. Pulcini, "An Exponential Reliability Growth Model in Multi-copy Testing Program," *IEEE Trans. Reliability*, vol. 50, num. 4, pp. 365-373, 2001.
[148] R. Hodge, J. Quigley, I. James and J. Marshall, "Integrating Reliability Improvement Modeling into Practice-Challenges and Pitfalls," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 2002, pp. 158-164.
[149] G. Pulcini, "Correction to: How to Model Reliability Growth when Times of Design Modifications are Known," *IEEE Trans. Reliability*, vol. 51, num. 2, pp. 252-253, 2002.
[150] L. H. Crow, "Methods for Reducing the Cost to Maintain a Fleet of Repairable Systems," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 2003, pp. 392-399.
[151] T. Gurunatha and R. P. Siegel, "Applying Quality Tools to Reliability: A 12-Step Six - Sigma Process to Accelerate Reliability Growth in Product Design," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 2003, pp. 562-567.
[152] O. P. Yadav, N Singh and P. S. Goel, "A Practical Approach to System Reliability Growth Modeling and Improvement," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 2003, pp. 351-359.
[153] J. Quigley and L. Walls, "Confidence Intervals for Reliability Growth Models with Small Sample Sizes," *IEEE Trans. Reliability*, vol. 52, num. 2, pp. 257-262, 2003.
[154] O. Gaudoin, B. Yang and M. Xie, "A Simple Goodness-of-Fit Test for the Power-Law Process, based on the Duane Plot," *IEEE Trans. Reliability*, vol. 52, num. 1, pp. 69-74, 2003.
[155] D. P. Gaver, P. A. Jacobs, K. D. Glazenbrook and E. A. Seglie, "Probability Models for Sequential-Stage System Reliability Growth via Failure Mode Removal," *International Journal of Reliability, Quality & Safety Engineering*, vol. 10, issue 1, pp. 15-40, 2003.
[156] P. M. Ellner and J. B. Hall, “AMPM-Stein,” *AMSAA*, Aberdeen Proving Ground, MD, Tech. Rep. 751, 2004.
[157] L. H. Crow, "An Extended Reliability Growth Model for Managing and Assessing Corrective Actions," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 2004, pp. 73-80.
[158] T. C. Smith, "Reliability Growth Planning under Performance based Logistics," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 2004, pp. 418-423.
[159] M. Krasich and J. Quigley, "Modeling Reliability Growth in the Product Design Process," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 2004, pp. 424-430.
[160] C. E. Ebeling, *An Introduction to Reliability and Maintainability Engineering*. Illinois: Waveland Press Inc., 2005.
[161] D. E. Mortin and P.M. Ellner, “Reliability Growth for Military Vehicles – Emerging Methodology and Remaining Challenges,” *Journal of Commercial Vehicles*, vol. 114, pp.28-34, 2005.
[162] P. M. Ellner and J. B. Hall, “AMSAA Maturity Projection Model based on Stein

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:37Z
Check the source to verify that this is the current version before use.

Estimation," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 2005, pp. 271-277.
[163] W. J. Broemm, "User Guide for the Visual Growth Suite of Reliability Growth Models," AMSAA, Aberdeen Proving Ground, MD, Tech. Rep. 764, 2005.
[164] Department of Defense, "DoD Guide for Achieving Reliability, Availability and Maintainability," Washington, DC, 2005.
[165] P. E. Acevedo, D. S. Jackson and R. W. Kotlowitz, "Reliability Growth and Forecasting for Critical Hardware through Accelerated Life Testing," *Bell Labs Technical Journal*, vol. 11, num. 3, pp. 121-135, 2006.
[166] P. M. Ellner and J. B. Hall, "An Approach to Reliability Growth Planning based on Failure Mode Discovery and Correction using AMSAA Projection Methodology," *IEEE Proceedings of the Annual Reliability and Maintainability Symposium*, 2005, pp. 271-277.
[167] A. M. Sarhan, F. M. Guess and J. S. Usher, "Estimators for Reliability Measures in Geometric Distribution Model Using Dependent Masked System Life Test Data," *IEEE Trans. Reliability*, vol. R-56, num. 2, 2007.
[168] AMSAA. (2003) Reliability Technology. [online]. Available: http://www.amsaa.army.mil/ReliabilityTechnology/RelGrowth.htm
[169] ReliaSoft Corporation. (1992) Reliability Software. [online]. Available: http://www.reliasoft.com/
[170] Relex Software Corporation. (2007) Reliability Software. [online]. Available: http://www.relex.com/
[171] "The Effects of Aging on Costs of Operating and Maintaining Military Equipment, *U.S. Congressional Budget Office*, 2001.

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Appendix D Derivations for OC Analysis Propositions

D.1 Proposition 1.

$$f_{obs} \le c \quad \Leftrightarrow \quad TR \le \ell_{\gamma}(f_{obs})$$

D.1.1 Proof.

To prove this relation, we use the equation below which follows directly from the definition of a 100 γ percent lower confidence bound when f_{obs} failures occur in a demonstration test of length T_{Dem}:

$$\sum_{i=0}^{f_{obs}} e^{-T_{Dem}/\ell} \frac{(T_{Dem}/\ell)^i}{i!} = 1-\gamma$$

where

$$\ell \quad \underline{\Delta} \quad \ell_{\gamma}(f_{obs}).$$

Let g be the function of x > 0 defined by the left-hand side of the equation above with ℓ replaced by x. Note g is a strictly increasing function of x > 0 since g(x) is the probability of obtaining f_{obs} or fewer failures when the constant configuration under test has MTBF x.

I. First we will show $f_{obs} \le c \Rightarrow TR \le \ell$.

Thus, let $f_{obs} \le c$. Suppose ℓ < TR. Then

$$g(\ell) < g(TR) = \sum_{i=0}^{f_{obs}} e^{-T_{Dem}/TR} \frac{(T_{Dem}/TR)^i}{i!}$$

$$\le \sum_{i=0}^{c} e^{-T_{Dem}/TR} \frac{(T_{Dem}/TR)^i}{i!}$$

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$$\leq \quad 1-\gamma$$

which is a contradiction since $g(\ell) = 1-\gamma$. Thus, $TR \leq \ell$.

II. Next we will show $TR \leq \ell \Rightarrow f_{obs} \leq c$. Thus, let $TR \leq \ell$. Suppose $f_{obs} > c$. Then

$$\sum_{i=0}^{f_{obs}} e^{-T_{Dem}/TR} \frac{(T_{Dem}/TR)^i}{i!} \quad = \quad g(TR) \quad \leq \quad g(\ell) \quad = \quad 1-\gamma.$$

Since $f_{obs} > c$, this contradicts the definition of c (see Equation (5) in Section 2.1.2). Thus, $f_{obs} \leq c$.

D.2 Proposition 2.

For each $\alpha<1$, $T>0$, and $M(T)>0$, the corresponding distribution function of $L_\gamma(N,S)$ satisfies the inequality

$$\text{Prob}\left(L_\gamma(N,S) \leq M(T)\right) \quad \geq \quad \gamma$$

D.2.1 Proof.

Let f_W denote the density function of W (defined by Equation (20) in Section 2.1.3) corresponding to $\alpha<1$, $T>0$, $M(T)>0$. By inequality (21) in Section 2.1.3,

$$\text{Prob}\left(L_\gamma(N,S) \quad \leq \quad M(T)\right)$$

$$= \int_0^\infty \left\{\text{Prob}\left(L_\gamma(N,S;w) \leq M(T)\right)\right\} f_w(w)\,dw$$

$$\geq \quad \gamma \int_0^\infty f_w(w)\,dw$$

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$$= \gamma$$

D.3 Proposition 3.

For each α<1, T>0, and M(T)>0,

$$\text{Prob}\left(L_\gamma \left(N, S \right) = x \right) \quad = \quad 0$$

for all real x.

D.3.1 Proof.

Let α<1, T>0, and M(T)>0. Clearly, $L_\gamma(N,S) \geq 0$. Thus, we need to consider $x \geq 0$.

Let $L_\gamma(n,S)$ denote $L_\gamma(N,S)$ conditioned on $N = n$. As shown in Appendix A, of]Statistical Precision and Robustness of the AMSAA Continuous Reliability Growth Estimators, Ziad, Tariq and Ellner, Paul, AMSAA TR-453, April 1988,

,

$$\frac{\hat{M}_n(T)}{M(T)} \quad \sim \quad \left(\frac{\lambda T^\beta}{2n^2} \right) \chi^2_{2n}$$

where χ^2_ν is the chi-square random variable with ν degrees of freedom.

Thus,

$$\hat{M}_n(T) \quad \sim \quad \left(\frac{1}{\lambda \beta T^{\beta-1}} \right) \left(\frac{\lambda T^\beta}{2n^2} \right) \chi^2_{2n}$$

$$= \left(\frac{T}{2\beta n^2} \right) \chi^2_{2n}$$

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Then, by (12) in Section 2.1.3,

$$L_\gamma(\mathrm{n,S}) \sim \left(\frac{2\mathrm{n}}{z_\gamma(n)}\right)^2 \left(\frac{T}{2\beta n^2}\right) \chi^2_{2\mathrm{n}}$$

i.e.,

$$L_\gamma(n,S) \sim \left(\frac{2T}{\beta}\right)\left(\frac{\chi^2_{2n}}{z^2_\gamma(n)}\right) \qquad (34)$$

Thus,

$$\mathrm{Prob}\left(\mathrm{L}_\gamma(n,S)=x\right) = \mathrm{Prob}\left(\chi^2_{2\mathrm{n}} = \frac{\beta z^2_\gamma(\mathrm{n})\mathrm{x}}{2\mathrm{T}}\right) = 0$$

It then follows that,

$$\mathrm{Prob}\left(\mathrm{L}_\gamma(\mathrm{N,S})=\mathrm{x}\right) =$$

$$\left[\mathrm{Prob}(N=0)\right]^{-1} \sum_{n=1}^{\infty}\left[\mathrm{Prob}\left(L_\gamma(n,S)=x\right)\right]\mathrm{Prob}(N=n)$$

$$= 0, \text{ since Prob (N=0)} > 0.$$

D.4 Proposition 4.

Type II = $\mathrm{Prob}\left(TR \le L_\gamma(N,S)\right) \le 1-\gamma$ for each $\alpha<1$ and $T>0$ where $M(T)=TR$.

D.4.1 Proof.

Let $\alpha<1$ and $T>0$ with $M(T)=TR$.

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Then

$$\text{Prob}\left(TR \le L_{\gamma}(N,S)\right) =$$

$$\text{Prob}\left(L_{\gamma}(N,S) = TR\right) \quad + \quad \text{Prob}\left(TR < L_{\gamma}(N,S)\right)$$

$$= \text{Prob}\left(TR < L_{\gamma}(N,S)\right), \text{ by Proposition 3,}$$

$$= 1 - \text{Prob}\left(L_{\gamma}(N,S) \le TR\right) \quad \le \quad 1-\gamma, \text{ by Proposition 2.}$$

D.5 Proposition 5.

For a growth curve with parameters (α, T, M(T)), the expected number of failures (E(N)) can be determined by

$$\text{E(N)} \quad = \quad \frac{\text{T}}{(1-\alpha)\,\text{M(T)}}$$

D.5.1 Proof.

The observed number of failures by test duration t, denoted by N(t), is a non-homogeneous Poisson process with $N(T) = N$ and intensity function

$$\rho(\text{t}) \quad = \quad \frac{1}{\text{M(t)}} \quad = \quad \lambda\beta\,\text{t}^{\beta-1}$$

This implies that N is Poisson distributed with expected value

$$\text{E(N)} \quad = \quad \int_0^{\text{T}} \rho(\text{t})\,\text{dt} \quad = \quad \lambda\text{T}^{\beta}$$

By Equation (18) in Section 2.1.3,

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$$E(N) = \frac{T^{\beta}}{(M(T))\beta T^{\beta-1}}$$

This yields

$$E(N) = \frac{T}{\beta M(T)} = \frac{T}{(1-\alpha)\ M(T)}.$$

D.6 Proposition 6.

For a growth curve with parameters (α, T, M(T)),

Prob (A; α, T, M(T)) =

$$(1-e^{-\mu})^{-1} \sum_{n=1}^{\infty} \left[\text{Prob}\left(\frac{\chi^2_{2n}}{z^2_{\gamma}(n)} \geq \frac{1}{2\mu d} \right) \right] e^{-\mu} \left(\frac{\mu^n}{n!} \right)$$

where $\mu \triangleq E(N)$ and $d \triangleq M(T)/TR$.

D.6.1 Proof.

From (23) in Section 2.1.3 and (34),

$$\text{Prob}(A;\alpha,T,M(T)) = \text{Prob}(L_{\gamma}(N,S) \geq TR)$$

$$= [1 - \text{Prob}(N=0)]^{-1} \sum_{n=1}^{\infty} [\text{Prob}(L\gamma(n,S) \geq TR)]\ \text{Prob}(N=n)$$

$$= [1-\text{Prob}(N=0)]^{-1} \sum_{n=1}^{\infty} \left[\text{Prob}\left(\left(\frac{2T}{\beta} \right) \left(\frac{\chi^2_{2n}}{z^2_{\gamma}(n)} \right) \geq TR \right) \right] \text{Prob}(N=n)$$

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$$= \quad [1 - \text{Prob}\,(N = 0)]^{-1} \sum_{n=1}^{\infty} \left[\text{Prob} \left(\frac{\chi^2_{2n}}{z^2_{\gamma}\,(x)} \geq \frac{\beta(TR)}{2T} \right) \right] \text{Prob}(N = n)$$

Letting $\mu \triangleq E(N)$ and $d \triangleq M(T)/TR$,

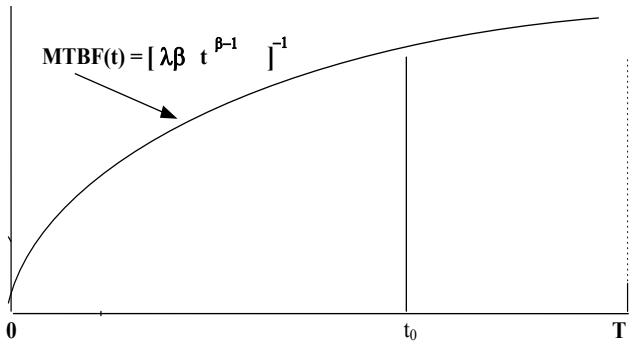
$$\text{Prob}\left(A; \alpha, T, M(T)\right) \quad =$$

$$(1 - e^{-\mu})^{-1} \sum_{n=1}^{\infty} \left[\text{Prob} \left(\frac{\chi^2_{2n}}{z^2_{\gamma}(n)} \geq (1/2) \left(\frac{\beta M(T)}{T} \right) \left(\frac{TR}{M(T)} \right) \right) \right] e^{-\mu} \left(\frac{\mu^n}{n!} \right)$$

$$= \quad (1 - e^{-\mu})^{-1} \sum_{n=1}^{\infty} \left[\text{Prob} \left(\frac{\chi^2_{2n}}{z^2_{\gamma}\,(n)} \geq \frac{1}{2\mu d} \right) \right] e^{-\mu} \left(\frac{\mu^n}{n!} \right).$$

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Appendix E Threshold Derivations



$\widehat{M}(t_0) = \frac{t_0}{n_o \hat{\beta}}$ where $n_0 = no.of\ failures\ over\ [0, t_0]$, and

$\hat{\beta} = \frac{n_0}{\sum_{i=1}^{n_0} ln\left(\frac{t_0}{x_i}\right)}$, where x_i = cumulative time to i^{th} failure, and

where $0<x_1<\cdots<x_{n_0} \leq t_0$.

Now,

$\frac{\hat{\beta}}{n_0} \sim \frac{2\beta n_0}{\chi^2(2n_0)}$

$$\therefore\ \widehat{M}(t_0)/n_0 \sim \left(\frac{t_0}{2\beta {n_0}^2}\right)\chi^2(2n_0)$$

and

$$\therefore \Pr\left\{\widehat{M}(t_0)|n_0\right\} \leq x = Pr\left\{\left(\frac{t_0}{2\beta {n_0}^2}\right)\chi^2(2n_0) \leq x\right\}$$
$$= Pr\left\{\chi^2(2n_0) \leq \frac{2\beta {n_0}^2 x}{t_0}\right\}$$

Thus,

$$Pr\left\{\widehat{M}(t_0) \leq x\right\} = \{1 - Pr(N_0 = 0)\}^{-1} \sum_{n_0=1}^{\infty} F_{n_0}\left(\frac{2\beta n_0^2 x}{t_0}\right) Pr(N_0 = n_0)$$

where F_{n_0} is the distribution function of $\chi^2(2n)$ and $N_0 \sim Poi\left(\lambda t_0^{\beta}\right)$.

One concludes

$$Pr\left\{\widehat{M}(t_0) \leq x\right\} = \left\{1 - e^{-\lambda {t_0}^{\beta}}\right\}^{-1} \sum_{n_0=1}^{\infty} F_{n_0}\left(\frac{2\beta n_0^2 x}{t_o}\right) e^{-\lambda t_0^{\beta}} \frac{\left(\lambda t_0^{\beta}\right)^{n_o}}{n_0!}$$

where χ^2 distribution with ν degrees of freedom is given by

$$f(x) = \frac{x^{\frac{\nu}{2}-1} e^{-\frac{x}{2}}}{\Gamma\left(\frac{\nu}{2}\right) 2^{\frac{\nu}{2}}}, \qquad 0 < x < \infty.$$

Let $\nu=2n_0$. Then

$f(x) = \frac{x^{n_0-1} e^{-\frac{x}{2}}}{\Gamma(n_0) 2^{n_0}} = \frac{x^{n_0-1} e^{-\frac{x}{2}}}{2^{n_0}(n_0-1)!}$, and it follows that

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$$F_{n_0}\left(\frac{2\beta n_0^2 x}{t_0}\right) = \int_0^{\frac{2\beta n_0^2 x}{t_0}} f(z)dz = \int_0^{\frac{2\beta n_0^2 x}{t_0}} \frac{z^{n_0-1}e^{-\frac{z}{2}}}{2^{n_0}(n_0-1)!}dz.$$

Therefore,

$$F_{n_0}\left(\frac{2\beta n_0^2 x}{t_0}\right) e^{-\lambda t_0^\beta} \frac{\left(\lambda t_0^\beta\right)^{n_0}}{n_0!} = e^{-\lambda t_0^\beta} \frac{\left(\lambda t_0^\beta\right)^{n_0}}{n_0!} \int_0^{\frac{2\beta n_0^2 x}{t_0}} \frac{z^{n_0-1}e^{-\frac{z}{2}}}{2^{n_0}(n_0-1)!}dz$$

$$= \int_0^{\frac{2\beta n_0^2 x}{t_0}} \frac{\lambda t_0^\beta \left(\lambda t_0^\beta z\right)^{n_0-1} e^{-\left(\lambda t_0^\beta + \frac{z}{2}\right)}}{2^{n_0}(n_0-1)!n_0!}\,\mathrm{dz}$$

Thus we have

$$Pr\left\{\widehat{M}(t_0) \le x\right\} = \left\{1 - e^{-\lambda t_0^\beta}\right\}^{-1} \sum_{n_0-1}^{\infty} \int_0^{\frac{2\beta n_0^2 x}{t_0}} \left\{ \frac{\lambda t_0^\beta \left(\lambda t_0^\beta z\right)^{n_0-1} e^{-\left(\frac{2\lambda t_0^\beta + z}{2}\right)}}{2^{n_0}(n_0-1)!n_0!} \right\} dz.$$

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Appendix F TABLES: Approximation Of The Probability Of Acceptance

The following tables provide approximations to the probability of acceptance, $\text{Prob}(A;\mu,d)$, where μ denotes the expected number of failures and d = M(T)/TR. The tabular entries were calculated using a modification to Equation (25). This modification entails (1) approximating $z_\gamma^2(n)$ by $4n\chi^2_{n+2,\gamma}$ and (2) conditioning on $N \geq 2$ instead of $N \geq 1$. Thus, in Equation (25) the expression $1 - e^{-\mu}$ is replaced by $1 - P(N \leq 1) = 1 - e^{-\mu} - \mu e^{-\mu}$ and the summation is over $N \geq 2$.

The approximation used for $z_\gamma^2(n)$ follows from the lower confidence bound approximation given by

$$\ell_\gamma(n,s) \cong (n/\chi^2_{n+2,\gamma})\hat{M}_n \qquad (1)$$

where $\hat{M}_n$ is the MLE of M(T) calculated from the observed data s = (t_1, t_2,...t_n). Here t_i denotes the cumulative operating time to the ith failure. This approximation was suggested by Dr. Larry Crow for conveniently approximating ℓ_γ(n, s). It has been our experience that the approximation in (1) results in slightly more conservative lower bounds on $M(T)$ than ℓ_γ(n, s). This implies that use of the corresponding approximation to $z_\gamma^2(n)$ would yield slightly smaller values of $\text{Prob}(A;\mu,d)$ than one would obtain by utilizing $z_\gamma^2(n)$. Based on our experience with $\text{Prob}(A;\mu,d)$ estimated by simulation, the approximating values appear to be within 0.01 of values obtained through simulation. We also observed that the approximation improves as n increases. The comparison between the lower confidence bound approximation given by (1) and the lower confidence bound using $z_\gamma^2(n)$ was based on Table 2 contained in Section 3. Since the entries in this table were for $n \geq 2$, the probability of acceptance, $\text{Prob}(A;\mu,d)$, was conditioned on $N \geq 2$. In most cases of interest for the model discussed in this report, $\text{Prob}(N \geq 2)$ will be close to one. In this situation, conditioning on $N \geq 2$ yields values of $\text{Prob}(A;\mu,d)$ that are, for practical purposes, essentially the same as those obtained by conditioning on $N \geq 1$.

The entries in these tables were calculated using the well-known relationship between the complement of a Chi-square distribution function and the cumulative Poisson sum. This relationship was applied to calculate

$$\text{Prob}\left(\chi^2_{2n} \geq \frac{z_\gamma^2(n)}{2\mu d}\right)$$

in the expression for $\text{Prob}(A;\mu,d)$ in Section 2.1.3 with $z_\gamma^2(n)$ replaced by its approximation, i.e., $4n\chi^2_{n+2,\gamma}$. In terms of the cumulative Poisson sum, this yields

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$$\text{Prob}\left(\chi^2_{2n} \geq \frac{2\,(n\chi^2_{n+2,\gamma}\,)}{\mu d} \right) \;=\; \sum_{x=0}^{n-1} e^{-w} \frac{w^x}{x!} \qquad (2)$$

where

$$w \;=\; (\,n\,\chi^2_{n+2,\gamma}\,)\,/\,(\,\mu d\,).$$

With additional computational effort, one can more precisely calculate $\text{Prob}(A;\mu,d)$ by iteratively solving for $z_\gamma(n)$ as the z-solution to Equation (13) of Section 2.1.3 over an appropriate range of n. Then Equation (2) can be utilized with $2n\chi^2_{n+2,\gamma}$ replaced by $z^2_\gamma(n)/2$.

The tables contained in this appendix are approximation values of $\text{Prob}(A;\mu,d)$ for three confidence levels; namely, for $\gamma = 0.70$, $\gamma = 0.80$, and $\gamma = 0.90$.

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TABLE F.I. FOR 70 PERCENT CONFIDENCE

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 70 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	5	6	7	8	9	10	11	12
1.00	0.131	0.150	0.163	0.173	0.180	0.186	0.191	0.195
1.05	0.150	0.171	0.187	0.199	0.208	0.216	0.224	0.230
1.10	0.169	0.194	0.212	0.226	0.238	0.249	0.258	0.266
1.15	0.189	0.217	0.238	0.255	0.269	0.282	0.294	0.304
1.20	0.209	0.240	0.264	0.284	0.301	0.316	0.330	0.343
1.25	0.231	0.264	0.291	0.314	0.334	0.351	0.368	0.383
1.30	0.252	0.289	0.319	0.344	0.367	0.387	0.405	0.422
1.35	0.274	0.314	0.347	0.375	0.400	0.422	0.443	0.462
1.40	0.296	0.339	0.375	0.405	0.432	0.457	0.479	0.500
1.45	0.318	0.364	0.402	0.435	0.465	0.491	0.515	0.538
1.50	0.340	0.389	0.430	0.465	0.496	0.525	0.550	0.574
1.55	0.362	0.414	0.457	0.494	0.527	0.557	0.584	0.609
1.60	0.384	0.438	0.484	0.523	0.557	0.588	0.616	0.642
1.65	0.406	0.462	0.510	0.550	0.586	0.618	0.647	0.673
1.70	0.427	0.486	0.535	0.577	0.614	0.647	0.676	0.703
1.75	0.448	0.509	0.560	0.603	0.641	0.674	0.704	0.730
1.80	0.469	0.531	0.583	0.628	0.666	0.700	0.729	0.756
1.85	0.489	0.553	0.606	0.651	0.690	0.724	0.754	0.780
1.90	0.509	0.575	0.628	0.674	0.713	0.746	0.776	0.802
1.95	0.529	0.595	0.650	0.695	0.734	0.768	0.797	0.822
2.00	0.548	0.615	0.670	0.716	0.754	0.787	0.816	0.840
2.05	0.566	0.634	0.689	0.735	0.773	0.806	0.833	0.857
2.10	0.584	0.652	0.708	0.753	0.791	0.823	0.849	0.872
2.15	0.601	0.670	0.725	0.770	0.807	0.838	0.864	0.885
2.20	0.618	0.687	0.742	0.786	0.823	0.853	0.877	0.898
2.25	0.634	0.703	0.758	0.802	0.837	0.866	0.890	0.909
2.30	0.650	0.719	0.773	0.816	0.850	0.878	0.901	0.919
2.35	0.665	0.733	0.787	0.829	0.863	0.889	0.911	0.928
2.40	0.679	0.747	0.800	0.841	0.874	0.900	0.920	0.936
2.45	0.693	0.761	0.813	0.853	0.884	0.909	0.928	0.943
2.50	0.706	0.774	0.825	0.864	0.894	0.917	0.936	0.950
2.55	0.719	0.786	0.836	0.874	0.903	0.925	0.942	0.955
2.60	0.732	0.797	0.846	0.883	0.911	0.932	0.948	0.960
2.65	0.743	0.808	0.856	0.892	0.918	0.938	0.954	0.965
2.70	0.755	0.818	0.865	0.900	0.925	0.944	0.958	0.969
2.75	0.766	0.828	0.874	0.907	0.932	0.950	0.963	0.973
2.80	0.776	0.837	0.882	0.914	0.937	0.954	0.967	0.976
2.85	0.786	0.846	0.889	0.920	0.943	0.959	0.970	0.978
2.90	0.795	0.855	0.896	0.926	0.947	0.963	0.973	0.981
2.95	0.804	0.862	0.903	0.932	0.952	0.966	0.976	0.983
3.00	0.813	0.870	0.909	0.937	0.956	0.969	0.979	0.985

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 70 PERCENT CONFIDENCE EXPECTED NUMBER OF FAILURES

M(T)/TR	13	14	15	16	17	18	19	20
1.00	0.199	0.202	0.206	0.208	0.211	0.213	0.215	0.217
1.05	0.235	0.241	0.245	0.250	0.254	0.258	0.262	0.265
1.10	0.274	0.281	0.288	0.294	0.300	0.306	0.311	0.316
1.15	0.314	0.323	0.332	0.340	0.348	0.355	0.363	0.369
1.20	0.355	0.366	0.377	0.387	0.397	0.406	0.415	0.423
1.25	0.397	0.410	0.423	0.435	0.446	0.457	0.467	0.477
1.30	0.438	0.454	0.468	0.481	0.494	0.507	0.519	0.530
1.35	0.480	0.496	0.512	0.527	0.542	0.555	0.568	0.581
1.40	0.520	0.538	0.555	0.572	0.587	0.602	0.616	0.629
1.45	0.559	0.578	0.597	0.614	0.630	0.646	0.660	0.674
1.50	0.596	0.617	0.636	0.654	0.671	0.687	0.702	0.716
1.55	0.632	0.653	0.673	0.691	0.709	0.725	0.740	0.754
1.60	0.666	0.687	0.707	0.726	0.743	0.759	0.774	0.788
1.65	0.697	0.719	0.739	0.758	0.775	0.791	0.805	0.819
1.70	0.727	0.749	0.769	0.787	0.804	0.819	0.833	0.846
1.75	0.754	0.776	0.796	0.813	0.829	0.844	0.857	0.869
1.80	0.780	0.801	0.820	0.837	0.852	0.866	0.879	0.890
1.85	0.803	0.823	0.842	0.858	0.873	0.886	0.897	0.908
1.90	0.824	0.844	0.861	0.877	0.891	0.903	0.913	0.923
1.95	0.843	0.862	0.879	0.894	0.906	0.918	0.927	0.936
2.00	0.861	0.879	0.895	0.908	0.920	0.930	0.939	0.947
2.05	0.877	0.894	0.908	0.921	0.932	0.941	0.949	0.956
2.10	0.891	0.907	0.921	0.932	0.942	0.951	0.958	0.964
2.15	0.903	0.919	0.931	0.942	0.951	0.959	0.965	0.970
2.20	0.915	0.929	0.941	0.950	0.959	0.965	0.971	0.976
2.25	0.925	0.938	0.949	0.958	0.965	0.971	0.976	0.980
2.30	0.934	0.946	0.956	0.964	0.970	0.976	0.980	0.984
2.35	0.942	0.953	0.962	0.969	0.975	0.980	0.984	0.987
2.40	0.949	0.959	0.967	0.974	0.979	0.983	0.987	0.989
2.45	0.955	0.965	0.972	0.978	0.982	0.986	0.989	0.991
2.50	0.961	0.969	0.976	0.981	0.985	0.989	0.991	0.993
2.55	0.966	0.973	0.979	0.984	0.988	0.990	0.993	0.994
2.60	0.970	0.977	0.982	0.987	0.990	0.992	0.994	0.995
2.65	0.974	0.980	0.985	0.989	0.991	0.993	0.995	0.996
2.70	0.977	0.983	0.987	0.990	0.993	0.995	0.996	0.997
2.75	0.980	0.985	0.989	0.992	0.994	0.996	0.997	0.998
2.80	0.982	0.987	0.991	0.993	0.995	0.996	0.997	0.998
2.85	0.984	0.989	0.992	0.994	0.996	0.997	0.998	0.998
2.90	0.986	0.990	0.993	0.995	0.996	0.997	0.998	0.999
2.95	0.988	0.992	0.994	0.996	0.997	0.998	0.999	0.999
3.00	0.990	0.993	0.995	0.996	0.998	0.998	0.999	0.999

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

WITH 70 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	21	22	23	24	25	26	27	28
1.00	0.219	0.221	0.223	0.224	0.225	0.227	0.228	0.229
1.05	0.269	0.272	0.275	0.278	0.280	0.283	0.286	0.288
1.10	0.321	0.326	0.331	0.335	0.339	0.343	0.347	0.351
1.15	0.376	0.382	0.388	0.394	0.400	0.406	0.411	0.417
1.20	0.432	0.440	0.447	0.455	0.462	0.469	0.476	0.482
1.25	0.487	0.496	0.505	0.514	0.523	0.531	0.539	0.547
1.30	0.541	0.552	0.562	0.572	0.581	0.590	0.599	0.608
1.35	0.593	0.605	0.616	0.626	0.637	0.646	0.656	0.665
1.40	0.642	0.654	0.666	0.677	0.688	0.698	0.708	0.717
1.45	0.688	0.700	0.712	0.723	0.734	0.745	0.755	0.764
1.50	0.729	0.742	0.754	0.765	0.776	0.786	0.796	0.805
1.55	0.767	0.780	0.792	0.803	0.813	0.823	0.832	0.841
1.60	0.801	0.813	0.825	0.835	0.845	0.854	0.863	0.871
1.65	0.831	0.843	0.854	0.863	0.873	0.881	0.889	0.897
1.70	0.858	0.868	0.878	0.888	0.896	0.904	0.911	0.918
1.75	0.881	0.891	0.900	0.908	0.916	0.923	0.929	0.935
1.80	0.900	0.910	0.918	0.925	0.932	0.939	0.944	0.949
1.85	0.917	0.926	0.933	0.940	0.946	0.951	0.956	0.961
1.90	0.931	0.939	0.946	0.952	0.957	0.962	0.966	0.970
1.95	0.944	0.950	0.956	0.961	0.966	0.970	0.973	0.977
2.00	0.954	0.960	0.965	0.969	0.973	0.977	0.980	0.982
2.05	0.962	0.967	0.972	0.976	0.979	0.982	0.984	0.986
2.10	0.969	0.974	0.977	0.981	0.984	0.986	0.988	0.990
2.15	0.975	0.979	0.982	0.985	0.987	0.989	0.991	0.992
2.20	0.980	0.983	0.986	0.988	0.990	0.992	0.993	0.994
2.25	0.984	0.986	0.989	0.991	0.992	0.994	0.995	0.996
2.30	0.987	0.989	0.991	0.993	0.994	0.995	0.996	0.997
2.35	0.989	0.991	0.993	0.994	0.995	0.996	0.997	0.998
2.40	0.991	0.993	0.995	0.996	0.996	0.997	0.998	0.998
2.45	0.993	0.995	0.996	0.997	0.997	0.998	0.998	0.999
2.50	0.995	0.996	0.997	0.997	0.998	0.998	0.999	0.999
2.55	0.996	0.997	0.997	0.998	0.998	0.999	0.999	0.999
2.60	0.996	0.997	0.998	0.998	0.999	0.999	0.999	0.999
2.65	0.997	0.998	0.998	0.999	0.999	0.999	0.999	1.000
2.70	0.998	0.998	0.999	0.999	0.999	0.999	1.000	1.000
2.75	0.998	0.999	0.999	0.999	0.999	1.000	1.000	1.000
2.80	0.999	0.999	0.999	0.999	1.000	1.000	1.000	1.000
2.85	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
2.90	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000
2.95	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000
3.00	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 70 PERCENT CONFIDENCE EXPECTED NUMBER OF FAILURES

M(T)/TR	29	30	31	32	33	34	35	36
1.00	0.231	0.232	0.233	0.234	0.235	0.236	0.236	0.237
1.05	0.291	0.293	0.295	0.297	0.300	0.302	0.304	0.306
1.10	0.355	0.359	0.362	0.366	0.369	0.373	0.376	0.379
1.15	0.422	0.427	0.432	0.437	0.441	0.446	0.450	0.455
1.20	0.489	0.495	0.501	0.507	0.513	0.519	0.525	0.530
1.25	0.554	0.562	0.569	0.576	0.582	0.589	0.596	0.602
1.30	0.616	0.624	0.632	0.640	0.648	0.655	0.662	0.669
1.35	0.674	0.683	0.691	0.699	0.707	0.715	0.722	0.729
1.40	0.727	0.735	0.744	0.752	0.760	0.767	0.775	0.782
1.45	0.773	0.782	0.790	0.798	0.806	0.813	0.820	0.827
1.50	0.814	0.822	0.830	0.838	0.845	0.852	0.859	0.865
1.55	0.849	0.857	0.864	0.871	0.878	0.884	0.890	0.896
1.60	0.879	0.886	0.893	0.899	0.905	0.911	0.916	0.921
1.65	0.904	0.910	0.916	0.922	0.927	0.932	0.936	0.941
1.70	0.924	0.930	0.935	0.940	0.944	0.948	0.952	0.956
1.75	0.941	0.946	0.950	0.954	0.958	0.961	0.965	0.968
1.80	0.954	0.958	0.962	0.965	0.969	0.971	0.974	0.976
1.85	0.965	0.968	0.971	0.974	0.977	0.979	0.981	0.983
1.90	0.973	0.976	0.978	0.981	0.983	0.985	0.986	0.988
1.95	0.979	0.982	0.984	0.986	0.987	0.989	0.990	0.991
2.00	0.984	0.986	0.988	0.990	0.991	0.992	0.993	0.994
2.05	0.988	0.990	0.991	0.992	0.993	0.994	0.995	0.996
2.10	0.991	0.992	0.994	0.994	0.995	0.996	0.997	0.997
2.15	0.993	0.994	0.995	0.996	0.997	0.997	0.998	0.998
2.20	0.995	0.996	0.997	0.997	0.998	0.998	0.998	0.999
2.25	0.996	0.997	0.998	0.998	0.998	0.999	0.999	0.999
2.30	0.997	0.998	0.998	0.999	0.999	0.999	0.999	0.999
2.35	0.998	0.998	0.999	0.999	0.999	0.999	0.999	1.000
2.40	0.999	0.999	0.999	0.999	0.999	1.000	1.000	1.000
2.45	0.999	0.999	0.999	0.999	1.000	1.000	1.000	1.000
2.50	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000
2.55	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

WITH 70 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	37	38	39	40	41	42	43	44
1.00	0.238	0.239	0.240	0.240	0.241	0.242	0.242	0.243
1.05	0.308	0.309	0.311	0.313	0.315	0.317	0.318	0.320
1.10	0.382	0.385	0.388	0.391	0.394	0.397	0.400	0.403
1.15	0.459	0.464	0.468	0.472	0.476	0.480	0.484	0.488
1.20	0.535	0.541	0.546	0.551	0.556	0.561	0.566	0.570
1.25	0.608	0.614	0.620	0.626	0.632	0.637	0.643	0.648
1.30	0.676	0.682	0.688	0.695	0.701	0.707	0.712	0.718
1.35	0.736	0.743	0.749	0.755	0.762	0.768	0.773	0.779
1.40	0.789	0.795	0.802	0.808	0.814	0.819	0.825	0.830
1.45	0.834	0.840	0.846	0.851	0.857	0.862	0.867	0.872
1.50	0.871	0.877	0.882	0.887	0.892	0.897	0.901	0.906
1.55	0.901	0.906	0.911	0.916	0.920	0.924	0.928	0.931
1.60	0.925	0.930	0.934	0.938	0.941	0.945	0.948	0.951
1.65	0.944	0.948	0.952	0.955	0.958	0.961	0.963	0.966
1.70	0.959	0.962	0.965	0.968	0.970	0.972	0.974	0.976
1.75	0.970	0.973	0.975	0.977	0.979	0.981	0.982	0.984
1.80	0.979	0.980	0.982	0.984	0.985	0.987	0.988	0.989
1.85	0.985	0.986	0.988	0.989	0.990	0.991	0.992	0.993
1.90	0.989	0.990	0.991	0.992	0.993	0.994	0.995	0.995
1.95	0.992	0.993	0.994	0.995	0.995	0.996	0.996	0.997
2.00	0.995	0.995	0.996	0.996	0.997	0.997	0.998	0.998
2.05	0.996	0.997	0.997	0.998	0.998	0.998	0.998	0.999
2.10	0.997	0.998	0.998	0.998	0.999	0.999	0.999	0.999
2.15	0.998	0.999	0.999	0.999	0.999	0.999	0.999	0.999
2.20	0.999	0.999	0.999	0.999	0.999	1.000	1.000	1.000
2.25	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
2.30	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

WITH 70 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	45	46	47	48	49	50	51	52
1.00	0.244	0.244	0.245	0.245	0.246	0.246	0.247	0.247
1.05	0.322	0.323	0.325	0.326	0.328	0.329	0.331	0.332
1.10	0.406	0.408	0.411	0.414	0.416	0.419	0.421	0.424
1.15	0.491	0.495	0.499	0.502	0.506	0.510	0.513	0.516
1.20	0.575	0.580	0.584	0.589	0.593	0.597	0.602	0.606
1.25	0.653	0.659	0.664	0.669	0.673	0.678	0.683	0.687
1.30	0.724	0.729	0.734	0.739	0.744	0.749	0.754	0.759
1.35	0.784	0.790	0.795	0.800	0.805	0.810	0.814	0.819
1.40	0.835	0.841	0.845	0.850	0.855	0.859	0.863	0.867
1.45	0.877	0.881	0.886	0.890	0.894	0.898	0.902	0.905
1.50	0.910	0.914	0.917	0.921	0.924	0.928	0.931	0.934
1.55	0.935	0.938	0.941	0.944	0.947	0.950	0.952	0.955
1.60	0.954	0.957	0.959	0.961	0.964	0.966	0.968	0.970
1.65	0.968	0.970	0.972	0.974	0.975	0.977	0.979	0.980
1.70	0.978	0.980	0.981	0.982	0.984	0.985	0.986	0.987
1.75	0.985	0.986	0.987	0.988	0.989	0.990	0.991	0.992
1.80	0.990	0.991	0.992	0.992	0.993	0.994	0.994	0.995
1.85	0.993	0.994	0.995	0.995	0.996	0.996	0.996	0.997
1.90	0.996	0.996	0.997	0.997	0.997	0.998	0.998	0.998
1.95	0.997	0.998	0.998	0.998	0.998	0.998	0.999	0.999
2.00	0.998	0.998	0.999	0.999	0.999	0.999	0.999	0.999
2.05	0.999	0.999	0.999	0.999	0.999	0.999	1.000	1.000
2.10	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

WITH 70 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	53	54	55	56	57	58	59	60
1.00	0.248	0.248	0.249	0.249	0.250	0.250	0.250	0.251
1.05	0.334	0.335	0.336	0.338	0.339	0.340	0.342	0.343
1.10	0.426	0.428	0.431	0.433	0.435	0.438	0.440	0.442
1.15	0.520	0.523	0.526	0.530	0.533	0.536	0.539	0.542
1.20	0.610	0.614	0.618	0.622	0.626	0.629	0.633	0.637
1.25	0.692	0.696	0.701	0.705	0.709	0.713	0.717	0.721
1.30	0.764	0.768	0.772	0.777	0.781	0.785	0.789	0.793
1.35	0.823	0.828	0.832	0.836	0.840	0.844	0.847	0.851
1.40	0.871	0.875	0.879	0.883	0.886	0.889	0.893	0.896
1.45	0.909	0.912	0.915	0.918	0.921	0.924	0.927	0.929
1.50	0.937	0.939	0.942	0.944	0.947	0.949	0.951	0.953
1.55	0.957	0.959	0.961	0.963	0.965	0.967	0.968	0.970
1.60	0.971	0.973	0.975	0.976	0.977	0.979	0.980	0.981
1.65	0.981	0.983	0.984	0.985	0.986	0.987	0.988	0.988
1.70	0.988	0.989	0.990	0.990	0.991	0.992	0.992	0.993
1.75	0.992	0.993	0.994	0.994	0.995	0.995	0.995	0.996
1.80	0.995	0.996	0.996	0.996	0.997	0.997	0.997	0.998
1.85	0.997	0.997	0.998	0.998	0.998	0.998	0.998	0.999
1.90	0.998	0.998	0.999	0.999	0.999	0.999	0.999	0.999
1.95	0.999	0.999	0.999	0.999	0.999	0.999	1.000	1.000
2.00	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

WITH 70 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	61	62	63	64	65	66	67	68
1.00	0.251	0.252	0.252	0.252	0.253	0.253	0.253	0.254
1.05	0.344	0.346	0.347	0.348	0.349	0.350	0.352	0.353
1.10	0.445	0.447	0.449	0.451	0.453	0.455	0.457	0.459
1.15	0.545	0.548	0.551	0.554	0.557	0.560	0.563	0.566
1.20	0.640	0.644	0.648	0.651	0.655	0.658	0.661	0.665
1.25	0.725	0.729	0.733	0.737	0.740	0.744	0.747	0.751
1.30	0.797	0.801	0.804	0.808	0.812	0.815	0.819	0.822
1.35	0.855	0.858	0.862	0.865	0.868	0.871	0.874	0.877
1.40	0.899	0.902	0.905	0.908	0.911	0.913	0.916	0.918
1.45	0.932	0.934	0.937	0.939	0.941	0.943	0.945	0.947
1.50	0.955	0.957	0.959	0.961	0.962	0.964	0.966	0.967
1.55	0.971	0.973	0.974	0.976	0.977	0.978	0.979	0.980
1.60	0.982	0.983	0.984	0.985	0.986	0.987	0.987	0.988
1.65	0.989	0.990	0.990	0.991	0.992	0.992	0.993	0.993
1.70	0.993	0.994	0.994	0.995	0.995	0.996	0.996	0.996
1.75	0.996	0.997	0.997	0.997	0.997	0.998	0.998	0.998
1.80	0.998	0.998	0.998	0.998	0.998	0.999	0.999	0.999
1.85	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
1.90	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
1.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

WITH 70 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	69	70	71	72	73	74	75	76
1.00	0.254	0.254	0.255	0.255	0.255	0.256	0.256	0.256
1.05	0.354	0.355	0.356	0.357	0.358	0.359	0.361	0.362
1.10	0.461	0.463	0.465	0.467	0.469	0.471	0.473	0.475
1.15	0.569	0.571	0.574	0.577	0.579	0.582	0.585	0.587
1.20	0.668	0.671	0.674	0.678	0.681	0.684	0.687	0.690
1.25	0.754	0.758	0.761	0.764	0.768	0.771	0.774	0.777
1.30	0.825	0.828	0.832	0.835	0.838	0.841	0.844	0.846
1.35	0.880	0.883	0.886	0.888	0.891	0.894	0.896	0.899
1.40	0.921	0.923	0.925	0.927	0.930	0.932	0.934	0.936
1.45	0.949	0.951	0.953	0.954	0.956	0.958	0.959	0.961
1.50	0.968	0.970	0.971	0.972	0.974	0.975	0.976	0.977
1.55	0.981	0.982	0.983	0.984	0.985	0.985	0.986	0.987
1.60	0.989	0.990	0.990	0.991	0.991	0.992	0.992	0.993
1.65	0.994	0.994	0.994	0.995	0.995	0.995	0.996	0.996
1.70	0.996	0.997	0.997	0.997	0.997	0.998	0.998	0.998
1.75	0.998	0.998	0.998	0.999	0.999	0.999	0.999	0.999
1.80	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
1.85	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 70 PERCENT CONFIDENCE EXPECTED NUMBER OF FAILURES

M(T)/TR	77	78	79	80	81	82	83	84
1.00	0.256	0.257	0.257	0.257	0.257	0.258	0.258	0.258
1.05	0.363	0.364	0.365	0.366	0.367	0.368	0.369	0.370
1.10	0.477	0.479	0.481	0.483	0.485	0.486	0.488	0.490
1.15	0.590	0.593	0.595	0.598	0.600	0.602	0.605	0.607
1.20	0.693	0.696	0.699	0.702	0.704	0.707	0.710	0.713
1.25	0.780	0.783	0.786	0.789	0.792	0.795	0.797	0.800
1.30	0.849	0.852	0.855	0.857	0.860	0.862	0.865	0.867
1.35	0.901	0.903	0.906	0.908	0.910	0.912	0.914	0.916
1.40	0.937	0.939	0.941	0.943	0.944	0.946	0.948	0.949
1.45	0.962	0.963	0.965	0.966	0.967	0.968	0.969	0.971
1.50	0.978	0.979	0.980	0.980	0.981	0.982	0.983	0.984
1.55	0.987	0.988	0.989	0.989	0.990	0.990	0.991	0.991
1.60	0.993	0.993	0.994	0.994	0.994	0.995	0.995	0.995
1.65	0.996	0.997	0.997	0.997	0.997	0.997	0.998	0.998
1.70	0.998	0.998	0.998	0.999	0.999	0.999	0.999	0.999
1.75	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
1.80	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT
WITH 70 PERCENT CONFIDENCE
EXPECTED NUMBER OF FAILURES

M(T)/TR	85	86	87	88	89	90	91	92
1.00	0.258	0.259	0.259	0.259	0.259	0.260	0.260	0.260
1.05	0.371	0.372	0.373	0.374	0.375	0.376	0.377	0.378
1.10	0.492	0.493	0.495	0.497	0.499	0.500	0.502	0.504
1.15	0.610	0.612	0.614	0.617	0.619	0.621	0.624	0.626
1.20	0.715	0.718	0.721	0.723	0.726	0.728	0.731	0.734
1.25	0.803	0.805	0.808	0.811	0.813	0.816	0.818	0.820
1.30	0.870	0.872	0.874	0.877	0.879	0.881	0.883	0.885
1.35	0.918	0.920	0.922	0.924	0.925	0.927	0.929	0.931
1.40	0.951	0.952	0.954	0.955	0.956	0.958	0.959	0.960
1.45	0.972	0.973	0.974	0.975	0.975	0.976	0.977	0.978
1.50	0.984	0.985	0.986	0.986	0.987	0.987	0.988	0.988
1.55	0.992	0.992	0.992	0.993	0.993	0.994	0.994	0.994
1.60	0.996	0.996	0.996	0.996	0.997	0.997	0.997	0.997
1.65	0.998	0.998	0.998	0.998	0.998	0.998	0.999	0.999
1.70	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
1.75	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

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WITH 70 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	93	94	95	96	97	98	99	100
1.00	0.260	0.260	0.261	0.261	0.261	0.261	0.261	0.262
1.05	0.379	0.379	0.380	0.381	0.382	0.383	0.384	0.385
1.10	0.506	0.507	0.509	0.510	0.512	0.514	0.515	0.517
1.15	0.628	0.630	0.633	0.635	0.637	0.639	0.641	0.643
1.20	0.736	0.738	0.741	0.743	0.746	0.748	0.750	0.753
1.25	0.823	0.825	0.827	0.830	0.832	0.834	0.836	0.839
1.30	0.887	0.889	0.891	0.893	0.895	0.897	0.899	0.901
1.35	0.932	0.934	0.935	0.937	0.938	0.940	0.941	0.942
1.40	0.961	0.962	0.963	0.964	0.965	0.966	0.967	0.968
1.45	0.979	0.979	0.980	0.981	0.982	0.982	0.983	0.984
1.50	0.989	0.989	0.990	0.990	0.991	0.991	0.991	0.992
1.55	0.994	0.995	0.995	0.995	0.995	0.996	0.996	0.996
1.60	0.997	0.997	0.998	0.998	0.998	0.998	0.998	0.998
1.65	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
1.70	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000
1.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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TABLE F.II. FOR 80 PERCENT CONFIDENCE

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

WITH 80 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	5	6	7	8	9	10	11	12
1.00	0.079	0.093	0.102	0.109	0.114	0.118	0.121	0.124
1.05	0.092	0.108	0.119	0.128	0.135	0.140	0.145	0.150
1.10	0.105	0.124	0.138	0.148	0.157	0.165	0.172	0.178
1.15	0.120	0.141	0.157	0.170	0.181	0.191	0.200	0.208
1.20	0.135	0.159	0.178	0.193	0.207	0.219	0.230	0.240
1.25	0.151	0.178	0.200	0.218	0.234	0.248	0.261	0.274
1.30	0.168	0.198	0.222	0.243	0.261	0.278	0.294	0.309
1.35	0.185	0.218	0.245	0.269	0.290	0.309	0.327	0.344
1.40	0.203	0.239	0.269	0.295	0.319	0.341	0.361	0.381
1.45	0.221	0.260	0.293	0.322	0.348	0.373	0.395	0.417
1.50	0.240	0.282	0.318	0.349	0.378	0.405	0.429	0.453
1.55	0.259	0.304	0.342	0.377	0.408	0.436	0.463	0.488
1.60	0.278	0.326	0.367	0.404	0.437	0.468	0.496	0.523
1.65	0.297	0.348	0.392	0.431	0.466	0.499	0.529	0.556
1.70	0.316	0.370	0.417	0.458	0.495	0.529	0.560	0.589
1.75	0.336	0.392	0.441	0.484	0.523	0.558	0.590	0.620
1.80	0.355	0.414	0.465	0.510	0.550	0.586	0.620	0.650
1.85	0.374	0.436	0.488	0.535	0.576	0.614	0.647	0.678
1.90	0.393	0.457	0.511	0.559	0.602	0.640	0.674	0.705
1.95	0.412	0.478	0.534	0.583	0.626	0.665	0.699	0.730
2.00	0.431	0.498	0.556	0.606	0.650	0.688	0.723	0.753
2.05	0.449	0.519	0.577	0.628	0.672	0.711	0.745	0.775
2.10	0.468	0.538	0.598	0.649	0.694	0.732	0.766	0.796
2.15	0.485	0.557	0.618	0.669	0.714	0.752	0.786	0.814
2.20	0.503	0.576	0.637	0.689	0.733	0.771	0.804	0.832
2.25	0.520	0.594	0.656	0.708	0.751	0.789	0.821	0.848
2.30	0.537	0.612	0.674	0.725	0.769	0.805	0.836	0.862
2.35	0.553	0.629	0.691	0.742	0.785	0.821	0.851	0.876
2.40	0.569	0.645	0.707	0.758	0.800	0.835	0.864	0.888
2.45	0.585	0.661	0.723	0.773	0.814	0.848	0.876	0.899
2.50	0.600	0.676	0.738	0.787	0.828	0.861	0.887	0.909
2.55	0.614	0.691	0.752	0.801	0.840	0.872	0.898	0.918
2.60	0.629	0.705	0.765	0.814	0.852	0.883	0.907	0.926
2.65	0.642	0.718	0.778	0.826	0.863	0.892	0.916	0.934
2.70	0.656	0.731	0.791	0.837	0.873	0.901	0.923	0.941
2.75	0.669	0.744	0.802	0.847	0.882	0.910	0.931	0.947
2.80	0.681	0.756	0.813	0.857	0.891	0.917	0.937	0.952
2.85	0.693	0.767	0.824	0.867	0.899	0.924	0.943	0.957
2.90	0.705	0.778	0.834	0.875	0.907	0.931	0.948	0.962
2.95	0.716	0.789	0.843	0.884	0.914	0.936	0.953	0.966
3.00	0.727	0.799	0.852	0.891	0.920	0.942	0.958	0.969

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 80 PERCENT CONFIDENCE EXPECTED NUMBER OF FAILURES

M(T)/TR	13	14	15	16	17	18	19	20
1.00	0.127	0.129	0.131	0.133	0.135	0.136	0.138	0.139
1.05	0.154	0.158	0.161	0.164	0.168	0.171	0.173	0.176
1.10	0.184	0.189	0.194	0.199	0.204	0.208	0.213	0.217
1.15	0.216	0.223	0.230	0.237	0.243	0.250	0.255	0.261
1.20	0.250	0.260	0.268	0.277	0.285	0.293	0.301	0.308
1.25	0.286	0.297	0.308	0.319	0.329	0.339	0.348	0.357
1.30	0.323	0.336	0.349	0.362	0.374	0.386	0.397	0.408
1.35	0.361	0.376	0.391	0.406	0.419	0.433	0.446	0.458
1.40	0.399	0.416	0.433	0.449	0.465	0.479	0.494	0.508
1.45	0.437	0.456	0.475	0.492	0.509	0.525	0.541	0.556
1.50	0.475	0.496	0.516	0.534	0.553	0.570	0.586	0.602
1.55	0.512	0.534	0.555	0.575	0.594	0.612	0.630	0.646
1.60	0.548	0.571	0.593	0.614	0.634	0.653	0.670	0.687
1.65	0.583	0.607	0.630	0.651	0.671	0.690	0.708	0.725
1.70	0.616	0.641	0.664	0.686	0.706	0.725	0.743	0.760
1.75	0.648	0.673	0.697	0.719	0.739	0.757	0.775	0.791
1.80	0.678	0.703	0.727	0.749	0.769	0.787	0.804	0.819
1.85	0.706	0.732	0.755	0.776	0.796	0.813	0.830	0.844
1.90	0.733	0.758	0.781	0.802	0.820	0.837	0.853	0.867
1.95	0.758	0.782	0.805	0.825	0.843	0.859	0.873	0.886
2.00	0.781	0.805	0.826	0.845	0.863	0.878	0.891	0.903
2.05	0.802	0.825	0.846	0.864	0.880	0.895	0.907	0.918
2.10	0.821	0.844	0.864	0.881	0.896	0.909	0.921	0.931
2.15	0.839	0.861	0.880	0.896	0.910	0.922	0.933	0.942
2.20	0.856	0.876	0.894	0.909	0.922	0.933	0.943	0.951
2.25	0.871	0.890	0.907	0.921	0.933	0.943	0.952	0.959
2.30	0.884	0.903	0.918	0.931	0.942	0.952	0.959	0.966
2.35	0.896	0.914	0.928	0.940	0.951	0.959	0.966	0.972
2.40	0.907	0.924	0.937	0.948	0.958	0.965	0.971	0.977
2.45	0.917	0.933	0.945	0.955	0.964	0.971	0.976	0.981
2.50	0.926	0.941	0.952	0.961	0.969	0.975	0.980	0.984
2.55	0.935	0.948	0.958	0.967	0.974	0.979	0.983	0.987
2.60	0.942	0.954	0.964	0.971	0.977	0.982	0.986	0.989
2.65	0.948	0.960	0.968	0.975	0.981	0.985	0.988	0.991
2.70	0.954	0.964	0.973	0.979	0.984	0.987	0.990	0.993
2.75	0.959	0.969	0.976	0.982	0.986	0.989	0.992	0.994
2.80	0.964	0.973	0.979	0.984	0.988	0.991	0.993	0.995
2.85	0.968	0.976	0.982	0.987	0.990	0.993	0.994	0.996
2.90	0.972	0.979	0.984	0.988	0.991	0.994	0.995	0.997
2.95	0.975	0.982	0.986	0.990	0.993	0.995	0.996	0.997
3.00	0.978	0.984	0.988	0.992	0.994	0.996	0.997	0.998

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

WITH 80 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	21	22	23	24	25	26	27	28
1.00	0.141	0.142	0.143	0.144	0.145	0.146	0.147	0.148
1.05	0.178	0.181	0.183	0.185	0.187	0.190	0.192	0.193
1.10	0.221	0.225	0.228	0.232	0.235	0.239	0.242	0.245
1.15	0.267	0.272	0.277	0.282	0.287	0.292	0.297	0.302
1.20	0.316	0.323	0.330	0.336	0.343	0.349	0.355	0.361
1.25	0.366	0.375	0.384	0.392	0.400	0.408	0.416	0.423
1.30	0.418	0.429	0.439	0.448	0.458	0.467	0.476	0.485
1.35	0.470	0.482	0.493	0.504	0.515	0.525	0.535	0.545
1.40	0.521	0.534	0.546	0.558	0.570	0.582	0.593	0.603
1.45	0.570	0.584	0.598	0.610	0.623	0.635	0.646	0.658
1.50	0.617	0.632	0.646	0.659	0.672	0.684	0.696	0.708
1.55	0.662	0.677	0.691	0.704	0.717	0.730	0.741	0.753
1.60	0.703	0.718	0.732	0.746	0.758	0.770	0.782	0.793
1.65	0.741	0.755	0.769	0.783	0.795	0.807	0.818	0.828
1.70	0.775	0.789	0.803	0.816	0.828	0.839	0.849	0.859
1.75	0.806	0.820	0.833	0.845	0.856	0.866	0.876	0.885
1.80	0.834	0.847	0.859	0.870	0.880	0.890	0.899	0.907
1.85	0.858	0.870	0.882	0.892	0.901	0.910	0.918	0.925
1.90	0.879	0.891	0.901	0.911	0.919	0.927	0.934	0.940
1.95	0.898	0.909	0.918	0.926	0.934	0.941	0.947	0.953
2.00	0.914	0.924	0.932	0.940	0.947	0.953	0.958	0.963
2.05	0.928	0.937	0.944	0.951	0.957	0.962	0.967	0.971
2.10	0.940	0.948	0.954	0.960	0.965	0.970	0.974	0.977
2.15	0.950	0.957	0.963	0.968	0.972	0.976	0.979	0.982
2.20	0.958	0.964	0.970	0.974	0.978	0.981	0.984	0.986
2.25	0.966	0.971	0.975	0.979	0.982	0.985	0.987	0.989
2.30	0.972	0.976	0.980	0.983	0.986	0.988	0.990	0.992
2.35	0.977	0.981	0.984	0.987	0.989	0.991	0.992	0.994
2.40	0.981	0.984	0.987	0.989	0.991	0.993	0.994	0.995
2.45	0.984	0.987	0.990	0.992	0.993	0.994	0.996	0.996
2.50	0.987	0.990	0.992	0.993	0.995	0.996	0.997	0.997
2.55	0.989	0.992	0.993	0.995	0.996	0.997	0.997	0.998
2.60	0.991	0.993	0.995	0.996	0.997	0.997	0.998	0.998
2.65	0.993	0.995	0.996	0.997	0.997	0.998	0.998	0.999
2.70	0.994	0.996	0.997	0.997	0.998	0.998	0.999	0.999
2.75	0.995	0.996	0.997	0.998	0.998	0.999	0.999	0.999
2.80	0.996	0.997	0.998	0.998	0.999	0.999	0.999	0.999
2.85	0.997	0.998	0.998	0.999	0.999	0.999	0.999	1.000
2.90	0.997	0.998	0.999	0.999	0.999	0.999	1.000	1.000
2.95	0.998	0.999	0.999	0.999	0.999	1.000	1.000	1.000
3.00	0.998	0.999	0.999	0.999	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 80 PERCENT CONFIDENCE EXPECTED NUMBER OF FAILURES

M(T)/TR	29	30	31	32	33	34	35	36
1.00	0.149	0.149	0.150	0.151	0.151	0.152	0.153	0.153
1.05	0.195	0.197	0.199	0.201	0.202	0.204	0.206	0.207
1.10	0.248	0.251	0.254	0.257	0.260	0.263	0.266	0.269
1.15	0.306	0.311	0.315	0.319	0.324	0.328	0.332	0.336
1.20	0.367	0.373	0.379	0.385	0.390	0.396	0.401	0.407
1.25	0.430	0.438	0.445	0.452	0.459	0.465	0.472	0.478
1.30	0.494	0.502	0.510	0.518	0.526	0.534	0.542	0.549
1.35	0.555	0.564	0.574	0.582	0.591	0.600	0.608	0.616
1.40	0.614	0.624	0.633	0.643	0.652	0.661	0.670	0.678
1.45	0.668	0.679	0.689	0.699	0.708	0.717	0.726	0.735
1.50	0.718	0.729	0.739	0.749	0.758	0.767	0.776	0.784
1.55	0.764	0.774	0.784	0.793	0.802	0.811	0.819	0.827
1.60	0.803	0.813	0.823	0.832	0.840	0.848	0.856	0.863
1.65	0.838	0.847	0.856	0.864	0.872	0.879	0.886	0.893
1.70	0.868	0.876	0.884	0.892	0.899	0.905	0.911	0.917
1.75	0.893	0.901	0.908	0.915	0.921	0.926	0.932	0.937
1.80	0.914	0.921	0.927	0.933	0.938	0.943	0.948	0.952
1.85	0.932	0.938	0.943	0.948	0.953	0.957	0.961	0.964
1.90	0.946	0.951	0.956	0.960	0.964	0.967	0.971	0.973
1.95	0.957	0.962	0.966	0.969	0.973	0.976	0.978	0.980
2.00	0.967	0.971	0.974	0.977	0.979	0.982	0.984	0.986
2.05	0.974	0.977	0.980	0.983	0.985	0.987	0.988	0.990
2.10	0.980	0.983	0.985	0.987	0.989	0.990	0.991	0.993
2.15	0.985	0.987	0.989	0.990	0.992	0.993	0.994	0.995
2.20	0.988	0.990	0.991	0.993	0.994	0.995	0.996	0.996
2.25	0.991	0.992	0.994	0.995	0.995	0.996	0.997	0.997
2.30	0.993	0.994	0.995	0.996	0.997	0.997	0.998	0.998
2.35	0.995	0.996	0.997	0.997	0.998	0.998	0.998	0.999
2.40	0.996	0.997	0.997	0.998	0.998	0.999	0.999	0.999
2.45	0.997	0.998	0.998	0.998	0.999	0.999	0.999	0.999
2.50	0.998	0.998	0.999	0.999	0.999	0.999	0.999	1.000
2.55	0.998	0.999	0.999	0.999	0.999	1.000	1.000	1.000
2.60	0.999	0.999	0.999	0.999	1.000	1.000	1.000	1.000
2.65	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
2.70	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000
2.75	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

WITH 80 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	37	38	39	40	41	42	43	44
1.00	0.154	0.154	0.155	0.156	0.156	0.157	0.157	0.157
1.05	0.209	0.210	0.212	0.213	0.214	0.216	0.217	0.218
1.10	0.271	0.274	0.276	0.279	0.281	0.284	0.286	0.289
1.15	0.340	0.344	0.347	0.351	0.355	0.359	0.362	0.366
1.20	0.412	0.417	0.422	0.427	0.432	0.437	0.441	0.446
1.25	0.485	0.491	0.497	0.503	0.509	0.515	0.521	0.526
1.30	0.556	0.563	0.570	0.577	0.584	0.591	0.597	0.604
1.35	0.624	0.632	0.639	0.647	0.654	0.661	0.668	0.675
1.40	0.687	0.695	0.703	0.710	0.718	0.725	0.732	0.739
1.45	0.743	0.751	0.759	0.766	0.773	0.781	0.787	0.794
1.50	0.792	0.800	0.807	0.814	0.821	0.828	0.834	0.841
1.55	0.834	0.842	0.848	0.855	0.861	0.867	0.873	0.879
1.60	0.870	0.876	0.883	0.888	0.894	0.899	0.905	0.909
1.65	0.899	0.905	0.910	0.915	0.920	0.925	0.929	0.933
1.70	0.923	0.928	0.932	0.937	0.941	0.945	0.948	0.952
1.75	0.941	0.946	0.949	0.953	0.957	0.960	0.963	0.965
1.80	0.956	0.960	0.963	0.966	0.969	0.971	0.973	0.976
1.85	0.967	0.970	0.973	0.975	0.977	0.979	0.981	0.983
1.90	0.976	0.978	0.980	0.982	0.984	0.986	0.987	0.988
1.95	0.982	0.984	0.986	0.987	0.989	0.990	0.991	0.992
2.00	0.987	0.989	0.990	0.991	0.992	0.993	0.994	0.995
2.05	0.991	0.992	0.993	0.994	0.995	0.995	0.996	0.996
2.10	0.994	0.994	0.995	0.996	0.996	0.997	0.997	0.998
2.15	0.995	0.996	0.997	0.997	0.997	0.998	0.998	0.998
2.20	0.997	0.997	0.998	0.998	0.998	0.999	0.999	0.999
2.25	0.998	0.998	0.998	0.999	0.999	0.999	0.999	0.999
2.30	0.998	0.999	0.999	0.999	0.999	0.999	0.999	1.000
2.35	0.999	0.999	0.999	0.999	0.999	1.000	1.000	1.000
2.40	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
2.45	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 80 PERCENT CONFIDENCE EXPECTED NUMBER OF FAILURES

M(T)/TR	45	46	47	48	49	50	51	52
1.00	0.158	0.158	0.159	0.159	0.160	0.160	0.160	0.161
1.05	0.220	0.221	0.222	0.224	0.225	0.226	0.227	0.228
1.10	0.291	0.294	0.296	0.298	0.300	0.303	0.305	0.307
1.15	0.369	0.373	0.376	0.380	0.383	0.386	0.390	0.393
1.20	0.451	0.455	0.460	0.464	0.469	0.473	0.478	0.482
1.25	0.532	0.538	0.543	0.548	0.554	0.559	0.564	0.569
1.30	0.610	0.616	0.622	0.628	0.634	0.640	0.645	0.651
1.35	0.681	0.688	0.694	0.701	0.707	0.713	0.718	0.724
1.40	0.745	0.752	0.758	0.764	0.770	0.776	0.782	0.787
1.45	0.800	0.807	0.813	0.818	0.824	0.829	0.835	0.840
1.50	0.846	0.852	0.858	0.863	0.868	0.873	0.878	0.882
1.55	0.884	0.889	0.894	0.899	0.903	0.907	0.911	0.915
1.60	0.914	0.918	0.922	0.926	0.930	0.934	0.937	0.940
1.65	0.937	0.941	0.944	0.947	0.950	0.953	0.956	0.959
1.70	0.955	0.958	0.960	0.963	0.965	0.968	0.970	0.972
1.75	0.968	0.970	0.972	0.974	0.976	0.978	0.980	0.981
1.80	0.978	0.979	0.981	0.983	0.984	0.985	0.986	0.988
1.85	0.984	0.986	0.987	0.988	0.989	0.990	0.991	0.992
1.90	0.989	0.990	0.991	0.992	0.993	0.994	0.994	0.995
1.95	0.993	0.994	0.994	0.995	0.995	0.996	0.996	0.997
2.00	0.995	0.996	0.996	0.997	0.997	0.997	0.998	0.998
2.05	0.997	0.997	0.998	0.998	0.998	0.998	0.999	0.999
2.10	0.998	0.998	0.998	0.999	0.999	0.999	0.999	0.999
2.15	0.999	0.999	0.999	0.999	0.999	0.999	0.999	1.000
2.20	0.999	0.999	0.999	0.999	1.000	1.000	1.000	1.000
2.25	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 80 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	53	54	55	56	57	58	59	60
1.00	0.161	0.161	0.162	0.162	0.162	0.163	0.163	0.163
1.05	0.230	0.231	0.232	0.233	0.234	0.235	0.236	0.237
1.10	0.309	0.311	0.313	0.316	0.318	0.320	0.322	0.324
1.15	0.396	0.399	0.403	0.406	0.409	0.412	0.415	0.418
1.20	0.486	0.490	0.494	0.498	0.502	0.506	0.510	0.514
1.25	0.574	0.579	0.584	0.589	0.593	0.598	0.603	0.607
1.30	0.656	0.662	0.667	0.672	0.677	0.682	0.687	0.692
1.35	0.730	0.735	0.740	0.746	0.751	0.756	0.761	0.766
1.40	0.793	0.798	0.803	0.808	0.813	0.818	0.822	0.827
1.45	0.845	0.850	0.854	0.859	0.863	0.867	0.872	0.876
1.50	0.887	0.891	0.895	0.899	0.902	0.906	0.910	0.913
1.55	0.919	0.922	0.926	0.929	0.932	0.935	0.938	0.941
1.60	0.943	0.946	0.949	0.951	0.954	0.956	0.958	0.961
1.65	0.961	0.963	0.965	0.967	0.969	0.971	0.973	0.974
1.70	0.974	0.975	0.977	0.979	0.980	0.981	0.982	0.984
1.75	0.983	0.984	0.985	0.986	0.987	0.988	0.989	0.990
1.80	0.989	0.990	0.990	0.991	0.992	0.993	0.993	0.994
1.85	0.993	0.993	0.994	0.994	0.995	0.995	0.996	0.996
1.90	0.995	0.996	0.996	0.997	0.997	0.997	0.998	0.998
1.95	0.997	0.997	0.998	0.998	0.998	0.998	0.999	0.999
2.00	0.998	0.998	0.999	0.999	0.999	0.999	0.999	0.999
2.05	0.999	0.999	0.999	0.999	0.999	0.999	1.000	1.000
2.10	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 80 PERCENT CONFIDENCE EXPECTED NUMBER OF FAILURES

M(T)/TR	61	62	63	64	65	66	67	68
1.00	0.163	0.164	0.164	0.164	0.164	0.165	0.165	0.165
1.05	0.238	0.239	0.240	0.241	0.242	0.243	0.244	0.245
1.10	0.326	0.328	0.330	0.332	0.334	0.335	0.337	0.339
1.15	0.421	0.424	0.427	0.430	0.433	0.435	0.438	0.441
1.20	0.518	0.522	0.526	0.529	0.533	0.537	0.540	0.544
1.25	0.612	0.616	0.620	0.625	0.629	0.633	0.637	0.641
1.30	0.697	0.701	0.706	0.710	0.715	0.719	0.724	0.728
1.35	0.770	0.775	0.779	0.784	0.788	0.792	0.797	0.801
1.40	0.831	0.835	0.840	0.844	0.848	0.851	0.855	0.859
1.45	0.879	0.883	0.887	0.890	0.894	0.897	0.900	0.903
1.50	0.916	0.919	0.922	0.925	0.928	0.931	0.933	0.936
1.55	0.943	0.946	0.948	0.950	0.953	0.955	0.957	0.959
1.60	0.963	0.964	0.966	0.968	0.970	0.971	0.973	0.974
1.65	0.976	0.977	0.979	0.980	0.981	0.982	0.983	0.984
1.70	0.985	0.986	0.987	0.988	0.988	0.989	0.990	0.991
1.75	0.991	0.991	0.992	0.992	0.993	0.994	0.994	0.994
1.80	0.994	0.995	0.995	0.996	0.996	0.996	0.997	0.997
1.85	0.997	0.997	0.997	0.997	0.998	0.998	0.998	0.998
1.90	0.998	0.998	0.998	0.999	0.999	0.999	0.999	0.999
1.95	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
2.00	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 80 PERCENT CONFIDENCE EXPECTED NUMBER OF FAILURES

M(T)/TR	69	70	71	72	73	74	75	76
1.00	0.165	0.166	0.166	0.166	0.166	0.167	0.167	0.167
1.05	0.246	0.247	0.248	0.249	0.250	0.251	0.252	0.253
1.10	0.341	0.343	0.345	0.347	0.348	0.350	0.352	0.354
1.15	0.444	0.447	0.449	0.452	0.455	0.457	0.460	0.463
1.20	0.548	0.551	0.555	0.558	0.561	0.565	0.568	0.571
1.25	0.645	0.649	0.653	0.657	0.661	0.665	0.668	0.672
1.30	0.732	0.736	0.740	0.744	0.748	0.752	0.756	0.759
1.35	0.805	0.809	0.812	0.816	0.820	0.823	0.827	0.830
1.40	0.862	0.866	0.869	0.873	0.876	0.879	0.882	0.885
1.45	0.906	0.909	0.912	0.915	0.917	0.920	0.923	0.925
1.50	0.938	0.940	0.943	0.945	0.947	0.949	0.951	0.953
1.55	0.960	0.962	0.964	0.965	0.967	0.968	0.970	0.971
1.60	0.975	0.977	0.978	0.979	0.980	0.981	0.982	0.983
1.65	0.985	0.986	0.987	0.988	0.988	0.989	0.990	0.990
1.70	0.991	0.992	0.992	0.993	0.993	0.994	0.994	0.995
1.75	0.995	0.995	0.996	0.996	0.996	0.997	0.997	0.997
1.80	0.997	0.997	0.998	0.998	0.998	0.998	0.998	0.998
1.85	0.998	0.999	0.999	0.999	0.999	0.999	0.999	0.999
1.90	0.999	0.999	0.999	0.999	0.999	0.999	1.000	1.000
1.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 80 PERCENT CONFIDENCE EXPECTED NUMBER OF FAILURES

M(T)/TR	77	78	79	80	81	82	83	84
1.00	0.167	0.167	0.168	0.168	0.168	0.168	0.168	0.169
1.05	0.254	0.254	0.255	0.256	0.257	0.258	0.259	0.260
1.10	0.355	0.357	0.359	0.361	0.362	0.364	0.366	0.367
1.15	0.465	0.468	0.471	0.473	0.476	0.478	0.481	0.483
1.20	0.575	0.578	0.581	0.584	0.588	0.591	0.594	0.597
1.25	0.676	0.679	0.683	0.687	0.690	0.693	0.697	0.700
1.30	0.763	0.767	0.770	0.774	0.777	0.781	0.784	0.787
1.35	0.834	0.837	0.840	0.844	0.847	0.850	0.853	0.856
1.40	0.888	0.891	0.894	0.896	0.899	0.901	0.904	0.906
1.45	0.927	0.930	0.932	0.934	0.936	0.938	0.940	0.942
1.50	0.954	0.956	0.958	0.959	0.961	0.962	0.964	0.965
1.55	0.972	0.974	0.975	0.976	0.977	0.978	0.979	0.980
1.60	0.984	0.985	0.985	0.986	0.987	0.988	0.988	0.989
1.65	0.991	0.991	0.992	0.992	0.993	0.993	0.994	0.994
1.70	0.995	0.995	0.996	0.996	0.996	0.996	0.997	0.997
1.75	0.997	0.997	0.998	0.998	0.998	0.998	0.998	0.998
1.80	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
1.85	0.999	0.999	0.999	0.999	0.999	1.000	1.000	1.000
1.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 80 PERCENT CONFIDENCE EXPECTED NUMBER OF FAILURES

M(T)/TR	85	86	87	88	89	90	91	92
1.00	0.169	0.169	0.169	0.169	0.169	0.170	0.170	0.170
1.05	0.260	0.261	0.262	0.263	0.264	0.265	0.265	0.266
1.10	0.369	0.371	0.372	0.374	0.376	0.377	0.379	0.380
1.15	0.486	0.488	0.491	0.493	0.495	0.498	0.500	0.502
1.20	0.600	0.603	0.606	0.609	0.612	0.615	0.618	0.621
1.25	0.704	0.707	0.710	0.713	0.716	0.720	0.723	0.726
1.30	0.790	0.794	0.797	0.800	0.803	0.806	0.809	0.812
1.35	0.859	0.861	0.864	0.867	0.870	0.872	0.875	0.877
1.40	0.909	0.911	0.913	0.916	0.918	0.920	0.922	0.924
1.45	0.944	0.945	0.947	0.949	0.950	0.952	0.953	0.955
1.50	0.966	0.968	0.969	0.970	0.971	0.972	0.973	0.974
1.55	0.981	0.982	0.982	0.983	0.984	0.985	0.985	0.986
1.60	0.989	0.990	0.990	0.991	0.991	0.992	0.992	0.993
1.65	0.994	0.995	0.995	0.995	0.996	0.996	0.996	0.996
1.70	0.997	0.997	0.997	0.998	0.998	0.998	0.998	0.998
1.75	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
1.80	0.999	0.999	0.999	0.999	0.999	1.000	1.000	1.000
1.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 80 PERCENT CONFIDENCE EXPECTED NUMBER OF FAILURES

M(T)/TR	93	94	95	96	97	98	99	100
1.00	0.170	0.170	0.170	0.170	0.171	0.171	0.171	0.171
1.05	0.267	0.268	0.269	0.269	0.270	0.271	0.272	0.272
1.10	0.382	0.384	0.385	0.387	0.388	0.390	0.391	0.393
1.15	0.505	0.507	0.509	0.512	0.514	0.516	0.519	0.521
1.20	0.624	0.626	0.629	0.632	0.635	0.638	0.640	0.643
1.25	0.729	0.732	0.735	0.738	0.741	0.743	0.746	0.749
1.30	0.815	0.818	0.820	0.823	0.826	0.828	0.831	0.834
1.35	0.880	0.882	0.885	0.887	0.889	0.891	0.893	0.896
1.40	0.926	0.928	0.929	0.931	0.933	0.935	0.936	0.938
1.45	0.956	0.958	0.959	0.960	0.961	0.963	0.964	0.965
1.50	0.975	0.976	0.977	0.978	0.979	0.980	0.980	0.981
1.55	0.987	0.987	0.988	0.988	0.989	0.989	0.990	0.990
1.60	0.993	0.993	0.994	0.994	0.994	0.995	0.995	0.995
1.65	0.997	0.997	0.997	0.997	0.997	0.997	0.998	0.998
1.70	0.998	0.998	0.999	0.999	0.999	0.999	0.999	0.999
1.75	0.999	0.999	0.999	0.999	0.999	0.999	1.000	1.000
1.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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TABLE F.III. FOR 90 PERCENT CONFIDENCE

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 90 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	5	6	7	8	9	10	11	12
1.00	0.036	0.044	0.049	0.053	0.055	0.057	0.059	0.060
1.05	0.043	0.052	0.059	0.064	0.067	0.071	0.073	0.076
1.10	0.050	0.062	0.070	0.076	0.081	0.085	0.089	0.093
1.15	0.059	0.072	0.082	0.089	0.096	0.102	0.107	0.112
1.20	0.068	0.083	0.095	0.104	0.113	0.120	0.127	0.134
1.25	0.078	0.095	0.109	0.120	0.130	0.140	0.149	0.157
1.30	0.088	0.108	0.124	0.137	0.150	0.161	0.172	0.182
1.35	0.099	0.121	0.140	0.156	0.170	0.184	0.197	0.209
1.40	0.111	0.136	0.156	0.175	0.192	0.208	0.223	0.238
1.45	0.124	0.151	0.174	0.195	0.214	0.233	0.250	0.267
1.50	0.136	0.166	0.192	0.216	0.238	0.258	0.278	0.298
1.55	0.150	0.183	0.211	0.237	0.262	0.285	0.307	0.329
1.60	0.164	0.199	0.231	0.260	0.287	0.312	0.337	0.361
1.65	0.178	0.217	0.251	0.282	0.312	0.340	0.367	0.393
1.70	0.193	0.234	0.271	0.305	0.337	0.368	0.397	0.425
1.75	0.208	0.252	0.292	0.329	0.363	0.396	0.427	0.456
1.80	0.223	0.270	0.313	0.352	0.389	0.424	0.457	0.488
1.85	0.238	0.289	0.334	0.376	0.415	0.451	0.486	0.518
1.90	0.254	0.307	0.355	0.399	0.440	0.479	0.515	0.548
1.95	0.270	0.326	0.376	0.423	0.465	0.505	0.543	0.578
2.00	0.286	0.345	0.398	0.446	0.490	0.532	0.570	0.606
2.05	0.302	0.364	0.419	0.469	0.515	0.557	0.596	0.633
2.10	0.318	0.382	0.439	0.491	0.539	0.582	0.622	0.658
2.15	0.334	0.401	0.460	0.513	0.562	0.606	0.646	0.683
2.20	0.351	0.419	0.480	0.535	0.585	0.629	0.670	0.706
2.25	0.367	0.437	0.500	0.556	0.606	0.652	0.692	0.729
2.30	0.383	0.456	0.520	0.577	0.628	0.673	0.714	0.749
2.35	0.399	0.473	0.539	0.597	0.648	0.694	0.734	0.769
2.40	0.415	0.491	0.557	0.616	0.668	0.713	0.753	0.787
2.45	0.430	0.508	0.576	0.635	0.686	0.732	0.771	0.805
2.50	0.446	0.525	0.593	0.653	0.705	0.749	0.788	0.821
2.55	0.461	0.541	0.611	0.670	0.722	0.766	0.803	0.835
2.60	0.476	0.558	0.627	0.687	0.738	0.782	0.818	0.849
2.65	0.491	0.573	0.644	0.703	0.754	0.796	0.832	0.862
2.70	0.505	0.589	0.659	0.719	0.769	0.810	0.845	0.874
2.75	0.520	0.604	0.674	0.733	0.783	0.824	0.857	0.885
2.80	0.534	0.618	0.689	0.748	0.796	0.836	0.868	0.895
2.85	0.547	0.633	0.703	0.761	0.809	0.847	0.879	0.904
2.90	0.561	0.646	0.717	0.774	0.821	0.858	0.889	0.913
2.95	0.574	0.660	0.730	0.786	0.832	0.868	0.897	0.920
3.00	0.587	0.673	0.742	0.798	0.842	0.878	0.906	0.927

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT
WITH 90 PERCENT CONFIDENCE
EXPECTED NUMBER OF FAILURES

M(T)/TR	13	14	15	16	17	18	19	20
1.00	0.062	0.063	0.064	0.065	0.066	0.066	0.067	0.068
1.05	0.078	0.080	0.082	0.083	0.085	0.087	0.088	0.090
1.10	0.096	0.099	0.102	0.105	0.108	0.111	0.113	0.116
1.15	0.117	0.121	0.126	0.130	0.134	0.138	0.142	0.146
1.20	0.140	0.146	0.152	0.158	0.164	0.169	0.174	0.180
1.25	0.165	0.173	0.181	0.188	0.196	0.203	0.210	0.217
1.30	0.192	0.202	0.212	0.221	0.230	0.239	0.248	0.257
1.35	0.221	0.233	0.245	0.256	0.267	0.278	0.289	0.300
1.40	0.252	0.266	0.280	0.293	0.306	0.319	0.331	0.344
1.45	0.284	0.300	0.316	0.331	0.346	0.361	0.375	0.389
1.50	0.317	0.335	0.353	0.370	0.387	0.403	0.419	0.435
1.55	0.350	0.370	0.390	0.409	0.428	0.446	0.463	0.480
1.60	0.384	0.406	0.427	0.448	0.468	0.488	0.507	0.525
1.65	0.418	0.442	0.465	0.487	0.508	0.529	0.549	0.568
1.70	0.451	0.477	0.502	0.525	0.548	0.569	0.590	0.610
1.75	0.485	0.512	0.537	0.562	0.585	0.608	0.629	0.649
1.80	0.517	0.546	0.572	0.598	0.622	0.644	0.666	0.686
1.85	0.549	0.578	0.606	0.632	0.656	0.679	0.701	0.721
1.90	0.580	0.610	0.638	0.664	0.689	0.711	0.733	0.753
1.95	0.610	0.640	0.669	0.695	0.719	0.742	0.763	0.782
2.00	0.639	0.669	0.697	0.723	0.747	0.770	0.790	0.809
2.05	0.666	0.696	0.725	0.750	0.774	0.795	0.815	0.833
2.10	0.692	0.722	0.750	0.775	0.798	0.819	0.837	0.854
2.15	0.716	0.746	0.773	0.798	0.820	0.840	0.858	0.873
2.20	0.739	0.769	0.795	0.819	0.840	0.859	0.876	0.891
2.25	0.761	0.790	0.816	0.838	0.858	0.876	0.892	0.906
2.30	0.781	0.809	0.834	0.856	0.875	0.892	0.906	0.919
2.35	0.800	0.827	0.851	0.872	0.890	0.905	0.919	0.930
2.40	0.818	0.844	0.866	0.886	0.903	0.917	0.930	0.940
2.45	0.834	0.859	0.881	0.899	0.915	0.928	0.940	0.949
2.50	0.849	0.873	0.893	0.911	0.925	0.938	0.948	0.957
2.55	0.863	0.886	0.905	0.921	0.935	0.946	0.955	0.963
2.60	0.875	0.897	0.915	0.930	0.943	0.953	0.962	0.969
2.65	0.887	0.908	0.925	0.939	0.950	0.960	0.967	0.974
2.70	0.898	0.917	0.933	0.946	0.957	0.965	0.972	0.978
2.75	0.907	0.926	0.941	0.953	0.962	0.970	0.976	0.981
2.80	0.916	0.934	0.947	0.958	0.967	0.974	0.980	0.984
2.85	0.924	0.941	0.953	0.964	0.972	0.978	0.983	0.987
2.90	0.932	0.947	0.959	0.968	0.975	0.981	0.985	0.989
2.95	0.938	0.952	0.963	0.972	0.979	0.984	0.988	0.991
3.00	0.944	0.958	0.968	0.975	0.981	0.986	0.989	0.992

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

WITH 90 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	21	22	23	24	25	26	27	28
1.00	0.068	0.069	0.070	0.070	0.071	0.071	0.072	0.072
1.05	0.091	0.093	0.094	0.095	0.096	0.098	0.099	0.100
1.10	0.118	0.121	0.123	0.125	0.127	0.130	0.132	0.134
1.15	0.150	0.153	0.157	0.160	0.164	0.167	0.170	0.174
1.20	0.185	0.190	0.195	0.200	0.205	0.209	0.214	0.219
1.25	0.224	0.230	0.237	0.243	0.250	0.256	0.262	0.269
1.30	0.266	0.274	0.282	0.290	0.298	0.306	0.314	0.322
1.35	0.310	0.320	0.330	0.340	0.349	0.359	0.368	0.378
1.40	0.356	0.368	0.379	0.391	0.402	0.413	0.424	0.434
1.45	0.403	0.416	0.429	0.442	0.455	0.467	0.479	0.491
1.50	0.450	0.465	0.479	0.494	0.507	0.521	0.534	0.547
1.55	0.497	0.513	0.528	0.544	0.558	0.573	0.587	0.600
1.60	0.542	0.560	0.576	0.592	0.607	0.622	0.637	0.650
1.65	0.587	0.604	0.621	0.638	0.654	0.669	0.683	0.697
1.70	0.629	0.647	0.664	0.681	0.697	0.712	0.726	0.740
1.75	0.668	0.687	0.704	0.721	0.736	0.751	0.765	0.779
1.80	0.706	0.724	0.741	0.757	0.772	0.787	0.800	0.813
1.85	0.740	0.758	0.774	0.790	0.805	0.818	0.831	0.843
1.90	0.771	0.789	0.805	0.820	0.833	0.846	0.858	0.869
1.95	0.800	0.816	0.832	0.846	0.859	0.871	0.882	0.892
2.00	0.826	0.841	0.856	0.869	0.881	0.892	0.902	0.911
2.05	0.849	0.864	0.877	0.889	0.900	0.910	0.919	0.927
2.10	0.869	0.883	0.896	0.907	0.917	0.926	0.934	0.941
2.15	0.888	0.900	0.912	0.922	0.931	0.939	0.946	0.952
2.20	0.904	0.915	0.926	0.935	0.943	0.950	0.956	0.962
2.25	0.918	0.928	0.938	0.946	0.953	0.959	0.965	0.969
2.30	0.930	0.940	0.948	0.955	0.961	0.967	0.972	0.976
2.35	0.940	0.949	0.957	0.963	0.968	0.973	0.977	0.981
2.40	0.950	0.957	0.964	0.970	0.974	0.978	0.982	0.985
2.45	0.957	0.964	0.970	0.975	0.979	0.983	0.985	0.988
2.50	0.964	0.970	0.975	0.980	0.983	0.986	0.988	0.990
2.55	0.970	0.975	0.980	0.983	0.986	0.989	0.991	0.993
2.60	0.975	0.979	0.983	0.986	0.989	0.991	0.993	0.994
2.65	0.979	0.983	0.986	0.989	0.991	0.993	0.994	0.995
2.70	0.982	0.986	0.989	0.991	0.993	0.994	0.996	0.996
2.75	0.985	0.988	0.991	0.993	0.994	0.996	0.996	0.997
2.80	0.988	0.990	0.992	0.994	0.995	0.996	0.997	0.998
2.85	0.990	0.992	0.994	0.995	0.996	0.997	0.998	0.998
2.90	0.991	0.993	0.995	0.996	0.997	0.998	0.998	0.999
2.95	0.993	0.995	0.996	0.997	0.998	0.998	0.999	0.999
3.00	0.994	0.996	0.997	0.998	0.998	0.999	0.999	0.999

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT
WITH 90 PERCENT CONFIDENCE
EXPECTED NUMBER OF FAILURES

M(T)/TR	29	30	31	32	33	34	35	36
1.00	0.072	0.073	0.073	0.074	0.074	0.074	0.075	0.075
1.05	0.101	0.102	0.103	0.104	0.105	0.106	0.107	0.108
1.10	0.136	0.138	0.140	0.142	0.144	0.146	0.147	0.149
1.15	0.177	0.180	0.183	0.186	0.189	0.192	0.195	0.198
1.20	0.223	0.228	0.232	0.237	0.241	0.246	0.250	0.254
1.25	0.275	0.281	0.287	0.292	0.298	0.304	0.310	0.315
1.30	0.329	0.337	0.344	0.352	0.359	0.366	0.373	0.380
1.35	0.387	0.396	0.404	0.413	0.422	0.430	0.439	0.447
1.40	0.445	0.455	0.465	0.475	0.485	0.495	0.504	0.513
1.45	0.503	0.514	0.525	0.536	0.547	0.557	0.568	0.578
1.50	0.559	0.571	0.583	0.595	0.606	0.617	0.628	0.638
1.55	0.613	0.626	0.638	0.650	0.662	0.673	0.684	0.695
1.60	0.664	0.677	0.689	0.701	0.713	0.724	0.735	0.745
1.65	0.711	0.723	0.736	0.748	0.759	0.770	0.780	0.790
1.70	0.753	0.766	0.778	0.789	0.800	0.810	0.820	0.829
1.75	0.791	0.803	0.815	0.825	0.835	0.845	0.854	0.863
1.80	0.825	0.836	0.847	0.857	0.866	0.875	0.883	0.891
1.85	0.854	0.865	0.874	0.883	0.892	0.900	0.907	0.914
1.90	0.880	0.889	0.898	0.906	0.913	0.920	0.927	0.933
1.95	0.901	0.910	0.918	0.925	0.931	0.937	0.943	0.948
2.00	0.919	0.927	0.934	0.940	0.946	0.951	0.956	0.960
2.05	0.935	0.941	0.947	0.953	0.958	0.962	0.966	0.970
2.10	0.947	0.953	0.958	0.963	0.967	0.971	0.974	0.977
2.15	0.958	0.963	0.967	0.971	0.975	0.978	0.980	0.983
2.20	0.967	0.971	0.974	0.978	0.981	0.983	0.985	0.987
2.25	0.973	0.977	0.980	0.983	0.985	0.987	0.989	0.991
2.30	0.979	0.982	0.985	0.987	0.989	0.990	0.992	0.993
2.35	0.984	0.986	0.988	0.990	0.992	0.993	0.994	0.995
2.40	0.987	0.989	0.991	0.992	0.994	0.995	0.996	0.996
2.45	0.990	0.992	0.993	0.994	0.995	0.996	0.997	0.997
2.50	0.992	0.994	0.995	0.996	0.996	0.997	0.998	0.998
2.55	0.994	0.995	0.996	0.997	0.997	0.998	0.998	0.999
2.60	0.995	0.996	0.997	0.998	0.998	0.998	0.999	0.999
2.65	0.996	0.997	0.998	0.998	0.999	0.999	0.999	0.999
2.70	0.997	0.998	0.998	0.999	0.999	0.999	0.999	0.999
2.75	0.998	0.998	0.999	0.999	0.999	0.999	1.000	1.000
2.80	0.998	0.999	0.999	0.999	0.999	1.000	1.000	1.000
2.85	0.999	0.999	0.999	0.999	1.000	1.000	1.000	1.000
2.90	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
2.95	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000
3.00	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

WITH 90 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	37	38	39	40	41	42	43	44
1.00	0.075	0.075	0.076	0.076	0.076	0.077	0.077	0.077
1.05	0.109	0.110	0.111	0.112	0.113	0.113	0.114	0.115
1.10	0.151	0.153	0.155	0.156	0.158	0.160	0.162	0.163
1.15	0.201	0.204	0.207	0.210	0.213	0.215	0.218	0.221
1.20	0.258	0.262	0.267	0.271	0.275	0.279	0.283	0.287
1.25	0.321	0.326	0.332	0.337	0.343	0.348	0.353	0.358
1.30	0.387	0.394	0.401	0.407	0.414	0.421	0.427	0.433
1.35	0.455	0.463	0.471	0.479	0.486	0.494	0.501	0.509
1.40	0.522	0.531	0.540	0.549	0.557	0.566	0.574	0.582
1.45	0.587	0.597	0.606	0.616	0.625	0.633	0.642	0.651
1.50	0.649	0.659	0.668	0.678	0.687	0.696	0.705	0.713
1.55	0.705	0.715	0.724	0.734	0.743	0.752	0.760	0.768
1.60	0.755	0.765	0.774	0.783	0.792	0.800	0.808	0.816
1.65	0.800	0.809	0.818	0.826	0.834	0.842	0.849	0.856
1.70	0.838	0.847	0.855	0.862	0.870	0.877	0.883	0.889
1.75	0.871	0.878	0.886	0.892	0.899	0.905	0.911	0.916
1.80	0.898	0.905	0.911	0.917	0.923	0.928	0.933	0.937
1.85	0.920	0.926	0.932	0.937	0.941	0.946	0.950	0.954
1.90	0.938	0.943	0.948	0.952	0.956	0.960	0.963	0.966
1.95	0.953	0.957	0.961	0.964	0.967	0.970	0.973	0.976
2.00	0.964	0.967	0.971	0.974	0.976	0.978	0.981	0.983
2.05	0.973	0.976	0.978	0.981	0.983	0.984	0.986	0.988
2.10	0.980	0.982	0.984	0.986	0.987	0.989	0.990	0.991
2.15	0.985	0.987	0.988	0.990	0.991	0.992	0.993	0.994
2.20	0.989	0.990	0.992	0.993	0.994	0.994	0.995	0.996
2.25	0.992	0.993	0.994	0.995	0.996	0.996	0.997	0.997
2.30	0.994	0.995	0.996	0.996	0.997	0.997	0.998	0.998
2.35	0.996	0.996	0.997	0.997	0.998	0.998	0.998	0.999
2.40	0.997	0.997	0.998	0.998	0.998	0.999	0.999	0.999
2.45	0.998	0.998	0.998	0.999	0.999	0.999	0.999	0.999
2.50	0.998	0.999	0.999	0.999	0.999	0.999	1.000	1.000
2.55	0.999	0.999	0.999	0.999	1.000	1.000	1.000	1.000
2.60	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
2.65	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT WITH 90 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	45	46	47	48	49	50	51	52
1.00	0.077	0.077	0.078	0.078	0.078	0.078	0.078	0.079
1.05	0.116	0.117	0.117	0.115	0.119	0.120	0.121	0.121
1.10	0.165	0.167	0.168	0.170	0.171	0.173	0.175	0.176
1.15	0.224	0.226	0.229	0.232	0.234	0.237	0.240	0.242
1.20	0.291	0.295	0.298	0.302	0.306	0.310	0.314	0.317
1.25	0.364	0.369	0.374	0.379	0.384	0.389	0.394	0.399
1.30	0.440	0.446	0.452	0.458	0.464	0.470	0.476	0.482
1.35	0.516	0.523	0.530	0.537	0.544	0.551	0.558	0.564
1.40	0.590	0.598	0.605	0.613	0.620	0.627	0.635	0.642
1.45	0.659	0.667	0.675	0.683	0.690	0.698	0.705	0.712
1.50	0.721	0.729	0.737	0.745	0.752	0.759	0.767	0.773
1.55	0.776	0.784	0.792	0.799	0.806	0.813	0.819	0.825
1.60	0.824	0.831	0.838	0.844	0.851	0.857	0.863	0.868
1.65	0.863	0.870	0.876	0.882	0.887	0.892	0.898	0.902
1.70	0.895	0.901	0.906	0.911	0.916	0.921	0.925	0.929
1.75	0.921	0.926	0.931	0.935	0.939	0.943	0.946	0.949
1.80	0.941	0.945	0.949	0.953	0.956	0.959	0.962	0.964
1.85	0.957	0.960	0.963	0.966	0.969	0.971	0.973	0.975
1.90	0.969	0.972	0.974	0.976	0.978	0.980	0.982	0.983
1.95	0.978	0.980	0.982	0.983	0.985	0.986	0.988	0.989
2.00	0.984	0.986	0.987	0.989	0.990	0.991	0.992	0.993
2.05	0.989	0.990	0.991	0.992	0.993	0.994	0.994	0.995
2.10	0.992	0.993	0.994	0.995	0.995	0.996	0.996	0.997
2.15	0.995	0.995	0.996	0.996	0.997	0.997	0.998	0.998
2.20	0.996	0.997	0.997	0.998	0.998	0.998	0.998	0.999
2.25	0.998	0.998	0.998	0.998	0.999	0.999	0.999	0.999
2.30	0.998	0.999	0.999	0.999	0.999	0.999	0.999	0.999
2.35	0.999	0.999	0.999	0.999	0.999	1.000	1.000	1.000
2.40	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT
WITH 90 PERCENT CONFIDENCE
EXPECTED NUMBER OF FAILURES

M(T)/TR	53	54	55	56	57	58	59	60
1.00	0.079	0.079	0.079	0.079	0.079	0.080	0.080	0.080
1.05	0.122	0.123	0.123	0.124	0.125	0.126	0.126	0.127
1.10	0.178	0.179	0.181	0.182	0.184	0.185	0.187	0.188
1.15	0.245	0.247	0.250	0.252	0.255	0.257	0.260	0.262
1.20	0.321	0.325	0.328	0.332	0.336	0.339	0.343	0.346
1.25	0.404	0.408	0.413	0.418	0.422	0.427	0.432	0.436
1.30	0.488	0.494	0.499	0.505	0.511	0.516	0.522	0.527
1.35	0.571	0.577	0.584	0.590	0.596	0.602	0.608	0.614
1.40	0.649	0.655	0.662	0.669	0.675	0.681	0.687	0.694
1.45	0.719	0.726	0.732	0.739	0.745	0.751	0.757	0.763
1.50	0.780	0.786	0.793	0.799	0.805	0.811	0.816	0.822
1.55	0.832	0.838	0.843	0.849	0.854	0.859	0.864	0.869
1.60	0.874	0.879	0.884	0.889	0.893	0.898	0.902	0.906
1.65	0.907	0.912	0.916	0.920	0.924	0.927	0.931	0.934
1.70	0.933	0.937	0.940	0.943	0.947	0.950	0.952	0.955
1.75	0.953	0.955	0.958	0.961	0.963	0.966	0.968	0.970
1.80	0.967	0.969	0.971	0.973	0.975	0.977	0.979	0.980
1.85	0.977	0.979	0.981	0.982	0.984	0.985	0.986	0.987
1.90	0.985	0.986	0.987	0.988	0.989	0.990	0.991	0.992
1.95	0.990	0.991	0.992	0.992	0.993	0.994	0.994	0.995
2.00	0.993	0.994	0.995	0.995	0.996	0.996	0.996	0.997
2.05	0.996	0.996	0.997	0.997	0.997	0.998	0.998	0.998
2.10	0.997	0.998	0.998	0.998	0.998	0.999	0.999	0.999
2.15	0.998	0.998	0.999	0.999	0.999	0.999	0.999	0.999
2.20	0.999	0.999	0.999	0.999	0.999	0.999	1.000	1.000
2.25	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

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WITH 90 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	61	62	63	64	65	66	67	68
1.00	0.080	0.080	0.080	0.081	0.081	0.081	0.081	0.081
1.05	0.128	0.128	0.129	0.130	0.130	0.131	0.131	0.132
1.10	0.190	0.191	0.192	0.194	0.195	0.197	0.198	0.200
1.15	0.265	0.267	0.270	0.272	0.274	0.277	0.279	0.281
1.20	0.350	0.353	0.357	0.360	0.364	0.367	0.371	0.374
1.25	0.441	0.445	0.450	0.454	0.459	0.463	0.467	0.472
1.30	0.532	0.538	0.543	0.548	0.553	0.558	0.563	0.568
1.35	0.620	0.626	0.631	0.637	0.642	0.648	0.653	0.659
1.40	0.700	0.705	0.711	0.717	0.722	0.728	0.733	0.739
1.45	0.769	0.775	0.780	0.786	0.791	0.796	0.801	0.806
1.50	0.827	0.832	0.837	0.842	0.847	0.852	0.856	0.860
1.55	0.874	0.878	0.883	0.887	0.891	0.895	0.899	0.902
1.60	0.910	0.914	0.918	0.921	0.924	0.928	0.931	0.934
1.65	0.938	0.941	0.944	0.946	0.949	0.951	0.954	0.956
1.70	0.958	0.960	0.962	0.964	0.966	0.968	0.970	0.972
1.75	0.972	0.974	0.975	0.977	0.978	0.980	0.981	0.982
1.80	0.982	0.983	0.984	0.985	0.986	0.987	0.988	0.989
1.85	0.988	0.989	0.990	0.991	0.991	0.992	0.993	0.993
1.90	0.993	0.993	0.994	0.994	0.995	0.995	0.996	0.996
1.95	0.995	0.996	0.996	0.997	0.997	0.997	0.997	0.998
2.00	0.997	0.997	0.998	0.998	0.998	0.998	0.999	0.999
2.05	0.998	0.998	0.999	0.999	0.999	0.999	0.999	0.999
2.10	0.999	0.999	0.999	0.999	0.999	0.999	1.000	1.000
2.15	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

PROBABILITY OF DEMONSTRATING TECHNICAL REQUIREMENT

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WITH 90 PERCENT CONFIDENCE

EXPECTED NUMBER OF FAILURES

M(T)/TR	69	70	71	72	73	74	75	76
1.00	0.081	0.081	0.081	0.082	0.082	0.082	0.082	0.082
1.05	0.133	0.133	0.134	0.135	0.135	0.136	0.136	0.137
1.10	0.201	0.202	0.204	0.205	0.206	0.208	0.209	0.210
1.15	0.284	0.286	0.289	0.291	0.293	0.295	0.298	0.300
1.20	0.377	0.381	0.384	0.387	0.391	0.394	0.397	0.400
1.25	0.476	0.480	0.484	0.488	0.493	0.497	0.501	0.505
1.30	0.573	0.578	0.583	0.588	0.592	0.597	0.602	0.606
1.35	0.664	0.669	0.674	0.679	0.684	0.689	0.694	0.698
1.40	0.744	0.749	0.754	0.759	0.763	0.768	0.773	0.777
1.45	0.811	0.816	0.820	0.825	0.829	0.833	0.837	0.842
1.50	0.865	0.869	0.873	0.877	0.880	0.884	0.888	0.891
1.55	0.906	0.909	0.913	0.916	0.919	0.922	0.925	0.928
1.60	0.937	0.939	0.942	0.944	0.947	0.949	0.951	0.953
1.65	0.958	0.960	0.962	0.964	0.966	0.968	0.969	0.971
1.70	0.973	0.975	0.976	0.978	0.979	0.980	0.981	0.982
1.75	0.983	0.984	0.985	0.986	0.987	0.988	0.989	0.990
1.80	0.990	0.991	0.991	0.992	0.992	0.993	0.993	0.994
1.85	0.994	0.994	0.995	0.995	0.996	0.996	0.996	0.997
1.90	0.996	0.997	0.997	0.997	0.998	0.998	0.998	0.998
1.95	0.998	0.998	0.998	0.998	0.999	0.999	0.999	0.999
2.00	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
2.05	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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EXPECTED NUMBER OF FAILURES

M(T)/TR	77	78	79	80	81	82	83	84
1.00	0.082	0.082	0.082	0.082	0.082	0.083	0.083	0.083
1.05	0.138	0.138	0.139	0.139	0.140	0.141	0.141	0.142
1.10	0.212	0.213	0.214	0.216	0.217	0.218	0.220	0.221
1.15	0.302	0.304	0.307	0.309	0.311	0.313	0.316	0.318
1.20	0.404	0.407	0.410	0.413	0.416	0.419	0.422	0.426
1.25	0.509	0.513	0.517	0.521	0.525	0.528	0.532	0.536
1.30	0.611	0.615	0.620	0.624	0.628	0.633	0.637	0.641
1.35	0.703	0.708	0.712	0.717	0.721	0.725	0.730	0.734
1.40	0.782	0.786	0.790	0.795	0.799	0.803	0.807	0.811
1.45	0.845	0.849	0.853	0.857	0.860	0.864	0.867	0.871
1.50	0.894	0.898	0.901	0.904	0.907	0.910	0.913	0.915
1.55	0.930	0.933	0.935	0.938	0.940	0.942	0.944	0.946
1.60	0.955	0.957	0.959	0.961	0.963	0.964	0.966	0.967
1.65	0.972	0.974	0.975	0.976	0.977	0.979	0.980	0.981
1.70	0.983	0.984	0.985	0.986	0.987	0.988	0.988	0.989
1.75	0.990	0.991	0.991	0.992	0.993	0.993	0.993	0.994
1.80	0.994	0.995	0.995	0.996	0.996	0.996	0.996	0.997
1.85	0.997	0.997	0.997	0.998	0.998	0.998	0.998	0.998
1.90	0.998	0.998	0.999	0.999	0.999	0.999	0.999	0.999
1.95	0.999	0.999	0.999	0.999	0.999	0.999	0.999	1.000
2.00	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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EXPECTED NUMBER OF FAILURES

M(T)/TR	85	86	87	88	89	90	91	92
1.00	0.083	0.083	0.083	0.083	0.083	0.083	0.083	0.083
1.05	0.142	0.143	0.143	0.144	0.144	0.145	0.146	0.146
1.10	0.222	0.224	0.225	0.226	0.227	0.229	0.230	0.231
1.15	0.320	0.322	0.324	0.326	0.329	0.331	0.333	0.335
1.20	0.429	0.432	0.435	0.438	0.441	0.444	0.447	0.450
1.25	0.540	0.544	0.547	0.551	0.555	0.558	0.562	0.565
1.30	0.645	0.649	0.653	0.657	0.661	0.665	0.669	0.673
1.35	0.738	0.742	0.746	0.750	0.754	0.758	0.762	0.765
1.40	0.815	0.818	0.822	0.826	0.829	0.833	0.836	0.840
1.45	0.874	0.877	0.880	0.884	0.887	0.889	0.892	0.895
1.50	0.918	0.920	0.923	0.925	0.928	0.930	0.932	0.934
1.55	0.948	0.950	0.952	0.954	0.956	0.957	0.959	0.961
1.60	0.969	0.970	0.971	0.973	0.974	0.975	0.976	0.977
1.65	0.982	0.983	0.984	0.984	0.985	0.986	0.987	0.987
1.70	0.990	0.990	0.991	0.991	0.992	0.992	0.993	0.993
1.75	0.994	0.995	0.995	0.995	0.996	0.996	0.996	0.997
1.80	0.997	0.997	0.997	0.998	0.998	0.998	0.998	0.998
1.85	0.998	0.999	0.999	0.999	0.999	0.999	0.999	0.999
1.90	0.999	0.999	0.999	0.999	0.999	1.000	1.000	1.000
1.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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EXPECTED NUMBER OF FAILURES

M(T)/TR	93	94	95	96	97	98	99	100
1.00	0.084	0.084	0.084	0.084	0.084	0.084	0.084	0.084
1.05	0.147	0.147	0.148	0.148	0.149	0.149	0.150	0.150
1.10	0.232	0.234	0.235	0.236	0.237	0.239	0.240	0.241
1.15	0.337	0.339	0.341	0.343	0.345	0.348	0.350	0.352
1.20	0.453	0.456	0.458	0.461	0.464	0.467	0.470	0.473
1.25	0.569	0.573	0.576	0.580	0.583	0.586	0.590	0.593
1.30	0.677	0.681	0.684	0.688	0.692	0.695	0.699	0.702
1.35	0.769	0.773	0.776	0.780	0.783	0.787	0.790	0.794
1.40	0.843	0.846	0.849	0.852	0.855	0.858	0.861	0.864
1.45	0.898	0.900	0.903	0.905	0.908	0.910	0.913	0.915
1.50	0.936	0.938	0.940	0.942	0.944	0.946	0.948	0.949
1.55	0.962	0.964	0.965	0.966	0.968	0.969	0.970	0.971
1.60	0.978	0.979	0.980	0.981	0.982	0.983	0.983	0.984
1.65	0.988	0.989	0.989	0.990	0.990	0.991	0.991	0.992
1.70	0.994	0.994	0.994	0.995	0.995	0.995	0.996	0.996
1.75	0.997	0.997	0.997	0.997	0.998	0.998	0.998	0.998
1.80	0.998	0.998	0.999	0.999	0.999	0.999	0.999	0.999
1.85	0.999	0.999	0.999	0.999	0.999	1.000	1.000	1.000
1.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.30	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.40	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.45	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.50	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.55	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.60	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.65	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.70	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.80	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.85	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.90	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3.00	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

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Appendix G ANNEX

G.1 Annex 1

We will show the following:

$$(1) \qquad E[I_i(t)] = 1 - e^{-\lambda_i t}$$

$$(2) \qquad \mu(t;\, \underline{\lambda}) = \sum_{i=1}^{K} (1 - e^{-\lambda_i t})$$

$$(3) \qquad h(t;\, \underline{\lambda}) = \sum_{i=1}^{K} \lambda_i e^{-\lambda_i t}$$

To show equation (1) observe that $I_i(t)$ is a random variable that only takes on the values zero and one. Thus

$$E[I_i(t)] = (0)\Pr(I_i(t) = 0) + (1)\Pr(I_i(t) = 1)$$
$$= \Pr(I_i(t) = 1) = 1 - e^{-\lambda_i t}$$

To show equation (2), let M(t) denote the number of distinct B-modes that occur by *t*. Then

$$M(t) = \sum_{i=1}^{K} I_i(t)$$

Thus

$$\mu(t;\, \underline{\lambda}) = E(M(t)) = \sum_{i=1}^{K} E[I_i(t)] = \sum_{i=1}^{K} (1 - e^{-\lambda_i t})$$

Note that equation (3) follows from equation (2) since

$$h(t;\, \underline{\lambda}) = \frac{d\mu(t;\, \underline{\lambda})}{dt} = \sum_{i=1}^{K} \lambda_i e^{-\lambda_i t}$$

G.2 Annex 2

Recall $\Lambda \sim \Gamma(\alpha, \beta)$. Let Ψ denote the moment generating function for Λ. Thus, by definition, $\Psi(x) = E(e^{x\Lambda})$ for all real x for which the expectation with respect to Λ exists. One can show that Ψ is defined for $x < \frac{1}{\beta}$ and $\Psi(x) = (1 - \beta x)^{-(\alpha+1)}$ (see e.g., Mood and Graybill [9]). We will utilize Ψ9x) to express $\lambda_{B,K}$, $\mu(t)$, $h(t)$, $\rho(t)$, and $\theta(t)$ in terms of K and the gamma parameters α and β. We summarize our results below:

$$(1) \quad \lambda_{B,K} = K\beta(\alpha + 1)$$

$$(2) \quad \mu(t) = K[1 - (1 + \beta t)^{-(\alpha+1)}]$$

$$(3) \quad h(t) = \frac{K\beta(\alpha + 1)}{(1 + \beta t)^{\alpha+2}} = \frac{d\mu(t)}{dt}$$

$$(4) \quad \rho(t) = \lambda_A + (1 - \mu_d)K\,\beta(\alpha + 1) + \frac{\mu_d K\beta(\alpha+1)}{(1+\beta t)^{\alpha+2}}$$

$$(5) \quad \theta(t) = 1 - (1 + \beta t)^{-(\alpha+2)}$$

To show (1), recall

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$$\lambda_{B,K} = E\left(\sum_{i=1}^{K} \Lambda_i\right)$$

Where $\underline{\Lambda} = (\Lambda_1, \cdots, \Lambda_K)$ is a random sample from Λ. Thus,

$$\lambda_{B,K} = K\,E(\Lambda) = K\left.\frac{d\Psi(x)}{dx}\right|_{x=0}$$
$$= K\beta(\alpha+1)$$

To demonstrate (2), note by Annex 1

$$\mu(t) = E[\mu(t;\,\underline{\Lambda})]$$
$$= E\left[\mu(t;\,\underline{\Lambda})\right]$$
$$= E\left[\sum_{i=1}^{K}(1-e^{-\Lambda_i t})\right]$$
$$= K - E\left\{\sum_{i=1}^{K} e^{-\Lambda_i t}\right\}$$
$$= K - KE[e^{-\Lambda t}]$$

Thus

$$\mu(t) = K\{1 - E[e^{-\Lambda t}]\}$$
$$= K\{1 - \Psi(-t)\}$$
$$= K\left\{1 - (1+\beta t)^{-(\alpha+1)}\right\}$$

To derive (3) we can utilize the expression for $h(t;\,\underline{\Lambda})$ in Annex 1. Doing so we arrive at

$$h(t) = E\left[h(t;\,\underline{\Lambda})\right]$$
$$= E\left[\sum_{i=1}^{K}\Lambda_i e^{-\Lambda_i t}\right]$$
$$= K\,E[\Lambda e^{-\Lambda t}]$$

Note

$$E[\Lambda e^{-\Lambda t}] = \left.\frac{d\Psi}{dx}\right|_{x=-t}$$
$$= \frac{\beta(\alpha+1)}{(1+\beta t)^{\alpha+2}}$$

This yields

$$h(t) = \frac{K\beta(\alpha+1)}{(1+\beta t)^{\alpha+2}}$$

Note by (2) above,

$$\frac{d\,\mu(t)}{d\,t} = \frac{K\beta(\alpha+1)}{(1+\beta t)^{\alpha+2}}$$

Thus, as expected,

$$\frac{d\,\mu(t)}{d\,t} = E[h(t;\,\Lambda)] = h(t)$$

To obtain (4) we recall the expression in (16) of Section 4.4.3 for $\rho(t)$:

$$\rho(t) = \lambda_A + (1-\mu_d)\lambda_{B,K} + \mu_d h(t)$$

Thus, (4) directly follows from (1) and (3) above.
Finally, recall by (23) of Section 4.4.3 we have

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$$\theta(t) = \frac{\lambda_{B,K} - h(t)}{\lambda_{B,K}}$$

By (1) and (3) above we note

$$h(t) = \frac{\lambda_{B,K}}{(1+\beta t)^{\alpha+2}}$$

Thus

$$\theta(t) = \frac{\lambda_{B,K} - \frac{\lambda_{B,K}}{(1+\beta t)^{\alpha+2}}}{\lambda_{B,K}} = 1 - (1+\beta t)^{-(\alpha+2)}$$

G.3 Annex 3

G.3.1 Maximum Likelihood Estimates for AMPM

To obtain maximum likelihood estimates (LME's) for our finite K and NHPP variants of the AMPM, assume m distinct B-modes first occur at test times $0 < t_1 \le t_2 \le \cdots \le t_m$ respectively over a test period of length T. Let n_A denote the number of A-mode failures that occur over test period T. We will denote an estimate of a model parameter by replacing the symbol "^" over the parameter. Thus, e.g., $\hat{\lambda}_A = {}^{n_A}/_{T}$ since λ_A is constant over test period T.

Let $\underline{t}$ be the vector of B-mode first occurrence times $(t_1, \cdots, t_m)$. Also, let ⟨K⟩ denote the set of positive integers less than or equal to K and let S_m denote the set of all subsets of ⟨K⟩ of size m. Then, conditioned on $\underline{\Lambda} = \underline{\lambda}$, the likelihood function for the test data $(m, \underline{t})$ is $L(m, \underline{t};\ \underline{\lambda})$ where

$$L(m, \underline{t};\ \underline{\lambda}) = m! \sum_{S \in S_m} \left[\prod_{i \in S} \lambda_i e^{-\lambda_i t_i} \prod_{i \in \langle K \rangle - S} e^{-\lambda_i T} \right] \qquad (1)$$

The summation in (1) is over all the mutually exclusive sets of exactly m distinct B-modes that can occur at first occurrence times $(t_1, \cdots, t_m)$.

Consider the corresponding likelihood random variable

$L(m, \underline{t}; \underline{\Lambda}) = m! \sum_{S \in S_m} \left[\prod_{i \in S} \Lambda_i e^{-\Lambda_i t_i} \prod_{i \in \langle K \rangle - S} e^{-\Lambda_i T} \right]$ (1)

Where $\underline{\Lambda} = (\Lambda_1, \cdots, \Lambda_K)$ is a random sample from $\Gamma(\alpha, \beta)$. Denote the expected value of $L(m, \underline{t};\ \underline{\Lambda})$ with respect to $\underline{\Lambda}$ by $L(m, \underline{t})$. Since Λ_i are independent and identically distributed for i=1,⋯, K we have

$$L(m, \underline{t}) = m! \binom{K}{m} \left[\prod_{i=1}^{m} E(\Lambda e^{-\Lambda t_i}) \right] [E(e^{-\Lambda T})]^{K-m} \qquad (2)$$

where $\Lambda \sim \Gamma(\alpha, \beta)$. We wish to find the point (α, β) that maximizes the $L(m, \underline{t})$. We will denote these values of α and β by $\hat{\alpha}_K$ and $\hat{\beta}_K$, respectively.

By direct calculation of $E[\Lambda^p e^{\mu\Lambda}]$, recalling the form of density function f_Λ given in Section 4.4.2, we can show

$$E[\Lambda^p e^{\mu\Lambda}] = \frac{(\alpha+p)!\,\beta^p}{\alpha!\,(1-\beta\mu)^{\alpha+1+p}} \qquad (3)$$

for $\mu < \beta^{-1}$ and $p > -(1+\alpha)$. From (2) and (3) we obtain

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$$L(m,\underline{t}) = K(K-1)\cdots(K-m+1)\left[\frac{\beta^m(\alpha+1)^m}{(1+\beta T)^{(\alpha+1)(K-m)}}\right]\prod_{i=1}^{m}(1+\beta t_i)^{-(\alpha+2)} \qquad (4)$$

Let $Z = \ln\{L(m,\underline{t})\}$. Then it follows that

$$\frac{\partial Z}{\partial \alpha} = \frac{m}{\alpha+1} - (K-m)\ln(1+\beta t) - \sum_{i=1}^{m}\ln(1+\beta t) \qquad (5)$$

and

$$\frac{\partial Z}{\partial \beta} = \frac{m}{\beta} - \frac{(\alpha+1)(K-m)T}{1+\beta T} - (\alpha+2)\sum_{i=1}^{m}\frac{t_i}{1+\beta t} \qquad (6)$$

Treating K as a positive real number we also obtain

$$\frac{\partial Z}{\partial K} = \sum_{i=0}^{m-1}\frac{1}{K-i} - (\alpha-1)\ln(1+\beta T) \qquad (7)$$

In Section 4.4 and this appendix we will not use (7) since we are only interested in obtaining $\hat{\alpha}_K$, $\hat{\beta}_K$ in terms of K and the test data. We will then hold the test data constant and let $K \to \infty$ to study the limiting behavior of our AMPM estimators. Let $\nu_K \triangleq (\alpha, \beta, K)$ and $\hat{\nu}_K \triangleq (\hat{\alpha}_K, \hat{\beta}_K, K)$. Then by (5) our maximum likelihood equation for α is

$$\left.\frac{\partial Z}{\partial \alpha}\right|_{\nu_K=\hat{\nu}_K} = 0 \Leftrightarrow (\hat{\alpha}_K+1)^{-1} = m^{-1}\left[K\ln(1+\hat{\beta}_K T) - \sum_{i=1}^{m}\left(\frac{1+\hat{\beta}_K T}{1+\hat{\beta}_K t_i}\right)\right] \qquad (8)$$

By (6) our maximum likelihood equation for β is

$$\left.\frac{\partial Z}{\partial \beta}\right|_{\nu_K=\hat{\nu}_K} = 0 \Leftrightarrow \hat{\alpha}_K + 1 = \frac{\frac{m}{\hat{\beta}_K} - \sum_{i=1}^{m}\frac{t_i}{1+\hat{\beta}_K t_i}}{\frac{(K-m)T}{1+\hat{\beta}_K T} + \sum_{i=1}^{m}\frac{t_i}{1+\hat{\beta}_K t_i}} \qquad (9)$$

Equating the expressions for $(\hat{\alpha}_K+1)^{-1}$ obtained from (8) and (9) we arrive at a linear equation for K. Solving for K we obtain

$$K = \frac{\sum_{i=1}^{m}\frac{1+\hat{\beta}_K T}{1+\hat{\beta}_K t_i}\sum_{i=1}^{m}\frac{1}{1+\hat{\beta}_K t_i} - \frac{m\hat{\beta}_K}{1+\beta_K T}\sum_{i=1}^{m}\frac{T-t_i}{1+\hat{\beta}_K t_i}}{\ln(1+\hat{\beta}_K T)\sum_{i=1}^{m}\frac{1}{1+\hat{\beta}_K t_i} - \frac{m\hat{\beta}_K}{1+\hat{\beta}_K T}T} \qquad (10)(10)$$

For a given K and data set $(m,\underline{t})$ generated over test period T we can solve (10) for $\hat{\beta}_K$. Then we can use either (8) or (9) to obtain $\hat{\alpha}_K$. Using $(\hat{\alpha}_K, \hat{\beta}_K, K)$ we can estimate all our finite K AMPM projection quantities where $\hat{\lambda}_A = {}^{n_A}/_{T}$ and μ_d is assessed as

$$\mu_d^* = \frac{1}{m}\sum_{i\epsilon obs} d_i^* \qquad (11)$$

In (11), the value of d_i^* will often be based largely on engineering judgment. The value of d_i^* should reflect several considerations: (1) how certain we are that the problem has been correctly identified; (2) the nature of the fix, e.g., its complexity; (3) past FEF experience and (4) any germane testing (including assembly level testing).

In practice, we do not know the value of K. We could try to develop an MLE for K based on (7) or by directly maximizing Z. We have found that a solution $(\hat{\alpha}, \hat{\beta}, \hat{K})$ to the maximum likelihood equations (5), (6), and (7) can be a saddle point of $L(m,\underline{t})$. This can occur even for a

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large data set that appears to fit the model well. We present graphs in Section 4.4.6 for such a data set that clearly illustrate the difficulty in obtaining a reasonable estimate for K. Thus we prefer to take the point of view that we should not attempt to assess K. However, by conducting a standard failure modes and effects criticality analysis (FMECA), we can place a lower bound on K, say K_l. Our experience with the AMPM is that if K is substantially higher than m, say, e.g., $K \geq 10m$, then our AMPM projection quantities will be insensitive to the value of K. We believe for a complex system or subsystem it will often be the case that $K_l \geq 10m$ or at least the unknown value of K will be 10m or higher. The factor of 10 may be larger than necessary. In practice, we suggest exercising the AMPM model with several plausible lower bound values for K and comparing the associated projections with those obtained in the limit as $K \to \infty$. This is illustrated for a data set in Section 4.4.6.

We now consider the behavior of our AMPM estimators as $K \to \infty$. To do so, let $\langle \hat{\beta}_K \rangle_{K>m}$ be a sequence satisfying (10) with limit $\hat{\beta}_\infty \in [0, \infty)$. We will assume that such a sequence exists for our data set $(m, \underline{t})$ generated over [0, T]. Then by (10) we have

$$\ln(1 + \hat{\beta}_\infty T) \sum_{i=1}^{m} \frac{1}{1 + \hat{\beta}_\infty t_i} - \frac{m\hat{\beta}_\infty T}{1 + \hat{\beta}_\infty T} = 0 \qquad (12)$$

Recall by Annex 2, $\lambda_{B,K} = K\beta(\alpha + 1)$, where we previously suppressed the subscript K. Thus, we define $\hat{\lambda}_{B,K}$ by

$$\hat{\lambda}_{B,K} \triangleq K\hat{\beta}(\hat{\alpha} + 1) \qquad (13)$$

By (8) we obtain

$$\hat{\lambda}_{B,K} = K\hat{\beta}(\hat{\alpha}_K + 1) = \frac{Km\hat{\beta}_K}{K\ln(1+\hat{\beta}_K T) - \sum_{i=1}^{m} \ln\frac{1+\hat{\beta}_K T}{1+\hat{\beta}_K t_i}} \qquad (14)$$

Taking the limit in (14) as $K \to \infty$ we arrive at

$$\hat{\lambda}_{B,K} \triangleq \lim_{K\to\infty} \hat{\lambda}_{B,K} = \frac{m\hat{\beta}_\infty}{\ln(1 + \hat{\beta}_\infty T)} \qquad (15)$$

provided $\hat{\beta}_\infty > 0$. If $\hat{\beta}_\infty = 0$, then we can show, by applying L'Hospital's rule, that the limit of the right hand side of (10) goes to a finite positive number as $K \to \infty$. This contradiction establishes that $\hat{\beta}_\infty > 0$. Since $K\hat{\beta}_K(\hat{\alpha}_K + 1) \to \hat{\lambda}_{B,\infty} \in (0, \infty)$ as $K \to \infty$ and $\hat{\beta}_\infty \in (0, \infty)$, we obtain

$$\hat{\alpha}_\infty = \lim_{K\to\infty} \hat{\alpha}_K = -1 \qquad (16)$$

We can now obtain our limiting AMPM estimates as $K \to \infty$. We first numerically solve (12) for $\hat{\beta}_\infty$ and then obtain $\hat{\lambda}_{B,\infty}$ from (15). From (16), the value of $\hat{\alpha}_\infty$ is -1. To go from the finite K AMPM estimate to the associated limiting estimate, we first consider $h_K(t)$ given by (3) in Annex 2, where we have suppressed the subscript K. Motivated by (3), we define

$$\hat{h}_K(t) \triangleq \frac{K\hat{\beta}_K(\hat{\alpha}_K+1)}{(1+\hat{\beta}_K t)^{\hat{\alpha}_K+2}} \qquad (17)$$

Then

$$\hat{h}_\infty(t) \triangleq \lim_{K\to\infty} \hat{h}_K(t) = \frac{\hat{\lambda}_{B,\infty}}{1 + \hat{\beta}_\infty t} \qquad (18)$$

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From (2) in Annex 2, we define

$$\hat{\mu}_K(t) \triangleq K\left[1-\left(1+\hat{\beta}_K t\right)^{-(\hat{\alpha}_K+1)}\right] \tag{19}$$

We can obtain $\hat{\mu}_\infty(t)$ more readily from (18) than from (19).

$$\hat{\mu}_\infty(t) \triangleq \lim_{K\to\infty}\hat{\mu}_K(t) = \lim_{K\to\infty}\int_0^t \hat{h}_K(x)dx = \int_0^t \lim_{K\to\infty}\hat{h}_K(x)\,dx$$

$$= \int_0^t \frac{\hat{\lambda}_{B,\infty}dx}{(1+\hat{\beta}_\infty x)} = \frac{\hat{\lambda}_{B,\infty}}{\hat{\beta}_\infty}\ln\left(1+\hat{\beta}_\infty t\right) \tag{20}$$

From (20), we can see that Equation (15) simply says

$$\hat{\mu}_\infty(T) = m \tag{21}$$

In accordance with (5) in Annex 2, we define

$$\hat{\theta}_K(t) \triangleq 1-\left(1+\hat{\beta}_K t\right)^{-(\hat{\alpha}_K+2)} \tag{22}$$

Then

$$\hat{\theta}_\infty(t) \triangleq \lim_{K\to\infty}\hat{\theta}_K = 1-\left(1+\hat{\beta}_\infty t\right)^{-1} = \frac{\hat{\beta}_\infty t}{1+\hat{\beta}_\infty t} \tag{23}$$

Finally, from (4) in Annex 2, we define

$$\hat{\rho}_K(t) \triangleq \hat{\lambda}_A + (1-\mu_d^*)K\hat{\beta}_K(\hat{\alpha}_K+1) + \mu_d^*\frac{K\hat{\beta}_K(\hat{\alpha}_K+1)}{\left(1+\hat{\beta}_K t\right)^{\hat{\alpha}_K}} \tag{24}$$

From (24) we have

$$\hat{\rho}_\infty \triangleq \lim_{K\to\infty}\hat{\rho}_K(t) = \hat{\lambda}_A + (1-\mu_d^*)\hat{\lambda}_{B,\infty} + \mu_d^*\left(\frac{\hat{\lambda}_{B,\infty}}{1+\hat{\beta}_\infty t}\right) \tag{25}$$

Recall in Section 4.4.4 we showed our finite K AMPM converged to a NHPP in the sense that the process $\{X_\infty(t), 0 \le t < \infty\}$ as K$\to\infty$. We also noted $\{X_\infty(t), 0 \le t < \infty\}$ has the mean value function μ_∞ given in 4.4.4. By so doing, one can show that these estimators are identical to the limiting AMPM estimators.

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Appendix H Bibliography

AMSAA. (13 February 1981). *MIL-HDBK-189, Reliability Growth Management.* APG.
Buele, D. M. (2000). *Engineering Design of Systems - Models and Methods.* John Wiley & Sons.
Crow, L. H. (March 1983). *AMSAA Discrete Reliability Growth Model.* APG: Methodology Office Note 1-83.
Crow, L. H. (2004). An Extended Reliability Growth Model for Managing and Assessing Corrective Actions. *Proceedings of RAMS 2004*, (pp. 73-80).
Crow, L. H. (June 1977). *Confidence Interval Procedures for Reliability Growth Analysis.* APG: AMSAA TR-197.
Crow, L. H. (1974). *Reliabilty Analysis for Complex Repairable Systems.* APG: AMSAA TR-138.
Ellner, P. M., & Hall, B. J. (March 2006). *Planning Model Based on Projection Methodology (PM2).* APG: AMSAA TR2006-09.
Ellner, P. M., & Mioduski, R. (August 1992). *Operating Characteristic Analysis for Reliability Growth Programs.* APG: AMSAA TR-524.
Ellner, P. M., & Trapnell, B. (1990). AMSAA Reliability Growth Data Study. *AMSAA Interim Note IN-R-184* .
Ellner, P. M., Broemm, W. J., & Woodworth, W. J. (September 2000). *AMSAA Reliability Growth Guide.* APG: AMSAA.
Ellner, P. M., Wald, L., & J., W. (1998). A Parametric Emprirical Bayes Approach to Reliability Projection. *Workshop on Reliability Growth Modeling: Objectives, Expectations and Approaches.* The Center for Reliability Engineering.
Hall, B. J. (11 November 2008 (Version 1)). *Reliability Growth Planning for Discrete Systems.* APG: ATEC Technical Note.
Hall, J. B. (2008a). Methods for Evaluating Reliability Growth Programs of Discrete Systems. *PhD Dissertation* . University of MD, College Park.
Hall, J. B., & Mosleh, A. (2008b). An Analytical Framework for Reliability Growth of One-Shot Systems. *vol. 93* , 1, 751-1, 760.
McCarthy, M., Mortin, D., Ellner, P., & Querido, D. (September 1994). *Developing Subsystem Reliability Growth Program Using Subsystem Reliability Growth Planning Model (SSPLAN).* APG: AMSAA TR-555.
Musa, J. (1975). A Theory of Software Reliability and Application. *IEEE Transactions on Software Engineering* , pp. vol. SE-1, No. 3.
Musa, J., & Okumoto, K. (1984). A Logarithmic Poisson Execution Time Model for Software Reliability Measurement. *7th International Conference on Software Engineering*, (pp. 230-238).
Musa, J., Iannino, A., & Okumoto, K. (1987). *Software Reliability Measurement, Prediction Application.* McGraw-Hillbook Company.
Rosner, N. (1961). System Analysis - Nonlinear Estimation Techniques. *National Symposium on Reliability and Quality Control* (pp. 203-207). New York: IRE.

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Concluding Material

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Review Activities

Army-CR

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